Ответы и официальные разборы

Том 3. Официальные оригиналы

Оригинальная вёрстка сохранена вместе с формулами, рисунками и авторскими пояснениями. В сборниках могут встречаться и условия, и ответы, и критерии оценивания.

Соответствие отдельной задаче проверяйте по олимпиаде, году, классу, номеру и варианту. Если этих данных недостаточно, документ не объявляется точным решением конкретной задачи.

Оглавление PDF содержит начала документов. Ссылки на источники и номера страниц находятся в каталоге библиотеки.

Национальная олимпиада по анализу данных DANO. 1 этап (вариант 3)

В заданиях с 1 по 9 необходимо выбрать 1 правильный вариант ответа. Каждое задание оценивается в 2 балла.

1. В стране A проживает N > 1 человек. Каждый из N–1 жителей зарабатывает 80 тыс д.е. в месяц, а один человек— 120 тыс д.е. Чему должно быть равно N, чтобы медианный ежемесячный доход в стране был равен среднему?
   1) 80
   2) 120
   3) 200
   4) 280
   5) Нет верного ответа

2. Библиотекари северного читального зала Величайшей библиотеки считают, что именно в этом зале чаще развивается аллергия на пыль. Они заметили, что среди 40 получивших аллергию на пыль 30 работали в северном зале. Руководитель библиотеки решил рассмотреть жалобы библиотекарей северного зала и посмотрел на всю статистику: северный зал и южный зал. Какой вывод о влиянии работы в северном зале на заболеваемость аллергией можно сделать на основе данных, если известно, что распределение людей в определенный зал случайно?

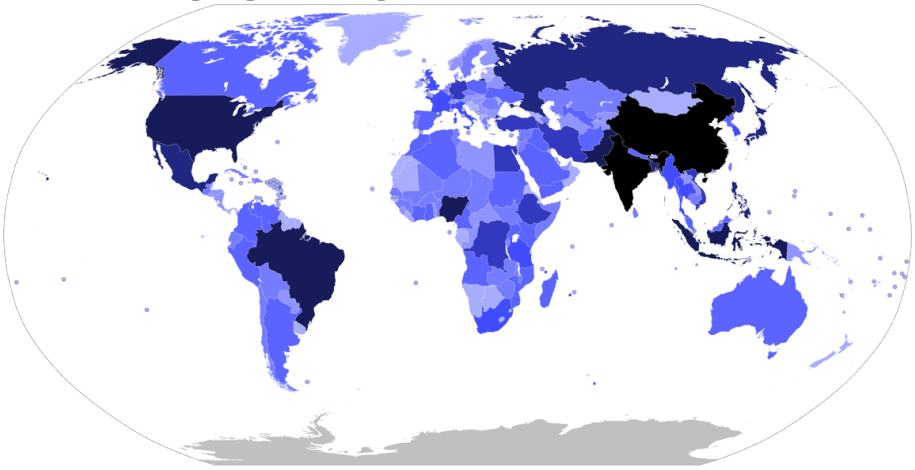
	Заболели	Не заболели
Северный зал	30	45
Южный зал	10	15

   1) Работа в северном зале увеличивает вероятность заболеть аллергией на пыль
   2) Работа в северном зале снижает вероятность заболеть аллергией на пыль
   3) Работа в северном зале не влияет на вероятность заболеть аллергией на пыль
   4) Доля заболевших среди библиотекарей в северном зале больше, чем среди библиотекарей южного зала

3. Средняя производительность (кол-во решенных задач) разработчиков одной крупной IT компании повысилась с переходом на дистанционный формат работы. Какой вывод о производительности из этого следует?
   1) Хотя бы один разработчик компании стал производительнее
   2) Все разработчики компании стали производительнее
   3) Хотя бы один разработчик компании стал менее производительным.
   4) Никакой из указанных в других вариантах выводов сделать нельзя

4. В местный шахматный клуб ходят 16 человек. Преподаватель в этом клубе заметила, что за 2 года средний возраст 8 самых младших участников вырос на X%, а старших 8 — на Y%. Коллектив за 2 года не изменялся. Разница в возрасте между самым старшим и самым младшим составляет 30 лет. Как соотносятся X и Y?
   1) X больше Y
   2) X меньше Y
   3) X в точности равняется Y
   4) Нет верных утверждений среди других вариантов

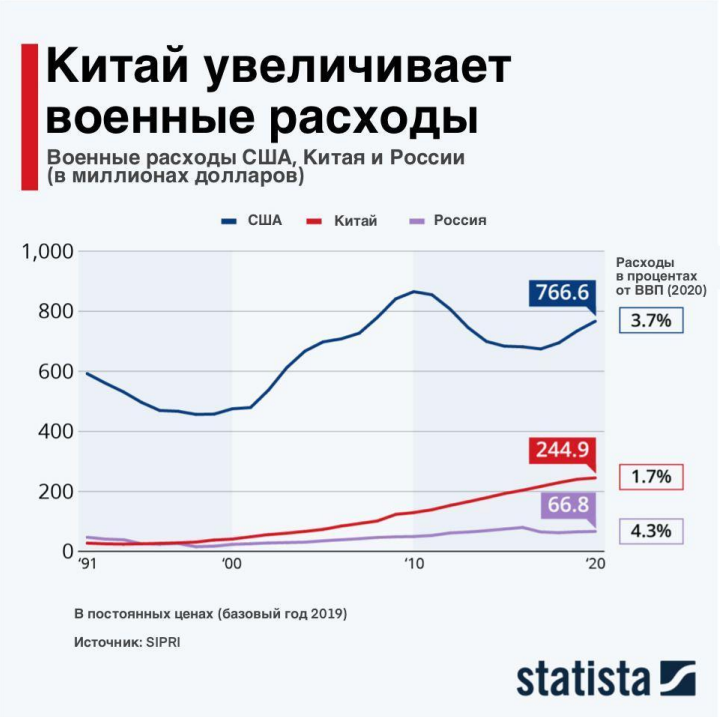
5. Перед вами карта мира, на которой цветом показан определенный показатель. Чем темнее цвет, тем выше уровень данного показателя; чем светлее — тем ниже уровень данного показателя. Например, в Нигерии и Индонезии этот показатель высокий, а в Мавритании и Парагвае — низкий. По какому показателю раскрашены страны?



   1) Количество змей на 1 квадратный километр
   2) Добыча алмазов в денежном выражении
   3) Численность населения
   4) Число IT компаний

6. Работодатель хочет провести опрос и выяснить, как наличие фруктов и печенек на кухне в офисе влияет на продуктивность работников. Какая из выборок будет наиболее репрезентативной (при одинаковом количестве респондентов)?
   1) Случайные работники, находящиеся на кухне офиса
   2) Случайно выбранные x% сотрудников офиса
   3) Работники, чьи рабочие места находятся ближе всего к кухне офиса
   4) Работники, придерживающиеся строгой диеты

7. На графике ниже отображены реальные (как и все остальные показатели, в ценах 2019 года) военные расходы США, Китая и России за 1991–2020 годы, также указаны доли военных расходов от ВВП стран в 2020 году.



Выберите верное утверждение, следующее из графика:

1) Минимальная величина военных расходов Китая наблюдалась в 1999 году
2) В 2020 году ВВП США был больше ВВП России.
3) На протяжении всего рассматриваемого периода США тратил на военные расходы меньше, чем Россия
4) В 2010 ВВП Китая был больше 20 триллионов долларов
5) Нет верного ответа среди других вариантов

8. Ученик Арсений переходит в 10 класс и выбирает новую школу. Он нашел интервью учителя и ученика об одной школе. В интервью учитель утверждает, что сейчас в среднем в 10-х классах по 18 человек в классе. Ученик же, зная, что вероятность оказаться в определенном классе пропорциональна количеству учеников в нем, утверждает, что математическое ожидание количества одноклассников у поступающего будет 21. Такая ситуация

1) Невозможна в любом случае
2) Возможна, если есть 4 класса с 10, 12, 20 и 30 учениками соответственно
3) Возможна, если есть 2 класса с 15 и 21 учениками соответственно
4) Возможна, если есть 3 класса с 9, 18 и 27 учениками соответственно

9. В городе М от станции А до станции Б целый день с разными интервалами ходят трамваи, других остановок в городе нет. Исследователь Вася решил измерить средний интервал между трамваями интересным способом: он целый день стоял у остановки Б и опрашивал случайные 25% пассажиров (для

простоты считайте, что количество пассажиров в трамвае всегда кратно 4), выходящих из каждого прибывшего трамвая, о том, сколько они ждали свой трамвай. Затем он посчитал среднее по всем полученным данным и умножил на 2. Будем считать, что пассажиры приходят равномерно по 1 человеку за 15 секунд, поэтому количество пассажиров в трамвае прямо пропорционально времени ожидания. Каким получился результат Васи по отношению к реальному среднему интервалу между двумя трамваями?

1) Результат Васи равен реальному среднему
2) Результат Васи завышен по отношению к реальному среднему
3) Результат Васи занижен по отношению к реальному среднему
4) Результат Васи может быть как завышен, так и занижен

В заданиях с 10 по 22 необходимо выбрать один или несколько правильных вариантов ответа. Каждое задание оценивается в 5 баллов.

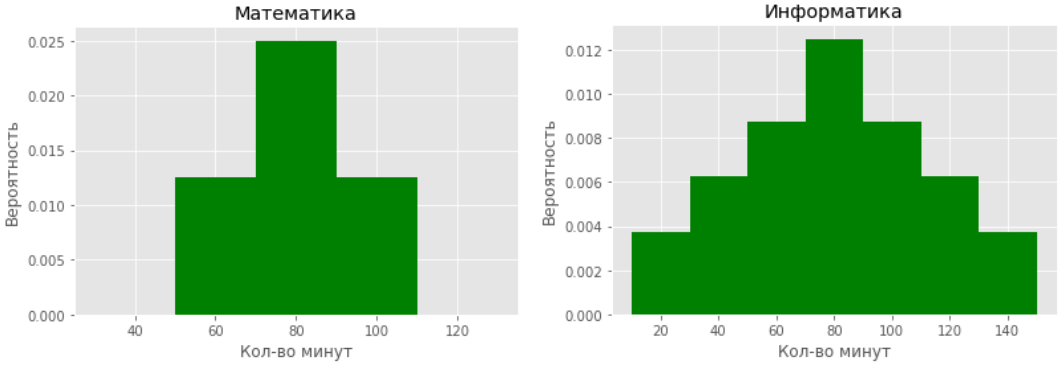
10. Выберите все верные утверждения:
    1) Медиана выборки не может быть равна среднему значению этой выборки.
    2) У выборки может быть несколько мод.
    3) У выборки может быть только одно срежнее значение.
    4) Для выборок работает правило «невозможной тройки», из которого следует, что мода, медиана и среднее значение выборки не могут быть одновременно равны друг другу.
    5) Если в выборку {16; 1; 5; 11; 0; 3} добавить число 4, то медиана выборки не изменится

11. В таблице представлено количество учеников, посмотревших лекции по анализу данных по каждой теме, и количество учеников, решивших задачи из листочков. Решать листочки можно только после просмотра лекций. Какие выводы не противоречат данным в таблице?

ТЕМЫ	Видео	Листочки
Исследовательский вопрос	388	178
Анкетирование	236	156
Статистика	218	81
Визуализация	130	83

1) Число уникальных учеников, смотревших видео, от 388 до 972
2) По теме «Статистика» доля учеников, решавших листочки после просмотра видео по этой теме, ниже, чем аналогичная доля по теме «Визуализация»
3) Между количеством посмотревших видео и количеством решавших листки наблюдается положительная взаимосвязь (корреляция)
4) Нет верного ответа среди остальных вариантов

12. На графиках представлено распределение количества минут, за которые Рита делает домашнее задание по математике и информатике.



Выберите все верные утверждения про эти распределения

1) В среднем домашнее задание по математике Рита делает быстрее, чем по информатике
2) Дисперсия времени на домашнее задание по информатике больше, чем по математике
3) Медианное время на домашнее задание по математике меньше, чем на домашнее задание по информатике
4) Если на выполнение домашнего задания есть 40 минут, то домашнее задание по информатике будет выполнено с большей вероятностью, чем по математике
5) Если домашнее задание необходимо выполнить за 120 минут, то домашнее задание по математике будет выполнено с большей вероятностью, чем по информатике

13. Коля скрывает от мамы свою оценку за недавно написанный тест. Оценка может составлять любое целое число от 1 до 100. Мама Коли знает, что среднее за прошлые три теста равно 50, при этом мода этих оценок была равна 35. Коля подсказал маме, что при добавлении новой оценки в выборку оценок Коли за тест медиана выборки не изменится. Какую оценку из предложенных мог получить Коля за тест?

1) 34
2) 48
3) 9
4) 22
5) 57

14. Университет ЭШВ практикует программу обмена с иностранными учебными заведениями. За рубеж ЭШВ отправляет лишь своих самых сильных студентов (с самым высоким уровнем образования среди всех). Но иностранные вузы, наоборот, отправляют студентов с самым низким уровнем образования. Предположим, осуществился обмен студента ЭШВ на студента из зарубежного университета. Как в таком случае могли поменяться средние уровни образования в ЭШВ и иностранном вузе, соответственно?

1) Увеличиться, увеличиться
2) Уменьшиться, уменьшиться
3) Увеличиться, уменьшиться

4) Уменьшиться, увеличиться
5) Не измениться, не измениться

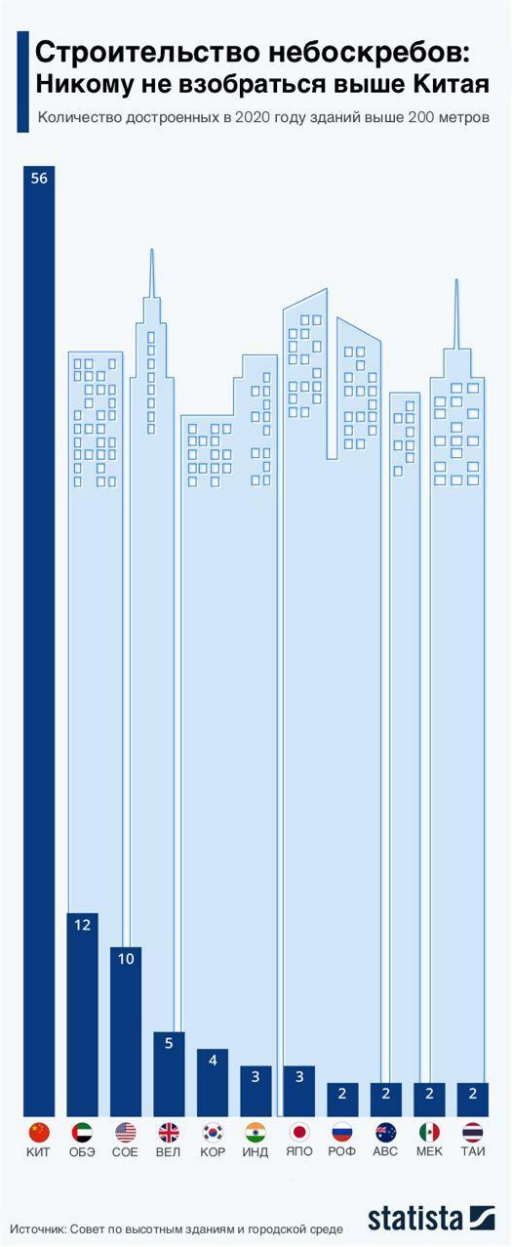
15. Перед Вами диаграмма, показывающая долю жителей, имеющих антитела к коронавирусу, для четырёх частей Соединенного королевства по состоянию на 17.05.2021.



Выберите все верные утверждения, о которых можно сделать вывод из графика:

1) В Англии проживает меньше людей с антителами, чем в Шотландии
2) Около 78% жителей Соединенного королевства имеют антитела
3) В среднем на каждые 500 человек в Англии приходится на 38–39 жителей с антителами больше, чем в Шотландии
4) Доля населения с антителами в Соединенном королевстве может составлять 72,6% и меньше
5) Нет верных утверждений среди других вариантов

16. Перед Вами диаграмма, отображающая количество зданий высотой более 200 метров (небоскребов), построенных в 2020 году в различных странах мира. Ира знает приведенную в задаче статистику по странам, а Катя и Маша знают только среднее и медианное значения. Ира предложила девушкам игру: она называет каждую из 11 приведенных стран, а Маша и Катя угадывают, сколько там небоскребов. Та, кто назовет число более близкое к настоящему, получает 1 балл. Маша решила всегда называть медиану, а Катя — среднее, округленное до целого числа по правилам арифметики.

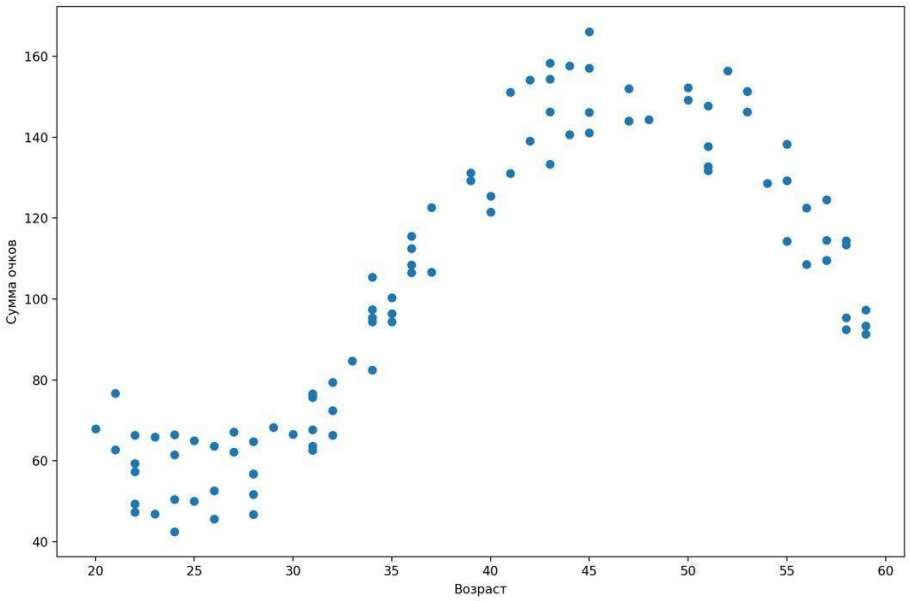


Выберите все верные утверждения, следующие из графика:

1) В такой игре Маша наберет больше баллов, чем Катя.

2) Из всех представленных в выборке стран самое низкое построенное здание находится в Тайланде
3) Всего в Китае и США есть 66 зданий выше 200 м
4) Если бы в США в 2020 году построили 25 небоскребов, то число построенных небоскребов в Китае было бы меньше суммарного количества построенных небоскребов во всех других странах, представленных в выборке
5) Округленное до целого числа среднее количество построенных в 2020 году небоскребов в странах равно 9

17. Исследователь Данил любит играть в боулинг. Но больше, чем играть в боулинг, Данил любит собирать статистику, поэтому в один день он собрал статистику, как другие играют в боулинг. Данил нарисовал диаграмму рассеяния в координатах (возраст, сумма очков), где каждая точка — показатели одного человека в этот день.



Выберите все верные утверждения

1) С увеличением возраста положительная взаимосвязь между суммой набранных баллов и возрастом сменяется отрицательной
2) Количество набранных баллов можно однозначно угадать, зная возраст
3) Возраст человека можно однозначно угадать, зная количество набранных баллов
4) Нет верных утверждений среди остальных вариантов

18. Мальчик Вова очень любит играть в компьютерные игры. Однако его мама считает, что это снижает его способности в целом, в том числе к программированию. Тогда Вова решил провести эксперимент: в течение недели каждый день после 2 часов игры Вова решал 10 задачек по программированию. Оказалось, что с каждым днем Вова решал задачки по программированию все

быстрее. Следующие 2 недели Вова только играл в игры, а затем еще в течение недели повторял эксперимент. Снова выяснилось, что с каждым днем Вова решал задачки все быстрее:

1) Время решения задачек Вовой в самый последний день эксперимента больше, чем в самый первый день
2) Время решения задачек Вовой в самый последний день эксперимента меньше, чем в самый первый день
3) Время решения задачек Вовой в самый последний день эксперимента может быть как больше, так и меньше, чем в самый первый день
4) Если Вова продолжит играть в компьютерные игры, то он будет решать задачи по программированию все быстрее
5) Нет верного ответа среди остальных вариантов

19. Из всех обращений в службу поддержки маркетплейса М 90% обращений составляют жалобы на доставку. Выберите выводы, которые **НЕ** следуют из этого факта.
    1) Большая часть обращений приходится на жалобы на доставку
    2) Большинство пользователей маркетплейса М жалуются на доставку
    3) Доставка в маркетплейсе М работает хуже, чем в других маркетплейсах
    4) Маркетплейсу стоит снизить время доставки, чтобы снизить число обращений в службу поддержки
    5) Не все пользователи обращаются в службу поддержки

20. Перед вами четыре обнаруженные корреляции между парами признаков. Выберите все случаи, когда корреляция, скорее всего, возникает из-за прямой причинно-следственной связи между признаками:
    1) Чем больше потребление органической еды, тем с большей вероятностью развивается аутизм
    2) Чем выше импорт лимонов из Мексики, тем больше количество смертей в автомобильных катастрофах
    3) Чем выше потребление маргарина, тем больше разводов на 10000 человек
    4) Чем выше образование, тем выше заработная плата сотрудника

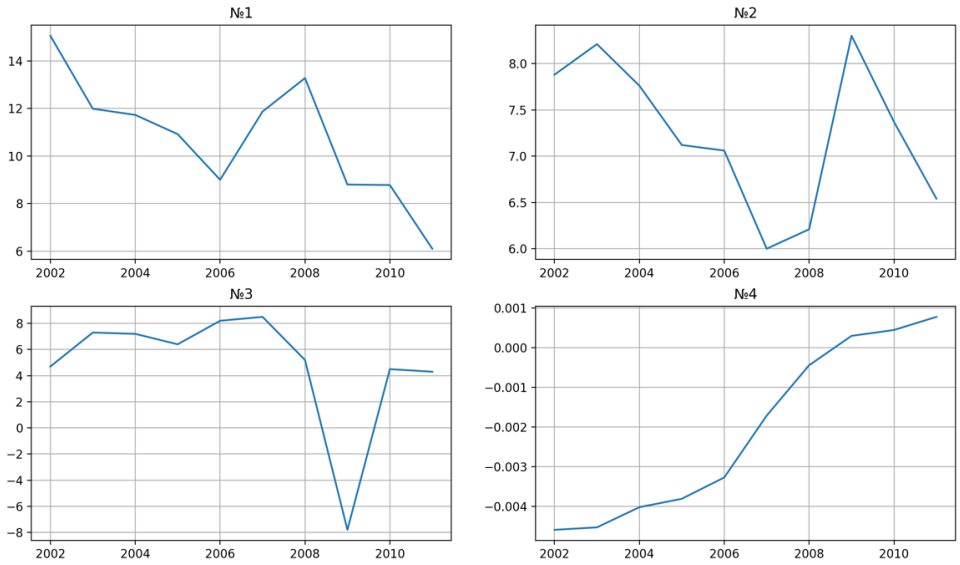
21. В одной из школ в Конго дети, которые воспитываются мусульманских семьях, учатся в среднем лучше тех, кто воспитывается в христианских. На какие генеральные совокупности можно распространить этот вывод?
    1) Городская школа в Конго
    2) Конго
    3) Африка
    4) Весь мир
    5) Ни на какую из них

22. Студенты проводят исследование, пытаясь ответить на вопрос: «Как прогулка в парке во время просмотра онлайн лекций влияет на успеваемость?» Какие из механизмов могут описывать гипотезу, выдвигаемую в качестве ответа на него?
    1) При прогулке в парке на свежем воздухе органы человека в том числе мозг насыщаются кислородом, поэтому успеваемость увеличивается

2) Во время прогулки теряется концентрация, внимание рассеивается, поэтому успеваемость снижается
3) Прогулка в парке никак не влияет на концентрацию, поэтому успеваемость не изменяется
4) Просмотр дополнительных лекций по время прогулок в парке повышает успеваемость
5) Нет верного ответа среди других вариантов

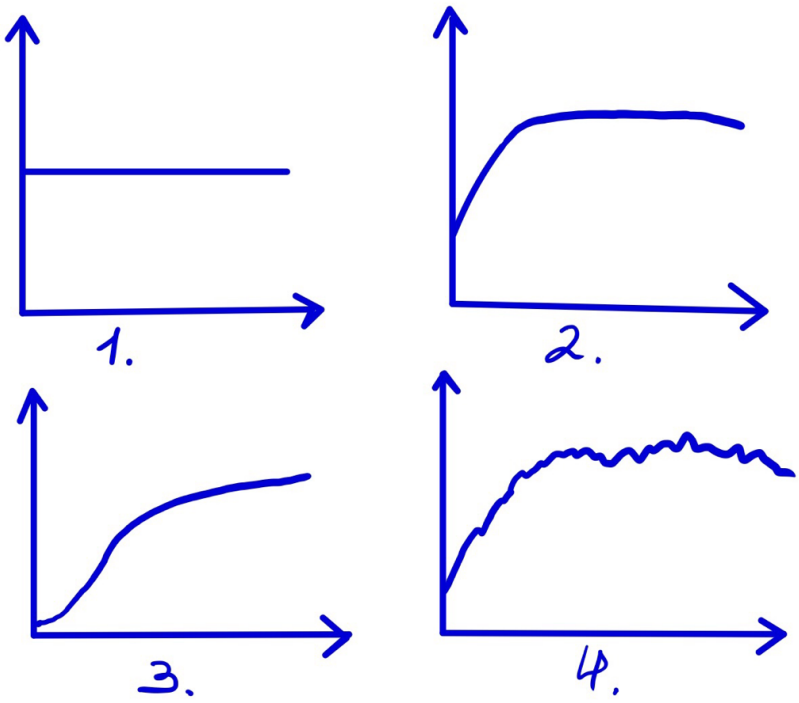
Открытые вопросы. Каждое задание оценивается в 5 баллов.

23. Перед Вами четыре графика, отражающих уровень безработицы (А), а также годовые темпы прироста ВВП (Б), населения (В) и уровня цен (Г) в России за 2002–2011 года (все показатели измерялись в %). Соотнесите показатели с соответствующими графиками их динамики и выпишите верный порядок букв, обозначающих показатели (пример ответа: АБВГ; такой ответ означает, что графику 1 ставится в соответствие показатель А, графику 2 — показатель Б и



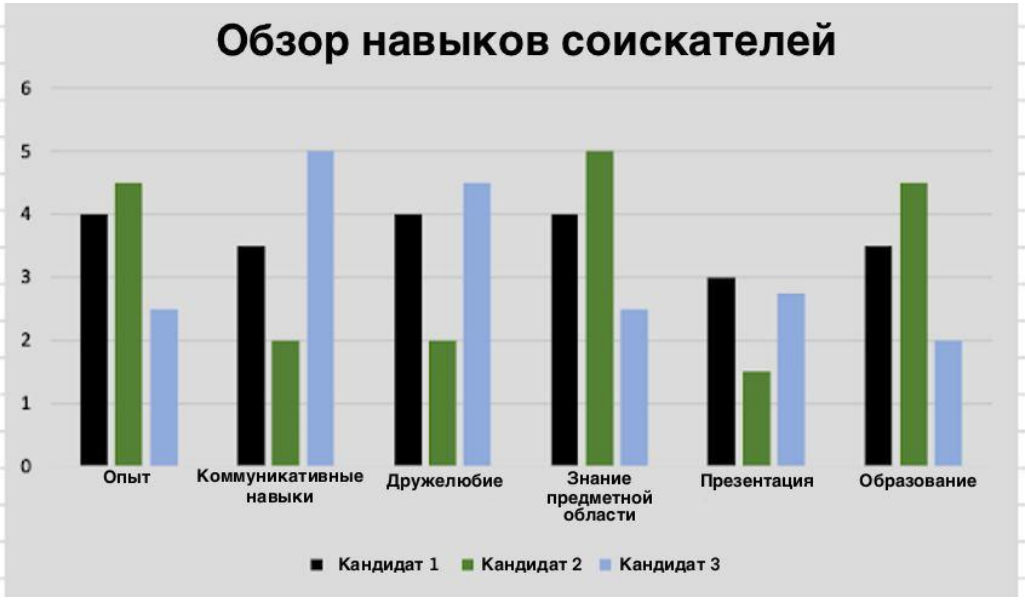
т.д.).
Ответ: ГАБВ

24. Исследователь Вася провёл четыре исследования о зависимости различных показателей от возраста, на основе которых от руки нарисовал графики найденных им функций для одного человека. Но забыл подписать графики и теперь не знает, какой рисунок относится к какому исследованию. Взгляните на графики, а затем соотнесите исследования с номерами соответствующих визуализаций. Названия исследований представлены в виде пары «возраст — признак зависимый», на графиках возраст всегда отображаются по оси абсцисс.



- «Возраст — длина позвоночника»: 2
- «Возраст — уровень IQ»: 3
- «Возраст — группа крови»: 1
- «Возраст — число подписчиков в социальных сетях»: 4

25. Работодатель занимается подбором кандидатов на вакансию, где дружелюбие в два раза важнее, чем каждая из характеристик: опыт, коммуникативные навыки и образование. Остальные навыки для вакансии не важны. У работодателя есть 3 кандидата, чьи способности отражены на графиках ниже. Работодатель ранжирует кандидатов, используя сумму всех баллов с учетом важности каждой характеристики, от большей суммы к меньшей. Приведите в ответе это ранжирование кандидатов (сначала самый привлекательный кандидат).



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Национальная олимпиада по анализу данных DANO. 2 этап (вариант 1)

Во всех заданиях с выбором ответа может быть один или несколько правильных вариантов ответа.

Задания с 1 по 3 оцениваются в 2 балла.

Вопрос 1

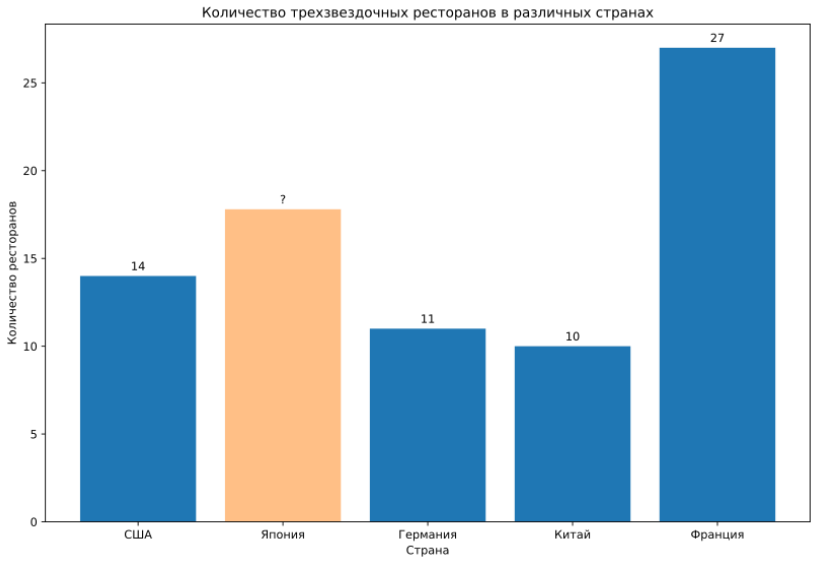
Как известно, Израиль является одной из лидирующих стран по темпам вакцинации. По данным на июнь 2021 г. в стране вакцинировано 60% граждан (85% взрослого населения). Однако среди заразившихся в этом же месяце (июне 2021 года), как признали власти Израиля, примерно половина была уже вакцинирована. **Что можно сказать об эффективности вакцины на основании этих данных?**

1) Данные не свидетельствуют об эффективности вакцины, т. к. вероятность заразиться составляет 50%, независимо от того, вакцинировался человек или нет
2) Данные не свидетельствуют об эффективности вакцины, т. к. среди вакцинированных есть заразившиеся
3) Данные свидетельствуют об эффективности вакцины, т. к. если бы она не работала, доля вакцинированных среди заболевших была бы равна доле вакцинированных среди всего населения страны
4) Данные свидетельствуют об эффективности вакцины, т. к. вакцинированные переносят болезнь в более легкой форме

Ответ: 3

Вопрос 2

Перед Вами диаграмма, показывающая количество ресторанов с тремя звездами Мишлен в пяти странах, лидирующих по этому показателю. Некоторые данные на диаграмме пропущены:



Высота столбца Японии на диаграмме может не отражать реальные данные.

Известно, что у рассматриваемой выборки медиана равна 14, а среднее значение числа трехзвездочных ресторанов в трех странах с наибольшими количествами отличается от аналогичного среднего у трех стран с наименьшими количествами в рассматриваемой выборке на 11,(3). **Чему равно количество ресторанов с тремя звездами Мишлен в Японии?**

1) 18
2) 23
3) Нет верного ответа
4) 28
5) 9

Ответ: 4

Вопрос 3

Министерство здравоохранения пытается оценить тяжесть ситуации с эпидемией в стране, для этого они используют число больных, поступивших в больницу с подтвержденным диагнозом Covid-19. КАКУЮ ОЦЕНКУ ЗАБОЛЕВАЕМОСТИ ОНИ МОГУТ ПОЛУЧИТЬ С ПОМОЩЬЮ ТАКОЙ СТАТИСТИКИ?

A) Оценку сверху

B) Оценку снизу

C) Точную оценку

D) Нельзя ответить на основе данной информации

Ответ: B

Задания с 4 по 10 оцениваются в 5 баллов

Вопрос 4

Вы хотите изучать влияние исторического распределения школ на текущий уровень человеческого капитала, который включает в себя образование, престижность профессии, продолжительность жизни и другие факторы. **Укажите, какие из графиков способствуют поиску ответа на поставленный вопрос:**

1) Дистанция до ближайшей школы в прошлом ~ текущий уровень грамотности, по индивидам
2) Текущее количество школ в регионе ~ текущий средний уровень образования, по регионам
3) Категория занятости (работники ВУЗов, профессиональные рабочие, низкоквалифицированный труд и др.) ~ дистанция до ближайшей школы в прошлом, по индивидам
4) Уровень образования по годам в изучаемом и соседнем регионе, по регионам
5) Нет верного ответа

Ответ: 1 и 3

Вопрос 5

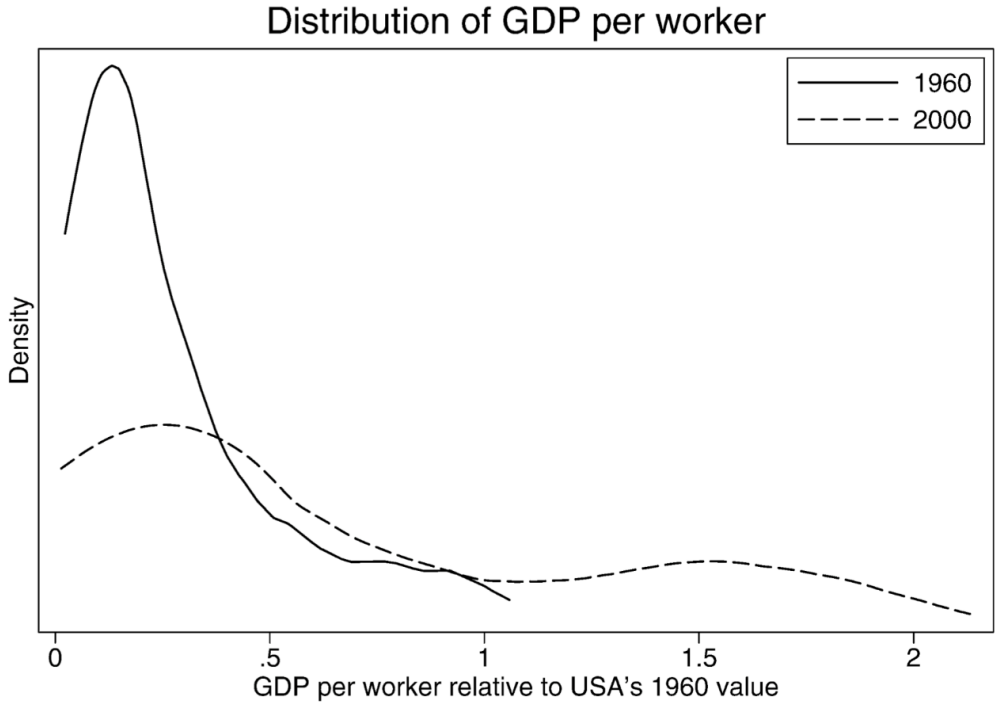
Современные технологии распознавания лиц позволяют правоохранительным органам вычислять преступников на улицах. Алгоритм разделяет людей на обычных граждан и подозреваемых. Подозреваемым считается тот, чью фотографию опознал алгоритм в базе. Предположим, что алгоритм никогда не отмечает преступников как обычных граждан. Известно, что все преступники из базы сейчас находятся на территории города и их разыскивает полиция. **Какие утверждения верны, если все население города попало на камеру за определенный период?**

1) Алгоритм может распознать как подозреваемых не меньше людей, чем есть в базе преступников.
2) Алгоритм распознает столько подозреваемых, сколько в базе преступников.
3) Алгоритм может распознать меньше подозреваемых, чем есть в базе преступников.
4) Алгоритм распознает половину граждан как подозреваемых.
5) Нет верного ответа

Ответ: 1

Вопрос 6

Группа ученых, исследующих экономический рост, посчитала за 1960 и 2000 года отношение ВВП на одного работника в разных странах к этому же показателю для США. На графике ниже представлена функция плотности данного соотношения. **Какие утверждения из приведенных ниже являются верными на основе приведенного графика?**



Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics.

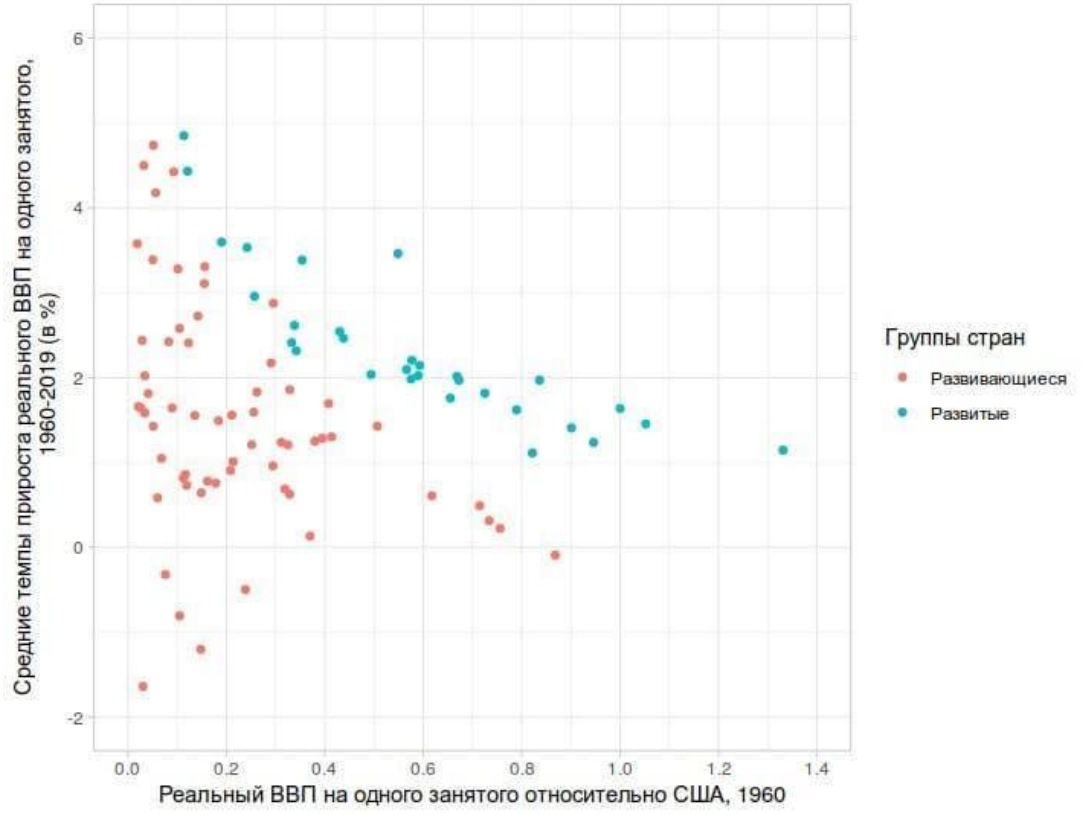
1) В 1960 году в большинстве стран ВВП на работника был ниже, чем в США
2) ВВП на работника в США не изменился за период 1960-2000гг.
3) С 1960 по 2000 год ВВП на работника в некоторых странах рос быстрее, чем США

4) В 2020 году были страны, ВВП на работника в которых превышал этот показатель для США примерно в 2 раза
5) Нет ни одного верного ответа

Ответ: 1 и 3

Вопрос 7

Конвергенция в теории экономического роста – это процесс, когда страны с меньшим уровнем доходов на душу населения догоняют страны с более высокими доходами. На графике ниже представлена диаграмма взаимосвязи темпов роста в 1960-2000 годах и реального ВВП на одного работника по отношению к ВВП на одного работника в США в 1960 году. Также на графике разными цветами выделены развитые и развивающиеся страны на основании методологии МВФ по состоянию на 2018 год.



Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics, построение авторов

Какие утверждения из приведенных ниже являются верными на основании приведенного графика?

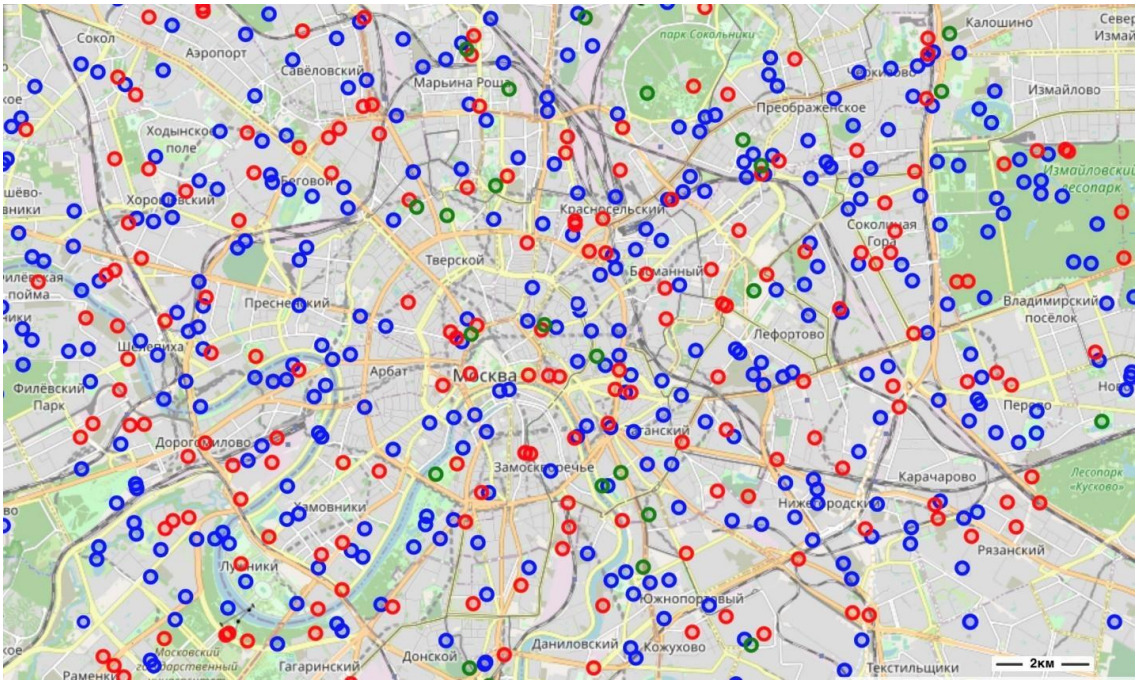
1) Наблюдается конвергенция в уровне доходов на одного работника в рассматриваемый промежуток времени, поскольку большое количество стран находятся близко друг к другу
2) В развитых странах и выпуск на работника, и темпы его роста оказываются выше, чем в развивающихся, поэтому конвергенция не наблюдается
3) В течение 60 лет выпуск на одного работника возрастает во всех странах, а значит конвергенция наблюдается

4) В большинстве стран темпы роста выпуска на одного работника ниже, чем в США, поэтому конвергенции не наблюдается
5) Нет ни одного верного утверждения

Ответ: 5

Вопрос 8

Предположим, что есть некая интернет-платформа, пункты выдачи заказов которой нанесены на карту по типам. Красные точки отображают собственные пункты выдачи заказов, синие точки отображают партнерские пункты, а зеленые – отделения почты, через которые также могут выдаваться интернет-заказы. **Какие выводы на основе данной карты можно сделать?**



1) компания в основном опирается на партнерские пункты выдачи
2) собственные пункты выдачи генерируют большую прибыль
3) отделения Почты преобладают в отдаленных районах города
4) человек, находящийся в любом месте на данной карте, сможет найти пункт выдачи заказов в радиусе 3 км
5) нет ни одного верного ответа

Ответ: 5

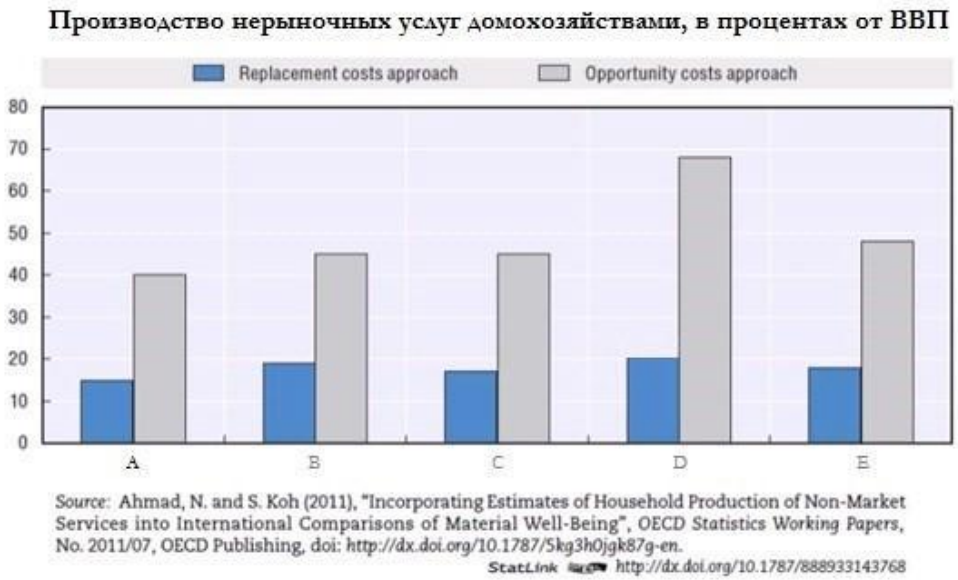
Вопрос 9

Экономисты часто задаются вопросом относительно учета в ВВП различных показателей, которые невозможно посчитать напрямую. Так, например, включили в ВВП проживание в собственной квартире исходя из цены аренды помещения на рынке. Еще один интересный и довольно значимый вклад в выпуск может внести работа людей на самих себя в рамках ведения своих ежедневных дел. Домохозяйка, которая сидит дома и занимается исключительно своими бытовыми заботами, по факту оказывает довольно большое количество услуг: уборщицы, повара, няни и т.п.

Экономисты предлагают два альтернативных способа оценить стоимость оказываемых услуг. Первый способ – replacement costs approach (подход на основе издержек замещения)

– это оценить стоимость услуг, которые домохозяйки оказывают своему домохозяйству, исходя из рыночных цен на аналогичные услуги (то есть сложить сколько стоит нанять няню, повара и горничную для выполнения тех же работ). Второй способ – opportunity costs approach (подход на основе альтернативных издержек) – это оценить, сколько бы в среднем женщина зарабатывала, выйдя на рынок труда. На графике ниже приведены оценки неучтенного ВВП из-за труда домохозяек для разных стран (A, B, C, D и E).

Какие выводы мы можем сделать, основываясь на данных графиках?



1) Самые большие зарплаты в стране D
2) В стране A домохозяйки оказывают сами себе меньше услуг, чем в стране E
3) Мы ничего не можем сказать о соотношении средних зарплат горничных со средней рыночной зарплатой в других профессиях рассматриваемой страны
4) Разница между среднерыночной зарплатой и зарплатой поваров, горничных и нянь во всех странах одинаковая
5) Нет ни одного верного ответа

Ответ: 3

Вопрос 10

Напитки, содержащие сахар, считаются вредными для здоровья, и поэтому в Беркли (в США) в начале 2015 года ввели налог на продажу этих напитков для уменьшения их потребления. Первая вертикальная линия на графике показывает момент объявления о введении налога, вторая – момент начала выплат налога продавцами. Однако, чтобы отследить чистый эффект налога, исследователи собрали данные из магазинов в соседних городах, где налог не был введен, и сравнили их с ценами на напитки в Беркли. Цены на сахар в указанный период времени не менялись. Данные представлены в виде графиков цен на напитки в Беркли и в других городах.



Какие свойства графика позволяют нам сделать вывод о воздействии налогов на цену напитков?

1) Цены на облагаемые налогом напитки до объявления о введении налога в Беркли и вне Беркли менялись примерно одинаково
2) Цены на облагаемые и необлагаемые налогом напитки в Беркли в первые месяцы после объявления о введении налога менялись по-разному
3) Цены на облагаемые налогом напитки в первые месяцы после объявления о введении налога в Беркли и вне Беркли менялись по-разному
4) Цены на облагаемые и необлагаемые налогом напитки вне Беркли в первые месяцы после объявления о введении налога менялись по-разному
5) Нет ни одного верного ответа

Ответ: 1, 2, 3, 4

Задания с 11 по 15 оцениваются в 8 баллов

Вопрос 11

В представленном файле (**Задача про футбол**) вы можете найти информацию о футбольных матчах Английской премьер-лиги сезона 2017-2018 года. **Из приведенных ниже утверждений выберите все верные.**

1) В домашних матчах команды в среднем забивают больше голов, чем в выездных матчах
2) В домашних матчах команды пропускают голов больше, чем в гостевых
3) Большинство матчей выигрывается в домашних играх
4) Максимальное количество мячей, забитых в домашних матчах, больше, чем в гостевых
5) Нет ни одного верного ответа

Ответ: 1, 3, 4

Вопрос 12

Выберите все верные утверждения на основании анализа данных, предложенных вам в файле (**Задача про машины**)

1) Более старые автомобили продаются в среднем по более низким ценам
2) В лизинг продаются в среднем более старые автомобили
3) Более старые автомобили имеют в среднем более низкий пробег
4) Автомобили с дизельным двигателем (Diesel) продаются в среднем дешевле, чем автомобили с бензиновым двигателем (Petrol)
5) Нет ни одного верного ответа

Ответ: 1

Вопрос 13

Индекс человеческого развития (ИЧР) — интегральный показатель, рассчитываемый ежегодно для межстранового сравнения и измерения уровня жизни, грамотности, образованности и долголетия как основных характеристик человеческого потенциала исследуемой территории. Он является стандартным инструментом при общем сравнении уровня жизни различных стран и регионов.

(Источник: Википедия)

Необходимо посчитать индексы человеческого развития (ИЧР) для представленных стран в файле (**Задача про ИЧР**) (Настоящая методология расчета ИЧР включает в себя некоторые поправки, которые в данном задании опущены. В связи с этим результаты могут отличаться от опубликованных значений ИЧР). Для вычисления ИЧР выполните следующие пункты:

а) Используя индикатор Ожидаемая продолжительность жизни при рождении (Life expectancy at birth), посчитайте Индекс здоровья по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

б) Посчитайте Индекс образования, используя два индикатора:

Средняя продолжительность обучения населения в годах и Ожидаемая продолжительность обучения населения, ещё получающего образование, в годах.

Для каждого индикатора вычислите свой индекс по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

в) Посчитайте Индекс доходов, используя показатель натуральный логарифм Валового национального дохода на душу населения (GNI per capita). Для его вычисления необходимо взять натуральный логарифм значений (этот показатель уже посчитан и приведен в файле с данными, Log GNI).

(Натуральный логарифм есть обратная функция для экспоненты. Функция является монотонно возрастающей. Ее часто применяют для преобразования различных экономических переменных, которые характеризуются постоянным ростом (например, для ВВП). Для его вычисления в Excel используется функция LN()).

Индекс считается по формуле:

$$\text{Индекс} = \frac{\text{значение логарифма показателя} - \text{минимальное значение логарифма показателя}}{\text{максимальное значение логарифма показателя} - \text{минимальное значение логарифма показателя}}$$

г) Рассчитайте Индекс человеческого развития, вычислив геометрическое среднее по трем индексам. Округлите значения до 3 знаков после запятой и расположите страны в порядке убывания индекса.

Выберите все верные утверждения в соответствии с Вашим решением.

1) Максимальное значение Индекса здоровья – единица
2) ИЧР равен нулю в двух странах
3) В Болгарии ИЧР выше, чем в России
4) Наибольший Индекс образования в Германии
5) Нет ни одного верного варианта ответа

Ответ: 1, 2, 3

Вопрос 14

В этом задании будет рассмотрено влияние появления нового мусоросжигательного завода на стоимость жилья в одном из городов США. Для этого мы будем использовать данные за два года (до и после строительства завода) о домах, находящихся рядом с заводом, и домах, находящихся далеко от завода. Данные вы можете найти в файле **«Задача про дома»**. Единицей наблюдения в нем является дом.

Отметим, что слухи о том, что в городе будет построен новый мусоросжигательный завод, появились после 1978 года, а строительство началось в 1981 году. Мы будем использовать данные о ценах на дома, проданные в 1978 году (период «до»), а также данные о ценах на дома, проданные в 1981 году (период «после»). Гипотеза заключается в том, что цены на дома, расположенные вблизи мусоросжигательного завода, упадут по сравнению с ценами на более удаленные дома.

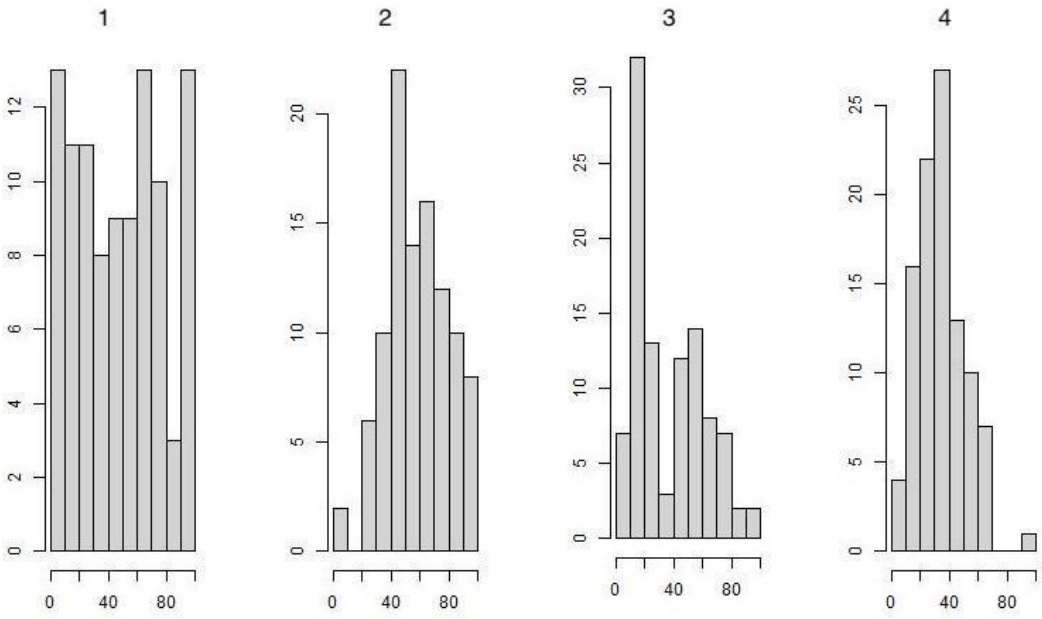
Выберите, какие из представленных ответов соответствуют данным и одновременно помогают обосновать представленную гипотезу.

1) Средняя цена на дома в городе снизилась в 1981 году по сравнению с 1978 годом
2) Средняя цена на дома, расположенные далеко от завода, выросла в 1981 году по сравнению с 1978 годом
3) В 1981 году по сравнению с 1978 годом средняя цена на дома, расположенные далеко от завода, выросла сильнее (в долл.), чем средняя цена на дома, расположенные вблизи к заводу
4) Эффект снижения средней цены на дома в связи со строительством завода составляет примерно 7 тыс.
5) Нет ни одного верного ответа

Ответ: 3

Вопрос 15

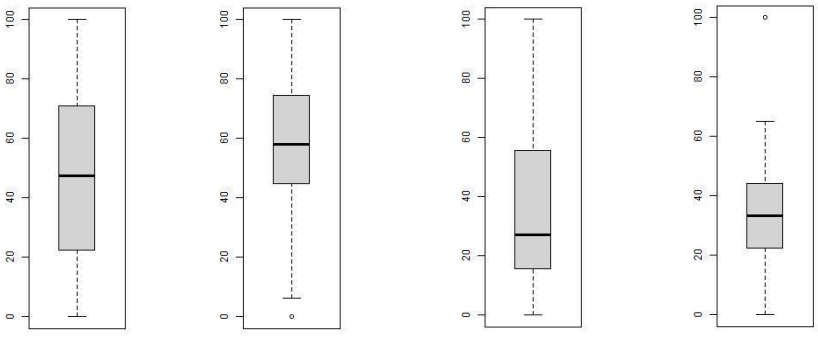
На представленных вам рисунках вы можете найти частотные диаграммы и графики box-plot.
Соотнесите разные типы диаграмм, которые отображают один и тот же набор данных.



В данном задании вы получаете 2 балла за каждое верное сочетание гистограммы и box plot. Неверное сочетание не штрафуется.

Ответ:

1) 2) 3) 4)



Вопрос 16 *(Задание оценивается в 10 баллов)*

В файле «**Задача про студентов**» вам дана таблица, где есть ученики, данные про них, и курсовые проекты, которые они выбрали по приоритетам. Если анкета заполнена

несколько раз, то учитывается последняя отправка. Если два студента хотят выбрать один и тот же проект, то в итоге на него будет записан студент с более высоким рейтингом.

Выберите верное распределение курсовых проектов между студентами, если для каждого нужно выбрать только один проект и на каждый проект может записаться не более одного студента.

В данном задании вы получаете 2 балла за каждое верное сочетание студента и курсового проекта. Неверное сочетание не штрафуется.

Ответ:

1) Современная литература
2) Программирование
3) Анализ данных
4) Микроэкономика
5) Введение в социологию

История России (лишнее)

Вопрос 17 *(Задание оценивается в 14 баллов)*

Заполните пропуски, чтобы получилось верное рассуждение на тему расчета индекса потребительских цен.

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

Инфляция обычно измеряется с помощью индекса потребительских цен. Для расчёта этого показателя берётся фиксированная(ый) [[1]], общая стоимость которой(ого) потом используется для расчёта индекса: стоимость [[5]] в текущий год сравнивается со стоимостью в [[9]] год и это изменение интерпретируется как инфляция.

Однако цены растут неравномерно и из-за эффекта замещения люди, выбирая между двумя очень похожими товарами (разными сортами яблок, к примеру), начинают покупать [[13]] тех, что медленнее дорожают. В результате, фактический состав [[16]] меняется. За счёт этого уровень инфляции, посчитанный по неизменной(ому) [[20]], оказывается [[24]] "настоящего" уровня инфляции.

Ответ:

1) Корзина

5) Корзины

9) базовый

13) больше

16) корзины

20) корзине

24) выше

Вопрос 18 *(Задание оценивается в 8 баллов)*

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. **Далее вам предлагается заполнить пропуски в тексте ниже, чтобы получить верный текст, который начинающий аналитик представляет коллегам в качестве результатов исследования.**

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

В своем исследовании мы задаемся вопросом о влиянии доходов семей на итоговые баллы по различным предметам. Основная гипотеза состоит в том, что [[1]]. Доход семьи мы будем измерять косвенным способом, для чего используем [[5]]. Проведенные нами расчеты говорят о том, что доход семьи и итоговый балл [[6]]. Данный результат [[10]] основную поставленную гипотезу.

Ответ:

1) доход семьи положительно влияет на общий балл по всем предметам

5) переменную lunch, поскольку скидка на питание предоставляется только семьям с невысоким доходо

6) связаны положительно

10) не отвергает

Задания 19 и 20 оцениваются в 1 балл

Вопрос 19

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее обеспеченных семей используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Средний суммарный балл по всем тестам среди более бедных семей составляет:

Ответ: 172,97

Вопрос 20

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее обеспеченных семей используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Средний суммарный балл по всем тестам среди более обеспеченных семей составляет:

Ответ: 199,85

*В представленном файле (**Задача про меню**) вы можете найти информацию о различных заказах, сделанных в кафе за месяц. Вас попросили сделать небольшую аналитику по имеющимся продажам. В задании 21-25 вам будет необходимо по очереди заполнить пропуски в предложенном ниже тексте. Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33). Каждое задание оценивается в 2 балла.*

Вопрос 21

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Всего за указанный период количество заказов составило:

Ответ: 1223

Вопрос 22

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

В среднем в каждом заказе (...) позиции.

Ответ: 2,51

Вопрос 23

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

В среднем в каждом заказе (ответ из предыдущего задания) позиции на общую сумму:

Ответ: 18,77

Вопрос 24

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
Всего за указанный период было заказано (...) уникальных позиций из нашего меню.

Ответ: 1365

Вопрос 25

В качестве ответа внесите в поле нужный item_name, представленный словом или словосочетанием, латинскими буквами, в той же форме, в которой оно написано в базе данных, которую вы анализируете.

Из всех видов позиций "item_name" наибольшую выручку в нашем кафе принесла:

Ответ: chicken bowl

Национальная олимпиада по анализу данных DANO. 2 этап (вариант 2)

Во всех заданиях с выбором ответа может быть один или несколько правильных вариантов ответа.

Задания с 1 по 3 оцениваются в 2 балла.

Вопрос 1

Как известно, Израиль является одной из лидирующих стран по темпам вакцинации. По данным на июнь 2021 г. в стране вакцинировано 60% граждан (85% взрослого населения). Однако среди заразившихся в этом же месяце (июне 2021 года), как признали власти Израиля, примерно половина была уже вакцинирована. **Что можно сказать об эффективности вакцины на основании этих данных?**

1) Данные не свидетельствуют об эффективности вакцины, т. к. вероятность заразиться составляет 50%, независимо от того, вакцинировался человек или нет
2) Данные не свидетельствуют об эффективности вакцины, т. к. среди вакцинированных есть заразившиеся
3) Данные свидетельствуют об эффективности вакцины, т. к. если бы она не работала, доля вакцинированных среди заболевших была бы равна доле вакцинированных среди всего населения страны
4) Данные свидетельствуют об эффективности вакцины, т. к. вакцинированные переносят болезнь в более легкой форме

Ответ: 3

Вопрос 2

Известно, что у рассматриваемой выборки медиана равна 14, а среднее значение числа трехзвездочных ресторанов в трех странах с наибольшими количествами отличается от аналогичного среднего у трех стран с наименьшими количествами в рассматриваемой выборке на 7,(3). **Чему равно количество ресторанов с тремя звездами Мишлен в Японии?**

1) 16
2) 9
3) Нет верного ответа
4) 28
5) 6
6) 6

Ответ: 1

Вопрос 3

Министерство здравоохранения пытается оценить тяжесть ситуации с эпидемией нового заболевания в стране. Эта болезнь характеризуется тем, что симптомы очень похожи на обычную простуду, однако у части людей может протекать совершенно бессимптомно. Также в данной стране все люди очень ответственны и при возникновении симптомов обращаются к врачу. Для этих целей министерство использует число всех зарегистрированных случаев простудных заболеваний. КАКУЮ ОЦЕНКУ

ЗАБОЛЕВАЕМОСТИ НОВОЙ БОЛЕЗНЬЮ ОНИ МОГУТ ПОЛУЧИТЬ С ПОМОЩЬЮ ТАКОЙ СТАТИСТИКИ?

A) Оценку сверху

B) Оценку снизу

C) Точную оценку

D) Нельзя ответить на основе данной информации

Ответ: D

Задания с 4 по 10 оцениваются в 5 баллов

Вопрос 4

Вы хотите изучать влияние исторического распределения школ на текущий уровень человеческого капитала, который включает в себя образование, престижность профессии, продолжительность жизни и другие факторы. **Укажите, какие из графиков способствуют поиску ответа на поставленный вопрос:**

1) Дистанция до ближайшей школы в прошлом ~ текущий уровень грамотности, по индивидам
2) Текущее количество школ в регионе ~ текущий средний уровень образования, по регионам
3) Категория занятости (работники ВУЗов, профессиональные рабочие, низкоквалифицированный труд и др.) ~ дистанция до ближайшей школы в прошлом, по индивидам
4) Уровень образования по годам в изучаемом и соседнем регионе, по регионам
5) Нет верного ответа

Ответ: 1 и 3

Вопрос 5

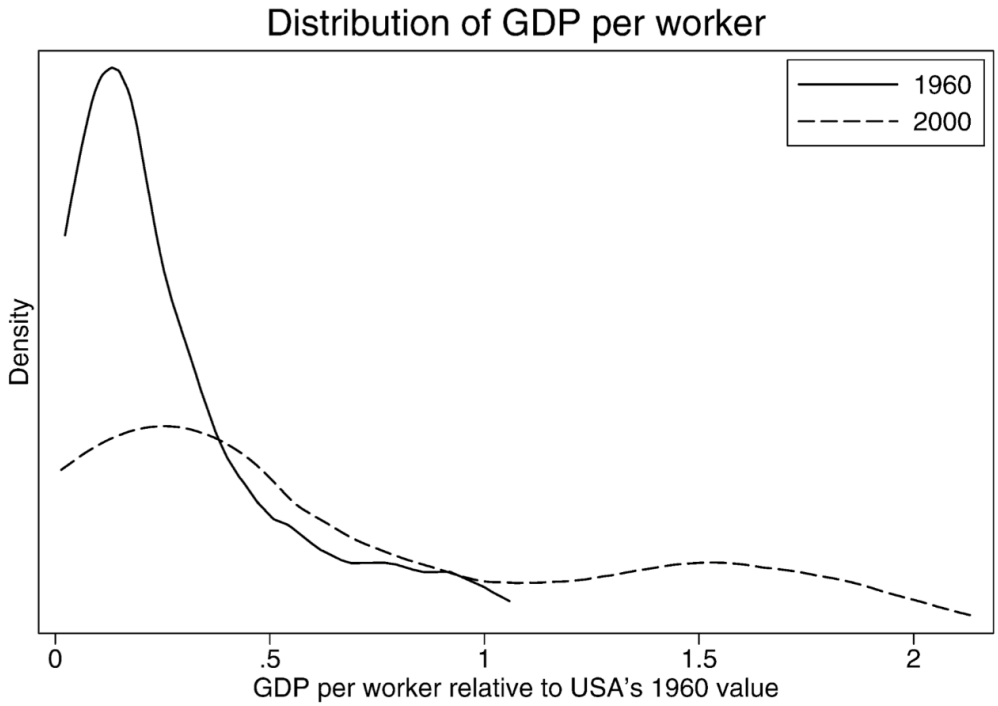
Современные технологии распознавания лиц позволяют правоохранительным органам вычислять преступников на улицах. Алгоритм разделяет людей на обычных граждан и подозреваемых. Подозреваемым считается тот, чью фотографию опознал алгоритм в базе. Предположим, что алгоритм никогда не отмечает преступников как обычных граждан. Известно, что все преступники из базы сейчас находятся на территории города и их разыскивает полиция. **Какие утверждения верны, если все население города попало на камеру за определенный период?**

1) Алгоритм может распознать как подозреваемых не меньше людей, чем есть в базе преступников.
2) Алгоритм распознает столько подозреваемых, сколько в базе преступников.
3) Алгоритм может распознать меньше подозреваемых, чем есть в базе преступников.
4) Алгоритм распознает половину граждан как подозреваемых.
5) Нет верного ответа

Ответ: 1

Вопрос 6

Группа ученых, исследующих экономический рост, посчитала за 1960 и 2000 года отношение ВВП на одного работника в разных странах к этому же показателю для США. На графике ниже представлена функция плотности данного соотношения. **Какие утверждения из приведенных ниже являются верными на основе приведенного графика?**



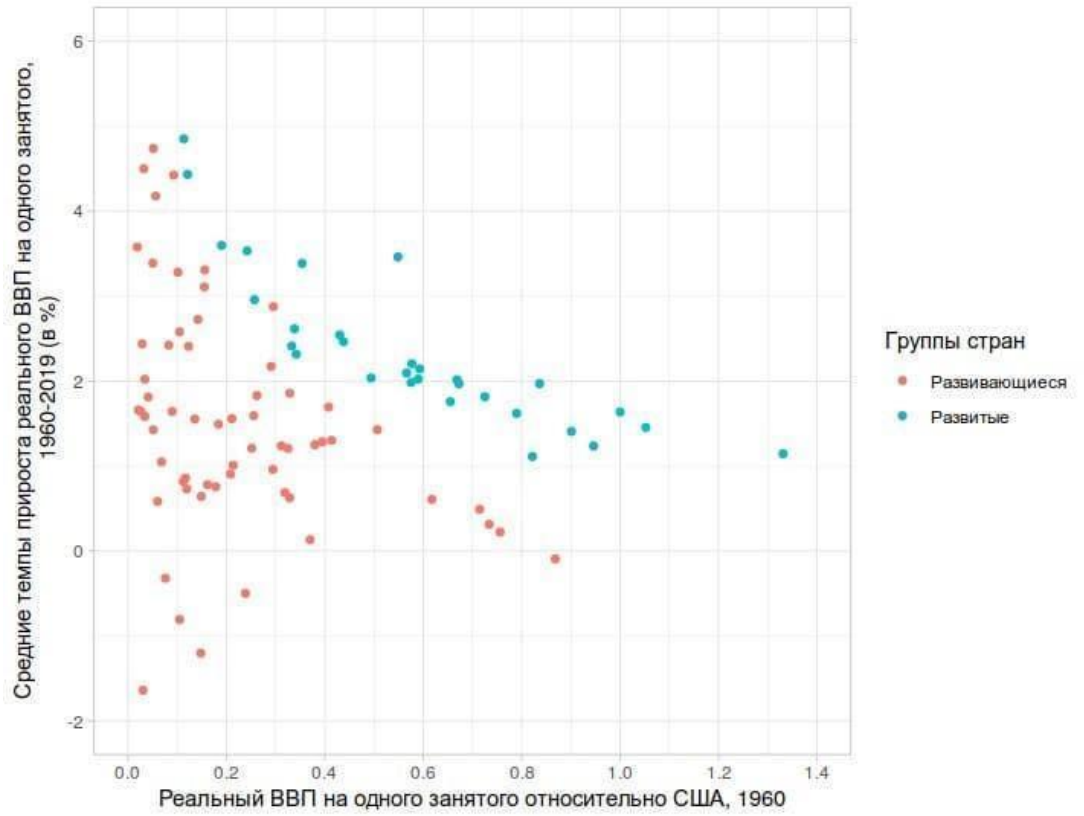
Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics.

1) В 2000 году в большинстве стран ВВП на работника был ниже, чем в США
2) ВВП на работника в США вырос за период 1960-2000гг.
3) С 1960 по 2000 год ВВП на работника во всех странах рос медленнее, чем США
4) В 1960 году были страны, ВВП на работника в которых превышал этот показатель для США больше, чем в 2 раза
5) Нет ни одного верного ответа

Ответ: 1

Вопрос 7

Конвергенция в теории экономического роста – это процесс, когда страны с меньшим уровнем доходов на душу населения догоняют страны с более высокими доходами. На графике ниже представлена диаграмма взаимосвязи темпов роста в 1960-2000 годах и реального ВВП на одного работника по отношению к ВВП на одного работника в США в 1960 году. Также на графике разными цветами выделены развитые и развивающиеся страны на основании методологии МВФ по состоянию на 2018 год.



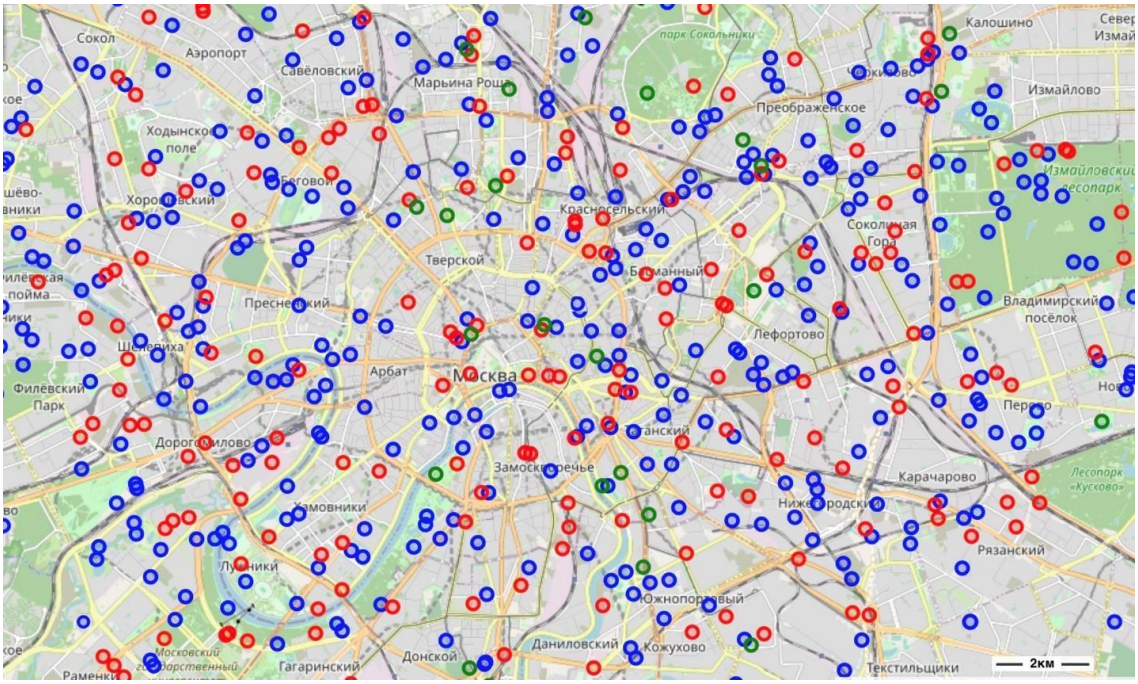
Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics, построение авторов

1) Наблюдается конвергенция в уровне доходов на одного работника в рассматриваемый промежуток времени только в развитых странах, поскольку эти точки демонстрируют отрицательную взаимосвязь показателей
2) В развивающихся странах и выпуск на работника, и темпы его роста оказываются ниже, чем в развитых, поэтому конвергенция не наблюдается
3) В течение 60 лет выпуск на одного работника возрастает не во всех странах, а значит конвергенция наблюдается
4) В большинстве развитых темпы роста выпуска на одного работника ниже, чем в США, поэтому конвергенции не наблюдается
5) Нет ни одного верного утверждения

Ответ: 1

Вопрос 8

Предположим, что есть некая интернет-платформа, пункты выдачи заказов которой нанесены на карту по типам. Красные точки отображают собственные пункты выдачи заказов, синие точки отображают партнерские пункты, а зеленые – отделения почты, через которые также могут выдаваться интернет-заказы. **Какие выводы на основе данной карты можно сделать?**



1) компания в основном опирается на собственные пункты выдачи
2) собственные пункты выдачи генерируют большую прибыль
3) отделения Почты распределены неравномерно по городу
4) человек, находящийся в любом месте на данной карте, сможет найти пункт выдачи заказов в радиусе 3 км
5) нет ни одного верного ответа

Ответ: 3

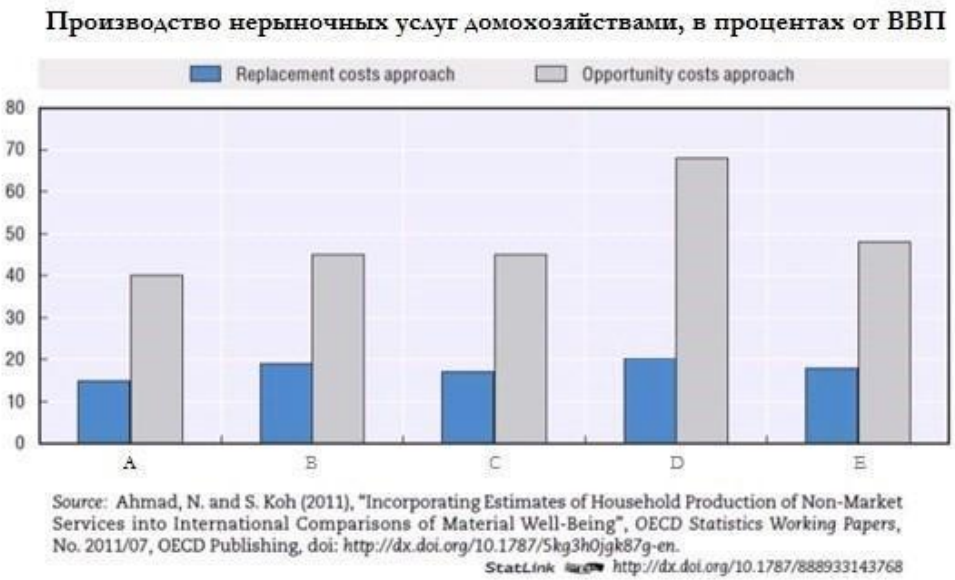
Вопрос 9

Экономисты часто задаются вопросом относительно учета в ВВП различных показателей, которые невозможно посчитать напрямую. Так, например, включили в ВВП проживание в собственной квартире исходя из цены аренды помещения на рынке. Еще один интересный и довольно значимый вклад в выпуск может внести работа людей на самих себя в рамках ведения своих ежедневных дел. Домохозяйка, которая сидит дома и занимается исключительно своими бытовыми заботами, по факту оказывает довольно большое количество услуг: уборщицы, повара, няни и т.п.

Экономисты предлагают два альтернативных способа оценить стоимость оказываемых услуг. Первый способ – replacement costs approach (подход на основе издержек замещения) – это оценить стоимость услуг, которые домохозяйки оказывают своему домохозяйству,

исходя из рыночных цен на аналогичные услуги (то есть сложить сколько стоит нанять няню, повара и горничную для выполнения тех же работ). Второй способ – opportunity costs approach (подход на основе альтернативных издержек) – это оценить, сколько бы в среднем женщина зарабатывала, выйдя на рынок труда. На графике ниже приведены оценки неучтенного ВВП из-за труда домохозяек для разных стран (A, B, C, D и E).

Какие выводы мы можем сделать, основываясь на данных графиках?



1) В стране A домохозяйки оказывают сами себе меньше услуг, чем в стране E
2) Во всех странах из приведённых на графике зарплаты нянь, горничных и поваров в среднем ниже, чем средние зарплаты в других профессиях
3) Мы ничего не можем сказать о соотношении средних рыночных зарплат в разных странах
4) Разница между среднерыночной зарплатой и зарплатой поваров, горничных и нянь больше в стране D, чем в стране A
5) Нет ни одного верного ответа

Ответ: 2 и 3

Вопрос 10

Напитки, содержащие сахар, считаются вредными для здоровья, и поэтому в Беркли (в США) в начале 2015 года ввели налог на продажу этих напитков для уменьшения их потребления. Первая вертикальная линия на графике показывает момент объявления о введении налога, вторая – момент начала выплат налога продавцами. Однако, чтобы отследить чистый эффект налога, исследователи собрали данные из магазинов в соседних городах, где налог не был введен, и сравнили их с ценами на напитки в Беркли. Цены на сахар в указанный период времени не менялись. Данные представлены в виде графиков цен на напитки в Беркли и в других городах.



Какие свойства графика позволяют нам сделать вывод о воздействии налогов на цену напитков?

1) Цены на облагаемые налогом напитки до введения налога в Беркли и вне Беркли должны меняться по-разному
2) Цены на облагаемые и необлагаемые налогом напитки в Беркли до введения налога должны меняться по-разному
3) Цены на облагаемые налогом напитки после введения налога в Беркли и вне Беркли должны меняться одинаково
4) Цены на необлагаемые налогом напитки после введения налога в Беркли и вне Беркли должны меняться одинаково
5) Нет ни одного верного ответа

Ответ: 4

Задания с 11 по 15 оцениваются в 8 баллов

Вопрос 11

В представленном файле (**Задача про футбол**) вы можете найти информацию о футбольных матчах Английской премьер-лиги сезона 2017-2018 года. **Из приведенных ниже утверждений выберите все верные.**

1) В домашних матчах команды в среднем забивают меньше голов, чем в выездных матчах
2) В выездном матче больше шансов сыграть вничью, чем выиграть матч
3) Большинство матчей выигрывается в гостевых играх
4) Вариация в количестве забитых мячей в домашних матчах выше, чем в гостевых
5) Нет ни одного верного ответа

Ответ: 4

Вопрос 12

Выберите все верные утверждения на основании анализа данных, предложенных вам в файле (**Задача про машины**).

1) Более старые автомобили продаются в среднем по более высоким ценам
2) Автомобили с большим пробегом продаются в среднем по более низким ценам
3) Более старые автомобили имеют в среднем более высокий пробег
4) Автомобили в хорошем состоянии продаются в среднем примерно на 67% дороже автомобилей в плохом состоянии
5) Нет ни одного верного ответа

Ответ: 2, 3, 4

Вопрос 13

Индекс человеческого развития (ИЧР) — интегральный показатель, рассчитываемый ежегодно для межстранового сравнения и измерения уровня жизни, грамотности, образованности и долголетия как основных характеристик человеческого потенциала исследуемой территории. Он является стандартным инструментом при общем сравнении уровня жизни различных стран и регионов.

(Источник: Википедия)

Необходимо посчитать индексы человеческого развития (ИЧР) для представленных стран в файле (**Задача про ИЧР**) (Настоящая методология расчета ИЧР включает в себя некоторые поправки, которые в данном задании опущены. В связи с этим результаты могут отличаться от опубликованных значений ИЧР). Для вычисления ИЧР выполните следующие пункты:

а) Используя индикатор Ожидаемая продолжительность жизни при рождении (Life expectancy at birth), посчитайте Индекс здоровья по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

б) Посчитайте Индекс образования, используя два индикатора:

Средняя продолжительность обучения населения в годах и Ожидаемая продолжительность обучения населения, ещё получающего образование, в годах.

Для каждого индикатора вычислите свой индекс по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

в) Посчитайте Индекс доходов, используя показатель натуральный логарифм Валового национального дохода на душу населения (GNI per capita). Для его вычисления необходимо взять натуральный логарифм значений (этот показатель уже посчитан и приведен в файле с данными, Log GNI).

(Натуральный логарифм есть обратная функция для экспоненты. Функция является монотонно возрастающей. Ее часто применяют для преобразования различных экономических переменных, которые характеризуются постоянным ростом (например, для ВВП). Для его вычисления в Excel используется функция LN()).

Индекс считается по формуле:

$$\text{Индекс} = \frac{\text{значение логарифма показателя} - \text{минимальное значение логарифма показателя}}{\text{максимальное значение логарифма показателя} - \text{минимальное значение логарифма показателя}}$$

г) Рассчитайте Индекс человеческого развития, вычислив геометрическое среднее по трем индексам. Округлите значения до 3 знаков после запятой и расположите страны в порядке убывания индекса.
Выберите все верные утверждения в соответствии с Вашим решением.

1) Максимальное значение Индекса образования – единица
2) ИЧР равен нулю в трех странах
3) В Казахстане ИЧР выше, чем в России
4) Наибольший Индекс образования в Австралии
5) Нет ни одного верного варианта ответа

Ответ: 3 и 4

Вопрос 14

В этом задании будет рассмотрено влияние появления нового мусоросжигательного завода на стоимость жилья в одном из городов США. Для этого мы будем использовать данные за два года (до и после строительства завода) о домах, находящихся рядом с заводом, и домах, находящихся далеко от завода. Данные вы можете найти в файле **«Задача про дома»**. Единицей наблюдения в нем является дом.

Отметим, что слухи о том, что в городе будет построен новый мусоросжигательный завод, появились после 1978 года, а строительство началось в 1981 году. Мы будем использовать данные о ценах на дома, проданные в 1978 году (период «до»), а также данные о ценах на дома, проданные в 1981 году (период «после»). Гипотеза заключается в том, что цены на дома, расположенные вблизи мусоросжигательного завода, упадут по сравнению с ценами на более удаленные дома.

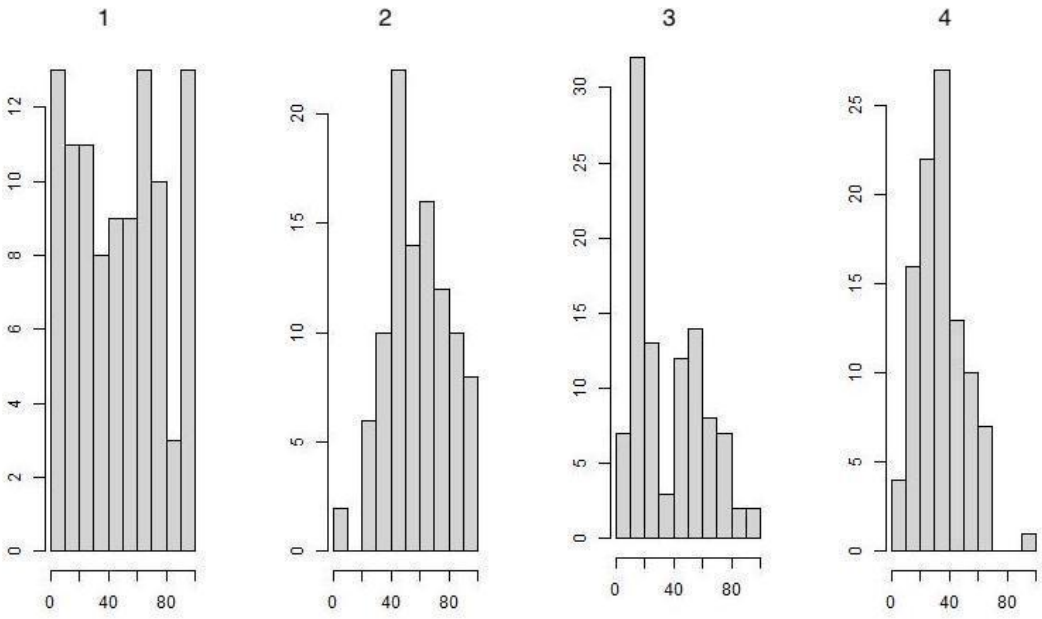
Выберите, какие из представленных ответов соответствуют данным и одновременно помогают обосновать представленную гипотезу.

1) Средняя цена на дома в 1981 году, расположенные далеко от завода, выше по сравнению с ценой домов, расположенных близко к заводу
2) Средняя цена на дома, расположенные далеко от завода, выросла в 1981 году по сравнению с 1978 годом
3) В 1981 году по сравнению с 1978 годом средняя цена на дома, расположенные вблизи к заводу, выросла сильнее, чем средняя цена на дома, расположенные далеко от завода
4) Эффект снижения средней цены на дома в связи со строительством завода составляет примерно 11 тыс.
5) Нет ни одного верного ответа

Ответ: 4

Вопрос 15

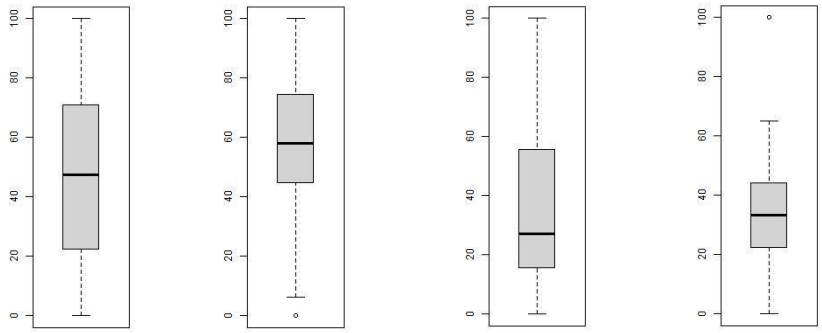
На представленных вам рисунках вы можете найти частотные диаграммы и графики box-plot.
Соотнесите разные типы диаграмм, которые отображают один и тот же набор данных.



В данном задании вы получаете 2 балла за каждое верное сочетание гистограммы и box plot. Неверное сочетание не штрафуется.

Ответ:

1) 2) 3) 4)



Вопрос 16 *(Задание оценивается в 10 баллов)*

В файле «**Задача про студентов**» вам дана таблица, где есть ученики, данные про них, и курсовые проекты, которые они выбрали по приоритетам. Если анкета заполнена

несколько раз, то учитывается последняя отправка. Если два студента хотят выбрать один и тот же проект, то в итоге на него будет записан студент с более высоким рейтингом.

Выберите верное распределение курсовых проектов между студентами, если для каждого нужно выбрать только один проект и на каждый проект может записаться не более одного студента.

В данном задании вы получаете 2 балла за каждое верное сочетание студента и курсового проекта. Неверное сочетание не штрафуется.

Ответ:

1) Современная литература
2) Анализ данных
3) Программирование
4) Микроэкономика
5) Введение в социологию
   История России (лишнее)

Вопрос 17 *(Задание оценивается в 14 баллов)*

Заполните пропуски, чтобы получилось верное рассуждение на тему расчета индекса потребительских цен.

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

Инфляция обычно измеряется с помощью индекса потребительских цен. Для расчёта этого показателя берётся фиксированная(ый) [[1]], общая стоимость которой(ого) потом используется для расчёта индекса: стоимость [[5]] в текущий год сравнивается со стоимостью в [[9]] год и это изменение интерпретируется как инфляция.

Однако цены растут неравномерно и из-за эффекта замещения люди, выбирая между двумя очень похожими товарами (разными сортами яблок, к примеру), начинают покупать [[13]] тех, что медленнее дорожают. В результате, фактический состав [[16]] меняется. За счёт этого уровень инфляции, посчитанный по неизменной(ому) [[20]], оказывается [[24]] "настоящего" уровня инфляции.

1) Корзина

5) Корзины

9) базовый

13) больше

16) корзины

20) корзине

24) выше

Вопрос 18 *(Задание оценивается в 8 баллов)*

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. **Далее вам предлагается заполнить пропуски в тексте ниже,**

чтобы получить верный текст, который начинающий аналитик представляет коллегам в качестве результатов исследования.

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

В своем исследовании мы хотим выяснить, учатся ли более ответственные ученики лучше. Основная гипотеза исследования состоит в том, что [[1]]. Ответственность учащихся мы будем измерять с помощью [[5]]. Проведенные нами расчеты говорят о том, что ответственность и итоговый балл [[6]]. Данный результат [[9]] основную поставленную гипотезу.

1) у более ответственных учеников суммарный балл за все тесты выше

5) переменной test preparation course, потому что более ответственные ученики скорее пройдут подготовительный курс до конца

6) связаны положительно

9) не отвергает

Задания 19 и 20 оцениваются в 1 балл

Вопрос 19

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее ответственных учеников используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Средний суммарный балл по всем тестам среди более ответственных учеников составляет:

Ответ: 210,19

Вопрос 20

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее ответственных учеников используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Средний суммарный балл по всем тестам среди менее ответственных учеников составляет:

Ответ: 177,64

*В представленном файле (**Задача про меню**) вы можете найти информацию о различных заказах, сделанных в кафе за месяц. Вас попросили сделать небольшую аналитику по имеющимся продажам. В задании 21-25 вам будет необходимо по очереди заполнить пропуски в предложенном ниже тексте. Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33). Каждое задание оценивается в 2 балла.*

Вопрос 21

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
Всего за указанный период количество заказов составило:

Ответ: 1223

Вопрос 22

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

В среднем в каждом заказе (...) позиции.

Ответ: 2,51

Вопрос 23

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
В среднем в каждом заказе (ответ из предыдущего задания) позиции на общую сумму:

Ответ: 18,77

Вопрос 24

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
Всего за указанный период было заказано (...) уникальных позиций из нашего меню.

Ответ: 1365

Вопрос 25

В качестве ответа внесите в поле нужный item_name, представленный словом или словосочетанием, латинскими буквами, в той же форме, в которой оно написано в базе данных, которую вы анализируете.

Из всех видов позиций "item_name" наибольшую выручку в нашем кафе принесла:

Ответ: chicken bowl

Национальная олимпиада по анализу данных DANO. 2 этап (вариант 3)

Во всех заданиях с выбором ответа может быть один или несколько правильных вариантов ответа.

Задания с 1 по 3 оцениваются в 2 балла.

Вопрос 1

Как известно, Израиль является одной из лидирующих стран по темпам вакцинации. По данным на июнь 2021 г. в стране вакцинировано 60% граждан (85% взрослого населения). Однако среди заразившихся в этом же месяце (июне 2021 года), как признали власти Израиля, примерно половина была уже вакцинирована. **Что можно сказать об эффективности вакцины на основании этих данных?**

1) Данные не свидетельствуют об эффективности вакцины, т. к. вероятность заразиться составляет 50%, независимо от того, вакцинировался человек или нет
2) Данные не свидетельствуют об эффективности вакцины, т. к. среди вакцинированных есть заразившиеся
3) Данные свидетельствуют об эффективности вакцины, т. к. если бы она не работала, доля вакцинированных среди заболевших была бы равна доле вакцинированных среди всего населения страны
4) Данные свидетельствуют об эффективности вакцины, т. к. вакцинированные переносят болезнь в более легкой форме

Ответ: 3

Вопрос 2

Известно, что у рассматриваемой выборки медиана равна 11, а среднее значение числа трехзвездочных ресторанов в трех странах с наибольшими количествами отличается от аналогичного среднего у трех стран с наименьшими количествами в рассматриваемой выборке на 7,(3). **Чему равно количество ресторанов с тремя звездами Мишлен в Японии?**

1) 18
2) 9
3) Нет верного ответа
4) 28
5) 6

Ответ: 2

Вопрос 3

Министерство здравоохранения пытается оценить тяжесть ситуации с эпидемией нового заболевания в стране. Эта болезнь характеризуется тем, что симптомы очень похожи на обычную простуду, однако нет случаев бессимптомного течения заболевания. Также в данной стране все люди очень ответственны и при возникновении симптомов обращаются к врачу. Для этих целей министерство использует число всех зарегистрированных случаев простудных заболеваний. КАКУЮ ОЦЕНКУ ЗАБОЛЕВАЕМОСТИ НОВОЙ БОЛЕЗНЬЮ ОНИ МОГУТ ПОЛУЧИТЬ С ПОМОЩЬЮ ТАКОЙ СТАТИСТИКИ?

A) Оценку сверху

B) Оценку снизу

C) Точную оценку

D) Нельзя ответить на основе данной информации

Ответ: A

Вопрос 4

Вы хотите изучать влияние исторического распределения школ на текущий уровень человеческого капитала, который включает в себя образование, престижность профессии, продолжительность жизни и другие факторы. **Укажите, какие из графиков способствуют поиску ответа на поставленный вопрос:**

1) Дистанция до ближайшей школы в прошлом ~ текущий уровень грамотности, по индивидам
2) Текущее количество школ в регионе ~ текущий средний уровень образования, по регионам
3) Категория занятости (работники ВУЗов, профессиональные рабочие, низкоквалифицированный труд и др.) ~ дистанция до ближайшей школы в прошлом, по индивидам
4) Уровень образования по годам в изучаемом и соседнем регионе, по регионам
5) Нет верного ответа

Ответ: 1 и 3

Вопрос 5

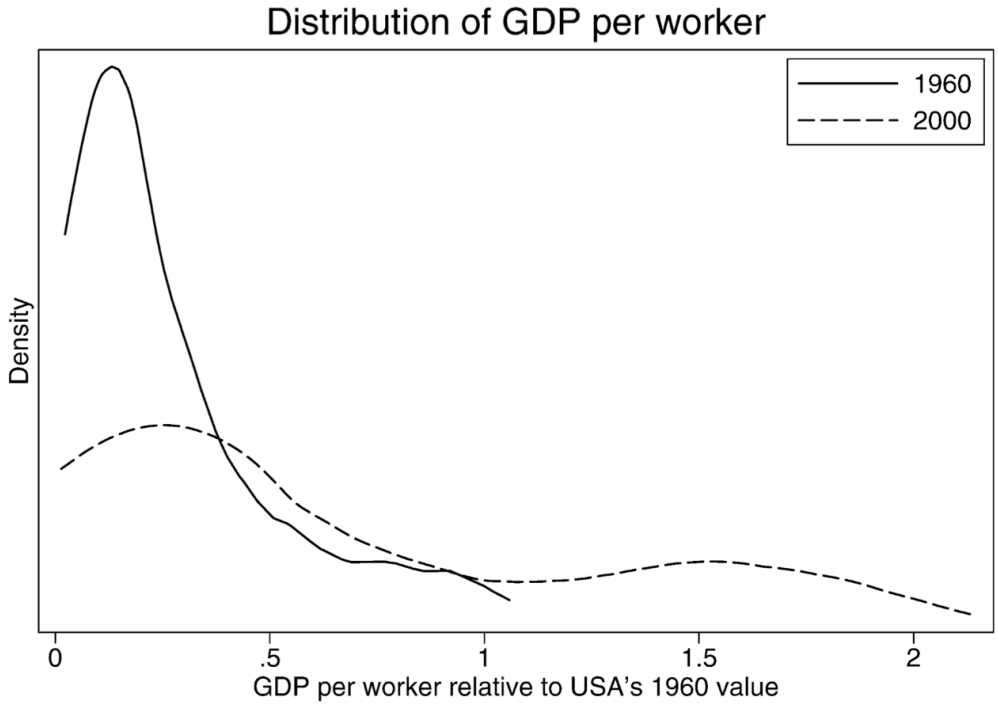
Современные технологии распознавания лиц позволяют правоохранительным органам вычислять преступников на улицах. Алгоритм разделяет людей на обычных граждан и подозреваемых. Подозреваемым считается тот, чью фотографию опознал алгоритм в базе. Предположим, что алгоритм никогда не отмечает преступников как обычных граждан. Известно, что все преступники из базы сейчас находятся на территории города и их разыскивает полиция. **Какие утверждения верны, если все население города попало на камеру за определенный период?**

1) Алгоритм может распознать как подозреваемых не меньше людей, чем есть в базе преступников.
2) Алгоритм распознает столько подозреваемых, сколько в базе преступников.
3) Алгоритм может распознать меньше подозреваемых, чем есть в базе преступников.
4) Алгоритм распознает половину граждан как подозреваемых.
5) Нет верного ответа

Ответ: 1

Вопрос 6

Группа ученых, исследующих экономический рост, посчитала за 1960 и 2000 года отношение ВВП на одного работника в разных странах к этому же показателю для США. На графике ниже представлена функция плотности данного соотношения. **Какие утверждения из приведенных ниже являются верными на основе приведенного графика?**



Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics.

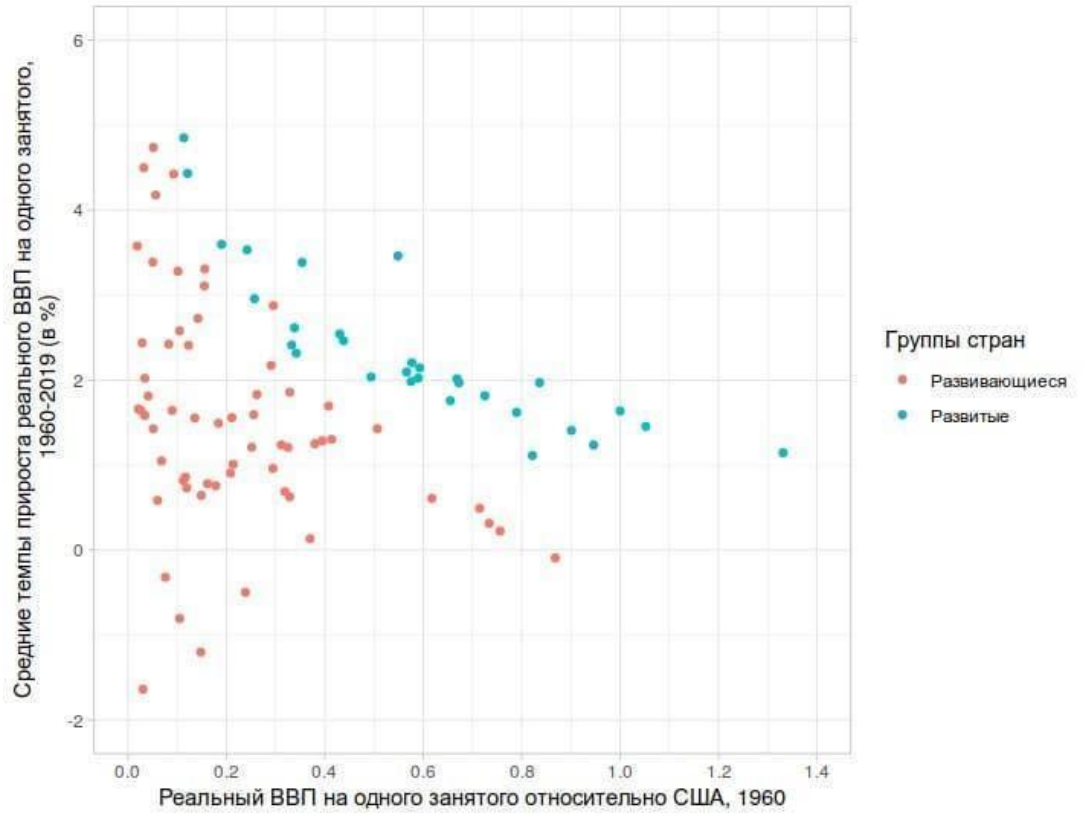
1) В 1960 году в большинстве стран ВВП на работника был выше, чем в США
2) ВВП на работника в США уменьшился за период 1960-2000гг.
3) С 1960 по 2000 год ВВП на работника во всех рос быстрее, чем США
4) В 2000 году количество стран, в которых ВВП на работника ниже этого показателя для США, уменьшилось по сравнению с 1960 годом.
5) Нет ни одного верного ответа

Ответ: 4

Вопрос 7

Конвергенция в теории экономического роста – это процесс, когда страны с меньшим уровнем доходов на душу населения догоняют страны с более высокими доходами. На графике ниже представлена диаграмма взаимосвязи темпов роста в 1960-2000 годах и реального ВВП на одного работника по отношению к ВВП на одного работника в США в 1960 году. Также на графике разными цветами выделены развитые и развивающиеся

страны на основании методологии МВФ по состоянию на 2018 год.



Источник: Durlauf, S. N., Johnson, P. A., & Temple, J. R. W. Growth econometrics, построение авторов

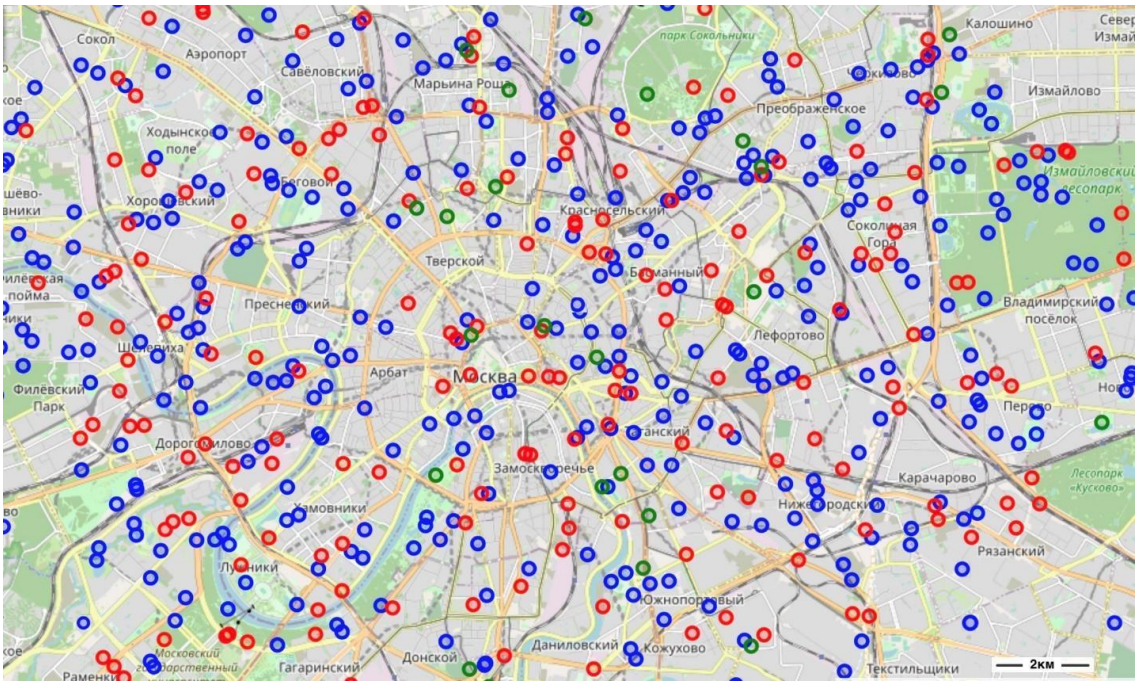
Какие утверждения из приведенных ниже являются верными на основании приведенного графика?

1) Наблюдается конвергенция в уровне доходов на одного работника в рассматриваемый промежуток времени только в развивающихся странах, поскольку эти точки демонстрируют отрицательную взаимосвязь показателей
2) В развивающихся странах и выпуск на работника, и темпы его роста оказываются ниже, чем в развитых, поэтому конвергенция наблюдается
3) В течение 60 лет в большинстве стран с низким доходом выпуск на одного работника рос небольшим или отрицательным темпом, поэтому конвергенция не наблюдается
4) В большом количестве развивающихся стран темпы роста выпуска на одного работника ниже, чем в США, поэтому конвергенции не наблюдается
5) Нет ни одного верного утверждения

Ответ: 4

Вопрос 8

Предположим, что есть некая интернет-платформа, пункты выдачи заказов которой нанесены на карту по типам. Красные точки отображают собственные пункты выдачи заказов, синие точки отображают партнерские пункты, а зеленые – отделения почты, через которые также могут выдаваться интернет-заказы. **Какие выводы на основе данной карты можно сделать?**



1) компания в основном опирается на партнерские пункты выдачи
2) партнерские пункты выдачи генерируют большую прибыль
3) отделения Почты распределены неравномерно по городу
4) отделения Почты преобладают в отдаленных районах города
5) нет ни одного верного ответа

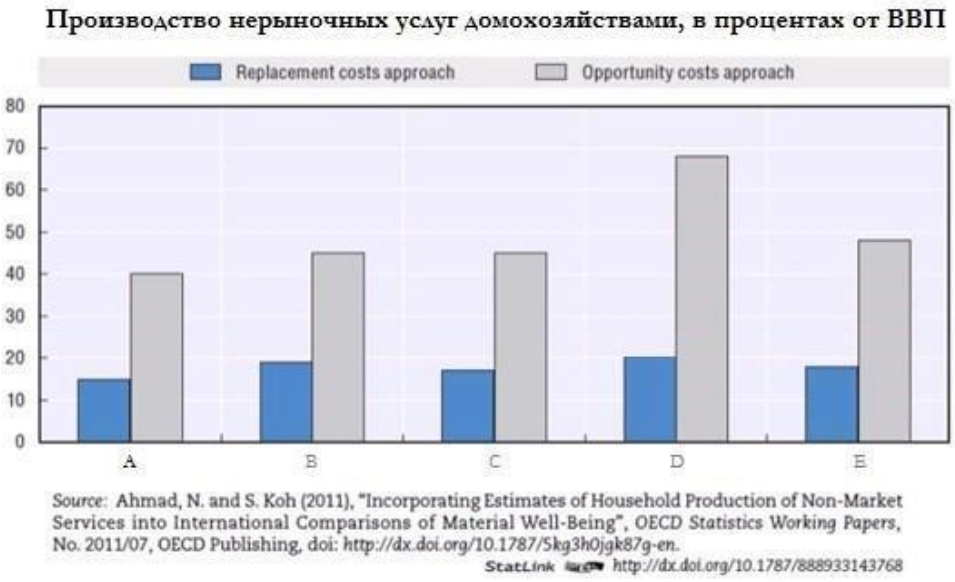
Ответ: 3

Вопрос 9

Экономисты часто задаются вопросом относительно учета в ВВП различных показателей, которые невозможно посчитать напрямую. Так, например, включили в ВВП проживание в собственной квартире исходя из цены аренды помещения на рынке. Еще один интересный и довольно значимый вклад в выпуск может внести работа людей на самих себя в рамках ведения своих ежедневных дел. Домохозяйка, которая сидит дома и занимается исключительно своими бытовыми заботами, по факту оказывает довольно большое количество услуг: уборщицы, повара, няни и т.п.

Экономисты предлагают два альтернативных способа оценить стоимость оказываемых услуг. Первый способ – replacement costs approach (подход на основе издержек замещения) – это оценить стоимость услуг, которые домохозяйки оказывают своему домохозяйству, исходя из рыночных цен на аналогичные услуги (то есть сложить сколько стоит нанять няню, повара и горничную для выполнения тех же работ). Второй способ – opportunity costs approach (подход на основе альтернативных издержек) – это оценить, сколько бы в среднем женщина зарабатывала, выйдя на рынок труда. На графике ниже приведены оценки неучтенного ВВП из-за труда домохозяек для разных стран (A, B, C, D и E).

Какие выводы мы можем сделать, основываясь на данных графиках?



1) В стране E больше пользуются услугами поваров, горничных и нянь, чем в стране A
2) Во всех странах из приведённых на графике зарплаты нянь, горничных и поваров в среднем ниже, чем зарплаты в других профессиях
3) Любая домохозяйка на рынке труда могла бы получать зарплату больше, чем стоимость тех услуг, которые она производит дома
4) Мы ничего не можем сказать о соотношении средних зарплат горничных со средней рыночной зарплатой в других профессиях рассматриваемой страны
5) Нет ни одного верного ответа

Ответ: 2

Вопрос 10

Напитки, содержащие сахар, считаются вредными для здоровья, и поэтому в Беркли (в США) в начале 2015 года ввели налог на продажу этих напитков для уменьшения их потребления. Первая вертикальная линия на графике показывает момент объявления о введении налога, вторая – момент начала выплат налога продавцами. Однако, чтобы отследить чистый эффект налога, исследователи собрали данные из магазинов в соседних городах, где налог не был введен, и сравнили их с ценами на напитки в Беркли. Цены на сахар в указанный период времени не менялись. Данные представлены в виде графиков цен на напитки в Беркли и в других городах.



Какие свойства графика позволяют нам сделать вывод о воздействии налогов на цену напитков?

1) Цены на облагаемые и необлагаемые налогом напитки в Беркли до введения налога должны меняться одинаково
2) Цены на облагаемые и необлагаемые налогом напитки в Беркли после введения налога должны меняться по-разному
3) Цены на необлагаемые налогом напитки после введения налога в Беркли и вне Беркли должны меняться по-разному
4) Цены на облагаемые и необлагаемые налогом напитки вне Беркли после введения налога должны меняться одинаково
5) Нет ни одного верного ответа

Ответ: 1, 2, 4

Задания с 11 по 15 оцениваются в 8 баллов

Вопрос 11

В представленном файле (**Задача про футбол**) вы можете найти информацию о футбольных матчах Английской премьер-лиги сезона 2017-2018 года. **Из приведенных ниже утверждений выберите все верные.**

1) В выездном матче меньше шансов сыграть вничью, чем выиграть матч
2) В гостевых матчах команды пропускают голов больше, чем в домашних
3) Вариация в количестве забитых мячей в гостевых матчах выше, чем в домашних
4) Максимальное количество мячей, забитых в гостевых матчах, больше, чем в домашних
5) Нет ни одного верного ответа

Ответ: 1 и 2

Вопрос 12

Выберите все верные утверждения на основании анализа данных, предложенных вам в файле (**Задача про машины**).

1) Автомобили с большим пробегом продаются в среднем по более высоким ценам
2) В лизинг продаются в среднем более новые автомобили
3) Автомобили в хорошем состоянии продаются в среднем примерно на 40% дороже автомобилей в плохом состоянии
4) Автомобили с дизельным двигателем (Diesel) продаются в среднем дороже, чем автомобили с бензиновым двигателем (Petrol)
5) Нет ни одного верного ответа

Ответ: 2 и 4

Вопрос 13

Индекс человеческого развития (ИЧР) — интегральный показатель, рассчитываемый ежегодно для межстранового сравнения и измерения уровня жизни, грамотности, образованности и долголетия как основных характеристик человеческого потенциала исследуемой территории. Он является стандартным инструментом при общем сравнении уровня жизни различных стран и регионов.

(Источник: Википедия)

Необходимо посчитать индексы человеческого развития (ИЧР) для представленных стран в файле (**Задача про ИЧР**) (Настоящая методология расчета ИЧР включает в себя некоторые поправки, которые в данном задании опущены. В связи с этим результаты могут отличаться от опубликованных значений ИЧР). Для вычисления ИЧР выполните следующие пункты:

а) Используя индикатор Ожидаемая продолжительность жизни при рождении (Life expectancy at birth), посчитайте Индекс здоровья по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

б) Посчитайте Индекс образования, используя два индикатора:

Средняя продолжительность обучения населения в годах и Ожидаемая продолжительность обучения населения, ещё получающего образование, в годах.

Для каждого индикатора вычислите свой индекс по представленной формуле:

$$\text{Индекс} = \frac{\text{значение показателя} - \text{минимальное значение показателя}}{\text{максимальное значение показателя} - \text{минимальное значение показателя}}$$

в) Посчитайте Индекс доходов, используя показатель натуральный логарифм Валового национального дохода на душу населения (GNI per capita). Для его вычисления необходимо взять натуральный логарифм значений (этот показатель уже посчитан и приведен в файле с данными, Log GNI).

(Натуральный логарифм есть обратная функция для экспоненты. Функция является монотонно возрастающей. Ее часто применяют для преобразования различных экономических переменных, которые характеризуются постоянным ростом (например, для ВВП). Для его вычисления в Excel используется функция LN()).

Индекс считается по формуле:

$$\text{Индекс} = \frac{\text{значение логарифма показателя} - \text{минимальное значение логарифма показателя}}{\text{максимальное значение логарифма показателя} - \text{минимальное значение логарифма показателя}}$$

г) Рассчитайте Индекс человеческого развития, вычислив геометрическое среднее по трем индексам. Округлите значения до 3 знаков после запятой и расположите страны в порядке убывания индекса.
Выберите все верные утверждения в соответствии с Вашим решением.

1) Максимальное значение Индекса доходов равно единице
2) Индекс здоровья в Центральной Африканской Республике равен нулю
3) ИЧР в Омане равен ИЧР в Коста Рике
4) Страна-лидер по ИЧР – Норвегия
5) Нет ни одного верного варианта ответа

Ответ: 1, 2, 3

Вопрос 14

В этом задании будет рассмотрено влияние появления нового мусоросжигательного завода на стоимость жилья в одном из городов США. Для этого мы будем использовать данные за два года (до и после строительства завода) о домах, находящихся рядом с заводом, и домах, находящихся далеко от завода. Данные вы можете найти в файле **«Задача про дома»**. Единицей наблюдения в нем является дом.

Отметим, что слухи о том, что в городе будет построен новый мусоросжигательный завод, появились после 1978 года, а строительство началось в 1981 году. Мы будем использовать данные о ценах на дома, проданные в 1978 году (период «до»), а также данные о ценах на дома, проданные в 1981 году (период «после»). Гипотеза заключается в том, что цены на дома, расположенные вблизи мусоросжигательного завода, упадут по сравнению с ценами на более удаленные дома.

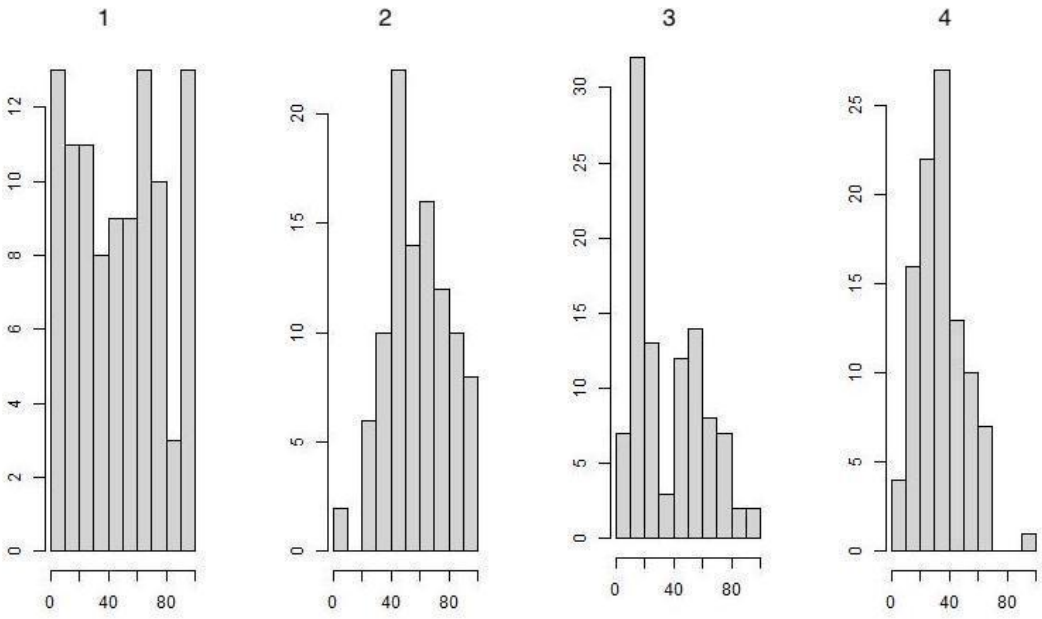
Выберите, какие из представленных ответов соответствуют данным и одновременно помогают обосновать представленную гипотезу.

1) Средняя цена на дома в городе снизилась в 1981 году по сравнению с 1978 годом
2) Средняя цена на дома, расположенные вблизи завода, снизилась в 1981 году по сравнению с 1978 годом
3) В 1981 году по сравнению с 1978 годом средняя цена на дома, расположенные далеко от завода, выросла сильнее, чем средняя цена на дома, расположенные вблизи к заводу
4) Эффект снижения средней цены на дома в связи со строительством завода составляет примерно 11 тыс.
5) Нет ни одного верного ответа

Ответ: 3 и 4

Вопрос 15

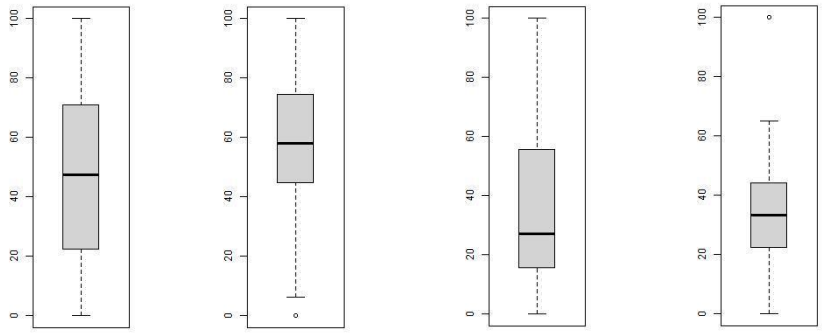
На представленных вам рисунках вы можете найти частотные диаграммы и графики box-plot.
Соотнесите разные типы диаграмм, которые отображают один и тот же набор данных.



В данном задании вы получаете 2 балла за каждое верное сочетание гистограммы и box plot. Неверное сочетание не штрафуется.

Ответ:

1) 2) 3) 4)



Вопрос 16 *(Задание оценивается в 10 баллов)*

В файле «**Задача про студентов**» вам дана таблица, где есть ученики, данные про них, и курсовые проекты, которые они выбрали по приоритетам. Если анкета заполнена несколько раз, то учитывается последняя отправка. Если два студента хотят выбрать один и тот же проект, то в итоге на него будет записан студент с более высоким рейтингом.

Выберите верное распределение курсовых проектов между студентами, если для каждого нужно выбрать только один проект и на каждый проект может записаться не более одного студента.

В данном задании вы получаете 2 балла за каждое верное сочетание студента и курсового проекта. Неверное сочетание не штрафуется.

Ответ:

1) Введение в социологию
2) История России
3) Программирование
4) Анализ данных
5) Современная литература
   Микроэкономика (лишнее)

Вопрос 17 *(Задание оценивается в 14 баллов)*

Заполните пропуски, чтобы получилось верное рассуждение на тему расчета индекса потребительских цен.

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

Инфляция обычно измеряется с помощью индекса потребительских цен. Для расчёта этого показателя берётся фиксированная(ый) [[1]], общая стоимость которой(ого) потом используется для расчёта индекса: стоимость [[5]] в текущий год сравнивается со стоимостью в [[9]] год и это изменение интерпретируется как инфляция.

Однако цены растут неравномерно и из-за эффекта замещения люди, выбирая между двумя очень похожими товарами (разными сортами яблок, к примеру), начинают покупать [[13]] тех, что медленнее дорожают. В результате, фактический состав [[16]] меняется. За счёт этого уровень инфляции, посчитанный по неизменной(ому) [[20]], оказывается [[24]] "настоящего" уровня инфляции.

1) Корзина

5) Корзины

9) базовый

13) больше

16) корзины

20) корзине

24) выше

Вопрос 18 *(Задание оценивается в 8 баллов)*

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. **Далее вам предлагается заполнить пропуски в тексте ниже, чтобы получить верный текст, который начинающий аналитик представляет коллегам в качестве результатов исследования.**

В данном задании вам необходимо выбрать верный вариант заполнения пропуска из предложенных вариантов. Каждый правильно заполненный пропуск оценивается в 2 балла. Неверный ответ не штрафуется.

В своем исследовании мы хотим выяснить, учатся ли более ответственные ученики лучше. Основная гипотеза исследования состоит в том, что [[1]]. Ответственность учащихся мы будем измерять с помощью [[5]]. Проведенные нами расчеты говорят о том, что ответственность и средний балл [[6]]. Данный результат [[9]] основную поставленную гипотезу.

1) у более ответственных учеников средний балл за все тесты выше

5) переменной test preparation course, потому что более ответственные ученики скорее пройдут подготовительный курс до конца

6) связаны положительно

9) не отвергает

Задания 19 и 20 оцениваются в 1 балл

Вопрос 19

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее ответственных учеников используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Среднее значение среднего балла по всем тестам среди более ответственных учеников составляет:

Ответ: 70,06

Вопрос 20

В представленном файле (**Задача про оценки**) вы можете найти данные о характеристиках различных учеников. В этом задании для ответа на вопрос вам будет необходимо посчитать некоторые показатели по данной базе.

Подсказка: для выделения более или менее ответственных учеников используйте наиболее подходящий из следующих трех показателей: lunch, test preparation course или gender.

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

Среднее значение среднего по всем тестам среди менее ответственных учеников составляет:

Ответ: 59,21

*В представленном файле (**Задача про меню**) вы можете найти информацию о различных заказах, сделанных в кафе за месяц. Вас попросили сделать небольшую аналитику по имеющимся продажам. В задании 21-25 вам будет необходимо по очереди заполнить пропуски в предложенном ниже тексте. Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33). Каждое задание оценивается в 2 балла.*

Вопрос 21

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
Всего за указанный период количество заказов составило:

Ответ: 1223

Вопрос 22

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

В среднем в каждом заказе (...) позиции.

Ответ: 2,51

Вопрос 23

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).

В среднем в каждом заказе (ответ из предыдущего задания) позиции на общую сумму:

Ответ: 18,77

Вопрос 24

Если ваш ответ представлен дробным числом, запишите его через запятую с округлением до 2 знаков после запятой (например, если вы получили 1/3, то в ответе необходимо указать 0,33).
Всего за указанный период было заказано (...) уникальных позиций из нашего меню.

Ответ: 1365

Вопрос 25

В качестве ответа внесите в поле нужный item_name, представленный словом или словосочетанием, латинскими буквами, в той же форме, в которой оно написано в базе данных, которую вы анализируете.

Из всех видов позиций "item_name" наибольшую выручку в нашем кафе принесла:

Ответ: chicken bowl

Критерии оценки исследовательских проектов на заключительном этапе Национальной олимпиады по анализу данных «DANO»

Оценка исследовательских проектов, подготовленных участниками, будет проводиться по трем группам критериев: Содержание проекта, Исполнение, Презентация. Максимально возможная общая оценка – 100 баллов.

Максимальный балл по каждому из критериев может быть выставлен жюри в случае, если проект в полной мере содержательно и технически отвечает приведенным требованиям, а команда участников в ходе презентации продемонстрировала высокий уровень владения методами анализа данных и понимания исследуемой области. Результаты проекта участники должны презентовать как связное изложение, а не как набор отдельных расчетов. Для максимального балла это должен быть именно рассказ, а не чтение с листа.

В случае частичного несоответствия проекта приведенным требованиям жюри снижает баллы по каждому из пунктов в соответствии с экспертной оценкой степени проработанности данной части проекта и представленной разбалловкой.

Жюри может попросить участников показать исходный скрипт (код) программы, использованной при анализе. Собственно качество исполнения кода или эффективность методов, содержащихся в скрипте, не оценивается, но участники должны продемонстрировать полное понимание того, что содержится в скрипте или в коде, в противном случае команда может лишиться всех баллов за несамостоятельное выполнение проекта.

1 – Содержание проекта. Максимум 45 баллов.

- 15 баллов: Исследовательский вопрос+Гипотеза+Механизм

Участники сформулировали исследовательский вопрос и гипотезу, понимают различие между ними (5 баллов). Гипотеза сформулирована корректно и отражает правильное направление связи (3 балла). Приведены примеры ситуаций, подтверждающих поставленную гипотезу (2 балла). В проекте описан механизм взаимосвязи между причиной и следствием, строго прослеживается логика причинно-следственной цепочки (5 баллов).
Предварительный анализ предметной области на основе других источников и опубликованных исследований не является обязательным, но может быть засчитан и отмечен жюри как преимущество.

- 15 баллов: Дизайн

Описание дизайна исследования. Участники должны объяснить логику проведения исследования, каким образом будет получен ответ на исследовательский вопрос, как будут проверяться выдвинутые гипотезы, какие методы будут применяться и почему (5 баллов). Почему для проведения исследования отобраны те или иные переменные, по какому принципу сформирована выборка (если исследование построено не на всех данных доступных в базе данных) или почему исследование можно проводить на всей базе данных (если участники ее не ограничивали) (5 баллов), какие связи будут проверятся и

зачем, как это поможет в проверке гипотез и получении ответа на исследовательский вопрос (5 баллов).

- 15 баллов: Ограничения + интерпретация

Участники способны интерпретировать и критически оценить полученные в ходе исследования результаты. Проанализированы возможные возникающие несостыковки результата с исходной гипотезой (5 баллов). Участники рассмотрели в какой мере результаты исследования могут быть распространены на более широкую совокупность (3 балла), отметили каких переменных может не хватать для дальнейшего анализа (2 балла) и обозначили направления для улучшения модели (2 балла). В работе присутствуют рекомендации по проведению социально-экономической политики, основанные на проведенном исследовании (3 балла).

2 – Исполнение проекта, в том числе техника исполнения. Максимум 35 баллов.

- 7 баллов: Подготовка данных

В презентации присутствует описание основных переменных с расшифровкой используемых обозначений (1 балл). Участники верно трактуют смысл каждой переменной и понимают ее особенности (2 балла). Проведен предварительный анализ данных: характеристика выборки (размер, критерий отбора наблюдений, какие-то особенности ее формирования) и описательная статистика (показатели центра и вариации основных переменных, графики их распределения) (4 балла).

- 14 баллов: Модель

При подготовке проекта участники использовали статистические методы и модели (1 балл). В работе представлено обоснование выбранных переменных для каждой используемой техники анализа (3 балла). Использованные методы анализа данных применены корректно, в том числе с точки зрения однородности выборки и работы с выбросами (5 баллов). Грамотная интерпретация каждого полученного количественного показателя (5 баллов).

- 14 баллов: Визуализация

В презентации присутствует графическое представление построенной модели (3 балла). Используемые диаграммы, графики и таблицы наглядно демонстрируют результаты исследования (3 балла). Все графические элементы несут необходимую для понимания результатов исследования информацию (3 балла), надлежащим образом оформлены и хорошо читаемы (5 баллов).

3 – Презентация. Максимум 20 баллов.

- 13 баллов: Ответы на вопросы

Участники ответили на все вопросы членов жюри в установленное время (2 балла). Ответы команды даны по существу каждого вопроса, по возможности опираясь на слайды презентации и (при необходимости) дополнительные материалы (6 баллов). Все участники

демонстрируют понимание всех аспектов проекта (жюри может задать любой вопрос любому участнику проекта) (5 баллов). При выставлении оценки жюри учитывают вовлеченность и участие в обсуждении всех членов команды.

- 7 баллов: Качество презентации

Презентация структурирована и имеет четкую логику изложения (2 балла). Использован единый стиль форматирования (заголовки, шрифт, отступы, нумерация), не нарушающий визуальное восприятие материала (2 балла). В ходе доклада участники четко объясняют свою мысль без уточняющих вопросов со стороны экспертов, используют точные формулировки в своих ответах (1 балл).
Текстовые элементы на презентации не повторяют, а выгодно дополняют устное выступление команды (1 балл). Выдержан тайминг выступления: в отведенное время рассказано все, что запланировано (1 балл).

Problem 1, EGMO 2026

P1

A 2026×2026 board is said to be *bordeaux* if at least one of its 2026^2 unit cells is coloured red. A rectangular region made of cells is *oddly-rectangular* if it contains an odd number of red cells. Determine the largest positive integer M such that, in every possible 2026×2026 bordeaux board, there exists an oddly-rectangular region of at least M cells.

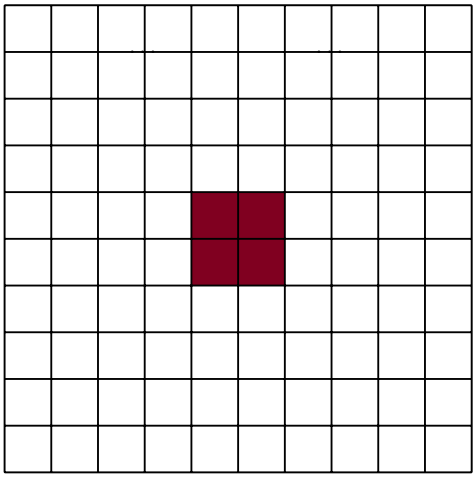
Note: A rectangular region has sides that are parallel to the sides of the board and contains all of its interior.

Solutions

Solution 1

Answer. 1013^2.

Example. Here is an example for a grid 10×10. The same can be done for the grid 2026×2026.



Colour the four central cells red. Any rectangle that contains at least two of them contains either 2 or 4 red cells. In order to contain exactly one red cell, the rectangle must have one of its vertices at the center of the square, and therefore its area is at most 1013^2.

Bound. Suppose that every rectangle of area at least 1013^2 contains an even number of red cells. Consider an arbitrary cell (x, y). WLOG assume that the cell is in the bottom right part of the board, i.e. $x, y \geq 1014$. We consider the rectangles

$$[1, x] \times [1, y], \quad [1, x-1] \times [1, y], \quad [1, x] \times [1, y-1], \quad [1, x-1] \times [1, y-1].$$

Each of these rectangles contains an even number of red cells. Every cell other than (x, y) belongs to an even number of these rectangles, and cell (x, y) belongs to exactly one of them. So by a double-counting

argument, it can't be red. Since (x, y) was arbitrary, we conclude that none of the cells are red, which is a contradiction.

Solution 2

See Solution 1 for a proof that $M \leq 1013^2$.

Lower Bound. Suppose that every rectangle of area at least 1013^2 contains an even number of red cells.

Consider a horizontal arbitrary $1 \times n$ rectangle with $n \geq 1013$. It is the extremity of a $1014 \times n$ included in the grid (above it if it is in the lower half of the grid, under it otherwise), which contains an even number of red cells as its area is at least 1013^2. The same rectangle without the $1 \times n$ rectangle also has area at least 1013^2 thus contains an even number of red cells. Thus the $1 \times n$ rectangle contains an even number of red cells.

Since every $1 \times n$ rectangle with $n \geq 1013$ contains an even number of cells, and each cell is the extremity of a 1×1014 rectangle, by the same reasoning, every cell is uncoloured, which is a contradiction.

Solution 3

See Solution 1 for a proof that $M \leq 1013^2$.

Lower Bound. Suppose by contradiction that every rectangle of area at least 1013^2 contains an even number of red cells.

Claim: Any two cells that differ by a horizontal shift of at least 1013 are either both red, or both not red.

Proof. Consider an arbitrary $n \times 1$ rectangle C, with $n \geq 1013$. Let R be a $n \times m$ rectangle, having C as its first or last column, such that $m \geq 1014$ (for m at least 1014 and at most the maximum distance from C to the vertical boundary + 1). Removing the first or last column of R gives two rectangles that have area at least 1013^2. They thus both have an even number of red cells. This shows that if we can map two columns of size at least 1013 by a horizontal shift of at least 1013, the parity of the number of red cells they contain is the same.

Consider two cells A and B that differ by a horizontal shift of at least 1013. Suppose WLOG they are in the lower half of the grid. Consider the 1014×1 columns C_1 and C_2 whose lowest cells are A and B. Then C_1 and C_2 have the same parity of the number of red cells, and so do $C_1 \setminus A$ and $C_2 \setminus B$. So A and B are either both red, or both not red.

We have thus indeed shown that any two cells that differ by a horizontal shift of at least 1013 are either both red, or both not red. This also works for vertical shifts.

□

Claim: Either all cells are red or none of them are red.

Proof. If we divide the grid in its four 1013×1013 sub grids, notice that any cell of a subgrid is linked to the opposite corner by two suitable shifts. Furthermore, all the corners are linked by suitable shifts, so either they are all red, or none of them are. □

This shows that all the cells have the same state, which has to be red because the board is bordeaux. Hence, any rectangle of area 1013^2 has an odd number of red cells, contradicting our original hypothesis.

Solution 4

See Solution 1 for a proof that $M \leq 1013^2$.

Lower Bound.

Suppose, for the sake of contradiction, that every rectangle of size at least 1013^2 contains an even number of red cells.

Claim 1. Every row and column contains an even number of red cells. **(*)**

Proof. Let $k \in \{1014, \ldots, 2026\}$. Note that $[1, 2026] \times [1, k]$ and $[1, 2026] \times [1, k-1]$ both have size greater than 1013^2, so they both contain an even number of red cells. Therefore their difference, the kth column $[1, 2026] \times [k, k]$, contains an even number of red cells.

It follows that all columns on the right side of the board contain an even number of red cells. By symmetry, we can analogously show that the same holds for columns on the left and for all rows. □

Now consider a red cell. Its row and its column both contain an even number of red cells. Therefore its cross (the union of the row and column) contains an odd number of red cells. Since the full 2026×2026 grid contains an even number of red cells, out of the four (possibly empty) remaining rectangular regions, either one or three have an odd number of red cells. Either way, at least one of these regions that contain a corner is oddly-rectangular.

Claim 2. If a rectangular region R containing a corner is oddly-rectangular, then there exists a rectangle with area at least 1013^2 that is oddly-rectangular. **(**)**

Proof. The lines of the sides of R cut the whole grid into 4 rectangular regions. By Claim 1, two rectangular regions adjacent to R have the same parity of the number of red cells, as their union is the disjoint union of some rows or columns. Hence, all 4 regions are oddly-rectangular, and at least one has area at least 1013^2, as their union is the whole grid. □

This completes the proof by contradiction.

Solution 5

See Solution 1 for a proof that $M \leq 1013^2$.

Lower Bound.

Suppose, for the sake of contradiction, that every rectangle of size at least 1013^2 contains an even number of red cells.

Claim 1. Same as Solution 4 ($*$).

Consider any red cell B, and a rectangular region containing B and a corner cell C as it's corners. If there is another red cell in that region, replace B by this new red cell. Doing so repeatedly allows us to find B and C such that the rectangular region between them doesn't contain any other red cells. In particular, this region is odd as it contains exactly one red cell.

Claim 2. Same as Solution 4 ($**$)

This completes the proof by contradiction.

Problem 2, EGMO 2026

P2

Given a positive integer n, Marie plays a game where she starts with the number 1 on a blackboard. As many times as she wants, she can choose an integer j such that $1 \leq j \leq n$ and replace the number V on the blackboard with the number $j \cdot R\left(\frac{V}{j}\right)$. Here, $R(x)$ denotes the nearest integer to x; if x is exactly halfway between two consecutive integers, it is rounded up. For example, $R(1.3) = 1$ and $R(1.5) = R(1.8) = 2$.

a) Prove that for each given n, there is a positive integer B such that Marie can never end up with a number larger than B on the blackboard.

b) For any given n, let $f(n)$ be the maximum number obtainable on the blackboard after finitely many replacements. Show that there exists a positive integer N such that for all $n \geq N$, we have that 2026 divides $f(n)$.

Solutions

Solution 1

Claim 1. Marie can never reach a number greater than $n!$.

Proof. We prove this by induction on the number of steps. Before the first step, Marie's number is 1, which is less or equal to $n!$.

If at some point, Marie's number V is at most $n!$ and she chooses $j \in \{1, \ldots, n\}$, then since $V/j \leq n!/j$ and the function R is non-decreasing, her resulting number will still be at most $j \cdot R(n!/j) = n!$ since $j \leq n$, which concludes the induction. □

Note that this finishes part (a) of the problem, since Marie can never end up with a number greater than $B = n!$.

Claim 2. For any positive integer k and $n \geqslant 2^k$, $f(n)$ is a multiple of 2^k.

Proof. Fix a positive integer k, and $n \geq 2^k$. Assume 2^k does not divide $f(n)$ and let $\ell = \nu_2(f(n))$. We have $\ell \in \{0, \ldots, k-1\}$. Since 2^ℓ divides n and $2^{\ell+1}$ does not divide n, we have $n \equiv 2^\ell \pmod{2^{\ell+1}}$. Since $\ell \leq k-1$, $2^\ell \leq n$, and $f(n)/2^{\ell+1}$ is a half-integer, if Marie chooses $j = 2^{\ell+1}$, she can increase the number on the board, which is a contradiction. □

Claim 3. For any odd positive integer A, for any integer k such that $A < 2^k$, if $n \geqslant 2^{2k}$, then $f(n)$ is a multiple of A.

Proof. Fix an odd positive integer A, an integer k such that $A \leqslant 2^k$, and n such that $n \geqslant 2^{2k}$. Note that $f(n)$ has a remainder between 0 and $2^{j-1}A$ (excluding $2^{j-1}A$) mod 2^jA for each $j \in \{0, \ldots, k\}$, otherwise Marie could choose 2^jA and increase her number, as $2^jA \leq 2^{2k} \leq n$.

Let r_j be the remainder of $f(n)$ mod 2^jA for all $j \in \{0, \ldots, k\}$, we thus have $0 \leq r_j \leq 2^{j-1}A$. For all $j \in \{0, \ldots, k-1\}$, we have $r_j \equiv f(n) \equiv r_{j+1} \pmod{2^jA}$, and since $-2^jA < r_{j+1} - r_j < 2^jA$, we have

$r_j = r_{j+1}$, thus $r_0 = r_k$. Since 2^k divides $f(n)$, it divides $r_k = r_0$, but $0 \leq r_0 < A \leq 2^k$, therefore $r_0 = 0$. We conclude that A divides $f(n)$. □

Choosing $A = 1013$ in Claim 3 and combining this with Claim 2, we have that $2026 \mid f(n)$ for all $n \geqslant 1024^2$.

Solution 2

Call a number *special* if the remainder it leaves modulo k is strictly less than $\frac{k}{2}$ for each $1 \leqslant k \leqslant n$. Clearly $n!$ is special.

Claim 1. Marie can never reach a number greater than the smallest positive special number.

Proof. Assume the number Marie has currently written on the board is v, and say she applies the transformation for some j; this operation will increase the number on the board if and only if v leaves a remainder at least $j/2$ modulo j.

Let M be the value of the smallest special number. If Marie could cross this value, then there exists $a \leqslant M < b$ such that Marie changes the number from a to b. Let's say this change was done using $j = t$ for some $1 \leqslant t \leqslant n$. Then a leaves a remainder at least $t/2$ when divided by t, and the next multiple of t is b. Since $a \leq M < b$, M also leaves a remainder of at least $t/2$ modulo t, which is a contradiction to it being special. □

Note that this finishes part (a) of the problem, since Marie can never write a number greater than $B = n!$.

Clearly $f(n)$ is special, otherwise Marie could increase her number.

Claim 2. For any $n \geqslant k^2$, all special numbers are multiples of k.

Proof. We prove that for $n \geqslant k^2$, all special numbers are multiples of i for each $1 \leqslant i \leqslant k$, and we do this by induction. The cases of $i = 1$ and $i = 2$ are clear. Now assume we know this for some i. Let M be a special number being considered. By the induction hypothesis, we know that $i \mid M$. Now consider simultaneously

$$M \equiv x \pmod{i+1} \quad \text{and} \quad M \equiv y \pmod{i(i+1)}.$$

Note that $i + 1 \leqslant k$, so $i(i + 1) \leqslant k^2 \leqslant n$. Since M is known to be 0 modulo i, the possible remainders for the latter are $0, i, 2i, 3i, \ldots$ which correspond to remainders $0, -1, -2, \ldots$ modulo $i+1$. We need $y < \frac{i(i+1)}{2}$, which forces the remainder modulo $i+1$ to be in $0, -1, \ldots, -\left\lfloor \frac{i}{2} \right\rfloor$; but the only remainder here which is also in the first half of remainders modulo $i + 1$ is 0, so we must have $i + 1 \mid M$. □

Choosing $k = 2026$ in Claim 2, we get that $2026 \mid f(n)$ for all $n \geqslant 2026^2$.

Solution 3

We provide a reformulation of the first solution of the second part of the problem.

Claim 1. A number $V > 0$ is special if and only if for all $1 \leqslant k \leqslant n$, we have $\lfloor \frac{2V}{k} \rfloor = 2 \cdot \lfloor \frac{V}{k} \rfloor$.

Proof. It suffices to notice that if $1 \leq k \leq n$, the remainder that V leaves modulo k is $V - k \cdot \left\lfloor \frac{V}{k} \right\rfloor$, and

$$\begin{aligned} & V - k \cdot \left\lfloor \frac{V}{k} \right\rfloor < \frac{k}{2} \\ \iff & 2k \cdot \left\lfloor \frac{V}{k} \right\rfloor \leqslant 2V < k + 2k \cdot \left\lfloor \frac{V}{k} \right\rfloor \\ \iff & \left\lfloor \frac{2V}{k} \right\rfloor = 2 \cdot \left\lfloor \frac{V}{k} \right\rfloor. \end{aligned}$$

□

Let us consider $v > 0$ such that $2^v \geqslant 2026$, and $n \geqslant 2^{2v}$. Let us denote $f(n)$ by V. Because V is special (or else Marie could increase her number using some j), Claim 1 shows that for all $1 \leqslant j \leqslant v$, one has $\left\lfloor \frac{2V}{2^j} \right\rfloor = 2 \cdot \left\lfloor \frac{V}{2^j} \right\rfloor$. A straightforward induction then shows that

$$V = \left\lfloor \frac{2V}{2} \right\rfloor = 2 \cdot \left\lfloor \frac{2V}{4} \right\rfloor = \cdots = 2^v \left\lfloor \frac{V}{2^v} \right\rfloor.$$

Similarly, applying Claim 1 to $2026 \cdot 2^j$ for all $1 \leqslant j \leqslant v$ (these are indeed at most n by construction) gives

$$\left\lfloor \frac{V}{2026} \right\rfloor = \left\lfloor \frac{2V}{2 \cdot 2026} \right\rfloor = \cdots = 2^v \left\lfloor \frac{V}{2^v \cdot 2026} \right\rfloor.$$

We have thus shown that $2^v \mid V$ and $2^v \mid \left\lfloor \frac{V}{2026} \right\rfloor \mid 2026 \cdot \left\lfloor \frac{V}{2026} \right\rfloor$. But

$$0 \leqslant V - 2026 \cdot \left\lfloor \frac{V}{2026} \right\rfloor < 2026 \leqslant 2^v,$$

therefore this multiple of 2^v is zero. We have thus shown that for all $n \geqslant 2^{2v}$, one has $2026 \mid f(n)$.

Solution 4

We give an alternative solution for part (b).

For a given n, let

$$f(n) = \sum_{i=1}^{\infty} a_i i!$$

with $0 \leq a_i \leq i$ and all but finitely many a_i being 0 (it is known that such a decomposition exists).We prove by induction on m that for every m, if n is large, we have $a_0 = \cdots = a_m = 0$.

- Base case: $f(n)$ is even for $n \geq 2$, so $j = 1$ works.
- Induction step: take n large enough and $m \geq 2$ such that $a_1 = \cdots = a_{m-1} = 0$. Assume that $n \geq (m+1)!$. Using the fact that $f(n)$ is special with $j = (m+1)!$, we have that $a_m m! < (m+1)!/2$, so $a_m < (m+1)/2$.

We now plug in for j numbers of the form $(m+1)!/k$ for some $k \mid (m+1)!$. Since $f(n)$ is special, we have that the remainder of $f(n)$ modulo $(m+1)!/k$ is less than $(m+1)!/2k$, which can be rewritten as

$$\left\{ \frac{a_m k}{m+1} \right\} < \frac{1}{2}.$$

In particular, for $k = 1$, we have $a_m < (m+1)/2$.

We now present 3 ways of concluding:

- We prove by induction on $\ell \in \{1, \ldots, m+1\}$ that $a_m < (m+1)/2\ell$. For $\ell = 1$, the statement is already proven.

  Induction step: if the inequality is true for some $\ell \in \{1, \ldots, m\}$, then we have that

  $$0 \leq \frac{a_m(\ell+1)}{m+1} < \frac{\ell+1}{2\ell} \leq 1.$$

  Since $\left\{ \frac{a_m(\ell+1)}{(m+1)} \right\} < \frac{1}{2}$, we have $\frac{a_m(\ell+1)}{(m+1)} < \frac{1}{2}$, as desired.

- By taking $k = m$, the fractional part of $a_m \cdot m/(m+1)$ is less than $1/2$, but it also is the fractional part of $a_m - a_m/(m+1)$, thus it is the fractional part of $1 - a_m/(m+1)$. Since $1 \geq 1 - a_m/(m+1) > 1/2$, we have $a_m = 0$.
- By taking $k = \lfloor m/a_m \rfloor$, we have $a_m k < m+1$, and $k > \frac{m}{a_m} - 1$. Therefore

$$a_m k \geq m + 1 - a_m > \frac{m+1}{2},$$

i.e. $\frac{1}{2} < \frac{a_m k}{m+1} < 1$ which is absurd.

Solutions

Problem 3

Problem 3. Let $\mathbb{R}$ be the set of real numbers. Determine all functions $f\colon \mathbb{R} \to \mathbb{R}$ such that for all real numbers x, y, the following holds:

$$f((f(x)+f(y))^2) = (x+y)f(x+y).$$

Solution

Answer: The solutions are $f(x) = x, f(x) = -x, f(x) = 0$.

Denote the equation by $(*)$. A straightforward check shows that all the above mentioned solutions satisfy the problem.

Step A.1: $f(0) = 0$

Taking $x = y = 0$ in $(*)$ we get that $f(4f(0)^2) = 0$.

Denote $T = f^{-1}(0)$ the set of all zeroes of f. In particular, $4f(0)^2 \in T$. Then putting $x = 0$, $y = 4f(0)^2$ in $(*)$ gives $f(f(0)^2) = 0$, hence $f(0)^2 \in T$. Now if $t_1, t_2 \in T$ we have that, by taking $y = t_1$ and $y = t_2$:

$$(x+t_1)f(x+t_1) = f(f(x)^2) = (x+t_2)f(x+t_2) \tag{1}$$

Plugging $x = -t_1$ in (1), we get that $t_2 - t_1 \in T$, so $4f(0)^2 - f(0)^2 = 3f(0)^2 \in T$, implying $3f(0)^2 - f(0)^2 = 2f(0)^2 \in T$.

Now put $x = y = f(0)^2$ in $(*)$ to obtain

$$f(0) = 2f(0)^2 f(2f(0)^2) = 0.$$

Step A.2: f is injective or $f(x) = 0$.

Take $y = 0$ in $(*)$ to arrive at:

$$f(f(x)^2) = xf(x) \tag{2}$$

In (2), replace $x \to f(x)^2$, so:

$$f(x^2 f(x)^2) = f(x)^2 f(f(x)^2) = xf(x)^3$$

Multiply both sides of $(*)$ by $(f(x)+f(y))^2$ to obtain:

$$(x+y)f(x+y)(f(x)+f(y))^2 = (f(x)+f(y))^2 f((f(x)+f(y))^2) = f(f((f(x)+f(y))^2)^2) =$$
$$= f((x+y)^2 f(x+y)^2) = (x+y)f(x+y)^3$$

So for all $x \neq -y$:

$$f(x+y)(f(x+y) - f(x) - f(y))(f(x+y) + f(x) + f(y)) = 0 \tag{3}$$

We now claim that $T = \{0\}$ or $T = \mathbb{R}$. Assume not and take a non-zero $t \in T$. By (1) we have that $(0-t) - t = -2t$ is also in T.

Plug $x \notin T$ and $y = -2t$ in (3):

i) If $f(x+y)=0$, then $x+y\in T$, so $x=x+y-y\in T$, absurd.

ii) If $f(x+y)=f(x)+f(y)=f(x)$, we use (1) to write $(x+y)f(x+y)=xf(x)$, which then becomes $xf(x)=(x+y)f(x)$. Hence $f(x)=0$, contradicting $x\notin T$.

iii) If $-f(x+y)=f(x)+f(y)=f(x)$, once again we use (1) to write $(x+y)f(x+y)=xf(x)$, so that $xf(x)=-(x+y)f(x)$, so either $2x=-y=2t$ or $f(x)=0$, contradicting $t\in T$ and $x\notin T$.

Hence the claim is proven. To conclude, assume f is not identically 0, so that $T=\{0\}$. Assume $f(a)=f(b)$ for some a,b. If a or b is zero, then since $T=\{0\}$, the other is also zero so $a=b=0$. Otherwise, we write that $af(a)=f(f(a)^2)=f(f(b)^2)=bf(b)$, so $(a-b)f(a)=0$, i.e. $a=b$ because $f(a)\neq 0$.

Step B: f is additive.

Using $(*)$ for (x,y) and $(x+y,0)$ gives

$$f((f(x)+f(y))^2)=(x+y)f(x+y)=f(f(x+y)^2)$$

since $f(0)=0$. So by injectivity, we have for all real x,y, that $f(x+y)^2=(f(x)+f(y))^2$, i.e.

$$f(x+y)=f(x)+f(y)\quad\text{or}\quad -f(x+y)=f(x)+f(y) \tag{4}$$

Taking $y=-x$, we obtain $-f(x)=f(-x)$ for all real numbers x.

Assume that for some real x,y:

$$f(x+y)\neq f(x)+f(y) \tag{5}$$

Then we must have $-f(x+y)=f(x)+f(y)$. Consider $x\to x+y, y\to -y$ in (4):

i) If $f(x+y-y)=f(x+y)-f(y)$, then $f(x)+f(y)=f(x+y)$, contradiction with (5).

ii) If $-f(x+y-y)=f(x+y)-f(y)$, then combining with $-f(x+y)=f(x)+f(y)$ gives $f(y)=0$, hence $y=0$, so $f(x)+f(y)=f(x+y)$, contradiction with (5).

Step C: $f(x)=x$ and $f(x)=-x$ are the remaining solutions.

We know that f is additive, injective and satisfies (2). We will show that the above mentioned solutions are the only ones that satisfy these three conditions.

By plugging in $x=1$ into (2), we get $f(f(1)^2)=f(1)$, so by injectivity $f(1)^2=1$. If f is a solution of the equation, then $-f$ is a solution as well, so without loss of generality, we can assume $f(1)=1$. Consider $x\to x+1$ in (2). Then:

$$f(f(x)^2+2f(x)+1)=xf(x)+f(x)+x+1$$

So by subtracting $f(f(x)^2)=xf(x)$ (equation (2)) we get

$$2f(f(x))=f(x)+x. \tag{6}$$

Switching $x\to f(2x)$ in (2) we obtain:

$$\begin{aligned} f\Big(f(f(2x))^2\Big)&=f(2x)f(f(2x))\\ f\Big((2f(f(x)))^2\Big)&=4f(x)f(f(x))\\ f(f(x)^2+2xf(x)+x^2)&=2f(x)^2+2xf(x)\\ f(f(x)^2)+2f(xf(x))+f(x^2)&=2f(x)^2+2xf(x)\end{aligned}$$

where we used (6) to go from the second to the third line. Using (2) and

$$2f(xf(x)) \overset{(2)}{=} 2f\Big(f(f(x)^2)\Big) \overset{(6)}{=} f(x)^2 + f(f(x)^2) \overset{(2)}{=} f(x)^2 + xf(x),$$

the previous equation becomes:

$$xf(x) + (f(x)^2 + xf(x)) + f(x^2) = 2f(x)^2 + 2xf(x)$$

and so

$$f(x^2) = f(x)^2 \geq 0$$

Hence f is non-negative for all non-negative values of x, so it is well-known it is linear. Combining with $f(1) = 1$ (and the fact that $-f$ is also a solution) we get the desired solution set.

Alternate solution

We give different solutions for each of the steps.

Alternate step A: Let us start by showing injectivity. Assume that $f(x) = f(x')$ and use $(*)$ with $(x, 0)$ and $(x', 0)$ to get

$$xf(x) = f((f(x) + f(0))^2) = x'f(x').$$

If $f(x) \neq 0$, one may divide by $f(x)$ on both sides to get $x = x'$ as desired. It therefore remains to consider the preimages of 0. Our goal will be to show that either there is at most one, or the function is identically zero. Therefore start by assuming that there are r, s distinct such that $f(r) = f(s) = 0$. Applying the condition on f to (r, y) and (s, y), one gets

$$(r + y)f(r + y) = (s + y)f(s + y)$$

so that $x \mapsto xf(x)$ is $(s - r)$-periodic. For $x = 0$ this function is clearly zero, so $(s - r)f(s - r) = 0$ and so $f(s - r) = 0$. Then repeating the argument with r, s replaced with s, r we get $f(r - s) = 0$. Using $(*)$ with $(s - r, r - s)$ gives that $f(0) = 0$.

Now assume by contradiction that there is some a such that $f(a) \neq 0$ (notably $a \neq 0$). Recall that $xf(x)$ is T-periodic for some $T > 0$. Apply $(*)$ to $(T, a - T)$ and $(0, a)$. Then both right-hand sides are equal and non zero so that by the "almost injectivity" of f proved at the start,

$$(f(T) + f(a - T))^2 = (f(0) + f(a))^2$$
$$f(a - T)^2 = f(a)^2$$

But since $(a - T)f(a - T) = af(a)$, this is a contradiction (unless $a = \frac{T}{2}$ in which case one simply runs the argument with the period T replaced by $2T$). Hence, f is identically 0.

Assume now that f is injective. Applying $(*)$ to $(0, 0)$ gives $f(4f(0)^2) = 0$. Then, applying the condition to $(0, 4f(0)^2)$ gives $f(f(0)^2) = 0$. By injectivity, one must have $f(0) = 0$ in this case as well. Note that equation (2) follows from $f(0) = 0$.

Alternate step B: As in the other proof of step B, one proves that for all x, y,

$$(f(x) + f(y))^2 = f(x + y)^2 \tag{4}$$

$$f(-x) = -f(x).$$

Take $x \neq 0$. Using the above equation with (x, x) gives $f(2x) = \pm 2f(x)$. Assume by contradiction that $f(2x) = -2f(x)$. Then applying (4) with $(2x, x)$ would give $f(3x) = \pm f(x)$, and reapplying (4) with $(3x, x)$ gives $f(4x) \in \{0, \pm 2f(x)\}$. However, applying (4) with $(2x, 2x)$ gives $f(4x) = \pm 4f(x)$ so this is impossible since $f(x) \neq 0$. Therefore, for all x, we have $f(2x) = 2f(x)$.

Now let x and y be arbitrary. If $y = 0$ or $y = -x$, the equation $f(x + y) = f(x) + f(y)$ holds because of $f(0) = 0$ or $f(-x) = -f(x)$. Otherwise, assume that one has $f(x+y) = -f(x) - f(y)$. Then using (4) with $(x + y, y)$ gives $f(x + 2y) = \pm f(x)$. But on the other hand, using (4) with $(x, 2y)$ gives $f(x+2y) = \pm(f(x)+2f(y))$ which implies either $f(y) = 0$ or $f(y) = -f(x)$. By injectivity, this can only hold if $y = 0$ or $y = -x$, which we already treated. Hence, f satisfies the Cauchy equation $f(x + y) = f(x) + f(y)$.

Second alternate step B: As in the other proofs of step B, we arrive at

$$f(x + y)^2 = (f(x) + f(y))^2 \tag{4}$$

and $f(-x) = -f(x)$. Applying this equation to the pairs $x \to x, y \to y - x$ and $x \to -x, y \to y + x$ gives

$$f(y)^2 = (f(x) + f(y - x))^2 = (f(-x) + f(y + x))^2.$$

Expanding the second equality gives

$$\begin{aligned} 2f(x)f(y - x) + f(y - x)^2 &= -2f(x)f(y + x) + f(y + x)^2 \\ 2f(x)(f(x + y) - f(x - y)) &= (f(x + y) + f(x - y))(f(x + y) - f(x - y)). \end{aligned}$$

Assuming $y \neq 0$, injectivity gives that $f(x + y) - f(x - y) \neq 0$ so

$$2f(x) = f(x + y) + f(x - y).$$

Applying this equation successively with $x \to \frac{x+y}{2}, y \to \frac{x-y}{2}$ and $x \to \frac{x+y}{2}, y \to \frac{x+y}{2}$ gives

$$f(x) + f(y) = 2f(\tfrac{x+y}{2}) = f(x + y) + f(0)$$

and so $f(x + y) = f(x) + f(y)$ for all x, y.

Alternate step C: As in the first proof of step C, we prove $f(1) = \pm 1$, and so without loss of generality, $f(1) = 1$. We also follow the previous proof to obtain

$$f(f(x)) = \tfrac{1}{2}(x + f(x)). \tag{6}$$

Using linearity, setting $r = f(x) - x$ gives $f(r) = -\frac{1}{2}r$. In order to prove that $f(x) = x$, it remains to show that $r = 0$. To do this, notice that (6) can be rewritten as $f(2f(x) - x) = x$, so that the inverse function of $f(x)$ is $2f(x) - x$. Plugging in $x = 2f(a) - a$ into (2) gives that for all a,

$$f(a^2) = a(2f(a) - a). \tag{7}$$

Plugging in $a + b$ to (7) and subtracting the two equations (7) with a and b gives the more general version

$$f(ab) = af(b) + bf(a) - ab.$$

from which plugging in $b = a^2$ and using (7) gives

$$f(a^3) = a^2(3f(a) - 2a). \tag{8}$$

It is now time to use our r such that $f(r) = -\frac{1}{2}r$. Plugging in $a = r$ to (7) gives

$$f(r^2) = r(-2 \cdot \tfrac{1}{2}r - r) = -2r^2$$

and then plugging in $a = r^{2/3}$ to the equation (8) gives

$$f(r^2) = r^{4/3}(3f(r^{2/3}) - 2r^{2/3}) = 3r^{4/3}f(r^{2/3}) - 2r^2.$$

Combining both equations gives $r^{4/3}f(r^{2/3}) = 0$ and hence $r = 0$. This concludes the proof.

Problem 4, EGMO 2026

P4

Let $1 = a_1 \geq a_2 \geq a_3 \geq \cdots$ be an infinite sequence of real numbers such that $a_n = a_{2n} + a_{2n+1}$ for all positive integers n. For $r = 2026^{2026}$, prove that

$$\frac{1}{r} \leq a_r \leq \frac{2}{r+1}.$$

Solutions

We first state and prove three auxiliary lemmas, which are used later in the proofs.

Lemma 1 (Lemma A)**.** *For every $n \in \mathbb{N}$, we have*

$$a_n + a_{n+1} + \ldots + a_{2n-1} = 1.$$

Proof. We consider the equalities $a_k = a_{2k} + a_{2k+1}$ for $k \in \{1, \ldots, n-1\}$. Taking their sum gives us

$$a_1 + a_2 + \ldots + a_{n-1} = a_2 + a_3 + \ldots + a_{2n-2} + a_{2n-1}.$$

Together with $a_1 = 1$, this implies

$$a_n + \cdots + a_{2n-1} = (a_2 + a_3 + \ldots + a_{2n-2} + a_{2n-1}) - (a_2 + \ldots + a_{n-1}) = 1.$$

□

Lemma 2 (Weaker version of Lemma A)**.** *For every $k = 2^n$, $n \in \mathbb{N}$,*

$$a_k + a_{k+1} + \ldots + a_{2k-1} = 1.$$

Remark : Showing Lemma A for a specific range $[\![n, 2n-1]\!]$ containing $r+1$ counts as a weaker version of Lemma A.

Proof. By the problem statement, the lemma holds for $k = 1$. We proceed by induction and assume that the statement holds for some $k = 2^l$ with

$$a_{2^l} + a_{2^l+1} + \ldots + a_{2^{l+1}-1} = 1.$$

We can expand each term in the left-hand side and obtain

$$(a_{2^{l+1}} + a_{2^{l+1}+1}) + (a_{2^{l+1}+2} + a_{2^{l+1}+3}) + \ldots + (a_{2^{l+2}-2} + a_{2^{l+2}-1}) = 1,$$

which is precisely the statement of the lemma for $k = 2^{l+1}$. □

Lemma 3 (Lemma B)**.** *For every $k \in \mathbb{N}$ and every $m \leq 2k$, we have*

$$a_m \geq \frac{1}{2} a_k.$$

Proof. Since $a_k = a_{2k} + a_{2k+1}$ and $a_{2k} \geq a_{2k+1}$, we get that

$$a_k \leq 2a_{2k}.$$

Therefore, for any $m \leq 2k$, we have

$$a_m \geq a_{2k} \geq \frac{a_k}{2}.$$

□

Proof of the lower bound

Using Lemma A for $n = r$, we get

$$a_r + \ldots + a_{2r-1} = 1.$$

Since the sequence is decreasing, each of the terms

$$a_{r+1}, \ldots, a_{2r-1}$$

is less than or equal to a_r, which implies

$$ra_r \geq a_r + \ldots + a_{2r-1} = 1.$$

Therefore,

$$a_r \geq \frac{1}{r}.$$

Proof of the upper bound

Using Lemma A for $n = \frac{r}{2} + 1$, we get

$$a_{r/2+1} + \cdots + a_{r+1} = 1.$$

We note that each of the terms

$$a_{r/2+1}, \ldots, a_{r-1}$$

is greater than or equal to a_r. This implies

$$\frac{r}{2}a_r + a_{r+1} = \left(\frac{r}{2} - 1\right) a_r + a_r + a_{r+1} \leq a_{r/2+1} + \cdots + a_{r+1} = 1.$$

Together with

$$a_{r+1} \geq \frac{1}{r+1},$$

which can be shown analogously to the proof of the lower bound for r above, we get

$$a_r \leq \frac{2}{r}\left(1 - \frac{1}{r+1}\right) = \frac{2}{r+1}.$$

Remark : Instead of using the lower bound for a_{r+1} in the last step, we can use $a_{r+1} \geq \frac{a_r}{2}$.

Proof of the upper bound - solution 2

Using Lemma A for $n = r$, we get

$$a_r + a_{r+1} + \cdots + a_{2r-1} = 1.$$

Using Lemma B, we get that each of the terms $a_{r+1}, \ldots, a_{2r-1}$ is greater than or equal to $\frac{a_r}{2}$. Summing all of these inequalities implies that

$$a_r + a_{r+1} + \cdots + a_{2r-1} \geq a_r + (r-1)\frac{a_r}{2} = \frac{r+1}{2}a_r.$$

This gives us

$$a_r \leq \frac{2}{r+1}.$$

Remark : Alternatively, we can use Lemma A starting at $r+1$

$$a_{r+1} + \cdots + a_{2r} + a_{2r+1} = 1.$$

Since the average of the terms on the left hand side of the sum is greater than or equal to the average of the two smallest terms (a_{2r} *and* a_{2r+1}*), and* $a_{2r} + a_{2r+1} = a_r$*, we get* $a_r \leq \frac{2}{r+1}$*.*

Proof using Lemma 2 - solution 3

Using the weaker version of Lemma A, we get

$$a_m + \cdots + a_{r-1} + a_r + a_{r+1} + \cdots + a_{2m-1} = 1 \tag{1}$$

for some integer m such that $m \leq r \leq 2m-1$ (for instance for m a power of two, but the proof works for any such m; see also the remark below). Note that each of the terms

$$a_{r+1}, \ldots, a_{2m-1}$$

is smaller than or equal to a_r. Additionally, since $m, \ldots, r-1$ are all greater than $\frac{r}{2}$, by applying Lemma B, we get that each of the terms

$$a_m, \ldots, a_{r-1}$$

is smaller than or equal to $2a_r$. This, together with (1), gives us

$$2(r-m)a_r + a_r + (2m-r-1)a_r \geq 1,$$

which implies

$$a_r \geq \frac{1}{r}.$$

The upper bound follows similarly using (1),

$$a_m, \ldots, a_{r-1} \geq a_r$$

and

$$a_{r+1}, \ldots, a_{2m-1} \geq \frac{a_r}{2}.$$

The latter follows from Lemma B since $r+1, \ldots, 2m-1 < 2r$.

Remark. *The proof above for* $m = r$ *and* $m = r/2 + 1$ *gives the previous proofs. In that case, the use of Lemma B is reduced to the minimum.*

Proof without any form of Lemma A - solution 4

For $r \in \mathbb{N}$, define a function f_r by

$$f_r(k) = \begin{cases} 1, & r \leq k \leq 2r, \\ f_r(2k) + f_r(2k+1), & k < r. \end{cases}$$

Note that for all $r \leq k \leq 2r$, $a_k \leq a_r = f_r(k)a_r$. Since $f_r(k)$ sums up in the same manner as the original sequence a_k, we have that $a_k \leq f_r(k)a_r$ for $k < r$. To show the lower bound, we aim to prove that $f_r(1) = r$, which would imply $1 = a_1 \leq ra_r$.

Let n be an integer such that $2^n \le r < 2^{n+1}$. We want to work with the binary representation of r and consider how $f_r(\cdot)$ behaves when we perform a bit-shift on r. More formally, let us define the sequences $(q_i)_{i\in[\![0,n]\!]}$ and $(s_i)_{i\in[\![0,n]\!]}$ as the quotient and the remainder of r by 2^i, $i \in [\![0,n]\!]$. Hence, for $i \in [\![0,n]\!]$, we have $r = q_i 2^i + s_i$ and (a) $0 \le s_i < 2^i$, (b) $s_i = 2^{i-1} + s_{i-1}$ if q_{i-1} is odd, (c) $s_i = s_{i-1}$ if q_{i-1} is even, and (d) $q_i = \left\lfloor \frac{q_{i-1}}{2} \right\rfloor$, that also implies the sequence q_i is strictly decreasing.

We now use descending induction on k from n to 1 to show that

$$f_r(k) = \begin{cases} 2^i + s_i, & k = q_i, \\ 2^i, & q_i < k < q_{i-1}. \end{cases}$$

Base. We have $q_0 = r$ and $s_0 = 0$, hence, $f(r) = 1 = 2^0 + s_0$.

Induction step. Assume that the hypothesis holds for all l in $[\![k+1, r]\!]$ and let us prove it for k. We distinguish three cases:

- $\mathbf{q_i < k < q_{i-1}}$. If $i = 1$, then $f_r(k) = f_r(2k) + f_r(2k+1) = 2^1$ as $k > q_i = \left\lfloor \frac{r}{2} \right\rfloor$. Otherwise, using the property (d), we get $q_{i-1} < 2k < 2k+1 < q_{i-2}$ and $f(k) = f(2k) + f(2k+1) = 2^{i-1} + 2^{i-1} = 2^i$.
- $\mathbf{k = q_i}$ and $\mathbf{q_{i-1}}$ **is odd**. Then, $f_r(k) = f_r(q_{i-1} - 1) + f_r(q_{i-1}) = 2^i + 2^{i-1} + s_{i-1} = 2^i + s_i$.
- $\mathbf{k = q_i}$ and $\mathbf{q_{i-1}}$ **is even**. Then, $f_r(x) = f_r(q_{i-1}) + f_r(q_{i-1} + 1) = 2^{i-1} + s_{i-1} + 2^{i-1} = 2^i + s_i$.

Since $q_n = 1$, then $s_n = r - 2^n$, and $f_r(1) = 2^n + r - 2^n = r$.

For the upper bound, we define a similar function g_r such that for all k, $a_k \ge g_r(k) a_r$. The initial conditions are $g_r(k) = 1$ for $\frac{r}{2} < k \le r$, and $g_r(k) = \frac{1}{2}$ for $r < k \le 2r$, which come from Lemma B. Additionally, we have the same summing pattern $g_r(k) = g_r(2k) + g_r(2k+1)$. We then notice that the functions $2 \cdot g_r$ and f_{r+1} (introduced analogously to f_r) are defined by the same initial condition and the same recurrent formula, thus, $2 \cdot g_r(1) = f_{r+1}(1) = \frac{1}{r+1}$. Therefore, $a_r \le \frac{2}{r+1}$.

Remark. *Both estimates in the problem are sharp. To construct a sequence attaining the lower bound $a_r = \frac{1}{r}$, set*

$$a_r = a_{r+1} = \cdots = a_{2r-1} = \frac{1}{r}$$

and extend the sequence to the left by setting $a_n = a_{2n} + a_{2n+1}$ and to the right by setting $a_{2n} = a_{2n+1} = \frac{a_n}{2}$. It is straightforward to check that the resulting sequence is non-increasing. Moreover, because $a_n = a_{2n} + a_{2n+1}$ again implies that $a_n + a_{n+1} + \ldots + a_{2n-1}$ is constant, it follows that $a_1 = a_r + a_{r+1} + \cdots + a_{2r-1} = 1$, so the sequence satisfies the problem conditions and has $a_r = \frac{1}{r}$ by construction.

To construct a sequence attaining the lower bound $a_r = \frac{2}{r+1}$, we similarly set

$$a_{r+1} = a_{r+2} = \cdots = a_{2r+1} = \frac{1}{r+1}$$

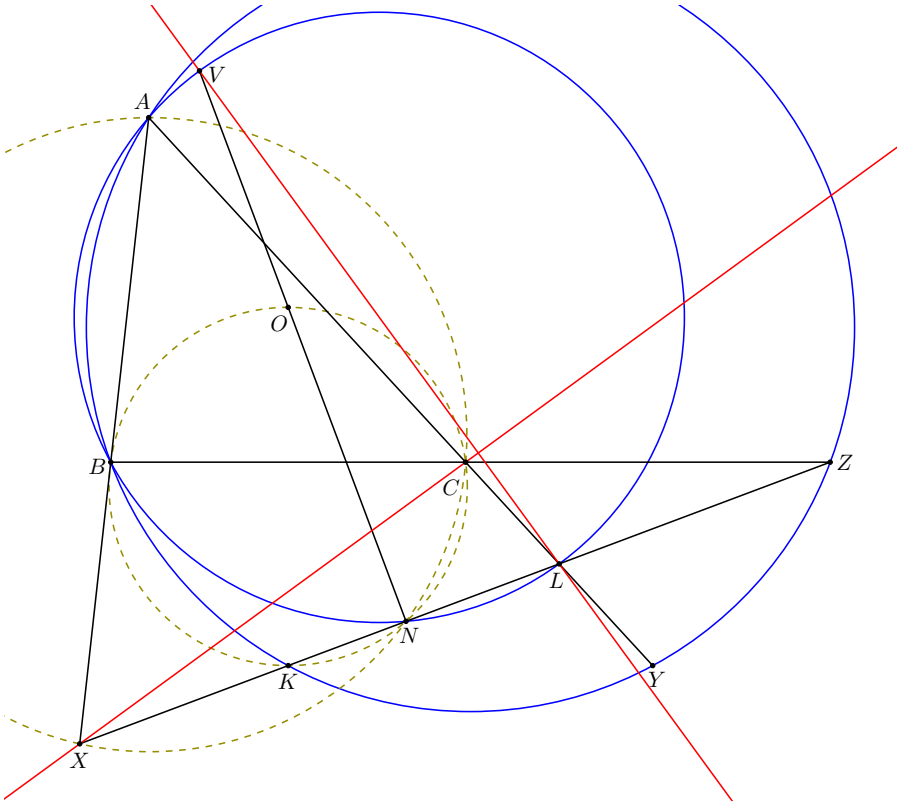
and extend it in the same way. The proof that this sequence satisfies the problem conditions is similar. Moreover, we have $a_r = a_{2r} + a_{2r+1} = 2 \cdot \frac{1}{r+1}$, as desired.

Problem 5, EGMO 2026

P5

Let ABC be an acute triangle with $AC > AB$. Denote by ω its circumcircle and by O its circumcentre. Let K be the intersection of the tangents to ω at B and C. Circle ABK intersects line BC again at $Z \neq B$. Let L be the midpoint of KZ. Let X be the intersection of lines KZ and AB. Let V be the point on circle ABL on the same side of BC as A such that OV is perpendicular to KZ. Prove that LV is perpendicular to CX.

Solutions



We split the proof in three claims; we show in turn.

➢ **Claim 1.** The point L is on the line AC.

➢ **Claim 2.** The foot of the altitude from O to KZ is on the circle ABL.

➢ **Claim 3.** The lines VL and CX are orthogonal.

Each claim relies on the previous ones. For each of the claim we give multiple proofs.

We denote by M the midpoint of BC, by N the foot of the perpendicular from O to KZ, by Y the intersection of the line AC with the circle ABK and by U the intersection of the line AK with the circle ABC.

Claim 1. *The point L is on the line AC.*

Proof 1 of Claim 1. Extend the line AC to meet the circle ABK again at Y. From the alternate segment theorem, we have

$$\angle ZYA = \angle ZBA = \angle KCY \text{ and } \angle AYK = 180^\circ - \angle KBA = \angle ACB.$$

Hence $ZY \parallel KC$ and $KY \parallel CZ$ so $KCZY$ is a parallelogram. In particular, as L is the midpoint of KZ, it is also the midpoint of CY. Hence L lies on AC. □

Proof 2 of Claim 1. Let AC intersect KZ at L'. Then we have

$$\angle AZL' = \angle AZK = 180^\circ - \angle KBA = \angle ACB = 180^\circ - \angle ZCA$$

and

$$\angle L'KA = \angle ZKA = \angle ZBA = 180^\circ - \angle ACK.$$

These two equalities show that the line KZ is tangent to both circles ACZ and ACK so, by power of point,

$$L'K^2 = L'C \cdot L'A = L'Z^2.$$

This implies $L'K = L'Z$ and thus L' is the midpoint of KZ. We have proved that $L' = L$. □

Proof 3 of Claim 1. Let L^* be the second intersection of the line AC with the circle MAZ. Then,

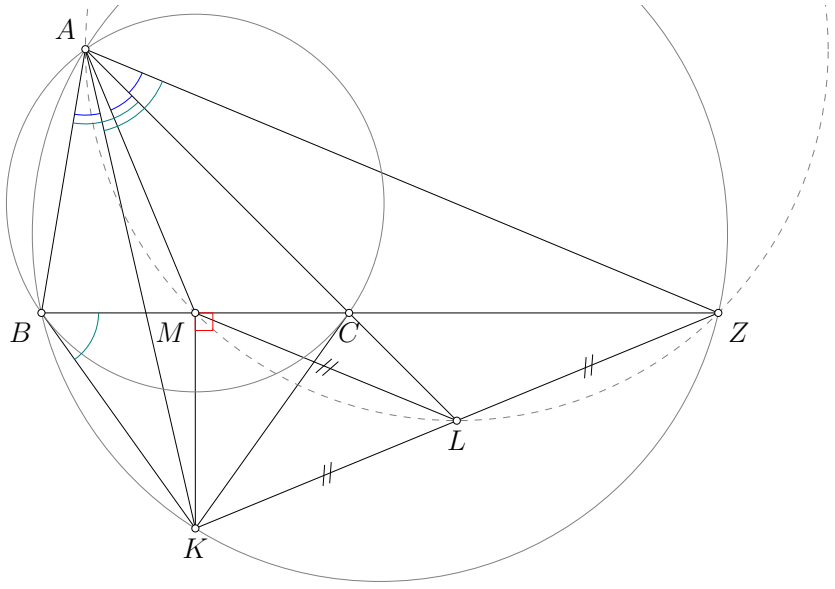
$$\angle BZL^* = \angle MAL^* = \angle BAK = \angle BZK,$$

where the second equality comes from the fact that AK is the A-symmedian of ABC. This means that K, Z and L^* are colinear.

Moreover, $\angle KAZ = \angle KBZ = \angle BAC$, so

$$\angle CAZ = \angle KAZ - \angle KAC = \angle BAC - \angle KAC = \angle BAK = \angle MAC.$$

Hence AL^* is the angle bisector of $\angle MAZ$, and $L^*M = L^*Z$. Since $KM \perp MZ$ we also have $KL^* = L^*Z$, so $L^* = L$.



□

Proof 4 of Claim 1. We prove that the triangles ABC and AKZ are similar. This can be proven in many ways but we give a simple solution by angle-chasing. We have indeed

$$\angle AKZ = \angle ABZ = \angle ABC \text{ and } \angle KAZ = \angle KBZ = \angle BAC.$$

Let M be the midpoint of BC. Since M and L are the midpoints of the sides BC and KZ this implies that $ABCM$ and $AKZL$ are similar and so $\angle MAC = \angle LAZ$.

We know that the line AK is the A-symmedian in ABC which means $\angle BAK = \angle MAC$. We have

$$\angle BAK + \angle KAZ = \angle BAZ = \angle BAC + \angle CAZ.$$

Since $\angle KAZ = \angle BAC$, this means that

$$\angle LAZ = \angle MAC = \angle BAK = \angle CAZ$$

and A, C and L are colinear. □

Proof 5 of Claim 1. As L is the midpoint of KZ, proving that A, C and L are colinear is equivalent to proving that the triangles AKC and ACZ have same area. We have now,

$$\begin{aligned} & [AKC] = [ACZ] \\ \Leftrightarrow \quad & \frac{AK \cdot AC \cdot \sin(\angle KAC)}{2} = \frac{AZ \cdot AC \cdot \sin(\angle CAZ)}{2} \\ \Leftrightarrow \quad & AK \cdot \sin(\angle KAC) = AZ \cdot \sin(\angle CAZ). \end{aligned}$$

We have $\angle KAZ = \angle KBZ = \angle KBC = \angle BAC$, so $\angle BAK = \angle CAZ$. This means that we want to prove that

$$\frac{AK}{AZ} = \frac{\sin(\angle BAK)}{\sin(\angle KAC)}.$$

We have

$$\angle AKZ = \angle ABZ = \angle ABC \text{ and } \angle KAZ = \angle KBZ = \angle BAC,$$

so the triangles ABC and AKZ are similar. This means that

$$\frac{AK}{AZ} = \frac{AB}{AC} = \frac{BU}{CU} = \frac{\sin(\angle BAU)}{\sin(\angle CAU)} = \frac{\sin(\angle BAK)}{\sin(\angle CAK)},$$

where the second equality comes from the harmonicity of the points A, B, U and C. □

Proof 6 of Claim 1. Since $\angle CUA = \angle CBA = \angle ZKA$, we have that the lines UC and KZ are parallel. Then, using the fact that the points A, U, B and C are harmonic we can project from C to the line KZ in order to obtain that the points K, Z, $CU \cap KZ$ and $AC \cap KZ$ are harmonic. From the parallelity, we have that $CU \cap KZ = \infty_{KZ}$ and thus AC intersect KZ in its midpoint. □

Proof 7 of Claim 1. One has that

$$\angle LKA = \angle ZKA = \angle ZBA = 180^\circ - \angle ACK,$$

hence, the line KZ is tangent to ACK and this means that $\angle CKA = \angle CAK$. Moreover,

$$\angle KCZ = 180^\circ - \angle BCK = 180^\circ - \angle BAK = 180^\circ - \angle KBC = 180^\circ - \angle KAZ.$$

We have thus proven that C is the A-Humpty point of triangle AZK which implies that the points A, C and L are colinear. □

Claim 2. *The foot N of the altitude from O to KZ is on the circle ABL.*

Proof 1 of Claim 2. Since $\angle OCK = \angle ONK = \angle KBO = 90°$, we have that $OCNKB$ is cyclic. From this, we get,

$$\angle BNK = \angle BCK = \angle BAC = \angle BAL,$$

where the last equality follows from the fact that L lies on the line AC (by Claim 1). □

Proof 2 of Claim 2. We start by proving the following claim.

Claim 2.1. *The circumcircles of ABC and AKL are tangent.*

Proof of Claim 2.1. Let t be the tangent to ABC at A. Then, $(t, AC) = \angle ABC = \angle AKL$. This finishes the proof of the claim. □

Now, we look at the inversion $\mathcal{I}$ of centre X and radius $r = \sqrt{XB \cdot XA}$. The inversion $\mathcal{I}$ exchanges B and A. Since, $ABKZ$ is cyclic, the inversion $\mathcal{I}$ also exchanges K and Z and in order to prove Claim 2, we need to show that the points L and N are also exchanged by $\mathcal{I}$. From Claim 2.1, and using the inversion $\mathcal{I}$, we just have to show that the circumcircles of BNZ and ABC are tangent.

This last fact follows from

$$KB^2 = KM \cdot KO = KN \cdot KZ,$$

where we used that $OMNZ$ is cyclic as $\angle OMZ = \angle ONZ = 90°$ in the last equality. □

Proof 3 of Claim 2. We prove the following.

Claim 2.2. *The points B, U and N are colinear.*

Proof 1 of Claim 2.2. We define N' as the intersection of BU with the perpendicular from O to KZ. We have first that $\angle CUA = \angle ABC = \angle AKZ$, which means that $UC \parallel KZ$ and $ON' \perp CU$. We prove now that $BKN'O$ is cyclic as

$$\angle BN'O = 90° - \angle CUN' = 90° - \angle BAC = 90° - \angle CBK = \angle OKB.$$

Moreover, $OBKN$ is also cyclic since $\angle OBK = \angle ONK = 90°$. Since O, N' and N are colinear we are done. □

Proof 2 of Claim 2.2. We have that $OBKN$ is cyclic since $\angle OBK = \angle ONK = 90°$, as is $OMNZ$ since $\angle OMZ = \angle ONZ = 90°$. We have

$$\angle NBK = \angle NOK = \angle NOM = \angle NZM = \angle BAK = \angle UBK.$$

This finishes the proof of the claim. □

Since U is on the line BN and A, U, B and C are harmonic, by projecting from B to the line KZ we have that the points X, N, K and Z are in harmonic configuration. From MacLaurin's identity we have that

$$XN \cdot XL = XK \cdot XZ = XA \cdot XB.$$

This finishes the proof of Claim 2. □

Proof 4 of Claim 2. We remark that we have that $OBKN$ is cyclic since $\angle OBK = \angle ONK = 90^\circ$, as is $OMNZ$ since $\angle OMZ = \angle ONZ = 90^\circ$.

Let t be the tangent line to the circle ABC in A. It is well-known that the lines t, AK, AB and AC are in harmonic configuration. However, we have

$$(t, AB) = \angle ACB = \angle XBK,\ \angle BAC = \angle KBC$$

and

$$\angle KBN = \angle NOK = \angle NOM = \angle NZM = \angle BAK,$$

which means that the lines angles between the quadruple of lines BX, BN, BK and BZ and the quadruple of lines t, AK, AB and AC are the same. Hence, the lines BX, BN, BK and BZ are also in harmonic configuration and by projecting on line KZ, that the points X, N, K and Z are in harmonic configuration. □

Remark. *The proof number 3 and 4 of Claim 2., that are very similar in spirit, do not use claim 1.*

Claim 3. *The lines VL and CX are orthogonal.*

Proof 1 of Claim 3. We start by proving the following claim.

Claim 3.1. *The quadrilateral $ACNX$ is cyclic.*

Proof 1 of Claim 3.1. Since $\angle KBO = \angle KCO = \angle KNO = 90°$, we have $OCNKB$ cyclic. So

$$\angle LNC = 180° - \angle CNK = \angle KBC = \angle BAC = \angle XAC$$

proving the claim. □

Proof 2 of Claim 3.1. We first notice that since

$$\angle LCZ = \angle ACB = 180° - \angle ABK = \angle AZL,$$

we have that the circle ACZ is tangent to the line ZK.

We have that $ABNL$ and $ABKZ$ are cyclic, hence by power of a point,

$$XN \cdot XL = XA \cdot XB = XK \cdot XZ.$$

Since L is the midpoint of segment KZ, from Newton's identity we have that X, N, K and Z are harmonic. Using now MacLaurin's identity and the above tangency, we have $LN \cdot LX = LZ^2 = LC \times LA$, which finishes the proof of the claim. □

We are now done as, from the above claims and $VN \perp KZ$,

$$\angle LXC = \angle NXC = \angle NAC = \angle NAL = \angle NVL = 90° - \angle VLN = 90° - \angle VLX$$

which gives $CX \perp LV$. □

Proof 2 of Claim 3. We prove a much needed cyclicity.

Claim 3.2. *The quadrilateral $MAZL$ is cyclic.*

Proof 1 of Claim 3.2. We have

$$\angle KZM = \angle KZB = \angle KAB = \angle CAM,$$

where the last equality comes from the fact that the line AK is the symmedian. □

Proof 2 of Claim 3.2. The third proof of Claim 1 already proves Claim 3.2. □

Now we can finish the proof of Claim 3 as follows,

$$CA \cdot CL = CM \cdot CZ$$

and

$$XB \cdot XA = XK \cdot XZ,$$

so that CX is the radical axis of the circles $ABNL$ and MKZ. Letting O' be the center of $ABNL$, we obtain that $O'L \perp CX$, and since VL is a diameter of $ABNL$, we finally obtain $VL \perp CX$. □

Proof 3 of Claim 3. We start by a small claim.

Claim 3.3. *The point L is the midpoint of CY.*

Proof 1 of Claim 3.3. We already know that L is on the line CY by Claim 1. We have thus,

$$\angle LYZ = \angle AYZ = \angle ABC = \angle KCL.$$

This implies that the lines CK and YZ are parallel and since $KL = LZ$, it implies that $CL = LY$. □

Proof 2 of Claim 3.3. Proof 1 of Claim 1 also proves Claim 3.3. □

Now we can finish the proof of Claim 3 as follows,

$$2CA \cdot CL = CA \cdot CY = CB \cdot CZ = 2CM \cdot CZ$$

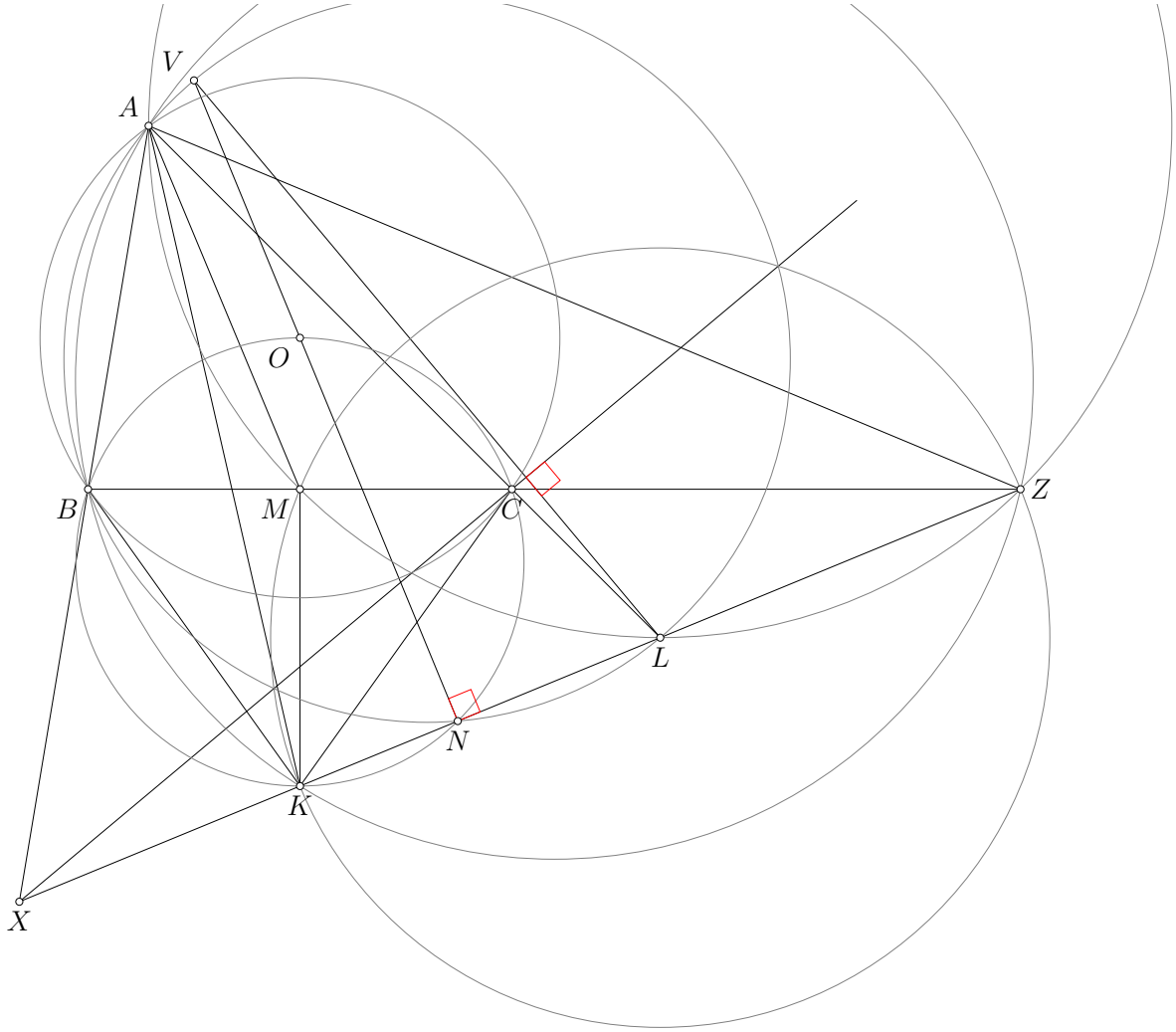
and

$$XB \cdot XA = XK \cdot XZ,$$

so that CX is the radical axis of the circles $ABNL$ and MKZ. Letting O' be the center of $ABNL$, we obtain that $O'L \perp CX$, and since VL is a diameter of $ABNL$, we finally obtain $VL \perp CX$.

□

Proof 4 of Claim 3. □



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Problem 6, EGMO 2026

P6

Let p be a prime number and let n be a positive integer such that p does **not** divide n. Denote by k the number of positive divisors of n, and by $1 = d_1 < d_2 < \cdots < d_k = n$ the positive divisors of n. For $i = 1, 2, \ldots, k$, let c_i be the number of positive divisors ℓ of d_i^2 such that $d_i - \ell$ is divisible by p. Prove that

$$(p-1)\,(c_1 + c_2 + \cdots + c_k) \geq k^2.$$

Solutions

Solution 1

Claim 1.1: The number of ordered pairs of divisors (d, d') such that $d \equiv d' \mod p$ (where d may be equal to d') at least $k^2/(p-1)$.

Proof of Claim 1.1: For each $t \in \{1, 2 \ldots p-1\}$, let N_t be the number of d congruent to t mod p. Consider the divisors of n modulo p. We know that $k = N_1 + N_2 + \ldots + N_{p-1}$, since $p \nmid n$, and the number of ordered pairs (d, d') such that $d \equiv d' \mod p$ is $N_1^2 + N_2^2 + \ldots + N_{p-1}^2$. By the arithmetic mean-quadratic mean inequality, we have

$$N_1^2 + N_2^2 + \ldots + N_{p-1}^2 \geq \frac{(N_1 + N_2 + \ldots + N_{p-1})^2}{p-1} = \frac{k^2}{p-1}.$$

■

Now it suffices to prove that the number of pairs (d_i, ℓ) is at least the number of ordered pairs of divisors (d, d') such that $d \equiv d' \mod p$. We do so by constructing for each such pair (d_i, d_j) a pair (a, b) with $a \mid n$, $b \mid a^2$ and $p \mid a - b$, such that $a|n$ and $b|a^2$ and all such pairs (a, b) are distinct.

For a pair (d_i, d_j) with $d_i \equiv d_j \mod p$ we define $a = \operatorname{lcm}(d_i, d_j)$ and $b = \frac{d_j^2}{\gcd(d_i, d_j)}$. Since $d_i, d_j \mid n$ we have $a = \operatorname{lcm}(d_i, d_j) \mid n$. Also, $b \mid d_j^2 \mid (\operatorname{lcm}(d_i, d_j))^2 = a^2$. Now note that $\operatorname{lcm}(d_i, d_j) = \frac{d_i d_j}{\gcd(d_i, d_j)}$ and hence $(a, b) = (d_i \cdot \frac{d_j}{\gcd(d_i, d_j)}, d_j \cdot \frac{d_j}{\gcd(d_i, d_j)})$. As $\frac{d_j}{\gcd(d_i, d_j)}$ is an integer and $p \mid d_i - d_j$, we also find $p \mid a - b$.

Now we just need to prove that all the pairs (a, b) we create in this way are distinct. Suppose (a, b) is created from both (d_i, d_j) and (d_i', d_j'). We have

$$\gcd(a, b) = \gcd\left(d_i \cdot \frac{d_j}{\gcd(d_i, d_j)}, d_j \cdot \frac{d_j}{\gcd(d_i, d_j)}\right) = \gcd(d_i, d_j) \cdot \frac{d_j}{\gcd(d_i, d_j)} = d_j,$$

while similarly $\gcd(a, b) = d_j'$, so $d_j = d_j'$. Since $\frac{d_i}{d_j} = \frac{a}{b} = \frac{d_i'}{d_j'}$ we now have $d_i = \frac{a}{b} \cdot d_j = \frac{a}{b} \cdot d_j' = d_i'$. We conclude that $(d_i, d_j) = (d_i', d_j')$. So all pairs (a, b) that we create are indeed distinct, which is the last part we had to prove. □

Remark. *The proof uses that* $(d_i, d_j) \mapsto (a, b) = \left(\text{lcm}(d_i, d_j), \frac{d_j^2}{\gcd(d_i, d_j)}\right)$ *defines an injection from the set of ordered pairs* (d_i, d_j) *with* $d_i, d_j \mid n$ *and* $p \mid d_i - d_j$ *to the set of ordered pairs* (a, b) *with* $a \mid n$, $b \mid a^2$ *and* $p \mid a - b$. *It is straightforward to show that this map is in fact a bijection, from which it follows that* $c_1 + \ldots + c_k = P$. *This also implies that equality occurs in* $(p-1)(c_1 + \ldots + c_k) \geq k^2$ *precisely when equality occurs in* $N_1^2 + \ldots + N_{p-1}^2 \geq (N_1 + \ldots + N_{p-1})^2/(p-1)$, *which happens when* $N_1 = \cdots = N_{p-1}$. *So equality holds in the desired inequality if and only if the divisors of* n *are equidistributed over the (nonzero) residue classes modulo* p.

Solution 2

Let $n = \prod_{j=1}^{s} q_j^{\alpha_j}$ be the prime factorisation of n. Then we can rephrase the conditions as follows:

- $d_i = \prod_{j=1}^{s} q_j^{\gamma_j}$ is s.t. $\gamma_j \leq \alpha_j$
- $\ell = \prod_{j=1}^{s} q_j^{\delta_j}$ is s.t. $\delta_j \leq 2\gamma_j$

Furthermore, let $r = d_i \ell^{-1} = \prod_{j=1}^{s} q_j^{\beta_j}$.

Claim 2.1: For any fixed r such that $r \equiv 1 \pmod{p}$, the number of pairs (d_i, ℓ) with d_i a divisor of n and ℓ a divisor of d_i^2 such that $r = \frac{d_i}{\ell}$ is the same as the number of pairs of divisors of n, (d, d') such that $rd = d'$.

Proof of Claim 2.1: We prove both are equal to $\prod_{j=1}^{s}(\alpha_j - |\beta_j| + 1)$. The number of pairs (d_i, ℓ), is the same as the number of d_i such that $0 \leq \gamma_i - \beta_i = \delta_i \leq 2\gamma_i$ and $\gamma_i \leq \alpha_i$ hence γ_j can take value in $\{|\beta_j|, \ldots \alpha_j\}$ hence $\prod_{j=1}^{s}(\alpha_j - |\beta_j| + 1)$ such pairs (d_i, ℓ) for each r.

The number of pairs (d, d') is the same as the number of $d = \prod_{j=1}^{s} q_j^{\gamma_j}$ such that for every j, $0 \leq \gamma_j, \gamma_j + \beta_j \leq \alpha_j$. Each γ_j can take any value in $\{0, \ldots \alpha_i - |\beta_i|\}$, hence $\prod_{j=1}^{s}(\alpha_j - |\beta_j| + 1)$ such pairs $(d_i, d_{i'})$ for each r. ■

Remark. *The claim holds without the congruence condition on* r.

Note that the sum $c_1 + \ldots + c_k$ is exactly the number of distinct pairs (d_i, ℓ) such that $r = d_i/\ell$ and $r \equiv 1 \pmod{p}$. By Claim 2.1, we know

$$c_1 + c_2 + \ldots + c_k = \text{Number of pairs of divisors } (d, d') \text{ such that } d \equiv d' \pmod{p}.$$

By Claim 1.1, we are done. □

Solution 3

This solution is similar in spirit to the previous two.

For any divisor d of n, let C_d denote the set of positive integers ℓ dividing d^2 such that $d \equiv \ell \pmod{p}$. We also define S_d as the set of pairs (u, v) of coprime divisors of d such that $u \equiv v \pmod{p}$. Finally, we let S denote the set of pairs of divisors (u', v') of n such that $u' \equiv v' \pmod{p}$.

Claim 3.1: There is a bijection between C_d and S_d.

Proof of Claim 3.1: We define $g_d : S_d \to C_d$ by $g_d((u, v)) = du/v$. This is well-defined as $v|d$ implies du/v is an integer, $u|d$ implies it divides d^2, and $du/v \equiv d \pmod{p}$ from the congruence condition on u and v. Reciprocally, $f_d : C_d \to S_d$ can be defined by sending $\ell \in C_d$ to the pair formed by the numerator and denominator of ℓ/d. ■

Remark. $g_d((u, v)) = du/v$ *is essentially the same as the map in Solution 1 the pairs of divisors* $d/u, d/v$, *is sent to* $d = lcm(d/u, d/v)$ *and* $\ell = du/v = g_d(u, v)$.

Claim 3.2: $c_1 + \ldots + c_k = \sum_{d|n} \#C_d = \sum_{d|n} \#S_d$.

Proof of Claim 3.2: This follows immediately from Claim 3.1 and the definition of the c_i. ■

Claim 3.3: $\sum_{d|n} \#S_d = \#S$.

Proof of Claim 3.3:

- We express the LHS in the following way. We fix a pair (u, v) of coprime divisors of n such that $u \equiv v$ and count the number of S_d in which it appears. Because of the coprimality condition, this is given by the number of divisors of $\frac{n}{uv}$, which we denote $\tau(\frac{n}{uv})$.
- Now look at the RHS. For each $(u', v') \in S$, note that defining $g := \gcd(u', v')$, $u = u'/g$ and $v = v'/g$, we have $(u, v) \in S_d$ for any d such that $uv|d$. The number of elements of S can again be counted by first looking at pairs (u, v) of possible coprime congruent divisors of n, which are obtained each for $\tau(\frac{n}{uv})$ possible values of d, which is exactly what was obtained previously.

■

We then conclude as in Solutions 1 and 2. □

Solution 4

Define $\alpha_j, \beta_j, \gamma_j, \delta_j$ as in Solution 2. Then, the bounds on $\alpha_j, \gamma_j, \delta_j$ are equivalent to $\alpha_j \geq \gamma_j \geq |\beta_j|$ for each j. $c_1 + \cdots + c_k$ is the number of such tuples such that $\prod_{j=1}^{s} q_j^{\beta_j} \equiv 1 \pmod p$.

First, we count the number of tuples without the last condition. Fixing each γ_j, there are $2\gamma_j + 1$ possible values of β_j. Hence, the number of such tuples is

$$\prod_j \sum_{\gamma_j=0}^{\alpha_j} (2\gamma_j + 1) = \prod_j (\alpha_j + 1)^2 = k^2. \tag{1}$$

Now, we show that out of such tuples, at least $\frac{1}{p-1}$ have $\prod_j q_j^{\beta_j} \equiv 1 \pmod p$. Let g be a primitive root modulo p. We count the number of tuples such that $\prod_j q_j^{\beta_j} \equiv g^a \pmod p$ for each $a = 0, \ldots, p-2$.

In the case where $n = q^\alpha$, where $q \equiv g^b \pmod p$, the possible values are $g^{b\beta}$, $\beta = -\alpha, -\alpha+1, \ldots, \alpha$, which have $1, 2, \ldots, \alpha+1, \ldots, 1$ possible values of γ respectively. Denote by $x^{a \pmod{p-1}}$ a term x^c, where $a \equiv c \pmod{p-1}$. Thus, the number of tuples such that $\prod_j q_j^{\beta_j} \equiv g^a \pmod p$ is the sum of the coefficients of the $x^{a \pmod{p-1}}$ terms in the expression

$$\left(x^{-\frac{\alpha b}{2}} + x^{-\frac{\alpha b}{2}+b} + \cdots + x^{\frac{\alpha b}{2}}\right)^2$$

(the sum in the bracket is a polynomial in $\sqrt{x}^{\pm}$, but either only has odd terms or even terms).

Hence, in the general case, the number of tuples such that $\prod_j q_j^{\beta_j} \equiv g^a \pmod p$ is the sum of the $x^{a \pmod{p-1}}$-coefficients of

$$\prod_j \left(\left(x^{-\frac{\alpha_j b_j}{2}} + x^{-\frac{\alpha_j b_j}{2}+b_j} + \cdots + x^{\frac{\alpha_j b_j}{2}}\right)^2\right) = \left(\prod_j \left(x^{-\frac{\alpha_j b_j}{2}} + x^{-\frac{\alpha_j b_j}{2}+b_j} + \cdots + x^{\frac{\alpha_j b_j}{2}}\right)\right)^2. \tag{2}$$

Expanding the term in the square, the coefficients are symmetric across 0. Thus, collapsing this modulo $p-1$, we are taking the sum of the $x^{a \pmod{p-1}}$-coefficients of some $\left(a_0 + a_1 x + \cdots + a_{p-2} x^{p-2}\right)^2$ or some $\left(a_{\frac{1}{2}} x^{\frac{1}{2}} + a_{\frac{3}{2}} x^{\frac{3}{2}} + \cdots + a_{p-\frac{3}{2}} x^{p-\frac{3}{2}}\right)^2$ where $a_m = a_{p-1-m}$ for each m.

When $a = 0$, this is (by QM-AM, and taking $a_{p-1} = a_0$)

$$\sum_m a_m a_{p-1-m} = \sum_m a_m^2 \geq \frac{1}{p-1}\left(\sum_m a_m\right)^2. \tag{3}$$

On the other hand, $\left(\sum_m a_m\right)^2$ is exactly the sum of all the coefficients, which is the total number of tuples, i.e. k^2. □

Remark. *This method also shows that the number of tuples such that* $r = \prod_{j=1}^{s} q_j^{\beta_j} \equiv 1 \pmod{p}$ *is at least the number of tuples such that* $r \equiv b \pmod{p}$ *for any other* b*: this boils down to showing* $\sum_m a_m^2 \geq \sum_m a_m a_{b-m}$ *(indices modulo* $p-1$*), which holds by rearrangement inequality.*

EGMO 2025

Problems and Solutions

Day 1

P1. For a positive integer N, let $c_1 < c_2 < \cdots < c_m$ be all the positive integers smaller than N that are coprime to N. Find all $N \geq 3$ such that

$$\gcd(N, c_i + c_{i+1}) \neq 1$$

for all $1 \leq i \leq m-1$.

Here $\gcd(a, b)$ *is the largest positive integer that divides both* a *and* b*. Integers* a *and* b *are coprime if* $\gcd(a, b) = 1$.

Solution 1. The answer is all even integers and all powers of 3. First we show that these work.

- When N is even, all c_i are odd, and so $2 \mid \gcd(N, c_i + c_{i+1})$ for every i.
- When N is a power of 3, the c_i are exactly the numbers in the range $1, 2, \ldots, N-1$ that are not divisible by 3. So, the sequence $c_1, \ldots, c_m$ alternates between numbers congruent to 1 (mod 3) and 2 (mod 3). Thus $3 \mid \gcd(N, c_i + c_{i+1})$ for every i.

Now we show that no other positive integer works. For the sake of contradiction, consider an odd candidate N that is not a power of 3, and suppose it satisfies the problem condition. Then, since $c_1 = 1$ and $c_2 = 2$, we have $\gcd(N, 1+2) \neq 1$, so $3 \mid N$. Thus, we can write N as $3^k M$, where $k \geq 1$ and $\gcd(M, 6) = 1$. We have $M \geq 5$ (as $M \neq 1$ by assumption) and $M + 2 < N$.

We now split into two cases based on M modulo 3.

Case 1. $M \equiv 1 \pmod 3$.

As $3 \mid M-1$ and $M \mid M$, neither $M-1$ nor M are in the sequence $\{c_i\}$. As M is odd and $M \equiv 1 \pmod 3$, $M-2$ and $M+1$ are congruent to 1 (mod 3), are coprime to M, and are smaller than N. Thus, both $M-2$ and $M+1$ are in the sequence, and they are consecutive terms. However, this means $\gcd(N, 2M-1) \neq 1$. This gives a contradiction, as both 3 and M are coprime to $2M-1$.

Case 2. $M \equiv 2 \pmod 3$.

This case is similar to Case 1. Neither M nor $M+1$ are in the sequence, but $M-1$ and $M+2$ are. We obtain a similar contradiction from $\gcd(N, 2M+1) \neq 1$.

Solution 2. We give an alternative way to show that any odd N has to be the power of 3. Suppose for contradiction that N has at least 2 distinct prime factors. Choose integers $a, b > 1$ such that

- $ab \mid N$;
- a and b are coprime;
- Every prime divisor of N divides ab.

We use the Chinese Remainder Theorem, to find an integer $n \in [1, ab]$ such that

$$\begin{cases} n \equiv 1 \pmod a, \\ n \equiv 2 \pmod b. \end{cases}$$

We claim that there is an index i such that $c_i = n - 3$ and $c_{i+1} = n$. Indeed, we note the following:

- n is consecutive to a multiple of a, so $\gcd(a, n) = 1$. In a similar way we get that $\gcd(b, n-3) = 1$.
- n is 2 away from a multiple of b, therefore as $2 \nmid b$, $\gcd(b, n) = 1$. In a similar way we get that $\gcd(a, n-3) = 1$.
- $n \leq ab \leq N$ and $n - 3 \geq (a+1) - 3 > 0$ as $a > 2$.

This implies that n and $n-3$ coprime with N, whereas neither $n-2$ nor $n-1$ are coprime with N, which proves our claim. Finally, we claim that $c_i + c_{i+1} = (n-3) + n = 2n - 3$ is coprime with N. Indeed we have $2n - 3 \equiv -1 \pmod{a}$ and $2n - 3 \equiv 1 \pmod{b}$. This implies that N is a prime power.

When N is odd, since $c_1 = 1$ and $c_2 = 2$, we have $\gcd(N, 1+2) \neq 1$, so $3 \mid N$, which implies that N is necessarily a power of 3. This concludes our proof.

P2. An infinite increasing sequence $a_1 < a_2 < a_3 < \cdots$ of positive integers is called *central* if for every positive integer n, the arithmetic mean of the first a_n terms of the sequence is equal to a_n.

Show that there exists an infinite sequence $b_1, b_2, b_3, \ldots$ of positive integers such that for every central sequence $a_1, a_2, a_3, \ldots$, there are infinitely many positive integers n with $a_n = b_n$.

Solution 1. We claim that the sequence b_1, b_2, $b_3, \ldots$ defined by $b_i = 2i - 1$ has this property.

Let $d_i = a_i - b_i = a_i - 2i + 1$. The condition $a_i < a_{i+1}$ now becomes $d_i + 2i - 1 < d_{i+1} + 2i + 1$, which can be rewritten as $d_{i+1} \geqslant d_i - 1$. Thus, if $d_{i+1} < d_i$, then d_{i+1} must be equal to $d_i - 1$. This implies in particular that if $d_{i_0} \geqslant 0$ but $d_{i_1} \leqslant 0$ for some indices $i_1 > i_0$, there must be some intermediate index $i_0 \leqslant i \leqslant i_1$ with $d_i = 0$.

Because the average of the first a_n terms of the sequence is equal to a_n, we know for all n that

$$\sum_{i=1}^{a_n} d_i = \sum_{i=1}^{a_n} (a_i - 2i + 1) = \sum_{i=1}^{a_n} a_i - \sum_{i=1}^{a_n} (2i - 1) = a_n^2 - a_n^2 = 0.$$

Because the sequence (a_n) is increasing, this implies that the sequence (d_i) contains infinitely many non-negative ($d_i \geqslant 0$) and infinitely many non-positive ($d_i \leqslant 0$) terms. In particular, we can find arbitrarily large indices $i_0 \leqslant i_1$ such that $d_{i_0} \geqslant 0$ and $d_{i_1} \leqslant 0$. By our earlier observation, it follows that there are infinitely many i such that $d_i = 0$, as desired.

Solution 2. We give an alternative proof that the sequence $b_i = 2i - 1$ works. This proof is by contradiction, so we assume that there are only finitely many a_i such that $a_i = 2i - 1$.

Let $S(n) = \sum_{i=1}^{n} a_i$. We have $S(a_n) = a_n^2$ and $S(a_{n+1}) = a_{n+1}^2$. If $a_{n+1} = a_n + 1$, then it follows that

$$S(a_{n+1}) - S(a_n) = a_{n+1}^2 - a_n^2 = a_{n+1}^2 - (a_{n+1} - 1)^2 = 2a_{n+1} - 1.$$

On the other hand, if $a_{n+1} = a_n + 1$, then $S(a_{n+1}) - S(a_n)$ is $a_{a_{n+1}}$, so it follows that $a_{a_{n+1}} = 2a_{n+1} - 1$. By assumption, this can only happen finitely many times, so for all sufficiently large n we must have $a_{n+1} \geqslant a_n + 2$.

For large enough n, we now know that $a_n > 2n - 1$ implies $a_{n+1} > (2n - 1) + 2 = 2(n + 1) - 1$. This means that there are two cases possible:

(A) For all sufficiently large n (say $n \geqslant N_A$) we have $a_n > 2n - 1$.

(B) For all sufficiently large n (say $n \geqslant N_B$) we have $a_n < 2n - 1$.

In case (A), we know for $m > N_A$ that

$$\begin{aligned} S(m) = S(N_A) + \sum_{i=N_A+1}^{m} a_i \geqslant S(N_A) + \sum_{i=N_A+1}^{m} 2i &= S(N_A) + m(m+1) - N_A(N_A+1) \\ &= m^2 + m + S(N_A) - N_A(N_A+1). \end{aligned}$$

For m large enough (e.g. $m > N_A(N_A + 1)$), this expression is always larger than m^2, contradicting $S(a_n) = a_n^2$ for all n.

Similarly, in case (B), we similarly know for $m > N_B$ that

$$\begin{aligned} S(m) = S(N_B) + \sum_{i=N_B+1}^{m} a_i \leqslant S(N_B) + \sum_{i=N_B+1}^{m} 2(i-1) &= S(N_B) + m(m-1) - N_B(N_B-1) \\ &= m^2 - m + S(N_B) - N_B(N_B-1). \end{aligned}$$

For m large enough (e.g. $m > S(N_B)$), this expression is always smaller than m^2, again contradicting $S(a_n) = a_n^2$ for all n.

Solution 3. We claim that the sequence $b_1, b_2, b_3, \ldots$ defined by $b_i = 2i - 1$ has this property.

Lemma. *If there are no terms a_j such that $a_j - a_{j-1} = 1$, then $a_j = a_{j-1} + 2$ for all j.*

Proof. Let c be such that $a_d = c$ for some d. Now

$$a_1 + a_2 + \cdots + a_c = c^2.$$

Equality holds for $a_i = 2i - 1$ for $1 \leq i \leq c$, so if any difference between two consecutive terms is greater, the left-hand side of the equation is greater than c^2, a contradiction. □

Lemma. *If both d and $d+1$ are terms of the sequence, i.e. $a_c = d$ and $a_{c+1} = d+1$ for some c, then $a_{d+1} = 2d + 1 = b_{d+1}$.*

Proof. We have $a_1 + a_2 + \cdots + a_d = d^2$ and $a_1 + a_2 + \cdots + a_{d+1} = (d+1)^2$. Hence $a_{d+1} = (d+1)^2 - d^2 = 2d + 1$. □

From the observations above, we see that we are done if there are infinitely many gaps of size 1. The only remaining case is one with finitely many gaps of size 1. This will be the subject of the following lemma.

Lemma. *If there are only finitely many indices j such that $a_{j+1} - a_j = 1$, then there is an index n_0 such that for all $k > n_0$, we have $a_k = 2k - 1$.*

Proof. Let r and s be indices such that for all the j satisfying $a_{j+1} - a_j = 1$, we have $j < r, s$. Furthermore, assume $s > r$ and that there are i_1 and i_2 such that $a_{i_1} = r$ and $a_{i_2} = s$. The first goal is to show that $a_s \geqslant 2s - 1$. If $a_r \geqslant 2r - 1$, this is clearly the case. Assume now $a_r < 2r - 1$. Now $a_r \geqslant 2r - 1 - m$, where m is the number of indices j with $a_{j+1} - a_j = 1$. Denote $a_{r+1} = 2r + 1 - m + \theta_1$, $a_{r+2} = 2r + 3 - m + \theta_2$, etc. Remember that $a_{r+j+1} - a_{r+j} \geqslant 2$ always. Now $0 \leqslant \theta_1 \leqslant \theta_2 \leqslant \ldots$. Furthermore, write $s = r + h$. Now

$$(r+h)^2 - r^2 = a_{r+1} + a_{r+2} + \cdots + a_{r+h} = \sum_{j=1}^{h} 2r - 1 + 2j - m + \theta_j.$$

From this we deduce

$$2rh + h^2 = 2rh - h - mh + h(h+1) + \sum_{j=1}^{h} \theta_j.$$

So we obtain $\sum_{j=1}^{h} \theta_j = mh$. Since the sequence θ_j is increasing, we have $\theta_h \geqslant m$. Hence, $a_s = a_{r+h} \geqslant 2r - 1 - m + 2h + m = 2r + 2h - 1 = 2s - 1$.

Now a_s is exactly the desired shape. If for any $t > s$, we have $a_t - a_{t-1} > 2$, then

$$a_s + a_{s+1} + \cdots + a_t > t^2 - s^2,$$

again a contradiction. □

Solution 4. Note that $a_1 = 1$ because if it is not the case, then $a_1^2 = a_1 + \cdots + a_{a_1} > a_1 + a_1 + \cdots + a_1 = a_1^2$.

Assume by contradiction that there are only finitely many indices k such that $a_k = 2k - 1$. Set i to be the largest integer such that $a_i = 2i - 1$ (which must exist as $a_1 = 1$). Assume that there exists $j \geqslant i$ such that $a_{j+1} - a_j = 1$. Then $2a_{j+1} - 1 = a_{j+1}^2 - a_j^2 = a_{a_{j+1}}$ and since $a_k \geqslant k$ for all k, we have $a_{j+1} \geqslant j + 1 > i$, which contradicts the definition of i. Thus for all $j \geqslant i$, we have $a_{j+1} \geqslant a_j + 2$, which implies by induction that $a_j \geqslant 2j - 1$ for $j \geqslant i$, and even $a_j \geq 2j$ if $j > i$.

There are two ways to finish the solution from here.

First way to finish the solution

For all n such that $a_n \geqslant i$, we have

$$\begin{aligned} a_{n+1}^2 - a_n^2 = a_{a_{n+1}} + a_{a_{n+1}-1} + \cdots + a_{a_n+1} &\geqslant 2a_{n+1} + 2(a_{n+1} - 1) + \cdots + 2(a_n + 1) \\ &= (a_{n+1} - a_n)(a_{n+1} + a_n + 1) \\ &> a_{n+1}^2 - a_n^2. \end{aligned}$$

This gives a contradiction.

Second way to finish the solution

For all n such that $a_n \geqslant i$, we introduce $x_n = a_{n+1} - a_n$. We have

$$x_n^2 + 2x_n a_n = a_{n+1}^2 - a_n^2 = a_{a_{n+1}} + a_{a_{n+1}-1} + \cdots + a_{a_n+1} \geqslant \sum_{j=1}^{x_n} (a_{a_n} + 2j) \geq x_n a_{a_n} + x_n(x_n + 1).$$

By simplifying, we get $a_{a_n} \leq 2a_n - 1$, which gives a contradiction.

Comment. Proving that $a_1 = 1$ is not necessary for this solution. If there exists no i such that $a_i = 2i - 1$, then the same argument implies that there exists no j such that $a_{j+1} - a_j = 1$, thus $a_j \geqslant 2j - 1$ for $j \geqslant 1$.

P3. Let ABC be an acute triangle. Points B, D, E, and C lie on a line in this order and satisfy $BD = DE = EC$. Let M and N be the midpoints of AD and AE, respectively. Let H be the orthocentre of triangle ADE. Let P and Q be points on lines BM and CN, respectively, such that D, H, M, and P are concyclic and E, H, N, and Q are concyclic. Prove that P, Q, N, and M are concyclic.

The orthocentre of a triangle is the point of intersection of its altitudes.

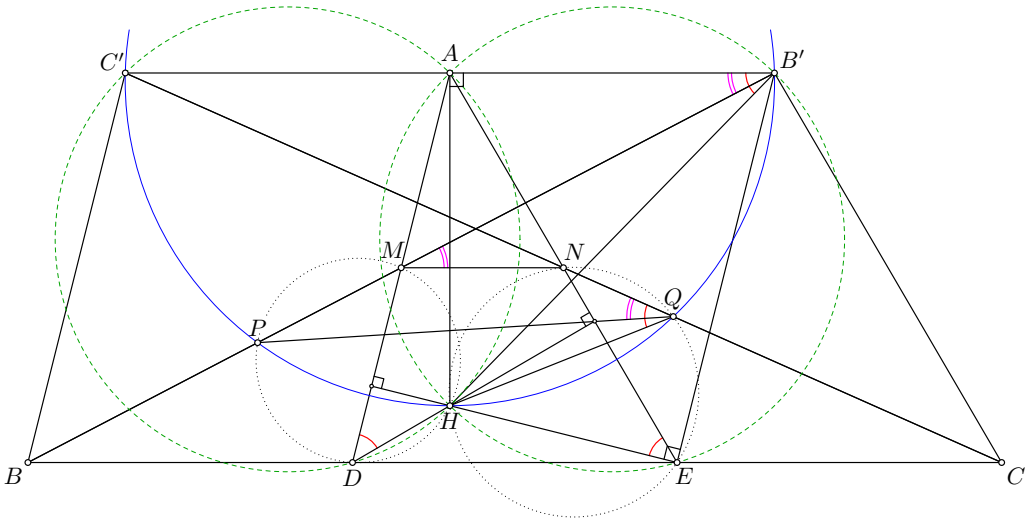
Solution 1. Denote by B' and C' the reflections of B and C in M and N, respectively. Points C', A, B' are clearly collinear and $DEB'A$ is a parallelogram. Since $EH \perp AD$, we have $EH \perp EB'$. Also $HA \perp AB'$, so points H, E, B', A are concyclic. This gives

$$\angle C'QH = \angle NQH = \angle NEH = \angle AEH = \angle AB'H = \angle C'B'H,$$

and so points C', B', Q, H are concyclic. Analogously points C', B', P, H are concyclic, and so all points B', C', P, Q, H are. Now we have

$$\angle NMB' = \angle AB'M = \angle C'B'P = \angle C'QP = \angle NQP,$$

which proves that P, Q, N, M are also concyclic.



Solution 1′. Introduce points B', C' as above. Also define $A' = B'E \cap C'D$, so that $EADA'$ is a parallelogram and ADE is the medial triangle of $A'B'C'$. It that follows that the orthocentre H of ADE is the circumcentre of $A'B'C'$, and in particular

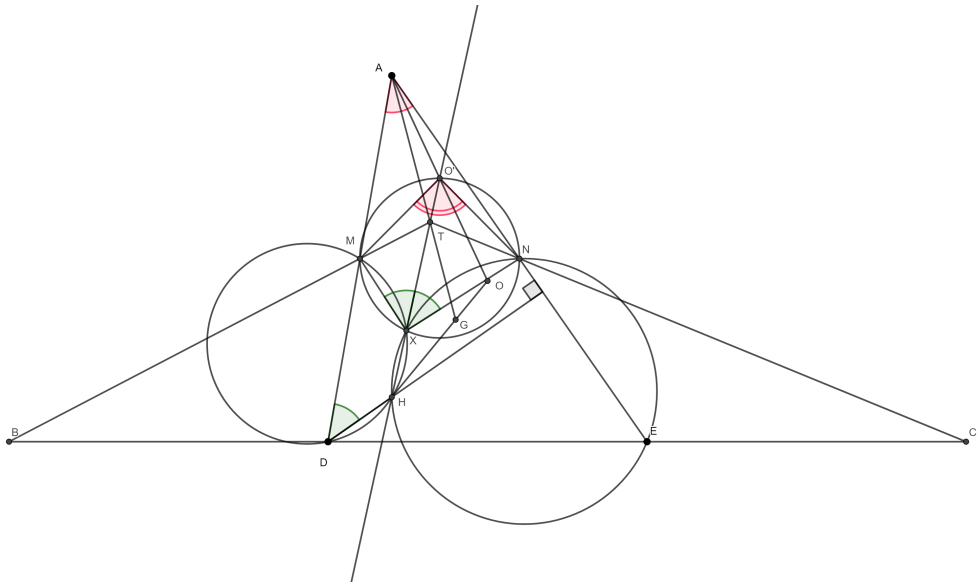
$$\angle C'B'H = 90^\circ - \angle B'A'C' = 90^\circ - \angle DAE = \angle AEH = \angle NEH = \angle NQH = \angle C'QH.$$

So again we have that C', B', Q, H are concyclic and conclude as in Solution 1.

Solution 2. Let X be the second intersection of (DHM) and (EHN) and let O' be the circumcentre of (AMN). Note that $\angle MXN = \angle MDH + \angle NEH = 180^\circ - 2\angle DAE$ and since $\angle MO'N = 2\angle DAE$ we have that X, M, O', N is cyclic and since $\angle MXH = \angle NXH$ it means that HX is the angle bisector of $\angle MXN$ but since $O'M = O'N$ it means that H, X, O' are collinear. Let BM and CN intersect at T and let K and L be the midpoints of MN and BC. Note that L is also the midpoint of DE. Since MN is parallel to BC it means that T, K, and L are collinear, but since A, K, and L are collinear we get that A, T, K, and L are collinear. Now, $\frac{TL}{TK} = \frac{BC}{MN} = 6$. Since $KL = KA$ it means that $\frac{AT}{TK} = \frac{TL-2TK}{TK} = 4$ so by the lemma below, T lies on HO'. Since HO' is the radical axis of (DHM) and (EHN) we finish the problem using the Radical Axes Theorem ($TM \cdot TP = TN \cdot TQ$).

Lemma. *Let A' be the reflection of A around the orthocentre H of $\triangle ABC$ and O and M be the circumcentre of $\triangle ABC$ and the midpoint of BC, respectively. Let T be the intersection of $A'O$ and AM. Then $\frac{AT}{TM} = 4$.*

Proof. Since $OM \parallel AA'$ we have $\frac{AT}{TM} = \frac{AA'}{OM} = \frac{2AH}{OM} = \frac{4OM}{OM} = 4$. We used here that $AH = 2OM$. □



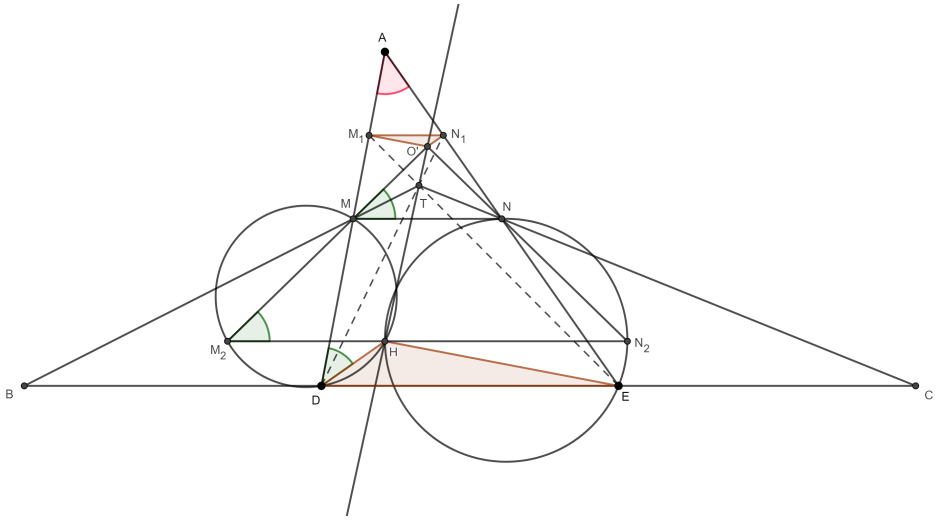
Solution 2′. As in solution 2, we will prove that O' is both on line HT and the radical axis of the circles, hence T is on the radical axis, from which we conclude the required concyclicity. We present alternative proofs of both facts, discovered by contestants.

Let M_1, N_1 be the midpoints of AM, AN, respectively, so that $AM_1 : M_1D = AN_1 : N_1E = 1 : 3$. It is easy to verify (e.g. by applying the theorems of Ceva or Menelaus, or by computing in barycentric coordinates as in Solution 3) that T lies on EM_1 and DN_1. Note that $M_1N_1 \parallel DE$, and also $M_1O' \parallel HE$ as both are perpendicular to AMD, and similarly $N_1O' \parallel HD$. It follows that DEH and N_1M_1O' are homothetically similar triangles, and the center of their (negative) homothety is $T = DN_1 \cap EM_1$. Therefore T also lies on HO', as claimed. (We also have that the homothety is by factor $\frac{M_1N_1}{ED} = -\frac{1}{4}$.)

Now, let M_2, N_2 be the second intersection points of $O'M$, $O'N$ with the circumcircles of HMD, HNE, respectively. To prove that O' is on the radical axis, it suffices to show that $O'M \cdot O'M_2 = O'N \cdot O'N_2$. But by definition of O' we have $O'M = O'N$, so we must show $O'M_2 = O'N_2$, which is equivalent to $M_2N_2 \parallel MN$. Angle chasing in circle MM_2DH gives

$$\angle OM_2H = \angle MM_2H = \angle MDH = \angle ADH = 90^\circ - \angle EAD = 90^\circ - \angle NAM = \angle O'MN,$$

from which it follows that $M_2H \parallel MN$. Similarly, we have $N_2H \parallel MN$, and the two facts together imply that M_2, H, N_2 are collinear and the line through them is parallel to MN, as claimed. □



Solution 3. We compute using linear combinations with respect to ADE. We have $B = 2D - E$, $C = 2E - D$, $M = \frac{A+D}{2}$, and $N = \frac{A+E}{2}$, from which we immediately obtain that the intersection of $T = BM \cap CN$ is $T = \frac{3A+D+E}{5} = \frac{6M-B}{5} = \frac{6N-C}{5}$. As in solution 2, we reduce to showing that T is on the radical axis of HDM and HEN, whence $TM \cdot TP = TN \cdot TQ$ proves the concyclicity of P, Q, N, M.

Synthetic finish, similar to Solution 2. Let O be the circumcentre of ADE and $O' = \frac{A+O}{2}$ be the circumcentre of AMN. As in Solution 2, we have that O' is on the desired radical axis, so it is enough to show $T \in O'H$. Let $G = \frac{A+D+E}{3}$ be the barycentre of ADE. By properties of the Euler line, we also have $G = \frac{H+2O}{3}$. Now using our known identities we find

$$T = \frac{3A + D + E}{5} = \frac{2A + 3G}{5} = \frac{2A + H + 2O}{5} = \frac{H + 4O'}{5}$$

and in particular $T \in HO'$, as we wanted to show.

Computational finish. Let $f(T)$ be the power difference at T w.r.t. DHM and EHN. We compute $f(A)$ and $f(L)$ where $L = \frac{D+E}{2}$. Since $T = \frac{3A+2L}{5}$, it is enough to show that $3P(A) + 2P(L) = 0$. In the following all lengths are directed. We compute trigonometrically: Let α, β, γ be the angles of ADE and assume the diameter of its circumcircle is 1. We have

$$f(A) = \frac{AD^2 - AE^2}{2} = \frac{\sin^2(\gamma) - \sin^2(\beta)}{2}.$$

To compute $P(L)$, let D', E' be the second intersection points of HMD, HNE with DE, and let M', N', F be the feet of the perpendiculars from H to AD, AE, DE, respectively. Note that $DM' = \sin\alpha\cos\beta$, thus

$$M'M = DM - DM' = \frac{\sin(\alpha+\beta)}{2} - \sin\alpha\cos\beta = \frac{\sin(\beta-\alpha)}{2}.$$

We also have $HM' = \cos\alpha\cos\beta$, $HF = \cos\beta\cos\gamma$, and $HM'M \sim HFD'$, therefore

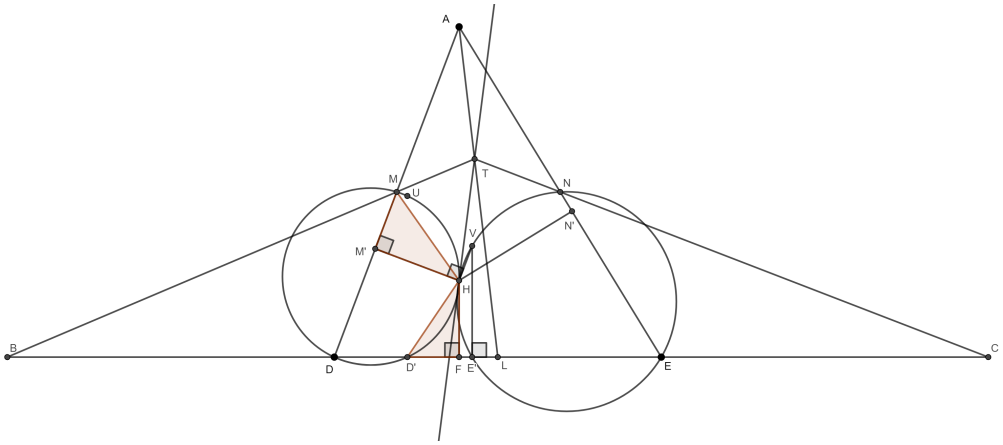
$$FD' = \frac{HF}{HM'}M'M = \frac{\cos\gamma}{\cos\alpha}\frac{\sin(\beta-\alpha)}{2}$$

and similarly

$$E'F = \frac{\cos\beta}{\cos\alpha}\frac{\sin(\gamma-\alpha)}{2}.$$

We also have the standard $FD = \sin\gamma\cos\beta$ and $EF = \sin\beta\cos\gamma$. We can finally compute

$$\begin{aligned} f(L) &= LD \cdot LD' - LE \cdot LE' = \frac{\sin\alpha}{2}(LD' + LE') = \frac{\sin\alpha}{2}(FD' + FE' - FD - FE) \\ &= \frac{\sin\alpha}{4\cos\alpha}(\cos\gamma\sin(\beta-\alpha) - \cos\beta\sin(\gamma-\alpha) - 2\cos\alpha(\sin\gamma\cos\beta - \sin\beta\cos\gamma)) \\ &= \frac{3\sin\alpha\sin(\beta-\gamma)}{4} = \frac{3}{4}(\sin(\beta+\gamma)\sin(\beta-\gamma)) = \frac{3}{8}(\cos(2\gamma) - \cos(2\beta)) \\ &= \frac{3}{4}(\sin^2(\beta) - \sin^2(\gamma)) = -\frac{3}{2}f(A). \quad \square \end{aligned}$$



Solution 4. Let $T = BM \cap CN$, let F be the foot of the altitude from A to BC, let O be the circumcentre of (ADE), let $D' \neq D$ and $E' \neq E$ be the second intersections of $(DHMP)$ and $(EHNQ)$ with line BC, and let U and V be the antipodes of D and E on $(DHMP)$ and $(EHNQ)$, respectively.

We begin with a bit of barycentric coordinates. Set barycentric coordinates in $\triangle ABC$, set so that $A = (1,0,0)$, $B = (0,1,0)$, and $C = (0,0,1)$. The definitions of D and E give $D = (0,2/3,1/3)$ and $E = (0,1/3,2/3)$, whence $M = (1/2,1/3,1/6)$ and $N = (1/2,1/6,1/3)$. This means that line BM is given by $(1/2 : y : 1/6)$ while line CN is given by $(1/2 : 1/6 : z)$. So their intersection T is $(1/2 : 1/6 : 1/6) = (3 : 1 : 1)$, giving $T = \frac{3A+B+C}{5} = \frac{3A+D+E}{5}$.

Our next tool is the linearity of the power of a point. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(Z) = \mathrm{Pow}_{(DHMP)}(Z) - \mathrm{Pow}_{(EHNQ)}(Z).$$

It suffices to show that $f(T) = 0$; from there, the required concyclicity will follow from $TM \cdot TP = TN \cdot TQ$. The key observation is that f is a linear function. In particular, $f(T) = \frac{3f(A)+f(D)+f(E)}{5}$. So, we need only show that $3f(A) + f(D) + f(E) = 0$. Pick an arbitrary one-dimensional coordinate system on the line BC and let τ be the map from a point on BC to its coordinate. We compute

$$\begin{aligned} f(A) &= AM \cdot AD - AN \cdot AE = \frac{AD^2 - AE^2}{2} = \frac{FD^2 - FE^2}{2} \\ &= (\tau(E) - \tau(D))\left(\tau(F) - \tau\left(\frac{D+E}{2}\right)\right), \\ f(D) &= -DE \cdot DE' = (\tau(E) - \tau(D))(\tau(D) - \tau(E')), \\ f(E) &= ED \cdot ED' = (\tau(E) - \tau(D))(\tau(E) - \tau(D')). \end{aligned}$$

Rearranging, it suffices to show that $3\tau(F) = \tau(D') + \tau(E') + \tau((D+E)/2)$. This can be rewritten as $3F = D' + E' + (D+E)/2$. By projecting down to line BC, it suffices to show that the displacement vector $H + 2A - (O + U + V)$ is perpendicular to line BC.

We do this using complex numbers. Let (ADE) be the unit circle with $A = a$, $D = d$, and $E = e$, so that $O = 0$ and $H = h = a + d + e$. Note that U satisfies $UM \perp AD$ and $UH \perp DH \perp AE$. Translating these conditions into equations, we have $u = ad\overline{u}$ and $u + ae\overline{u} = h + ae\overline{h}$. Rearranging gives

$$(d+e)u = du + e(ad\overline{u}) = d(h + ae\overline{h}) \implies u = \frac{dh + ade\overline{h}}{d+e}.$$

Computing v similarly, we get that

$$v := H + 2A - (O + U + V) = h + 2a - \frac{(d+e)h + 2ade\overline{h}}{d+e} = 2\left(a - \frac{ade\overline{h}}{d+e}\right) = -\frac{2de}{d+e}.$$

This displacement vector v satisfies $v = de\overline{v}$ and so it is orthogonal to line DE, as desired.

Solution 5. This solution uses almost exclusively complex numbers. As in other solutions, we reduce to showing that H, $S := (DHM) \cap (EHN)$, and $T := BM \cap CN$ are collinear; this is all of the synethetic information we shall use. (If one computes $T = \frac{3A+D+E}{5}$ using means other than complex numbers, the solution becomes much shorter.)

We use complex numbers with $A = a$, $D = d$, and $E = e$ on the unit circle. Note that $H = a + d + e$, $M = \frac{a+d}{2}$, and $B = 2d - e$. We will make great use of the "arbitrary line intersection formula," which says that the intersection between lines WX and YZ can be written explicitly as

$$\frac{(\overline{w}x - w\overline{x})(y-z) - (w-x)(\overline{y}z - y\overline{z})}{(\overline{w} - \overline{x})(y-z) - (w-x)(\overline{y} - \overline{z})}.$$

We first use this to find $T = t$. We compute

$$\begin{aligned}
b - m &= (2d - e) - \frac{a+d}{2} = \frac{3d - a - 2e}{2} \\
\overline{b - m} &= \frac{3ae - de - 2ad}{2ade} \\
\overline{b}m - b\overline{m} &= \frac{2e - d}{de} \cdot \frac{a+d}{2} - (2d - e) \cdot \frac{a+d}{2ad} \\
&= \frac{a(a+d)(2e-d) - (2d-e)(a+d)e}{2ade} = \frac{(a+d)(2ae + e^2 - ad - 2de)}{2ade}.
\end{aligned}$$

If $\mathcal{E}$ is some expression, we use the notation $\mathcal{E} - \{\sim\}$ to denote $\mathcal{E}$ minus the expression formed by swapping d and e in $\mathcal{E}$. We now have

$$\begin{aligned}
t &= \frac{(\overline{b}m - b\overline{m})(c-n) - \{\sim\}}{(\overline{b} - \overline{m})(c-n) - \{\sim\}} \\
&= \frac{(a+d)(2ae + e^2 - ad - 2de)(3e - a - 2d) - \{\sim\}}{(3ae - de - 2ad)(3e - a - 2d) - \{\sim\}} \\
&= \frac{\left[a^3(d-2e) + a^2(3d^2 - 7de + 5e^2) + a(2d^3 - d^2e - 3de^2 + 3e^3) + de(4d^2 - 8de + 3e^2)\right] - \{\sim\}}{\left[a^2(2d-3e) + a(4d^2 - 11de + 9e^2) + de(2d - 3e)\right] - \{\sim\}} \\
&= \frac{a^3(3(d-e)) - a^2(2(d^2 - e^2)) + a(-(d^3 - e^3) + 2de(d-e)) + de(d^2 - e^2)}{a^2(5(d-e)) - a(5(d^2 - e^2)) + de(5(d-e))} \\
&= \frac{3a^3 - 2a^2(d+e) - a(d^2 - de + e^2) + de(d+e)}{5\left(a^2 - a(d+e) + de\right)} \\
&= \frac{(a-d)(a-e)(3a+d+e)}{5(a-d)(a-e)} = \frac{3a+d+e}{5}.
\end{aligned}$$

(The factorization in the last line can be motivated by noting that the expression, while cubic in a, is only quadratic in d. When written out as a polynomial in d, each coefficient is divisible by $a - e$; by symmetry, the numerator is divisible by $a - d$ as well, and the factorization follows.)

Computing $S = s$ is slightly harder, as it is the intersection of two circles rather than of two lines. We get around this by noting that $\{h, d, m, s\}$ are concyclic if and only if $\{0, h-d, h-m, h-s\}$ are concyclic, which happens if and only if $\{\frac{1}{h-d}, \frac{1}{h-m}, \frac{1}{h-s}\}$ are collinear. (One can see this by inversion, or just by writing out the cross-ratio in the special case when one of the points is zero.) Thus $\frac{1}{h-s}$ is the intersection of the line through $w := \frac{1}{h-d}$ and $x := \frac{1}{h-m}$ and the line through $y := \frac{1}{h-e}$ and $z := \frac{1}{h-n}$. We compute

$$\begin{aligned}
w - x &= \frac{1}{a+e} - \frac{1}{\frac{a+d}{2}+e} = -\frac{a-d}{(a+e)(a+d+2e)} \\
\overline{w-x} &= \frac{ae^2(a-d)}{(a+e)(2ad+ae+de)} \\
\overline{w}x - w\overline{x} &= \frac{ae}{a+e}\cdot\frac{2}{a+d+2e} - \frac{1}{a+e}\cdot\frac{2ade}{2ad+ae+de} \\
&= 2ae\frac{(2ad+ae+de)-d(a+d+2e)}{(a+e)(a+d+2e)(2ad+ae+de)} \\
&= \frac{2ae(a-d)(d+e)}{(a+e)(a+d+2e)(2ad+ae+de)}.
\end{aligned}$$

Using the line intersection formula, we have

$$\begin{aligned}
\frac{1}{h-s} &= \frac{(\overline{w}x - w\overline{x})(y-z) - \{\sim\}}{(\overline{w}-\overline{x})(y-z) - \{\sim\}} \\
&= \frac{\left[-\frac{2ae(a-d)(d+e)}{(a+e)(a+d+2e)(2ad+ae+de)}\cdot\frac{(a-e)}{(a+d)(a+2d+e)}\right] - \{\sim\}}{\left[-\frac{ae^2(a-d)}{(a+e)(2ad+ae+de)}\cdot\frac{(a-e)}{(a+d)(a+2d+e)}\right] - \{\sim\}} \\
&= 2(d+e)\frac{\left[e(ad+2ae+de)\right] - \{\sim\}}{\left[e^2(a+d+2e)(ad+2ae+de)\right] - \{\sim\}} \\
&= 2(d+e)\frac{\left[a(de+2e^2)+de^2\right] - \{\sim\}}{\left[a^2(de^2+2e^3)+a(d^2e^2+5de^3+4e^4)+de(de^2+2e^3)\right] - \{\sim\}} \\
&= \frac{2(d+e)(a(2(e^2-d^2))+de(e-d))}{(a^2+de)(2(e^3-d^3)+de(e-d))+a(4(e^4-d^4)+5de(e^2-e^2))} \\
&= \frac{2(d+e)(2a(d+e)+de)}{(a^2+de)(2d^2+3de+2e^2)+a(d+e)(4d^2+5de+4e^2)}.
\end{aligned}$$

Since $h - t = \frac{2a+4d+4e}{5}$, this gives us

$$\frac{h-s}{h-t} = \frac{5}{4}\cdot\frac{(a^2+de)(2d^2+3de+2e^2)+a(d+e)(4d^2+5de+4e^2)}{(d+e)(2ad+2ae+de)(a+2d+2e)}.$$

It is easy to see that this is real by using the symmetry of the expressions (both the numerator and denominator satisfy $\mathcal{E} = a^2d^3e^3\overline{\mathcal{E}}$. We conclude that H, S, and T are collinear, as desired.

Solution 6. As usual, we reduce to proving $T = (3A + D + E)/5$ is on the radical axis and compute; this time in Cartesian coordinates.

Let $A(0, h)$, $D(b, 0)$, $E(c, 0)$ be coordinates for ADE. Then $H(0, -\frac{bc}{h})$, $M(\frac{b}{2}, \frac{h}{2})$, $N(\frac{c}{2}, \frac{h}{2})$ and $T(\frac{b+c}{5}, \frac{3h}{5})$. We compute $O_D(x_D, y_D)$ the circumcentre of DMH and obtain O_E by symmetry. We then have to verify that $O_DO_E \perp HT$, which can be done by comparing slopes.

The centre O_D can be given by the intersection of perpendicular bisectors of DM and DH. This gives the following system of equations on x_D, y_D:

$$h \cdot x_D + c \cdot y_D = \frac{bh}{2} - \frac{bc^2}{2h}$$
$$-b \cdot x_D + h \cdot y_D = -\frac{3b^2}{4} + \frac{h^2}{4}$$

Solving the system gives

$$(h^2 + bc)x_D = \frac{3b^2c}{4} - \frac{bc^2}{2} + \frac{h^2b}{2} - \frac{h^2c}{4}$$
$$(h^2 + bc)y_D = -\frac{hb^2}{4} + \frac{h^3}{4} - \frac{b^2c^2}{2h}$$

The formulas for x_E, y_E will be the same, swapping $b \leftrightarrow c$ by symmetry; thus $y_D - y_E$ and $x_D - x_E$ will be antisymmetric in b, c and divisible by $b - c$, and explicitly:

$$(h^2 + bc)(x_D - x_E) = \frac{b-c}{4}(5bc + 3h^2)$$
$$(h^2 + bc)(y_D - y_E) = -\frac{b-c}{4}h(b + c)$$

So $\frac{y_D - y_E}{x_D - x_E} = -\frac{h(b+c)}{5bc+3h^2}$.
The other slope is more immediate:

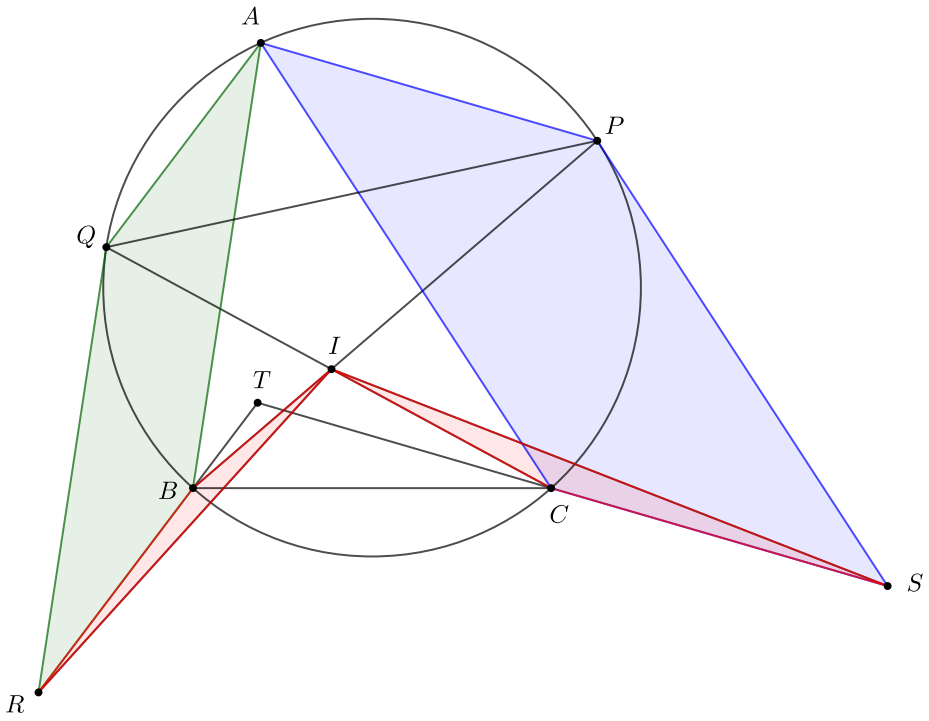
$$\frac{y_T - y_H}{x_T - x_H} = \frac{3h/5 + bc/h}{(b+c)/5} = \frac{5bc + 3h^2}{h(b+c)} = -\frac{x_D - x_E}{y_D - y_E}$$

so indeed the two slopes correspond to perpendicular lines.

Day 2

P4. Let ABC be an acute triangle with incentre I and $AB \neq AC$. Let lines BI and CI intersect the circumcircle of ABC at $P \neq B$ and $Q \neq C$, respectively. Consider points R and S such that $AQRB$ and $ACSP$ are parallelograms (with $AQ \parallel RB$, $AB \parallel QR$, $AC \parallel SP$, and $AP \parallel CS$). Let T be the point of intersection of lines RB and SC. Prove that points R, S, T, and I are concyclic.

Solution 1. We will prove that $\triangle BIR \sim \triangle CIS$, since the statement then follows from $\angle TRI = \angle BRI = \angle CSI = \angle TSI$.



Step 1. Let us prove $\angle RBI = \angle SCI$. We will use directed angles:

$$\begin{aligned}(BR, BI) &= (BR, AB) + (AB, BI) = (AQ, AB) + (BI, BC) \\ &= (CQ, BC) + (BI, BC) = (CI, CB) + (BI, BC),\end{aligned}$$

which is symmetric in B, C. Therefore, analogously we would obtain the same expression for (CS, CI).

Step 2. Let us prove $BR/BI = CS/CI$. Clearly $BR = AQ$ and $CS = AP$. Angle chasing gives $\angle ICB = \angle QCB = \angle APQ$, and similarly $\angle PQA = \angle CBI$, and so $\triangle IBC \sim \triangle AQP$, from which the desired $AQ/BI = AP/CI$ follows. This finishes the solution.

Remark. In the alternative solutions below, the notation (ABC) refers to the circle passing through points A, B, and C. D will refer to the midpoint of arc BC not containing point A, unless stated otherwise.

Let I_A be the A-excenter of $\triangle ABC$. It is well-known that points B, C, I, and I_A lie on a circle with diameter II_A. The center of this circle is point D. It is clear that A, I and D are collinear.

In some of the solutions, some of the facts below may be used:

- B, T, I, C are concyclic. Indeed, $\angle BTC = \angle QAP = A + \frac{B+C}{2} = 90^\circ + \frac{A}{2} = \angle BIC$, so B, T, I, C are concyclic.
- $\angle QAP = \angle BTC = \angle RTS$. Indeed, since $QA \parallel BT$ and $PA \parallel CT$, then $\angle QAP = \angle BTC = \angle RTS$.
- RQ and PS are tangents to (ABC). Indeed, $\angle RQB = \angle QBA = \angle QAB$ and similarly for PS.
- $\triangle AQP \sim \ IBC$ (see Solution 1 for the proof).
- QP is the perpendicular bisector of AI. Indeed, $QA = QI$ and $PA = PI$ so $PQ \perp AI$ and PQ bisects AI.

Solution 2. We use complex numbers, with (ABC) as the unit circle. Set $D = d$, $P = p$, $Q = q$, so that $A = a = -\frac{pq}{d}$, $B = b = -\frac{dq}{p}$, and $C = c = -\frac{dp}{q}$. Write $z \sim w$ if z/w is a nonzero real number. We observe that

$$\frac{R-T}{S-T} \sim \frac{R-B}{S-C} = \frac{Q-A}{P-A} = \frac{q+\frac{pq}{d}}{p+\frac{pq}{d}} = \frac{(d+p)q}{(d+q)p}.$$

So, it suffices to show that $\frac{I-R}{I-S} \sim \frac{(d+p)q}{(d+q)p}$. Indeed,

$$I - R = (d+p+q) - (Q+B-A) = d+p+\frac{dq}{p} - \frac{pq}{d} = (d+p)\left(1 + \frac{(d-p)q}{dp}\right) = \frac{(d+p)(dp+dq-pq)}{dp},$$

so

$$\frac{I-R}{I-S} = \frac{\frac{d+p}{dp}}{\frac{d+q}{dq}} = \frac{(d+p)q}{(d+q)p}.$$

Solution 3. In the following, all segment notations denote vectors.

As mentioned above, we find $\triangle AQP \sim \triangle IBC$, and by definitions of the parallelograms we have $BR = AQ$ and $CS = AP$ as well as $\angle RTS = \angle QAP$, so it suffices to show $\angle RIS = \angle QAP$. From the similarity $\triangle AQP \sim \triangle IBC$, we have a spiral map λ such that $IB = \lambda AQ$ and $IC = \lambda AP$. It follows that $IR = IB + BR = (\lambda + 1)AQ$ and $IS = IC + CS = (\lambda + 1)AP$. Because $\lambda + 1$ is also a spiral map, we have $\triangle IRS \sim \triangle AQP$ and in particular $\angle RIS = \angle QAP$, as we wanted to show.

Remark. This solution is deeply related to the complex numbers solution; indeed, the vectors can be interpreted as complex numbers and the spiral map as a complex scalar multiplication. But it only relies on the additive structure of the complex numbers as a real plane and the linear map acting on them (rather than, e.g., multiplying two points together), making vectors a slightly more natural language for the claims.

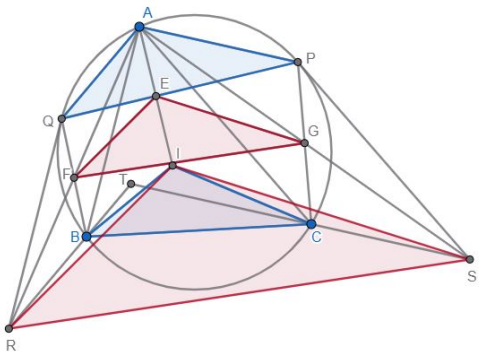
Remark. The number 1 in the solution above represents the identity map.

Solution 4. Let E, F, G be the midpoints of AI, BQ, CP. As in **Solution 1**, angle chase shows that $\triangle AQP \sim \triangle IBC$.

Note that by the Mean Geometry Theorem we have that $\frac{1}{2}AQP + \frac{1}{2}IBC = EFG$ is similar to $\triangle IBC$. Homothety with center A and scale-factor 2 maps EFG to IRS. Hence $\angle RIS = \angle FEG = \angle QAP = \angle BTC = \angle RTS$, so R, T, I, S are concyclic.

Remark. As shown above, E lies on QP and $AI \perp PQ$. One can prove that $\angle FEG = \angle BIC$ in another way. Let J be the midpoint of PQ. Then $\angle BIC = \angle FJG$ by midlines and $\angle FJG = \angle FEG$ by the lemma below applied in $BCPQ$.

Lemma. *Let $ABCD$ is a cyclic quadrilateral and E is the intersection of its diagonals. Then the midpoints of AB, BC, CD and the foot of the perpendicular from E to BC are concyclic.*



Solution 5. Let O be the circumcenter of (ABC). Let M, N, and L be the midpoints of OD, PC, and QB, respectively.

Claim 1. $\triangle OPQ$ and $\triangle DCB$ are directly similar.
Proof. Clearly $DB = DC$ and $OQ = OP$. Also note that $\angle QOP = 2\angle QDP = 2\angle QDA + 2\angle PDA = \angle BDA + \angle CDA = \angle BDC$. So the two triangles are directly similar by SAS.

Claim 2. $ML = MN$ and $\angle LMN = 180^\circ - \angle BAC$.
Proof. Note that since $\triangle OQP \sim \triangle DBC$ by the Mean Geometry Theorem, we have that the average of the two triangles is also similar to them, therefore $\triangle MLN \sim \triangle DBC \Rightarrow ML = MN$ and $\angle LMN = \angle BDC = 180^\circ - \angle BAC$.

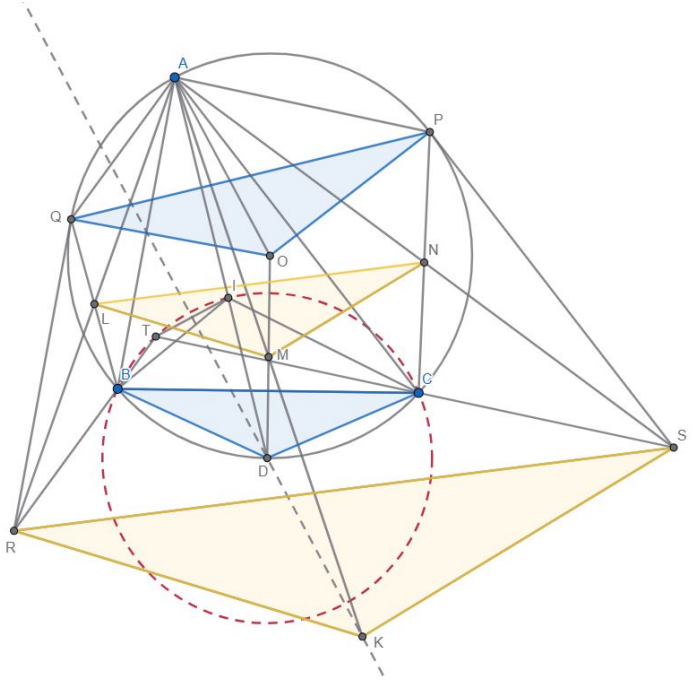
Let K be the reflection of A over M.

Claim 3. K is the circumcenter of $\triangle RTS$.
Proof. Note that since $AQRB$ and $APSC$ are parallelograms we have that $A - L - R$ are collinear and that $A - N - S$ are collinear. The homothety centered at A with scale-factor 2 maps $\triangle LMN$ to $\triangle RKS$, therefore $KR = KS$ and $\angle RKS = \angle LMN = \angle BDC = 2(180^\circ - \angle RTS)$ (and K and T are in opposite sides of RS), implying that K is the circumcenter of $\triangle RTS$.

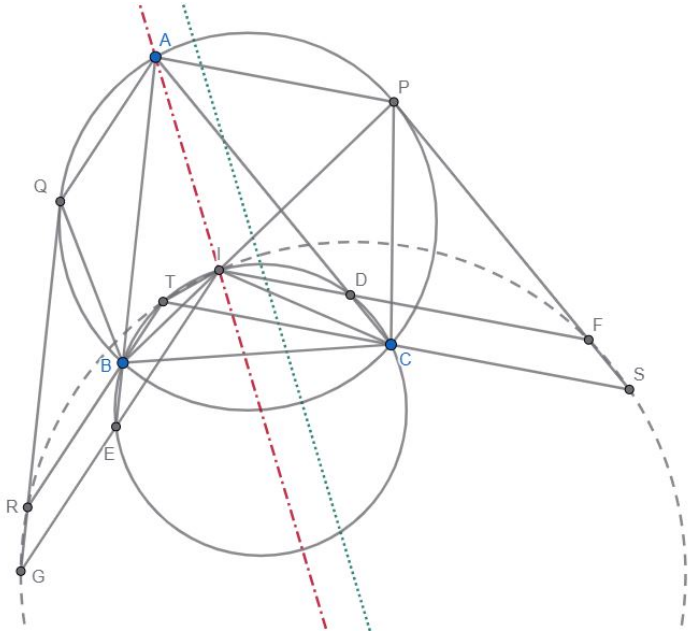
Claim 4. $KT = KI$.
Proof. Note that $AOKD$ is a parallelogram. Let BT intersect the (ABC) again at point G. Since $\angle ABG = \angle ABT = \angle QAB = \angle QCA \Rightarrow AQ = AG$ and also $OQ = OG$ hence $AO \perp QG$. Then by Reim's theorem we have that $QG \parallel TI$ and also that $AO \parallel DK$, so $DK \perp TI$. Since $DI = DT$, it means that KD is the perpendicular bisector of TI, therefore $KT = KI$.

This means that $RTIS$ is cyclic with center K.

Remark. When K and T are on the same side of RS, it can be shown that $\angle RKS = 2\angle RTS$.

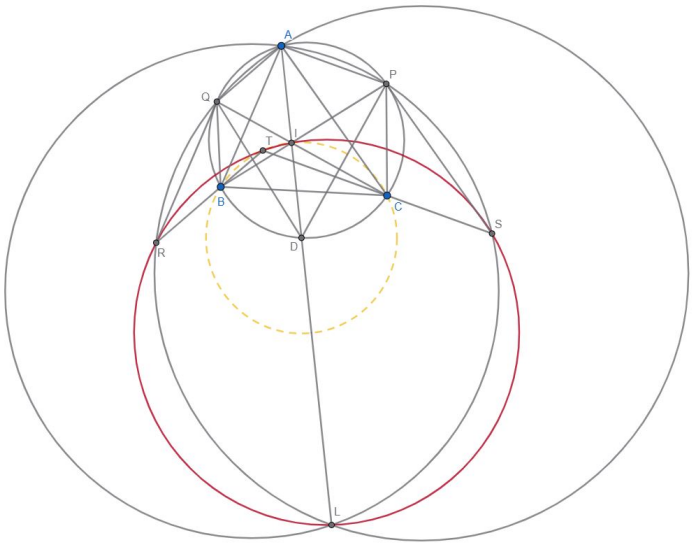


Solution 6. As shown above, we have that $BTIC$ is cyclic. Let D and E be the second intersections of AC and AB with this circle, respectively. Since the center of this circle lies on AI (by symmetry about AI), we have that $AB = AD$ and $AC = AE$, therefore $BE = CD$. Note that since $C - I - Q$ and $A - B - E$ are collinear, by Reim's theorem we have that $AQ \parallel EI$ and since $AQ \parallel BT$, we have that $BT \parallel EI$. Similarly, we get $CT \parallel DI$. Let F and G be the intersections of ID and IE with PS and QR, respectively. Clearly, $RGEB$ and $FSCD$ are parallelograms. Since $RGEB$ is parallelogram and $BEIT$ is isosceles trapezoid, we have that $RGIT$ is isosceles trapezoid. Similarly, $SFIT$ is isosceles trapezoid. Hence, both of them are cyclic. Note also that $QR = AB = AD = PF$ and $QG = AE = AC = PS$. Since QR and PS are tangents to the circumcircle of $\triangle ABC$ we have that R and F are symmetric (reflections) about the perpendicular bisector of PQ. Similarly, G and S are symmetric about the perpendicular bisector of PQ. This gives us that $QP \parallel RF \parallel GS$ and that $RFSG$ is an isosceles trapezoid, hence a cyclic quadrilateral with $\angle RGS = 180^\circ - \angle GQP = 180^\circ - \angle QAP = 180^\circ - \angle RTS \Rightarrow R, G, S, T$ are concylic. Combining all the facts about the cyclic quadrilaterals we proved above, we have that R, G, S, F, I, T are concylcic. Therefore R, T, I, S lie on a circle.

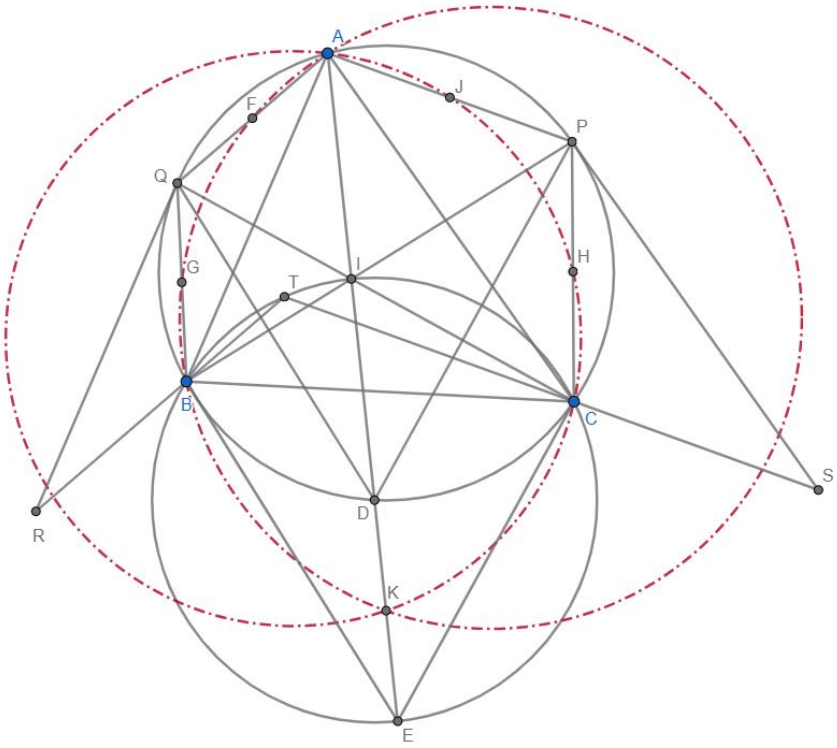


Solution 7. Let E be the A-excenter of $\triangle ABC$. Let the midpoints of AQ, QB, CP, PA be the points F, G, H, J, respectively. Both PD and EC are perpendicular to CI, hence $PD \parallel CE$.

Since $PA = PC$ we have that $AJHC$ is an isosceles trapezoid so it is cyclic. Let K be the second intersection of $(AJHC)$ and AI. Then $\angle ADP = \angle ACP = \angle ACH = \angle AKH \Rightarrow DP \parallel KH$. So KH is a line passing through the midpoint of the side CP of trapezoid $DPCE$ and parallel to the bases, hence K is the midpoint of DE. Similarly, we show that the circle $(AFGB)$ passes through the midpoint of DE. Homothety centered at A with scale-factor 2 maps (AJH) to (APS), (AFG) to (AQR), and line AI to line AI. This means that the circles (AQR) and (APS) intersect on AI, call it point L.



Now, $\angle ILR = 180° - \angle AQR = \angle QAB = \angle QCB = 180° - \angle ITB = 180° - \angle ITR$, therefore R, L, I, T are concyclic. Similarly, we get that S, L, T, I are concyclic. Combining these, it means that R and S belong to the circle (LIT). The conclusion follows.



P5. Let $n > 1$ be an integer. In a *configuration* of an $n \times n$ board, each of the n^2 cells contains an arrow, either pointing up, down, left, or right. Given a starting configuration, Turbo the snail starts in one of the cells of the board and travels from cell to cell. In each move, Turbo moves one square unit in the direction indicated by the arrow in her cell (possibly leaving the board). After each move, the arrows in all of the cells rotate 90° counterclockwise. We call a cell *good* if, starting from that cell, Turbo visits each cell of the board exactly once, without leaving the board, and returns to her initial cell at the end. Determine, in terms of n, the maximum number of good cells over all possible starting configurations.

Solution. We will show that the maximum number of good cells over all possible starting configurations is

$$\begin{cases} \dfrac{n^2}{4} & \text{if } n \text{ is even and} \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$

Odd n

First, we will prove that there are no good cells if n is an odd number.

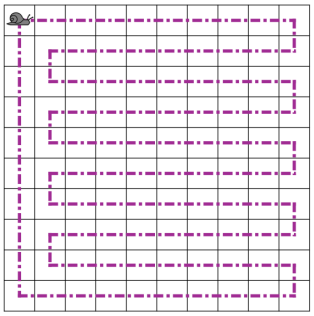
For Turbo to reach her goal, she must return to her initial cell after visiting every cell exactly once. Consider the chessboard coloring of the board. Without loss of generality, we assume that Turbo starts in a black cell. Since, at every step, Turbo moves to a cell of a different color; she will be in a white cell after $n^2 \equiv 1 \mod 2$ moves. Thus, it is impossible for Turbo to come back to her initial black cell on her n^2-th move, which is a contradiction. Thus there are no good cells.

Lower bound for even n

We will now construct a starting configuration with $\frac{n^2}{4}$ good cells for even n.

Let (i, j) denote the cell in row i and column j. Consider the following cycle

$$\begin{aligned} &(1,1) \to (1,2) \to (1,3) \to \ldots \to (1,n) \\ &\quad \to (2,n) \to (2,n-1) \to \ldots \to (2,2) \\ &\quad \ldots \\ &\quad \to (2i-1,2) \to (2i-1,3) \to \ldots \to (2i-1,n) \\ &\quad \to (2i,n) \to (2i,n-1) \to \ldots \to (2i,2) \\ &\quad \ldots \\ &\quad \to (n,n) \to (n,n-1) \to \ldots \to (n,2) \\ &\quad \to (n,1) \to (n-1,1) \to \ldots \to (2,1) \to (1,1). \end{aligned}$$



Note that the cycle returns to the initial cell after visiting every cell exactly once. To prove that $(1, 1)$ is good, we need to find a starting configuration such that Turbo traverses this cycle.

Let c_i be the $(i-1)$-th cell on the cycle: so we have $c_0 = (1, 1)$, $c_2 = (1, 2)$, $\ldots$, $c_{n^2-1} = (2, 1)$. For every i, we draw an arrow in cell c_i pointing towards cell c_{i+1} (or pointing towards c_0 if $i = n^2 - 1$) and then rotate this arrow i times 90° in the clockwise direction. After i moves, the arrow in c_i will have rotated i times 90° counterclockwise and be in the same direction as on the path defined above. Thus, Turbo will traverse the cycle $c_0, c_1, c_2, \ldots, c_{n^2-1}, c_0$ and $(1, 1)$ is good.

Every four moves, all arrows point in the same direction as in the beginning. Moreover, the board will return to its initial configuration after traversing the full cycle, since n^2, the length of the cycle, is divisible by 4. Therefore Turbo can also start at any c_i with $4 \mid i$ and follow the same route. Hence the cells $c_0, c_4, c_8, \ldots, c_{n^2-4}$ are good and there are $\frac{n^2}{4}$ of such cells.

Upper bound for even n

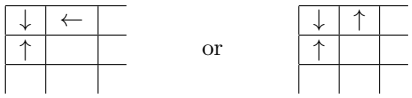
We will prove that for even n and any start configuration there are at most $\frac{n^2}{4}$ good cells.

Let a_0 be a good cell. Let $a_0, a_1, a_2, \ldots, a_{n^2-1}, a_{n^2} = a_0$ be the sequence of cells that Turbo visits when she starts at a_0. Now suppose there is another good cell b_0 and let $b_0, b_1, b_2, \ldots, b_{n^2-1}, b_{n^2} = b_0$ be the sequence of cells that Turbo visits when she starts at b_0.

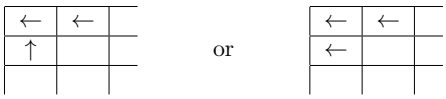
Note that, since $4 \mid n^2$, the arrows are back to their initial configuration after n^2 steps. Thus, if Turbo keeps walking after returning to her initial cell, she would just traverse the same cycle over and over again.

Consider the upper left corner of the board. With standard row and column numbering, the corner cell is $(1,1)$. This cell has only two neighbours, so both the a-route and the b-route must have cells $(2,1),(1,1),(1,2)$ in that order or $(1,2),(1,1),(2,1)$ in that order. Without loss of generality, $a_{i-1} = (2,1)$, $a_i = (1,1)$ and $a_{i+1} = (1,2)$ for some i. Let j be such that $b_j = (1,1)$. If $b_{j-1} = (2,1) = a_{i-1}$, then the arrow in cell $(2,1)$ must be pointed in the same direction after $i-1$ steps and after $j-1$ steps, so $i \equiv j \mod 4$. But then the arrow in cell $b_j = (1,1) = a_i$ must also be pointed in the same direction after i and after j steps, so Turbo moves to $b_{j+1} = a_{i+1}$ in both cases, and again finds the arrow pointed in the same direction in both cases. Continuing, we find that the b-route is actually identical to $a_{4t}, a_{4t+1}, \ldots, a_{n^2} = a_0, a_1, \ldots, a_{4t-1}, a_{4t}$ for some t, as any other starting point would have the arrows in the wrong direction initially.

Now suppose instead that $b_{j+1} = (2,1) = a_{i-1}$. Considering the a-route, the arrows in the upper left corner after $i-1$ steps must be like this:



Considering the b-route instead, the arrows after $j-1$ steps must be like this:



From the arrows in cell $(1,1)$ we see that $i \equiv j+1 \mod 4$. However, for the cells $(2,1)$ and $(1,2)$ this gives a contradiction.

We conclude that the only possible good cells are a_{4t} for $t = 0, 1, \ldots, \frac{n^2}{4} - 1$, which gives at most $\frac{n^2}{4}$ good cells.

P6. In each cell of a 2025×2025 board, a nonnegative real number is written in such a way that the sum of the numbers in each row is equal to 1, and the sum of the numbers in each column is equal to 1. Define r_i to be the largest value in row i, and let $R = r_1 + r_2 + \cdots + r_{2025}$. Similarly, define c_i to be the largest value in column i, and let $C = c_1 + c_2 + \cdots + c_{2025}$.

What is the largest possible value of $\frac{R}{C}$?

Solution 1. *Answer:* $\frac{2025}{89}$.

In general, if the table is $m^2 \times m^2$, the answer is $\frac{m^2}{2m-1}$.

The example is as follows: label rows and columns from 1 to m^2, from top to bottom and left to right. For the first m columns, write $\frac{1}{m}$ in all squares whose coordinates have the same residue modulo m and place 0 everywhere else. For the remaining $m^2 - m$ columns, place $\frac{1}{m^2}$ everywhere. Then $R = m^2 \cdot \frac{1}{m} = m$, and $C = m \cdot \frac{1}{m} + (m^2 - m) \cdot \frac{1}{m^2} = 2 - \frac{1}{m}$. So the ratio is as claimed.

$\frac{1}{2}$	0	$\frac{1}{4}$	$\frac{1}{4}$
0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
$\frac{1}{2}$	0	$\frac{1}{4}$	$\frac{1}{4}$
0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

In particular, when $n := m^2 = 2025$, we get $\frac{2025}{89}$. Now we need to show that $\frac{R}{C} \le \frac{n}{2\sqrt{n}-1}$.

For each row, select one cell having the largest value appearing in said row and colour it red. Then, without loss of generality, we may rearrange the columns such that the red cells appear in the first k columns from the left, and each such column contains at least one red cell, for some $k \le n$.

For the jth column, for all $1 \le j \le k$, let p_j and n_j denote the sum and number of red cells in it, respectively. We observe that c_j, the biggest number in the jth column, is at least $\frac{p_j}{n_j}$, for all $1 \le j \le k$. For all other columns, the largest value they contain is at least $\frac{1}{n}$, as their sum is 1. Thus, $C \ge \frac{p_1}{n_1} + \frac{p_2}{n_2} + \cdots + \frac{p_k}{n_k} + \frac{n-k}{n}$.

We can also observe that $R = p_1 + p_2 + \cdots + p_k$.

Therefore we have to show that:

$$p_1 + p_2 + \cdots + p_k \le \frac{n}{2\sqrt{n}-1} \cdot \left(\frac{p_1}{n_1} + \frac{p_2}{n_2} + \cdots + \frac{p_k}{n_k} + \frac{n-k}{n} \right). \tag{$*$}$$

By construction, $n_1 + n_2 + \cdots + n_k = n$, and, as the numbers in every column are nonnegative, we see that $p_j \le 1$ for every j. Also, since each number in a red cell is at least $\frac{1}{n}$, we also have $p_j \ge \frac{n_j}{n}$.

Since our inequality is linear in each p_j, it suffices to prove it when each variable equals one of its two critical values. By relabeling, we may assume that $p_j = \frac{n_j}{n}$ for $1 \le j \le t$, and $p_j = 1$ for $t+1 \le j \le k$, for an integer $0 \le t \le k$.

First, if $t = k$, we observe that $p_1 + p_2 + \cdots + p_k = \frac{n_1}{n} + \frac{n_2}{n} + \cdots + \frac{n_k}{n} = 1$, and that $\frac{p_1}{n_1} + \frac{p_2}{n_2} + \cdots + \frac{p_k}{n_k} = \frac{k}{n}$, so the inequality becomes $1 \le \frac{n}{2\sqrt{n}-1}$, which is true.

From now on we may assume that $t < k$. We need to show that:

$$\frac{n_1 + \cdots + n_t}{n} + k - t \le \frac{n}{2\sqrt{n}-1} \cdot \left(\frac{t}{n} + \frac{1}{n_{t+1}} + \cdots + \frac{1}{n_k} + \frac{n-k}{n} \right).$$

By Cauchy–Schwarz inequality we have that:

$$\frac{1}{n_{t+1}} + \cdots + \frac{1}{n_k} \ge \frac{(k-t)^2}{n_{t+1} + \cdots + n_k} = \frac{(k-t)^2}{n - (n_1 + \cdots + n_t)}. \tag{CS}$$

Let $n_1 + \cdots + n_t = n \cdot q$, where $0 \le q < 1$. Thus, it is now enough to show that:

$$q + k - t \le \frac{n}{2\sqrt{n} - 1} \left(\frac{t}{n} + \frac{(k-t)^2}{n - nq} + \frac{n-k}{n} \right).$$

Let $k - t = \ell \ge 1$. The inequality becomes:

$$q + \ell \le \frac{1}{2\sqrt{n} - 1} \cdot \left(n - \ell + \frac{\ell^2}{1-q} \right).$$

Rearranging this we get:

$$n + q + \frac{\ell^2}{1-q} \ge 2(q + \ell)\sqrt{n}.$$

If $q = 0$ the inequality is trivially true by $AM - GM$. Suppose now that $0 < q < 1$. Then, by Cauchy-Schwarz, we have:

$$q + \frac{\ell^2}{1-q} = \frac{q^2}{q} + \frac{\ell^2}{1-q} \ge (q + \ell)^2.$$

It is therefore enough to show that $n + (q + \ell)^2 \ge 2(q + \ell)\sqrt{n}$, which is true by $AM - GM$, completing the proof.

Remark. Another way to show $n + q + \frac{\ell^2}{1-q} \ge 2(q + \ell)\sqrt{n}$ is to split it into two $AM - GM$'s, as:

$$\left(n(1-q) + \frac{\ell^2}{1-q} \right) + (nq + q) \ge 2\ell\sqrt{n} + 2q\sqrt{n}.$$

Solution 1′. We prove the main inequality $(*)$ in a slightly different manner. Instead of the strong lower bound $p_j \ge \frac{n_j}{n}$, we use the weaker, simpler and more immediate lower bound $p_j \ge 0$ (thus proving the inequality in a larger regime).

As in Solution 1, suppose $p_j = 0$ for $1 \le j \le t$ and $p_j = 1$ for $t + 1 \le j \le k$, with $\ell = k - t$. We also denote by $m = n_{t+1} + \cdots + n_k$ and note that $m \le n - t$, since $n_j \ge 1$ for each $i \le t$. We need to prove that:

$$\ell \le \frac{n}{2\sqrt{n} - 1} \cdot \left(\frac{1}{n_{t+1}} + \cdots + \frac{1}{n_k} + \frac{n-k}{n} \right).$$

Rearranging and using the same Cauchy-Schwarz (CS) as in Solution 1, we see it suffices to show that:

$$(2\sqrt{n} - 1)\ell \le n - k + \frac{n\ell^2}{m},$$

or equivalently, that:

$$2\sqrt{n}\ell \le n - t + \frac{n\ell^2}{m}.$$

But since $n - t \ge m$ this immediately follows from $2\sqrt{n}\ell \le m + \frac{n\ell^2}{m}$, which is a simple application of $AM - GM$.

(Note that in Solution 1 the case $t = k$, which is equivalent to $m = \ell = 0$, was dealt with separately, to avoid the appearance of $\frac{0}{0}$ terms such as $\frac{n\ell^2}{m}$. It is easy to verify that the corresponding term should in fact be 0 and the final $AM - GM$ replaced with $0 \le 0$, and all transitions are valid. Alternatively, the case can be argued directly by simply noting that $(*)$ evaluates to $0 \le \frac{n-k}{2\sqrt{n}-1}$, which is obvious.)

Solution 1″. This is an alternative way of getting the upper bound on $\frac{R}{C}$ from

$$\frac{R}{C} \le \frac{p_1 + p_2 + \ldots + p_k}{\frac{p_1}{n_1} + \frac{p_2}{n_2} + \ldots + \frac{p_k}{n_k} + \frac{n-k}{n}}.$$

Using the fact that $\sum_{j=1}^{k} n_j = n$, we can rewrite the above right hand side as follows:

$$\frac{\sum_{j=1}^{k} p_j}{\sum_{j=1}^{k}\left(\frac{p_j}{n_j}+\frac{n_j-1}{n}\right)}.$$

We notice that this is a quotient of affine functions in the p_j's, for which the denominator does not vanish over the set defined by $0 \le p_j \le 1$. Therefore the maximum of this function is attained when a certain number of p_j's are 1 and the others are 0. Without loss of generality we may assume that the first t are equal to 1 and the other $k-t$ are 0 for some $0 \le t \le k$. Then one has that the previous expression is at most

$$\frac{t}{\sum_{1\le j\le t}\left(\frac{1}{n_j}+\frac{n_j-1}{n}\right)+\sum_{t<j\le k}\frac{n_j-1}{n}}.$$

We now lower bound the denominator by observing that the second sum is non negative, while each term of the first sum can be bounded by $AM-GM$ as follows:

$$\frac{1}{n_j}+\frac{n_j-1}{n} \ge \frac{2}{\sqrt{n}}-\frac{1}{n}.$$

We therefore have

$$\frac{R}{C} \le \max_{1\le t\le k}\frac{t}{\sum_{1\le j\le t}\left(\frac{2}{\sqrt{n}}-\frac{1}{n}\right)} = \frac{n}{2\sqrt{n}-1},$$

which finishes the proof.

Solution 1‴. We follow the same notation as above. First, we apply Cauchy-Schwarz as follows :

$$\left(\sum_{i=1}^{k}\frac{p_i}{n_i}\right)\left(\sum_{i=1}^{k}p_i n_i\right) \ge \left(\sum_{i=1}^{k}p_i\right)^2 = R^2.$$

We now write $z_i = 1-p_i$ for all $1 \le i \le k$, and observe that all z_i are positive. Moreover, we have that $\sum_{i=1}^{k} z_i = k-R$, and $\sum_{i=1}^{k} p_i n_i = n - \sum_{i=1}^{k} p_i z_i$. Thus, from our last inequality we get

$$\frac{p_1}{n_1}+\cdots+\frac{p_k}{n_k} \ge \frac{R^2}{n-\sum_{i=1}^{k} n_i z_i}.$$

As before we have $C \ge \frac{p_1}{n_1}+\cdots+\frac{p_k}{n_k}+\frac{n-k}{n}$, and so

$$C \ge \frac{R^2}{n-\sum_{i=1}^{k} n_i z_i}+\frac{n-k}{n} = \frac{R^2}{n-\sum_{i=1}^{k} n_i z_i}+\frac{n-R-\sum_{i=1}^{k} z_i}{n}.$$

Putting everything together we get

$$\frac{R}{C} \le \frac{R}{\frac{R^2}{n-\sum_{i=1}^{k} n_i z_i}+\frac{n-R-\sum_{i=1}^{k} z_i}{n}} = \frac{n}{\frac{Rn}{n-\sum_{i=1}^{k} n_i z_i}+\frac{n-\sum_{i=1}^{k} z_i}{R}-1}.$$

Applying $AM-GM$ to the denominator we get $\frac{R}{C} \le \frac{n}{2\sqrt{n\frac{n-\sum_{i=1}^{k} z_i}{n-\sum_{i=1}^{k} n_i z_i}}-1}$, which finishes the proof by noting that $\frac{n-\sum_{i=1}^{k} z_i}{n-\sum_{i=1}^{k} n_i z_i} \ge 1$.

Solution 2. This is an alternative approach that goes via an intermediary quantity in order to establish to the upper bound on R/C.

Let x_{ij} be the entry in row i and column j. Let $n = 2025$. The key idea is to analyze the expression:

$$T := \sum_{i,j} x_{ij} \max\left(x_{ij}, \frac{1}{n}\right).$$

On one hand, since $c_j \geq 1/n$, we have $x_{ij} \max(x_{ij}, \frac{1}{n}) \leq x_{ij} c_j$ for every (i, j). So

$$T \leq \sum_j \sum_i x_{ij} c_j = \sum_j c_j = C.$$

On the other hand, let j_i be one of the indices for which $r_i = x_{ij_i}$. We therefore have:

$$\begin{aligned} T &= \sum_i \left(x_{ij_i} \max\left(x_{ij_i}, \frac{1}{n}\right) + \sum_{j \neq j_i} x_{ij} \max\left(x_{ij}, \frac{1}{n}\right) \right) \\ &\geq \sum_i \left(r_i^2 + \frac{1}{n} \sum_{j \neq j_i} x_{ij} \right) \\ &= \sum_i \left(r_i^2 - \frac{1}{n} r_i + \frac{1}{n} \right) \geq \sum_i \left(\frac{2}{\sqrt{n}} r_i - \frac{1}{n} r_i \right) = \left(\frac{2}{\sqrt{n}} - \frac{1}{n} \right) R. \end{aligned}$$

This gives the claimed result.

EGMO 2024 - Problems and Solutions

Problems

Problem 1. Two different integers u and v are written on a board. We perform a sequence of steps. At each step we do one of the following two operations:

(i) If a and b are different integers on the board, then we can write $a+b$ on the board, if it is not already there.

(ii) If a, b and c are three different integers on the board, and if an integer x satisfies $ax^2+bx+c=0$, then we can write x on the board, if it is not already there.

Determine all pairs of starting numbers (u, v) from which any integer can eventually be written on the board after a finite sequence of steps.

Problem 2. Let ABC be a triangle with $AC > AB$, and denote its circumcircle by Ω and incentre by I. Let its incircle meet sides BC, CA, AB at D, E, F respectively. Let X and Y be two points on minor arcs $\widehat{DF}$ and $\widehat{DE}$ of the incircle, respectively, such that $\angle BXD = \angle DYC$. Let line XY meet line BC at K. Let T be the point on Ω such that KT is tangent to Ω and T is on the same side of line BC as A. Prove that lines TD and AI meet on Ω.

Problem 3. We call a positive integer n *peculiar* if, for any positive divisor d of n, the integer $d(d+1)$ divides $n(n+1)$. Prove that for any four different peculiar positive integers A, B, C and D, the following holds:

$$\gcd(A, B, C, D) = 1.$$

Here $\gcd(A, B, C, D)$ *is the largest positive integer that divides all of A,B,C and D.*

Problem 4. For a sequence $a_1 < a_2 < \cdots < a_n$ of integers, a pair (a_i, a_j) with $1 \le i < j \le n$ is called interesting if there exists a pair (a_k, a_l) of integers with $1 \le k < l \le n$ such that

$$\frac{a_l - a_k}{a_j - a_i} = 2.$$

For each $n \ge 3$, find the largest possible number of interesting pairs in a sequence of length n.

Problem 5. Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that the following assertions are true for every pair of positive integers (x, y):

- x and $f(x)$ have the same number of positive divisors.

- If x does not divide y and y does not divide x, then

$$\gcd(f(x), f(y)) > f(\gcd(x, y)).$$

Here $\gcd(m, n)$ *is the largest positive integer that divides both* m *and* n*.*

Problem 6. Find all positive integers d for which there exists a degree d polynomial P with real coefficients such that there are at most d different values among $P(0), P(1), \ldots, P(d^2 - d)$.

Solutions

Problem 1 (SVK)

Problem. Two different integers u and v are written on a board. We perform a sequence of steps. At each step we do one of the following two operations:

(i) If a and b are different integers on the board, then we can write $a + b$ on the board, if it is not already there.

(ii) If a, b and c are three different integers on the board, and if an integer x satisfies $ax^2 + bx + c = 0$, then we can write x on the board, if it is not already there.

Determine all pairs of starting numbers (u, v) from which any integer can eventually be written on the board after a finite sequence of steps.

Solution 1. We will show that the answer are the integer pairs (u, v) such that $u \neq 0$, $v \neq 0$, $\{u, v\} \neq \{-1, 1\}$ and $u > 0$ or $v > 0$.

If $u = 0$ or $v = 0$, then (i) will never yield a new number and we cannot use (ii) with only two numbers. Hence, if $u = 0$ or $v = 0$, we cannot reach every possible y. From now on, assume $u \neq 0$ and $v \neq 0$.

If both numbers u, v were negative, we will show that there can only be negative numbers on the board. With negative numbers a, b, operation (i) will only yield a negative number. The same holds for operation (ii), because for a non-negative x and negative a, b, c, we have $ax^2 + bx + c < 0$. Hence, if both $u < 0$ and $v < 0$, then we cannot reach every possible y. From now on, assume that at least one of u, v is positive. Without loss of generality take $u < v$, and so $v > 0$.

After one step, we can have the numbers $u, v, u + v$, which are mutually distinct due to u, v being nonzero. Notice that the equation $ux^2 + (u + v)x + v = 0$ has a root -1, and so we can have -1 on the board.

We now check two cases: case $v = 1$, and case $v > 1$.

If $v = 1$, then $u < 0$. Further split the case of $v = 1$ based on whether $u = -1$ or $u < -1$.

If $v = 1$ and $u = -1$, we can only additionally write number 0 on the board using operation (i); and no additional numbers using (ii) because setting $\{a, b, c\}$ to $\{-1, 0, 1\}$ in any order only has solutions for x in $\{-1, 0, 1\}$. Hence, if $\{u, v\} = \{-1, 1\}$, then we cannot reach every possible y.

If $v = 1$ and $u < -1$, we can use operation (i) on numbers $u, -1$ (and then repeat choosing the obtained result and -1) to get any negative number smaller than u, and operation (i) on numbers $(u, 1)$ (and then repeat choosing the obtained result and 1) to get any negative number larger than u, as well as 0. Then, we set $(a, b, c) = (0, 1, -2)$ and apply operation (ii) to additionally get number 2. Applying (i) on $(2, 1)$ (and then repeat choosing the obtained result and 1), we can get all the remaining integers too.

From now on, assume $v > 1$. Recall that we can make $u + v$ and -1.

We will now apply operation (i). First, $(v, -1)$ gives $v - 1$. Next, $(v, v - 1)$ gives $2v - 1$. Since $v > 1$, we know $v \neq 2v - 1$, so we can apply operation (i) on $(v, 2v - 1)$ to get $3v - 1$, and then continue adding $(v, kv - 1)$ to get $(k + 1)v - 1$ for any positive k. Since $v > 1$, we can get an arbitrarily large integer by repeating this.

If a is any positive number on the board, applying (i) to $(a, -1)$ gives $a - 1$. By repeating this, we have that we can get all numbers smaller than a and larger than or equal to -1. Together with previously having found a way to get an arbitrarily large integer, we have that we can get any integer $l \geq -1$ on the board.

Now, we set $(a, b, c) = (0, 1, 2)$ and apply operation (ii) to additionally get number -2. Then we can repeat operation (i) on $(-1, -2)$ (and afterwards on -1 and the obtained result) to get any negative number.

Therefore, if $u \neq 0$, $v \neq 0$, $\{u, v\} \neq \{-1, 1\}$ and $u > 0$ or $v > 0$, we can write every integer on the board.

Remark. There are of course many other ways to create all integers in case of (u, v) that satisfies all the requirements. Basically you will always use (i) to create arbitrarily large numbers and then (ii) to create a number in a range that you didn't have before.

Solution 2. If $u = 0$ or $v = 0$ then we can only get $\{u, v\}$.

Proof. If $u = 0$ or $v = 0$, then (i) will never yield a new number and we cannot use (ii) with only two numbers. □

If $\max(u, v) < 0$, we cannot get non-negative numbers.

Proof. For $a, b, c < 0$:

(i) cannot generate a non-negative number as $a + b < 0$.

(ii) cannot generate a non-negative number as for $x \geq 0 : ax^2 + bx + c \leq c < 0$. □

If $u + v \neq 0$, $u, v \neq 0$ and $max(u, v) > 0$, we can get every number.

Proof. $u \neq v \to u + v$ can be written. $u \neq 0$, so $u + v \neq v \to u + 2v$ can be written. $v \neq 0$, so $u + 2v \neq u$, meaning that $2u + 2v$ can be written. If for $n > 1$, $n(u + v)$ can be written then $(n + 1)(u + v) = n(u + v) + (u + v)$ can also be written, because $u + v \neq 0$, so $n(u + v) \neq (u + v)$. Therefore, by induction for all $n > 0$ the number $n(u + v)$ can be written.

Taking $n = 2, 3, \ldots$ and $(u+v)(x+1)(x+n) = (u+v)x^2+(u+v)(n+1)x+(u+v)n = 0$ gives $x = -1$ and $x = -n \to$ for all $n > 0$ we can get $-n$ (we can get all the negative numbers). Additionally, we can get $u + (-u) = 0$.

For $n \geq 1$ we can take $0 \cdot x^2 + ux + (-un) = 0$ as $u \neq 0 \to u, -nu, 0$ are all distinct, therefore we can get n. Thus, we can get all the numbers. □

If $u + v = 0$ and $\max(u, v) > 1$ we can get all the numbers.

Proof. We can get $0 = u + v$ as $u \neq v$. Take $0 \cdot x^2 + ux - u = 0$ so we can get 1 written on the board. Take $(u', v') = (1, \max(u, v))$ and then use the result from claim 3. As $0 < u' = 1 < \max(u, v) = v'$ and $u' + v' > 0$ we can get all the numbers. □

Remaining case: If $v = 1$ and $u = -1$, we can only additionally write number 0 on the board using operation (i); and no additional numbers using (ii) because setting $\{a, b, c\}$ to $\{-1, 0, 1\}$ in any order only has solutions for x in $\{-1, 0, 1\}$. Hence, if $\{u, v\} = \{-1, 1\}$, then we cannot reach every possible y.

Solution 3. We show none of the initial number can be 0, as in Solution 1. Then, we split into three cases: initial numbers having different signs, both being positive, and both being negative.

Case 1. Suppose we have two numbers u, v with different signs such that $\gcd(u, v) = k$ for some $k \in \mathbb{Z}^+$. Without loss of generality assume that $u > 0$ and $v < 0$.

Case 1.1. $v < -k$:

We can generate all numbers y such that $k \mid y$ and $y \leq u$.

Proof. Define $u' = \frac{u}{k}$, $v' = \frac{v}{k}$. Note that by definition, $\gcd(u', v') = 1$. If starting with numbers u', v', we can make a sequence of applications of only the (i) rule to write a number $y = p \cdot u' + q \cdot v'$ on the board, then we can apply the same sequence of moves to u, v to write $k \cdot y = p \cdot u + q \cdot v$ on the board.

Therefore, we will prove instead that if $v' < -1$, we can write all numbers $y < u'$ on the board, which is equivalent to the lemma.

We attempt to write numbers $u' + q \cdot v'$ for $q \in \mathbb{Z}^+$ on the board by repeatedly adding v' to u'. This process can only ever halt if we reach a point where $u' + q \cdot v' = v'$. That cannot occur, as it would imply $u' = (1 - q) \cdot v'$. Taking into account that $u', v' \neq 0$, that implies $\gcd(u', v') = |v'|$. That is a contradiction, as $|v'| > 1$.

Therefore, we can write all numbers of the form $u' + q \cdot v'$ on the board for $q \in \mathbb{Z}^+$. We will use these numbers to construct an arbitrary integer y.

If we want to write a number $y < u'$ on the board, and we already have a number $n < y$ on the board such that $y \equiv n \pmod{u'}$, then we can construct y by repeatedly adding u' to n until we reach y, skipping all numbers that are already on the board. As $y < u'$, none of the numbers we attempt to add u' to will be equal to u'.

Suppose we fix any number $y < u'$. As $\gcd(u', v') = 1$, $q \cdot v'$ takes all residues modulo u' as q runs through the positive integers. Therefore, we will always be able to find a number n of the form $u' + q \cdot v'$ such that $y \equiv n \mod u'$. We can generate an arbitrarily small n' by taking $n' = n + l \cdot u' \cdot v' = u' + (q + l \cdot u') \cdot v'$ for large enough l, making both $n' < y$ and $n' \equiv y \mod u'$ true.

Therefore, we can write all numbers $y < u'$ on the board. Thus, starting from u and v, we can get all numbers $y \leq u$ s.t. $k \mid y$. □

The numbers k, 0 and all negative multiples of k are a subset of the integers $y \leq u$ s.t. $k \mid y$. Therefore, we have all of those numbers on the board.

We can now get an arbitrary nonzero number by applying (ii) to the polynomial $k \cdot (x - n)(x + n) = k \cdot x^2 + 0 \cdot x - n^2 \cdot k$. The coefficients of this polynomial are distinct for all integers $n \neq 0$, and they are from the set $\{k, 0\} \cup \{q \cdot k \mid q < 0\}$, which we have on the board. Therefore the rule application is valid.

As this works for all integers $n \in \mathbb{Z} \setminus \{0\}$, and 0 is already on the board, we have proven that we can write all integers on the board.

Case 1.2.: $v = -k$:

Case 1.2.1.: $k \neq 1$.

If $k \neq 1$, we can generate -1 from the polynomial $u \cdot x^2 + (u + v)x + v$. We can now repeatedly add -1 to $v = -k$ until we reach $-2k$ on the board. Now, we can appeal to Case 1.1.

Case 1.2.2. $k = 1$.

To restate the conditions of this sub-case, $v = -k = -1$, and u is an arbitrary positive number.

Case 1.2.2.1. $u = 1$.

If $u = 1$, we are only ever able to construct the numbers $-1, 0, 1$, no matter how we apply the rules.

Case 1.2.2.2. $u \neq 1$.

We can subtract 1 from u until we reach 0. This procedure also generates 1. We now add 1 to u until we get all positive numbers.

Now, we find an arbitrary polynomial with different positive coefficients that have a negative root smaller than -1. An example is $(x + 3)^2$.

We will now keep adding -1 to -3 to get all the negative numbers $y \leq -3$. To get -2, we can add -3 to 1, generating all integers.

Case 2 Suppose that both of u, v are positive.

We can now use any method from the previous solutions to generate a negative number, and then appeal to Case 1.

Case 3 Suppose $u < 0, v < 0$.

Then there can only be negative numbers on the board. With negative numbers a, b, operation (i) will only yield a negative number. The same holds for operation (ii), because for a non-negative x and negative a, b, c, we have $ax^2 + bx + c < 0$. Hence, if both $u < 0$ and $v < 0$, then we cannot reach every possible y.

Problem 2 (UNK)

Problem. Let ABC be a triangle with $AC > AB$, and denote its circumcircle by Ω and incentre by I. Let its incircle meet sides BC, CA, AB at D, E, F respectively. Let X and Y be two points on minor arcs $\overset{\frown}{DF}$ and $\overset{\frown}{DE}$ of the incircle, respectively, such that $\angle BXD = \angle DYC$. Let line XY meet line BC at K. Let T be the point on Ω such that KT is tangent to Ω and T is on the same side of line BC as A. Prove that lines TD and AI meet on Ω.

Solution 1. By the alternate segment theorem we have that:

$$180^\circ = \angle DCY + \angle CYD + \angle YDC = \angle DCY + \angle DXB + \angle YXD = \angle DCY + \angle YXB.$$

Therefore opposite angle of $BXYC$ are supplementary and so $CYXB$ cyclic.

One can apply power of a point at K:

$$KT^2 = KB \cdot KC = KX \cdot KY = KD^2 \Longrightarrow KT = KD.$$

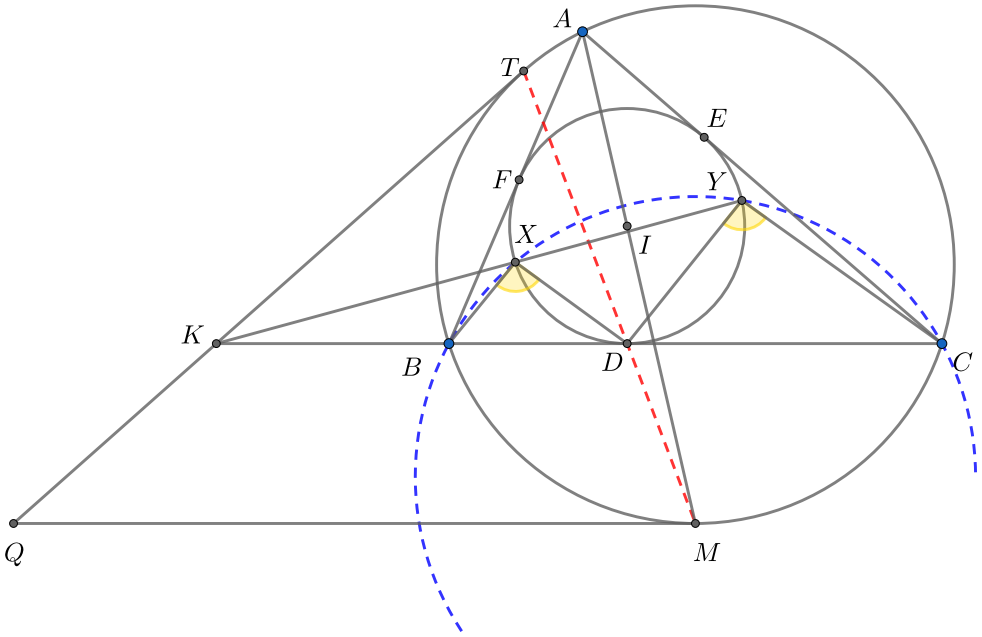


Figure 1: The proposer's solution using a new point Q

(Alternatively you can sidestep power of a point by observing that K is the radical centre of the incircle DEF, the circumcircle Γ and the circle $CYXB$ and so $KT^2 = KD^2$.)

Now let AI meet Ω at M, the midpoint of $\widehat{BC}$ not containing A. Let the tangent at M meet KT at Q. Observe that $QM \parallel KD$ so $\angle TKD = \angle TQM$ and also $KT = KD$, $QT = QM$ hence $\triangle TKD \sim \triangle TQM$. As T, K, Q are collinear, this means that T, D, M are collinear so TD and AI meet at M which lies on Ω.

Solution 2. The role of X and Y in this problem is secondary. Draw any circle through B and C which meets the incircle at X and Y and you determine the same point K. This is because K is the radical centre of the three circles in play. Therefore K is the intersection of BC with the radical axis of the circumcircle Ω and the incircle DEF, and so is independent of the choice of circle through B and C which gives rise to X and Y. In the problem as posed, this is disguised, with angle properties of X and Y giving rise the the circle $BXYC$.

You can now work with the simplified diagram shown in Figure 2.

Somehow we must use a characterization of M in order to finish. In the proposer's solution, we used the tangent to the circumcircle Ω at M being parallel to BC. In this alternative, we use the fact that the internal angle bisector of angle $\angle CTB$ meets Ω again at M.

Triangle TKD is isosceles with apex K so

$$\angle KDT = \angle DTK = \angle DTB + \angle BTK.$$

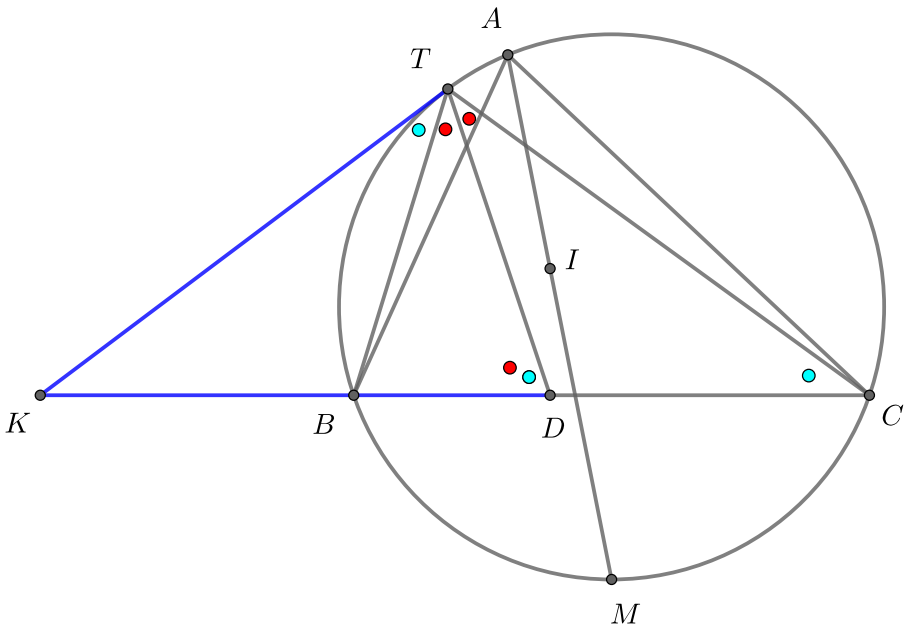


Figure 2: Illustration of an angle chase

By the alternate segment theorem, $\angle BTK = \angle BCT = \angle DCT$. Now angle $\angle KDT$ is an exterior angle of triangle DCT so $\angle CTD = \angle DTB$.

Therefore the line TD is the internal angle bisector of angle $\angle CTB$ and so must pass through M, the midpoint of the arc $\widehat{BC}$ of Ω which does not contain T.

Observation This affords a construction of the radical axis of the incircle and circumcircle of triangle ABC: M and D determine T and the tangent at T meets BC at a point on the required radical axis. See Figure 3. Now do a cyclic change of letters of ABC and find a point on CA which is also on the radical axis (the dashed magenta line).

Solution 3. The centre of direct enlargement from the incircle to the circumcircle gives another way to finish the proof. This enlargement carries D to M since the tangent lines to their associated circles are parallel. The centre of enlargement therefore lies on the line MD. Let MD meet the incircle again at U and the circumcircle again at V. Draw the tangent to the incircle at U to meet BC at W so triangle WDU is isosceles with apex W and has equal base angles $\angle WDU$ and $\angle DUW$. The enlargement carries the line WU to the tangent line to the circumcircle at V which meets BC at S. Enlargements carry lines to parallel lines, so $\angle DVK' = \angle DUW = \angle WDU = \angle K'DV$. Therefore triangle $K'DV$ is isosceles with apex K'.

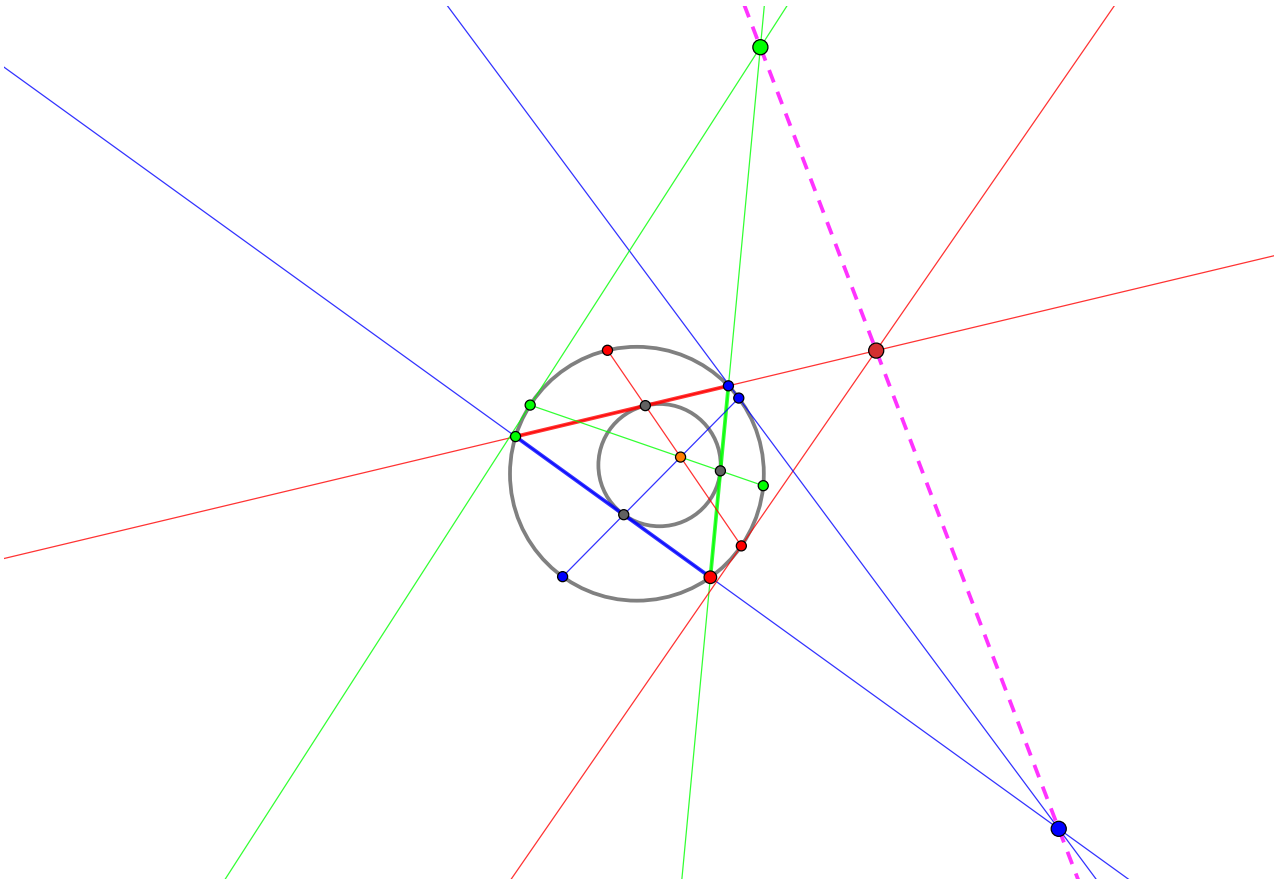


Figure 3: The three points K on the radical axis.

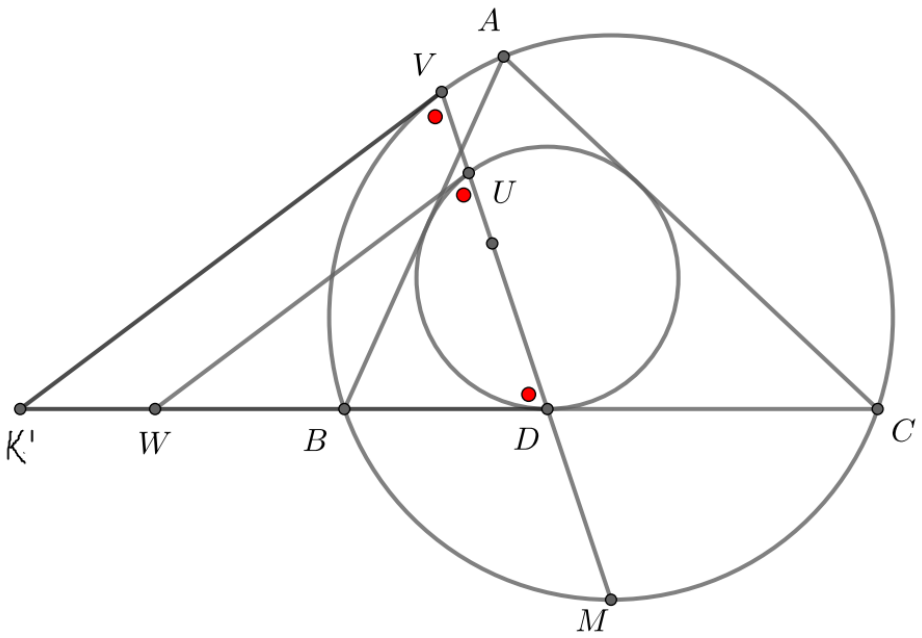


Figure 4: The enlargement of W gives K'.

This identifies S as the intersection of the radical axis of the two circles with BC, so $K' = K$ and $V = T$ and the proof is complete.

Solution 4. Let ϕ be the inversion with center K and radius KD. Note that this inversion maps incirlce of ABC to itself and K, X, Y are collinear, hence $\phi(X) = Y$. Also $\phi(D) = D$, so ϕ maps circle XBD to cicle YDB', where $B' := \phi(B)$ is the point on BC different from B such that $\angle BXD = \angle DYB'$, hence $\phi(B) = C$. From $\phi(B) = C$ we get that $KD = KT$ as $KD^2 = KB \cdot KC = KT^2$ since KD is the radius of inversion.

The rest of the solutions is the same as in the other solutions.

Problem 3 (NLD)

Problem. We call a positive integer n *peculiar* if, for any positive divisor d of n, the integer $d(d+1)$ divides $n(n+1)$. Prove that for any four different peculiar positive integers A, B, C and D, the following holds:

$$\gcd(A, B, C, D) = 1.$$

Here $\gcd(A, B, C, D)$ *is the largest positive integer that divides all of A,B,C and D.*

First part of the solution, option 1

Observe that $n = 1$ is peculiar and that every prime is peculiar. Consider $\frac{n}{d}$, where $d > 1$

is a divisor of n. Then $\frac{n}{d}\left(\frac{n}{d}+1\right)$ divides $n(n+1)$, equivalent to $n+d$ dividing $d^2(n+1)$. Since $n \equiv -d \pmod{n+d}$, we obtain that $n+d$ divides d^3-d^2, hence $n+d \leq d^3-d^2$ and in particular $n < d^3$. Now taking d to be the smallest prime divisor of n, we deduce that n cannot have more than two prime factors, counting multiplicities. Hence from now on we work with $n = pq$, where p and q are primes. In fact, $p \neq q$, as otherwise $p+1 \mid p^2-p$, i.e. $p+1 \mid 2$, contradiction.

First part of the solution, option 2

Let us first prove that n is square-free. Assume that $n = p^k h$, where $\gcd(p,h) = 1$. Now $p^{k-1}h(p^{k-1}h+1) \mid p^k h(k^k h+1)$, which is equivalent to $(p^{k-1}h+1) \mid p(p^k h+1)$. Since $\gcd(p, p^{k-1}h+1) = 1$, we must have

$$(p^{k-1}h+1) \mid (p^k h+1) = p^k h + p - p + 1,$$

so $(p^{k-1}h+1) \mid (p-1)$. If $k \geq 2$, this is a contradiction. Hence, n is square-free.

Let us now show that n has at most 2 different prime factors. Write $n = p_1 p_2 \cdots p_k$. Now we must have

$$p_2 \cdots p_k (p_2 \cdots p_k + 1) \mid p_1 \cdots p_k (p_1 \cdots p_k + 1),$$

which is equivalent to

$$(p_2 \cdots p_k + 1) \mid p_1 (p_1 \cdots p_k + 1).$$

Now

$$p_1(p_1 \cdots p_k + 1) - p_1^2 (p_2 \cdots p_k + 1) = p_1 - p_1^2.$$

Hence,

$$(p_2 \cdots p_k + 1) \mid p_1^2 - p_1.$$

Pick p_1 to be the smallest prime factor of n. Then $p_2 \cdots p_k > p_1^2$ if $k > 2$, which is a contradiction.

Second part of the solution, option 1

The only relations of concern are $p(p+1) \mid n(n+1)$ and $q(q+1) \mid n(n+1)$. The former yields $p(p+1) \mid pq(pq+1)$, i.e. $p+1 \mid q(pq+1)$, that is, $p+1 \mid q(q-1)$; analogously $q+1 \mid p(p-1)$. Without loss of generality assume $p > q$. If q does not divide $p+1$, then since $p+1$ and q are relatively prime, we obtain from $p+1 \mid q(q-1)$ that $p+1$ divides $q-1$ and hence $p < p+1 \leq q-1 < q$, contradiction. Hence q divides $p+1$. Also, p does not divide $q+1$ due to $p > q$ (unless $p = q+1$, i.e. $p = 3$, $q = 2$, but then $p+1 \nmid q(q-1)$) and now from $q+1 \mid p(p-1)$ we deduce that $q+1$ divides $p-1$.

Write $p+1 = mq$ for some positive integer m. Then $q+1$ divides $p-1 = mq-2 = m(q+1)-m-2$, so $q+1$ divides $m+2$. On the other hand, $p+1 \mid q(q-1)$ implies that m divides $q-1$. If the ratio of the latter is greater than 1, then $m \leq \frac{q-1}{2}$, but $q+1$ dividing $m+2$ from above insists on $q \leq m+1$, so $m \leq \frac{q-1}{2} \leq \frac{m}{2}$, contradiction. Hence $m = q-1$ and we have proven the following: if $n > 1$ is composite and peculiar, then $n = pq$, where $p > q$ are primes and $p = q^2 - q - 1$.

Second part of the solution, option 2

Hence from now on we work with $n = ab$, with $a < b$ primes. The only relations of concern are $a(a+1) \mid n(n+1)$ and $b(b+1) \mid n(n+1)$. The former yields $a(a+1) \mid ab(ab+1)$, i.e. $a+1 \mid b(ab+1)$, that is, $a+1 \mid b(b-1)$; analogously $b+1 \mid a(a-1)$. If a does not divide $b+1$, then since $b+1$ and a are relatively prime, we obtain from $b+1 \mid a(a-1)$ that $b+1$ divides $a-1$ and hence $b < b+1 \leq a-1 < a$, contradiction. Hence a divides $b+1$. Writing $b+1 = ac$, from $b+1 \mid a(a-1)$ we have that c divides $a-1$.

Now from $b+1 = ac$ with $c \mid a-1$ and $a+1 \mid b(b-1)$ we derive that $b = a^2 - a - 1$ in a different (and longer) manner, without using the primality of b until the final step. Note that since $b = ac - 1 \equiv -c - 2 \pmod{a+1}$, we have $a+1 \mid (c+1)(c+2)$. Since $c \mid a-1$, we may write $a - 1 = cd$, and note that $cd \equiv -2 \pmod{a+1}$. We then have

$$0 \equiv -d(c+1)(c+2) = -c^2 d - 3cd - 2d \equiv 2(c-d) + 6 \pmod{a+1} \qquad (1)$$

Since $1 \leq c, d \leq a-1$, we have $|2(c-d)+6| \leq 2(a-2)+6 = 2a+2$. Furthermore, we have $c \geq 2$ (otherwise $b = ac - 1 < a$, contradiction), and so $d \leq \frac{a-1}{2}$, thus $2(c-d)+6 \geq 11 - a > -(a+1)$. From these inequalities and equivalence (1), we obtain $2(c-d)+6 \in \{0, a+1, 2(a+1)\}$. It is immediate to verify that these three possible values correspond to the following three cases, respectively:

(i) $d = c+3$ and $a = c^2 + 3c + 1$;

(ii) $d = 2$, $c = \frac{a-1}{2}$, $a = 2c+1$; and

(iii) $d = 1$, $c = a - 1$.

Substituting these values in $b = ac - 1$, we obtain in case (i), $b = c^3 + 3c^2 + c - 1 = (c+1)(c^2+2c-1)$, and in case (ii), $b = 2c^2 + c - 1 = (c+1)(2c-1)$. In both cases these decompositions contradict the primality of b, as $c \geq 2$. Thus we are left only with case (iii), where $b = a^2 - a + 1$, as claimed.

End of the solution (joint for different approaches)

In conclusion, a prime p can only divide the peculiar numbers p, $p(p^2-p-1)$ (if p^2-p-1 is prime) and pq where $p = q^2 - q - 1$ (if q is prime). In particular, the greatest common divisor of more than three peculiar numbers must be 1. (Note that three peculiar numbers may indeed have a common divisor, e.g. 5, $15 = 3 \cdot 5$ and $95 = 5 \cdot 19$ are all peculiar.)

Problem 4 (UKR)

Problem. For a sequence $a_1 < a_2 < \cdots < a_n$ of integers, a pair (a_i, a_j) with $1 \leq i < j \leq n$ is called interesting if there exists a pair (a_k, a_l) of integers with $1 \leq k < l \leq n$ such that

$$\frac{a_l - a_k}{a_j - a_i} = 2.$$

For each $n \geq 3$, find the largest possible number of interesting pairs in a sequence of length n.

Answer. $\frac{1}{2}(n-1)(n-2)+1$.

Solution. Consider the numbers $a_i = 2^i$ for $2 \le i \le n$ and $a_1 = 0$, and choose any pair (i, j) with $1 \le i < j \le n$.

If $i = 1$, the pair $(1, j)$ is interesting for all $2 \le j \le n-1$, as $\dfrac{a_{j+1} - a_1}{a_j - a_1} = \dfrac{2^{j+1}}{2^j} = 2$.

If $i \ge 2$, the pair (i, j) is also interesting for all j such that $i+1 \le j \le n-1$, since $\dfrac{a_{j+1} - a_{i+1}}{a_j - a_i} = \dfrac{2^{j+1} - 2^{i+1}}{2^j - 2^i} = 2$.

In addition, the pair $(n-1, n)$ is interesting, as $\dfrac{a_n - a_0}{a_n - a_{n-1}} = 2$.

Hence, all pairs in which $j \le n-1$, and additionally the pair $(n-1, n)$, are interesting, giving $\frac{1}{2}(n-1)(n-2)+1$ pairs in total.

Now, let's show that there can't be more interesting pairs. To this end we show that at least $\frac{1}{2}n(n-1) - \left(\frac{1}{2}(n-1)(n-2)+1\right) = n-2$ pairs are not interesting. Clearly, the pair $(1, n)$ is not interesting.

Let's make the following observation: if a pair (i, j) is interesting and it's not $(1, n)$, then $a_j - a_i$ can't exceed $\frac{1}{2}(a_n - a_1)$. (Since, if (i, j) is interesting then $a_l - a_k = 2(a_j - a_i)$ for some pair (k, l). If then $2(a_j - a_i) > a_n - a_1$ we would have $a_l - a_k > a_n - a_1$, which is not possible.)

Finally, for any $2 \le i \le n-1$ consider pairs $(1, i)$ and (i, n). If both of them are interesting, then, following the observation above, the only possibility is $a_i - a_1 = a_n - a_i = \dfrac{a_n - a_1}{2}$. Clearly, this is possible for at most one i, so for all other $n-3$ values of i, at least one of pairs $(1, i)$ and (i, n) is not interesting. Combining with the pair $(1, n)$, we get at least $n-2$ not interesting pairs in total, as desired.

Comment: The originally proposed formulation considered a sequence $a_1 < a_2 < \cdots < a_n$ of n real numbers, rather than integers. The conclusion is however the same and the solution is identical.

Problem 5 (HRV)

Problem. Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that the following assertions are true for every pair of positive integers (x, y):

- x and $f(x)$ have the same number of positive divisors.
- If x does not divide y and y does not divide x, then

$$\gcd(f(x), f(y)) > f(\gcd(x, y)).$$

Here $\gcd(m, n)$ *is the largest positive integer that divides both* m *and* n*.*

Solution 1. Throughout this proof, when we write divisors we mean positive divisors.

Let f be a function satisfying these properties. From the first one, it follows that $f(1) = 1$ and for each prime number p, $f(p) = q_p$ for some prime number q_p. Assume $p \neq 2$, the pair $(2, p)$ in property 2 gives

$$\gcd(q_2, q_p) = \gcd(f(2), f(p)) > f(\gcd(2, p)) = f(1) = 1,$$

so $q_p = q_2$ for all p. Set $q = q_2$.

Denote by $d(n)$ the number of divisors of a positive integer n.

We will prove the following claim by induction on $k \geq 0$, the number of distinct prime divisors of n: For a positive integer n, $f(n)$ is a power of q.

The claim, together with the first property, imply that $f(n) = q^{d(n)-1}$.

For $k = 0$ the claim is already proved.

For $k = 1$, we need to prove that $f(p^m)$ is a power of q for all positive m. The case $m = 1$ was already proved. Assume now the result being proved for all positive integers $\leq m$. As $d(p^t) = t + 1$, we then know $f(p^t) = q^t$ for all $t \leq m$. Now we consider $t = m + 1$.

Let $r \neq p$ be a prime. Plug the pair $(p^{m-1}r, p^m)$ into the second property:

$$\gcd(f(p^{m-1}r), q^m) = \gcd(f(p^{m-1}r), f(p^m)) > f(p^{m-1}) = q^{m-1}.$$

This implies q^m divides $f(p^{m-1}r)$. Since $f(p^{m-1}r)$ has $2m$ divisors, and $v_q(f(p^{m-1}r)) \geq m$, it follows that $f(p^{m-1}r)$ does not have prime divisors other than q, since it would then have at least $2 \cdot (v_q(f(p^{m-1}r)) + 1) > 2m$ divisors. Thus, $f(p^{m-1}r)$ is a power of q. And since it has $2m$ divisors, we must have $f(p^{m-1}r) = q^{2m-1}$.

Now, plug the pair $(p^{m+1}, p^{m-1}r)$ into the second property. We have

$$\gcd(f(p^{m+1}), q^{2m-1}) = \gcd(f(p^{m+1}), f(p^{m-1}r)) > f(p^{m-1}) = q^{m-1}.$$

Then $f(p^{m+1})$ is divisible by q^m. If $f(p^{m+1})$ had a prime factor other than q, it would have at least $2(m + 1)$ divisors, but it has $m + 2$ divisors and $2(m + 1) > m + 2$ since $m \geq 1$. Hence, $f(p^{m+1})$ must also be a power of q. By induction, the proof is finished for $k = 1$.

Suppose the claim is true for all integers n with at most k distinct prime factors, for some $k \geq 1$. In order to prove the claim for integers with $k+1$ distinct prime factors, it suffices

to show that for every positive integer N with k distinct prime factors, every positive integer m and every prime p which does not divide N, the number $f(Np^m)$ is a power of q. We will prove this by induction on m. The case $m = 0$ follows from the previous induction hypothesis. Now suppose it has already been proved for a certain $m \geq 0$.

Let s be a prime divisor of N. Consider the numbers (Np^{m+1}, Nsp^m). By the second property,

$$\gcd(f(Np^{m+1}), q^{d(Nsp^m)-1}) = \gcd(f(Np^{m+1}), f(Nsp^m)) > f(Np^m) = q^{d(Np^m)-1}.$$

Then $q^{d(Np^m)}$ divides $f(Np^{m+1})$. If $f(Np^{m+1})$ has a prime factor other than q, it would have at least $2(d(Np^m) + 1) = 2((m + 1)d(N) + 1) = (2m + 2)d(N) + 2$ divisors, but it has only $(m + 2)d(N)$ divisors. Hence, $f(Np^{m+1})$ must also be a power of q.

Finally, let us check that the function $f(n) = q^{d(n)-1}$ does satisfy the properties of the problem for any prime number q. The first property obviously holds. To check the second one, note that

$$\gcd(f(x), f(y)) = q^{\min(d(x),d(y))} > q^{d(\gcd(x,y))} = f(\gcd(x, y))$$

whenever $x \nmid y$ and $y \nmid x$.

Solution 2. Firstly, one can prove that for any prime p and any $m \geq 0$, $f(p^m) = q^m$, in the same way as in the first solution.

We will prove the claim this time by induction on $k = d(n)$. We notice again that the claim implies that $f(n) = q^{d(n)-1}$.

The cases $k = 1, 2, 3$ are then already proved. Now suppose that for some $k \geq 3$ we have proved the claim for all positive integers n with at most k divisors.

Consider a positive integer n with $k + 1$ divisors. If n is a power of a prime, then it is already proved that $f(n)$ is a power of q. Suppose n is not a power of a prime. Consider distinct prime numbers p_1 and p_2 which divide n, such that $v_{p_1}(n) \leq v_{p_2}(n)$. Consider the number $\frac{np_2}{p_1}$: its number of divisors is $\frac{v_{p_1}(n)(v_{p_2}(n)+2)}{(v_{p_1}(n)+1)(v_{p_2}(n)+1)}d(n)$, which is strictly less than the number of divisors of n. Thus, by induction hypothesis, $f(\frac{np_2}{p_1})$ is a power of q.

By applying the second property to the pair $(n, \frac{np_2}{p_1})$, we have

$$\gcd(f(n), q^{d\left(\frac{np_2}{p_1}\right)-1}) = \gcd(f(n), f\left(\frac{np_2}{p_1}\right)) > f\left(\frac{n}{p_1}\right) = q^{d(\frac{n}{p_1})-1}.$$

This implies that $q^{d(\frac{n}{p_1})}$ divides $f(n)$. Again, if other primes than q divided $f(n)$, it would have at least $2(d(\frac{n}{p_1})+1)$ divisors. But $2(d(\frac{n}{p_1})+1) = 2(\frac{v_{p_1}(n)}{v_{p_1}(n)+1}d(n)+1) > d(n)$. So $f(n)$ is a power of q and the Claim is proved.

We finally check as in the first solution that $f(n) = q^{d(n)-1}$ satisfies the 2 properties for all prime q.

Problem 6 (LUX+BEL)

Problem. Find all positive integers d for which there exists a degree d polynomial P with real coefficients such that there are at most d different values among $P(0), P(1), \ldots, P(d^2-d)$.

Remark 1 You can add a constant to a polynomial satisfying the conditions and obtain another polynomial satisfying the conditions.

Remark 2 This problem is equivalent to: Find all positive integers d for which there exists a polynomial P of degree d that takes at most d different values at members of a an arithmetic progression with d^2-d+1 (different) terms.
The answer to this EGMO problem implies a solution to this more general problem by scaling and translating, and the general result solves the EGMO problem as a special case. This means that instead of working with $0, 1, \ldots, d^2-d$ we have the option of working with a region that is symmetric about the origin. Irrespective of the parity of d we can use the d^2-d+1 integers which run from $-(d^2-d)/2$ to $(d^2-d)/2$ inclusive.

The advantage of using an arithmetic progression which is symmetric about the 0 is that $P(X)$ is a polynomial which works for the problem if, and only if, $P(-X)$ is a polynomial which works for the problem. This means that working at one end of the run of integers is the reflection of the activity at the other end. T Remarks dismissing the cases at the other end of the run as similar now become obviously correct rather than having to be studied separately.

Solution. We claim that such polynomials exist if and only if $d \leq 3$. The following examples show that such polynomials do exist for $d \leq 3$:

$$d=1: \quad d^2-d=0, \quad P_1(x)=x, \qquad P(0)=0;$$

$$d=2: \quad d^2-d=2, \quad P_2(x)=x(x-1), \qquad \begin{cases} P(0)=P(1)=0, \\ P(2)=2; \end{cases}$$

$$d=3: \quad d^2-d=6, \quad P_3(x)=x(x-4)(x-5), \qquad \begin{cases} P(0)=P(4)=P(5)=0, \\ P(1)=P(2)=P(6)=12, \\ P(3)=6. \end{cases}$$

We can make more examples by adding constants.

Now we will show that there are no examples of degree greater than 3.

From now on we assume (without loss of generality) that the leading coefficient of our polynomial P is positive and that all values $P(i)$ are positive (by adding a constant if necessary) for integers i in the range $0 \leq i \leq d^2-d+1$.

Assume (for contradiction) that P is a polynomial of degree $d \geq 4$ that satisfies the conditions of the problem and let $P(0), \ldots, P(d^2-d)$ take values among $p_1 < \cdots < p_d$. For $i = 1, \ldots, d$, let $n_i \geq 0$ be the number of appearances of p_i among $P(0), \ldots, P(d^2-d)$.

By definition $n_1+\cdots+n_d = d^2-d+1$. Since P has degree d, $n_i \le d$. The key observation is now the following.

Claim 1 Make the convention that $n_0 = n_{d+1} = 0$. If $n_i = d$ for some i in the range $1 \le i \le d$, then $n_{i\pm1} \le d-2$.

Proof. Up to scaling and hence without loss of generality, P has leading coefficient $+1$. Since $n_i = d$, there exist non-negative integers $a_{1,i} < \cdots < a_{d,i} \le d^2 - d$ such that

$$P(X) = (X - a_{1,i}) \cdots (X - a_{d,i}) + p_i.$$

By construction, each of the $d-1$ intervals $I_j = [a_{j,i}, a_{j+1,i}]$ contains at least one local extremum of P, so contains exactly one such extremum because P, having degree d, has at most $d-1$ such extrema. Suppose that $i \le d-1$ and that $P(m) = p_{i+1} > p_i$ for some $m \in \{0, \ldots, d^2-d\}$. Since P has positive leading coefficient,

$$m \in (a_{d,i}, \infty) \cup (a_{d-2,i}, a_{d-1,i}) \cup \cdots \cup (a_{i,1}, a_{2,i})$$

if d is odd or

$$m \in (a_{d,i}, \infty) \cup (a_{d-2,i}, a_{d-1,i}) \cup \cdots \cup (-\infty, a_{1,i})$$

if d is even.

Suppose that $a_{j,i} < m < a_{j+1,i}$, for some $j \in \{1, \ldots, d-1\}$. If $a_{j,i} + 1 < m < a_{j+1,i} - 1$, then, because I_j contains exactly one local extremum (which is a maximum),

Either $p_{i+1} = P(m) > P(a_{j,i}+1)$ or $p_{i+1} = P(m) > P(a_{j+1,i}-1)$. Since $P(a_{j,i}+1) > P(a_{j,i}) = p_i$ and $P(a_{j,i}-1) > P(a_{j+1,i}) = p_i$, this contradicts the requirement that $P(a_{j,i}+1), P(a_{j+1,i}-1) \in \{p_1, \ldots, p_d\}$. Hence $m = a_{j,i}+1$ or $m = a_{j,i}-1$. Similarly, if $m > a_{d,i}$, then $m = a_{d,i}+1$, but if $m < a_{i,1}$ (which may arise when d is even), then $m = a_{i,1}-1$. This shows that m belongs to this list:

$$a_{d,i}+1, a_{d-1,i}-1, \ldots, a_{2,i}+(-1)^d, a_{1,i}-(-1)^d.$$

This list contains at most d different integers. It follows in particular that, if $n_{i+1} > d-2$, then either

$$P(a_{d,i}+1) = p_{i+1} = P(a_{d-1,i}-1)$$

or

$$P(a_{2,i}+(-1)^d) = p_{i+1} = P(a_{1,i}-(-1)^d)$$

with, additionally, $a_{2,i}+(-1)^d \ne a_{1,i}-(-1)^d$.

We have

$$|P(a_{1,i} \pm 1) - p_i| = 1 \cdot |a_{1,i} \pm 1 - a_{2,i}| \cdot \prod_{j=3}^{d} |a_{1,i} \pm 1 - a_{j,i}|$$

and

$$|P(a_{2,i} \mp 1) - p_i| = |a_{2,i} \mp 1 - a_{1,i}| \cdot 1 \cdot \prod_{j=3}^{d} |a_{2,i} \mp 1 - a_{j,i}|.$$

As $a_{1,i} < a_{2,i} < a_{3,i} < \ldots < a_{d,i}$ we have $|a_{1,i} \pm 1 - a_{j,i}| \geq |a_{2,i} \mp 1 - a_{j,i}|$ with equality possible only if $a_{1,i} + 1 = a_{2,i} - 1$. We also have $|a_{1,i} \pm 1 - a_{2,i}| = |a_{2,i} \mp 1 - a_{1,i}|$, which can be zero only if $a_{1,i} + 1 = a_{2,i} - 1$. We conclude that $|P(a_{1,i} \pm 1) - p_i| > |P(a_{2,i} \mp 1) - p_i|$ or $a_{1,i} + 1 = a_{2,i} - 1$.

Looking at the other end of the list of $(a_{j,i})$ as j varies, we have

$$|P(a_{d-1,i} - 1) - p_i| = 1 \cdot |a_{d-1,i} - 1 - a_{d,i}| \cdot \prod_{j=1}^{d-2} |a_{d-1,i} - 1 - a_{j,i}|$$

and

$$|P(a_{d,i} + 1) - p_i| = |a_{d,i} + 1 - a_{d-1,i}| \cdot 1 \cdot \prod_{j=1}^{d-2} |a_{d,i} + 1 - a_{j,i}|.$$

In these two formulas the shared factor outside the product is at least 2 and so is not 0. Now look at the factors behind the product symbols. As $a_{1,i} < a_{2,i} < a_{3,i} < \ldots < a_{d,i}$, for $j \leq d-2$ we have $|a_{d,i} + 1 - a_{j,i}| > |a_{d-1,i} - 1 - a_{j,i}|$. We conclude that $|P(a_{d,i} + 1) - p_i| > |P(a_{d-1,i} - 1) - p_i|$. Claim 1 is proved. ■

For each $i \in \{1, \ldots, d-1\}$, there are three possibilities:

- $n_i, n_{i+1} \leq d - 1$
- $n_i = d$ and $n_{i+1} \leq d - 2$
- $n_{i+1} = d$ and $n_i \leq d - 2$.

In all three cases, $n_i + n_{i+1} \leq 2(d-1)$. If n is even, this leads to the contradiction

$$d^2 - d + 1 = (n_1 + n_2) + \cdots + (n_{d-1} + n_d) \leq (d/2)[2(d-1)] = d^2 - d.$$

This is an important staging point in the argument because we have eliminated the possibility of a polynomial of even degree d satisfying the conditions of the problem if $d \geq 4$.

From now on we assume that $d \geq 5$ is odd, and

$$d^2 - d + 1 = (n_1 + n_2) + \cdots + (n_{d-2} + n_{d-1}) + n_d \leq [(d-1)/2][2(d-1)] + d = d^2 - d + 1.$$

Equality must therefore hold throughout. Since the sum can also be grouped as

$$n_1 + (n_2 + n_3) + \cdots + (n_{d-1} + n_d),$$

this requires $n_1 = n_d = d$, $n_i + n_{i+1} = 2(d-1)$ for $i = 1, 2, \ldots, d-1$ i.e. $n_i = d$ for odd i and $n_i = d - 2$ for even i.

We are interested in the degree of P being $d \geq 5$ and odd, and showing that no polynomial P satisfying the conditions of the problem can exist. There are $d - 1 \geq 4$ extremal points points which alternate between local maxima and minima (in that order) as you read from left to right (we normalize so that P is monic). For any p_i with i odd, the line $y = p_i$ (with

i odd) crosses the graph of P in d places with x-coordinates in the real closed interval $[0, d^2 - d]$ at points (z, p_i) so each z must be an integer. Suppose that J is a real interval on the x-axis ending at adjacent local extrema. The function defined by P is monotonic on each J. The line $y = p_i$ (i odd) meets the graph at most once on J. Therefore it meets the graph of P exactly once in the interior of each J (there are $d - 2$ such intervals) and at the only two possible places outside the union of these intervals.

Now consider p_j when j is even (so $n_j = d - 2$). These $d - 2$ intervals J afford $d - 2$ real values at which P will take p_j as a value where j is fixed and even. The question is, are the corresponding arguments integers? The proof of Claim 1 tells us that in the middle of the run $\{0, 1, \ldots, d^2 - d + 1\}$ all is well: the polynomial is assuming the value p_j at an integer where the polynomial assumes the values p_{j-1} and p_{j+1} at adjacent integers in some order. The problem is at the ends of the run where $|P(a_{1,i} + 1) - p_i| > |P(a_{2,i} - 1) - p_i|$ and $|P(a_{d,i} + 1) - p_i| > |P(a_{d-1,i} - 1) - p_i|$. When j is even, two of the roots of $P(x) - p_j$ are not integers, and we now know approximately where this trouble is happening (at the ends).

At this point we could finish if $d \geq 7$, because the run of regular behaviour in the middle is sufficiently long that we could obtain a contradiction. However we have to work a little harder to include the case $d = 5$. We now show that the run of regular behaviour is slightly longer than we have currently established. We do this using Claim 2.

Claim 2. If d is odd, and $n_i = d$, $n_{i\pm1} = d - 2$ for some i, then P attains $p_{i\pm1}$ precisely at the $d - 2$ integers

$$a_{2,i} - 1, a_{3,i} + 1, \cdots, a_{d-2,i} - 1, a_{d-1,i} + 1.$$

Proof Suppose (for contradiction) $a_{1,i} + 1 \neq a_{2,i} - 1$ and $P(a_{1,i} + 1) = p_{i+1}$. Now $a_{1,i} < a_{2,i}$ so either $a_{1,i} + 1 < a_{2,i} - 1$ or $a_{1,i} = a_{2,i} - 1$. In the latter case, $P(a_{1,i} + 1) = P(a_{2,i}) = p_i$, a contradiction. In the former case, the proof of Claim 1 shows that

$$|P(a_{2,i} - 1) - p_i| < |P(a_{1,i} + 1) - p_i| = |p_{i+1} - p_i| = p_{i+1} - p_i$$

so $P(a_{2,i} - 1) < p_{i+1}$. The polynomial is decreasing on the interval $(a_{2,d}, a_{2,1})$ so $p_i < P(a_{2,i} - 1) < p_{i+1}$ which is absurd because $P(a_{2,i} - 1) = p_j$ for some j. Therefore $P(a_{2,i} - 1) = p_{i+1}$ for all odd i. A similar argument shows that $P(a_{d-1,i} + 1) = p_{i+1}$ so Claim 2 is established. ■

Now we have n sequence of alternating falling then rising then falling etc. full runs starting at $(a_{2,d}, p_d)$ and ending at $(a_{d-1,1}, p_1)$ so the initial run of $3d+1$ terms of this run of values is

$$p_d, p_{d-1}, \cdots p_1, p_1, p_2, \ldots, p_d, p_d, p_{d-1}, \cdots p_1, p_1 \qquad (*)$$

which starts at $(a_{2,d}, p_d)$ and ends at $(a_{4,1} + 1, p_1)$ which is fine because $4 \leq d - 1$.

There are now various ways we can finish.

(a) Consider the run of length $2d$ consecutive values

$$p_d, p_{d-1}, \cdots p_1, p_1, p_2, \ldots, p_d.$$

The first $d+1$ points determine $P(x)$. The last $d+1$ values also determine P but the values are in the reverse order, so $P(X) = P(c - X)$ for some constant c. However, the coefficients of X^d are have opposite signs (d is odd) so this is absurd.

(b) The idea in (a) can be expressed in terms of Lagrange interpolation to obtain essentially the same contradiction. Construct P in two ways using Lagrange interpolation on both the first $d+1$ and the last $d+1$ points. The symmetry in the data forces the graph of P to have a vertical axis of symmetry. This is absurd because the degree of P is odd.

(c) The initial fragment length $3d+1$ mentioned above at $(*)$ includes two identical runs of values of P (in the same order) of length $d+1$. The polynomial P is determined by each of them and so $P(X) = P(X+c)$ for some constant c and so the polynomial defines a bounded function which is absurd.

■

EGMO 2023 – Problems and solutions

Problems

Problem 1. There are $n \geqslant 3$ positive real numbers $a_1, a_2, \ldots, a_n$. For each $1 \leqslant i \leqslant n$ we let $b_i = \frac{a_{i-1}+a_{i+1}}{a_i}$ (here we define a_0 to be a_n and a_{n+1} to be a_1). Assume that for all i and j in the range 1 to n, we have $a_i \leqslant a_j$ if and only if $b_i \leqslant b_j$.

Prove that $a_1 = a_2 = \cdots = a_n$.

Problem 2. We are given an acute triangle ABC. Let D be the point on its circumcircle such that AD is a diameter. Suppose that points K and L lie on segments AB and AC, respectively, and that DK and DL are tangent to circle AKL.

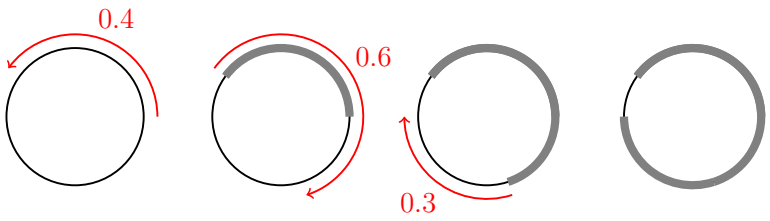
Show that line KL passes through the orthocentre of ABC.

The altitudes of a triangle meet at its orthocentre.

Problem 3. Let k be a positive integer. Lexi has a dictionary $\mathcal{D}$ consisting of some k-letter strings containing only the letters A and B. Lexi would like to write either the letter A or the letter B in each cell of a $k \times k$ grid so that each column contains a string from $\mathcal{D}$ when read from top-to-bottom and each row contains a string from $\mathcal{D}$ when read from left-to-right.

What is the smallest integer m such that if $\mathcal{D}$ contains at least m different strings, then Lexi can fill her grid in this manner, no matter what strings are in $\mathcal{D}$?

Problem 4. Turbo the snail sits on a point on a circle with circumference 1. Given an infinite sequence of positive real numbers $c_1, c_2, c_3, \ldots$, Turbo successively crawls distances $c_1, c_2, c_3, \ldots$ around the circle, each time choosing to crawl either clockwise or counterclockwise.
For example, if the sequence $c_1, c_2, c_3, \ldots$ is $0.4, 0.6, 0.3, \ldots$, then Turbo may start crawling as follows:



Determine the largest constant $C > 0$ with the following property: for every sequence of positive real numbers $c_1, c_2, c_3, \ldots$ with $c_i < C$ for all i, Turbo can (after studying the sequence) ensure that there is some point on the circle that it will never visit or crawl across.

Problem 5. We are given a positive integer $s \geqslant 2$. For each positive integer k, we define its *twist* k' as follows: write k as $as + b$, where a, b are non-negative integers and $b < s$, then $k' = bs + a$. For the positive integer n, consider the infinite sequence $d_1, d_2, \ldots$ where $d_1 = n$ and d_{i+1} is the twist of d_i for each positive integer i.

Prove that this sequence contains 1 if and only if the remainder when n is divided by $s^2 - 1$ is either 1 or s.

Problem 6. Let ABC be a triangle with circumcircle Ω. Let S_b and S_c respectively denote the midpoints of the arcs AC and AB that do not contain the third vertex. Let N_a denote the midpoint of arc BAC (the arc BC containing A). Let I be the incentre of ABC. Let ω_b be the circle that is tangent to AB and internally tangent to Ω at S_b, and let ω_c be the circle that is tangent to AC and internally tangent to Ω at S_c. Show that the line IN_a, and the line through the intersections of ω_b and ω_c, meet on Ω.

The incentre of a triangle is the centre of its incircle, the circle inside the triangle that is tangent to all three sides.

Solutions

Problem 1. There are $n \geqslant 3$ positive real numbers $a_1, a_2, \ldots, a_n$. For each $1 \leqslant i \leqslant n$ we let $b_i = \frac{a_{i-1}+a_{i+1}}{a_i}$ (here we define a_0 to be a_n and a_{n+1} to be a_1). Assume that for all i and j in the range 1 to n, we have $a_i \leqslant a_j$ if and only if $b_i \leqslant b_j$.

Prove that $a_1 = a_2 = \cdots = a_n$.

Solution 1. Suppose that not all a_i are equal. Consider an index i such that a_i is maximal and $a_{i+1} < a_i$. Then

$$b_i = \frac{a_{i-1} + a_{i+1}}{a_i} < \frac{2a_i}{a_i} = 2.$$

But since a_i is maximal, b_i is also maximal, so we must have $b_j < 2$ for all $j \in \{1, 2, \ldots, n\}$.

However, consider the product $b_1 b_2 \ldots b_n$. We have

$$\begin{aligned} b_1 b_2 \ldots b_n &= \frac{a_n + a_2}{a_1} \cdot \frac{a_1 + a_3}{a_2} \cdot \ldots \cdot \frac{a_{n-1} + a_1}{a_n} \\ &\geqslant 2^n \frac{\sqrt{a_n a_2}\sqrt{a_1 a_3} \ldots \sqrt{a_{n-1} a_1}}{a_1 a_2 \ldots a_n} \\ &= 2^n, \end{aligned}$$

where we used the inequality $x + y \geqslant 2\sqrt{xy}$ for $x = a_{i-1}, y = a_{i+1}$ for all $i \in \{1, 2, \ldots, n\}$ in the second row.

Since the product of all b_i is at least 2^n, at least one of them must be greater than 2, which is a contradiction with the previous conclusion.

Thus, all a_i must be equal. □

Solution 2. This is a version of Solution 1 without use of proof by contradiction.

Taking a_i such that it is maximal among $a_1, \ldots, a_n$, we obtain $b_i \leqslant 2$. Thus $b_i \leqslant 2$ for all $j \in \{1, 2, \ldots, n\}$.

The second part of Solution 1 then gives $2^n \geqslant b_1 \cdots b_n \geqslant 2^n$, which together with $b_i \leqslant 2$ for all $j \in \{1, 2, \ldots, n\}$ implies that $b_j = 2$ for all $j \in \{1, 2, \ldots, n\}$. Since we have $b_1 = b_2 = \cdots = b_n$, the condition that $a_i \leqslant a_j \iff b_i \leqslant b_j$ gives that $a_1 = a_2 = \cdots = a_n$. □

Solution 3. We first show that $b_j \leqslant 2$ for all j as in Solution 2. Then

$$\begin{aligned} 2n \geqslant & b_1 + \cdots + b_n = \frac{a_n}{a_1} + \frac{a_2}{a_1} + \frac{a_1}{a_2} + \frac{a_3}{a_2} + \cdots + \frac{a_{n-1}}{a_n} + \frac{a_1}{a_n} \\ \geqslant & 2n \sqrt[n]{\frac{a_n}{a_1} \cdot \frac{a_2}{a_1} \cdot \frac{a_1}{a_2} \cdot \frac{a_3}{a_2} \cdots \frac{a_{n-1}}{a_n} \cdot \frac{a_1}{a_n}} = 2n \cdot 1 = 2n, \end{aligned}$$

where we used the AM-GM inequality.

It follows that all b_j's are equal which as in Solution 2 gives $a_1 = a_2 = \cdots = a_n$. □

Solution 4. By assumption $a_ib_i = a_{i-1} + a_{i+1}$ for $i = \{1, 2, \ldots, n\}$, hence,

$$\sum_{i=1}^{n} a_ib_i = 2\sum_{i=1}^{n} a_i.$$

Since $a_i \leqslant a_j$ if and only if $b_i \leqslant b_j$, the Chebyshev's inequality implies

$$\left(\sum_{i=1}^{n} a_i\right) \cdot \left(\sum_{i=1}^{n} b_i\right) \leqslant n \cdot \sum_{i=1}^{n} a_ib_i = 2n \cdot \sum_{i=1}^{n} a_i$$

and so $\sum_{i=1}^{n} b_i \leqslant 2n$. On the other hand, we have

$$\sum_{i=1}^{n} b_i = \sum_{i=1}^{n} \frac{a_{i-1}}{a_i} + \sum_{i=1}^{n} \frac{a_{i+1}}{a_i} = \sum_{i=1}^{n} \frac{a_{i-1}}{a_i} + \sum_{i=1}^{n} \frac{a_i}{a_{i-1}} = \sum_{i=1}^{n} \left(\frac{a_{i-1}}{a_i} + \frac{a_i}{a_{i-1}}\right),$$

so we can use the AM-GM inequality to estimate

$$\sum_{i=1}^{n} b_i \geqslant \sum_{i=1}^{n} 2\sqrt{\frac{a_{i-1}}{a_i} \cdot \frac{a_i}{a_{i-1}}} = 2n.$$

We conclude that we must have equalities in all the above, which implies $\frac{a_{i-1}}{a_i} = \frac{a_i}{a_{i-1}}$ and consequently $a_i = a_{i-1}$ for all positive integers i. Hence, all a's are equal. □

Solution 5. As in Solution 4 we show that $\sum_{i=1}^{n} b_i \leqslant 2n$ and as in Solution 1 we show that $\prod_{i=1}^{n} b_i \geqslant 2^n$. We now use the AM-GM inequality and the first inequality to get

$$\prod_{i=1}^{n} b_i \leqslant \left(\frac{1}{n}\sum_{i=1}^{n} b_i\right)^n \leqslant \left(\frac{1}{n} \cdot 2n\right)^n = 2^n.$$

This implies that we must have equalities in all the above. In particular, we have equality in the AM-GM inequality, so all b's are equal and as in Solution 2 then all a's are equal. □

Solution 6. Let a_i to be minimal and a_j maximal among all a's. Then

$$b_j = \frac{a_{j-1} + a_{j+1}}{a_j} \leqslant \frac{2a_j}{a_j} = 2 = \frac{2a_i}{a_i} \leqslant \frac{a_{i-1} + a_{i+1}}{a_i} = b_i$$

and by assumption $b_i \leqslant b_j$. Hence, we have equalities in the above so $b_j = 2$ so $a_{j-1} + a_{j+1} = 2a_j$ and therefore $a_{j-1} = a_j = a_{j+1}$. We have thus shown that the two neighbors of a maximal a are also maximal. By an inductive argument all a's are maximal, hence equal. □

Solution 7. Choose an arbitrary index i and assume without loss of generality that $a_i \leqslant a_{i+1}$. (If the opposite inequality holds, reverse all the inequalities below.) By induction we will show that for each $k \in \mathbb{N}_0$ the following two inequalities hold

$$a_{i+1+k} \geqslant a_{i-k} \tag{1}$$

$$a_{i+1+k}a_{i+1-k} \geqslant a_{i-k}a_{i+k} \tag{2}$$

(where all indices are cyclic modulo n). Both inequalities trivially hold for $k = 0$.

Assume now that both inequalities hold for some $k \geqslant 0$. The inequality $a_{i+1+k} \geqslant a_{i-k}$ implies $b_{i+1+k} \geqslant b_{i-k}$, so

$$\frac{a_{i+k} + a_{i+2+k}}{a_{i+1+k}} \geqslant \frac{a_{i-1-k} + a_{i+1-k}}{a_{i-k}}.$$

We may rearrange this inequality by making a_{i+2+k} the subject so

$$a_{i+2+k} \geqslant \frac{a_{i+1+k}a_{i-1-k}}{a_{i-k}} + \frac{a_{i+1-k}a_{i+1+k} - a_{i+k}a_{i-k}}{a_{i-k}} \geqslant \frac{a_{i+1+k}a_{i-1-k}}{a_{i-k}},$$

where the last inequality holds by (2). It follows that

$$a_{(i+1)+(k+1)}a_{(i+1)-(k+1)} \geqslant a_{i+(k+1)}a_{i-(k+1)},$$

i.e. the inequality (2) holds also for $k + 1$. Using (1) we now get

$$a_{(i+1)+(k+1)} \geqslant \frac{a_{i+k+1}}{a_{i-k}}a_{i-(k+1)} \geqslant a_{i-(k+1)},$$

i.e. (1) holds for $k + 1$.

Now we use the inequality (1) for $k = n - 1$. We get $a_i \geqslant a_{i+1}$, and since at the beginning we assumed $a_i \leqslant a_{i+1}$, we get that any two consecutive a's are equal, so all of them are equal. □

Solution 8. We first prove the following claim by induction:

Claim 1: If $a_k a_{k+2} < a_{k+1}^2$ and $a_k < a_{k+1}$, then $a_j a_{j+2} < a_{j+1}^2$ and $a_j < a_{j+1}$ for all j.

We assume that $a_i a_{i+2} < a_{i+1}^2$ and $a_i < a_{i+1}$, and then show that $a_{i-1}a_{i+1} < a_i^2$ and $a_{i-1} < a_i$.

Since $a_i \leq a_{i+1}$ we have that $b_i \leq b_{i+1}$. By plugging in the definition of b_i and b_{i+1} we have that

$$a_{i+1}a_{i-1} + a_{i+1}^2 \leq a_i^2 + a_{i+2}a_i. \tag{3}$$

Using $a_i a_{i+2} < a_{i+1}^2$ we get that

$$a_{i+1}a_{i-1} < a_i^2. \tag{4}$$

Since $a_i < a_{i+1}$ we have that $a_{i-1} < a_i$, which concludes the induction step and hence proves the claim.

We cannot have that $a_j < a_{j+1}$ for all indices j. Similar as in the above claim, one can prove that if $a_k a_{k+2} < a_{k+1}^2$ and $a_{k+2} < a_{k+1}$, then $a_{j+1} < a_j$ for all j, which also cannot be the case. Thus we have that $a_k a_{k+2} \geq a_{k+1}^2$ for all indices k.

Next observe (e.g. by taking the product over all indices) that this implies $a_k a_{k+2} = a_{k+1}^2$ for all indices k, which is equivalent to $b_k = b_{k+1}$ for all k and hence $a_{k+1} = a_k$ for all k. □

Solution 9. Define $c_i := \frac{a_i}{a_{i+1}}$, then $b_i = c_{i-1} + 1/c_i$. Assume that not all c_i are equal to 1. Since, $\prod_{i=1}^{n} c_i = 1$ there exists a k such that $c_k \geqslant 1$. From the condition given in the problem statement for $(i, j) = (k, k+1)$ we have

$$c_k \geqslant 1 \iff c_{k-1} + \frac{1}{c_k} \geqslant c_k + \frac{1}{c_{k+1}} \iff c_{k-1}c_kc_{k+1} + c_{k+1} \geqslant c_k^2 c_{k+1} + c_k. \tag{5}$$

Now since $c_{k+1} \leqslant c_k^2 c_{k+1}$, it follows that

$$c_{i-1}c_ic_{i+1} \geqslant c_i \Longrightarrow (c_{i-1} \geqslant 1 \text{ or } c_{i+1} \geqslant 1)\,. \tag{6}$$

So there exist a set of at least 2 consecutive integers, such that the corresponding c_i are greater or equal to one. By the innitial assumption there must exist an index ℓ, such that $c_{\ell-1}, c_\ell \geqslant 1$ and $c_{\ell+1} < 1$. We distinguish two cases:

Case 1: $c_\ell > c_{\ell-1} \geqslant 1$
From $c_{\ell-1}c_\ell c_{\ell+1} < c_\ell^2 c_{\ell+1}$ and the inequality (5), we get that $c_{\ell+1} > c_\ell \geqslant 1$, which is a contradiction to our choice of ℓ.

Case 2: $c_{\ell-1} \geqslant c_\ell \geqslant 1$
Once again looking at the inequality (5) we can find that

$$c_{\ell-2}c_{\ell-1}c_\ell \geqslant c_{\ell-1}^2 c_\ell \Longrightarrow c_{\ell-2} \geqslant c_{\ell-1}. \tag{7}$$

Note that we only needed $c_{\ell-1} \geqslant c_\ell \geqslant 1$ to show $c_{\ell-2} \geqslant c_{\ell-1} \geqslant 1$. So using induction we can easily show $c_{\ell-s-1} \geqslant c_{\ell-s}$ for all s.
So

$$c_1 \leqslant c_2 \leqslant \cdots \leqslant c_n \leqslant c_1 \tag{8}$$

a contradiction to our innitial assumption.

So our innitial assumtion must have been wrong, which implies that all the a_i must have been equal from the start. □

Problem 2. We are given an acute triangle ABC. Let D be the point on its circumcircle such that AD is a diameter. Suppose that points K and L lie on segments AB and AC, respectively, and that DK and DL are tangent to circle AKL.

Show that line KL passes through the orthocentre of ABC.

The altitudes of a triangle meet at its orthocentre.

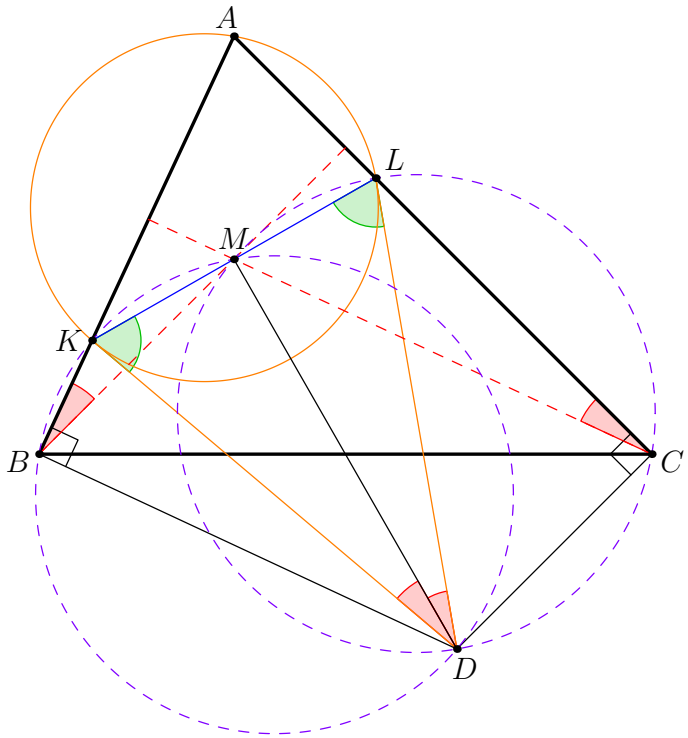


Figure 1: Diagram to solution 1

Solution 1. Let M be the midpoint of KL. We will prove that M is the orthocentre of ABC. Since DK and DL are tangent to the same circle, $|DK| = |DL|$ and hence $DM \perp KL$. The theorem of Thales in circle ABC also gives $DB \perp BA$ and $DC \perp CA$. The right angles then give that quadrilaterals $BDMK$ and $DMLC$ are cyclic.

If $\angle BAC = \alpha$, then clearly $\angle DKM = \angle MLD = \alpha$ by angle in the alternate segment of circle AKL, and so $\angle MDK = \angle LDM = \frac{\pi}{2} - \alpha$, which thanks to cyclic quadrilaterals gives $\angle MBK = \angle LCM = \frac{\pi}{2} - \alpha$. From this, we have $BM \perp AC$ and $CM \perp AB$, and so M indeed is the orthocentre of ABC. □

Solution 2. Preliminaries

Let ABC be a triangle with circumcircle Γ. Let X be a point in the plane. The Simson line (Wallace-Simson line) is defined via the following theorem. Drop perpendiculars from X to each of the three side lines of ABC. The feet of these perpendiculars are collinear (on

the Simson line of X) if and only if X lies of Γ. The Simson line of X in the circumcircle bisects the line segment XH where H is the orthocentre of triangle ABC. See Figure 2

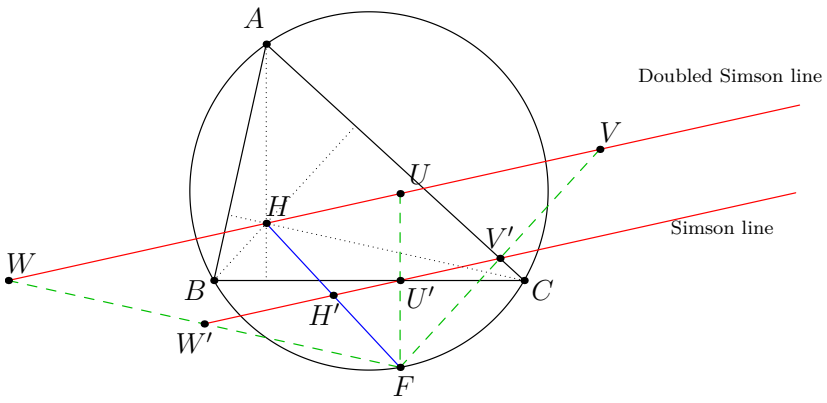


Figure 2: The Wallace-Simson configuration

When X is on Γ, we can enlarge from X with scale factor 2 (a homothety) to take the Simson line to the *doubled* Simson line which passes through the orthocentre H and contains the reflections of X in each of the three sides of ABC.

Solution of the problem

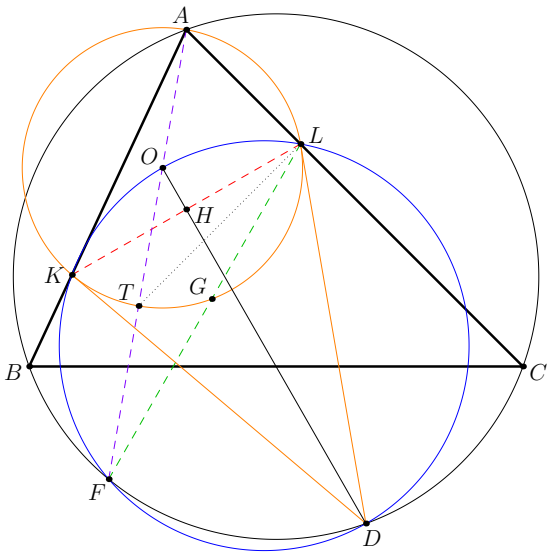


Figure 3: Three circles do the work

Let Γ be the circle ABC, Σ be the circle AKL with centre O, and Ω be the circle on diameter OD so K and L are on this circle by converse of Thales. Let Ω and Γ meet at D and F. By Thales in both circles, $\angle AFD$ and $\angle OFD$ are both right angles so AOF is a line. Let AF meet Σ again at T so AT (containing O) is a diameter of this circle and by Thales, $TL \perp AC$.

Let G (on Σ) be the reflection of K in AF. Now AT is the internal angle bisector of $\angle GAK$ so, by an upmarket use of angles in the same segment (of Σ), TL is the internal

angle bisector of $\angle GLK$. Thus the line GL is the reflection of the line KL in TL, and so also the reflection of KL in the line AC (internal and external angle bisectors).

Our next project is to show that LGF are collinear. Well $\angle FLK = \angle FOK$ (angles in the same segment of Ω) and $\angle GLK = \angle GAK$ (angles in the same segment of Σ) $= 2\angle OAK$ (AKG is isosceles with apex A) $= \angle TOK$ (since OAK is isosceles with apex O, and this is an external angle at O). The point T lies in the interior of the line segment FO so $\angle TOK = \angle FOK$. Therefore $\angle FLK = \angle GLK$ so LGF is a line.

Now from the second paragraph, F is on the reflection of KL in AC. By symmetry, F is also on the reflection of KL in AB. Therefore the reflections of F in AB and AC are both on KL which must therefore be the doubled Wallace-Simson line of F. Therefore the orthocentre of ABC lies on KL. □

Solution 3. Let H be the orthocentre of triangle ABC and Σ the circumcircle of AKL with centre O. Let Ω be the circle with diameter OD, which contains K and L by Thales, and let Γ be the circumcircle of ABC containing D. Denote the second intersection of Ω and Γ by F. Since OD and AD are diameters of Ω and Γ we have $\angle OFD = \frac{\pi}{2} = \angle AFD$, so the points A, O, F are collinear. Let M and N be the second intersections of CH and BH with Γ, respectively. It is well-known that M and N are the reflections of H in AB and AC, respectively (because $\angle NCA = \angle NBA = \angle ACM = \angle ABM$). By collinearity of A, O, F and the angles in Γ we have

$$\angle NFO = \angle NFA = \angle NBA = \frac{\pi}{2} - \angle BAC = \frac{\pi}{2} - \angle KAL.$$

Since DL is tangent to Σ we obtain

$$\angle NFO = \frac{\pi}{2} - \angle KLD = \angle LDO,$$

where the last equality follows from the fact that OD is bisector of $\angle LDK$ since LD

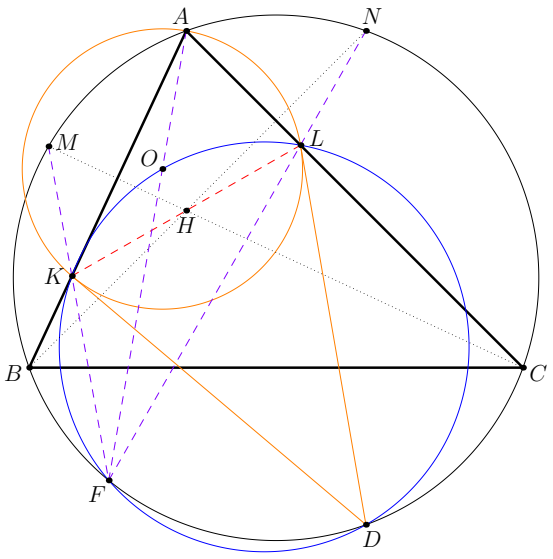


Figure 4: Diagram to Solution 3

and KD are tangent to Σ. Furthermore, $\angle LDO = \angle LFO$ since these are angles in Ω. Hence, $\angle NFO = \angle LFO$, which implies that points N, L, F are collinear. Similarly points M, K, F are collinear. Since N and M are reflections of H in AC and AB we have

$$\angle LHN = \angle HNL = \angle BNF = \angle BMF = \angle BMK = \angle KHB.$$

Hence,

$$\angle LHK = \angle LHN + \angle NHK = \angle KHB + \angle NHK = \pi$$

and the points L, H, K are collinear. □

Solution 4. As in Solution 3 let M and N be the reflections of the orthocentre in AB and AC. Let $\angle BAC = \alpha$. Then $\angle NDM = \pi - \angle MAN = \pi - 2\alpha$.

Let MK and NL intersect at F. See Figure 3.

Claim. $\angle NFM = \pi - 2\alpha$, so F lies on the circumcircle.

Proof. Since KD and LD are tangents to circle AKL, we have $|DK| = |DL|$ and $\angle DKL = \angle KLD = \alpha$, so $\angle LDK = \pi - 2\alpha$.

By definition of M, N and D, $\angle MND = \angle AND - \angle ANM = \frac{\pi}{2} - (\frac{\pi}{2} - \alpha) = \alpha$ and analogously $\angle DMN = \alpha$. Hence $|DM| = |DN|$.

From $\angle NDM = \angle LDK = \pi - 2\alpha$ if follows that $\angle LDN = \angle KDM$. Since $|DK| = |DL|$ and $|DM| = |DN|$, triangles MDK and NDL are related by a rotation about D through angle $\pi - 2\alpha$, and hence the angle between MK and NL is $\pi - 2\alpha$, which proved the claim. □

We now finish as in Solution 3:

$$\angle MHK = \angle KMH = \angle FMC = \angle FAC,$$

$$\angle LHN = \angle HNL = \angle BNF = \angle BAF.$$

As $\angle BAF + \angle FAC = \alpha$, we have $\angle LHK = \alpha + \angle NHM = \alpha + \pi - \alpha = \pi$, so H lies on KL. □

Solution 5. Since AD is a diameter, it is well known that $DBHC$ is a parallelogram (indeed, both BD and CH are perpendicular to AB, hence parallel, and similarly for $DC \parallel BH$). Let B', C' be the reflections of D in lines AKB and ALC, respectively; since ABD and ACD are right angles, these are also the factor-2 homotheties of B and C with respect to D, hence H is the midpoint of $B'C'$. We will prove that $B'KC'L$ is a parallelogram: it will then follow that the midpoint of $B'C'$, which is H, is also the midpoint of KL, and in particular is on the line, as we wanted to show.

We will prove $B'KC'L$ is a parallelogram by showing that $B'K$ and $C'L$ are the same length and direction. Indeed, for lengths we have $KB' = KD = LD = LC'$, where the first and last equalities arise from the reflections defining B' and C', and the middle one

is equality of tangents. For directions, let α, β, γ denote the angles of triangle AKL. Immediate angle chasing in the circle AKL, and the properties of the reflections, yield

$$\angle C'LC = \angle CLD = \angle AKL = \beta$$
$$\angle BKB' = \angle DKB = \angle KLA = \gamma$$
$$\angle LDK = 2\alpha - \pi$$

and therefore in directed angles (mod 2π) we have

$$\angle(C'L, B'K) = \angle C'LC + \angle CLD + \angle LDK + \angle DKB + \angle BKB' = 2\alpha + 2\beta + 2\gamma - \pi = \pi$$

and hence $C'L$ and $B'K$ are parallel and in opposite directions, i.e. $C'L$ and KB' are in the same direction, as claimed. □

Comment. While not necessary for the final solution, the following related observation motivates how the fact that H is the midpoint of KL (and therefore $B'KC'L$ is a parallelogram) was first conjectured. We have $AB' = AD = AC'$ by the reflections, i.e. $B'AC'$ is an isosceles triangle with H being the midpoint of the base. Thus AH is the median, altitude and angle bisector in $B'AC'$, thus $\angle B'AK + \angle KAH = \angle HAL + \angle LAC'$. Since from the reflections we also have $\angle B'AK = \angle KAD$ and $\angle DAL = \angle LAC'$ it follows that $\angle HAL = \angle KAD$ and $\angle KAH = \angle DAL$. Since D is the symmedian point in AKL, the angle conjugation implies AH is the median line of KL. Thus, if H is indeed on KL (as the problem assures us), it can only be the midpoint of KL.

Solution 6. There are a number of "phantom point" arguments which define K' and L' in terms of angles and then deduce that these points are actually K and L.

Note: In these solutions it is necessary to show that K and L are uniquely determined by the conditions of the problem. One example of doing this is the following:

To prove uniqueness of K and L, let us consider that there exist two other points K' and L' that satisfy the same properties (K' on AB and L' on AC such that DK' and DL' are tangent to the circle $AK'L'$).
Then, we have that $DK = DL$ and $DK' = DL'$. We also have that $\angle KDL = \angle K'DL' = \pi - 2\angle A$. Hence, we deduce $\angle KDK' = \angle KDL - \angle K'DL = \angle K'DL' - \angle K'DL = \angle LDL'$ Thus we have that $\triangle KDK' \equiv \triangle LDL'$, so we deduce $\angle DKA = \angle DKK' = \angle DLL' = \pi - \angle ALD$. This implies that $AKDL$ is concyclic, which is clearly a contradiction since $\angle KAL + \angle KDL = \pi - \angle BAC$. □

Solution 7. We will use the usual complex number notation, where we will use a capital letter (like Z) to denote the point associated to a complex number (like z). Consider $\triangle AKL$ on the unit circle. So, we have $a \cdot \bar{a} = k \cdot \bar{k} = l \cdot \bar{l} = 1$ As point D is the intersection of the tangents to the unit circle at K and L, we have that

$$d = \frac{2kl}{k+l} \text{ and } \bar{d} = \frac{2}{k+l}$$

Defining B as the foot of the perpendicular from D on the line AK, and C as the foot of the perpendicular from D on the line AL , we have the formulas:

$$b = \frac{1}{2}\left(d + \frac{(a-k)\bar{d} + \bar{a}k - a\bar{k}}{\bar{a} - \bar{k}}\right)$$

$$c = \frac{1}{2}\left(d + \frac{(a-l)\bar{d} + \bar{a}l - a\bar{l}}{\bar{a} - \bar{l}}\right)$$

Simplyfing these formulas, we get:

$$b = \frac{1}{2}\left(d + \frac{(a-k)\frac{2}{k+l} + \frac{k}{a} - \frac{a}{k}}{\frac{1}{a} - \frac{1}{k}}\right) = \frac{1}{2}\left(d + \frac{\frac{2(a-k)}{k+l} + \frac{k^2-a^2}{ak}}{\frac{k-a}{ak}}\right)$$

$$b = \frac{1}{2}\left(\frac{2kl}{k+l} - \frac{2ak}{k+l} + (a+k)\right) = \frac{k(l-a)}{k+l} + \frac{1}{2}(k+a)$$

$$c = \frac{1}{2}\left(d + \frac{(a-l)\frac{2}{k+l} + \frac{l}{a} - \frac{a}{l}}{\frac{1}{a} - \frac{1}{l}}\right) = \frac{1}{2}\left(d + \frac{\frac{2(a-l)}{k+l} + \frac{l^2-a^2}{al}}{\frac{l-a}{al}}\right)$$

$$c = \frac{1}{2}\left(\frac{2kl}{k+l} - \frac{2al}{k+l} + (a+l)\right) = \frac{l(k-a)}{k+l} + \frac{1}{2}(l+a)$$

Let O be the the circumcenter of triangle $\triangle ABC$. As AD is the diameter of this circle, we have that:

$$o = \frac{a+d}{2}$$

Defining H as the orthocentre of the $\triangle ABC$, we get that:

$$h = a + b + c - 2 \cdot o = a + \left(\frac{k(l-a)}{k+l} + \frac{1}{2}(k+a)\right) + \left(\frac{l(k-a)}{k+l} + \frac{1}{2}(l+a)\right) - (a+d)$$

$$h = a + \frac{2kl}{k+l} - \frac{a(k+l)}{k+l} + \frac{1}{2}k + + \frac{1}{2}l + + a - \left(a + \frac{2kl}{k+l}\right)$$

$$h = \frac{1}{2}(k+l)$$

Hence, we conclude that H is the midpoint of KL, so H, K, L are collinear. □

Solution 8. Let us employ the barycentric coordinates. Set $A(1,0,0)$, $K(0,1,0)$, $L(0,0,1)$.

The tangent at K of (AKL) is $a^2z + c^2x = 0$, and the tangent of of L at (AKL) is $a^2y + b^2x = 0$. Their intersection is

$$D(-a^2 : b^2 : c^2).$$

Since $B \in AK$, we can let $B(1-t, t, 0)$. Solving for $\overrightarrow{AB} \cdot \overrightarrow{BD} = 0$ gives

$$t = \frac{3b^2 + c^2 - a^2}{2(b^2 + c^2 - a^2)} \implies B = \left(\frac{-a^2 - b^2 + c^2}{2(b^2 + c^2 - a^2)}, \frac{-a^2 + 3b^2 + c^2}{2(b^2 + c^2 - a^2)}, 0\right).$$

Likewise, C has the coordinate

$$C = \left(\frac{-a^2 + b^2 - c^2}{2(b^2 + c^2 - a^2)}, 0, \frac{-a^2 + b^2 + 3c^2}{2(b^2 + c^2 - a^2)}\right).$$

The altitude from B for triangle ABC is

$$-b^2\left(x-z-\frac{-a^2-b^2+c^2}{2(b^2+c^2-a^2)}\right)+(c^2-a^2)\left(y-\frac{-a^2+3b^2+c^2}{2(b^2+c^2-a^2)}\right)=0.$$

Also the altitude from C for triangle ABC is

$$-c^2\left(x-y-\frac{-a^2+b^2-c^2}{2(b^2+c^2-a^2)}\right)+(a^2-b^2)\left(z-\frac{-a^2+b^2+3c^2}{2(b^2+c^2-a^2)}\right)=0.$$

The intersection of these two altitudes, which is the orthocenter of triangle ABC, has the barycentric coordinate

$$H=(0,1/2,1/2),$$

which is the midpoint of the segment KL. □

Problem 3. Let k be a positive integer. Lexi has a dictionary $\mathcal{D}$ consisting of some k-letter strings containing only the letters A and B. Lexi would like to write either the letter A or the letter B in each cell of a $k \times k$ grid so that each column contains a string from $\mathcal{D}$ when read from top-to-bottom and each row contains a string from $\mathcal{D}$ when read from left-to-right.

What is the smallest integer m such that if $\mathcal{D}$ contains at least m different strings, then Lexi can fill her grid in this manner, no matter what strings are in $\mathcal{D}$?

Solution. We claim the minimum value of m is 2^{k-1}.

Firstly, we provide a set $\mathcal{S}$ of size $2^{k-1}-1$ for which Lexi cannot fill her grid. Consider the set of all length-k strings containing only As and Bs which end with a B, and remove the string consisting of k Bs. Clearly there are 2 independent choices for each of the first $k-1$ letters and 1 for the last letter, and since exactly one string is excluded, there must be exactly $2^{k-1}-1$ strings in this set.

Suppose Lexi tries to fill her grid. For each row to have a valid string, it must end in a B. But then the right column would necessarily contain k Bs, and not be in our set. Thus, Lexi cannot fill her grid with our set, and we must have $m \geqslant 2^{k-1}$.

Now, consider any set $\mathcal{S}$ with at least 2^{k-1} strings. Clearly, if $\mathcal{S}$ contained either the uniform string with k As or the string with k Bs, then Lexi could fill her grid with all of the relevant letters and each row and column would contain that string.

Consider the case where $\mathcal{S}$ contains neither of those strings. Among all 2^k possible length-k strings with As and Bs, each has a complement which corresponds to the string with Bs in every position where first string had As and vice-versa. Clearly, the string with all As is paired with the string with all Bs. We may assume that we do not take the two uniform strings and thus applying the pigeonhole principle to the remaining set of strings, we must have two strings which are complementary.

Let this pair of strings be $\ell, \ell' \in \mathcal{S}$ in some order. Define the set of indices $\mathcal{J}$ corresponding to the As in ℓ and thus the Bs in ℓ', and all other indices (not in $\mathcal{J}$) correspond to Bs in ℓ (and thus As in ℓ'). Then, we claim that Lexi puts an A in the cell in row r, column c if $r, c \in \mathcal{J}$ or $r, c \notin \mathcal{J}$, and a B otherwise, each row and column contains a string in $\mathcal{S}$.

We illustrate this with a simple example: If $k = 6$ and we have that $AAABAB$ and $BBBABA$ are both in the dictionary, then Lexi could fill the table as follows:

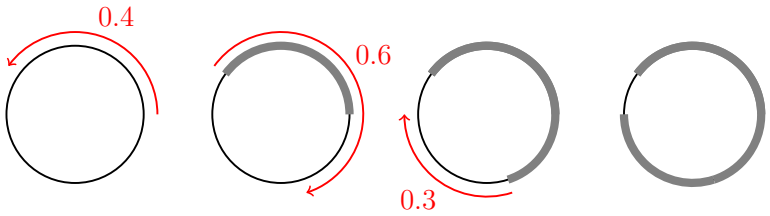
A	A	A	B	A	B
A	A	A	B	A	B
A	A	A	B	A	B
B	B	B	A	B	A
A	A	A	B	A	B
B	B	B	A	B	A

Suppose we are looking at row i or column i for $i \in \mathcal{J}$. Then by construction the string in this row/column contains As at indices k with $k \in \mathcal{J}$ and Bs elsewhere, and thus is precisely ℓ. Suppose instead we are looking at row i or column i for $i \notin \mathcal{J}$. Then again

by construction the string in this row/column contains As at indices k with $k \notin \mathcal{J}$ and Bs elsewhere, and thus is precisely ℓ'. So each row and column indeed contains a string in $\mathcal{S}$.

Thus, for any $\mathcal{S}$ with $|\mathcal{S}| \geqslant 2^{k-1}$, Lexi can definitely fill the grid appropriately. Since we know $m \geqslant 2^{k-1}$, 2^{k-1} is the minimum possible value of m as claimed. □

Problem 4. Turbo the snail sits on a point on a circle with circumference 1. Given an infinite sequence of positive real numbers $c_1, c_2, c_3, \ldots$, Turbo successively crawls distances $c_1, c_2, c_3, \ldots$ around the circle, each time choosing to crawl either clockwise or counterclockwise.
For example, if the sequence $c_1, c_2, c_3, \ldots$ is $0.4, 0.6, 0.3, \ldots$, then Turbo may start crawling as follows:



Determine the largest constant $C > 0$ with the following property: for every sequence of positive real numbers $c_1, c_2, c_3, \ldots$ with $c_i < C$ for all i, Turbo can (after studying the sequence) ensure that there is some point on the circle that it will never visit or crawl across.

Solution 1. The largest possible C is $C = \frac{1}{2}$.

For $0 < C \leqslant \frac{1}{2}$, Turbo can simply choose an arbitrary point P (different from its starting point) to avoid. When Turbo is at an arbitrary point A different from P, the two arcs AP have total length 1; therefore, the larger of the two the arcs (or either arc in case A is diametrically opposite to P) must have length $\geqslant \frac{1}{2}$. By always choosing this larger arc (or either arc in case A is diametrically opposite to P), Turbo will manage to avoid the point P forever.

For $C > \frac{1}{2}$, we write $C = \frac{1}{2} + a$ with $a > 0$, and we choose the sequence

$$\frac{1}{2}, \quad \frac{1+a}{2}, \quad \frac{1}{2}, \quad \frac{1+a}{2}, \quad \frac{1}{2}, \quad \ldots$$

In other words, $c_i = \frac{1}{2}$ if i is odd and $c_i = \frac{1+a}{2} < C$ when i is even. We claim Turbo must eventually visit all points on the circle. This is clear when it crawls in the same direction two times in a row; after all, we have $c_i + c_{i+1} > 1$ for all i. Therefore, we are left with the case that Turbo alternates crawling clockwise and crawling counterclockwise. If it, without loss of generality, starts by going clockwise, then it will always crawl a distance $\frac{1}{2}$ clockwise followed by a distance $\frac{1+a}{2}$ counterclockwise. The net effect is that it crawls a distance $\frac{a}{2}$ counterclockwise. Because $\frac{a}{2}$ is positive, there exists a positive integer N such that $\frac{a}{2} \cdot N > 1$. After $2N$ crawls, Turbo will have crawled a distance $\frac{a}{2}$ counterclockwise N times, therefore having covered a total distance of $\frac{a}{2} \cdot N > 1$, meaning that it must have crawled over all points on the circle. □

Note: Every sequence of the form $c_i = x$ if i is odd, and $c_i = y$ if i is even, where $0 < x, y < C$, such that $x + y \geqslant 1$, and $x \neq y$ satisfies the conditions with the same argument. There might be even more possible examples.

Solution 2. Alternative solution (to show that $C \leqslant \frac{1}{2}$)

We consider the following related problem:

We assume instead that the snail Chet is moving left and right on the real line. Find the size M of the smallest (closed) interval, that we cannot force Chet out of, using a sequence of real numbers d_i with $0 < d_i < 1$ for all i.

Then $C = 1/M$. Indeed if for every sequence $c_1, c_2, \ldots$, with $c_i < C$ there exists a point that Turbo can avoid, then the circle can be cut open at the avoided point and mapped to an interval of size M such that Chet can stay inside this interval for any sequence of the from $c_1/C, c_2/C, \ldots$, see Figure 5. However, all sequences $d_1, d_2, \ldots$ with $d_i < 1$ can be written in this form. Similarly if for every sequence $d_1, d_2, \ldots$, there exists an interval of length smaller or equal M that we cannot force Chet out of, this projects to a subset of the circle, that we cannot force Turbo out of using any sequence of the form $d_1/M, d_2/M, \ldots$. These are again exactly all the sequences with elements in $[0, C)$.

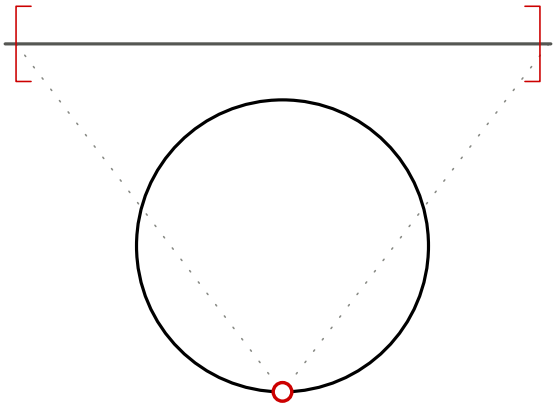


Figure 5: Chet and Turbo equivalence

Claim: $M \geqslant 2$.

Proof. Suppose not, so $M < 2$. Say $M = 2 - 2\varepsilon$ for some $\varepsilon > 0$ and let $[-1+\varepsilon, 1-\varepsilon]$ be a minimal interval, that Chet cannot be forced out of. Then we can force Chet arbitrarily close to $\pm(1-\varepsilon)$. In partiular, we can force Chet out of $[-1+\frac{4}{3}\varepsilon, 1-\frac{4}{3}\varepsilon]$ by minimality of M. This means that there exists a sequence $d_1, d_2, \ldots$ for which Chet has to leave $[-1+\frac{4}{3}\varepsilon, 1-\frac{4}{3}\varepsilon]$, which means he ends up either in the interval $[-1+\varepsilon, -1+\frac{4}{3}\varepsilon)$ or in the interval $(1-\frac{4}{3}\varepsilon, 1-\varepsilon]$.

Now consider the sequence,

$$d_1,\ 1-\frac{7}{6}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{7}{6}\varepsilon,\ d_2,\ 1-\frac{7}{6}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{7}{6}\varepsilon,\ d_3,\ \ldots$$

obtained by adding the sequence $1-\frac{7}{6}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{7}{6}\varepsilon$ in between every two steps. We claim that this sequence forces Chet to leave the larger interval $[-1+\varepsilon, 1-\varepsilon]$. Indeed no two consecutive elements in the sequence $1-\frac{7}{6}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{2}{3}\varepsilon,\ 1-\frac{7}{6}\varepsilon$ can

have the same sign, because the sum of any two consecutive terms is larger than $2-2\varepsilon$ and Chet would leave the interval $[-1+\varepsilon, 1-\varepsilon]$. It follows that the $(1-\frac{7}{6}\varepsilon)$'s and the $(1-\frac{2}{3}\varepsilon)$'s cancel out, so the position after d_k is the same as before d_{k+1}. Hence, the positions after each d_k remain the same as in the original sequence. Thus, Chet is also forced to the boundary in the new sequence.

If Chet is outside the interval $[-1+\frac{4}{3}\varepsilon, 1-\frac{4}{3}\varepsilon]$, then Chet has to move $1-\frac{7}{6}\varepsilon$ towards 0, and ends in $[-\frac{1}{6}\varepsilon, \frac{1}{6}\varepsilon]$. Chet then has to move by $1-\frac{2}{3}\varepsilon$, which means that he has to leave the interval $[-1+\varepsilon, 1-\varepsilon]$. Indeed the absolute value of the final position is at least $1-\frac{5}{6}\varepsilon$. This contradicts the assumption, that we cannot force Chet out of $[-1+\varepsilon, 1-\varepsilon]$. Hence $M \geqslant 2$ as needed. □

Problem 5. We are given a positive integer $s \geqslant 2$. For each positive integer k, we define its *twist* k' as follows: write k as $as + b$, where a, b are non-negative integers and $b < s$, then $k' = bs + a$. For the positive integer n, consider the infinite sequence $d_1, d_2, \ldots$ where $d_1 = n$ and d_{i+1} is the twist of d_i for each positive integer i.

Prove that this sequence contains 1 if and only if the remainder when n is divided by $s^2 - 1$ is either 1 or s.

Solution 1. First, we consider the difference $k - k''$. If $k = as + b$ as in the problem statement, then $k' = bs + a$. We write $a = ls + m$ with m, l non-negative numbers and $m \leq s - 1$. This gives $k'' = ms + (b + l)$ and hence $k - k'' = (a - m)s - l = l(s^2 - 1)$.

We conclude

Fact 1.1. $k \geq k''$ for every every $k \geq 1$

Fact 1.2. $s^2 - 1$ divides the difference $k - k''$.

Fact 1.2 implies that the sequences $d_1, d_3, d_5, \ldots$ and $d_2, d_4, d_6, \ldots$ are constant modulo $s^2 - 1$. Moreover, Fact 1.1 says that the sequences are (weakly) decreasing and hence eventually constant. In other words, the sequence $d_1, d_2, d_3, \ldots$ is 2-periodic modulo $s^2 - 1$ (from the start) and is eventually 2-periodic.

Now, assume that some term in the sequence is equal to 1. The next term is equal to $1' = s$ and since the sequence is 2-periodic from the start modulo $s^2 - 1$, we conclude that d_1 is either equal to 1 or s modulo $s^2 - 1$. This proves the first implication.

To prove the other direction, assume that d_1 is congruent to 1 or s modulo $s^2 - 1$. We need the observation that once one of the sequences $d_1, d_3, d_5, \ldots$ or $d_2, d_4, d_6, \ldots$ stabilises, then their value is less than s^2. This is implied by the following fact.

Fact 1.3. If $k = k''$, then $k = k'' < s^2$.

Proof. We use the expression for $k - k''$ found before. If $k = k''$, then $l = 0$, and so $k'' = ms + b$. Both m and b are reminders after division by s, so they are both $\leq s - 1$. This gives $k'' \leq (s - 1)s + (s - 1) < s^2$. □

Using Fact 1.2, it follows that the sequence $d_1, d_3, d_5, \ldots$ is constant to 1 or s modulo $s^2 - 1$ and stabilises to 1 or s by Fact 1.3. Since $s' = 1$, we conclude that the sequence contains a 1. □

Solution 2. We make a number of initial observations. Let k be a positive integer.

Fact 2.1. If $k \geq s^2$, then $k' < k$.

Proof. Write $k = as + b$, as in the problem statement. If $k \geq s^2$, then $a \geq s$ because $b < s$. So, $k' = bs + a \leq (s - 1)s + a \leq as \leq as + b = k$. Moreover, we cannot have equality since that would imply $s - 1 = b = 0$. □

Fact 2.2. If $k \leq s^2 - 1$, then $k' \leq s^2 - 1$ and $k'' = k$.

Proof. Write $k = as + b$, as in the problem statement. If $k < s^2$, then it must hold $1 \leq a, b < s$, hence $k' = bs + a < s^2$ and $k'' = (bs + a)' = as + b = k$. □

Fact 2.3. We have $k' \equiv sk \pmod{s^2 - 1}$ (or equivalently $k \equiv sk' \pmod{s^2 - 1}$).

Proof. We write $k = as + b$, as in the problem statement. Now,

$$sk - k' = s(as + b) - (bs + a) = a(s^2 - 1) \equiv 0 \pmod{s^2 - 1},$$

as desired. □

Combining Facts 2.1 and 2.2, we find that the sequence $d_1, d_2, d_3 \ldots$ is eventually periodic with period 2, starting at the first value less than s^2. From Fact 2.3, it follows that

$$k'' \equiv sk' \equiv s^2 k \equiv k \pmod{s^2 - 1}$$

and hence the sequence is periodic modulo $s^2 - 1$ from the start with period 2.

Now, if the sequence contains 1, the sequence eventually alternates between 1 and s since the twist of 1 is s and vice versa. Using periodicity modulo $s^2 - 1$, we must have $n \equiv 1, s \pmod{s^2 - 1}$. Conversely, if $n \equiv 1, s \pmod{s^2 - 1}$ then the eventual period must contain at least one value congruent to either 1 or s modulo $s^2 - 1$. Since these values must be less than s^2, this implies that the sequence eventually alternates between 1 and s, showing that it contains a 1. □

Solution 3. We give an alternate proof of the direct implication: if the sequence contains a 1, then the first term is 1 or s modulo $s^2 - 1$. We prove the following fact, which is a combination of Facts 2.1 and 2.3.

Fact 3.1. For all $k \geq s^2$, we have $(k - s^2 + 1)' \in \{k', k' - s^2 + 1\}$.

Proof. We write $k = as + b$, as in the problem statement. Since $k \geq s^2$, we have $a \geq s$. If $b < s - 1$, then

$$(k - s^2 + 1)' = \Big((a - s)s + (b + 1)\Big)' = (b + 1)s + (a - s) = bs + a = k'.$$

On the other hand, if $b = s - 1$, then

$$(k - s^2 + 1)' = \Big((a - s + 1)s + 0\Big)' = 0s + (a - s + 1) = a - s + 1 = k' - s^2 + 1.$$

□

Now assume $n \geq s^2$ and the sequence $d_1, d_2, \ldots$ contains a 1. Denote by $e_1, e_2, \ldots$ the sequence constructed as in the problem statement, but with initial value $e_1 = n - s^2 + 1$. Using the above fact, we deduce that $e_i \equiv d_i \pmod{s^2 - 1}$ and $e_i \leq d_i$ for all $i \geq 1$ by induction on i. Hence, the sequence $e_1, e_2, \ldots$ also contains a 1.

Since the conclusion we are trying to reach only depends on the residue of d_1 modulo $s^2 - 1$, we conclude that without loss of generality we can assume $n < s^2$.

Using Fact 2.2, it now follows that the sequence $d_1, d_2, \ldots$ is periodic with period two. Since 1 and s are twists of each other, it follows that if this sequence contains a 1, it must be alternating between 1 and s. Hence, $d_1 \equiv 1, s \pmod{s^2-1}$ as desired.

For the other direction we can make a similar argument, observing that the second of the two cases in the proof of Fact 3.1 can only apply to finitely many terms of the sequence $d_1, d_2, d_3, \ldots$, allowing us to also go the other way. □

Solution 4. First assume that $d_k = 1$ for some k. Let k be the smallest such index. If $k = 1$ then $n = 1$, so we may assume $k \geqslant 2$.

Then $d_{k-1} = as + b$ for some non-negative integers a, b satisfying $b < s$ and $bs + a = 1$. The only solution is $b = 0, a = 1$, so $d_{k-1} = s$. So, if $k = 2$, then $n = s$, so we may assume $k \geqslant 3$.

Then there exist non-negative integers c, d satisfying $d_{k-2} = cs + d$, $d < s$ and $ds + c = s$. We have two solutions: $d = 0, c = s$ and $d = 1, c = 0$. However, in the second case we get $d_{k-2} = 1$, which contradicts the minimality of k. Hence, $d_{k-2} = s^2$. If $k = 3$, then $n = d_1 = s^2$, which gives remainder 1 when divided by $s^2 - 1$.

Assume now that $k \geqslant 4$. We will show that for each $m \in \{3, 4, \ldots, k-1\}$ there exist $b_1, b_2, \ldots, b_{m-2} \in \{0, 1, \ldots, s-1\}$ such that

$$d_{k-m} = s^m - \sum_{i=1}^{m-2} b_i \left(s^{m-i} - s^{m-i-2}\right). \tag{9}$$

We will prove this equality by induction on m. If $m = 3$, then $d_{k-3} = a_1 s + b_1$ for some non-negative integers a_1, b_1 satisfying $b_1 < s$ and $b_1 s + a_1 = d_{k-2} = s^2$. Then $a_1 = s^2 - b_1 s$, so $d_{k-3} = s^3 - b_1(s^2 - 1)$, which proves (9) for $m = 3$.

Assume that (9) holds for some m and consider $d_{k-(m+1)}$. There exist non-negative integers a_{m-1}, b_{m-1} such that $d_{k-(m+1)} = a_{m-1}s + b_{m-1}$, $b_{m-1} < s$ and $d_{k-m} = b_{m-1}s + a_{m-1}$. Using the inductive assumption we get

$$a_{m-1} = d_{k-m} - b_{m-1}s = s^m - \sum_{i=1}^{m-2} b_i \left(s^{m-i} - s^{m-i-2}\right) - b_{m-1}s,$$

therefore

$$\begin{aligned} d_{k-(m+1)} =& a_{m-1}s + b_{m-1} = s^{m+1} - \sum_{i=1}^{m-2} b_i \left(s^{m-i} - s^{m-i-2}\right) s - b_{m-1}s^2 + b_{m-1} \\ =& s^{m+1} - \sum_{i=1}^{m-1} b_i \left(s^{m+1-i} - s^{m-i-1}\right), \end{aligned}$$

which completes the proof of (9). In particular, for $m = k - 1$ we get

$$d_1 = s^{k-1} - \sum_{i=1}^{k-3} b_i \left(s^{k-i-1} - s^{k-i-3}\right).$$

The above sum is clearly divisible by $s^2 - 1$, and it is clear that the remainder of s^{k-1} when divided by $s^2 - 1$ is 1 when k is odd, and s when k is even. It follows that the remainder when $n = d_1$ is divided by $s^2 - 1$ is either 1 or s.

To prove the other implication, assume that n gives remainder 1 or s when divided by s^2-1. If $n \in \{1, s, s^2\}$, then one of the numbers d_1, d_2 and d_3 is 1. We therefore assume that $n > s^2$. Since the reminder when a power of s is divided by s^2-1 is either 1 or s, there exists a positive integer m such that $s^m - n$ is non-negative and divisible by s^2-1. By our assumption $m \geqslant 3$. We also take the smallest such m, so that $n > s^{m-2}$. The quotient $\frac{s^m-n}{s^2-1}$ is therefore smaller than s^{m-2}, so there exist $b_1, \ldots, b_{m-2} \in \{0, 1, \ldots, s-1\}$ such that $\frac{s^m-n}{s^2-1} = \sum_{i=1}^{m-2} b_i s^{i-1}$. It follows that

$$n = s^m - \sum_{i=1}^{m-2} b_i \left(s^{i+1} - s^{i-1}\right).$$

We now show that

$$d_j = s^{m+1-j} - \sum_{i=1}^{m-1-j} b_i \left(s^{i+1} - s^{i-1}\right) \tag{10}$$

for $j = 1, 2, \ldots, m-2$ by induction on j. For $j = 1$ this follows from $d_1 = n$. Assume now that (10) holds for some $j < m-2$. Then

$$d_j = \left(s^{m-j} - \sum_{i=2}^{m-1-j} b_i \left(s^i - s^{i-2}\right) - b_1 s\right) s + b_1.$$

As d_j is positive and $b_1 \in \{0, 1, \ldots, s-1\}$, the expression $s^{m-j} - \sum_{i=2}^{m-1-j} b_i \left(s^i - s^{i-2}\right) - b_1 s$ has to be non-negative, so we can compute the twist of d_j as

$$d_{j+1} = b_1 s + s^{m-j} - \sum_{i=2}^{m-1-j} b_i \left(s^i - s^{i-2}\right) - b_1 s = s^{m-j} - \sum_{i=1}^{m-2-j} b_i \left(s^{i+1} - s^{i-1}\right),$$

which finishes the induction.

Now we use (10) for $j = m-2$ and get $d_{m-2} = s^3 - b_1(s^2-1) = (s^2 - b_1 s) + b_1$. Then $d_{m-1} = b_1 s + s^2 - b_1 s = s^2 = s \cdot s + 0$, $d_m = 0 \cdot s + s = s = 1 \cdot s + 0$ and $d_{m+1} = 0 \cdot s + 1 = 1$. $\square$

Problem 6. Let ABC be a triangle with circumcircle Ω. Let S_b and S_c respectively denote the midpoints of the arcs AC and AB that do not contain the third vertex. Let N_a denote the midpoint of arc BAC (the arc BC containing A). Let I be the incentre of ABC. Let ω_b be the circle that is tangent to AB and internally tangent to Ω at S_b, and let ω_c be the circle that is tangent to AC and internally tangent to Ω at S_c. Show that the line IN_a, and the line through the intersections of ω_b and ω_c, meet on Ω.

The incentre of a triangle is the centre of its incircle, the circle inside the triangle that is tangent to all three sides.

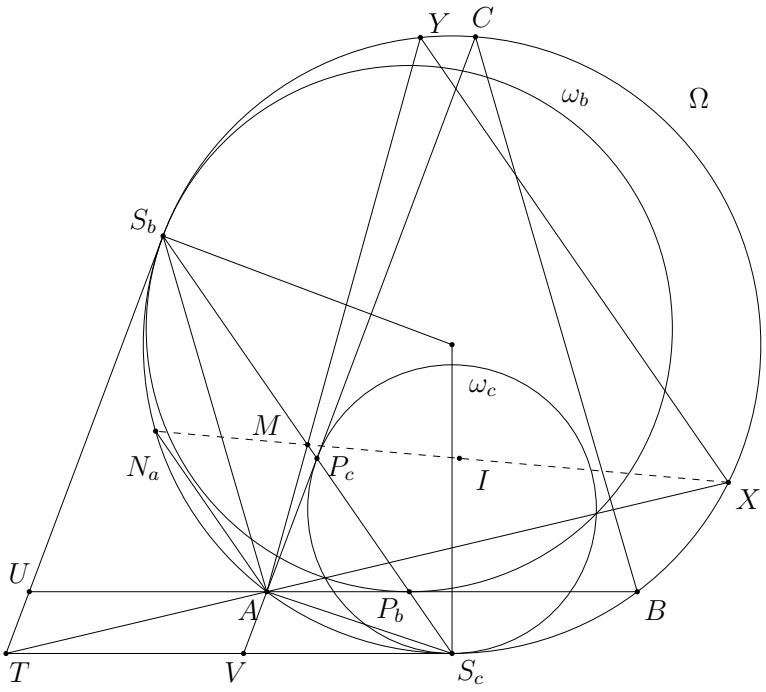


Figure 6: Diagram to Solution 1

Solution 1. Part I: First we show that A lies on the radical axis of ω_b and ω_c.

We first note that the line through the intersections of two circles is the radical line of the two circles. Let the tangents to Ω at S_b and S_c intersect at T. Clearly T is on the radical axis of ω_b and ω_c (and in fact is the radical centre of ω_b, ω_c and Ω).

We next show that A lies on the radical axis of ω_b and ω_c. Let P_b denote the point of tangency of ω_b and AB, and let P_c denote the point of tangency of ω_c and AC. Furthermore, let U be the intersection of the tangent to Ω at S_b with the line AB, and let V be the intersection of the tangent to Ω at S_c with the line AC. Then $TVAU$ is parallelogram. Morover, due to equality of tangent segments we have $|US_b| = |UP_b|$, $|VP_c| = |VS_c|$ and $|TS_b| = |TS_c|$. It follows that

$$\begin{aligned} |AP_b| &= |UP_b| - |UA| = |US_b| - |TV| = |TS_b| - |TU| - |TV| \\ &= |TS_s| - |TV| - |TU| = |VS_c| - |AV| = |VP_c| - |VA| = |AP_c|. \end{aligned} \tag{11}$$

But $|AP_b|$, $|AP_c|$ are exactly the square roots of powers of A with respect to ω_b and ω_c, hence A is indeed on their radical axis.

Thus, the radical axis of ω_b, ω_c is AT.

Part II: Consider the triangle AS_bS_c. Note that since T is the intersection of the tangents at S_b and S_c to the circumcircle of AS_bS_c, it follows that AT is the symmedian of A in this triangle. Let X denote the second intersection of the symmedian AT with Ω. We wish to show that X is also on IN_a.

Note that AN_a is the external angle bisector of angle A, and therefore it is parallel to S_bS_c. Let M denote the midpoint of S_bS_c, and let Y be the second intersection of AM with Ω. Since in AS_bS_c, AXT is the symmedian and AMY is the median, it follows that XY is also parallel to S_bS_c. Thus, reflecting in the perpendicular bisector of S_bS_c sends the line AMY to line N_aMX.

Next, consider the quadrilateral AS_bIS_c. From the trillium theorem we have $|S_bA| = |S_bI|$ and $|S_cA| = |S_cI|$, thus the quadrilateral is a kite, from which it follows that the reflection of the line AM in S_bS_c is the line IM. But previously we have seen that this is also the line N_aMX. Thus M, I, N_a and X are collinear, as we wanted to show. □

Solution 2. This is a variation of Solution 1 which avoids the theory of the symmedian point.

We begin by showing that the radical axis of ω_b, ω_c is AT as in Solution 1.

Part II: We introduce the point S_a with the obvious meaning. Observe that the incentre I of ABC is the orthocentre of $S_aS_bS_c$ either because this is well-known, or because of an angle argument that A reflects in S_bS_c to I (and similar results by cyclic change of letters). Therefore AS_a is perpendicular to S_bS_c.

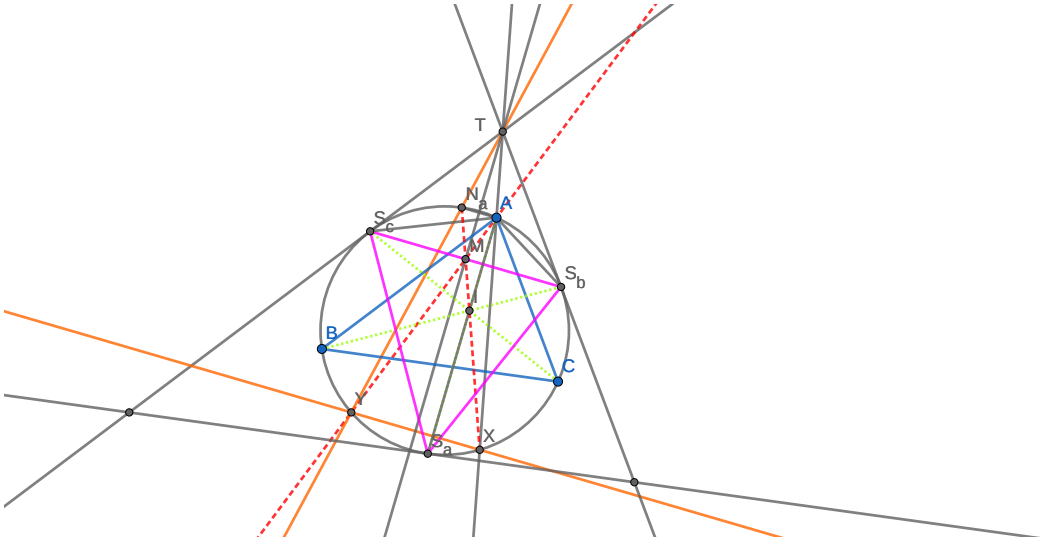


Figure 7: A reflections argument for Solution 2

Let M denote the midpoint of S_bS_c. Then A is the reflection of S_a in the diameter parallel to S_bS_c, so the reflection of A in the diameter perpendicular to S_bS_c is N_a, the antipode

of S_a. Let the reflection of X in TM be Y, so TY passes through N_a and is the reflection of TX in TM.

Now S_bS_c is the polar line of T with respect to Ω, so AY and N_aX meet on this line, and by symmetry at its midpoint M. The line N_aMX is therefore the reflection of the line YMA in S_bS_c, and so N_aMX passes through I (the reflection of A in S_bS_c). □

The triangle AS_cS_b can be taken as generic, and from the argument above we can extract the fact that the symmedian point and the centroid are isogonal conjugates in that triangle.

Solution 3. Assume the notation from Solution 1, part I of Solution 1, and let O be the centre of Ω.

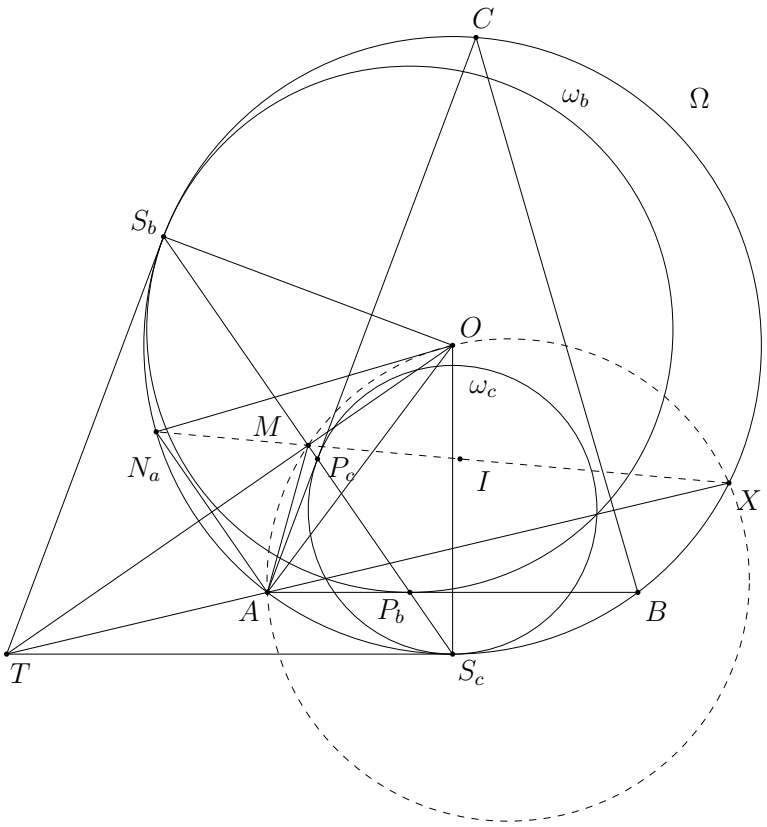


Figure 8: Diagram to Solution 3

Part II: As in Solution 1, by the trillelium theorem, S_cS_b bisects AI, and since $N_aA \parallel S_bS_c$, then OT is a bisector of AN_a. This implies $|MN_a| = |MA| = |MI|$, since M is the midpoint of S_cS_b and lies also on OT. Hence, M is the circumcentre of triangle IAN_a. But this triangle has a right angle at A (since AI and AN_a are the inner and outer angle bisector at A), hence M lies on IN_a.

Again, let X be the second intersection of TA and Ω. By the above, it suffices to prove that X lies on the line N_aM. From the power of point T with respect to Ω we get $|TA| \cdot |TX| = |TS_c|^2$. Since M is the foot of the altitude of right triangle TS_cO, we obtain

$|TS_c|^2 = |TM| \cdot |TO|$. Hence, $|TA| \cdot |TX| = |TM| \cdot |TO|$ so the points O, M, A, X are concyclic. It follows that $\angle MXA = \angle MOA = \frac{1}{2}\angle N_aOA = \angle N_aXA$. Hence, X lies on the line N_aM.

Remark. To show that $OMAX$ is cyclic, one can also invert the line TAX in the circumcircle of the triangle ABC. □

Solution 4. Part I is done as in solution 1.

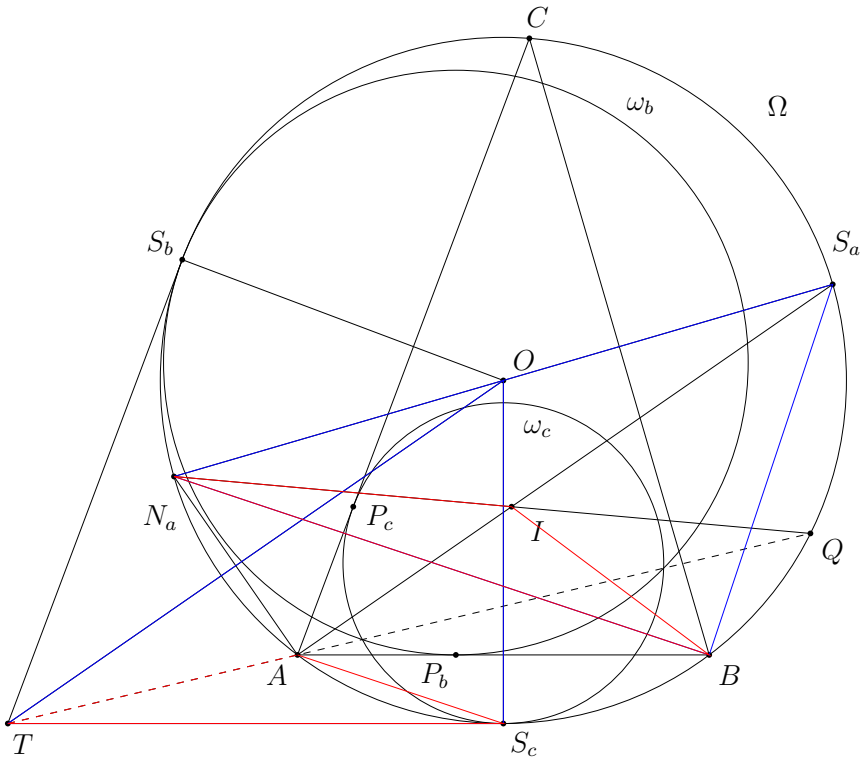


Figure 9: Diagram to Solution 4

Part II: as in Solution 1 we show that $AN_a || S_bS_c$. In particular, $\angle N_aOT = \angle TOA$. The conclusion of the problem trivially holds if $|AB| = |AC|$, therefore we assume without loss of generality that $|AC| > |AB|$. Let S_a be the midpoint of the arc BC which does not contain A. Then N_aS_a is a diameter, so $\angle S_aBN_a = \frac{\pi}{2} = \angle OS_cT$. We also compute $\angle BN_aS_a = \angle BAS_a = \frac{1}{2}\angle BAC = \frac{1}{2}\angle S_cTS_b = \angle S_cTO$. It follows that the triangles TS_cO and N_aBS_a are similar. In particular,

$$\frac{|N_aB|}{|TS_c|} = \frac{|S_aB|}{|OS_c|}. \tag{12}$$

Next we compute

$$\angle IS_aB = \angle N_aS_aB - \angle N_aS_aI = \angle TOS_c - \frac{1}{2}\angle N_aOA = \angle TOS_c - \angle TOA = \angle AOS_c \tag{13}$$

and

$$\angle IBN_a = \angle CBN_a - \angle CBI = \frac{1}{2}(\pi - \angle BN_aC) - \frac{1}{2}\angle CBA = \frac{1}{2}\angle ACB = \angle ACS_c = \angle AS_cT, \tag{14}$$

hence

$$\angle S_aBI = \frac{\pi}{2} - \angle IBN_a = \frac{\pi}{2} - \angle AS_cT = \angle OS_cA.$$

Together with (13) it follows that the triangles IBS_a and AS_cO are similar, so $\frac{|S_aB|}{|OS_c|} = \frac{|IB|}{|AS_c|}$, and (12) implies $\frac{|N_aB|}{|TS_c|} = \frac{|IB|}{|AS_c|}$. Consequently, by (14) the triangles TS_cA and N_aBI are similar and therefore $\angle S_cTA = \angle BN_aI$. Now let Q be the second intersection of N_aI with Ω. Then $\angle BN_aI = \angle BN_aQ = \angle BAQ$, so $\angle S_cTA = \angle BAQ$. Since AB is parallel to TS_c, we get $AQ||TA$, i.e. A, T, Q are collinear. □

Remark. After proving similarity of triangles TS_cO and N_aBS_a one can use spiral symmetry to show similarity of triangles TS_cA and N_aBI.

Solution 5. Part I: First we show that A lies on the radical axis between ω_b and ω_c.

Let T be the radical center of the circumcircle, ω_b and ω_c; then TS_b and TS_c are common tangents of the circles, as shown in Figure 5a. Moreover, let $P_b = AB \cap S_bS_c$ and $P_c = AC \cap S_bS_c$. The triangle TS_cS_b is isosceles: $AB \parallel TS_c$ and $AC \parallel TS_b$ so

$$\angle AP_bP_c = \angle TS_cS_b = \angle S_cS_bT = \angle P_bP_cA.$$

From these angles we can see that ω_b passes through P_b, ω_c passes through P_c, and finally AP_b and AP_c are equal tangents to ω_b and ω_c, so A lies on the radical axis.

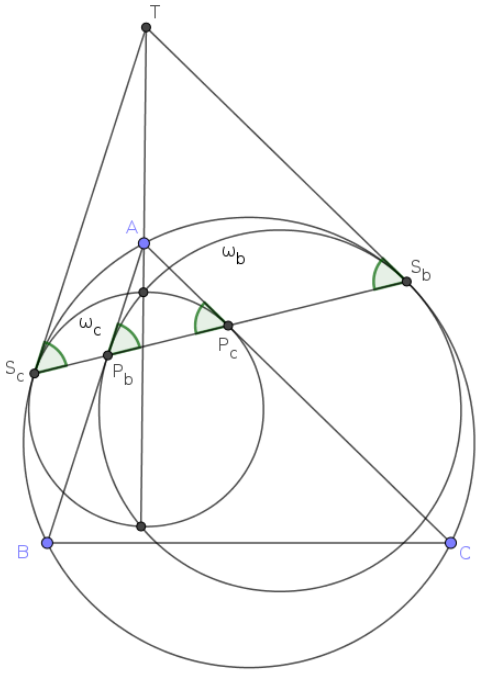


Figure 5a

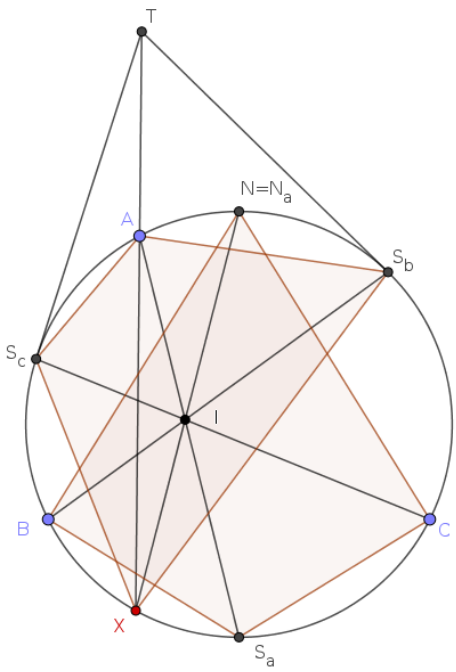


Figure 5b

Part II. Let the radical axis TA meet the circumcircle again at X, let S_a be the midpoint of the arc BC opposite to A, and let XI meet the circumcirlce again at N. (See Figure 2.) For solving the problem, we have prove that $N_a = N$.

The triples of points A, I, S_a; B, I, S_b and C, I, S_c are collinear because they lie on the angle bisectors of the triangle ABC.

Notice that the quadrilateral AS_cXS_b is harmonic, because the tangents at S_b and S_c, and the line AX are concurrent at T. This quadrilateral can be projected (or inverted) to the quadrilateral S_aCNB through I. So, S_aCNB also is a harmonic quadrilateral. Due to $S_aB = S_aC$, this implies $NB = NC$, so $N = N_a$. Done.

Remark. Instead of mentioning inversion and harmonic quadrilaterals, from the similar triangles $\triangle TS_cA \sim \triangle TXS_c$ and $\triangle TAS_b \sim \triangle TS_bX$ we can get

$$\frac{AS_c}{S_cX} = \frac{AS_b}{S_bX}.$$

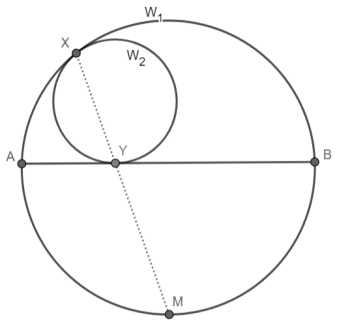
Then, we can apply the trigonometric form of Ceva's theorem to triangle BCX

$$\frac{\sin\angle BXN_a}{\sin\angle N_aXC} \cdot \frac{\sin\angle CBS_b}{\sin\angle S_bBX} \cdot \frac{\sin\angle XCS_c}{\sin\angle S_cCB} = \frac{BN_a}{N_aC} \cdot \frac{-CS_b}{S_bX} \cdot \frac{XS_c}{-S_cB} = 1 \cdot \frac{S_bA}{S_bN_a} \cdot \frac{N_aS_c}{S_cB} = 1,$$

so the Cevians BS_b, CS_c and XN_a are concurrent. □

Solution 6. Part I: First let's show that this is equivalent to proving that TA and N_aI intersect in Ω.

Lemma: Let's recall that if we have two circles ω_1 and ω_2 which are internally tangent at point X and if we have a line AB tangent to ω_2 at Y. Let M be the midpoint of the arc AB not containing Z. We have that Z, Y, M are collinear.



Solution 6: Lemma

Let $P_b = AB \cap \omega_b$ and $P_c = AC \cap \omega_c$. We can notice by the lemma that S_b, P_b and S_c are collinear, and similarly S_c, P_c and S_b are also collinear. Therefore S_c, P_b, P_c, and S_b are collinear, and since $\angle AP_bP_c = \frac{\angle ABC}{2} + \frac{\angle ACB}{2} = \angle AP_cP_b$ then $AP_b = AP_c$ so A is on the radical axis of ω_b and ω_c. Let T be the intersection of the tangent lines of Ω through S_c and S_b. Since $TS_c = TS_b$ then AT is the radical axis between ω_b and ω_c.

Part II: TA and N_aI intersect in Ω.

Let ω_a the A-mixtilinear incircle (that is, the circle internally tangent to Ω, and tangent to both AB and AC), and let $X = \Omega \cap \omega_a$. It is known that N_a, I, X are collinear.

Let M_c and M_b be the tangent points of ω_A to AB and AC respectively, then by the lemma X, M_c, S_c are collinear and X, M_b, S_b are collinear. We can see that S_cTS_b and M_cAM_b are homothetic with respect to X; therefore T and A are homothetic with respect to X, impying that T, A, X are collinear. □

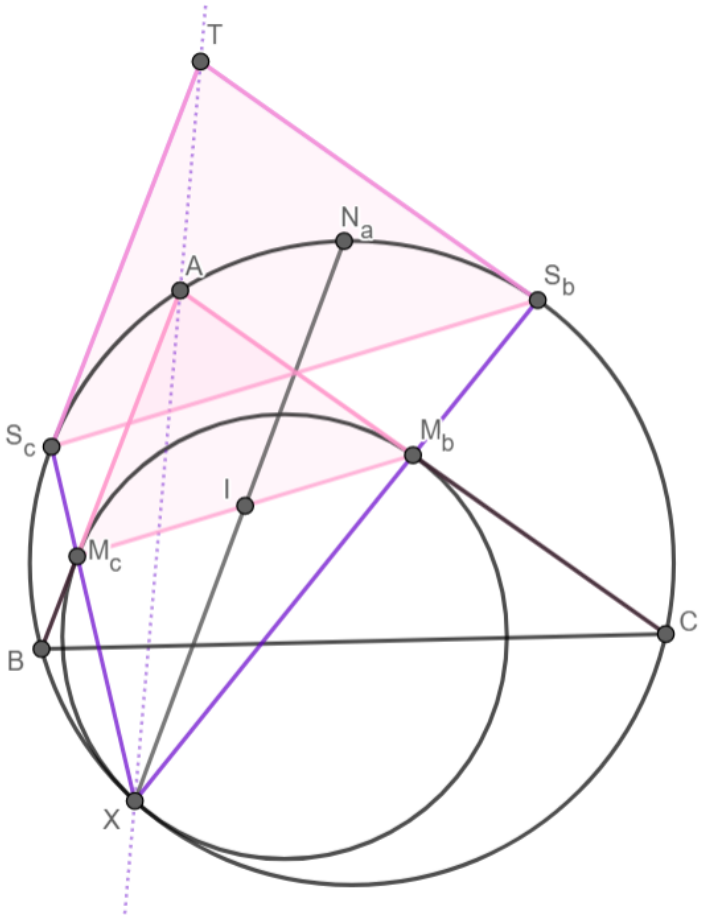


Figure 10: Diagram to Solution 6

EGMO2022 Problems

DAY 1

Problem 1.
Let ABC be an acute-angled triangle in which $BC < AB$ and $BC < CA$. Let point P lie on segment AB and point Q lie on segment AC such that $P \neq B$, $Q \neq C$ and $BQ = BC = CP$. Let T be the circumcentre of triangle APQ, H the orthocentre of triangle ABC, and S the point of intersection of the lines BQ and CP. Prove that T, H and S are collinear.

Problem 2.
Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ be the set of all positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for any positive integers a and b, the following two conditions hold:

(1) $f(ab) = f(a)f(b)$, and

(2) at least two of the numbers $f(a)$, $f(b)$ and $f(a + b)$ are equal.

Problem 3.
An infinite sequence of positive integers $a_1, a_2, \ldots$ is called *good* if

(1) a_1 is a perfect square, and

(2) for any integer $n \geq 2$, a_n is the smallest positive integer such that

$$na_1 + (n-1)a_2 + \ldots + 2a_{n-1} + a_n$$

is a perfect square.

Prove that for any good sequence $a_1, a_2, \ldots$, there exists a positive integer k such that $a_n = a_k$ for all integers $n \geq k$.

EGMO2022 Problems

DAY 2

Problem 4.
Given a positive integer $n \geq 2$, determine the largest positive integer N for which there exist $N+1$ real numbers $a_0, a_1, \ldots, a_N$ such that

(1) $a_0 + a_1 = -\dfrac{1}{n}$, and

(2) $(a_k + a_{k-1})(a_k + a_{k+1}) = a_{k-1} - a_{k+1}$ for $1 \leq k \leq N-1$.

Problem 5.
For all positive integers n, k, let $f(n, 2k)$ be the number of ways an $n \times 2k$ board can be fully covered by nk dominoes of size 2×1. (For example, $f(2,2) = 2$ and $f(3,2) = 3$.)
Find all positive integers n such that for every positive integer k, the number $f(n, 2k)$ is odd.

Problem 6.
Let $ABCD$ be a cyclic quadrilateral with circumcentre O. Let the internal angle bisectors at A and B meet at X, the internal angle bisectors at B and C meet at Y, the internal angle bisectors at C and D meet at Z, and the internal angle bisectors at D and A meet at W. Further, let AC and BD meet at P. Suppose that the points X, Y, Z, W, O and P are distinct.

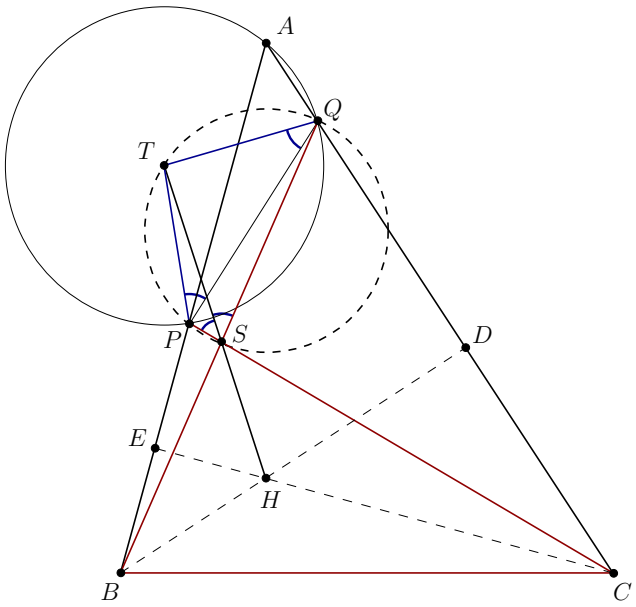
Prove that O, X, Y, Z and W lie on the same circle if and only if P, X, Y, Z and W lie on the same circle.

EGMO2022 Problems – Solutions

Problem 1.
Let ABC be an acute-angled triangle in which $BC < AB$ and $BC < CA$. Let point P lie on segment AB and point Q lie on segment AC such that $P \neq B$, $Q \neq C$ and $BQ = BC = CP$. Let T be the circumcentre of triangle APQ, H the orthocentre of triangle ABC, and S the point of intersection of the lines BQ and CP. Prove that T, H and S are collinear.

Proposed by: Netherlands

Solution 1.



We show that T and H are both on the angle bisector ℓ of $\angle BSC$.

We first prove that $H \in \ell$. The altitude CH in triangle ABC is also the altitude in isosceles triangle PBC with $CP = CB$. Therefore, CH is also the angle bisector of $\angle PCB$ and hence also of $\angle SCB$. Analogously, BH is the angle bisector of $\angle SBC$. We conclude that H, as the intersection of two angle bisectors of $\triangle BSC$, is also on the third angle bisector, which is ℓ.

We now prove that $T \in \ell$.

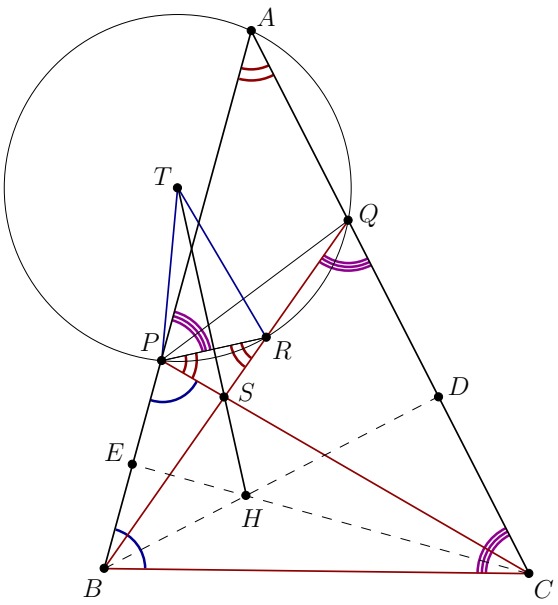
Variant 1. In the isosceles triangles $\triangle BCP$ and $\triangle CBQ$ we see that $\angle BCP = 180° - 2\angle B$ and $\angle CBQ = 180° - 2\angle C$. This yields $\angle PSQ = \angle BSC = 180° - (180° - 2\angle B) - (180° - 2\angle C) = 180° - 2\angle A$. Furthermore, $\angle PTQ = 2\angle PAQ = 2\angle A$ (T being circumcentre of $\triangle APQ$). Now $\angle PTQ + \angle PSQ = 180°$, so $PTQS$ is a cyclic quadrilateral. From $PT = TQ$ we then obtain that $\angle PST = \angle PQT = \angle QPT = \angle QST$, so T is on the angle bisector $\angle PSQ$, which is also ℓ.

We conclude that T, S and H are collinear.

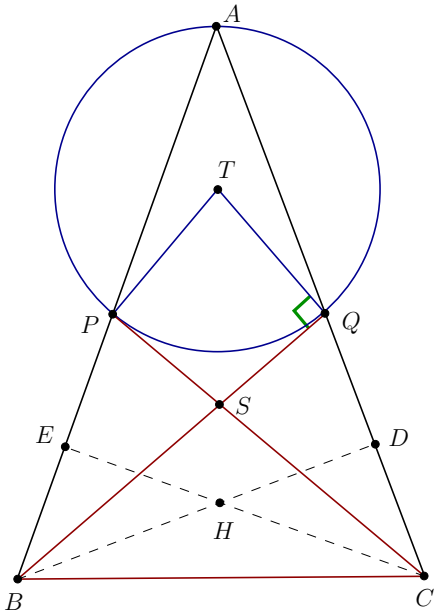
Variant 2. Let R be the second intersection of BQ and $\odot APQ$.

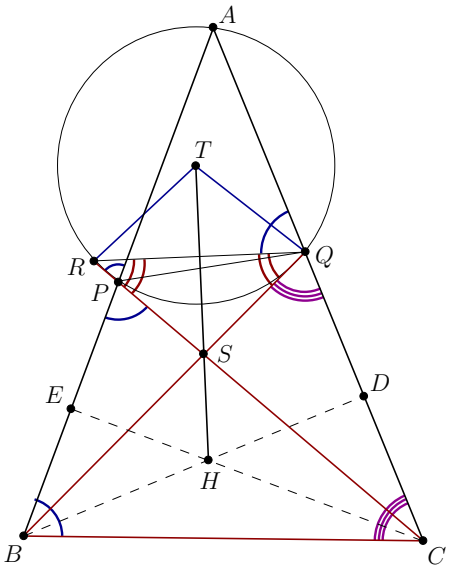
$APRQ$ is cyclic quadrilateral, so $\angle PRS = \angle A$, $\angle APR = \angle BQC = \angle C$. On the other hand $\angle BPC = \angle B$. Therefore $\angle RPS = 180° - \angle B - \angle C = \angle A$. Hence, the triangle PRS is isosceles with $SP = SR$; then ℓ is the perpendicular bisector of the chord PR in the circle that passes through T.

Note that if BQ is tangent to $\odot APQ$, then CP is also tangent to $\odot APQ$ and triangle ABC is isosceles, so T, S and H lie on the altitude from A.

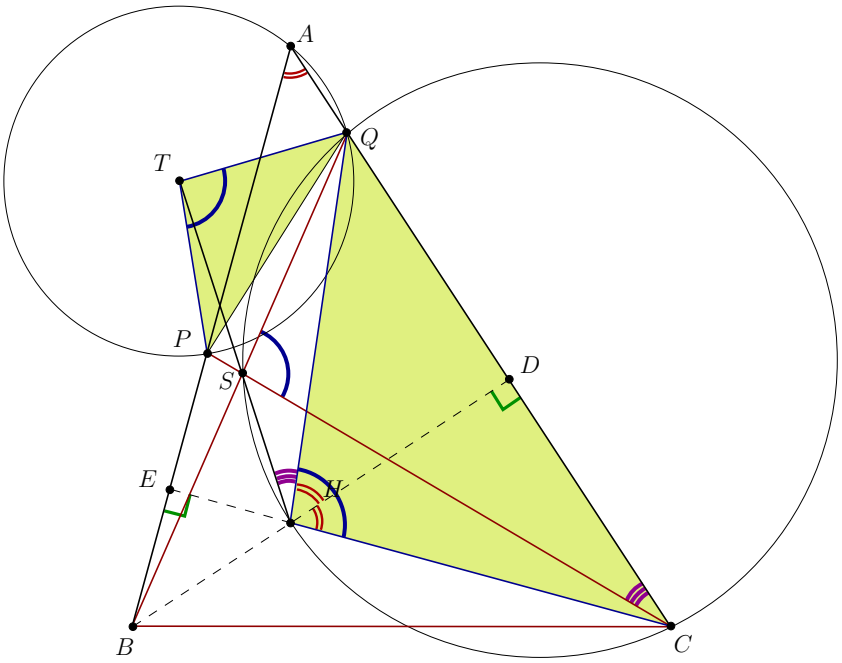


Remark. The fact that CP is also tangent to $\odot APQ$ could be shown with $PTQS$ beeing a cyclic quadrilateral like in the first variant of the solution. Otherwise we can consider R the second intersection of CP and $\odot APQ$ and prove that triangle QRS is isosceles.





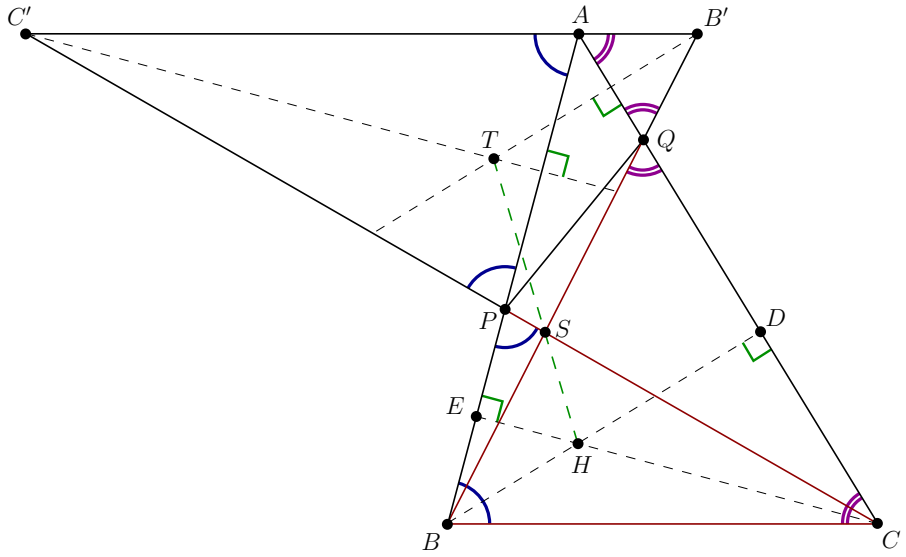
Solution 2.



In the same way as in the previous solution, we see that $\angle PSQ = 180° - 2\angle A$, so $\angle CSQ = 2\angle A$. From the cyclic quadrilateral $AEHD$ (with E and D feet of the altitudes CH and BH) we see that $\angle DHC = \angle DAE = \angle A$. Since BH is the perpendicular bisector of CQ, we have $\angle DHQ = \angle A$ as well, so $\angle CHQ = 2\angle A$. From $\angle CHQ = 2\angle A = \angle CSQ$, we see $CHSQ$ is a cyclic quadrilateral. This means $\angle QHS = \angle QCS$.

Since triangles PTQ and CHQ are both isosceles with apex $2\angle A$, we get $\triangle PTQ \sim \triangle CHQ$. We see that one can be obtained from the other by a spiral similarity centered at Q, so we also obtain $\triangle QTH \sim \triangle QPC$. This means that $\angle QHT = \angle QCP$. Combining this with $\angle QHS = \angle QCS$, we see that $\angle QHT = \angle QCP = \angle QCS = \angle QHS$. So $\angle QHT = \angle QHS$, which means that T, S and H are collinear.

Solution 3.



Let us draw a parallel f to BC through A. Let $B' = BQ \cap f$, $C' = CP \cap f$. Then $AQB' \sim CQB$ and $APC' \sim BPC$, therefore both QAB' and APC' will be isosceles.

Also, $BCS \sim B'C'S$ with respect to the similarity with center S, therefore if we take the image of the line BH (which is the perpendicular bisector of the segment CQ) through this transformation, it will go through the point B', and be perpendicular to AQ (as the image of CQ is parallel to CQ). As AQB' is isosceles, this line is the perpendicular bisector of the segment AQ. This means it goes through T, the circumcenter of APQ. Similarly on the other side the image of CH also goes through T. This means that the image of H with respect to the similarity though S will be T, so T, S, H are collinear.

Problem 2.
Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ be the set of all positive integers. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for any positive integers a and b, the following two conditions hold:

(1) $f(ab) = f(a)f(b)$, and

(2) at least two of the numbers $f(a)$, $f(b)$ and $f(a + b)$ are equal.

Proposed by: Fedir Yudin, Ukraine

Answer: $f(n) = a^{v_p(n)}$, where $a \in \mathbb{N}$, p is a prime and $v_p(n)$ is the largest exponent of p dividing n.

Solution 1. First, all such functions f satisfy the conditions, as $v_p(a) \neq v_p(b)$ implies $v_p(a + b) = \min(v_p(a), v_p(b))$. Plugging $a, b = 1$ into (1) gives $f(1) = 1$. Also, a simple induction gives that $f(\prod_{i=1}^{k} p_i^{a_i}) = \prod_{i=1}^{k} f(p_i)^{a_i}$. Let S be the set of all primes p such that $f(p) \neq 1$. If $|S| = 1$, then $S = \{p\}$ and $f(p) = a$ for some $a \neq 1$, and thus $f(n) = a^{v_p(n)}$ for all n. If $S = \emptyset$ then $f(n) = 1$ for all n, which can be also written in the above form for $a = 1$.

Now suppose that S contains at least two primes. Let $p < q$ be the two smallest elements of S. Since all prime divisors of $q - p$ are smaller than q and not equal to p, $f(q - p) = 1$. Using (2) for p and $q - p$ gives us that some two of $f(p) \neq 1$, $f(p - q) = 1$ and $f(q) \neq 1$ are equal, meaning that $f(p) = f(q)$. Let $t \geq 2$ be the smallest integer such that $p^t > q$, and let $p^t = aq + b$ with $0 \leq b < q$. Since $q \geq p^{t-1}$ and q is not divisible by p, we have $q > p^{t-1}$, so $a < p$, and thus $f(a) = 1$. Therefore aq is not divisible by p, thus neither is b, and since $b < q$, $f(b) = 1$. Now using (2) for aq and b we get that some two of $f(aq) = f(a)f(q) = f(q) = f(p)$, $f(b) = 1$ and $f(aq + b) = f(p^t) = f(p)^t$ are equal, which is a contradiction.

Solution 1B (for the main part).
Suppose that there are two primes, $p < q$ such that $f(p) > 1$ and $f(q) > 1$, and these are the two smallest such primes.

$1 < p - q < q$ and it is not divisible by p and q, therefore its prime factors are smaller than q, so we conclude $f(q - p) = 1$. By (2), two of $f(q - p) = 1, f(q - p) > 1$ and $f(q) > 1$ are equal, so $f(q) = f(p) = a$.

If $p^2 < q$ then $q - p^2$ is not divisible by p and q, so we conclude $f(q - p^2) = 1$ in the same way. Then $f(p^2) = f(p)^2 = a^2$, $f(q - p^2) = 1$ and $f(q) = a$ are distinct, contradicting (2). Hence, $q < p^2$.

Let $c = \lfloor p^2/q \rfloor$; then $0 < c < p$, so $f(c) = 1$. By $0 < p^2 - cq < q$, we get $f(p^2 - cq) = 1$. Now $f(p^2 - cq) = 1$, $f(p^2) = a^2$ and $f(cq) = a$ are distinct, contradicting (2) again.

Solution 2 (for the main part).
Suppose there exists a positive integer n such that $f(n) \neq 1$ and $f(n + 1) \neq 1$. We know that $f(1) = 1$, so by (2) two of $f(n) \neq 1, f(n + 1) \neq 1$ and $f(1) = 1$ are equal, therefore $f(n) = f(n + 1)$.

$f(n^2) = f(n)^2 \neq 1$ and $f(n^2 - 1) = f(n + 1)f(n - 1) = f(n)f(n - 1) \neq 1$, but by (2) two of $f(1) = 1, f(n^2 - 1) \neq 1$ and $f(n^2) \neq 1$ are equal, so $f(n^2 - 1) = f(n^2)$ and thus $f(n)f(n - 1) = f(n)^2$, therefore $f(n - 1) = f(n)$.

Then for $n - 1$, it is also true that $f(n - 1) \neq 1$ and $f(n - 1 + 1) \neq 1$, so the same proof applies, $f(n - 1 - 1) = f(n - 1)$ and so on. By induction we get that $f(n) = f(1)$, which is a contradiction, as $f(1) = 1$ and we assumed that $f(n) \neq 1$. Therefore, there can't be any n for which $f(n) \neq 1$ and $f(n + 1) \neq 1$.

Now suppose there are two primes, $p \neq q \in S$ such that $f(p) \neq 1$ and $f(q) \neq 1$. Then by the Chinese remainder theorem, there exists a number n such that $p|n$ and $q|n + 1$, so $f(p)|f(n)$ and $f(q)|f(n + 1)$ thus $f(n) \neq 1$ and $f(n + 1) \neq 1$. This is a contradiction, so S can't have two distinct elements.

Solution 3 (Joseph Myers) (for the second part).

Claim 3.1. If $f(m) > 1$ for some $m \in \mathbb{N}$, then there are less than m different prime numbers p_i with $f(p_i) > 1$.

Proof. On the one hand, if $f(m) = c > 1$, then for any $n \in \mathbb{N}$, there is some $k \in \{n+1, n+2 \ldots, n+m\}$ such that $f(k) \in \{1, c\}$. To show this, it is enough to see that condition (2) implies that for any $n \in \mathbb{N}$

- if $f(n) = c$, then $f(n+1) \in \{1, c\}$;
- if $f(n) = 1$, then $f(n+m) \in \{1, c\}$.

On the other hand, if we suppose that there are m different prime numbers $p_1, p_2, \ldots, p_m$, then we know (by the Chinese Remainder Theorem) that for any number $t \in \mathbb{N}$, there exists a number $n_t \in \mathbb{N}$ such that $p_k^t \mid n_t + k$ for each $k \in \{1, 2, \ldots, m\}$. Hence for each k, we have

$$f(n_t + k) \geq f(p_i^t) = f(p_i)^t \geq 2^t.$$

If $2^t > c$, then it means that $f(k) > c$ for all $k \in \{n_t + 1, n_t + 2 \ldots, n_t + m\}$. □

Now we can suppose that for some $m \geq 2$, the set of primes with $f(p) > 1$ is $S = \{p_1, p_2, \ldots, p_m\}$. Now let S_1 and S_2 be two disjoint nonempty subsets such that $S_1 \cup S_2 = S$. Let $a = \prod_{p_i \in S_1} p_i$ and $b = \prod_{p_j \in S_2} p_j$.

Now for any $k, \ell \in \mathbb{N}$, $a^k + b^\ell$ is coprime to all primes in S, hence $f(a^k + b^\ell) = 1$. It is easy to see that we can choose k and ℓ such that $f(a^k) \neq f(b^\ell)$, which contradicts condition (2).

Solution 4 (for the main part). Let p the minimal (prime) number with $f(p) = c > 1$.

Claim 4.1. For any $k \in \mathbb{N}$, $f(1 + p + p^2 + \ldots + p^k) \in \{1, c, c^2, \ldots, c^k\}$.

Proof of the claim. Use condition (2) recursively. □

Now suppose that there exists some prime $q > p$ with $f(q) > 1$, and consider the following three values of f:

- $f(1) = 1$.
- $f(p^{q-1} - 1) = f(p-1)f(p^{q-2} + p^{q-3} + \ldots + p + 1) \in \{1, c, c^2, \ldots, c^{q-2}\}$ using our claim, (note that we know $f(p-1) = 1$ by the minimality of p).
  But we also know (by Fermat's little theorem) that $q \mid p^{q-1} - 1$, hence $f(q) \mid f(p^{q-1} - 1)$.
  Therefore we have that: $f(p^{q-1} - 1) \in \{c, c^2, \ldots, c^{q-2}\}$.
- $f(p^{q-1}) = c^{q-1}$.

These are three different values, hence we have a contradiction by condition (2).

Solution 4B (for the main part). Let p the minimal (prime) number with $f(p) = c > 1$.

Lemma 4B.1. For any $k \in \mathbb{N}$, $f(1 + p + p^2 + \ldots + p^k) = 1$.

Proof of the Lemma, by induction. $k = 0$ is trivial.

$k = 1$ is also easy, as $p + 1$ cannot be a prime, except in the case of $p = 2$, which is also easy. (If $f(2) = c > 1$, then $f(3) \in \{1, c\}$ by $3 = 1 + 2$ and $f(3) \in \{1, c^2\}$ by $2^2 = 1 + 3$.)

From now we suppose $k \geq 2$ and $f(1 + p + \ldots + p^{k-1}) = 1$. By condition (2), we have that:

- As $f(1 + p + \ldots + p^{k-1}) = 1$ and $f(p^k) = c^k$,
  we have that $f(1 + p + \ldots + p^{k-1}) \in \{1, c^k\}$
- As $f(1) = 1$ and $f(p + \ldots + p^k) = f(p)f(1 + p + \ldots + p^{k-1}) = c$,
  we also have that $f(1 + p + \ldots + p^{k-1}) \in \{1, c\}$.

If $k > 1$, $c \neq c^k$, therefore $f(1 + p + \ldots + p^k) = 1$. □

Now suppose that there exists some prime $q > p$ with $f(q) > 1$. Then by condition (1),

$$f(p^{q-1} - 1) = f(p-1)f(1 + p + \ldots + p^{q-2})$$

here $f(p-1) = 1$ (by the minimality of p) and $f(1 + p + \ldots + p^{q-2}) = 1$ by the lemma. But we also know (by Fermat's little theorem) that $q \mid p^{q-1} - 1$, hence $1 < f(q) \leq f(p^{q-1} - 1)$, which is a contradiction.

Solution 5 (for the main part).

Lemma 5.1. If $f(m) \neq f(n)$, then $f(n+m) = f(n-m)$ (for all $n > m \in \mathbb{N}$).

Proof of the Lemma. By condition (2), $\{f(n+m), f(n-m)\} \subseteq \{f(n), f(m)\}$. We also have that:

$$f(n-m)f(n+m) = f(n^2 - m^2) \in \{f(n)^2, f(m)^2\}$$

which is only possible if $f(n-m) = f(n+m) = f(n)$ or $f(n-m) = f(n+m) = f(m)$. □

Now let p be the smallest (prime) number with $f(p) > 1$.

Claim 5.2. If p does not divide n, then $f(n) = 1$.

Proof of the Claim. Let n be the minimal counterexample, and and let $n = dp + r$ with $0 < r < p$. Here $d \geq 1$ by the minimality of p.

Now $1 < f(d)f(p) = f(dp) \neq f(r) = 1$, hence by our lemma, $f(dp + r) = f(dp - r) = 1$, contradiction.

Problem 3.
An infinite sequence of positive integers $a_1, a_2, \ldots$ is called *good* if

(1) a_1 is a perfect square, and

(2) for any integer $n \geq 2$, a_n is the smallest positive integer such that

$$na_1 + (n-1)a_2 + \ldots + 2a_{n-1} + a_n$$

is a perfect square.

Prove that for any good sequence $a_1, a_2, \ldots$, there exists a positive integer k such that $a_n = a_k$ for all integers $n \geq k$.

Proposed by: Joe Benton and Dominic Yeo, United Kingdom

Solution. Define the following auxiliary sequences:

$$b_1 = a_1, \quad b_n = a_1 + a_2 + \cdots + a_n,$$

$$c_1 = b_1, \quad c_n = b_1 + b_2 + \cdots + b_n.$$

Observe that

$$c_n - c_{n-1} = b_n,$$

$$b_n - b_{n-1} = (c_n - c_{n-1}) - (c_{n-1} - c_{n-2}) = a_n.$$

Therefore,

$$c_n = (2c_{n-1} - c_{n-2}) + a_n$$

is the smallest square greater than $2c_{n-1} - c_{n-2}$.

Claim. $\sqrt{c_n} - \sqrt{c_{n-1}} \leq \sqrt{c_{n-1}} - \sqrt{c_{n-2}}$

Proof 1. Let $c_n = m_n^2$, $c_{n-1} = m_{n-1}^2$, $c_{n-2} = m_{n-2}^2$, where the value of c_n is momentarily permitted to exceed its minimum. Any value is permitted for which

$$b_n = c_n - c_{n-1} > b_{n-1} = c_{n-1} - c_{n-2}.$$

Factoring,

$$(m_n - m_{n-1})(m_n + m_{n-1}) > (m_{n-1} - m_{n-2})(m_{n-1} + m_{n-2}).$$

In the case that $m_n - m_{n-1} = m_{n-1} - m_{n-2}$, this inequality clearly holds, as $m_n + m_{n-1} > m_{n-1} + m_{n-2}$. Thus, the minimal possible value of m_n satisfies the claimed inequality.

Proof 2. Denote $c_{n-1} = x^2$, $c_{n-2} = (x-d)^2$. Then

$$2c_{n-1} - c_{n-2} = 2x^2 - (x-d)^2 = x^2 + 2dx - d^2 = (x+d)^2 - 2d^2 < (x+d)^2.$$

It follows that $c_n \leq (x+d)^2$. ■

And so the sequence of positive integers $\sqrt{c_n} - \sqrt{c_{n-1}}$ is decreasing. Any such sequence is eventually constant.

As a corollary, $c_n = (x+nd)^2$ for sufficiently large n, with fixed integers x and d. Along the lines above, it becomes clear that $a_n = c_n - 2c_{n-1} + c_{n-2} = 2d^2$, so the sequence (a_n) is constant.

Solution. We write:

$$s_n^2 = S_n = a_1 + (a_1 + a_2) + \ldots + (a_1 + \ldots + a_n).$$

So, setting $b_n := a_1 + \ldots + a_n$, we have $S_n = b_1 + b_2 + \ldots + b_n$ and, in particular $S_{n+1} = S_n + b_{n+1}$. Now, we study the quantity $S_n b_n = +b_1 + b_2 + \ldots + b_n + b_n$ in two different ways. Since b_{n+1} is the smallest integer strictly greater than b_n such that $b_1 + \ldots + b_n + b_{n+1}$ is a perfect square, we must have

$$b_1 + b_2 + \ldots + b_n + b_n \geq (s_{n+1} - 1)^2.$$

However, we also have

$$b_1 + b_2 + \ldots + b_n + b_n = S_n + b_n = 2S_n - S_{n-1}.$$

Combining, we obtain

$$s_n^2 \;\geq\; \frac{s_{n-1}^2 + (s_{n+1} - 1)^2}{2} \;>\; \left(\frac{s_{n-1} + s_{n+1} - 1}{2}\right)^2.$$

where the final inequality is strict since the sequence (s_k) is strictly increasing. Taking a square root, and noting that all the (s_n) are integers, one obtains $s_{n+1} - s_n \leq s_n - s_{n-1}$.
Now we focus on the sequence $d_n = s_{n+1} - s_n$. (s_k) is strictly increasing thus (d_k) is positive. However we proved that $d_{n+1} \leq d_n$, so the sequence (d_k) is eventually constant, so eventually $s_n = bn + c$ and $S_n = (bn + c)^2$ with some numbers b,c; then

$$a_{n+2} = S_{n+2} - 2S_{n+1} + S_n = (b(n+2) + c)^2 - 2(b(n+1) + c)^2 + (bn + c)^2 = 2b^2.$$

Remark. The key idea is to use the D difference operator on the space of sequences. That is a function which takes a sequence as an input and outputs another sequence the following way: if (x) is the input sequence, then

$$D(x) = D\left((x_1, x_2, x_3, x_4, ...)\right) = (x_2 - x_1, x_3 - x_2, x_4 - x_3, ...).$$

Here $DD(S) = (a)$ (with some shift), $D(s) = (d)$. This D operator has some nice properties; if someone studied calculus, these properties will be familiar. $D(x + y) = D(x) + D(y)$; $D(\lambda x) = \lambda D(x)$; (x) constant iff Dx zero ; (x) is linear iff Dx constant ; in general (x) is a polynomial with degree $l + 1$ iff $D(x)$ is a polynomial with degree l.

First we proved that sequence (d) is eventually constant, thus $D(d) = DD(s)$ is eventually zero. Therefore the sequence (s) is eventually a linear function, hence $s^2 = S$ is eventually a quadratic polynomial, therefore $0 = DDD(S) = D(a)$, so (a) is constant eventually.

Problem 4.
Given a positive integer $n \geq 2$, determine the largest positive integer N for which there exist $N+1$ real numbers $a_0, a_1, \ldots, a_N$ such that

(1) $a_0 + a_1 = -\dfrac{1}{n}$, and

(2) $(a_k + a_{k-1})(a_k + a_{k+1}) = a_{k-1} - a_{k+1}$ for $1 \leq k \leq N-1$.

Proposed by: Romania

Solution 1. $(a_k+a_{k-1})(a_k+a_{k+1}) = a_{k-1}-a_{k+1}$ is equivalent to $(a_k+a_{k-1}+1)(a_k+a_{k+1}-1) = -1$. Let $b_k = a_k + a_{k+1}$. Thus we need $b_0, b_1, ...$ the following way: $b_0 = -\frac{1}{n}$ and $(b_{k-1}+1)(b_k-1) = -1$. There is a proper sequence $b_0, b_1, ..., b_{N-1}$ if and only if there is proper sequence $a_0, a_1, ..., a_N$, because from a a proper (a_k) sequence we can get a proper (b_k) sequence with $b_k = a_k + a_{k+1}$ for $k = 0, 1, ..., N-1$ and from a proper (b_k) sequence we can get a proper (a_k) sequence by arbitrarily setting a_0 and then inductively defining $a_k = b_{k-1} - a_{k-1}$ for $k = 1, 2, ..., N$.

We prove by induction that $b_k = -\frac{1}{n-k}$ for $k < n$. This is true for $k = 0$, as $b_0 = -\frac{1}{n}$ and

$$b_k = 1 - \frac{1}{b_{k-1}+1} = 1 - \frac{1}{1-\frac{1}{n-k+1}} = -\frac{1}{n-k}$$

for $k < n$. Thus there is a proper sequence $b_0, b_1, ..., b_{n-1}$, but it can't be continued, because $b_{n-1}+1 = 0$ so there is no b_n for which $(b_{n-1}+1)(b_n-1) = -1$.

Therefore the longest proper sequence (b_k) is n-long, so the longest proper sequence (a_k) is $n+1$ long, so $N = n$.

Solution 2. The required maximum is $N = n$.

To rule out the case $N \geq n+1$, it is clearly sufficient to rule out the case $N = n+1$.

Assume for contradiction that $a_0, a_1, \ldots, a_{n+1}$ are real numbers satisfying both conditions in the statement. It is sufficient to show that $a_k + a_{k+1} = 0$ for some $k \leq n$, because then $a_{k-1} - a_{k+1} = 0$ so $a_{k+1} = a_{k-1}$, therefore $a_{k-1} + a_k = 0$ and so on, by backwards recursion we get that $a_j + a_{j+1} = 0$ for all $0 \leq j \leq k$, but this is a contradiction with $a_0 + a_1 = -\frac{1}{n}$.

To prove that $a_k + a_{k+1} = 0$ for some $k \leq n$, assume that $a_k + a_{k+1} \neq 0$ for all $k \leq n$, to rewrite the second condition in the statement in the form

$$\frac{1}{a_k + a_{k+1}} - \frac{1}{a_{k-1} + a_k} = 1, \quad k = 1, \ldots, n.$$

and sum both sides over the full range from $k = 1$ to n. This gives

$$\frac{1}{a_n + a_{n+1}} - \frac{1}{a_0 + a_1} = n.$$

As $a_0 + a_1 = -\frac{1}{n}$, this means that $\frac{1}{a_n+a_{n+1}} = 0$, which is a contradiction. Consequently, $N \leq n$.

To provide $n+1$ real numbers satisfying both conditions in the statement, fix a_0 and go through the telescoping procedure above to obtain

$$a_k = (-1)^k a_0 + \sum_{j=1}^{k} \frac{(-1)^{k-j+1}}{n-j+1}, \quad k = 1, \ldots, n.$$

This concludes the proof.

Problem 5.
For all positive integers n, k, let $f(n, 2k)$ be the number of ways an $n \times 2k$ board can be fully covered by nk dominoes of size 2×1. (For example, $f(2, 2) = 2$ and $f(3, 2) = 3$.)
Find all positive integers n such that for every positive integer k, the number $f(n, 2k)$ is odd.

Proposed by: U.S.A.

Solution. The integers n with the requested property are exactly the ones of the form $2^k - 1$.

In what follows, let $f(m, n)$ denote the number of domino tilings of an $m \times n$ grid. (For convenience, we also allow m and n to be 0, in which case $f(m, n) = 1$.)

Claim. $f(m, 2n + 1) \equiv f(m, n) \pmod 2$ for all n and even m

Proof. Consider reflecting $m \times (2n + 1)$ tilings across the central column. In this way, the tilings are grouped into singletons and pairs, so modulo 2, $f(m, 2n+1)$ is congruent to the number of singletons.

If a tiling is invariant under reflection in the central column, then every domino with one cell in that column must have both cells in that column.

In other words, the central column is filled with $\frac{m}{2}$ vertical dominoes, splitting the remainder of the grid into two $m \times n$ grids. Obeying the symmetry constraint, there are $f(m, n)$ tilings: each of the tilings of the left grid fixes the tiling of the right grid.

Claim. $f(n, n) \equiv 0 \pmod 2$ for all even $n \geq 2$. (Recall that $f(0, 0) = 1$ is odd.)

Proof. Consider reflecting $n \times n$ tilings across the diagonal. This groups the tilings into pairs, (no tiling is grouped with itself). Hence the number of $n \times n$ tilings is even.

We are ready to complete the solution.

- If n is odd, the first claim shows that n satisfies the property if and only if $\frac{1}{2}(n - 1)$ does.
- If $n \geq 2$ is even, the second claim shows that n does not satisfy the property.
- If $n = 0$, then n satisfies the property, as $f(m, 0) = 1$ always.

This concludes the proof that the sought numbers are the ones of the form $2^k - 1$.

Solution 2.

Color the board as a chessboard. Consider the bipartite graph whose vertices are the squares and the neighbors are connected by an edge. Notice that a domino tiling described in the problem corresponds to a perfect matching in this bipartite graph, so we are interested in the number of perfect matchings.

And that is the permanent of the (bipartite) adjacency matrix. However we need to compute it mod 2 (over F_2), so that is the determinant of that matrix.

So our goal is to decide whether the determinant is 0 or 1 mod 2, or in other world, that matrix is singular or not. From now we compute everything mod 2. Now we understand what does it mean, that a vector v is in the nullspace (has eigenvalue 0). But each entry of v is $0 - 1$, so in other words we choose a subset of the black squares (v is the characteristic vector of that subset). So in the language of the board, there is a nontrivial v in the nullspace iff we have a subspace of black square such a way that each white square has even number of black chosen neighbors.

If n is even, then we can choose a subset such a way in the $n \times n$ square: the diagonal.

Now we prove similar connection between n and $2n + 1$: there is a construction for n iff we have one for $2n + 1$. k is always even. If we have a construction in $n \times k$ then we have one in $(2n + 1) \times k$: we mirror the construction to the middle row.

Now we prove that from a construction in $(2n+1) \times k$ we can make a construction in $n \times k$. First, we

don't have any chosen black square in the middle row. Then the chosen black squares in the upper (or lower) $n \times k$ rectangle is a construction for $n \times k$. Second, we have some chosen black square in the middle row. Because we can find a white square in the middle row such that is has only one chosen black neighbor in the middle row. Thus this construction isn't symmetric to the horizontal line the symmetric difference of the construction and its reflection (to the vertical line) is a nontrivial vector in the nullspace, thus we can use the first case.

Problem 6.
Let $ABCD$ be a cyclic quadrilateral with circumcentre O. Let the internal angle bisectors at A and B meet at X, the internal angle bisectors at B and C meet at Y, the internal angle bisectors at C and D meet at Z, and the internal angle bisectors at D and A meet at W. Further, let AC and BD meet at P. Suppose that the points X, Y, Z, W, O and P are distinct.

Prove that O, X, Y, Z and W lie on the same circle if and only if P, X, Y, Z and W lie on the same circle.

Proposed by: Ethan Tan, Australia

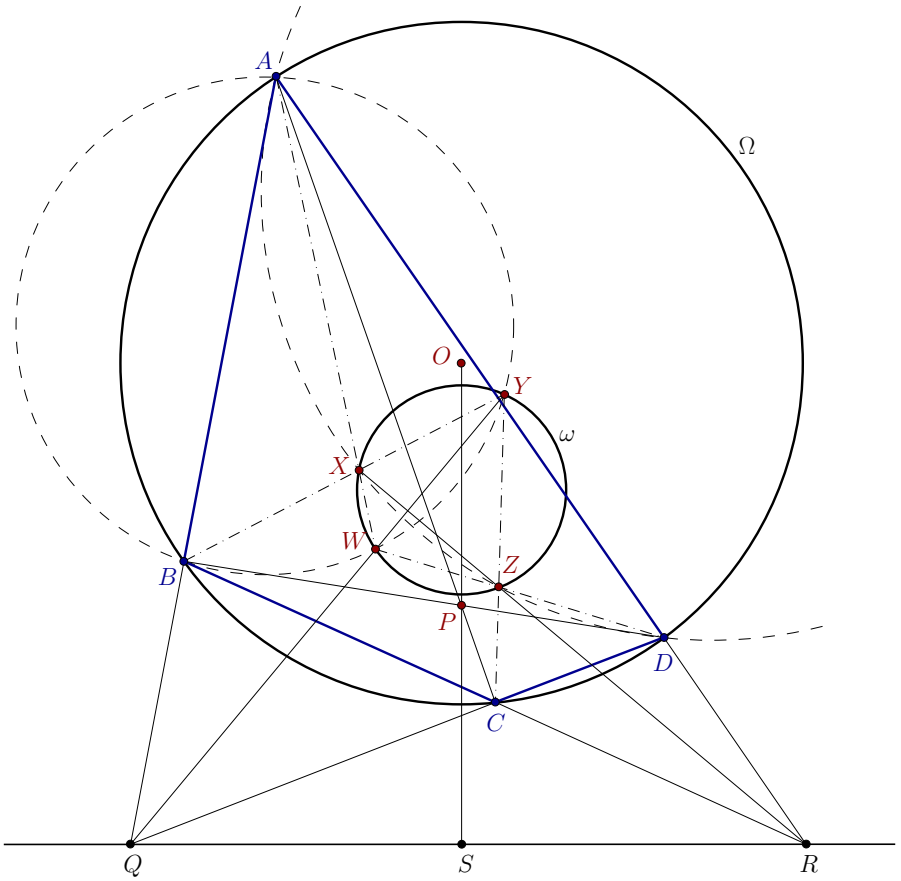
Solution 1. Let Ω be the circumcircle of the quadrilateral $ABCD$ and let r be its radius.

First, notice that the points X, Y, Z, W are concyclic. Indeed, using oriented (modulo 180°) angles,

$$\angle(XW, XY) + \angle(ZY, ZW) = \angle(XA, XB) + \angle(ZC, ZD) = -\frac{\angle A + \angle B}{2} - \frac{\angle C + \angle D}{2} = 0.$$

Let ω be the circle passing through these four points. Our goal is to prove that $O \in \omega$ occurs if and only if $P \in \omega$.

Next, we rule out the case when $ABCD$ is a trapezium. Suppose that if $ABCD$ is an isosceles trapezium; without loss of generality say $AB \parallel CD$. By symmetry, points X, Z, O and P lie on the symmetry axis (that is, the common perpendicular bisector of AB and CD), therefore they are collinear. By the conditions of the problem, these four points are distinct and X and Z are on ω, so neither O, nor P can lie on ω; therefore the problem statement is obvious. From now on we assume that $AB \nparallel CD$ and $BC \nparallel AD$.



Let AB and CD meet at Q, and let BC and AD meet at R. Without loss of generality, suppose that B lies between A and Q, and D lies between A and R. Now we show that line QR is the radical axis between circles Ω and ω.

Point W is the intersection point of the bisectors of $\angle A$ and $\angle D$, so in triangle ADQ, point W is the incentre. Similary, BY and CY are the external bisectors of $\angle QBC$ and $\angle BCQ$, so in triangle BCQ, point Y is the excentre opposite to Q. Hence, both Y and W lie on the bisector of $\angle DQA$.

By $\angle DQA = 180^\circ - \angle D - \angle A = \angle B - \angle A$ we get

$$\angle BYQ = \angle YBA - \angle YQB = \frac{\angle B}{2} - \frac{\angle DQA}{2} = \frac{\angle B}{2} - \frac{\angle B - \angle A}{2} = \frac{\angle A}{2} = \angle BAW,$$

so the points A, B, Y, W are concyclic, and therefore $QA \cdot QB = QY \cdot QW$; hence, Q has equal power with respect to Ω and ω. It can be seen similarly that R has equal power with respect to the circles Ω and ω. Hence, the radical axis of Ω and ω is line QR.

Let lines OP and QR meet at point S. It is well-known that with respect to circle $ABCD$, the diagonal triangle PQR is autopolar. As consequences we have $OP \perp QR$, and the points P and S are symmetric to circle $ABCD$, so $OS \cdot OP = r^2$.

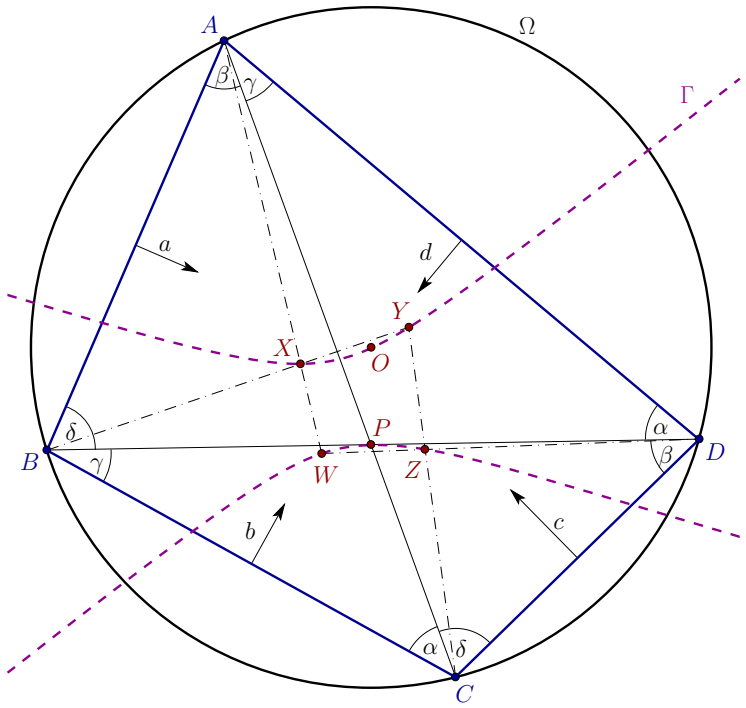
Notice that P and O lie inside Ω, so the polar line QR lies entirely outside, so S is different from P and O. Moreover,

$$SO \cdot SP = OS \cdot (OS - OP) = OS^2 - OS \cdot OP = SO^2 - r^2,$$

so $SO \cdot SP$ is equal to the power of S with respect to Ω. Since S lies on the radical axis QR, it has equal power with respect to the two circles; therefore, $SO \cdot SP$ is equal to the power of S with respect to ω. From this, it follows that $O \in \omega$ if and only if $P \in \omega$.

Solution 2. We will prove that the points X, Y, Z, W, O, P lie on a conic section Γ. Five distinct points uniquely determine a conic section, therefore

$$X, Y, Z, W, O \text{ are concyclic} \quad \Leftrightarrow \quad \Gamma \text{ is a circle} \quad \Leftrightarrow \quad X, Y, Z, W, P \text{ are concyclic.}$$



Let $\alpha = \angle ACB = \angle ADB$, $\beta = \angle BDC = \angle BAC$, $\gamma = \angle CAD = \angle CBD$ and $\delta = \angle DBA = \angle DCA$, where of course $\alpha+\beta+\gamma+\delta = 180°$. For every point ξ in the plane, let $a(\xi)$, $b(\xi)$, $c(\xi)$ and $d(\xi)$ be the signed distances between ξ and the lines AB, BC, CD and DA, respectively, such that the quadrilateral lies on the positive sides of these lines.

The equations of the bisectors of $\angle AXW$, $\angle BXY$, $\angle CYZ$ and $\angle DZW$ are $a(\xi) - d(\xi) = 0$, $b(\xi) - a(\xi) = 0$, $c(\xi) - b(\xi) = 0$ and $d(\xi) - c(\xi) = 0$, respectively. Notice that the points X, Y, Z, W satisfy the quadratic equations $\big(a(\xi) - d(\xi)\big)\big(b(\xi) - c(\xi)\big) = 0$ and $\big(a(\xi) - b(\xi)\big)\big(c(\xi) - d(\xi)\big) = 0$, so

$$a(\xi)b(\xi) + c(\xi)d(\xi) = a(\xi)c(\xi) + b(\xi)d(\xi) = a(\xi)d(\xi) + b(\xi)c(\xi) \quad \text{for } \xi = X, Y, Z, W. \tag{1}$$

Without loss of generality, suppose that $ABCD$ is inscribed in a unit circle. Then

$$a(O) = \cos\alpha, \quad b(O) = \cos\beta, \quad c(O) = \cos\gamma, \quad d(O) = \cos\delta. \tag{2a}$$

By $\dfrac{a(P)}{b(P)} = \dfrac{BP\cdot\sin\delta}{BP\cdot\sin\gamma} = \dfrac{1/\sin\gamma}{1/\sin\delta}$ and the analogous relations we can see that

$$a(P) = \frac{k}{\sin\gamma}, \quad b(P) = \frac{k}{\sin\delta}, \quad c(P) = \frac{k}{\sin\alpha}, \quad d(P) = \frac{k}{\sin\beta} \tag{2b}$$

with some positive number k.

Now let $s = \dfrac{\sin\alpha\sin\beta\sin\gamma\sin\delta}{k^2}$; then, by (2$a$) and (2$b$),

$$\begin{aligned} a(O)b(O) - s\cdot a(P)b(P) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta = \cos(\alpha+\beta) \\ c(O)d(O) - s\cdot c(P)d(P) &= \cos(\gamma+\delta) = -\cos(\alpha+\beta) \\ a(O)b(O) + c(O)d(O) &= s\cdot\big(a(P)b(P) + c(P)d(P)\big); \end{aligned} \tag{3a}$$

analogous calculation provides

$$a(O)c(O) + b(O)d(O) = s\cdot\big(a(P)c(P) + b(P)d(P)\big) \tag{3b}$$

$$a(O)d(O) + b(O)c(O) = s\cdot\big(a(P)d(P) + b(P)c(P)\big). \tag{3c}$$

In order to find the equation of the curve Γ, choose real numbers u, v, w, not all zero, such that

$$u + v + w = 0; \tag{4}$$

$$u\cdot\big(a(P)b(P) + c(P)d(P)\big) + v\cdot\big(a(P)c(P) + b(P)d(P)\big) + w\cdot\big(a(P)d(P) + b(P)c(P)\big) = 0. \tag{5}$$

This is always possible, because this a system of two homogeneous linear equations with three variables. Then the equation of Γ will be

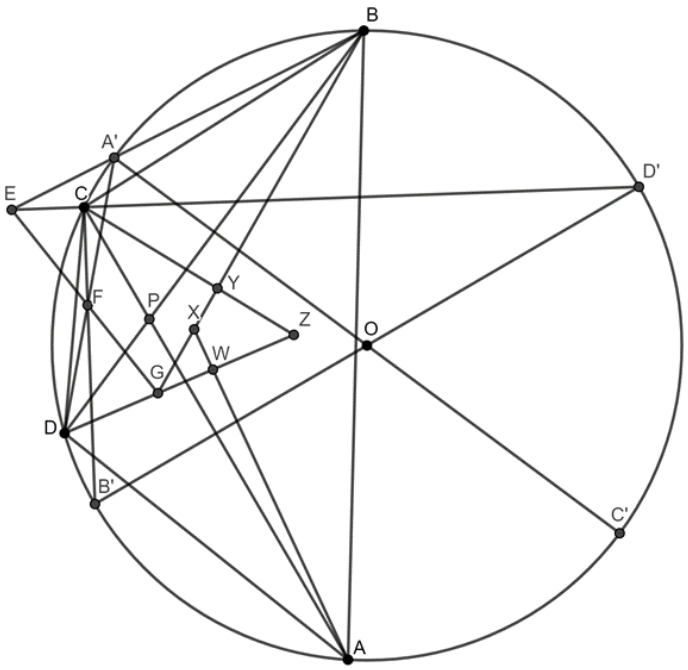
$$f(\xi) = u\cdot\big(a(\xi)b(\xi) + c(\xi)d(\xi)\big) + v\cdot\big(a(\xi)c(\xi) + b(\xi)d(\xi)\big) + w\cdot\big(a(\xi)d(\xi) + b(\xi)c(\xi)\big) = 0. \tag{$*$}$$

As can be seen, $f(X) = f(Y) = f(Z) = f(W) = 0$ follows from (1) and (4), $f(P) = 0$ follows from (5), then $f(O) = s\cdot f(P) = 0$ follows from (3a − 3c). So, the points X, Y, Z, W, O, P all satisfy the equation $f(\xi) = 0$.

Notice that $f(B) = u\cdot c(B)d(B)$ and $f(A) = w\cdot b(A)c(A)$; since at least one of u and w is nonzero, either A or B does not satisfy ($*$). Therefore, the equation cannot degenerate to an identity.

Hence, the equation $f(\xi) = 0$ is at most quadratic, it is not an identity, but satisfied by X, Y, Z, W, O, P, so these six points lie on a conic section.

Solution 3. Let A', B', C' and D' be the second intersection of the angle bisectors of $\angle DAC$, $\angle ABC$, $\angle BCD$ and $\angle CDA$ with the circumcircle of $ABCD$, respectively. Notice that A' and C' are the midpoints of the two arcs of BD and B' and D' are the midpoints of arcs corresponding to AC so the lines $A'C'$ and $B'D'$ intersect in O.



We will prove that the points X, Y, Z, W, P and O lie on a conic section. This is a more general statement because five points uniquely determine a conic sections so if X, Y, Z, W and P lie on a circle then this circle is the conic section so O also lies on this circle. In the same way if X, Y, Z, W and O are on a circle then it also contains P.

Let E be the intersection of the lines $A'B$ and CD'. Let F be the intersection of the lines $A'D$ and $B'C$. Finally, let G be the intesection of XY and ZW.

Now we use Pascal's theorem five times:

1. From $BDD'CAA'$ we see that the points P, W and E are collinear.
2. From $BB'D'CC'A'$ we see that the points Y, O and E are collinear.
3. From $B'BDA'AC$ we see that the points X, P and F are collinear.
4. From $B'D'DA'C'C$ we see that the points O, Z and F are collinear.
5. From $D'CB'BA'D$ we see that the points E, F and G are collinear.

So the opposite sides of the hexagon $PXYOZW$ intersect at points F, G and E, which are collinear, so by the converse of Pascal's theorem the points X, Y, Z, W, P and O indeed lie on a conic section.

Solutions of EGMO 2021

Problem 1. According to Anna, the number 2021 is *fantabulous*. She states that if any element of the set $\{m, 2m+1, 3m\}$ is fantabulous for a positive integer m, then they are all fantabulous. Is the number 2021^{2021} fantabulous?

(Australia, Angelo Di Pasquale)

Answer: Yes

Solution 1.

Consider the sequence of positive integers $m, 3m, 6m+1, 12m+3, 4m+1, 2m$. Since each number in the sequence is fantabulous if and only if the next one is, we deduce that m is fantabulous if and only if $2m$ is fantabulous.

Combined with the fact that m is fantabulous if and only if $2m+1$ is fantabulous, this implies that $m > 1$ is fantabulous if and only if $f(m) = \left[\frac{m}{2}\right]$ is fantabulous. We can apply f sufficiently many times to any positive integer n to conclude that n is fantabulous if and only if 1 is fantabulous. Therefore, the fact that 2021 is fantabulous implies that 1 is fantabulous, which in turn implies that 2021^{2021} is fantabulous.

Solution 2.

Let $m > 1$ be a fantabulous number. Note that at least one of the following four cases must hold.

■ *Case 1.* The number m is odd;
We have $m = 2a+1$ for some positive integer a, so $a < m$ is also fantabulous.

■ *Case 2.* The number m is a multiple of 3;
We have $m = 3a$ for some positive integer a, so $a < m$ is also fantabulous.

■ *Case 3.* The number m is 4 modulo 6;
We have $m = 6a-2$ for some positive integer a. We have the sequence of fantabulous numbers

$$(6a-2) \to (12a-3) \to (4a-1),$$

so $4a-1 < m$ is also fantabulous.

■ *Case 4.* The number m is 2 modulo 6;
We have $m = 6a + 2$ for some positive integer a. We have the sequence of fantabulous numbers

$$(6a + 2) \to (12a + 5) \to (36a + 15) \to (18a + 7) \to (9a + 3) \to (3a + 1),$$

so $3a + 1 < m$ is also fantabulous.

In all cases, we see that there is another fantabulous number less than m. Since 2021 is fantabulous, it follows that 1 is fantabulous.

Observe that a number m is not fantabulous if and only if all of the elements of the set $\{m, 2m + 1, 3m\}$ are not fantabulous. So, the argument above shows that if there exists a positive integer that is not fantabulous, then 1 would not be fantabulous either. This is a contradiction, so all positive integers are fantabulous and, in particular, 2021^{2021} is fantabulous.

Solution 3.

The following transformations show that a is fantabulous if and only if $3a$, $3a + 1$ or $3a + 2$ are fantabulous.

$a \to 3a$
$a \to 2a + 1 \to 6a + 3 \to 3a + 1$
$a \to 2a + 1 \to 4a + 3 \to 12a + 9 \to 36a + 27 \to 18a + 13 \to 9a + 6 \to 3a + 2$

This implies that $a \geq 3$ is fantabulous if and only if $f(a) = \left[\frac{a}{3}\right]$ is fantabulous. We can use this to deduce that 1 and 2 are fantabulous from the fact that 2021 is fantabulous in the following way:

$$2021 \to 673 \to 224 \to 74 \to 24 \to 8 \to 2 \to 5 \to 1$$

We can apply f sufficiently many times to any positive integer n to arrive at the number 1 or 2. It follows that every positive integer is fantabulous, so 2021^{2021} is fantabulous.

Problem 2. Find all functions $f: \mathbb{Q} \to \mathbb{Q}$ such that the equation

$$f(xf(x) + y) = f(y) + x^2$$

holds for all rational numbers x and y.
Here, $\mathbb{Q}$ denotes the set of rational numbers.

(Slovakia, Patrik Bak)

Answer: $f(x) = x$ and $f(x) = -x$.

Solution. Denote the equation from the statement by (1). Let $xf(x) = A$ and $x^2 = B$. The equation (1) is of the form

$$f(A + y) = f(y) + B$$

Also, if we put $y \to -A + y$, we have $f(A - A + y) = f(-A + y) + B$. Therefore

$$f(-A + y) = f(y) - B$$

We can easily show that for any integer n we even have

$$f(nA + y) = f(y) + nB \tag{2}$$

Indeed, it's trivially true for $n = 0$ and if this holds true for some integer n, then

$$f\big((n+1)A + y\big) = f(A + y + nA) = f(ny + A) + B = f(y) + nB + B = f(y) + (n+1)B$$

and

$$f\big((n-1)A + y\big) = f(-A + nA + y) = f(nA + y) - B = f(y) + nB - B = f(y) + (n-1)B.$$

So, equation (2) follows from the induction on n.

Now we can say that for any integer k it holds

$$f(nxf(x) + y) = f(y) + nx^2 \tag{3}$$

If y is given, then $f(y) + nx^2$ can be any rational number, since nx^2 can be any rational number. If it is supposed to be $\frac{p}{q}$, where $q \neq 0$, then we may take $n = pq$, and $x = \frac{1}{q}$. Therefore f is surjective on $\mathbb{Q}$. So there's a rational number c such that $f(c) = 0$. Be putting $x = c$ into (1) we immediately get $c = 0$, i.e. $f(0) = 0$. Therefore, $f(x) = 0$ if and only if $x = 0$.

For any integer n and for any rational x, y it holds

$$f(n^2xf(x) + y) = f(y) + n^2x^2 = f(y) + (nx)^2 = f(nxf(nx) + y) \tag{4}$$

After taking $y = -nxf(nx)$ in (4), the right-hand side becomes 0, therefore

$$n^2xf(x) - nxf(nx) = 0.$$

This simplifies into $nf(x) = f(nx)$ for $x \neq 0$, but it also holds for $x = 0$. Therefore, for any rational number $x = \frac{p}{q}$ we have,

$$f(x) = f\left(\frac{p}{q}\right) = f\left(p \cdot \frac{1}{q}\right) = p \cdot f\left(\frac{1}{p}\right) = p \cdot \frac{f\left(q \cdot \frac{1}{q}\right)}{q} = \frac{p}{q} \cdot f(1) = xf(1)$$

So, we have $f(x) = kx$, for some rational number k. Let's put this answer in (1) and we get $k(xkx + y) = ky + x^2$, thus $k^2 = 1$. Therefore $f(x) = x$ and $f(x) = -x$ are solutions.

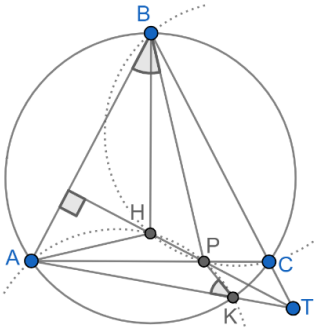
Problem 3. Let ABC be a triangle with an obtuse angle at A. Let E and F be the intersections of the external bisector of angle A with the altitudes of ABC through B and C respectively. Let M and N be the points on the segments EC and FB respectively such that $\angle EMA = \angle BCA$ and $\angle ANF = \angle ABC$. Prove that the points E, F, N, M lie on a circle.

(Ukraine, Anton Trygub)

Solution 1.

The first solution is based on the main Lemma. We present this Lemma with two different proofs.

Lemma: Let ABC be an acute triangle with $AB = BC$. Let P be any point on AC. Line passing through P perpendicular to AB, intersects ray BC in point T. If the line AT intersects the circumscribed circle of the triangle ABC the second time at point K, then $\angle AKP = \angle ABP$.



Proof 1:

Let H be the orthocenter of the triangle ABP. Then

$$\angle BHP = 180^0 - \angle BAC = 180^0 - \angle BCP.$$

So $BHPC$ is cyclic. Then we get

$$TK \cdot TA = TC \cdot TB = TP \cdot TH.$$

So, $AHPK$ is also cyclic. But then

$$\angle AKP = 180^0 - \angle AHP = \angle ABP.$$

Proof 2:

Consider the symmetric points B' and C' of B and C, respectively, with respect to the line PT. It is clear that

$$TC' \cdot TB' = TC \cdot TB = TK \cdot TA.$$

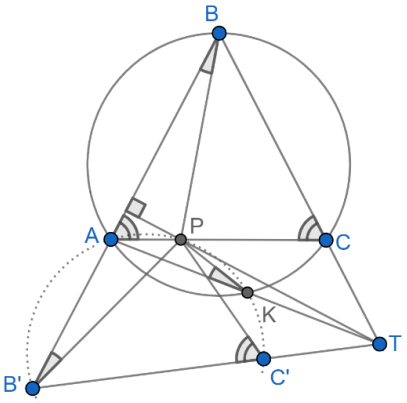
So $B'C'KA$ is cyclic. Also, because of the symmetry we have

$$\angle PC'B' = \angle PCB = \angle PAB.$$

So $B'C'PA$ is also cyclic. Therefore, the points B', C', K, P and A all lie on the common circle. Because of this fact and because of the symmetry again we have

$$\angle PKA = \angle PB'A = \angle PBA.$$

So, lemma is proved and now return to the problem.



Let H be intersection point of the altitudes at B and C. Denote by M' and N' the intersection points of the circumcircle of the triangle HEF with the segments EC and FB, respectively. We are going to show that $M = M'$ and $N = N'$ and it will prove the points E, F, N, M lie on a common circle.

Of course, A is an orthocenter of the triangle BCH. Therefore $\angle BHA = \angle BCA$, $\angle CHA = \angle CBA$ and $\angle HBA = \angle HCA$. Thus

$$\angle HEF = \angle HBA + \angle EAB = \angle HCA + \angle FAC = \angle HFE.$$

So, the triangle HEF is isosceles, $HE = HF$.

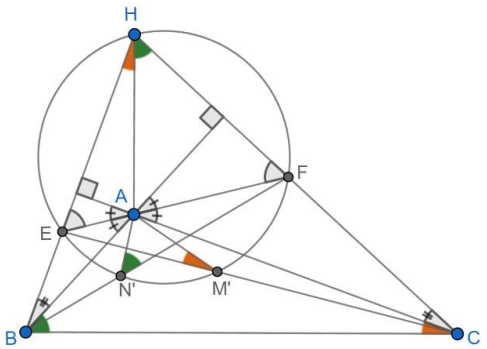
By using lemma, we get

$$\angle AM'E = \angle AHE = \angle ACB,$$

and

$$\angle AN'F = \angle AHF = \angle ABC.$$

Therefore $M = M'$ and $N = N'$ and we are done.



Solution 2.

Let X, Y be projections of B on AC, and C on AB, respectively. Let ω be circumcircle of $BXYC$. Let Z be intersection of EC and ω and D be projection of E on BA.

$$\angle MAC = \angle AME - \angle MCA = \angle XCB - \angle XCE = \angle ZCB = \angle ZXB$$

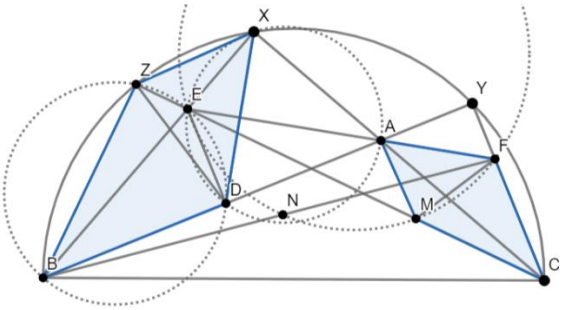
Since $BXYC$ is cyclic $\angle ACY = \angle XBA$, and since $DEXA$ is cyclic

$$\angle EXD = \angle EAD = \angle FAC.$$

Therefore, we get that the quadrangles $BZXD$ and $CMAF$ are similar. Hence $\angle FMC = \angle DZB$. Since $ZEDB$ is cyclic,

$$\angle DZB = \angle DEB = \angle XAB.$$

Thus $\angle FMC = \angle XAB$. Similarly, $\angle ENB = \angle YAC$. We get that $\angle FMC = \angle ENB$ and it implies that the points E, F, N, M lie on a circle.



Problem 4. Let ABC be a triangle with incentre I and let D be an arbitrary point on the side BC. Let the line through D perpendicular to BI intersect CI at E. Let the line through D perpendicular to CI intersect BI at F. Prove that the reflection of A in the line EF lies on the line BC.

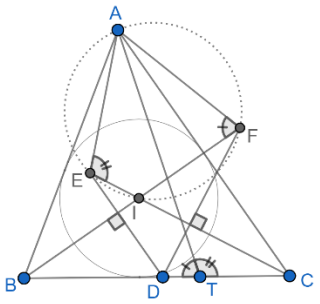
(Australia, Sampson Wong)

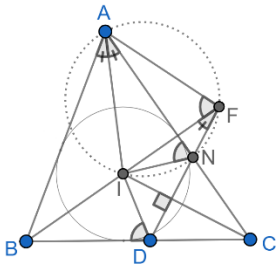
Solution 1.

Let us consider the case when I lies inside of triangle EFD. For the other cases the proof is almost the same only with the slight difference.

We are going to prove that the intersection point of the circumcircles of AEC and AFB (denote it by T) lies on the line BC and this point is the symmetric point of A with respect to EF. First of all we prove that $AEIF$ is cyclic, which implies that T lies on the line BC, because

$$\angle ATB + \angle ATC = \angle AFB + \angle AEC = \angle AFI + \angle AEI = 180^0.$$





Denote by N an intersection point of the lines DF and AC. Of course N is the symmetric point with respect to CI. Thus, $\angle INA = \angle IDB$. Also,

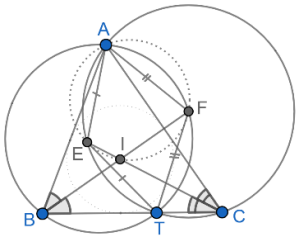
$$\angle IFD = \angle NDC - \angle IBC = 90^0 - \angle ICB - \angle IBC = \angle IAN.$$

So, we get that A, I, N and F lie on a common circle. Therefore, we have $\angle AFI = \angle INA = \angle IDB$. Analogously, $\angle AEI = \angle IDC$ and we have

$$\angle AFI + \angle AEI = \angle IDB + \angle IDC$$

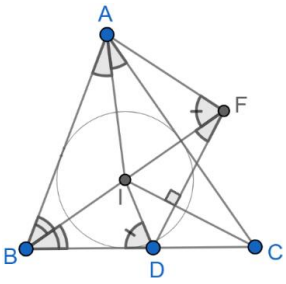
So $\angle AFI + \angle AEI = 180^0$, thus $AEIF$ is cyclic and T lies on the line BC.

Because EC bisects the angle ACB and $AETC$ is cyclic we get $EA = ET$. Because of the similar reasons we have $FA = FT$. Therefore T is the symmetric point of A with respect to the line EF and it lies on the line BC.



Solution 2.

Like to the first solution, consider the case when I lies inside of triangle EFD. we need to prove that $AEIF$ is cyclic. The finish of the proof is the same.



first note that $\triangle FDB \sim \triangle AIB$, because $\angle FBD = \angle ABI$, and

$$\angle BFD = \angle FDC - \angle IBC = 90^0 - \angle ICD - \angle IBC = \angle IAB.$$

Because of the similarity we have $\frac{AB}{AF} = \frac{BI}{BD}$. This equality of the length ratios with $\angle IBD = \angle ABF$ implies that $\triangle ABF \sim \triangle IBD$. Therefore, we have $\angle IDB = \angle AFB$. Analogously, we can get $\angle IDC = \angle AEC$, thus

$$\angle AFI + \angle AEI = \angle IDB + \angle IDC = 180^0.$$

So, $AEIF$ is cyclic and we are done.

Problems 5. A plane has a special point O called the origin. Let P be a set of 2021 points in the plane, such that

(i) no three points in P lie on a line and
(ii) no two points in P lie on a line through the origin.

A triangle with vertices in P is *fat*, if O is strictly inside the triangle. Find the maximum number of *fat* triangles.

(Austria, Veronika Schreitter)

Answer: $2021 \cdot 505 \cdot 337$

Solution

We will count minimal number of triangles that are not fat. Let F set of fat triangles, and S set of triangles that are not fat. If triangle $XYZ \in S$, we call X and Z *good* vertices if OY is located between OX and OZ. For $A \in P$ let $S_A \subseteq S$ be set of triangles in S for which A is one of the good vertex.

It is easy to see that

$$2|S| = \sum_{A \in P} |S_A| \qquad (1)$$

For $A \in P$, let $R_A \subset P$ and $L_A \subset P$ be parts of $P\backslash\{A\}$ divided by AO. Suppose for $AXY \in S$ vertex A is good, then clearly $X, Y \in R_A$ or $X, Y \in L_A$. On the other hand, if $X, Y \in R_A$ or $X, Y \in L_A$ then clearly $AXY \in S$ and A is its good vertex. Therefore,

$$|S_A| = \binom{|R_A|}{2} + \binom{|L_A|}{2} \tag{2}$$

It is easy to show following identity:

$$\frac{x(x-1)}{2} + \frac{y(y-1)}{2} - 2 \cdot \frac{\frac{x+y}{2}\left(\frac{x+y}{2} - 1\right)}{2} = \frac{(x-y)^2}{4} \tag{3}$$

By using (2) and (3) we get

$$|S_A| \geq 2 \cdot \binom{\frac{|R_A| + |L_A|}{2}}{2} = 2 \cdot \binom{1010}{2} = 1010 \cdot 1009 \tag{4}$$

and the equality holds when $|R_A| = |L_A| = 1010$. Hence

$$|S| = \frac{\sum_{A \in P} |S_A|}{2} \geq \frac{2021 \cdot 1010 \cdot 1009}{2} = 2021 \cdot 505 \cdot 1009. \tag{5}$$

Therefore,

$$|F| = \binom{2021}{3} - |S| \leq 2021 \cdot 1010 \cdot 673 - 2021 \cdot 505 \cdot 1009 = 2021 \cdot 505 \cdot 337. \tag{6}$$

For configuration of points on regular 2021-gon which is centered at O, inequalities in (4), (5), (6) become equalities. Hence $2021 \cdot 505 \cdot 337$ is indeed the answer.

Problems 6. Does there exist a nonnegative integer a for which the equation

$$\left\lfloor \frac{m}{1} \right\rfloor + \left\lfloor \frac{m}{2} \right\rfloor + \left\lfloor \frac{m}{3} \right\rfloor + \cdots + \left\lfloor \frac{m}{m} \right\rfloor = n^2 + a$$

has more than one million different solutions (m, n) where m and n are positive integers?
(*The expression* $\lfloor x \rfloor$ *denotes the integer part (or floor) of the real number* x. *Thus* $\left\lfloor \sqrt{2} \right\rfloor = 1, \lfloor \pi \rfloor = \left\lfloor \frac{22}{7} \right\rfloor = 3, \lfloor 42 \rfloor = 42$ *and* $\lfloor 0 \rfloor = 0$)

(Austria, Veronika Schreitter)

Answer: Yes.

Solution.

Denote the equation from the statement by (1). The left hand side of (1) depends only on m, and will throughout be denoted by $L(m)$. Fix an integer $q > 10^7$ and note that for $m = q^3$

$$L(q^3) = \sum_{k=1}^{q^3} \left[\frac{q^3}{k}\right] \le \sum_{k=1}^{q^3} \frac{q^3}{k} \le q^3 \cdot \sum_{k=1}^{q^3} \frac{1}{k} \le q^3 \cdot q = q^4. \quad (2)$$

Indeed, the first inequality results from $[x] \le x$. The second inequality can be seen (for instance) as follows. We divide the terms in the sum $\sum_{k=1}^{q^3} \frac{1}{k}$ into several groups: For $j \ge 0$, the j-th group contains the 2^j consecutive terms $\frac{1}{2^j}, \ldots, \frac{1}{2^{j+1}-1}$. Since every term in the j-th group is bounded by $\frac{1}{2^j}$, the overall contribution of the j-th group to the sum is at most 1. Since the first q groups together would contain $2^q - 1 > q^3$ terms, the number of groups does not exceed q, and hence the value of the sum under consideration is indeed bounded by q.

Call an integer m *special,* if it satisfies $1 \le L(m) \le q^4$. Denote by $g(m) \ge 1$ the largest integer whose square is bounded by $L(m)$; in other words $g^2(m) \le L(m) < (g(m) + 1)^2$. Note that $g(m) \le q^2$ for all special m, which implies

$$0 \le L(m) - g^2(m) < (g(m) + 1)^2 - g^2(m) = 2g(m) + 1 \le 2q^2 + 1. \quad (3)$$

Finally, we do some counting. Inequality (2) and the monotonicity of $L(m)$ imply that there exist at least q^3 special integers. Because of (3), every special integer m has $0 \le L(m) - g^2(m) \le 2q^2 + 1$. By averaging, at least $\frac{q^3}{2q^2+2} > 10^6$ special integers must yield the same value $L(m) - g^2(m)$. This frequently occurring value is our choice for α, which yields more than 10^6 solutions $\big(m, g(m)\big)$ to equation (1). Hence, the answer to the problem is YES.

Solutions of EGMO 2020

Problem 1. The positive integers a_0, a_1, a_2, ..., a_{3030} satisfy

$$2a_{n+2} = a_{n+1} + 4a_n \quad \text{for } n = 0, 1, 2, \ldots, 3028.$$

Prove that at least one of the numbers a_0, a_1, a_2, ..., a_{3030} is divisible by 2^{2020}.

Problem 2. Find all lists $(x_1, x_2, \ldots, x_{2020})$ of non-negative real numbers such that the following three conditions are all satisfied:

(i) $x_1 \leq x_2 \leq \ldots \leq x_{2020}$;

(ii) $x_{2020} \leq x_1 + 1$;

(iii) there is a permutation $(y_1, y_2, \ldots, y_{2020})$ of $(x_1, x_2, \ldots, x_{2020})$ such that

$$\sum_{i=1}^{2020} \big((x_i + 1)(y_i + 1)\big)^2 = 8 \sum_{i=1}^{2020} x_i^3.$$

A permutation of a list is a list of the same length, with the same entries, but the entries are allowed to be in any order. For example, $(2, 1, 2)$ is a permutation of $(1, 2, 2)$, and they are both permutations of $(2, 2, 1)$. Note that any list is a permutation of itself.

Problem 3. Let $ABCDEF$ be a convex hexagon such that $\angle A = \angle C = \angle E$ and $\angle B = \angle D = \angle F$ and the (interior) angle bisectors of $\angle A$, $\angle C$, and $\angle E$ are concurrent.

Prove that the (interior) angle bisectors of $\angle B$, $\angle D$, and $\angle F$ must also be concurrent.

Note that $\angle A = \angle FAB$. The other interior angles of the hexagon are similarly described.

Problem 4. A permutation of the integers 1, 2, ..., m is called *fresh* if there exists no positive integer $k < m$ such that the first k numbers in the permutation are 1, 2, ..., k in some order. Let f_m be the number of fresh permutations of the integers 1, 2, ..., m.

Prove that $f_n \geq n \cdot f_{n-1}$ for all $n \geq 3$.

For example, if $m = 4$, then the permutation $(3, 1, 4, 2)$ is fresh, whereas the permutation $(2, 3, 1, 4)$ is not.

Problem 5. Consider the triangle ABC with $\angle BCA > 90°$. The circumcircle Γ of ABC has radius R. There is a point P in the interior of the line segment AB such that $PB = PC$ and the length of PA is R. The perpendicular bisector of PB intersects Γ at the points D and E.

Prove that P is the incentre of triangle CDE.

Problem 6. Let $m > 1$ be an integer. A sequence $a_1, a_2, a_3, \ldots$ is defined by $a_1 = a_2 = 1$, $a_3 = 4$, and for all $n \geq 4$,

$$a_n = m(a_{n-1} + a_{n-2}) - a_{n-3}.$$

Determine all integers m such that every term of the sequence is a square.

Solutions to Problem 1

There are different ways of solving the problem. All of these use some induction argument. Most of these proofs use one of the following two lemmas. In many places, they can be used interchangeably.

Lemma. If a, b, c, d are integers with $2c = b + 4a$ and $2d = c + 4b$, then $4 \mid b$.

Proof: From $2d = c + 4b$ we have that c is even, and then from $2c = b + 4a$ it follows that b is divisible by 4.

Lemma'. For $0 \le n \le 3030$, denote by v_n the largest integer such that 2^{v_n} divides a_n. We claim the following:

$$(*) \qquad v_{n+1} \ge \min(v_n + 2, v_{n+2} + 1) \qquad \text{for } n = 0, 1, \ldots, 3028.$$

Proof: Let $0 \le n \le 3028$ and let $s = \min(v_n + 2, v_{n+2} + 1)$. Then $s \le v_n + 2$ implies $2^s \mid 4a_n$ and $s \le v_{n+2} + 1$ implies $2^s \mid 2a_{n+2}$. It follows that $a_{n+1} = 2a_{n+2} - 4a_n$ is also divisible by 2^s, hence $s \le v_{n+1}$, which proves $(*)$.

Here are different ways of working out the induction argument that is crucial in the proofs.

Induction part, alternative A.

Statement: For $k = 0, 1, \ldots, 1010$, the terms $a_k, a_{k+1}, \ldots, a_{3030-2k}$ are all divisible by 2^{2k}.

Reformulation of the statement using notation v_n is the largest integer such that 2^{v_n} divides a_n: we have $v_n \ge k$ for any n satisfying $\lceil \frac{1}{2}k \rceil \le n \le 3030 - k$. Here $\lceil x \rceil$ denotes the smallest integer not smaller than x.

Proof 1: We proceed by induction on k. For $k = 0$ the statement is obvious, so, for the inductive step, suppose that $a_k, a_{k+1}, \ldots, a_{3030-2k}$ are all divisible by 2^{2k}. Apply the Lemma with

$$(a, b, c, d) = \left(\frac{a_{i-1}}{2^{2k}}, \frac{a_i}{2^{2k}}, \frac{a_{i+1}}{2^{2k}}, \frac{a_{i+2}}{2^{2k}} \right)$$

for $i = k + 1, k + 2, \ldots, 3030 - 2k - 2$. We obtain that $\frac{a_i}{2^{2k}}$ is divisible by 4 (and hence a_i is divisible by 2^{2k+2}) for $i = k + 1, k + 2, \ldots, i = 3030 - 2k - 2$. This completes the induction. For $k = 1010$ we obtain that a_{1010} is divisible by 2^{2020}, and the solution is complete. □

Remark. Remark, notice that by replacing 1010 with n (and hence, $2n$ with 2020 and $3n$ with 3030, this argument works, too. Then the claim is the following: if a_0, a_1, …, a_{3n} are integers that satisfy the recursion in the problem, then a_n is divisible by 2^{2n}.

Proof 2: We will show by two step induction to $k \ge 0$: we have $v_n \ge k$ for any n satisfying $\lceil \frac{1}{2}k \rceil \le n \le 3030 - k$. Here $\lceil x \rceil$ denotes the smallest integer not smaller than x. Plugging in $k = 2020$ and $n = 1010$ will give the desired result.
The case $k = 0$ is trivial, since the v_n are non-negative. For the case $k = 1$, let $1 \le n \le 3029$, then $v_n \ge \min(v_{n-1} + 2, v_{n+1} + 1) \ge 1$.
Suppose we have it proven for some $k \ge 1$ and for $k - 1$. Let $\lceil \frac{1}{2}(k + 1) \rceil \le n \le 3030 - (k + 1)$. Then $\lceil \frac{1}{2}(k - 1) \rceil \le n - 1 \le 3030 - (k - 1)$ and also $\lceil \frac{1}{2}k \rceil \le n + 1 \le 3030 - k$. By induction hypothesis we have $v_{n-1} \ge k - 1$ and $v_{n+1} \ge k$. Then $v_n \ge \min(v_{n-1} + 2, v_{n+1} + 1) \ge k$, finishing the induction. □

Proof 3: The notation is the same as in alternative A1, but the induction step has two steps. We use the lemma stated in the beginning as the first step. Assume now that $2^{2k} \mid a_k, \ldots, a_{3030-2k}$ for some $k \ge 1$. We claim that then $2^{2k+2} \mid a_{k+1}, \ldots, a_{3030-2(k+1)}$. Notice first that any i on the interval $[k + 1, 3030 - 2k - 1]$ satisfies the equation $2a_{i+1} = a_i + 4a_{i-2}$, we have $2^{2k+1} \mid a_i$. Furthermore, if $i \in [k + 1, 3030 - 2k - 2]$, we have $2^{2k+2} \mid a_i$ since $2^{2k+1} \mid a_{i+1}$ and hence

$2^{2k+2} \mid 2a_{i+1}$ and $2^{2k+2} \mid 4a_{i-1}$. □

Induction part, alternative B.

Statement: if a_0, a_1, …, a_{3n} are integers that satisfy the recursion in the problem, then a_n is divisible by 2^{2n}. The problem statement follows for $n = 1010$.

Proof: The base case of the induction is exactly the Lemma we just proved. Now, for the inductive step, suppose that the statement holds for $n = k - 1$, and consider integers a_0, a_1, …, a_{3k} that satisfy the recursion. By applying the induction hypothesis to the four sequences $(a_0, a_1, \ldots, a_{3k-3})$, $(a_1, a_2, \ldots, a_{3k-2})$, $(a_2, a_3, \ldots, a_{3k-1})$ and $(a_3, a_4, \ldots, a_{3k})$, we find that a_{k-1}, a_k, a_{k+1} and a_{k+2} are all divisible by 2^{2k-2}. If we now apply the lemma to $a_{k-1}/2^{2k-2}$, $a_k/2^{2k-2}$, $a_{k+1}/2^{2k-2}$ and $a_{k+2}/2^{2k-2}$, we find that $a_k/2^{2k-2}$ is divisible by 4, so a_k is divisible by 2^{2k}, as desired. □

Induction part, alternative C.

Statement: Given positive integers a_0, a_1, a_2, …, a_{3k} such that

$$2a_{n+2} = a_{n+1} + 4a_n \quad \text{for } n = 0, 1, 2, \ldots, 3k-2,$$

then 2^{2k} divides at least one of the numbers a_0, a_1, a_2, …, a_{3k}.

Proof: The case $k = 1$ is obtained from Lemma.

Suppose that for some $k \geq 1$, our claim is true for any sequence of $3k + 1$ positive integers that satisfy similar defining relations. Then consider a sequence of $3(k + 1) + 1 = 3k + 4$ positive integers a_0, a_1, a_2, …, a_{3k+3} such that

$$2a_{n+2} = a_{n+1} + 4a_n \quad \text{for } n = 0, 1, 2, \ldots, 3k+1.$$

Then, we see that all numbers a_1, a_2,…, a_{3k+2} are even, and so we may set $a_i = 2b_i$ for some positive integers b_i for $i = 1, 2, \ldots, 3k + 2$. Then $2a_2 = a_1 + 4a_0$ becomes

$$2b_2 = b_1 + 2a_0.$$

Note that

$$2b_{n+2} = b_{n+1} + 4b_n \quad \text{for } n = 1, 2, \ldots, 3k.$$

Hence the positive integers b_1, b_2, …, b_{3k+1} are even, and so we may write $b_i = 2c_i$ for some positive integers c_i for $i = 1, 2, \ldots, 3k + 1$. Then $2b_2 = b_1 + 2a_0$ becomes $2c_2 = c_1 + a_0$, which does not give us anything. However, we have

$$2c_{n+2} = c_{n+1} + 4c_n \quad \text{for } n = 1, 2, \ldots, 3k-1.$$

So the sequence of positive integers c_1, c_2, …, c_{3k+1} is a sequence of $3k+1$ positive integers that satisfy the defining relations. By the inductive hypothesis, 2^{2k} divides at least one of the numbers c_1, c_2, …, c_{3k+1}. Since $a_i = 4c_i$ for all $i = 1, 2, \ldots, 3k+1$, it follows that $4 \cdot 2^{2k} = 2^{2(k+1)}$ divides at least one of the numbers $a_1, \ldots, a_{3k+1}$. This completes the induction, and hence the proof. □

Induction part, alternative D.

Statement: If v_0, v_1, …, v_{3030} is a sequence of non-negative integers satisfying $(*)$, there must be a k such that $v_k \geq 2020$. In fact, we will show that $v_{1010} \geq 2020$.

These proofs use Lemma'.

Proof 1: We will show by induction on k that $v_n \geq 2k$ for $k \leq n \leq 3030-2k$ and $v_n \geq 2k+1$ for $k+1 \leq n \leq 3030-2k-1$. For $k = 1010$, the first statement implies that $v_{1010} \geq 2020$.
For $k = 0$ the first statement $v_n \geq 0$ is obvious, and the second statement follows using $(*)$: we have $v_n \geq v_{n+1} + 1 \geq 1$ for $1 \leq n \leq 3029$.
Now suppose that the inductive hypothesis holds for $k = \ell$: we have $v_n \geq 2\ell$ for $\ell \leq n \leq 3030-2\ell$ and $v_n \geq 2\ell+1$ for $\ell+1 \leq n \leq 3030-2\ell-1$. For the first statement, consider an n with $\ell+1 \leq n \leq 3030-2\ell-2$. Then using $(*)$ and the inductive hypothesis, we obtain

$$v_n \geq \min(v_{n-1}+2, v_{n+1}+1) \geq \min(2\ell+2, 2\ell+2) = 2\ell+2$$

because $v_{n-1} \geq 2\ell$ (as $\ell \leq n-1 \leq 3030-2\ell$) and $v_{n+1} \geq 2\ell+1$ (as $\ell+1 \leq n+1 \leq 3030-2\ell-1$). Similarly, for $\ell+2 \leq n \leq 3030-2\ell-3$ we find

$$v_n \geq \min(v_{n-1}+2, v_{n+1}+1) \geq \min(2\ell+3, 2\ell+3) = 2\ell+3$$

because $v_{n-1} \geq 2\ell+1$ (as $\ell+1 \leq n-1 \leq 3030-2\ell-1$) and $v_{n+1} \geq 2\ell+2$ (as $\ell+1 \leq n \leq 3030-2\ell-2$). This completes the induction. □

Proof 2: The inequality $v_{1010} \geq \min(v_{1009}+2, v_{1011}+1)$ gives us two cases to consider: either $v_{1010} \geq v_{1009}+2$ or $v_{1010} \geq v_{1011}+1$.
Suppose first that $v_{1010} \geq v_{1009}+2$. In this case we will show by induction on k that $v_{1010-k} \geq v_{1009-k}+2$ for $0 \leq k \leq 1009$. The case $k = 0$ is assumed, so suppose that $v_{1010-\ell} \geq v_{1009-\ell}+2$ holds for some $0 \leq \ell < 1009$. We obtain

$$v_{1009-\ell} \geq \min(v_{1010-\ell}+1, v_{1008-\ell}+2) \geq \min(v_{1009-\ell}+3, v_{1008-\ell}+2).$$

Because $v_{1009-\ell} < v_{1009-\ell}+3$ we must have $v_{1009-\ell} \geq v_{1008-\ell}+2$, completing the induction. We conclude that

$$v_{1010} \geq v_{1009}+2 \geq v_{1008}+4 \geq \cdots \geq v_0+2020 \geq 2020.$$

In the second case, we show by induction on k that $v_{1010+k} \geq v_{1011+k}+1$ for $0 \leq k \leq 2019$. Again, the base case $k = 0$ is assumed, so suppose that $v_{1010+\ell} \geq v_{1011+\ell}+1$ holds for some $0 \leq \ell < 2019$. We obtain

$$v_{1011+\ell} \geq \min(v_{1010+\ell}+2, v_{1012+\ell}+1) \geq \min(v_{1011+\ell}+3, v_{1012+\ell}+1).$$

Because $v_{1011+\ell} < v_{1011+\ell}+3$, we must have $v_{1011+\ell} \geq v_{1012+\ell}+1$, completing the induction. We conclude that

$$v_{1010} \geq v_{1011}+1 \geq v_{1012}+2 \geq \cdots \geq v_{3030}+2020 \geq 2020,$$

as desired. □

Alternative E.

This solution is different from the other solutions. Shift the sequence so that it starts at a_{-1010} and ends at a_{2020}. We will show that a_0 is divisible by 2^{2020}. Consider a_0. It either has at least as many factors 2 as $2a_1$, or at least as many factors 2 as $4a_{-1}$ (this follows from $2a_1 = a_0 + 4a_{-1}$). Consider the first case, so $e_2(a_0) \geq e_2(2a_1)$. By multiplying the original recursion by 2^{n-1}, we note that $b_n = 2^n a_n$ satisfies the recursion $b_n = b_{n-1} + 8b_{n-2}$. Furthermore, we assumed that $e_2(b_0) \geq e_2(b_1)$. This implies that $e_2(b_n)$ is constant for $n \geq 1$. Furthermore, clearly $2^{2020} \mid b_{2020}$, so $2^{2020} \mid b_1$ and hence $2^{2020} \mid a_0$. The case where $e_2(a_0) \geq e_2(4a_{-1})$ is similar; we then look at $b_n = 4^n a_{-n}$ which satisfies $b_n = 8b_{n-2} - b_{n-1}$ and use that $2^{2020} \mid b_{1010}$. The rest of the argument is the same. □

Solutions to Problem 2

Answer. There are two solutions: $(\underbrace{0,0,\ldots,0}_{1010},\underbrace{1,1,\ldots,1}_{1010})$ and $(\underbrace{1,1,\ldots,1}_{1010},\underbrace{2,2,\ldots,2}_{1010})$.

Solution A. We first prove the inequality

$$((x+1)(y+1))^2 \geq 4(x^3+y^3) \tag{1}$$

for real numbers $x, y \geq 0$ satisfying $|x-y| \leq 1$, with equality if and only if $\{x,y\} = \{0,1\}$ or $\{x,y\} = \{1,2\}$.
Indeed,

$$\begin{aligned} 4(x^3+y^3) &= 4(x+y)(x^2-xy+y^2) \\ &\leq ((x+y)+(x^2-xy+y^2))^2 \\ &= (xy+x+y+(x-y)^2)^2 \\ &\leq (xy+x+y+1)^2 \\ &= ((x+1)(y+1))^2, \end{aligned}$$

where the first inequality follows by applying the AM-GM inequality on $x+y$ and x^2-xy+y^2 (which are clearly nonnegative). Equality holds in the first inequality precisely if $x+y = x^2-xy+y^2$ and in the second one if and only if $|x-y|=1$. Combining these equalities we have $x+y = (x-y)^2+xy = 1+xy$ or $(x-1)(y-1)=0$, which yields the solutions $\{x,y\} = \{0,1\}$ or $\{x,y\} = \{1,2\}$. Now, let $(x_1,x_2,\ldots,x_{2020})$ be any sequence satisfying conditions (i) and (ii) and let $(y_1,y_2,\ldots,y_{2020})$ be any permutation of $(x_1,x_2,\ldots,x_{2020})$. As $0 \leq \min(x_i,y_i) \leq \max(x_i,y_i) \leq \min(x_i,y_i)+1$, we can applying inequality (1) to the pair (x_i,y_i) and sum over all $1 \leq i \leq 2020$ to conclude that

$$\sum_{i=1}^{2020}((x_i+1)(y_i+1))^2 \geq 4\sum_{i=1}^{2020}(x_i^3+y_i^3) = 8\sum_{i=1}^{2020}x_i^3.$$

Therefore, in order to satisfy condition (iii), every inequality must be an equality. Hence, for every $1 \leq i \leq 2020$ we must have $\{x_i,y_i\} = \{0,1\}$ or $\{x_i,y_i\} = \{1,2\}$. By condition (ii) ,we see that either $\{x_i,y_i\} = \{0,1\}$ for all i or $\{x_i,y_i\} = \{1,2\}$ for all i.
If $\{x_i,y_i\} = \{0,1\}$ for every $1 \leq i \leq 2020$, this implies that the sequences $(x_1,x_2,\ldots,x_{2020})$ and $(y_1,y_2,\ldots,y_{2020})$ together have 2020 zeroes and 2020 ones. As $(y_1,y_2,\ldots,y_{2020})$ is a permutation of $(x_1,x_2,\ldots,x_{2020})$ this implies that $(x_1,x_2,\ldots,x_{2020}) = (0,0,\ldots,0,1,1,\ldots,1)$ with 1010 zeroes and 1010 ones. Conversely, note that this sequence satisfies conditions (i), (ii), and (iii) (in (iii), we take $(y_1,y_2,\ldots,y_{2020}) = (x_{2020},x_{2019},\ldots,x_1)$), showing that this sequence indeed works. The same reasoning holds for the case that $\{x_i,y_i\} = \{1,2\}$ for all i. □

Comment. There are multiple ways to show the main inequality (1):

- Write

$$\begin{aligned} ((x+1)(y+1))^2 &\geq ((x+1)(y+1))^2 - ((x-1)(y-1))^2 \\ &= 4(x+y)(xy+1) \\ &\geq 4(x+y)(xy+(x-y)^2) \\ &= 4(x^3+y^3), \end{aligned}$$

  where equality holds precisely if $|x-y|=1$ and $(x-1)(y-1)=0$.

- Write

$$(x+1)^2(y+1)^2 - 4x^3 - 4y^3 = (x-1)^2(y-1)^2 + 4(x+y)\big(1-(x-y)^2\big) \geq 0,$$

  where equality holds precisely if $(x-1)(y-1)=0$ and $(x-y)^2=1$.

- One can rewrite the difference between the two sides as a sum of nonnegative expressions. One such way is to assume that $y \geq x$ and then to rewrite the difference as
$$x^2(y-2)^2 + (x+1-y)(4y^2 - 4x^2 + 2xy + x + 3y + 1),$$
where $x^2(y-2)^2 \geq 0$, $x+1-y \geq 0$ and $4y^2 - 4x^2 + 2xy + x + 3y + 1 > 0$, so in the equality case we must have $x + 1 - y = 0$ and $x(y-2) = 0$.

- Again assume $y \geq x$; substitute $y = x + u$ with $0 \leq u \leq 1$ and rewrite the difference as
$$x^2(x+u-2)^2 + x^2(2-2u) + (4+6u-10u^2)x + (1+2u+u^2-4u^3),$$
of which each summand is nonnegative, with equality case $u = 1$ and $x \in \{0, 1\}$.

- Fix $x \geq 0$. We aim to show that the function
$$f(y) = ((x+1)(y+1))^2 - 4(x^3+y^3),$$
viewed as polynomial in y, is nonnegative on the interval $[x, x+1]$. First note that for $y = x$ and $y = x + 1$ the function equals
$$(x^2-2x)^2 + 2x^2 + 4x + 1 > 0 \qquad \text{and} \qquad (x^2-x)^2 \geq 0$$
respectively. The derivative of f with respect to y equals
$$2(x+1)^2 + 2(x+1)^2 y - 12y^2,$$
which is a quadratic with negative leading coefficient that evaluates as $2(x+1)^2 > 0$ for $y = 0$. Therefore, this quadratic has one positive and one negative root. Therefore, on $[0, \infty)$, the function f will be initially increasing and eventually decreasing, hence the minimum on the interval $[x, x+1]$ will be achieved on one of the endpoints. To have equality, we must have $x^2 - x = 0$, hence $x \in \{0, 1\}$.

- Observe that $((x+1)(y+1))^2 - 4(x^3+y^3)$ is the discriminant of
$$p(z) = (x^2 - xy + y^2)z^2 - (x+1)(y+1)z + (x+y).$$
Note that the leading coefficient $x^2 - xy + y^2 = (x-y)^2 + xy$ is positive unless $x = y = 0$, in which case $((0+1)^2(0+1)^2) > 4 \cdot 0^3 + 4 \cdot 0^3$. Substituting $z = 1$, we get
$$p(1) = (x^2 - xy + y^2) - (x+1)(y+1) + (x+y) = (x-y)^2 - 1 \leq 0.$$
It follows that the discriminant is non-negative. It equals zero if and only if $|x-y| = 1$ and $p(z)$ attains its minimum at $z = 1$. Without loss of generality $y = x + 1$. The minimum is attained at
$$1 = \frac{(x+1)(y+1)}{2(x^2 - xy + y^2)} = \frac{x^2+3x+2}{2x^2+2x+2},$$
which reduces to $x^2 = x$. Therefore, the only critical points are $(x, y) = (0, 1)$ and $(x, y) = (1, 2)$.

- Without loss of generality $y \geq x$. Let $z = \frac{x+y}{2}$ and $a = y - z$. Note that $z \geq 0$ and $0 \leq a \leq \min\{\frac{1}{2}, z\}$. We can rewrite the inequality in terms of z and a.
$$\begin{aligned}((x+1)(y+1))^2 - 4x^3 - 4y^3 &= ((z+1+a)(z+1-a))^2 - 4(z-a)^3 - 4(z+a)^3 \\ &= ((z+1)^2 - a^2)^2 - 8z^3 - 24za^2 \\ &= a^4 - (24z + 2(z+1)^2)a^2 + ((z+1)^4 - 8z^3)\end{aligned}$$
This is a quadratic in a^2. It attains its minimum at $a^2 = 12z + (z+1)^2 \geq 1$. Therefore it is strictly decreasing in a on the interval $[0, \min\{\frac{1}{2}, z\}]$. If $a = \frac{1}{2}$, then $y = x + 1$. It follows that
$$((x+1)(y+1))^2 - 4x^3 - 4y^3 = ((x+1)(x+2))^2 - 4x^3 - 4(x+1)^3 = x^2(x-1)^2 \geq 0$$
with equality if and only if $x = 0$ or $x = 1$. If $a = z$, then $x = 0$. It follows that
$$((x+1)(y+1))^2 - 4x^3 - 4y^3 = (y+1)^2 - 4y^3 = y^2(1-y) + 2y(1-y^2) + (1-y^3) \geq 0$$
with equality if and only if $y = 1$. Therefore $(0, 1)$ and $(1, 2)$ are the only critical points.

Solutions to Problem 3

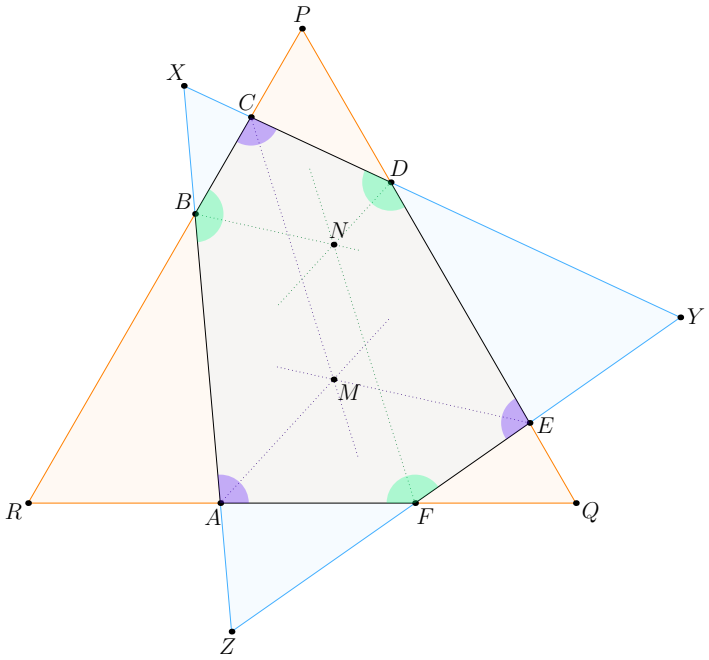
Solution A. Denote the angle bisector of A by a and similarly for the other bisectors. Thus, given that a, c, e have a common point M, we need to prove that b, d, f are concurrent. We write $\angle(x, y)$ for the value of the directed angle between the lines x and y, i.e. the angle of the counterclockwise rotation from x to y (defined $\pmod{180^\circ}$).
Since the sum of the angles of a convex hexagon is 720°, from the angle conditions we get that the sum of any two consecutive angles is equal to 240°. In particular, it now follows that $\angle(b, a) = \angle(c, b) = \angle(d, c) = \angle(e, d) = \angle(f, e) = \angle(a, f) = 60^\circ$ (assuming the hexagon is clockwise oriented).
Let $X = AB \cap CD$, $Y = CD \cap EF$ and $Z = EF \cap AB$. Similarly, let $P = BC \cap DE$, $Q = DE \cap FA$ and $R = FA \cap BC$. From $\angle B + \angle C = 240^\circ$ it follows that $\angle(ZX, XY) = \angle(BX, XC) = 60^\circ$. Similarly we have $\angle(XY, YZ) = \angle(YZ, ZX) = 60^\circ$, so triangle XYZ (and similarly triangle PQR) is equilateral. We see that the hexagon $ABCDEF$ is obtained by intersecting the two equilateral triangles XYZ and PQR.
We have $\angle(AM, MC) = \angle(a, c) = \angle(a, b) + \angle(b, c) = 60^\circ$, and since

$$\angle(AM, MC) = \angle(AX, XC) = \angle(AR, RC) = 60^\circ,$$

A, C, M, X, R are concyclic. Because M lies on the bisector of angle $\angle XAR$, we must have $MR = MX$, so triangle MRX is isosceles. Moreover, we have $\angle(MR, MX) = \angle(CR, CX) = \angle(BC, CD)$, which is angle C of the hexagon. We now see that the triangles MRX, MPY and MQZ are isosceles and similar. This implies that there is a rotation centered at M that sends X, Y and Z to R, P and Q respectively. In particular, the equilateral triangles XYZ and PQR are congruent.
It follows that there also exists a rotation sending X, Y, Z to P, Q, R respectively. Define N as the center of this rotation. Triangles NXZ and NPR are congruent and equally oriented, hence N is equidistant from XZ and PR and lies on the inner bisector b of $\angle B$ (we know N lies on the inner, not the outer bisector because the rotation centered at N is clockwise). In the same way we can show that N is on d and on f, so b, d, f are concurrent at N. □

Remark. The key observation (a rotation centered at M sends $\triangle XYZ$ to $\triangle RQP$) can be established in slightly different ways. E.g., since A, M, R, X are concyclic and A, M, Q, Z are concyclic, M is the Miquel point of the lines XZ, RQ, XR, ZQ, hence it is the center of similitude s sending $\vec{XZ}$ to $\vec{RQ}$. Repeating the same argument for the other pairs of vectors, we obtain that s sends $\triangle XYZ$ to $\triangle RQP$. Moreover, s is a rotation, since M is equidistant from XZ and RQ.

Remark. The reverse argument can be derived in a different way, e.g., defining N as the common point of the circles $BXPD$, $DYQF$, $FZRB$, and showing that $\triangle NXZ = \triangle NPR$, etc.



Solution B. As in Solution A, we prove that the hexagon $ABCDEF$ is the intersection of the equilateral triangles PQR and XYZ.
Let $d(S, AB)$ denote the signed distance from the point S to the line AB, where the negative sign is taken if AB separates S and the hexagon. We define similarly the other distances ($d(S, BC)$, etc). Since $M \in a$, we have $d(M, ZX) = d(M, QR)$. In the same way, we have $d(M, XY) = d(M, RP)$ and $d(M, YZ) = d(M, PQ)$. Therefore $d(M, ZX) + d(M, XY) + d(M, YZ) = d(M, QR) + d(M, RP) + d(M, PQ)$.
We now use of the following well-known lemma (which can be easily proved using areas) to deduce that triangles PQR and XYZ are congruent.

Lemma. The sum of the signed distances from any point to the sidelines of an equilateral triangle (where the signs are taken such that all distances are positive inside the triangle) is constant and equals the length of the altitude.

For $N = b \cap d$ we now find $d(N, ZX) = d(N, RP)$ and $d(N, XY) = d(N, PQ)$. Using again the lemma for the point N, we get $d(N, ZX) + d(N, XY) + d(N, YZ) = d(N, QR) + d(N, RP) + d(N, PQ)$. Therefore $d(N, YZ) = d(N, QR)$, thus $N \in f$. □

Remark. Instead of using the lemma, it is possible to use some equivalent observation in terms of signed areas.

Solution C. We use the same notations as in Solution A. We will show that a, c and e are concurrent if and only if

$$AB + CD + EF = BC + DE + FA,$$

which clearly implies the problem statement by symmetry.
Let $\vec{a}$ be the vector of unit length parallel to a directed from A towards the interior of the hexagon. We define analogously $\vec{b}$, etc. The angle conditions imply that opposite bisectors of the hexagon are parallel, so we have $\vec{a} \parallel \vec{d}$, $\vec{b} \parallel \vec{e}$ and $\vec{c} \parallel \vec{f}$. Moreover, as in the previous solutions, we know that $\vec{a}$, $\vec{c}$ and $\vec{e}$ make angles of 120° with each other. Let $M_A = c \cap e$, $M_C = e \cap a$ and $M_E = a \cap c$. Then M_A, M_C, M_E form an equilateral triangle with side length denoted by s. Note that the case $s = 0$ is equivalent to a, c and e being concurrent.
Projecting $\vec{M_EA} + \vec{AB} = \vec{M_EB} = \vec{M_EC} + \vec{CB}$ onto $\vec{e} = -\vec{b}$, we obtain

$$\vec{AB} \cdot \vec{b} - \vec{CB} \cdot \vec{b} = \vec{M_EC} \cdot \vec{b} - \vec{M_EA} \cdot \vec{b} = \vec{M_EA} \cdot \vec{e} - \vec{M_EC} \cdot \vec{e}.$$

Writing $\varphi = \frac{1}{2}\angle B = \frac{1}{2}\angle D = \frac{1}{2}\angle F$, we know that $\vec{AB}\cdot\vec{b} = -AB\cdot\cos(\varphi)$, and similarly $\vec{CB}\cdot\vec{b} = -CB\cdot\cos(\varphi)$. Because M_EA and M_EC intersect e at 120° angles, we have $\vec{M_EA}\cdot\vec{e} = \frac{1}{2}M_EA$ and $\vec{M_EC}\cdot\vec{e} = \frac{1}{2}M_EC$. We conclude that

$$2\cos(\varphi)(AB - CB) = M_EC - M_EA.$$

Adding the analogous equalities $2\cos(\varphi)(CD - ED) = M_AE - M_AC$ and $2\cos(\varphi)(EF - AF) = M_CA - M_CE$, we obtain

$$2\cos(\varphi)(AB + CD + EF - CB - ED - AF) = M_EC - M_EA + M_AE - M_AC + M_CA - M_CE.$$

Because M_A, M_C and M_E form an equilateral triangle with side length s, we have $M_EC - M_AC = \pm s$, $M_CA - M_EA = \pm s$, and $M_AE - M_CE = \pm s$. Therefore, the right hand side $M_EC - M_EA + M_AE - M_AC + M_CA - M_CE$ equals $\pm s \pm s \pm s$, which (irrespective of the choices of the $\pm$-signs) is 0 if and only if $s = 0$. Because $\cos(\varphi) \neq 0$, we conclude that

$$AB + CD + EF = CB + ED + AF \iff s = 0 \iff a,\ c,\ e \text{ concurrent},$$

as desired. □

Remark. Equalities used in the solution could appear in different forms, in particular, in terms of signed lengths.

Remark. Similar solutions could be obtained by projecting onto the line perpendicular to b instead of b.

Solution D. We use the the same notations as in previous solutions and the fact that $a \parallel d$, $b \parallel e$ and $c \parallel f$ make angles of 120°. Also, we may assume that E and C are not symmetric in a (if they are, the entire figure is symmetric and the conclusion is immediate).
We consider two mappings: the first one $s : a \to BC \to d$ sending $A' \mapsto B' \mapsto S$ is defined such that $A'B' \parallel AB$ and $B'S \parallel b$, and the second one $t : a \to EF \to d$ sending $A' \mapsto F' \mapsto T$ is defined such that $A'F' \parallel AF$ and $F'T \parallel f$. Both maps are affine linear since they are compositions of affine tranformations. We will prove that they coincide by finding two distinct points $A', A'' \in a$ for which $s(A') = t(A')$ and $s(A'') = t(A'')$. Then we will obtain that $s(A) = t(A)$, which by construction implies that the bisectors of $\angle B$, $\angle D$ and $\angle F$ are concurrent.
We will choose A' to be the reflection of C in e and A'' to be the reflection of E in c. They are distinct since otherwise C and E would be symmetric in a. Applying the above maps $a \to BC$ and $a \to EF$ to A', we get points B' and F' such that $A'B'CDEF'$ satisfies the problem statement. However, this hexagon is symmetric in e, hence the bisectors of $\angle B'$, $\angle D$, $\angle F'$ are concurrent and $s(A') = t(A')$. The same reasoning yields $s(A'') = t(A'')$, which finishes the solution. □

Remark. This solution is based on the fact that that two specific affine linear maps coincide. Here it was proved by exhibiting two points where they coincide. One could prove it in another way, exhibiting one such point and proving that the 'slopes' are equal.

Remark. There are similar solutions where claims and proofs could be presented in more 'elementary' terms. For example, an elementary reformulation of the 'slopes' being equal is: if b' passes through B' parallel to b, and f' passes through F' parallel to f, then the line through $b \cap f$ and $b' \cap f'$ is parallel to a (which is parallel to d).

Solution E. We use the same notations as in previous solutions.
Since the sum of the angles of a convex hexagon is 720°, from the angle conditions we get $\angle B + \angle C = 720°/3 = 240°$. From $\angle B + \angle C = 240°$ it follows that the angle between c and b equals 60°. The same is analogously true for other pairs of bisectors of neighboring angles.
Consider the points $O_a \in a$, $O_c \in c$, $O_e \in e$, each at the same distance d' from M, where $d' > \max\{MA, MC, ME\}$, and such that the rays AO_a, CO_c, EO_e point out of the hexagon. By

construcion, O_a and O_c are symmetrical in e, hence $O_aO_c \perp b$. Similarly, $O_cO_e \perp d$, $O_eO_a \perp f$. Thus it suffices to prove that perpendiculars from B, D, F to the sidelines of $\triangle O_aO_cO_e$ are concurrent. By a well-known criteria, this condition is equivalent to equality

$$O_aB^2 - O_cB^2 + O_cD^2 - O_eD^2 + O_eF^2 - O_aF^2 = 0. \qquad (*)$$

To prove $(*)$ consider a circle ω_a centered at O_a and tangent to AB and AF and define circles ω_c and ω_e in the same way. Rewrite O_aB^2 as $r_a^2 + B_aB^2$, where r_a is the radius of ω_a, and B_a is the touch point of ω_a with AB. Using similar notation for the other tangent points, transform $(*)$ into

$$B_aB^2 - B_cB^2 + D_cD^2 - D_eD^2 + F_eF^2 - F_aF^2 = 0. \qquad (**)$$

Furthermore, $\angle O_cO_aB_a = \angle MO_aB_a + \angle O_cO_aM = (90^\circ - \varphi) + 30^\circ = 120^\circ - \varphi$, where $\varphi = \frac{1}{2}\angle A$. (Note that $\varphi > 30^\circ$, since $ABCDEF$ is convex.) By analogous arguments, $\angle O_aO_cB_c = \angle O_eO_cD_c = \angle O_cO_eD_e = \angle O_aO_eF_e = \angle O_eO_aF_a = 120^\circ - \varphi$. It follows that rays O_aB_a and O_cB_c (being symmetrical in e) intersect at $U_e \in e$ forming an isosceles triangle $\triangle O_aU_eO_c$. Similarly define $\triangle O_cU_aO_e$ and $\triangle O_eU_cO_a$. These triangles are congruent (equal bases and corresponding angles). Therefore we have $O_aU_c = U_cO_e = O_eU_a = U_aO_c = O_cU_e = U_eO_a$. Moreover, we also have $B_aU_e = O_aU_e - r_a = O_aU_c - r_a = F_aU_c = x$, and thus similarly $D_cU_a = B_cU_e = y$, $F_eU_c = D_eU_a = z$.

Now from quadrilateral $BB_aU_eB_c$ with two opposite right angles $B_aB^2 - B_cB^2 = B_cU_e^2 - B_aU_e^2 = y^2 - x^2$. Similarly $D_cD^2 - D_eD^2 = D_eU_a^2 - D_cU_a^2 = z^2 - y^2$ and $F_eF^2 - F_aF^2 = F_aU_c^2 - F_eU_c^2 = x^2 - z^2$. Finally, we substitute this into $(**)$, and the claim is proved. □

Remark. Circles ω_a, ω_c and ω_e could be helpful in some other solutions. In particular, the movement of A along a in Solution D is equivalent to varying r_a.

Solutions to Problem 4

Solution A. Let $\sigma = (\sigma_1, \ldots, \sigma_{n-1})$ be a fresh permutation of the integers $1, 2, \ldots, n-1$. We claim that for any $1 \leq i \leq n-1$ the permutation

$$\sigma^{(i)} = (\sigma_1, \ldots, \sigma_{i-1}, n, \sigma_i, \ldots, \sigma_{n-1})$$

is a fresh permutation of the integers $1, 2, \ldots, n$. Indeed, let $1 \leq k \leq n-1$. If $k \geq i$ then we have $n \in \{\sigma_1^{(i)}, \ldots, \sigma_k^{(i)}\}$, but $n \notin \{1, 2, \ldots, k\}$. And if $k < i$ we have $k < n-1$, and $\{\sigma_1^{(i)}, \ldots, \sigma_k^{(i)}\} = \{\sigma_1, \ldots, \sigma_k\} \neq \{1, 2, \ldots, k\}$, since σ is fresh. Moreover, it is easy to see that when we apply this construction to all fresh permutations of $1, 2, \ldots, n-1$, we obtain $(n-1) \cdot f_{n-1}$ distinct fresh permutations of $1, 2, \ldots, n$.
Note that a fresh permutation of $1, 2, \ldots, n-1$ cannot end in $n-1$, and hence none of the previously constructed fresh permutations of $1, 2, \ldots, n$ will end in $n-1$ either. Therefore, we will complete the proof by finding f_{n-1} fresh permutations of $1, 2, \ldots, n$ that end in $n-1$. To do this, let $\sigma = (\sigma_1, \ldots, \sigma_{n-1})$ be a fresh permutation of $1, 2, \ldots, n-1$ and let j be such that $\sigma_j = n-1$. Define

$$\sigma' = (\sigma_1, \ldots, \sigma_{j-1}, n, \sigma_{j+1}, \ldots, \sigma_{n-1}, n-1),$$

then clearly σ' is a permutation of $1, 2, \ldots, n$ that ends in $n-1$. We show that σ' is fresh, so let $1 \leq k \leq n-1$. If $k \geq j$ then $n \in \{\sigma_1', \ldots, \sigma_k'\}$ but $n \notin \{1, 2, \ldots, k\}$; if $k < j$, then $k < n-1$ and $\{\sigma_1', \ldots, \sigma_k'\} = \{\sigma_1, \ldots, \sigma_k\} \neq \{1, 2, \ldots, k\}$, since σ is fresh. So we have constructed f_{n-1} additional fresh permutations of 1, 2, …, n (which again are all different), and the total number f_n of fresh permutations of 1, 2, …, n must at least be $(n-1)f_{n-1} + f_{n-1} = nf_{n-1}$, as required. □

Comment. A similar way to construct n fresh permutations of 1, 2, …, n for each fresh permutation σ of 1, 2, …, $n-1$ is as follows: increase all entries of σ by 1, and then add the number 1 anywhere; all these permutations are fresh, except the ones where 1 is added at the front, which can be made fresh by swapping the 1 and the 2. It is again straightforward to check that we obtain nf_{n-1} permutations that are fresh and distinct, although a little extra care is needed to account for the fact that we increased the entries of our original permutation.

Solution B. Assuming $n \geq 3$, we construct $f_{n-1} \cdot n$ different fresh permutations.
Consider a fresh permutation of the $n-1$ numbers $1, 3, 4, \ldots, n$ (the number 2 has been removed), by which we mean a permutation $(x_1, \ldots, x_{n-1})$ such that $x_1 \neq 1$ and $\{x_1, \ldots, x_k\} \neq \{1, 3, \ldots, k+1\}$ for all k with $2 \leq k \leq n-2$. There are exactly f_{n-1} such permutations.
By inserting 2 anywhere in such a permutation, i.e. before or after all entries, or between two entries x_i, x_{i+1}, we generate the following list of n distinct permutations of $1, 2, \ldots, n$, which we claim are all fresh:

$$(2, x_1, \ldots, x_{n-1}), \quad \ldots, \quad (x_1, \ldots, x_{i-1}, 2, x_i, \ldots, x_{n-1}), \quad \ldots, \quad (x_1, \ldots, x_{n-1}, 2).$$

In order to verify freshness, suppose that some permutation above is not fresh, that is, for some $1 \leq k \leq n-1$, the first k elements are $1, \ldots, k$. If $k = 1$ then the first element is 1; but the first element is either 2 or x_1 and $x_1 \neq 2$, so this is not possible. If $k \geq 2$ then the first k elements must contain 2 and $x_1, \ldots, x_{k-1}$, so $\{x_1, \ldots, x_{k-1}\} = \{1, 3, \ldots, k\}$, in contradiction to the fact that $(x_1, \ldots, x_{n-1})$ is fresh.
We have thus constructed $f_{n-1} \cdot n$ different fresh permutations, so $f_n \geq f_{n-1} \cdot n$. □

Comment. Note that the construction from solution B can also be performed by placing $n-1$ instead of 2, or indeed any k between 2 and $n-1$ (in which case it will work for $n \geq k+1$).
Moreover, similar arguments can be made while choosing to work with non-fresh permutations rather than fresh permutations. Indeed, in order to show that $n! - f_n \leq n((n-1)! - f_{n-1})$ for all $n \geq 3$, which is equivalent to the problem statement, one can argue as follows. Given a non-fresh permutation $(\sigma_1, \ldots, \sigma_n)$ of $1, 2, \ldots, n$, erasing $n-1$ from it and changing n to $n-1$ yields a permutation which cannot be fresh. Moreover, each non-fresh permutation of $1, 2, \ldots, n-1$ can be obtained in this way from at most n different non-fresh permutations of $1, 2, \ldots, n$. This type of argument would essentially mirror the one carried out in Solution B.

Solution C. By considering all permutations of $1, 2, \ldots, n$ and considering the smallest k for which the first k numbers are $1, 2, \ldots, k$ in some order, one can deduce the recursion

$$\sum_{k=1}^{n}(n-k)! \cdot f_k = n!,$$

which holds for any $n \geq 1$.
Therefore, if $n \geq 3$ we have

$$\begin{aligned}
0 &= n! - (n+1)(n-1)! + (n-1)(n-2)! \\
&= \sum_{k=1}^{n}(n-k)! \cdot f_k - (n+1)\sum_{k=1}^{n-1}(n-1-k)! \cdot f_k + (n-1)\sum_{k=1}^{n-2}(n-2-k)! \cdot f_k \\
&= f_n - nf_{n-1} + \sum_{k=1}^{n-2}(n-2-k)! \cdot ((n-k)(n-k-1) - (n+1)(n-k-1) + (n-1)) \cdot f_k \\
&= f_n - nf_{n-1} + \sum_{k=1}^{n-2}(n-2-k)! \cdot k(k+2-n) \cdot f_k \leq f_n - nf_{n-1},
\end{aligned}$$

which rewrites as $f_n \geq nf_{n-1}$. □

Solution D. We first show that for any $n \geq 2$ we have

$$f_n = \sum_{k=1}^{n-1} k(n-1-k)! \cdot f_k.$$

To deduce this relation, we imagine obtaining permutations of $1, 2, \ldots, n$ by inserting n into a permutation of the numbers $1, 2, \ldots, n-1$. Specifically, let $(\sigma_1, \ldots, \sigma_{n-1})$ be *any* permutation of $1, 2, \ldots, n-1$, and let $1 \leq k \leq n-1$ be minimal with $\{\sigma_1, \ldots, \sigma_k\} = \{1, 2, \ldots, k\}$. Note that there are $(n-1-k)! \cdot f_k$ such permutations. Furthermore, inserting n in this permutation will give a fresh permutation if and only if n is inserted before σ_k, so there are k options to do so. Consequently, if $n \geq 3$,

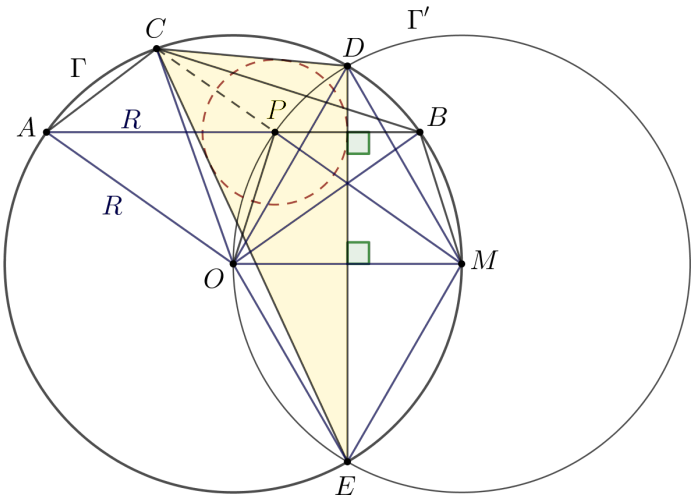
$$\begin{aligned}
f_n &= \sum_{k=1}^{n-1} k(n-1-k)! \cdot f_k \\
&= (n-1) \cdot f_{n-1} + \sum_{k=1}^{n-2} k(n-1-k)! \cdot f_k \\
&\geq (n-1) \cdot f_{n-1} + \sum_{k=1}^{n-2} k(n-2-k)! \cdot f_k \\
&= (n-1) \cdot f_{n-1} + f_{n-1} = n \cdot f_{n-1},
\end{aligned}$$

completing the proof. □

Comment. Fresh permutations are known as *indecomposable permutations* or *irreducible permutations* in the literature. The problem asks to prove that the probability that a randomly chosen permutation of 1, 2, …, n is indecomposable is a non-decreasing function of n. In fact, it turns out that this probability goes to 1 as $n \to \infty$: for large n, almost all permutations of 1, 2, …, n are indecomposable. More can be found in:
Y. Koh & S. Ree. Connected permutation graphs. *Discrete Mathematics* **307** (21):2628–2635, 2007.

Solutions to Problem 5

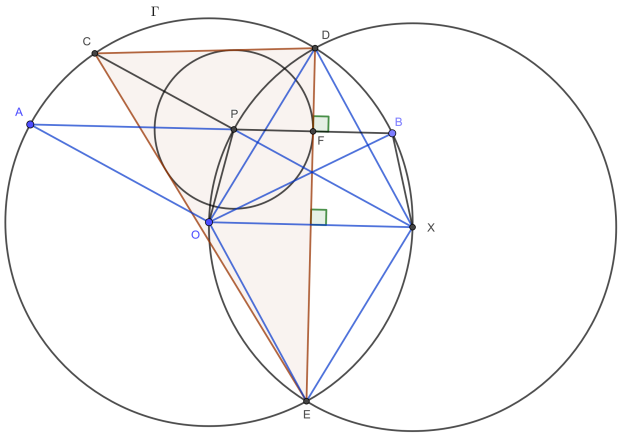
Solution A. The angle bisector of $\angle ECD$ intersects the circumcircle of CDE (which is Γ) at the midpoint M of arc DBE. It is well-known that the incentre is the intersection of the angle bisector segment CM and the circle with centre at M and passing through D, E. We will verify this property for P.



By the conditions we have $AP = OA = OB = OC = OD = OE = R$. Both lines OM and APB are perpendicular to ED, therefore $AP \| OM$; in the quadrilateral $AOMP$ we have $AP = OA = AM = R$ and $AP \| OM$, so $AOMP$ is a rhombus and its fourth side is $PM = R$. In the convex quadrilateral $OMBP$ we have $OM \| PB$, so $OMBP$ is a symmetric trapezoid; the perpendicular bisector of its bases AO and PB coincide. From this symmetry we obtain $MD = OD = R$ and $ME = OE = R$. (Note that the triangles OEM ad OMD are equilateral.) We already have $MP = MD = ME = R$, so P indeed lies on the circle with center M and passing through D, E. (Notice that this circle is the reflection of Γ about DE.)

From $PB = PC$ and $OB = OC$ we know that B and C are symmetrical about OP; from the rhombus $AOMP$ we find that A and M are also symmetrical about OP. By reflecting the collinear points B, P, A (with P lying in the middle) we get that C, P, M are collinear (and P is in the middle). Hence, P lies on the line segments CM. □

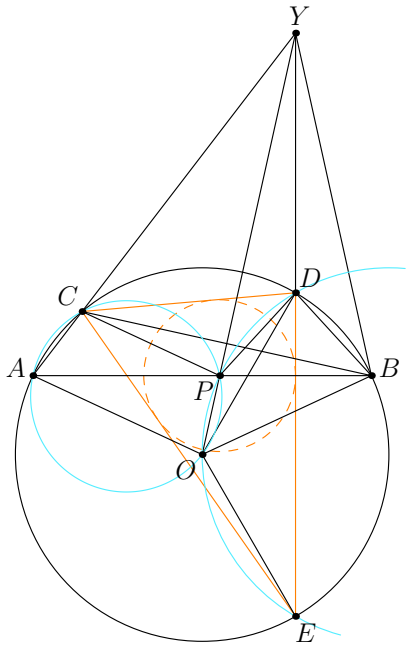
Solution B. Let X be the second intersection of CP with Γ. Using the power of the point P in the circle Γ and the fact that $PB = PC$, we find that $PX = PA = R$. The quadrilateral $AOXP$ has four sides of equal length, so it is a rhombus and in particular OX is parallel to AP. This proves that $OXBP$ is a trapezoid, and because the diagonals PX and OB have equal length, this is even an isosceles trapezoid. Because of that, DE is not only the perpendicular bisector of PB, but also of OX.



In particular we have $XD = XP = XO = XE = R$, which proves that X is the middle of the arc DE and P belongs to the circle with center X going through D and E. These properties, together with the fact that C, P, X are collinear, determine uniquely the incenter of CDE.

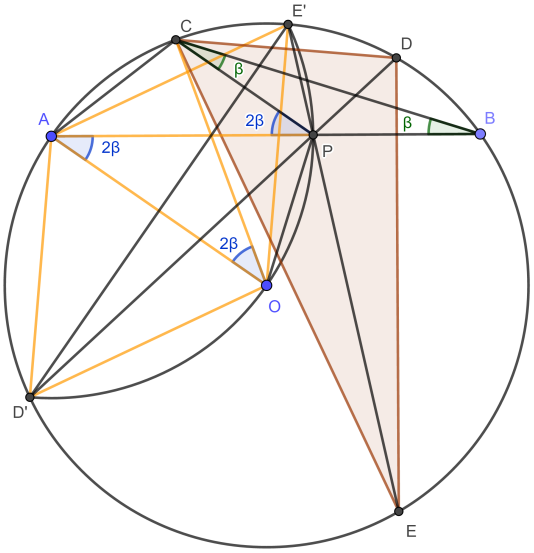
□

Solution C. Let Y be the circumcenter of triangle BPC. Then from $YB = YP$ it follows that Y lies on DE (we assume D lies in between Y and E), and from $YB = YC$ it follows that Y lies on OP, where O is the center of Γ.
From $\angle AOC = 2\angle ABC = \angle APC$ (because $\angle PBC = \angle PCB$) we deduce that $AOPC$ is a cyclic quadrilateral, and from $AP = R$ it follows that $AOPC$ is an isosceles trapezoid. We now find that $\angle YCP = \angle YPC = 180° - \angle OPC = 180° - \angle ACP$, so Y lies on AC.



Power of a point gives $YO \cdot YP = YC \cdot YA = YD \cdot YE$, so D, P, O and E are concyclic. It follows that $2\angle DAE = \angle DOE = \angle DPE = \angle DBE = 180° - \angle DAE$, so $\angle DAE = 60°$. We can now finish the proof by angle chasing.
From $AB \perp DE$ we have $\angle AOD + \angle BOE = 180°$ and from $\angle DOE = 2\angle DAE = 120°$ it follows that $\angle BOD + \angle BOE = 120°$. It follows that $\angle AOD - \angle BOD = 180° - 120° = 60°$. Let $\angle OAB = \angle OBA = 2\beta$; then $\angle AOD + \angle BOD = \angle AOB = 180° - 4\beta$. Together with $\angle AOD - \angle BOD = 60°$, this yields $\angle AOD = 120° - 2\beta$ and $\angle BOD = 60° - 2\beta$. We now find $\angle AED = \frac{1}{2}\angle AOD = 60° - \beta$, which together with $\angle DAE = 60°$ yields $\angle ADE = 60° + \beta$. From the isosceles trapezoid $AOPC$ we have $\angle CDA = \angle CBA = \frac{1}{2}\angle CPA = \frac{1}{2}\angle PAO = \beta$, so $\angle CDE = \angle CDA + \angle ADE = \beta + 60° + \beta = 60° + 2\beta$.
From $\angle BOD = 60° - 2\beta$ we deduce that $\angle BED = 30° - \beta$; together with $\angle DBE = 120°$ this yields $\angle EDB = 30° + \beta$. We now see that $\angle PDE = \angle BDE = 30° + \beta = \frac{1}{2}\angle CDE$, so P is on the angle bisector of $\angle CDE$. Similarly, P lies on the angle bisector of $\angle CED$, so P is the incenter of CDE. □

Solution D. We draw the lines DP and EP and let D' resp. E' be the second intersection point with Γ.



The triangles APD' and DPB are similar, and the triangles APE' and EPB are also similar, hence they are all isosceles and it follows that E', O, P, D' lie on a circle with center A. In particular AOD' and AOE' are equilateral triangles. Angle chasing gives

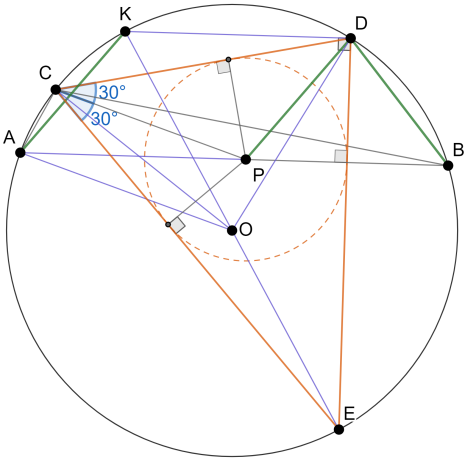
$$\angle CDP = \angle CDD' = \frac{1}{2}\angle COD' = \frac{1}{2}(60^\circ + \angle COA)$$

$$\angle EDP = \angle EDD' = \angle EE'D' = \angle PE'D' = \frac{1}{2}\angle PAD' = \frac{1}{2}(60^\circ + \angle PAO)$$

Similarly we prove $\angle CEP = \frac{1}{2}(60^\circ - \angle COA)$ and $\angle DEP = \frac{1}{2}(60^\circ - \angle PAO)$ so if we can prove that $\angle COA = \angle PAO$, we will have proven that P belongs to the angle bisector of $\angle CED$ and to the angle bisector of $\angle CDE$, which is enough to prove that P is the incenter of the triangle CDE.
Let $\beta = \angle ABC = \angle PCB$. We have $\angle APC = 2\beta$ and $\angle AOC = 2\beta$, so $AOPC$ is an inscribed quadrilateral. Moreover, since the diagonals AP and CO have equal length, this is actually an isosceles trapezoid, and hence $\angle PAO = \angle CPA = \angle COA$ which concludes the proof.

□

Solution E. Without loss of generality we assume that D and C are in the same half-plane regarding line AB.
Since $PC = BP$ and $ABCD$ is inscribed quadrilateral we have $\angle PCB = \angle CBP = \angle CEA = \alpha$.
As in the other solutions, $AOPC$ is an isosceles trapezoid and $2\alpha = \angle CPA = \angle PCO$

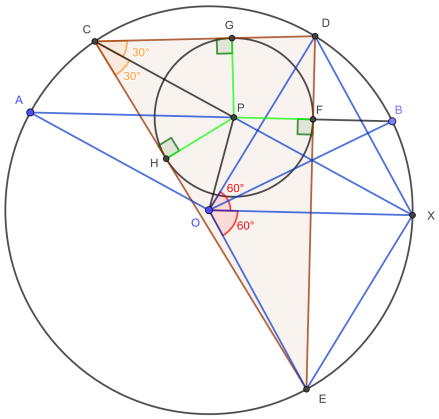


Let K be intersection of EO and Γ. Then $\angle KDE = 90^o$, $AB \| DK$ and $KDBA$ is isosceles trapezoid. We obtain $DP = BD = AK$, which implies that $DPAK$ is a parallelogram and hence $DK = PA = R = OD = OK$. We see that DOK is an equilateral triangle. Then $\angle ECD = \angle EKD = 60^o$.
Further we prove that PC bisects $\angle ECD$ using $\angle DCB = \angle DKB = \angle KDA$ (from isosceles trapeziod $DKAB$) and that $\angle KEC = \angle OEC = \angle OCE$ (from isosceles triangle OCE):

$$\angle DCP = \angle DCB + \angle BCP = \angle KDA + \alpha = \angle KEC + \angle CEA + \alpha = \angle OCE + 2\alpha = \angle PCE$$

Further by $\angle DCP = \angle PCE = \frac{1}{2}\angle ECD = 30°$ distances between P and sides $\triangle CDE$ are $\frac{1}{2}PC = \frac{1}{2}PB$ (as ratio between cathetus and hypotenuse in right triangle with angles 60° and 30°). So, we have found incentre. □

Solution F. Let F, G, H be the projections of P on the sides DE, DC resp. CE. If P is indeed the incenter, then the three line segments PF, PG, PH have the same length. This means that the problem is equivalent to proving that $PG = PH = PF = \frac{1}{2}PB = \frac{1}{2}PC$ and thus trigonometry in the right-angled triangles CPG and CPH tells us that it is enough to prove that $\angle DCP = \angle ECP = 30°$.



We introduce the point X as the second intersection of the line CP with Γ. Because O is the center of Γ we can reduce the problem to proving that $\angle XOD = \angle XOE = 60°$, or equivalently that XOD and XOE are equilateral triangles. This last condition is equivalent to X being the reflection of O on the line DE. Following the chain of equivalences, we see therefore that in order to solve the problem it is enough to prove that X is the reflection of O on DE. We prove this property as in Solution B, using the fact that $OXBP$ is an isosceles trapezoid.

□

Solution G. Assume AB is parallel to the horizontal axis, and that Γ is the unit circle. Write $f(\theta)$ for the point $(\cos(\theta), \sin(\theta))$ on Γ. As in Solution C, assume that $\angle AOB = 180° - 4\beta$; then we can take $B = f(2\beta)$ and $A = f(180° - 2\beta)$. As in Solution C, we observe that $AOPC$ is an isosceles trapezoid, which we use to deduce that $\angle ABC = \frac{1}{2}\angle APC = \frac{1}{2}\angle OAB = \beta$. We now know that $C = f(180° - 4\beta)$.
The point P lies on AB with $AP = R = 1$, so $P = (\cos(180° - 2\beta) + 1, \sin(2\beta)) = (1 - \cos(2\beta), \sin(2\beta))$. The midpoint of BP therefore has coordinates $(\frac{1}{2}, \sin(2\beta))$, so D and E have x-coordinate $\frac{1}{2}$. Without loss of generality, we take $D = f(60°)$ and $E = f(-60°)$.
We have now obtained coordinates for all points in the problem, with one free parameter (β). To show that P is the incenter of CDE, we will show that P lies on the bisector of $\angle CDE$; analogously, one can show that P lies on the bisector of angle CED. The bisector of angle CDE passes through the midpoint M of the arc CE not containing DE; because $C = f(180° - 4\beta)$ and $E = f(-60°)$, we have $M = f(240° - 2\beta)$.
It remains to show that $P = (1 - \cos(2\beta), \sin(2\beta))$ lies on the line connecting the points $D = (\cos(60°), \sin(60°))$ and $M = (\cos(240° - 2\beta), \sin(240° - 2\beta)) = (-\cos(60° - 2\beta), -\sin(60° - 2\beta))$. The equation for the line DM is

$$(Y + \sin(60° - 2\beta))(\cos(60°) + \cos(60° - 2\beta)) = (\sin(60°) + \sin(60° - 2\beta))\,(X + \cos(60° - 2\beta))\,,$$

which, using the fact that $\cos(60°) + \cos(60° - 2\beta) = 2\cos(\beta)\cos(60° - \beta)$ and $\sin(60°) + \sin(60° - 2\beta) = 2\cos(\beta)\sin(60° - \beta)$, simplifies to

$$(Y + \sin(60° - 2\beta))\cos(60° - \beta) = (X + \cos(60° - 2\beta))\sin(60° - \beta).$$

Because $\cos(60° - 2\beta)\sin(60° - \beta) - \sin(60° - 2\beta)\cos(60° - \beta) = \sin(\beta)$, this equation further simplifies to

$$Y\cos(60° - \beta) - X\sin(60° - \beta) = \sin(\beta).$$

Plugging in the coordinates of P, i.e., $X = 1 - \cos(2\beta)$ and $Y = \sin(2\beta)$, shows that P is on this line: for this choice of X and Y, the left hand side equals $\sin(60° + \beta) - \sin(60° - \beta) = 2\cos(60°)\sin(\beta)$, which is indeed equal to $\sin(\beta)$. So P lies on the bisector DM of $\angle CDE$, as desired. □

Solution H. Let Γ be the complex unit circle and let AB be parallel with the real line and $0 < \varphi = \arg b < \frac{\pi}{2}$. Then

$$|b| = 1, \quad a = -\overline{b}, \quad p = a + 1 = 1 - \overline{b}.$$

From $\operatorname{Re} d = \operatorname{Re} e = \operatorname{Re} \frac{p+b}{2} = \frac{1}{2}$ we get that $d = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $e = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ are conjugate 6th roots of unity; $d^3 = e^3 = -1$, $d + e = 1$, $d^2 = -e$, $e^2 = -d$ etc.
Point C is the reflection of B in line OP. From $\arg p = \arg(1 - \overline{b}) = \frac{1}{2}(\pi - \varphi)$, we can get $\arg c = 2\arg p - \arg b = \pi - 2\varphi$, so $c = -\overline{b}^2$.
Now we can verify that EP bisects $\angle CED$. This happens if and only if $(p-e)^2(\overline{c} - \overline{e})(\overline{d} - \overline{e})$ is real. Since $\overline{d} - \overline{e} = -\sqrt{3}i$, this is equivalent with $\operatorname{Re}\left[(p-e)^2(\overline{c} - \overline{e})\right] = 0$. Here

$$\begin{aligned}(p-e)^2(\overline{c} - \overline{e}) &= (1 - \overline{b} - e)^2(-b^2 - d) = (d - \overline{b})^2(-b^2 - d) \\ &= -|b|^4 + 2d|b|^2 b - d^2 b^2 - d\overline{b}^2 + 2d^2\overline{b} - d^3 \\ &= -1 - 2db + \overline{d}b^2 - d\overline{b}^2 - 2\overline{db} + 1 \\ &= -2(db - \overline{db}) + (\overline{d}b^2 - d\overline{b}^2),\end{aligned}$$

whose real part is zero. It can be proved similarly that DP bisects $\angle EDC$. □

Solutions to Problem 6

Answer. The only such m are $m = 2$ and $m = 10$.

Solution A. Consider an integer $m > 1$ for which the sequence defined in the problem statement contains only perfect squares. We shall first show that $m - 1$ is a power of 3.
Suppose that $m-1$ is even. Then $a_4 = 5m-1$ should be divisible by 4 and hence $m \equiv 1 \pmod 4$. But then $a_5 = 5m^2 + 3m - 1 \equiv 3 \pmod 4$ cannot be a square, a contradiction. Therefore $m - 1$ is odd.
Suppose that an odd prime $p \neq 3$ divides $m - 1$. Note that $a_n - a_{n-1} \equiv a_{n-2} - a_{n-3} \pmod p$. It follows that modulo p the sequence takes the form 1, 1, 4, 4, 7, 7, 10, 10, ...; indeed, a simple induction shows that $a_{2k} \equiv a_{2k-1} \equiv 3k - 2 \pmod p$ for $k \geq 1$. Since $\gcd(p, 3) = 1$ we get that the sequence $a_n \pmod p$ contains all the residues modulo p, a contradiction since only $(p+1)/2$ residues modulo p are squares. This shows that $m - 1$ is a power of 3.
Let h, k be integers such that $m = 3^k + 1$ and $a_4 = h^2$. We then have $5 \cdot 3^k = (h - 2)(h + 2)$. Since $\gcd(h - 2, h + 2) = 1$, it follows that $h - 2$ equals either $1, 3^k$ or 5, and $h + 2$ equals either $5 \cdot 3^k, 5$ or 3^k, respectively. In the first two cases we get $k = 0$ and in the last case we get $k = 2$. This implies that either $m = 2$ or $m = 10$.
We now show the converse. Suppose that $m = 2$ or $m = 10$. Let $t = 1$ or $t = 3$ so that $m = t^2+1$. Let $b_1, b_2, b_3, \ldots$ be a sequence of integers defined by $b_1 = 1, b_2 = 1, b_3 = 2$, and

$$b_n = tb_{n-1} + b_{n-2}\,, \quad \text{for all } n \geq 4\,.$$

Clearly, $a_n = b_n^2$ for $n = 1, 2, 3$. Note that if $m = 2$ then $a_4 = 9$ and $b_4 = 3$, and if $m = 10$ then $a_4 = 49$ and $b_4 = 7$. In both the cases we have $a_4 = b_4^2$.
If $n \geq 5$ then we have

$$b_n^2 + b_{n-3}^2 = (tb_{n-1} + b_{n-2})^2 + (b_{n-1} - tb_{n-2})^2 = (t^2 + 1)(b_{n-1}^2 + b_{n-2}^2) = m(b_{n-1}^2 + b_{n-2}^2)\,.$$

Therefore, it follows by induction that $a_n = b_n^2$ for all $n \geq 1$. This completes the solution. □

Solution B. We present an alternate proof that $m = 2$ and $m = 10$ are the only possible values of m with the required property.
Note that

$$\begin{aligned} a_4 &= 5m - 1\,, \\ a_5 &= 5m^2 + 3m - 1\,, \\ a_6 &= 5m^3 + 8m^2 - 2m - 4\,. \end{aligned}$$

Since a_4 and a_6 are squares, so is a_4a_6. We have

$$4a_4a_6 = 100m^4 + 140m^3 - 72m^2 - 72m + 16\,.$$

Notice that

$$\begin{aligned} (10m^2 + 7m - 7)^2 &= 100m^4 + 140m^3 - 91m^2 - 98m + 49 < 4a_4a_6\,, \\ (10m^2 + 7m - 5)^2 &= 100m^4 + 140m^3 - 51m^2 - 70m + 25 > 4a_4a_6\,, \end{aligned}$$

so we must have

$$4a_4a_6 = (10m^2 + 7m - 6)^2 = 100m^4 + 140m^3 - 71m^2 - 84m + 36\,.$$

This implies that $m^2 - 12m + 20 = 0$, so $m = 2$ or $m = 10$. □

Day 1. Solutions

Problem 1 (Netherlands). Find all triples (a, b, c) of real numbers such that $ab + bc + ca = 1$ and

$$a^2b + c = b^2c + a = c^2a + b.$$

Solution 1. First suppose that $a = 0$. Then we have $bc = 1$ and $c = b^2c = b$. So $b = c$, which implies $b^2 = 1$ and hence $b = \pm 1$. This leads to the solutions $(a, b, c) = (0, 1, 1)$ and $(a, b, c) = (0, -1, -1)$. Similarly, $b = 0$ gives the solutions $(a, b, c) = (1, 0, 1)$ and $(a, b, c) = (-1, 0, -1)$, while $c = 0$ gives $(a, b, c) = (1, 1, 0)$ and $(a, b, c) = (-1, -1, 0)$.

Now we may assume that $a, b, c \neq 0$. We multiply $ab + bc + ca = 1$ by a to find $a^2b + abc + ca^2 = a$, hence $a^2b = a - abc - a^2c$. Substituting this in $a^2b + c = b^2c + a$ yields $a - abc - a^2c + c = b^2c + a$, so $b^2c + abc + a^2c = c$. As $c \neq 0$, we find $b^2 + ab + a^2 = 1$.

Analogously we have $b^2 + bc + c^2 = 1$ and $a^2 + ac + c^2 = 1$. Adding these three equations yields $2\left(a^2 + b^2 + c^2\right) + ab + bc + ca = 3$, which implies $a^2 + b^2 + c^2 = 1$. Combining this result with $b^2 + ab + a^2 = 1$, we get $1 - ab = 1 - c^2$, so $c^2 = ab$.

Analogously we also have $b^2 = ac$ and $a^2 = bc$. In particular we now have that ab, bc and ca are all positive. This means that a, b and c must all be positive or all be negative. Now assume that $|c|$ is the largest among $|a|$, $|b|$ and $|c|$, then $c^2 \geq |ab| = ab = c^2$, so we must have equality. This means that $|c| = |a|$ and $|c| = |b|$. Since (a, b, c) must all have the same sign, we find $a = b = c$. Now we have $3a^2 = 1$, hence $a = \pm\frac{1}{3}\sqrt{3}$. We find the solutions $(a, b, c) = \left(\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}\right)$ and $(a, b, c) = \left(-\frac{1}{3}\sqrt{3}, -\frac{1}{3}\sqrt{3}, -\frac{1}{3}\sqrt{3}\right)$.

We conclude that all possible triples (a, b, c) are $(0, 1, 1)$, $(0, -1, -1)$, $(1, 0, 1)$, $(-1, 0, -1)$, $(1, 1, 0)$, $(-1, -1, 0)$, $\left(\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}\right)$ and $\left(-\frac{1}{3}\sqrt{3}, -\frac{1}{3}\sqrt{3}, -\frac{1}{3}\sqrt{3}\right)$.

Solution 2. From the problem statement $ab = 1 - bc - ca$ and thus $b^2c + a = a^2b + c = a - abc - a^2c + c$, $c\left(b^2 + a^2 + ab - 1\right) = 0$. If $c = 0$ then $ab = 1$ and $a^2b = b$, which implies $a = b = \pm 1$. Otherwise $b^2 + a^2 + ab = 1$. Cases $a = 0$ and $b = 0$ are completely analogous to $c = 0$, so we may suppose that $a, b, c \neq 0$. In this case we end up with

$$\begin{cases} a^2 + b^2 + ab = 1, \\ b^2 + c^2 + bc = 1, \\ c^2 + a^2 + ca = 1, \\ ab + bc + ca = 1. \end{cases}$$

Adding first three equations and subtracting the fourth yields $2(a^2 + b^2 + c^2) = 2 = 2(ab + bc + ca)$. Consequently, $(a - b)^2 + (b - c)^2 + (c - a)^2 = 0$. Now we can easily conclude that $a = b = c = \pm\frac{1}{\sqrt{3}}$.

Solution by Achilleas Sinefakopoulos, Greece. We have

$$c(1 - b^2) = a(1 - ab) = a(bc + ca) = c(ab + a^2),$$

and so

$$c(a^2 + ab + b^2 - 1) = 0.$$

Similarly, we have

$$b(a^2 + ac + c^2 - 1) = 0 \quad \text{and} \quad a(b^2 + bc + c^2 - 1) = 0.$$

If $c = 0$, then we get $ab = 1$ and $a^2b = a = b$, which give us $a = b = 1$, or $a = b = -1$. Similarly, if $a = 0$, then $b = c = 1$, or $b = c = -1$, while if $b = 0$, then $a = c = 1$, or $a = c = -1$.

So assume that $abc \neq 0$. Then

$$a^2 + ab + b^2 = b^2 + bc + c^2 = c^2 + ca + a^2 = 1.$$

Adding these gives us

$$2(a^2 + b^2 + c^2) + ab + bc + ca = 3,$$

and using the fact that $ab + bc + ca = 1$, we get

$$a^2 + b^2 + c^2 = 1 = ab + bc + ca.$$

Hence

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 2(a^2 + b^2 + c^2) - 2(ab + bc + ca) = 0$$

and so $a = b = c = \pm\dfrac{1}{\sqrt{3}}$.

Therefore, the solutions (a, b, c) are $(0, 1, 1)$, $(0, -1, -1)$, $(1, 0, 1)$, $(-1, 0, -1)$, $(1, 1, 0)$, $(-1, -1, 0)$, $(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$, $(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$

Solution by Eirini Miliori (HEL2). It is $ab + bc + ca = 1$ and

$$a^2b + c = b^2c + a = c^2a + b. \tag{1}$$

We have

$$\begin{aligned} a^2b + c = b^2c + a &\iff a^2b - a = b^2c - c \\ &\iff a(ab - 1) = c(b^2 - 1) \\ &\iff a(-bc - ac) = c(b^2 - 1) \\ &\iff -ac(a + b) = c(b^2 - 1) \end{aligned} \tag{2}$$

First, consider the case where one of a, b, c is equal to 0. Without loss of generality, assume that $a = 0$. Then $bc = 1$ and $b = c$ from (1), and so $b^2 = 1$ giving us $b = 1$ or -1. Hence $b = c = 1$ or $b = c = -1$.

Therefore, (a, b, c) equals one of the triples $(0, 1, 1)$, $(0, -1, -1)$, as well as their rearrangements $(1, 0, 1)$ and $(-1, 0, -1)$ when $b = 0$, or $(1, 1, 0)$ and $(-1, -1, 0)$ when $c = 0$.

Now consider the case where $a \neq 0$, $b \neq 0$ and $c \neq 0$. Then (2) gives us

$$-a(a + b) = b^2 - 1 \iff -a^2 - ab = b^2 - 1 \iff a^2 + ab + b^2 - 1 = 0.$$

The quadratic $P(x) = x^2 + bx + b^2 - 1$ has $x = a$ as a root. Let x_1 be its second root (which could be equal to a in the case where the discriminant is 0). From Vieta's formulas we get

$$\begin{cases} x_1 + a = -b & \iff x_1 = -b - a, \text{ and} \\ x_1 a = b^2 - 1 & \iff x_1 = \frac{b^2 - 1}{a}. \end{cases}$$

Using $a^2b+c=c^2a+b$ we obtain $b(a^2-1)=c(ac-1)$ yielding $a^2+ac+c^2-1=0$ in a similar way. The quadratic $Q(x)=x^2+cx+c^2-1$ has $x=a$ as a root. Let x_2 be its second root (which could be equal to a in the case where the discriminant is 0). From Vieta's formulas we get

$$\begin{cases} x_2+a=-c & \iff x_2=-c-a, \text{ and} \\ x_2a=c^2-1 & \iff x_2=\frac{c^2-1}{a}. \end{cases}$$

Then

$$\begin{cases} x_1+x_2=-b-a-c-a, \text{ and} \\ x_1+x_2=\frac{b^2-1}{a}+\frac{c^2-1}{a}, \end{cases}$$

which give us

$$\begin{aligned} -(2a+b+c)=\frac{b^2-1}{a}+\frac{c^2-1}{a} &\iff -2a^2-ba-ca=b^2+c^2-2 \\ &\iff bc-1-2a^2=b^2+c^2-2 \\ &\iff 2a^2+b^2+c^2=1+bc. \end{aligned} \tag{3}$$

By symmetry, we get

$$2b^2+a^2+c^2=1+ac, \text{ and} \tag{4}$$
$$2c^2+a^2+b^2=1+bc \tag{5}$$

Adding equations (3), (4), and (5), we get

$$4(a^2+b^2+c^2)=3+ab+bc+ca \iff 4(a^2+b^2+c^2)=4 \iff a^2+b^2+c^2=1.$$

From this and (3), since $ab+bc+ca=1$, we get

$$a^2=bc=1-ab-ac \iff a(a+b+c)=1.$$

Similarly, from (4) we get

$$b(a+b+c)=1,$$

and from (4),

$$c(a+b+c)=1.$$

Clearly, it is $a+b+c\neq 0$ (for otherwise it would be $0=1$, a contradiction). Therefore,

$$a=b=c=\frac{1}{a+b+c},$$

and so $3a^2=1$ giving us $a=b=c=\pm\frac{1}{\sqrt{3}}$.

In conclusion, the solutions (a,b,c) are $(0,1,1)$, $(0,-1,-1)$, $(1,0,1)$, $(-1,0,-1)$, $(1,1,0)$, $(-1,-1,0)$, $(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}})$, and $(-\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}})$.

Solution by ISR5. First, homogenize the condition $a^2b+c=b^2c+a=c^2a+b$ by replacing c by $c(ab+bc+ca)$ (etc.), yielding

$$a^2b+c=a^2b+abc+bc^2+c^2a=abc+\sum_{cyc}a^2b+(c^2b-b^2c)=abc+\sum_{cyc}a^2b+bc(c-b).$$

Thus, after substracting the cyclicly symmetric part $abc + \sum_{cyc} a^2b$ we find the condition is eqivalent to

$$D := bc(c-b) = ca(a-c) = ab(b-a).$$

Ending 1. It is easy to see that if e.g. $a = 0$ then $b = c = \pm 1$, and if e.g. $a = b$ then either $a = b = c = \pm\frac{1}{\sqrt{3}}$ or $a = b = \pm 1, c = 0$, and these are indeed solutions. So, to show that these are *all* solutions (up to symmetries), we may assume by contradiction that a, b, c are pairwise different and non-zero. All conditions are preserved under cyclic shifts and under simultaenously switching signs on all a, b, c, and by applying these operations as necessary we may assume $a < b < c$. It follows that $D^3 = a^2b^2c^2(c-b)(a-c)(b-a)$ must be negative (the only negative term is $a - c$, hence D is negative, i.e. $bc, ab < 0 < ac$. But this means that a, c have the same sign and b has a different one, which clearly contradicts $a < b < c$! So, such configurations are impossible.

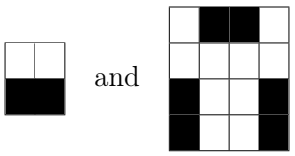
Ending 2. Note that $3D = \sum c^2b - \sum b^2c = (c-b)(c-a)(b-a)$ and $D^3 = a^2b^2c^2(c-b)(a-c)(b-a) = -3a^2b^2c^2D$. Since $3D$ and D^3 must have the same sign, and $-3a^2b^2c^2$ is non-positive, necessarily $D = 0$. Thus (up to cyclic permutation) $a = b$ and from there we immediately find either $a = b = \pm 1, c = 0$ or $a = b = c = \pm\frac{1}{\sqrt{3}}$.

Problem 2 (Luxembourg). Let n be a positive integer. Dominoes are placed on a $2n \times 2n$ board in such a way that every cell of the board is adjacent to exactly one cell covered by a domino. For each n, determine the largest number of dominoes that can be placed in this way.

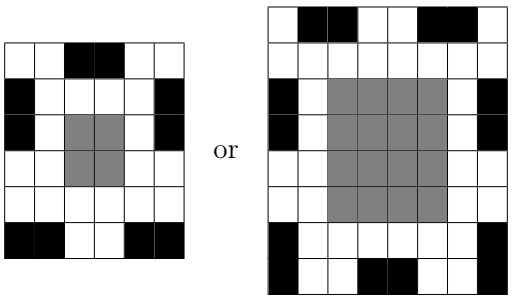
(A *domino* is a tile of size 2×1 or 1×2. Dominoes are placed on the board in such a way that each domino covers exactly two cells of the board, and dominoes do not overlap. Two cells are said to be *adjacent* if they are different and share a common side.)

Solution 1. Let M denote the maximal number of dominoes that can be placed on the chessboard. We claim that $M = n(n+1)/2$. The proof naturally splits into two parts: we first prove that $n(n+1)/2$ dominoes can be placed on the board, and then show that $M \leq n(n+1)/2$ to complete the proof.

We construct placings of the dominoes by induction. The base cases $n = 1$ and $n = 2$ correspond to the placings



Next, we add dominoes to the border of a $2n \times 2n$ chessboard to obtain a placing of dominoes for the $2(n+2) \times 2(n+2)$ board,



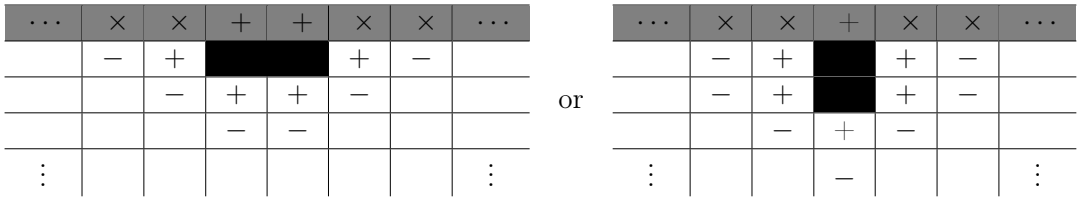
depending on whether n is odd or even. In these constructions, the interior square is filled with the placing for the $2n \times 2n$ board. This construction adds $2n + 3$ dominoes, and therefore places, in total,

$$\frac{n(n+1)}{2} + (2n+3) = \frac{(n+2)(n+3)}{2}$$

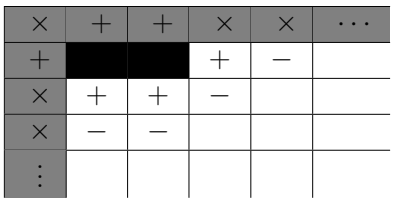
dominoes on the board. Noticing that the contour and the interior mesh together appropriately, this proves, by induction, that $n(n+1)/2$ dominoes can be placed on the $2nn$ board.

To prove that $M \leq n(n+1)/2$, we border the $2n \times 2n$ square board up to a $(2n+2) \times (2n+2)$ square board; this adds $8n + 4$ cells to the $4n^2$ cells that we have started with. Calling a cell covered if it belongs to a domino or is adjacent to a domino, each domino on the $2n \times 2n$ board is seen to cover exactly 8 cells of the $(2n+2) \times (2n+2)$ board (some of which may belong to the border). By construction, each of the $4n^2$ cells of the $2n \times 2n$ board is covered by precisely one domino.

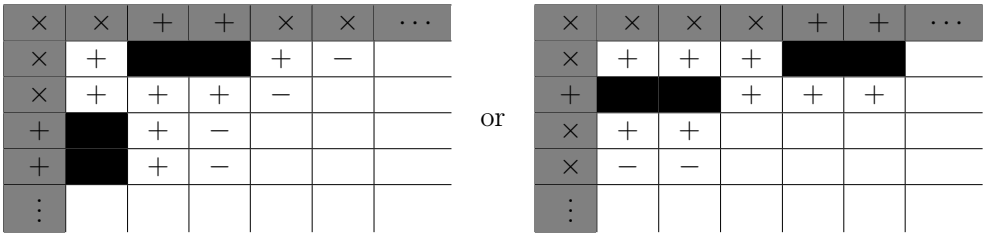
If two adjacent cells on the border, away from a corner, are covered, then there will be at least two uncovered cells on both sides of them; if one covered cell lies between uncovered cells, then again, on both sides of it there will be at least two uncovered cells; three or more adjacent cells cannot be all covered. The following diagrams, in which the borders are shaded, $\times$ marks an uncovered cell on the border, $+$ marks a covered cell not belonging to a domino, and $-$ marks a cell which cannot belong to a domino, summarize the two possible situations,



Close to a corner of the board, either the corner belongs to some domino,



or one of the following situations, in which the corner cell of the original board is not covered by a domino, may occur:

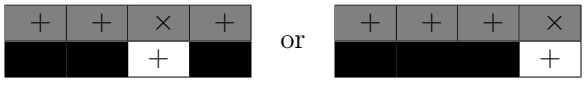


It is thus seen that at most half of the cells on the border, i.e. $4n+2$ cells, may be covered, and hence

$$M \leq \left[\frac{4n^2+(4n+2)}{8}\right] = \left[\frac{n(n+1)}{2}+\frac{1}{2}\right] = \frac{n(n+1)}{2},$$

which completes the proof of our claim.

Solution 2. We use the same example as in Solution 1. Let M denote the maximum number of dominoes which satisfy the condition of the problem. To prove that $M \leq n(n+1)/2$, we again border the $2n \times 2n$ square board up to a $(2n+2) \times (2n+2)$ square board. In fact, we shall ignore the corner border cells as they cannot be covered anyway and consider only the $2n$ border cells along each side. We prove that out of each four border cells next to each other at most two can be covered. Suppose three out of four cells A, B, C, D are covered. Then there are two possibilities below:



The first option is that A, B and D are covered (marked with + in top row). Then the cells inside the starting square next to A, B and D are covered by the dominoes, but the cell in between them has now two adjacent cells with dominoes, contradiction. The second option is that A, B and C are covered. Then the cells inside the given square next to A, B and C are covered by the dominoes. But then the cell next to B has two adjacent cells with dominoes, contradiction.

Now we can split the border cells along one side in groups of 4 (leaving one group of 2 if n is odd). So when n is even, at most n of the $2n$ border cells along one side can be covered, and when n is odd, at most $n+1$ out of the $2n$ border cells can be covered. For all four borders together, this gives a contribution of $4n$ when n is even and $4n+4$ when n is odd. Adding $4n^2$ and dividing by 8 we get the desired result.

Solution (upper bound) by ISR5. Consider the number of pairs of adjacent cells, such that one of them is covered by a domino. Since each cell is adjacent to one covered cell, the number of such pairs is exactly $4n^2$. On the other hand, let n_2 be the number of covered corner cells, n_3 the number of covered edge cells (cells with 3 neighbours), and n_4 be the number of covered interior cells (cells with 4 neighbours). Thus the number of pairs is $2n_2 + 3n_3 + 4n_4 = 4n^2$, whereas the number of dominoes is $m = \frac{n_2+n_3+n_4}{2}$.

Considering only the outer frame (of corner and edge cells), observe that every covered cell dominates two others, so at most half of the cells are ccovered. The frame has a total of $4(2n-1)$ cells, i.e. $n_2+n_3 \leq 4n-2$. Additionally $n_2 \leq 4$ since there are only 4 corners, thus

$$8m = 4n_2+4n_3+4n_4 = (2n_2+3n_3+4n_4)+(n_2+n_3)+n_2 \leq 4n^2+(4n-2)+4 = 4n(n+1)+2$$

Thus $m \leq \frac{n(n+1)}{2} + \frac{1}{4}$, so in fact $m \leq \frac{n(n+1)}{2}$.

Solution (upper and lower bound) by ISR5. We prove that this is the upper bound (and also the lower bound!) by proving that any two configurations, say A and B, must contain exactly the same number of dominoes.

Colour the board in a black and white checkboard colouring. Let W be the set of white cells covered by dominoes of tiling A. For each cell $w \in W$ let N_w be the set of its adjacent (necessarily black) cells. Since each black cell has exactly one neighbour (necessarily white) covered by a domino of tiling A, it follows that each black cell is contained in exactly one N_w, i.e. the N_w form a partition of the black cells. Since each white cell has exactly one (necessarily black) neighbour covered by a tile of B, each B_w contains exactly one black tile covered by a domino of B. But, since each domino covers exactly one white and one black cell, we have

$$|A| = |W| = |\{N_w : w \in W\}| = |B|$$

as claimed.

Problem 3 (Poland). Let ABC be a triangle such that $\angle CAB > \angle ABC$, and let I be its incentre. Let D be the point on segment BC such that $\angle CAD = \angle ABC$. Let ω be the circle tangent to AC at A and passing through I. Let X be the second point of intersection of ω and the circumcircle of ABC. Prove that the angle bisectors of $\angle DAB$ and $\angle CXB$ intersect at a point on line BC.

Solution 1. Let S be the intersection point of BC and the angle bisector of $\angle BAD$, and let T be the intersection point of BC and the angle bisector of $\angle BXC$. We will prove that both quadruples A, I, B, S and A, I, B, T are concyclic, which yields $S = T$.

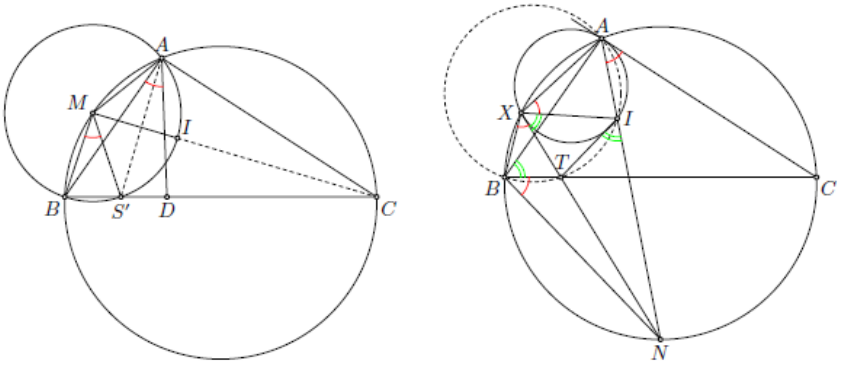
Firstly denote by M the middle of arc AB of the circumcenter of ABC which does not contain C. Consider the circle centered at M passing through A, I and B (it is well-known that $MA = MI = MB$); let it intersect BC at B and S'. Since $\angle BAC > \angle CBA$ it is easy to check that S' lies on side BC. Denoting the angles in ABC by α, β, γ we get

$$\angle BAD = \angle BAC - \angle DAC = \alpha - \beta.$$

Moreover since $\angle MBC = \angle MBA + \angle ABC = \frac{\gamma}{2} + \beta$, then

$$\angle BMS' = 180^\circ - 2\angle MBC = 180^\circ - \gamma - 2\beta = \alpha - \beta.$$

It follows that $\angle BAS' = 2\angle BMS' = 2\angle BAD$ which gives us $S = S'$.



Secondly let N be the middle of arc BC of the circumcenter of ABC which does not contain A. From $\angle BAC > \angle CBA$ we conclude that X lies on the arc AB of circumcircle of ABC not containing C. Obviously both AI and XT are passing through N. Since $\angle NBT = \frac{\alpha}{2} = \angle BXN$ we obtain $\triangle NBT \sim \triangle NXB$, therefore

$$NT \cdot NX = NB^2 = NI^2.$$

It follows that $\triangle NTI \sim \triangle NIX$. Keeping in mind that $\angle NBC = \angle NAC = \angle IXA$ we get

$$\angle TIN = \angle IXN = \angle NXA - \angle IXA = \angle NBA - \angle NBC = \angle TBA.$$

It means that A, I, B, T are concyclic which ends the proof.

Solution 2. Let $\angle BAC = \alpha$, $\angle ABC = \beta$, $\angle BCA = \gamma$ $\angle ACX = \phi$. Denote by W_1 and W_2 the intersections of segment BC with the angle bisectors of $\angle BXC$ and $\angle BAD$ respectively. Then $BW_1/W_1C = BX/XC$ and $BW_2/W_2D = BA/AD$. We shall show that $BW_1 = BW_2$.

Since $\angle DAC = \angle CBA$, triangles ADC and BAC are similar and therefore

$$\frac{DC}{AC} = \frac{AC}{BC}.$$

By the Law of sines

$$\frac{BW_2}{W_2D} = \frac{BA}{AD} = \frac{BC}{AC} = \frac{\sin\alpha}{\sin\beta}.$$

Consequently

$$\frac{BD}{BW_2} = \frac{W_2D}{BW_2} + 1 = \frac{\sin\beta}{\sin\alpha} + 1,$$

$$\frac{BC}{BW_2} = \frac{BC}{BD} \cdot \frac{BD}{BW_2} = \frac{1}{1 - DC/BC} \cdot \frac{BD}{BW_2} = \frac{1}{1 - AC^2/BC^2} \cdot \frac{BD}{BW_2} =$$

$$\frac{sin^2\alpha}{\sin^2\alpha - \sin^2\beta} \cdot \frac{\sin\beta + \sin\alpha}{\sin\alpha} = \frac{\sin\alpha}{\sin\alpha - \sin\beta}.$$

Note that $AXBC$ is cyclic and so $\angle BXC = \angle BAC = \alpha$. Hence, $\angle XBC = 180^\circ - \angle BXC - \angle BCX = 180^\circ - \alpha - \phi$. By the Law of sines for the triangle BXC, we have

$$\frac{BC}{W_1B} = \frac{W_1C}{W_1B} + 1 = \frac{CX}{BX} + 1 = \frac{\sin\angle CBX}{\sin\phi} + 1 =$$

$$\frac{\sin(\alpha + \phi)}{\sin\phi} + 1 = \sin\alpha\cot\phi + \cos\alpha + 1.$$

So, it's enough to prove that

$$\frac{\sin\alpha}{\sin\alpha - \sin\beta} = \sin\alpha\cot\phi + \cos\alpha.$$

Since AC is tangent to the circle AIX, we have $\angle AXI = \angle IAC = \alpha/2$. Moreover $\angle XAI = \angle XAB + \angle BAI = \phi + \alpha/2$ and $\angle XIA = 180^\circ - \angle XAI - \angle AXI = 180^\circ - \alpha - \phi$. Applying the Law of sines again XAC, XAI, IAC we obtain

$$\frac{AX}{\sin(\alpha + \phi)} = \frac{AI}{\sin\alpha/2},$$

$$\frac{AX}{\sin(\gamma - \phi)} = \frac{AC}{\sin\angle AXC} = \frac{AC}{\sin\beta},$$

$$\frac{AI}{\sin\gamma/2} = \frac{AC}{\sin(\alpha/2 + \gamma/2)}.$$

Combining the last three equalities we end up with

$$\frac{\sin(\gamma - \phi)}{\sin(\alpha + \phi)} = \frac{AI}{AC} \cdot \frac{\sin\beta}{\sin\alpha/2} = \frac{\sin\beta}{\sin\alpha/2} \cdot \frac{\sin\gamma/2}{\sin(\alpha/2 + \gamma/2)},$$

$$\frac{\sin(\gamma - \phi)}{\sin(\alpha + \phi)} = \frac{\sin\gamma\cot\phi - \cos\gamma}{\sin\alpha\cot\phi + \cos\alpha} = \frac{2\sin\beta/2\sin\gamma/2}{\sin\alpha/2},$$

$$\frac{\sin\alpha\sin\gamma\cot\phi-\sin\alpha\cos\gamma}{\sin\gamma\sin\alpha\cot\phi+\sin\gamma\cos\alpha}=\frac{2\sin\beta/2\cos\alpha/2}{\cos\gamma/2}$$

Subtracting 1 from both sides yields

$$\frac{-\sin\alpha\cos\gamma-\sin\gamma\cos\alpha}{\sin\gamma\sin\alpha\cot\phi+\sin\gamma\cos\alpha}=\frac{2\sin\beta/2\cos\alpha/2}{\cos\gamma/2}-1=$$

$$\frac{2\sin\beta/2\cos\alpha/2-\sin(\alpha/2+\beta/2)}{\cos\gamma/2}=\frac{\sin\beta/2\cos\alpha/2-\sin\alpha/2\cos\beta/2}{\cos\gamma/2},$$

$$\frac{-\sin(\alpha+\gamma)}{\sin\gamma\sin\alpha\cot\phi+\sin\gamma\cos\alpha}=\frac{\sin(\beta/2-\alpha/2)}{\cos\gamma/2},$$

$$\frac{-\sin\beta}{\sin\alpha\cot\phi+\cos\alpha}=2\sin\gamma/2\sin(\beta/2-\alpha/2)=$$

$$2\cos(\beta/2+\alpha/2)\sin(\beta/2-\alpha/2)=\sin\beta-\sin\alpha,$$

and the result follows. We are left to note that none of the denominators can vanish.

Solution by Achilleas Sinefakopoulos, Greece. We first note that

$$\angle BAD=\angle BAC-\angle DAC=\angle A-\angle B.$$

Let CX and AD meet at K. Then $\angle CXA=\angle ABC=\angle KAC$. Also, we have $\angle IXA=\angle A/2$, since ω is tangent to AC at A. Therefore,

$$\angle DAI=|\angle B-\angle A/2|=|\angle KXA-\angle IXA|=\angle KXI,$$

(the absolute value depends on whether $\angle B\geq\angle A/2$ or not) which means that $XKIA$ is cyclic, i.e. K lies also on ω.

Let IK meet BC at E. (If $\angle B=\angle A/2$, then IK degenerates to the tangent line to ω at I.) Note that $BEIA$ is cyclic, because

$$\angle EIA=180^\circ-\angle KXA=180^\circ-\angle ABE.$$

We have $\angle EKA=180^\circ-\angle AXI=180^\circ-\angle A/2$ and $\angle AEI=\angle ABI=\angle B/2$. Hence

$$\begin{aligned}\angle EAK&=180^\circ-\angle EKA-\angle AEI\\&=180^\circ-(180^\circ-\angle A/2)-\angle B/2\\&=(\angle A-\angle B)/2\\&=\angle BAD/2.\end{aligned}$$

This means that AE is the angle bisector of $\angle BAD$. Next, let M be the point of intersection of AE and BI. Then

$$\angle EMI=180^\circ-\angle B/2-\angle BAD/2=180^\circ-\angle A/2,$$

and so, its supplement is

$$\angle AMI = \angle A/2 = \angle AXI,$$

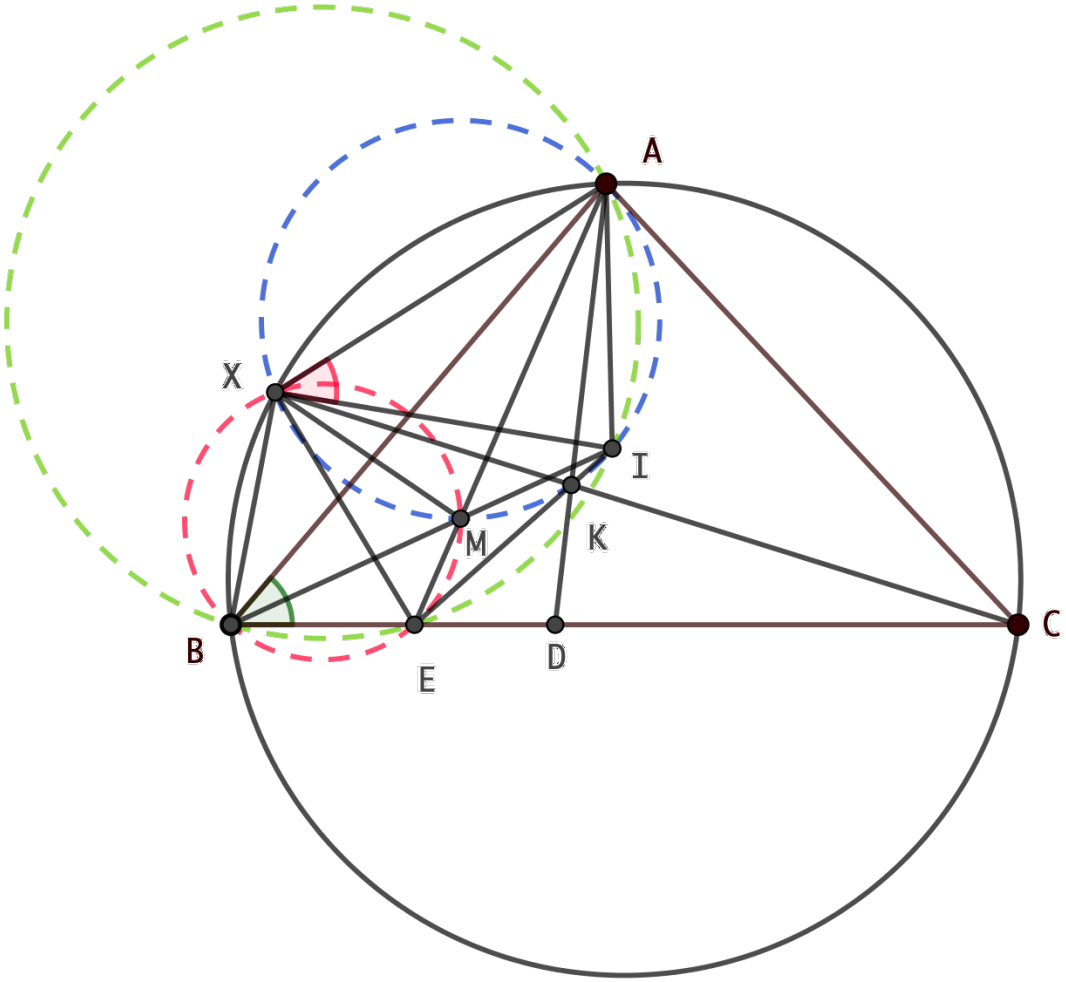
so X, M, K, I, A all lie on ω. Next, we have

$$\begin{aligned}\angle XMA &= \angle XKA \\ &= 180^\circ - \angle ADC - \angle XCB \\ &= 180^\circ - \angle A - \angle XCB \\ &= \angle B + \angle XCA \\ &= \angle B + \angle XBA \\ &= \angle XBE,\end{aligned}$$

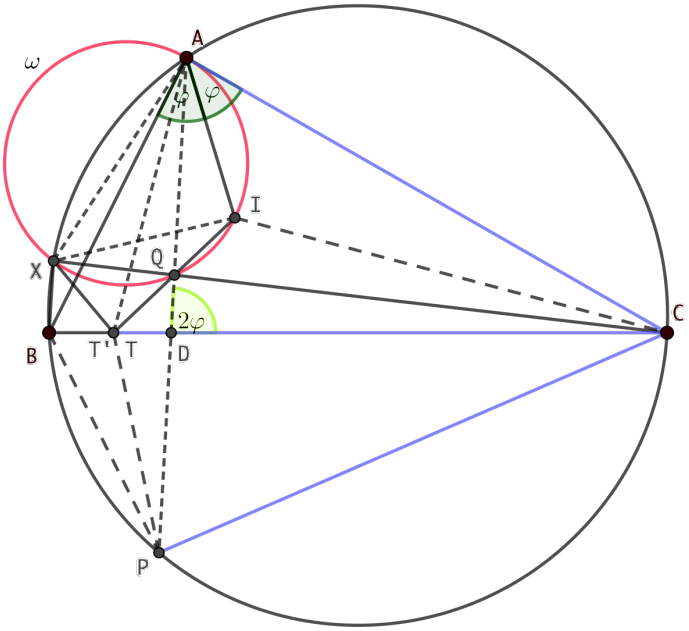
and so X, B, E, M are concyclic. Hence

$$\begin{aligned}\angle EXC &= \angle EXM + \angle MXC \\ &= \angle MBE + \angle MAK \\ &= \angle B/2 + \angle BAD/2 \\ &= \angle A/2 \\ &= \angle BXC/2.\end{aligned}$$

This means that XE is the angle bisector of $\angle BXC$ and so we are done!



Solution based on that by Eirini Miliori (HEL2), edited by A. Sinefakopoulos, Greece. It is $\angle ABD = \angle DAC$, and so $\overline{AC}$ is tangent to the circumcircle of $\triangle BAD$ at A. Hence $CA^2 = CD \cdot CB$.



Triangle $\triangle ABC$ is similar to triangle $\triangle CAD$, because $\angle C$ is a common angle and $\angle CAD = \angle ABC$, and so $\angle ADC = \angle BAC = 2\varphi$.

Let Q be the point of intersection of $\overline{AD}$ and $\overline{CX}$. Since $\angle BXC = \angle BAC = 2\varphi$, it follows that $BDQX$ is cyclic.Therefore, $CD \cdot CB = CQ \cdot CX = CA^2$ which implies that Q lies on ω.

Next let P be the point of intersection of $\overline{AD}$ with the circumcircle of triangle $\triangle ABC$. Then $\angle PBC = \angle PAC = \angle ABC = \angle APC$ yielding $CA = CP$. So, let T be on the side $\overline{BC}$ such that $CT = CA = CP$. Then

$$\angle TAD = \angle TAC - \angle DAC = \left(90^\circ - \frac{\angle C}{2}\right) - \angle B = \frac{\angle A - \angle B}{2} = \frac{\angle BAD}{2},$$

that is, line $\overline{AT}$ is the angle bisector of $\angle BAD$. We want to show that $\overline{XT}$ is the angle bisector of $\angle BXC$. To this end, it suffices to show that $\angle TXC = \varphi$.

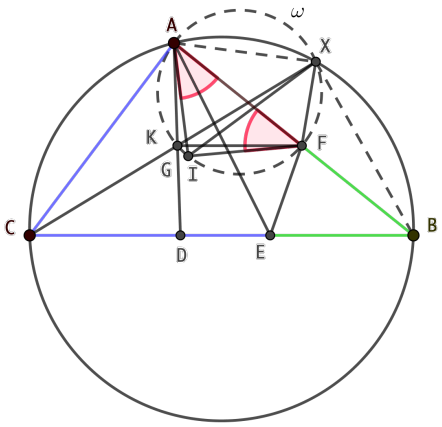
It is $CT^2 = CA^2 = CQ \cdot CX$, and so $\overline{CT}$ is tangent to the circumcircle of $\triangle XTQ$ at T. Since $\angle TXQ = \angle QTC$ and $\angle QDC = 2\varphi$, it suffices to show that $\angle TQD = \varphi$, or, in other words, that I, Q, and T are collinear.

Let T' is the point of intersection of $\overline{IQ}$ and $\overline{BC}$. Then $\triangle AIC$ is congruent to $\triangle T'IC$, since they share $\overline{CI}$ as a common side, $\angle ACI = \angle T'CI$, and

$$\angle IT'D = 2\varphi - \angle T'QD = 2\varphi - \angle IQA = 2\varphi - \angle IXA = \varphi = \angle IAC.$$

Therefore, $CT' = CA = CT$, which means that T coincides with T' and completes the proof.

Solution based on the work of Artemis-Chrysanthi Savva (HEL4), completed by A. Sinefakopoulos, Greece. Let G be the point of intersection of $\overline{AD}$ and $\overline{CX}$. Since the quadrilateral $AXBC$ is cyclic, it is $\angle AXC = \angle ABC$.



Let the line $\overline{AD}$ meet ω at K. Then it is $\angle AXK = \angle CAD = \angle ABC$, because the angle that is formed by a chord and a tangent to the circle at an endpoint of the chord equals the inscribed angle to that chord. Therefore, $\angle AXK = \angle AXC = \angle AXG$. This means that the point G coincides with the point K and so G belongs to the circle ω.

Let E be the point of intersection of the angle bisector of $\angle DAB$ with $\overline{BC}$. It suffices to show that

$$\frac{CE}{BE} = \frac{XC}{XB}.$$

Let F be the second point of intersection of ω with $\overline{AB}$. Then we have $\angle IAF = \frac{\angle CAB}{2} = \angle IXF$, where I is the incenter of $\triangle ABC$, because $\angle IAF$ and $\angle IXF$ are inscribed in the same arc of ω. Thus $\triangle AIF$ is isosceles with $AI = IF$. Since I is the incenter of $\triangle ABC$, we have $AF = 2(s-a)$, where $s = (a+b+c)/2$ is the semiperimeter of $\triangle ABC$. Also, it is $CE = AC = b$ because in triangle $\triangle ACE$, we have

$$\begin{aligned}\angle AEC &= \angle ABC + \angle BAE \\ &= \angle ABC + \frac{\angle BAD}{2} \\ &= \angle ABC + \frac{\angle BAC - \angle ABC}{2} \\ &= 90^\circ - \frac{\angle ACE}{2},\end{aligned}$$

and so $\angle CAE = 180^\circ - \angle AEC - \angle ACE = 90^\circ - \frac{\angle ACE}{2} = \angle AEC$. Hence

$$BF = BA - AF = c - 2(s-a) = a - b = CB - CE = BE.$$

Moreover, triangle $\triangle CAX$ is similar to triangle $\triangle BFX$, because $\angle ACX = \angle FBX$ and

$$\angle XFB = \angle XAF + \angle AXF = \angle XAF + \angle CAF = \angle CAX.$$

Therefore

$$\frac{CE}{BE} = \frac{AC}{BF} = \frac{XC}{XB},$$

as desired. The proof is complete.

Solution by IRL1 and IRL 5. Let ω denote the circle through A and I tangent to AC. Let Y be the second point of intersection of the circle ω with the line AD. Let L

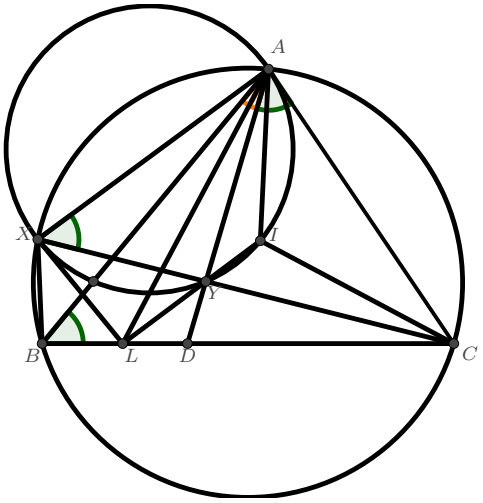
be the intersection of BC with the angle bisector of $\angle BAD$. We will prove $\angle LXC = 1/2\angle BAC = 1/2\angle BXC$.

We will refer to the angles of $\triangle ABC$ as $\angle A, \angle B, \angle C$. Thus $\angle BAD = \angle A - \angle B$.

On the circumcircle of $\triangle ABC$, we have $\angle AXC = \angle ABC = \angle CAD$, and since AC is tangent to ω, we have $\angle CAD = \angle CAY = \angle AXY$. Hence C, X, Y are collinear.

Also note that $\triangle CAL$ is isosceles with $\angle CAL = \angle CLA = \frac{1}{2}(\angle BAD) + \angle ABC = \frac{1}{2}(\angle A + \angle B)$ hence $AC = CL$. Moreover, CI is angle bisector to $\angle ACL$ so it's the symmetry axis for the triangle, hence $\angle ILC = \angle IAC = 1/2\angle A$ and $\angle ALI = \angle LIA = 1/2\angle B$. Since AC is tangent to ω, we have $\angle AYI = \angle IAC = 1/2\angle A = \angle LAY + \angle ALI$. Hence L, Y, I are collinear.

Since AC is tangent to ω, we have $\triangle CAY \sim \triangle CXA$ hence $CA^2 = CX \cdot CY$. However we proved $CA = CL$ hence $CL^2 = CX \cdot CY$. Hence $\triangle CLY \sim \triangle CXL$ and hence $\angle CXL = \angle CLY = \angle CAI = 1/2\angle A$.



Solution by IRL 5. Let M be the midpoint of the arc BC. Let ω denote the circle through A and I tangent to AC. Let N be the second point of intersection of ω with AB and L the intersection of BC with the angle bisector of $\angle BAD$. We know $\frac{DL}{LB} = \frac{AD}{AB}$ and want to prove $\frac{XB}{XC} = \frac{LB}{LC}$.

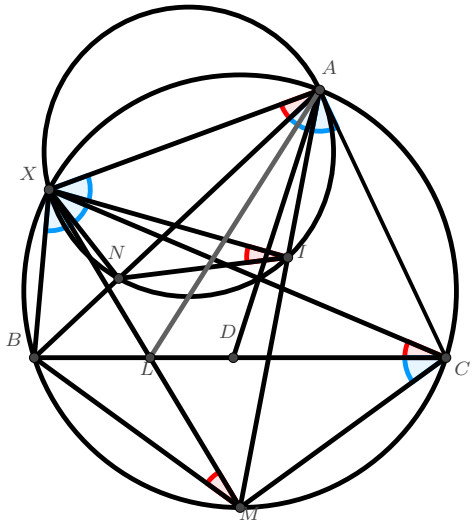
First note that $\triangle CAL$ is isosceles with $\angle CAL = \angle CLA = \frac{1}{2}(\angle BAD) + \angle ABC$ hence $AC = CL$ and $\frac{LB}{LC} = \frac{LB}{AC}$.

Now we calculate $\frac{XB}{XC}$:

Comparing angles on the circles ω and the circumcircle of $\triangle ABC$ we get $\triangle XIN \sim \triangle XMB$ and hence also $\triangle XIM \sim \triangle XNB$ (having equal angles at X and proportional adjoint sides). Hence $\frac{XB}{XM} = \frac{NB}{IM}$.

Also comparing angles on the circles ω and the circumcircle of $\triangle ABC$ and using the tangent AC we get $\triangle XAI \sim \triangle XCM$ and hence also $\triangle XAC \sim \triangle XIM$. Hence $\frac{XC}{XM} = \frac{AC}{IM}$.

Comparing the last two equations we get $\frac{XB}{XC} = \frac{NB}{AC}$. Comparing with $\frac{LB}{LC} = \frac{LB}{AC}$, it remains to prove $NB = LB$.



We prove $\triangle INB \equiv \triangle ILB$ as follows:

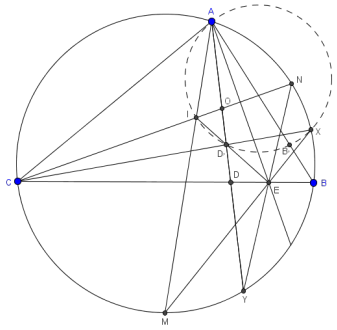
First, we note that I is the circumcentre of $\triangle ALN$. Indeed, CI is angle bisector in the isosceles triangle ACL so it's perpendicular bisector for AL. As well, $\triangle IAN$ is isosceles with $\angle INA = \angle CAI = \angle IAB$ hence I is also on the perpendicular bisector of AN.

Hence $IN = IL$ and also $\angle NIL = 2\angle NAL = \angle A - \angle B = 2\angle NIB$ (the last angle is calculated using that the exterior angle of $\triangle NIB$ is $\angle INA = \angle A/2$. Hence $\angle NIB = \angle LIB$ and $\triangle INB \equiv \triangle ILB$ by SAS.

Solution by ISR5 (with help from IRL5). Let M, N be the midpoints of arcs BC, BA of the circumcircle ABC, respectively. Let Y be the second intersection of AD and circle ABC. Let E be the incenter of triangle ABY and note that E lies on the angle bisectors of the triangle, which are the lines YN (immediate), BC (since $\angle CBY = \angle CAY = \angle CAD = \angle ABC$) and the angle bisector of $\angle DAB$; so the question reduces to showing that E is also on XM, which is the angle bisector of $\angle CXB$.

We claim that the three lines CX, ADY, IE are concurrent at a point D'. We will complete the proof using this fact, and the proof will appear at the end (and see the solution by HEL5 for an alternative proof of this fact).

To show that XEM are collinear, we construct a projective transformation which projects M to X through center E. We produce it as a composition of three other projections. Let O be the intersection of lines $AD'DY$ and CIN. Projecting the points $YNCM$ on the circle ABC through the (concyclic) point A to the line CN yields the points $ONCI$. Projecting these points through E to the line AY yields $OYDD'$ (here we use the facts that D' lies on IE and AY). Projecting these points to the circle ABC through C yields $NYBX$ (here we use the fact that D' lies on CX). Composing, we observe that we found a projection of the circle ABC to itself sending $YNCM$ to $NYBX$. Since the projection of the circle through E also sends YNC to NYB, and three points determine a projective transformation, the projection through E also sends M to X, as claimed.



Let B', D' be the intersections of AB, AD with the circle AXI, respectively. We wish to show that this D' is the concurrency point defined above, i.e. that $CD'X$ and $ID'E$ are collinear. Additionally, we will show that I is the circumcenter of $AB'E$.

Consider the inversion with center C and radius CA. The circles AXI and ABD are tangent to CA at A (the former by definition, the latter since $\angle CAD = \angle ABC$), so they are preserved under the inversion. In particular, the inversion transposes D and B and preserves A, so sends the circle CAB to the line AD. Thus X, which is the second intersection of circles ABC and AXI, is sent by the inversion to the second intersection of AD and circle AXI, which is D'. In particular $CD'X$ are collinear.

In the circle AIB', AI is the angle bisector of $B'A$ and the tangent at A, so I is the midpoint of the arc AB', and in particular $AI = IB'$. By angle chasing, we find that ACE is an isosceles triangle:

$$\angle CAE = \angle CAD + \angle DAE = \angle ABC + \angle EAB = \angle ABE + \angle EAB = \angle AEB = \angle AEC,$$

thus the angle bisector CI is the perpendicular bisector of AE and $AI = IE$. Thus I is the circumcenter of $AB'E$.

We can now show that $ID'E$ are collinear by angle chasing:

$$\angle EIB' = 2\angle EAB' = 2\angle EAB = \angle DAB = \angle D'AB' = \angle D'IB'.$$

Solution inspired by ISR2. Let W be the midpoint of arc BC, let D' be the second intersection point of AD and the circle ABC. Let P be the intersection of the angle bisector XW of $\angle CXB$ with BC; we wish to prove that AP is the angle bisector of DAB. Denote $\alpha = \frac{\angle CAB}{2}$, $\beta = \angle ABC$.

Let M be the intersection of AD and XC. Angle chasing finds:

$$\begin{aligned}\angle MXI &= \angle AXI - \angle AXM = \angle CAI - \angle AXC = \angle CAI - \angle ABC = \alpha - \beta \\ &= \angle CAI - \angle CAD = \angle DAI = \angle MAI\end{aligned}$$

And in particular M is on ω. By angle chasing we find

$$\angle XIA = \angle IXA + \angle XAI = \angle ICA + \angle XAI = \angle XAC = \angle XBC = \angle XBP$$

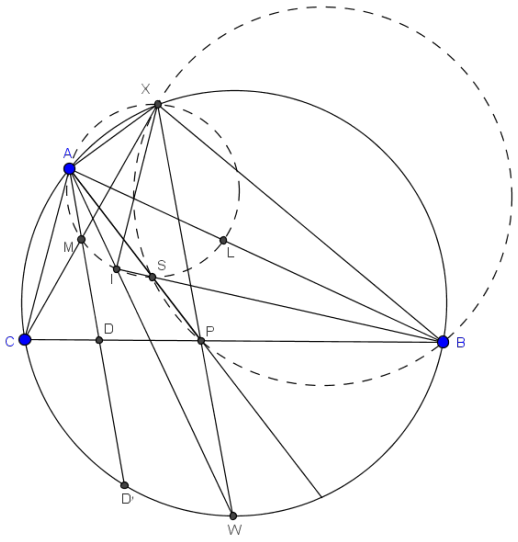
and $\angle PXB = \alpha = \angle CAI = \angle AXI$, and it follows that $\triangle XIA \sim \triangle XBP$. Let S be the second intersection point of the cirumcircles of XIA and XBP. Then by the spiral map lemma (or by the equivalent angle chasing) it follows that ISB and ASP are collinear.

Let L be the second intersection of ω and AB. We want to prove that ASP is the angle bisector of $\angle DAB = \angle MAL$, i.e. that S is the midpoint of the arc ML of ω. And this follows easily from chasing angular arc lengths in ω:

$$\begin{aligned}\widehat{AI} &= \angle CAI = \alpha \\ \widehat{IL} &= \angle IAL = \alpha \\ \widehat{MI} &= \angle MXI = \alpha - \beta \\ \widehat{AI} - \widehat{SL} &= \angle ABI = \tfrac{\beta}{2}\end{aligned}$$

And thus

$$\widehat{ML} = \widehat{MI} + \widehat{IL} = 2\alpha - \beta = 2(\widehat{AI} - \tfrac{\beta}{2}) = 2\widehat{SL}.$$



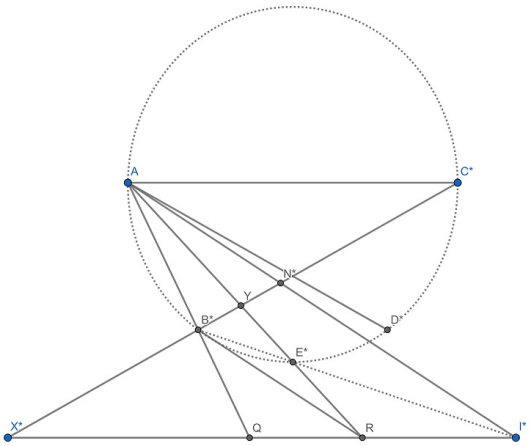
Solution by inversion, by JPN Observer A, Satoshi Hayakawa. Let E be the intersection of the bisector of $\angle BAD$ and BC, and N be the middle point of arc BC of the circumcircle of ABC. Then it suffices to show that E is on line XN.

We consider the inversion at A. Let P^* be the image of a point denoted by P. Then A, B^*, C^*, E^* are concyclic, X^*, B^*, C^* are colinear, and X^*I^* and AC^* are parallel. Now it suffices to show that A, X^*, E^*, N^* are concyclic. Let Y be the intersection of B^*C^* and AE^*. Then, by the power of a point, we get

$$\begin{aligned}A, X^*, E^*, N^* \text{ are concyclic} &\iff YX^* \cdot YN^* = YA \cdot YE^* \\ &\iff YX^* \cdot YN^* = YB^* \cdot YC^*. \\ &\qquad (A, B^*, C^*, E^* \text{ are concyclic})\end{aligned}$$

Here, by the property of inversion, we have

$$\angle AI^*B^* = \angle ABI = \frac{1}{2}\angle ABC = \frac{1}{2}\angle C^*AD^*.$$



Define Q, R as described in the figure, and we get by simple angle chasing

$$\angle QAI^* = \angle QI^*A, \quad \angle RAI^* = \angle B^*I^*A.$$

Especially, B^*R and AI^* are parallel, so that we have

$$\frac{YB^*}{YN^*} = \frac{YR}{YA} = \frac{YX^*}{YC^*},$$

and the proof is completed.

Day 2. Solutions

Problem 4 (Poland). Let ABC be a triangle with incentre I. The circle through B tangent to AI at I meets side AB again at P. The circle through C tangent to AI at I meets side AC again at Q. Prove that PQ is tangent to the incircle of ABC.

Solution 1. Let QX, PY be tangent to the incircle of ABC, where X, Y lie on the incircle and do not lie on AC, AB. Denote $\angle BAC = \alpha$, $\angle CBA = \beta$, $\angle ACB = \gamma$.

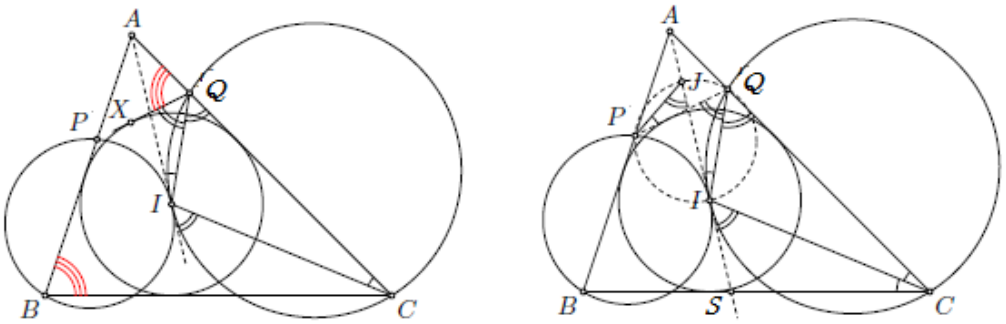
Since AI is tangent to the circumcircle of CQI we get $\angle QIA = \angle QCI = \frac{\gamma}{2}$. Thus

$$\angle IQC = \angle IAQ + \angle QIA = \frac{\alpha}{2} + \frac{\gamma}{2}.$$

By the definition of X we have $\angle IQC = \angle XQI$, therefore

$$\angle AQX = 180^\circ - \angle XQC = 180^\circ - \alpha - \gamma = \beta.$$

Similarly one can prove that $\angle APY = \gamma$. This means that Q, P, X, Y are collinear which leads us to the conclusion that $X = Y$ and QP is tangent to the incircle at X.



Solution 2. By the power of a point we have

$$AD \cdot AC = AI^2 = AP \cdot AB, \quad \text{which means that} \quad \frac{AQ}{AP} = \frac{AB}{AC}$$

and therefore triangles ADP, ABC are similar. Let J be the incenter of AQP. We obtain

$$\angle JPQ = \angle ICB = \angle QCI = \angle QIJ,$$

thus J, P, I, Q are concyclic. Let S be the intersection of AI and BC. It follows that

$$\angle IQP = \angle IJP = \angle SIC = \angle IQC.$$

This means that IQ is the angle bisector of $\angle CQP$, so QP is indeed tangent to the incircle of ABC.

Comment. The final angle chasing from the Solution 2 may simply be replaced by the observation that since J, P, I, Q are concyclic, then I is the A-excenter of triangle APQ.

Solution 3. Like before, notice that $AQ \cdot AC = AP \cdot AB = AI^2$. Consider the positive inversion Ψ with center A and power AI^2. This maps P to B (and vice-versa), Q to C

(and vice-versa), and keeps the incenter I fixed. The problem statement will follow from the fact that the image of the incircle of triangle ABC under Ψ is the so-called mixtilinear incircle of ABC, which is defined to be the circle tangent to the lines AB, AC, and the circumcircle of ABC. Indeed, since the image of the line QP is the circumcircle of ABC, and inversion preserves tangencies, this implies that QP is tangent to the incircle of ABC.

We justify the claim as follows: let γ be the incircle of ABC and let Γ_A be the A-mixtilinear incircle of ABC. Let K and L be the tangency points of γ with the sides AB and AC, and let U and V be the tangency points of Γ_A with the sides AB and AC, respectively. It is well-known that the incenter I is the midpoint of segment UV. In particular, since also $AI \perp UV$, this implies that $AU = AV = \frac{AI}{\cos\frac{A}{2}}$. Note that $AK = AL = AI \cdot \cos\frac{A}{2}$. Therefore, $AU \cdot AK = AV \cdot AL = AI^2$, which means that U and V are the images of K and L under Ψ. Since Γ_A is the unique circle simultaneously tangent to AB at U and to AC at V, it follows that the image of γ under Ψ must be precisely Γ_A, as claimed.

Solution by Achilleas Sinefakopoulos, Greece. From the power of a point theorem, we have

$$AP \cdot AB = AI^2 = AQ \cdot AC.$$

Hence $PBCQ$ is cyclic, and so, $\angle APQ = \angle BCA$. Let K be the circumcenter of $\triangle BIP$ and let L be the circumcenter of $\triangle QIC$. Then $\overline{KL}$ is perpendicular to $\overline{AI}$ at I.

Let N be the point of intersection of line $\overline{KL}$ with $\overline{AB}$.Then in the right triangle $\triangle NIA$, we have $\angle ANI = 90^\circ - \frac{\angle BAC}{2}$ and from the external angle theorem for triangle $\triangle BNI$, we have $\angle ANI = \frac{\angle ABC}{2} + \angle NIB$. Hence

$$\angle NIB = \angle ANI - \frac{\angle ABC}{2} = \left(90^\circ - \frac{\angle BAC}{2}\right) - \frac{\angle ABC}{2} = \frac{\angle BCA}{2}.$$

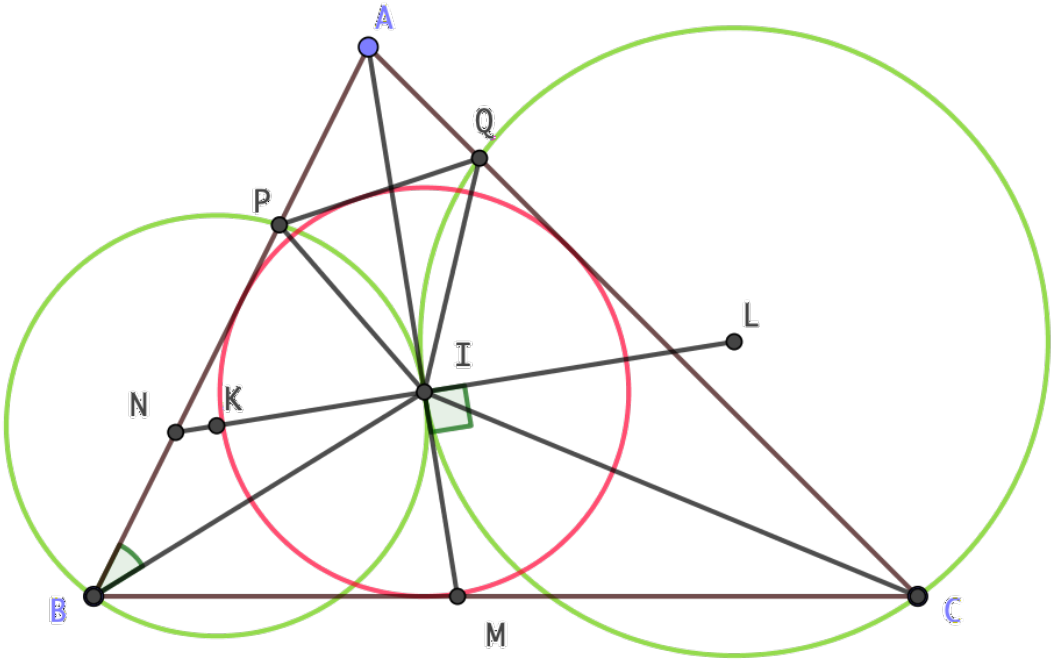
Since MI is tangent to the circumcircle of $\triangle BIP$ at I, we have

$$\angle BPI = \angle BIM = \angle NIM - \angle NIB = 90^\circ - \frac{\angle BCA}{2}.$$

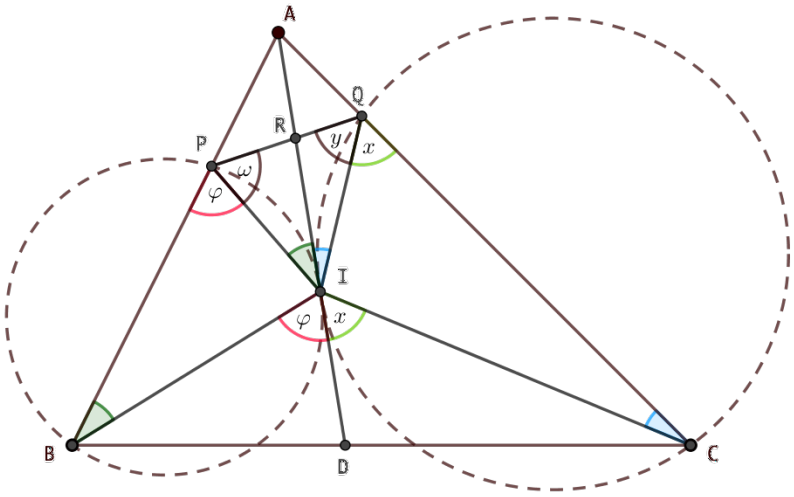
Also, since $\angle APQ = \angle BCA$, we have

$$\angle QPI = 180^\circ - \angle APQ - \angle BPI = 180^\circ - \angle BCA - \left(90^\circ - \frac{\angle BCA}{2}\right) = 90^\circ - \frac{\angle BCA}{2},$$

as well. Hence I lies on the angle bisector of $\angle BPQ$, and so it is equidistant from its sides $\overline{PQ}$ and $\overline{PB}$. Therefore, the distance of I from $\overline{PQ}$ equals the inradius of $\triangle ABC$, as desired.



Solution by Eirini Miliori (HEL2). Let D be the point of intersection of $\overline{AI}$ and $\overline{BC}$ and let R be the point of intersection of $\overline{AI}$ and $\overline{PQ}$. We have $\angle RIP = \angle PBI = \frac{\angle B}{2}$, $\angle RIQ = \angle ICQ = \frac{\angle C}{2}$, $\angle IQC = \angle DIC = x$ and $\angle BPI = \angle BID = \varphi$, since $\overline{AI}$ is tangent to both circles.



From the angle bisector theorem, we have

$$\frac{RQ}{RP} = \frac{AQ}{AP} \quad \text{and} \quad \frac{AC}{AB} = \frac{DC}{BD}.$$

Since $\overline{AI}$ is tangent to both circles at I, we have $AI^2 = AQ \cdot AC$ and $AI^2 = AP \cdot AB$. Therefore,

$$\frac{RQ}{RP} \cdot \frac{DC}{BD} = \frac{AQ \cdot AC}{AB \cdot AP} = 1. \tag{1}$$

From the sine law in triangles $\triangle QRI$ and $\triangle PRI$, it follows that $\dfrac{RQ}{\sin \frac{\angle C}{2}} = \dfrac{RI}{\sin y}$ and $\dfrac{RP}{\sin \frac{\angle B}{2}} = \dfrac{RI}{\sin \omega}$, respectively. Hence

$$\frac{RQ}{RP} \cdot \frac{\sin \frac{\angle B}{2}}{\sin \frac{\angle C}{2}} = \frac{\sin \omega}{\sin y}. \tag{2}$$

Similarly, from the sine law in triangles $\triangle IDC$ and $\triangle IDB$, it is $\dfrac{DC}{\sin x} = \dfrac{ID}{\sin \frac{\angle C}{2}}$ and $\dfrac{BD}{\sin \varphi} = \dfrac{ID}{\sin \frac{\angle B}{2}}$, and so

$$\frac{DC}{BD} \cdot \frac{\sin \varphi}{\sin x} = \frac{\sin \frac{\angle B}{2}}{\sin \frac{\angle C}{2}}. \tag{3}$$

By multiplying equations (2) with (3), we obtain $\dfrac{RQ}{RP} \cdot \dfrac{DC}{BD} \cdot \dfrac{\sin \varphi}{\sin x} = \dfrac{\sin \omega}{\sin y}$, which combined with (1) and cross-multiplying yields

$$\sin \varphi \cdot \sin y = \sin \omega \cdot \sin x. \tag{4}$$

Let $\theta = 90^\circ + \frac{\angle A}{2}$. Since I is the incenter of $\triangle ABC$, we have $x = 90^\circ + \frac{\angle A}{2} - \varphi = \theta - \phi$. Also, in triangle $\triangle PIQ$, we see that $\omega + y + \frac{\angle B}{2} + \frac{\angle C}{2} = 180^\circ$, and so $y = \theta - \omega$.

Therefore, equation (4) yields

$$\sin \varphi \cdot \sin(\theta - \omega) = \sin \omega \cdot \sin(\theta - \varphi),$$

or

$$\frac{1}{2}\left(\cos(\varphi - \theta + \omega) - \cos(\varphi + \theta - \omega)\right) = \frac{1}{2}\left(\cos(\omega - \theta + \varphi) - \cos(\omega + \theta - \varphi)\right),$$

which is equivalent to

$$\cos(\varphi + \theta - \omega) = \cos(\omega + \theta - \varphi).$$

So

$$\varphi + \theta - \omega = 2k \cdot 180^\circ \pm (\omega + \theta - \varphi), \quad (k \in \mathbb{Z}.)$$

If $\varphi + \theta - \omega = 2k \cdot 180^\circ + (\omega + \theta - \varphi)$, then $2(\varphi - \omega) = 2k \cdot 180^\circ$, with $|\varphi - \omega| < 180^\circ$ forcing $k = 0$ and $\varphi = \omega$. If $\varphi + \theta - \omega = 2k \cdot 180^\circ - (\omega + \theta - \varphi)$, then $2\theta = 2k \cdot 180^\circ$, which contradicts the fact that $0^\circ < \theta < 180^\circ$. Hence $\varphi = \omega$, and so PI is the angle bisector of $\angle QPB$.

Therefore the distance of I from $\overline{PQ}$ is the same with the distance of I from AB, which is equal to the inradius of $\triangle ABC$. Consequently, $\overline{PQ}$ is tangent to the incircle of $\triangle ABC$.

Problem 5 (Netherlands).

Let $n \geq 2$ be an integer, and let $a_1, a_2, \ldots, a_n$ be positive integers. Show that there exist positive integers $b_1, b_2, \ldots, b_n$ satisfying the following three conditions:

1. $a_i \leq b_i$ for $i = 1, 2, \ldots, n$;
2. the remainders of $b_1, b_2, \ldots, b_n$ on division by n are pairwise different; and
3. $b_1 + \cdots + b_n \leq n\left(\frac{n-1}{2} + \left\lfloor \frac{a_1 + \cdots + a_n}{n} \right\rfloor\right)$.

(Here, $\lfloor x \rfloor$ denotes the integer part of real number x, that is, the largest integer that does not exceed x.)

Solution 1. We define the b_i recursively by letting b_i be the smallest integer such that $b_i \geq a_i$ and such that b_i is not congruent to any of $b_1, \ldots, b_{i-1}$ modulo n. Then $b_i - a_i \leq i - 1$, since of the i consecutive integers $a_i, a_i + 1, \ldots, a_i + i - 1$, at most $i - 1$ are congruent to one of $b_1, \ldots, b_{i-1}$ modulo n. Since all b_i are distinct modulo n, we have $\sum_{i=1}^n b_i \equiv \sum_{i=1}^n (i-1) = \frac{1}{2}n(n-1)$ modulo n, so n divides $\sum_{i=1}^n b_i - \frac{1}{2}n(n-1)$. Moreover, we have $\sum_{i=1}^n b_i - \sum_{i=1}^n a_i \leq \sum_{i=1}^n (i-1) = \frac{1}{2}n(n-1)$, hence $\sum_{i=1}^n b_i - \frac{1}{2}n(n-1) \leq \sum_{i=1}^n$. As the left hand side is divisible by n, we have

$$\frac{1}{n}\left(\sum_{i=1}^{n} b_i - \frac{1}{2}n\,(n-1)\right) \leq \left[\frac{1}{n}\sum_{i=1}^{n} a_i\right]$$

which we can rewrite as

$$\sum_{i=1}^{n} b_i \leq n\left(\frac{n-1}{2} + \left[\frac{1}{n}\sum_{i=1}^{n} a_i\right]\right)$$

as required.

Solution 2. Note that the problem is invariant under each of the following operations:

- adding a multiple of n to some a_i (and the corresponding b_i);
- adding the same integer to all a_i (and all b_i);
- permuting the index set $1, 2, \ldots, n$.

We may therefore remove the restriction that our a_i and b_i be positive.

For each congruence class $\overline{k}$ modulo n ($\overline{k} = \overline{0}, \ldots, \overline{n-1}$), let $h(k)$ be the number of i such that a_i belongs to $\overline{k}$. We will now show that the problem is solved if we can find a $t \in \mathbb{Z}$ such that

$$\begin{aligned} h(t) &\geq 1, \\ h(t) + h(t+1) &\geq 2, \\ h(t) + h(t+1) + h(t+2) &\geq 3, \\ &\vdots \end{aligned}$$

Indeed, these inequalities guarantee the existence of elements $a_{i_1} \in \overline{t}$, $a_{i_2} \in \overline{t} \cup \overline{t+1}$, $a_{i_3} \in \overline{t} \cup \overline{t+1} \cup \overline{t+2}$, et cetera, where all i_k are different. Subtracting appropriate

multiples of n and reordering our elements, we may assume $a_1 = t$, $a_2 \in \{t, t+1\}$, $a_3 \in \{t, t+1, t+2\}$, et cetera. Finally subtracting t from the complete sequence, we may assume $a_1 = 0$, $a_2 \in \{0, 1\}$, $a_3 \in \{0, 1, 2\}$ et cetera. Now simply setting $b_i = i - 1$ for all i suffices, since $a_i \leq b_i$ for all i, the b_i are all different modulo n, and

$$\sum_{i=1}^{n} b_i = \frac{n(n-1)}{2} \leq \frac{n(n-1)}{2} + n \left[\frac{\sum_{i=1}^{n} a_i}{n} \right].$$

Put $x_i = h(i) - 1$ for all $i = 0, \ldots, n-1$. Note that $x_i \geq -1$, because $h(i) \geq 0$. If we have $x_i \geq 0$ for all $i = 0, \ldots, n-1$, then taking $t = 0$ completes the proof. Otherwise, we can pick some index j such that $x_j = -1$. Let $y_i = x_i$ where $i = 0, \ldots, j-1, j+1, \ldots, n-1$ and $y_j = 0$. For sequence $\{y_i\}$ we have

$$\sum_{i=0}^{n-1} y_i = \sum_{i=0}^{n-1} x_i + 1 = \sum_{i=0}^{n-1} h(i) - n + 1 = 1,$$

so from Raney's lemma there exists index k such that $\sum_{i=k}^{k+j} y_i > 0$ for all $j = 0, \ldots, n-1$ where $y_{n+j} = y_j$ for $j = 0, \ldots, k-1$. Taking $t = k$ we will have

$$\sum_{t=k}^{k+i} h(t) - (i+1) = \sum_{t=k}^{k+i} x(t) \geq \sum_{t=k}^{k+i} y(t) - 1 \geq 0,$$

for all $i = 0, \ldots, n-1$ and we are done.

Solution 3. Choose a random permutation $c_1, \ldots, c_n$ of the integers $1, 2, \ldots, n$. Let $b_i = a_i + f(c_i - a_i)$, where $f(x) \in \{0, \ldots, n-1\}$ denotes a remainder of x modulo n. Observe, that for such defined sequence the first two conditions hold. The expected value of $B := b_1 + \ldots + b_n$ is easily seen to be equal to $a_1 + \ldots + a_n + n(n-1)/2$. Indeed, for each i the random number $c_i - a_i$ has uniform distribution modulo n, thus the expected value of $f(c_i - a_i)$ is $(0 + \ldots + (n-1))/n = (n-1)/2$. Therefore we may find such c that $B \leq a_1 + \ldots + a_n + n(n-1)/2$. But $B - n(n-1)/2$ is divisible by n and therefore $B \leq n[(a_1 + \ldots + a_n)/n] + n(n-1)/2$ as needed.

Solution 4. We will prove the required statement for all sequences of non-negative integers a_i by induction on n.

Case $n = 1$ is obvious, just set $b_1 = a_1$.

Now suppose that the statement is true for some $n \geq 1$; we shall prove it for $n + 1$.

First note that, by subtracting a multiple of $n + 1$ to each a_i and possibly rearranging indices we can reduce the problem to the case where $0 \leq a_1 \leq a_2 \leq \cdots \leq a_n \leq a_{n+1} < n + 1$.

Now, by the induction hypothesis there exists a sequence $d_1, d_2, \ldots, d_n$ which satisfies the properties required by the statement in relation to the numbers $a_1, \ldots, a_n$. Set $I = \{i | 1 \leq i \leq n \text{ and } d_i \bmod n \geq a_i\}$ and construct b_i, for $i = 1, \ldots, n+1$, as follows:

$$b_i = \begin{cases} d_i \bmod n, \text{ when } i \in I, \\ n + 1 + (d_i \bmod n), \text{ when } i \in \{1, \ldots, n\} \setminus I, \\ n, \text{ for } i = n + 1. \end{cases}$$

Now, $a_i \le d_i \bmod n \le b_i$ for $i \in I$, while for $i \notin I$ we have $a_i \le n \le b_i$. Thus the sequence $(b_i)_{i=1}^{n+1}$ satisfies the first condition from the problem statement.

By the induction hypothesis, the numbers $d_i \bmod n$ are distinct for $i \in \{1, \ldots, n\}$, so the values $b_i \bmod (n+1)$ are distinct elements of $\{0, \ldots, n-1\}$ for $i \in \{1, \ldots, n\}$. Since $b_{n+1} = n$, the second condition is also satisfied.

Denote $k = |I|$. We have

$$\sum_{i=1}^{n+1} b_i = \sum_{i=1}^{n} b_i + n = \sum_{i=1}^{n} d_i \bmod n + (n-k)(n+1) + n =$$

$$\frac{n(n+1)}{2} + (n-k)(n+1),$$

hence we need to show that

$$\frac{n(n+1)}{2} + (n-k)(n+1) \le \frac{n(n+1)}{2} + (n+1)\left[\frac{\sum_{i=1}^{n+1} a_i}{n+1}\right];$$

equivalently, that

$$n - k \le \left[\frac{\sum_{i=1}^{n+1} a_i}{n+1}\right].$$

Next, from the induction hypothesis we have

$$\frac{n(n-1)}{2} + n\left[\frac{\sum_{i=1}^{n} a_i}{n}\right] \ge \sum_{i=1}^{n} d_i = \sum_{i \in I} d_i + \sum_{i \notin I} d_i \ge$$

$$\sum_{i \in I} d_i \bmod n + \sum_{i \notin I} (n + d_i \bmod n) = \frac{n(n-1)}{2} + (n-k)n$$

or

$$n - k \le \left[\frac{\sum_{i=1}^{n} a_i}{n}\right].$$

Thus, it's enough to show that

$$\frac{\sum_{i=1}^{n} a_i}{n} \le \frac{\sum_{i=1}^{n+1} a_i}{n+1}$$

because then

$$n - k \le \left[\frac{\sum_{i=1}^{n} a_i}{n}\right] \le \left[\frac{\sum_{i=1}^{n+1} a_i}{n+1}\right].$$

But the required inequality is equivalent to $\sum_{i=1}^{n} a_i \le n a_{n+1}$, which is obvious.

Solution 5. We can assume that all $a_i \in \{0, 1, \ldots, n-1\}$, as we can deduct n from both a_i and b_i for arbitrary i without violating any of the three conditions from the problem statement. We shall also assume that $a_1 \le \ldots \le a_n$.

Now let us provide an algorithm for constructing $b_1, \ldots, b_n$.

We start at step 1 by choosing $f(1)$ to be the maximum i in $\{1,\ldots,n\}$ such that $a_i \leq n-1$, that is $f(1)=n$. We set $b_{f(1)}=n-1$.

Having performed steps 1 through j, at step $j+1$ we set $f(j+1)$ to be the maximum i in $\{1,\ldots,n\}\setminus\{f(1),\ldots,f(j)\}$ such that $a_i \leq n-j-1$, if such an index exists. If it does, we set $b_{f(j+1)}=n-j-1$. If there is no such index, then we define $T=j$ and assign to the terms b_i, where $i \notin f(\{1,\ldots,j\})$, the values $n, n+1\ldots, 2n-j-1$, in any order, thus concluding the run of our algorithm.

Notice that the sequence $(b_i)_{i=1}^n$ satisfies the first and second required conditions by construction. We wish to show that it also satisfies the third.

Notice that, since the values chosen for the b_i's are those from $n-T$ to $2n-T-1$, we have

$$\sum_{i=1}^{n} b_i = \frac{n(n-1)}{2} + (n-T)n.$$

It therefore suffices to show that

$$\left[\frac{a_1+\ldots+a_n}{n}\right] \geq n-T,$$

or (since the RHS is obviously an integer) $a_1+\ldots+a_n \geq (n-T)n$.

First, we show that there exists $1 \leq i \leq T$ such that $n-i=b_{f(i)}=a_{f(i)}$.

Indeed, this is true if $a_n=n-1$, so we may suppose $a_n<n-1$ and therefore $a_{n-1}\leq n-2$, so that $T\geq 2$. If $a_{n-1}=n-2$, we are done. If not, then $a_{n-1}<n-2$ and therefore $a_{n-2}\leq n-3$ and $T\geq 3$. Inductively, we actually obtain $T=n$ and necessarily $f(n)=1$ and $a_1=b_1=0$, which gives the desired result.

Now let t be the largest such index i. We know that $n-t=b_{f(t)}=a_{f(t)}$ and therefore $a_1\leq\ldots\leq a_{f(t)}\leq n-t$. If we have $a_1=\ldots=a_{f(t)}=n-t$, then $T=t$ and we have $a_i\geq n-T$ for all i, hence $\sum_i a_i \geq n(n-T)$. Otherwise, $T>t$ and in fact one can show $T=t+f(t+1)$ by proceeding inductively and using the fact that t is the *last* time for which $a_{f(t)}=b_{f(t)}$.

Now we get that, since $a_{f(t+1)+1}\geq n-t$, then $\sum_i a_i \geq (n-t)(n-f(t+1)) = (n-T+f(t+1))(n-f(t+1)) = n(n-T)+nf(t+1)-f(t+1)(n-T+f(t+1)) = n(n-T)+tf(t+1)\geq n(n-T)$.

Greedy algorithm variant 1 (ISR). Consider the residues $0,\ldots,n-1$ modulo n arranged in a circle clockwise, and place each a_i on its corresponding residue; so that on each residue there is a stack of all a_is congruent to it modulo n, and the sum of the sizes of all stacks is exactly n. We iteratively flatten and spread the stacks forward, in such a way that the a_is are placed in the nearest available space on the circle clockwise (skipping over any already flattened residue or still standing stack). We may choose the order in which the stacks are flattened. Since the total amount of numbers equals the total number of spaces, there is always an available space and at the end all spaces are covered. The b_is are then defined by adding to each a_i the number of places it was moved forward, which clearly satifies (i) and (ii), and we must prove that they satisfy (iii) as well.

Suppose that we flatten a stack of k numbers at a residue i, causing it to overtake a stack of l numbers at residue $j\in(i,i+k)$ (we can allow j to be larger than n and identify it

with its residue modulo n). Then in fact in fact in whichever order we would flatten the two stacks, the total number of forward steps would be the same, and the total sum of the corresponding b_t (such that $a_t \bmod n \in \{i, j\}$) would be the same. Moreover, we can merge the stacks to a single stack of $k + l$ numbers at residue i, by replacing each $a_t \equiv j \pmod n$ by $a'_t = a_t - (j - i)$, and this stack would be flattened forward into the same positions as the separate stacks would have been, so applying our algorithm to the new stacks will yield the same total sum of $\sum b_i$ – but the a_is are strictly decreased, so $\sum a_i$ is decreased, so $\left\lfloor \frac{\sum a_i}{n} \right\rfloor$ is not increased – so by merging the stacks, we can only make the inequality we wish to prove tighter.

Thus, as long as there is some stack that when flattened will overtake another stack, we may merge stacks and only make the inequality tighter. Since the amount of numbers equals the amount of places, the merging process terminates with stacks of sizes $k_1, \ldots, k_m$, such that the stack j, when flattened, will exactly cover the interval to the next stack. Clearly the numbers in each such stack were advanced by a total of $\sum_{t=1}^{k_j-1} = \frac{k_j(k_j-1)}{2}$, thus $\sum b_i = \sum a_i + \sum_j \frac{k_j(k_j-1)}{2}$. Writing $\sum a_i = n \cdot r + s$ with $0 \le s < n$, we must therefore show

$$s + \sum_j \frac{k_j(k_j - 1)}{2} \le \frac{n(n-1)}{2}.$$

Ending 1. Observing that both sides of the last inequality are congruent modulo n (both are congruent to the sum of all different residues), and that $0 \le s < n$, the inequality is eqivalent to the simpler $\sum_j \frac{k_j(k_j-1)}{2} \le \frac{n(n-1)}{2}$. Since $x(x-1)$ is convex, and k_j are non-negative integers with $\sum_j k_j = n$, the left hand side is maximal when $k_{j'} = n$ and the rest are 0, and then eqaulity is achieved. (Alternatively it follows easily for any non-negative reals from AM-GM.)

Ending 2. If $m = 1$ (and $k_1 = n$), then all numbers are in a single stack and have the same residue, so $s = 0$ and equality is attained. If $m \ge 2$, then by convexity $\sum_j \frac{k_j(k_j-1)}{2}$ is maximal for $m = 2$ and $(k_1, k_2) = (n-1, 1)$, where it equals $\frac{(n-1)(n-2)}{2}$. Since we always have $s \le n - 1$, we find

$$s + \sum_j \frac{k_j(k_j - 1)}{2} \le (n-1) + \frac{(n-1)(n-2)}{2} = \frac{n(n-1)}{2}$$

as required.

Greedy algorithm variant 1' (ISR). We apply the same algorithm as in the previous solution. However, this time we note that we may merge stacks not only when they overlap after flattening, but also when they merely touch front-to-back: That is, we relax the condition $j \in (i, i+k)$ to $j \in (i, i+k]$; the argument for why such merges are allowed is exactly the same (But note that this is now sharp, as merging non-touching stacks can cause the sum of b_is to decrease).

We now observe that as long as there at least two stacks left, at least one will spread to touch (or overtake) the next stack, so we can perform merges until there is only one stack left. We are left with verifying that the inequality indeed holds for the case of only one stack which is spread forward, and this is indeed immediate (and in fact equality is achieved).

Greedy algorithm variant 2 (ISR). Let $c_i = a_i \bmod n$. Iteratively define $b_i = a_i + l_i$ greedily, write $d_i = c_i + l_i$, and observe that $l_i \le n-1$ (since all residues are present in $a_i, \dots, a_i + n - 1$), hence $0 \le d_i \le 2n-2$. Let $I = \{i \in I : d_i \ge n\}$, and note that $d_i = b_i \bmod n$ if $i \notin I$ and $d_i = (b_i \bmod n) + n$ if $i \in I$. Then we must show

$$\begin{aligned} &\sum (a_i + l_i) = \sum b_i \le \frac{n(n-1)}{2} + n\left\lfloor \frac{\sum a_i}{n} \right\rfloor \\ \iff &\sum (c_i + l_i) \le \sum (b_i \bmod n) + n\left\lfloor \frac{\sum c_i}{n} \right\rfloor \\ \iff &n|I| \le n\left\lfloor \frac{\sum c_i}{n} \right\rfloor \iff |I| \le \left\lfloor \frac{\sum c_i}{n} \right\rfloor \iff |I| \le \frac{\sum c_i}{n} \end{aligned}$$

Let $k = |I|$, and for each $0 \le m < n$ let $J_m = \{i : c_i \ge n - m\}$. We claim that there must be some m for which $|J_m| \ge m + k$ (clearly for such m, at least k of the sums d_j with $j \in J_m$ must exceed n, i.e. at least k of the elements of J_m must also be in I, so this m is a "witness" to the fact $|I| \ge k$). Once we find such an m, then we clearly have

$$\sum c_i \ge (n-m)|J_m| \ge (n-m)(k+m) = nk + m(n-(k+m)) \ge nk = n|I|$$

as required. We now construct such an m explicitly.

If $k = 0$, then clearly $m = n$ works (and also the original inequality is trivial). Otherwise, there are some d_is greater than n, and let $r + n = \max d_i$, and suppose $d_t = r + n$ and let $s = c_t$. Note that $r < s < r + n$ since $l_t < n$. Let $m \ge 0$ be the smallest number such that $n - m - 1$ is not in $\{d_1, \dots, d_t\}$, or equivalently m is the largest such that $[n-m, n) \subset \{d_1, \dots, d_t\}$. We claim that this m satisfies the required property. More specifically, we claim that $J'_m = \{i \le t : d_i \ge n - m\}$ contains exactly $m + k$ elements and is a subset of J_m.

Note that by the greediness of the algorithm, it is impossible that for $[c_i, d_i)$ to contain numbers congruent to d_j mod n with $j > i$ (otherwise, the greedy choice would prefer d_j to d_i at stage i). We call this the *greedy property*. In particular, it follows that all i such that $d_i \in [s, d_t) = [c_t, d_t)$ must satisfy $i < t$. Additionally, $\{d_i\}$ is disjoint from $[n + r + 1, 2n)$ (by maximality of d_t), but does intersect every residue class, so it contains $[r+1, n)$ and in particular also $[s, n)$. By the greedy property the latter can only be attained by d_i with $i < t$, thus $[s, n) \subset \{d_1, \dots, d_t\}$, and in particular $n - m \le s$ (and in particular $m \ge 1$).

On the other hand $n - m > r$ (since $r \notin \{d_i\}$ at all), so $n - m - 1 \ge r$. It follows that there is a time $t' \ge t$ for which $d_{t'} \equiv n - m - 1 \pmod n$: If $n - m - 1 = r$ then this is true for $t' = t$ with $d_t = n + r = 2n - m - 1$; whereas if $n - m - 1 \in [r+1, n)$ then there is some t' for which $d_{t'} = n - m - 1$, and by the definition of m it satisfies $t' > t$.

Therefore for all $i < t \le t'$ for which $d_i \ge n - m$, necessarily also $c_i \ge n - m$, since otherwise $d_{t'} \in [c_i, d_i)$, in contradiction to the greedy property. This is also true for $i = t$, since $c_t = s \ge n - m$ as previously shown. Thus, $J'_m \subset J_m$ as claimed.

Finally, since by definition of m and greediness we have $[n-m, n) \cup \{d_i : i \in I\} \subset \{d_1, \dots, d_t\}$, we find that $\{d_j : j \in J'_m\} = [n-m, n) \cup \{d_i : i \in I\}$ and thus $|J'_m| = |[n-m, n)| + |I| = m + k$ as claimed.

Problem 6 (United Kingdom).

On a circle, Alina draws 2019 chords, the endpoints of which are all different. A point is considered *marked* if it is either

(i) one of the 4038 endpoints of a chord; or

(ii) an intersection point of at least two chords.

Alina labels each marked point. Of the 4038 points meeting criterion (i), Alina labels 2019 points with a 0 and the other 2019 points with a 1. She labels each point meeting criterion (ii) with an arbitrary integer (not necessarily positive).

Along each chord, Alina considers the segments connecting two consecutive marked points. (A chord with k marked points has $k-1$ such segments.) She labels each such segment in yellow with the sum of the labels of its two endpoints and in blue with the absolute value of their difference.

Alina finds that the $N+1$ yellow labels take each value $0, 1, \ldots, N$ exactly once. Show that at least one blue label is a multiple of 3.

(A *chord* is a line segment joining two different points on a circle.)

Solution 1. First we prove the following:

Lemma: *if we color all of the points white or black, then the number of white-black edges, which we denote E_{WB}, is equal modulo 2 to the number of white (or black) points on the circumference, which we denote C_W, resp. C_B.*

Observe that changing the colour of any interior point does not change the parity of E_{WB}, as each interior point has even degree, so it suffices to show the statement holds when all interior points are black. But then $E_{WB} = C_W$ so certainly the parities are equal.

Now returning to the original problem, assume that no two adjacent vertex labels differ by a multiple of three, and three-colour the vertices according to the residue class of the labels modulo 3. Let E_{01} denote the number of edges between 0-vertices and 1-vertices, and C_0 denote the number of 0-vertices on the boundary, and so on.

Then, consider the two-coloring obtained by combining the 1-vertices and 2-vertices. By applying the lemma, we see that $E_{01} + E_{02} \equiv C_0 \bmod 2$.

$$\text{Similarly } E_{01} + E_{12} \equiv C_1, \quad \text{and } E_{02} + E_{12} \equiv C_2, \quad \bmod 2.$$

Using the fact that $C_0 = C_1 = 2019$ and $C_2 = 0$, we deduce that either E_{02} and E_{12} are even and E_{01} is odd; or E_{02} and E_{12} are odd and E_{01} is even.

But if the edge labels are the first N non-negative integers, then $E_{01} = E_{12}$ unless $N \equiv 0$ modulo 3, in which case $E_{01} = E_{02}$. So however Alina chooses the vertex labels, it is not possible that the multiset of edge labels is $\{0, \ldots, N\}$.

Hence in fact two vertex labels must differ by a multiple of 3.

Solution 2. As before, colour vertices based on their label modulo 3.

Suppose this gives a valid 3-colouring of the graph with 2019 0s and 2019 1s on the

circumference. Identify pairs of 0-labelled vertices and pairs of 1-labelled vertices on the circumference, with one 0 and one 1 left over. The resulting graph has even degrees except these two leaves. So the connected component $\mathcal{C}$ containing these leaves has an Eulerian path, and any other component has an Eulerian cycle.

Let E^*_{01} denote the number of edges between 0-vertices and 1-vertices in $\mathcal{C}$, and let E'_{01} denote the number of such edges in the other components, and so on. By studying whether a given vertex has label congruent to 0 modulo 3 or not as we go along the Eulerian path in $\mathcal{C}$, we find $E^*_{01} + E^*_{02}$ is odd, and similarly $E^*_{01} + E^*_{12}$ is odd. Since neither start nor end vertex is a 2-vertex, $E^*_{02} + E^*_{12}$ must be even.

Applying the same argument for the Eulerian cycle in each other component and adding up, we find that $E'_{01} + E'_{02}$, $E'_{01} + E'_{12}$, $E'_{02} + E'_{12}$ are all even. So, again we find $E_{01} + E_{02}$, $E_{01} + E_{12}$ are odd, and $E_{02} + E_{12}$ is even, and we finish as in the original solution.

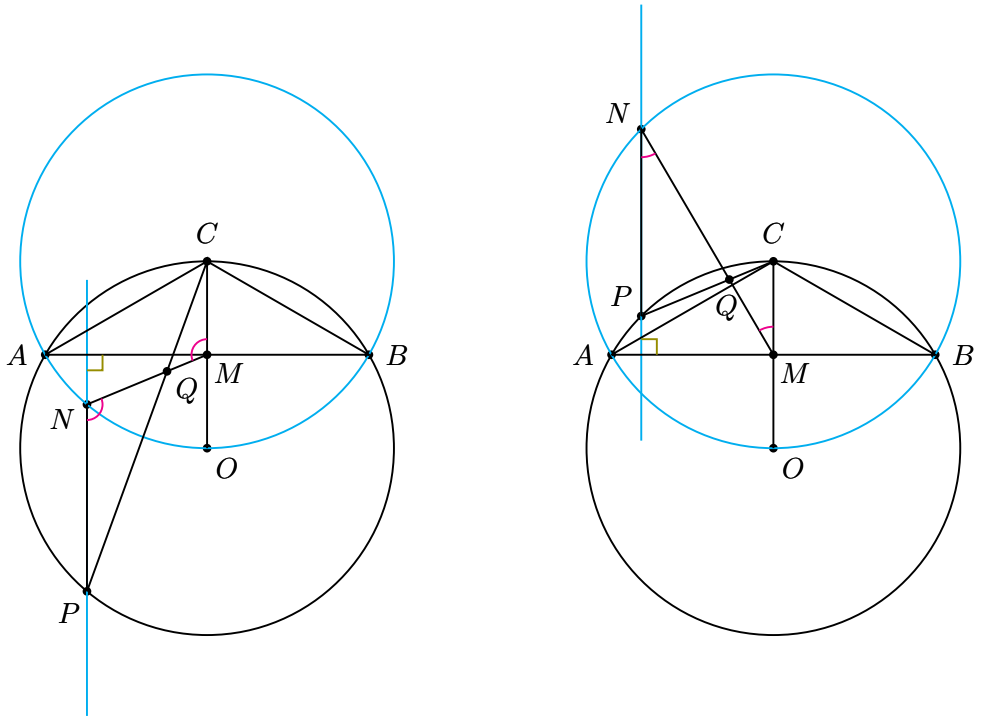
Problem 1 Let ABC be a triangle with $CA = CB$ and $\angle ACB = 120°$, and let M be the midpoint of AB. Let P be a variable point on the circumcircle of ABC, and let Q be the point on the segment CP such that $QP = 2QC$. It is given that the line through P and perpendicular to AB intersects the line MQ at a unique point N.

Prove that there exists a fixed circle such that N lies on this circle for all possible positions of P.

(Velina Ivanova, Bulgaria)

Solution Let O be the circumcenter of ABC. From the assumption that $\angle ACB = 120°$ it follows that M is the midpoint of CO.

Let ω denote the circle with center in C and radius CO. This circle in the image of the circumcircle of ABC through the translation that sends O to C. We claim that N lies on ω.



Let us consider the triangles QNP and QMC. The angles in Q are equal. Since NP is parallel to MC (both lines are perpendicular to AB), it turns out that $\angle QNP = \angle QMC$, and hence the two triangles are similar. Since $QP = 2QC$, it follows that

$$NP = 2MC = CO,$$

which proves that N lies on ω.

Comment The possible positions of N are all the points of ω with the exception of the two points lying on the line CO. Indeed, P does not lie on the line CO because otherwise the point N is not well-defined, and therefore also N does not lie on the same line.

Conversely, let N be any point on ω and not lying on the line CO. Let P be the corresponding point on the circumcircle of ABC, namely such that NP is parallel and equal to CO. Let Q be

the intersection of CP and NM. As before, the triangles QNP and QMC are similar, and now from the relation $NP = 2MC$ we deduce that $QP = 2QC$. This proves that N can be obtained from P through the construction described in the statement of the problem.

Alternative solution Let M' denote the symmetric of M with respect to O.

Let us consider the quadrilateral $MM'PN$. The lines MM' and NP are parallel by construction. Also the lines PM' and NM are parallel (homothety from C with coefficient 3). It follows that $MM'PN$ is a parallelogram, and hence $PN = MM' = OC$.

Computational solution There are many computation approaches to this problem. For example, we can set Cartesian coordinates so that

$$A = \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \qquad B = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), \qquad C = (0, 1), \qquad M = \left(0, \frac{1}{2}\right).$$

Setting $P = (a, b)$, we obtain that $Q = (a/3, (2 + b)/3)$. The equation of the line through P and perpendicular to AB is $x = a$. The equation of the line MQ (if $a \neq 0$) is

$$y - \frac{1}{2} = \frac{x}{a}\left(\frac{1}{2} + b\right).$$

The intersection of the two lines is therefore

$$N = (a, 1 + b) = P + (0, 1).$$

This shows that the map $P \to N$ in the translation by the vector $(0, 1)$. This result is independent of the position of P (provided that $a \neq 0$, because otherwise N is not well-defined).

When P lies on the circumcircle of ABC, with the exception of the two points with $a = 0$, then necessarily N lies on the translated circle (which is the circle with center in C and radius 1).

Problem 2 Consider the set

$$A = \left\{ 1 + \frac{1}{k} : k = 1, 2, 3, \ldots \right\}.$$

(a) Prove that every integer $x \geq 2$ can be written as the product of one or more elements of A, which are not necessarily different.

(b) For every integer $x \geq 2$, let $f(x)$ denote the minimum integer such that x can be written as the product of $f(x)$ elements of A, which are not necessarily different.

Prove that there exist infinitely many pairs (x, y) of integers with $x \geq 2$, $y \geq 2$, and

$$f(xy) < f(x) + f(y).$$

(Pairs (x_1, y_1) and (x_2, y_2) are different if $x_1 \neq x_2$ or $y_1 \neq y_2$).

(Mihail Baluna, Romania)

Solution Every integer $x \geq 2$ can be written as the telescopic product of $x - 1$ elements of A as

$$x = \left(1 + \frac{1}{x-1}\right) \cdot \left(1 + \frac{1}{x-2}\right) \cdot \ldots \cdot \left(1 + \frac{1}{2}\right) \cdot \left(1 + \frac{1}{1}\right),$$

which is enough to establish part (a). We now consider part (b). Notice that for any positive integer k we have

$$f(2^k + 1) \leq k + 1,$$

because $2^k + 1 = \left(1 + \frac{1}{2^k}\right) \cdot 2^k$ is a representation of $2^k + 1$ as a product of $k + 1$ elements of A. We claim that all the pairs (x, y) of the form

$$x = 5, \qquad y = \frac{2^{4k+2} + 1}{5}$$

satisfy the required inequality. Notice that y is an integer for any positive value of k, because $2^{4k+2} + 1 \equiv 16^k \cdot 4 + 1 \equiv 5 \equiv 0 \pmod 5$. Furthermore, $f(xy) = f(2^{4k+2} + 1) \leq 4k + 3$ (and $f(x) = f(2^2 + 1) \leq 3$) by the above. We now need some lower bounds on the values of f. Notice that no element of A exceeds 2, and therefore the product of at most k elements of A does not exceed 2^k: it follows that

$$f(n) \geq \lceil \log_2(n) \rceil, \tag{Q2.1}$$

and in particular that

$$f(5) = f(2^2 + 1) \geq \lceil \log_2(5) \rceil = 3.$$

We have thus proven $f(x) = f(5) = 3$. We want to show $f(xy) < f(x) + f(y)$, and since we know $f(xy) \leq 4k + 3$ and $f(x) = 3$ we are reduced to showing $f(y) > 4k$. Since $y > 2^{4k-1}$, from (Q2.1) we already know that $f(y) \geq 4k$, and hence we just need to exclude that $f(y) = 4k$. Let us assume that we can represent y in the form $a_1 \cdot \ldots \cdot a_{4k}$ with every a_i in A. At least one of the a_i is not 2 (otherwise the product would be a power of 2, while y is odd), and hence it is less than or equal to $3/2$. It follows that

$$a_1 \cdot \ldots \cdot a_{4k} \leq 2^{4k-1} \cdot \frac{3}{2} = 15 \cdot \frac{2^{4k-2}}{5} < \frac{2^{4k+2}}{5} < y,$$

which contradicts the fact that $a_1 \cdot \ldots \cdot a_{4k}$ is a representation of y.

Note. Using a similar approach one can also prove that all pairs of the form

$$\left(3, \frac{2^{2k+1}+1}{3}\right) \quad \text{and} \quad \left(11, \frac{2^{10k+5}+1}{11}\right)$$

satisfy the required inequality.

Second solution As in the previous solution we obtain the lower bound (Q2.1).

Now we claim that all the pairs of the form

$$x = 2^k + 1, \qquad y = 4^k - 2^k + 1$$

satisfy the required inequality when k is large enough. To begin with, it is easy to see that

$$2^k + 1 = \frac{2^k+1}{2^k} \cdot \underbrace{2 \cdot \ldots \cdot 2}_{k \text{ terms}} \qquad \text{and} \qquad 2^{3k} + 1 = \frac{2^{3k}+1}{2^{3k}} \cdot \underbrace{2 \cdot \ldots \cdot 2}_{3k \text{ terms}},$$

which shows that $f(2^k+1) \le k+1$ and $f(2^{3k}+1) \le 3k+1$. On the other hand, from (Q2.1) we deduce that the previous inequalities are actually equalities, and therefore

$$f(x) = k+1 \qquad \text{and} \qquad f(xy) = 3k+1.$$

Therefore, it remain to show that $f(y) > 2k$. Since $y > 2^{2k-1}$ (for $k \ge 1$), from (Q2.1) we already know that $f(y) \ge 2k$, and hence we just need to exclude that $f(y) = 2k$. Let us assume that we can represent y in the form $a_1 \cdot \ldots \cdot a_{2k}$. At least one of the factors is not 2, and hence it is less than or equal to $3/2$. Thus when k is large enough it follows that

$$a_1 \cdot \ldots \cdot a_{2k} \le 2^{2k-1} \cdot \frac{3}{2} = \frac{3}{4} \cdot 2^{2k} < 2^{2k} - 2^k < y,$$

which contradicts the fact that $a_1 \cdot \ldots \cdot a_{2k}$ is a representation of y.

Third solution Let's start by showing that $(x, y) = (7, 7)$ satisfies $f(xy) < f(x) + f(y)$. We have $f(7) \ge 4$ since 7 cannot be written as the product of 3 or fewer elements of A: indeed $2^3 > 7$, and any other product of at most three elements of A does not exceed $2^2 \cdot \frac{3}{2} = 6 < 7$. On the other hand, $f(49) \le 7$ since $49 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot \frac{3}{2} \cdot \frac{49}{48}$.

Suppose by contradiction that there exist only finitely many pairs (x, y) that satisfy $f(xy) < f(x) + f(y)$. This implies that there exists M large enough so that whenever $a > M$ or $b > M$ holds we have $f(ab) = f(a) + f(b)$ (indeed, it is clear that the reverse inequality $f(ab) \le f(a) + f(b)$ is always satisfied).

Now take any pair (x, y) that satisfies $f(xy) < f(x) + f(y)$ and let $n > M$ be any integer. We obtain

$$f(n) + f(xy) = f(nxy) = f(nx) + f(y) = f(n) + f(x) + f(y),$$

which contradicts $f(xy) < f(x) + f(y)$.

Problem 3 The n contestants of an EGMO are named $C_1, \ldots, C_n$. After the competition they queue in front of the restaurant according to the following rules.

- The Jury chooses the initial order of the contestants in the queue.
- Every minute, the Jury chooses an integer i with $1 \leq i \leq n$.
  - If contestant C_i has at least i other contestants in front of her, she pays one euro to the Jury and moves forward in the queue by exactly i positions.
  - If contestant C_i has fewer than i other contestants in front of her, the restaurant opens and the process ends.

(a) Prove that the process cannot continue indefinitely, regardless of the Jury's choices.

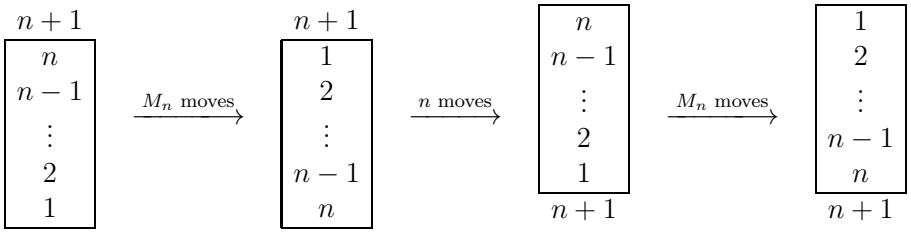
(b) Determine for every n the maximum number of euros that the Jury can collect by cunningly choosing the initial order and the sequence of moves.

(Hungary)

Solution The maximal number of euros is $2^n - n - 1$.

To begin with, we show that it is possible for the Jury to collect this number of euros. We argue by induction. Let us assume that the Jury can collect M_n euros in a configuration with n contestants. Then we show that the Jury can collect at least $2M_n + n$ moves in a configuration with $n+1$ contestants. Indeed, let us begin with all the contestants lined up in reverse order. In the first M_n moves the Jury keeps C_{n+1} in first position and reverses the order of the remaining contestants, then in the next n moves all contestants C_1, …, C_n (in this order) jump over C_{n+1} and end up in the first n positions of the line in reverse order, and finally in the last M_n moves the Jury rearranges the first n positions.

Since $M_1 = 0$ and $M_{n+1} \geq 2M_n + n$, an easy induction shows that $M_n \geq 2^n - n - 1$.



Let us show now that at most $2^n - n - 1$ moves are possible. To this end, let us identify a line of contestants with a permutation σ of $\{1, \ldots, n\}$. To each permutation we associate the set of *reverse pairs*

$$R(\sigma) := \{(i, j) : 1 \leq i < j \leq n \text{ and } \sigma(i) > \sigma(j)\},$$

and the nonnegative integer

$$W(\sigma) := \sum_{(i,j) \in R(\sigma)} 2^i,$$

which we call the *total weight* of the permutation. We claim that the total weight decreases after any move of the contestants. Indeed, let us assume that C_i moves forward in the queue, let σ be the permutation before the move, and let σ' denote the permutation after the move. Since C_i jumps over exactly i contestants, necessarily she jumps over at least one contestant C_j with index

$j > i$. This means that the pair (i, j) is reverse with respect to σ but not with respect to σ', and this yields a reduction of 2^i in the total weight. On the other hand, the move by C_i can create new reverse pairs of the form (k, i) with $k < i$, but their total contribution is at most

$$2^0 + 2^1 + \ldots + 2^{i-1} = 2^i - 1.$$

In conclusion, when passing from σ to σ', at least one term 2^i disappears from the computation of the total weight, and the sum of all the new terms that might have been created is at most $2^i - 1$. This shows that $W(\sigma') \leq W(\sigma) - 1$.

We conclude by observing that the maximum possible value of $W(\sigma)$ is realized when all pairs are reverse, in which case

$$W(\sigma) = \sum_{i=1}^{n} (i-1)2^i = 2^n - n - 1.$$

This proves that the number of moves is less than or equal to $2^n - n - 1$, and in particular it is finite.

Alternative solution As in the previous solution, the fundamental observation is again that, when a contestant C_i moves forward, necessarily she has to jump over at least one contestant C_j with $j > i$.

Let us show now that the process ends after a finite number of moves. Let us assume that this is not the case. Then at least one contestant moves infinitely many times. Let i_0 be the largest index such that C_{i_0} moves infinitely many times. Then necessarily C_{i_0} jumps infinitely many times over some fixed C_{j_0} with $j_0 > i_0$. On the other hand, we know that C_{j_0} makes only a finite number of moves, and therefore she can precede C_{i_0} in the line only a finite number of times, which is absurd.

In order to estimate from above the maximal number of moves, we show that the contestant C_i can make at most $2^{n-i} - 1$ moves. Indeed, let us argue by "backward extended induction". To begin with, we observe that the estimate is trivially true for C_n because she has no legal move.

Let us assume now that the estimate has been proved for C_i, C_{i+1}, ..., C_n, and let us prove it for C_{i-1}. When C_{i-1} moves, at least one contestant C_j with $j > i - 1$ must precede her in the line. The initial configuration can provide at most $n - i$ contestants with larger index in front of C_{i-1}, which means at most $n - i$ moves for C_{i-1}. All other moves are possible only if some contestant in the range C_i, C_{i+1}, ..., C_n jumps over C_{i-1} during her moves. As a consequence, the total number of moves of C_{i-1} is at most

$$n - i + \sum_{k=i}^{n} (2^{n-k} - 1) = 2^{n-i+1} - 1.$$

Summing over all indices we obtain that

$$\sum_{i=1}^{n} (2^{n-i} - 1) = 2^n - n - 1,$$

which gives an estimate for the total number of moves.

The same example of the first solution shows that this upper bound can actually be achieved.

Comment In every move of the example, the moving contestant jumps over exactly one contestant with larger index (and as a consequence over all contestants with smaller index).

Problem 4 A *domino* is a 1×2 or 2×1 tile.

Let $n \geq 3$ be an integer. Dominoes are placed on an $n \times n$ board in such a way that each domino covers exactly two cells of the board, and dominoes do not overlap.

The *value* of a row or column is the number of dominoes that cover at least one cell of this row or column. The configuration is called *balanced* if there exists some $k \geq 1$ such that each row and each column has a value of k.

Prove that a balanced configuration exists for every $n \geq 3$, and find the minimum number of dominoes needed in such a configuration.

(Merlijn Staps, The Netherlands)

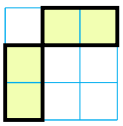
Solution The minimal number of dominoes required in a balanced configuration is $2n/3$ if n is a multiple of 3, and $2n$ otherwise.

In order to show that this number is necessary, we count in two different ways the number of elements of the set S of all pairs (ℓ, d), where ℓ is a row or a column of the board, and d is a domino that covers at least one cell of that row or column. On the one hand, since each row or column intersects the same number k of dominoes, the set S has $2nk$ elements. On the other hand, since each domino intersects 3 rows/columns, the set S has $3D$ elements, where D is the total number of dominoes on the board. This leads to the equality

$$2nk = 3D.$$

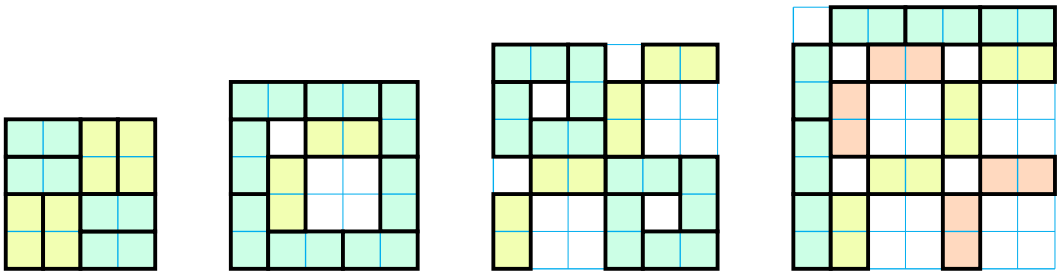
If n is a multiple of 3, from the trivial inequality $k \geq 1$ we obtain that $D \geq 2n/3$. If n is not a multiple of 3, then k is a multiple of 3, which means that $k \geq 3$ and hence $D \geq 2n$.

Now we need to exhibit a balanced configuration with this number of dominoes. The following diagram shows a balanced configuration with $n = 3$ and $k = 1$.



If n is any multiple of 3, we can obtain a balanced configuration with $k = 1$ by using $n/3$ of these 3×3 blocks along the principal diagonal of the board.

The following diagrams show balanced configurations with $k = 3$ and $n \in \{4, 5, 6, 7\}$.



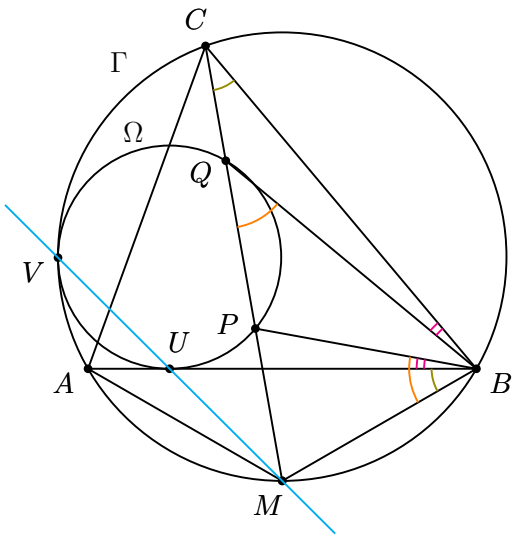
Any $n \geq 8$ can be written in the form $4A + r$ where A is a positive integer and $r \in \{4, 5, 6, 7\}$. Therefore, we can obtain a balanced configuration with $n \geq 8$ and $k = 3$ by using one block with size $r \times r$, and A blocks with size 4×4 along the principal diagonal of the board. In particular, this construction covers all the cases where n is not a multiple of 3.

Problem 5 Let Γ be the circumcircle of triangle ABC. A circle Ω is tangent to the line segment AB and is tangent to Γ at a point lying on the same side of the line AB as C. The angle bisector of $\angle BCA$ intersects Ω at two different points P and Q.

Prove that $\angle ABP = \angle QBC$.

(Dominika Regiec, Poland)

Solution 1 Let M be the midpoint of the arc AB that does not contain C, let V be the intersection of Ω and Γ, and let U be the intersection of Ω and AB.



The proof can be divided in two steps:

1. *Proving that* $MP \cdot MQ = MB^2$.

   It is well-known that V, U and M are collinear (indeed the homothety with center in V that sends Ω to Γ sends U to the point of Γ where the tangent to Γ is parallel to AB, and this point is M), and

   $$MV \cdot MU = MA^2 = MB^2.$$

   This follows from the similitude between the triangles $\triangle MAV$ and $\triangle MUA$. Alternatively, it is a consequence of the following well-known lemma: *Given a circle* Γ *with a chord* AB*, let* M *be the middle point of one of the two arcs* AB*. Take a line through* M *which intersects* Γ *again at* X *and* AB *at* Y*. Then* $MX \cdot MY$ *is independent of the choice of the line.*

   Computing the power of M with respect to Ω we obtain that

   $$MP \cdot MQ = MU \cdot MV = MB^2.$$

2. *Conclude the proof given that* $MP \cdot MQ = MB^2$.

   The relation $MP \cdot MQ = MB^2$ in turn implies that triangle $\triangle MBP$ is similar to triangle $\triangle MQB$, and in particular $\angle MBP = \angle MQB$. Keeping into account that $\angle MCB = \angle MBA$, we finally conclude that

   $$\angle QBC = \angle MQB - \angle MCB = \angle MBP - \angle MBA = \angle PBA,$$

   as required.

Solution 2 The second solution is in fact a different proof of the first part of Solution 1.

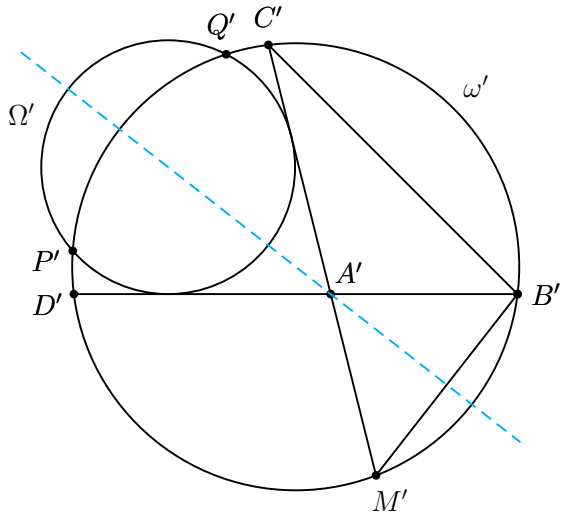
Let us consider the inversion with respect to circle with center M and radius $MA = MB$. This inversion switches AB and Γ, and fixes the line passing through M, U, V. As a consequence, it keeps Ω fixed, and therefore it switches P and Q. This is because they are the intersections between the fixed line MC and Ω, and the only fixed point on the segment MC is its intersection with the inversion circle (thus P and Q are switched). This implies that $MP \cdot MQ = MB^2$.

Solution 3 This solution is instead a different proof of the second step of Solution 1.

Let I and J be the incenter and the C-excenter of $\triangle ABC$ respectively. It is well-known that $MA = MI = MJ$, therefore the relation $MP \cdot MQ = MA^2$ implies that $(P, Q, I, J) = -1$.

Now observe that $\angle IBJ = 90°$, thus BI is the angle bisector of $\angle PBQ$ as it is well-known from the theory of harmonic pencils, and this leads easily to the conclusion.

Solution 4 Let D denote the intersection of AB and CM. Let us consider an inversion with respect to B, and let us use primes to denote corresponding points in the transformed diagram, with the gentlemen agreement that $B' = B$.



Since inversion preserves angles, it turns out that

$$\angle A'B'M' = \angle A'M'B' = \angle ACB,$$

and in particular triangle $A'B'M'$ is isosceles with basis $B'M'$.

The image of CM is the circumcircle of $B'C'M'$, which we denote by ω'. It follows that the centers of both ω' and the image Ω' of Ω lie on the perpendicular bisector of $B'M'$. Therefore, the whole transformed diagram is symmetric with respect to the perpendicular bisector of $B'M'$, and in particular the arcs $D'P'$ and $Q'C'$ of ω' are equal.

This is enough to conclude that $\angle D'B'P' = \angle Q'B'C'$, which implies the conclusion.

Problem 6

(a) Prove that for every real number t such that $0 < t < \frac{1}{2}$ there exists a positive integer n with the following property: for every set S of n positive integers there exist two different elements x and y of S, and a *non-negative* integer m (i.e. $m \geq 0$), such that

$$|x - my| \leq ty.$$

(b) Determine whether for every real number t such that $0 < t < \frac{1}{2}$ there exists an infinite set S of positive integers such that

$$|x - my| > ty$$

for every pair of different elements x and y of S and every *positive* integer m (i.e. $m > 0$).

(Merlijn Staps, The Netherlands)

Solution

Part (a) Let n be any positive integer such that

$$(1+t)^{n-1} \geq \frac{1}{t} \tag{Q6.1}$$

(this inequality is actually true for every large enough n due to Bernoulli's inequality).

Let S be any set of n distinct positive integers, which we denote by

$$s_1 < s_2 < \ldots < s_n.$$

We distinguish two cases.

- If $s_{i+1} \leq (1+t)s_i$ for some $i \in \{1, \ldots, n-1\}$, then

  $$|s_{i+1} - s_i| = s_{i+1} - s_i \leq ts_i,$$

  and therefore the required inequality is satisfied with $x = s_{i+1}$, $y = s_i$, and $m = 1$.

- If $s_{i+1} > (1+t)s_i$ for every $i \in \{1, \ldots, n-1\}$, then by induction we obtain that

  $$s_n > (1+t)^{n-1}s_1.$$

  As a consequence, from (Q6.1) it follows that

  $$|s_1| = s_1 < \frac{1}{(1+t)^{n-1}} \cdot s_n \leq ts_n,$$

  and therefore the required inequality is satisfied with $x = s_1$, $y = s_n$, and $m = 0$.

Part (b) (Explicit formula) We claim that an infinite set with the required property exists. To this end, we rewrite the required condition in the form

$$\left|\frac{x}{y} - m\right| > t.$$

This is equivalent to saying that the distance between the ratio x/y and the set of positive integers is greater than t.

Now we construct an increasing sequence s_n of odd coprime positive integers satisfying

$$\frac{1}{2} - \frac{1}{2s_n} > t \qquad \forall n \geq 1, \tag{Q6.2}$$

and such that for every $j > i$ it turns out that

$$\frac{s_i}{s_j} < \frac{1}{2} \qquad \text{and} \qquad t < \left\{\frac{s_j}{s_i}\right\} < \frac{1}{2}, \tag{Q6.3}$$

where $\{\alpha\}$ denotes the fractional part of α. This is enough to show that the set $S := \{s_n : n \geq 1\}$ has the required property.

To this end, we consider the sequence defined recursively by

$$s_{n+1} = \frac{(s_1 \cdot \ldots \cdot s_n)^2 + 1}{2},$$

with s_1 large enough. An easy induction shows that this is an increasing sequence of odd positive integers. For every $i \in \{1, \ldots, n\}$ it turns out that

$$\frac{s_i}{s_{n+1}} \leq \frac{2}{s_i} \leq \frac{2}{s_1} < \frac{1}{2}$$

because s_1 is large enough, which proves the first relation in (Q6.3). Moreover, it turns out that

$$\frac{s_{n+1}}{s_i} = \frac{(s_1 \cdot \ldots \cdot s_n)^2}{2s_i} + \frac{1}{2s_i}.$$

The first term is a positive integer plus $1/2$, from which it follows that the distance of s_{n+1}/s_i from the positive integers is greater than or equal to

$$\frac{1}{2} - \frac{1}{2s_i} \geq \frac{1}{2} - \frac{1}{2s_1},$$

which is greater than t if s_1 is large enough. This proves the second relation in (Q6.3).

Part (b) (Arithmetic approach) We produce an increasing sequence s_n of odd and coprime positive integers that satisfies (Q6.3) every $j > i$. As in the previous solution, this is enough to conclude.

We argue by induction. To begin with, we choose s_1 to be any odd integer satisfying the inequality in (Q6.2). Let us assume now that s_1, …, s_n have already been chosen, and let us choose s_{n+1} in such a way that

$$s_{n+1} \equiv \frac{s_i - 1}{2} \pmod{s_i} \qquad \forall i \in \{1, \ldots, n\}.$$

We can solve this system because the previously chosen integers are odd and coprime. Moreover, any solution of this system is coprime with $s_1, \ldots, s_n$. Indeed, for every $1 \leq i \leq n$ it turns out that

$$s_{n+1} = \frac{s_i - 1}{2} + k_i s_i$$

for some positive integer k_i. Therefore, any prime p that divides both s_{n+1} and s_i divides also $(2k_i + 1)s_i - 2s_{n+1} = 1$, which is absurd. Finally, we observe that we can assume that s_{n+1} is odd and large enough. In this way we can guarantee that

$$\frac{s_i}{s_{n+1}} < \frac{1}{2} \qquad \forall i \in \{1, \ldots, n\},$$

which is the first requirement in (Q6.3), and

$$k_i + t < k_i + \frac{1}{2} - \frac{1}{2s_i} = \frac{s_{n+1}}{s_i} < k_i + \frac{1}{2} \qquad \forall i \in \{1, \ldots, n\},$$

which implies the second requirement in (Q6.3).

Part (b) (Algebraic approach) Again we produce an increasing sequence s_n of positive integers that satisfies (Q6.3) every $j > i$.

To this end, for every positive integer x, we define its *security region*

$$S(x) := \bigcup_{n \geq 1} \left((n + t)x, (n + \tfrac{1}{2})x \right).$$

The security region $S(x)$ is a periodic countable union of intervals of length $(\frac{1}{2} - t)x$, whose left-hand or right-hand endpoints form an arithmetic sequence. It has the property that

$$t < \left\{ \frac{y}{x} \right\} < \frac{1}{2} \qquad \forall y \in S(x).$$

Now we prove by induction that we can choose a sequence s_n of positive integers satisfying (Q6.3) and in addition the fact that every interval of the security region $S(s_n)$ contains at least one interval of $S(s_{n-1})$.

To begin with, we choose s_1 large enough so that the length of the intervals of $S(s_1)$ is larger than 1. This guarantees that any interval of $S(s_1)$ contains at least a positive integer. Now let us choose a positive integer $s_2 \in S(s_1)$ that is large enough. This guarantees that s_1/s_2 is small enough, that the fractional part of s_2/s_1 is in $(t, 1/2)$, and that every interval of the security region $S(s_2)$ contains at least one interval of $S(s_1)$, and hence at least one positive integer.

Let us now assume that $s_1, \ldots, s_n$ have been already chosen with the required properties. We know that every interval of $S(s_n)$ contains at least one interval of $S(s_{n-1})$, which in turn contains an interval in $S(s_{n-2})$, and so on up to $S(s_1)$. As a consequence, we can choose a large enough positive integer s_{n+1} that lies in $S(s_k)$ for every $k \in \{1, \ldots, n\}$. Since s_{n+1} is large enough, we are sure that

$$\frac{s_k}{s_{n+1}} < t \qquad \forall k \in \{1, \ldots, n\}.$$

Moreover, we are sure also that all the intervals of $S(s_{n+1})$ are large enough, and therefore they contain at least one interval of $S(s_n)$, which in turn contains at least one interval of $S(s_{n-1})$, and so on. Finally, the condition

$$t < \left\{ \frac{s_{n-1}}{s_n} \right\} < \frac{1}{2}$$

is guaranteed by the fact that s_{n+1} was chosen in an interval that is contained in $S(s_k)$ for every $k \in \{1, \ldots, n\}$. This completes the induction.

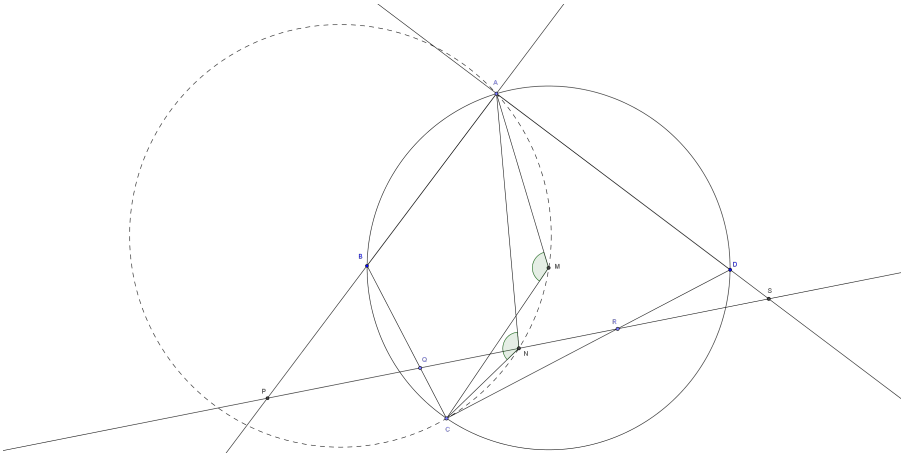
Problem 1

Let $ABCD$ be a convex quadrilateral with $\angle DAB = \angle BCD = 90^\circ$ and $\angle ABC > \angle CDA$. Let Q and R be points on the segments BC and CD, respectively, such that the line QR intersects lines AB and AD at points P and S, respectively. It is given that $PQ = RS$. Let the midpoint of BD be M and the midpoint of QR be N. Prove that M, N, A and C lie on a circle.

Mark Mordechai Etkind, Israel

Solution 1

Note that N is also the midpoint of PS. From right-angled triangles PAS and CQR we obtain $\angle ANP = 2\angle ASP$, $\angle CNQ = 2\angle CRQ$, hence $\angle ANC = \angle ANP + \angle CNQ = 2(\angle ASP + \angle CRQ) = 2(\angle RSD + \angle DRS) = 2\angle ADC$.
Similarly, using right-angled triangles BAD and BCD, we obtain $\angle AMC = 2\angle ADC$.
Thus $\angle AMC = \angle ANC$, and the required statement follows.



Solution 2

In this proof we show that we have $\angle NCM = \angle NAM$ instead. From right-angled triangles BCD and QCR we get $\angle DRS = \angle CRQ = \angle RCN$ and $\angle BDC = \angle DCM$. Hence $\angle NCM = \angle DCM - \angle RCN$. From right-angled triangle APS we get $\angle PSA = \angle SAN$. From right-angled triangle BAD we have $\angle MAD = \angle BDA$. Moreover, $\angle BDA = \angle DRS + \angle RSD - \angle RDB$.
Therefore $\angle NAM = \angle NAS - \angle MAD = \angle CDB - \angle DRS = \angle NCM$, and the required statement follows.

Solution 3

As N is also the midpoint of PS, we can shrink triangle APS to a triangle A_0QR (where P is sent to Q and S is sent to R). Then A_0, Q, R and C lie on a circle with center N. According to the shrinking the line A_0R is parallel to the line AD. Therefore $\angle CNA = \angle CNA_0 = 2\angle CRA_0 = 2\angle CDA = \angle CMA$. The required statement follows.

Problem 2

Find the smallest positive integer k for which there exist a colouring of the positive integers $\mathbb{Z}_{>0}$ with k colours and a function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ with the following two properties:

(i) For all positive integers m, n of the same colour, $f(m+n) = f(m) + f(n)$.

(ii) There are positive integers m, n such that $f(m+n) \neq f(m) + f(n)$.

In a colouring of $\mathbb{Z}_{>0}$ with k colours, every integer is coloured in exactly one of the k colours. In both (i) and (ii) the positive integers m, n are not necessarily different.

Merlijn Staps, the Netherlands

Solution 1:

The answer is $k = 3$.
First we show that there is such a function and coloring for $k = 3$. Consider $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ given by $f(n) = n$ for all $n \equiv 1$ or 2 modulo 3, and $f(n) = 2n$ for $n \equiv 0$ modulo 3. Moreover, give a positive integer n the i-th color if $n \equiv i$ (3).
By construction we have $f(1+2) = 6 \neq 3 = f(1) + f(2)$ and hence f has property (ii).
Now let n, m be positive integers with the same color i. If $i = 0$, then $n+m$ has color 0, so $f(n+m) = 2(n+m) = 2n + 2m = f(n) + f(m)$. If $i = 1$, then $n+m$ has color 2, so $f(n+m) = n+m = f(n) + f(m)$. Finally, if $i = 2$, then $n+m$ has color 1, so $f(n+m) = n+m = f(n) + f(m)$. Therefore f also satisfies condition (i).

Next we show that there is no such function and coloring for $k = 2$.
Consider any coloring of $\mathbb{Z}_{>0}$ with 2 colors and any function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ satisfying conditions (i) and (ii). Then there exist positive integers m and n such that $f(m+n) \neq f(m) + f(n)$. Choose m and n such that their sum is minimal among all such m, n and define $a = m + n$. Then in particular for every $b < a$ we have $f(b) = bf(1)$ and $f(a) \neq af(1)$.
If a is even, then condition (i) for $m = n = \frac{a}{2}$ implies $f(a) = f(\frac{a}{2}) + f(\frac{a}{2}) = f(1)a$, a contradiction. Hence a is odd. We will prove two lemmas.

Lemma 1. *Any odd integer $b < a$ has a different color than a.*

Proof. Suppose that $b < a$ is an odd integer, and that a and b have the same color. Then on the one hand, $f(a+b) = f(a) + bf(1)$. On the other hand, we also have $f(a+b) = f(\frac{a+b}{2}) + f(\frac{a+b}{2}) = (a+b)f(1)$, as $\frac{a+b}{2}$ is a positive integer smaller than a. Hence $f(a) = f(a+b) - bf(1) = (a+b)f(1) - bf(1) = af(1)$, which is again a contradiction. Therefore all odd integers smaller than a have a color different from that of a. □

Lemma 2. *Any even integer $b < a$ has the same color as a*

Proof. Suppose $b < a$ is an even integer, and that a and b have different colors. Then $a - b$ is an odd integer smaller than a, so it has the same color as b. Thus $f(a) = f(a-b) + f(b) = (a-b)f(1) + bf(1) = af(1)$, a contradiction. Hence all even integers smaller than a have the same color as a. □

Suppose now $a+1$ has the same color as a. As $a > 1$, we have $\frac{a+1}{2} < a$ and therefore $f(a+1) = 2f(\frac{a+1}{2}) = (a+1)f(1)$. As $a-1$ is an even integer smaller than a, we have by Lemma 2 that $a-1$ also has the same color as a. Hence $2f(a) = f(2a) = f(a+1) + f(a-1) = (a+1)f(1) + (a-1)f(1) = 2af(1)$, which implies that $f(a) = af(1)$, a contradiction. So a and $a+1$ have different colors.
Since $a-2$ is an odd integer smaller than a, by Lemma 1 it has a color different from that of a, so $a-2$ and $a+1$ have the same color. Also, we have seen by Lemma 2 that $a-1$ and a have the same color. So $f(a) + f(a-1) = f(2a-1) = f(a+1) + f(a-2) = (a+1)f(1) + (a-2)f(1) = (2a-1)f(1)$, from which it follows that $f(a) = (2a-1)f(1) - f(a-1) = (2a-1)f(1) - (a-1)f(1) = af(1)$, which contradicts our choice of a and finishes the proof.

Solution 2:

We prove that $k \leq 3$ just as in first solution.
Next we show that there is no such function and coloring for $k = 2$.
Consider any coloring of $\mathbb{Z}_{>0}$ with 2 colors and any function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ satisfying conditions (i) and (ii).
We first notice with $m = n$ that $f(2n) = 2f(n)$.

Lemma 3. *For every $n \in \mathbb{Z}_{>0}$, $f(3n) = 3f(n)$ holds.*

Proof. Define $c = f(n), d = f(3n)$. Then we have the relations

$$f(2n) = 2c, \qquad f(4n) = 4c, \qquad f(6n) = 2d.$$

- If n and $2n$ have the same color, then $f(3n) = f(n) + f(2n) = 3c = 3f(n)$.
- If n and $3n$ have the same color, then $4c = f(4n) = f(n) + f(3n) = c + f(3n)$, so $f(3n) = 3f(n)$.
- If $2n$ and $4n$ have the same color, then $2d = f(6n) = f(2n) + f(4n) = 2c + 4c = 6c$, so $f(3n) = d = 3c$.
- Otherwise n and $4n$ have the same color, and $2n$ and $3n$ both have the opposite color to n. Therefore we compute $5c = f(n) + f(4n) = f(5n) = f(2n) + f(3n) = 2c + f(3n)$ so $f(3n) = 3f(n)$.

Consequently, for $k = 2$ we necessarily have $f(3n) = 3f(n)$. □

Now let a be the smallest integer such that $f(a) \neq af(1)$. In particular a is odd and $a > 3$. Consider the three integers a, $\frac{a-3}{2}$, $\frac{a+3}{2}$. By pigeonhole principle two of them have the same color.

- If $\frac{a-3}{2}$ and $\frac{a+3}{2}$ have the same color, then $f(a) = \frac{a-3}{2}f(1) + \frac{a+3}{2}f(1) = af(1)$.
- If a and $\frac{a-3}{2}$ have the same color, then $3\frac{a-1}{2}f(1) = 3f(\frac{a-1}{2}) = f(\frac{3a-3}{2}) = f(a) + f(\frac{a-3}{2}) = f(a) + \frac{a-3}{2}f(1)$, so $f(a) = af(1)$.
- If a and $\frac{a+3}{2}$ have the same color, then $3\frac{a+1}{2}f(1) = 3f(\frac{a+1}{2}) = f(\frac{3a+3}{2}) = f(a) + f(\frac{a+3}{2}) = f(a) + \frac{a+3}{2}f(1)$, so $f(a) = af(1)$.

In the three cases we find a contradiction with $f(a) \neq af(1)$, so it finishes the proof.

Solution 3:

As before we prove that $k \leq 3$ and for any such function and colouring we have $f(2n) = 2f(n)$.
Now we show that there is no such function and coloring for $k = 2$.
Consider any coloring of $\mathbb{Z}_{>0}$ with 2 colors and any function $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ satisfying conditions (i) and (ii).
Say the two colors are white (W) and black (B). Pick m, n any two integers such that $f(m+n) = f(m)+f(n)$.
Without loss of generality we may assume that $m + n, m$ are black and n is white.

Lemma 4. *For all $l \in \mathbb{Z}_{>0}$ and every x whose color is black, we have $x + lm$ is black and $f(x + lm) = f(x) + lf(m)$.*

Proof. We proceed by induction. It is clearly true for $l = 0$. If $x + lm$ is black and satisfies $f(x + lm) = f(x) + lf(m)$, then $f(x + (l + 1)m) = f(x + lm) + f(m) = f(x) + (l + 1)f(m)$ and $f(x + (l + 1)m + n) = f(x + lm) + f(m + n) = f(x) + lf(m) + f(m + n) \neq f(x) + (l + 1)f(m) + f(n) = f(x + (l + 1)m) + f(n)$, so $x + (l + 1)m$ is not the same color of n, therefore $x + (l + 1)m$ is black. Thjs completes the induction. □

In particular we then must have that $2^l n$ is white for every l, because otherwise since $2^l m$ is black we would have $2^l f(m + n) = f(2^l m + 2^l n) = f(2^l m) + f(2^l n) = 2^l(f(m) + f(n))$, and consequently $f(m + n) = f(m) + f(n)$.

Lemma 5. *For every $l \geq 1$, $2^l m + 2^{l-1} n$ is black.*

Proof. On the one hand we have $2^l f(m+n) = f(2^l m + 2^l n) = f(2^{l-1}(2m+n) + 2^{l-1}n)$. On the other hand we have

$$2^l f(m+n) = 2^{l-1} \cdot 2f(m+n) \neq 2^{l-1}(f(m+n)+f(m)+f(n)) = 2^{l-1}(f(2m+n)+f(n)) = f(2^l m + 2^{l-1}n)) + f(2^{l-1}n).$$

Therefore $2^l m + 2^{l-1}n$ and $2^{l-1}n$ have different color, which means $2^l m + 2^{l-1}n$ is black. □

Combining the two lemmas give $jm + 2^{l-1}n$ is black for all $j \geq 2^l$ and every $l \geq 1$.
Now write $m = 2^{l-1}m'$ with m' odd. Let t be a number such that $\frac{2^t-1}{m'}$ is an integer and $j = \frac{2^t-1}{m'}n \geq 2^l$, i.e. t is some multiple of $\phi(m')$. Then we must have that $jm + 2^{l-1}n$ is black, but by definition $jm + 2^{l-1}n = (2^t - 1)2^{l-1}n + 2^{l-1}n = 2^{t+l-1}n$ is white. This is a contradiction, so $k = 2$ is impossible.

Problem 3

There are 2017 lines in a plane such that no 3 of them go through the same point. Turbo the snail can slide along the lines in the following fashion: she initially moves on one of the lines and continues moving on a given line until she reaches an intersection of 2 lines. At the intersection, she follows her journey on the other line turning left or right, alternating the direction she chooses at each intersection point she passes. Can it happen that she slides through a line segment for a second time in her journey but in the opposite direction as she did for the first time?

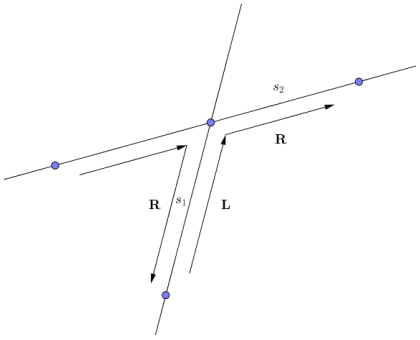
Márk Di Giovanni, Hungary

1. Solution

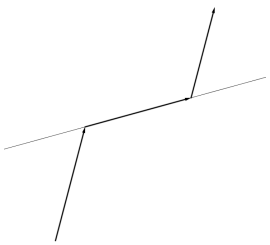
We show that this is not possible.
The lines divide the plane into disjoint regions. We claim that there exists an *alternating* 2-coloring of these regions, that is each region can be colored in black or white, such that if two regions share a line segment, they have a different color. We show this inductively.
If there are no lines, this is obvious. Consider now an arrangement of n lines in the plane and an alternating 2-coloring of the regions. If we add a line g, we can simply switch the colors of all regions in one of the half planes divided by g from white to black and vice versa. Any line segment not in g will still be between two regions of different color. Any line segment in g cuts a region determined by the n lines in two, and since we switched colors on one side of g this segment will also lie between two regions of different color.
Now without loss of generality we may assume, that Turbo starts on a line segment with a white region on her left and a black one on her right. At any intersection, if she turns right, she will keep the black tile to her right. If she turns left, she will keep the white tile to her left. Thus wherever she goes, she will always have a white tile on her left and a black tile on her right. As a consequence, she can cross every line segment only in the direction where she has a white tile to her left, and never the opposite direction where she would have a black tile to the left.

2. Solution

Suppose the assumption is true.
Let's label each segment in the snail's path with **L** or **R** depending on the direction that Turbo chose at the start point of this segment (one segment can have several labels if it has been visited several times).
Consider the first segment that has been visited twice in different directions, name this segment s_1. Assume without loss of generality that it is labeled with **L**. Then next segment must be labeled with **R**, name this one s_2.
Let's look at the label which s_1 can get on the second visit. If it gets **L** then the previous segment in the path must be s_2. But in this case s_1 is not the first segment that has been visited twice in different directions because s_2 has been visited earlier. So the second label of s_1 must be **R**, and Turbo must have come from the opposite side of s_2.



Since Turbo alters directions at each point, labels in her path also alter. And because two labels of s_1 are different, the number of visited segments between these two visits must be even.
Now let's make the following observation: each segment in the path corresponds to exactly one line, and its previous and next segments are on opposite sides of this line.



Again consider the path between two visits of s_1.
Each line intersecting this path must be crossed an even number of times because Turbo has to return to the initial side of each line. Therefore, an even number of segments of Turbo's path are contained on each of these lines. But the line containing s_1 must be crossed an odd number times. Since each crossing corresponds to exactly one segment in the path, the number of segments must be odd.
Here we get the contradiction. Therefore, the assumption is false.

3. Solution

Suppose that the snail always slides slightly to the right of the line segments on her path. When turning to the right, she does not cross any line, whereas when turning to the left, she crosses exactly two lines. This means that at any time of her journey, she has crossed an even number of lines.
Assuming that at some point she slides along a segment for the second time, but in the opposite direction, we argue that she needs to cross an odd number of lines. Let ℓ be the line on which the revisit happens. In order to get to the other side of ℓ, the snail has to cross ℓ an odd number of times. To visit the same segment of ℓ, she must cross every other line an even number of times.

4. Solution

Let us color in red all intersection points of the given lines and let us choose one of two possible directions on each segment (draw an arrow on each segment). Consider a red point R where two given lines a and b meet, and the four segments a_1, a_2, b_1, b_2 with endpoint R (so that $a_i \subset a$, $b_j \subset b$). R is called a *saddle* if on a_1, a_2 the arrows go out of R while on b_1, b_2 the arrows enter R, or visa versa, on b_1, b_2 the arrows go out of R while on a_1, a_2 the arrows enter R. The set of arrows (chosen on all segments) is said to be *good* if all red points are saddles. It is sufficient to prove that there exists a good set of arrows. Indeed, if initially Turbo is moving along (or opposite) the arrow, then this condition holds after she turns at a red point.
The given lines cut the plane into *regions*. Further we need the following *property* of the good set of arrows (this property directly follows from the definition): the boundary of any bounded region is a directed cycle of arrows; the boundary of any unbounded region is a directed chain of arrows.
We construct a good set of arrows by induction on n with trivial base $n = 1$. Now erase one of n given lines and assume we have a good set of arrows for remaining $n-1$ lines. Now restore the n-th line ℓ, assume that ℓ is horizontal. Denote by $A_1, \ldots, A_{n-1}$ all *new* red points on ℓ from the left to the right. Each of A_i belongs to some *old* segment m_i of the line ℓ_i. Let us call A_i *ascending* if the arrow on m_i goes up, and *descending* if the arrow on m_i goes down. Consider the region containing the segment A_iA_{i+1}. By the *property*, A_i and A_{i+1} can not be both ascending or both descending. Thus we can choose arrows on all pieces of ℓ so that each arrow goes from a descending to an ascending vertex.

Each of points A_i cuts m_i into two new pieces; the direction of new pieces supposed to be the same as on m_i. Now simultaneously change the direction of arrows on all pieces below the line ℓ. It is easy to see that $A_1, \ldots, A_{n-1}$ become saddles, while the other red points remain saddles. This completes the induction step.

Problem 4

Let $n \geq 1$ be an integer and let $t_1 < t_2 < \ldots < t_n$ be positive integers. In a group of $t_n + 1$ people, some games of chess are played. Two people can play each other at most once. Prove that it is possible for the following conditions to hold at the same time:

i) The number of games played by each person is one of $t_1, t_2, \ldots, t_n$,

ii) For every i with $1 \leq i \leq n$, there is someone who has played exactly t_i games of chess.

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Comment

In graph theory terms the problem is to prove that for any finite nonempty set $\mathcal{T}$ of positive integers there exists a graph of size $\max \mathcal{T} + 1$ such that the degree set of the graph is equal to $\mathcal{T}$.

Among graph theory specialists a generalization of this problem is known [1]. Nevertheless, the problem still suited the contest.

1. Solution (see also [2])

Let $\mathcal{T} = \{t_1, \ldots, t_n\}$. The proof proceeds by induction on $n = |\mathcal{T}|$. If $n = 1$ and $\mathcal{T} = \{t\}$, choose a group of $t + 1$ people and let every pair of two persons play against each other. Then every person has played t games and the conditions of the problem are satisfied.

In the inductive step, suppose that $\mathcal{T}$ has $n \geq 2$ elements $t_1 < t_2 < \cdots < t_n$. Consider the set

$$\mathcal{T}' = \{t_n - t_{n-1}, t_n - t_{n-2}, \ldots, t_n - t_1\}.$$

By the inductive hypothesis, there exists a group G' of $t_n - t_1 + 1$ people that satisfies the conditions of the problem for T'.

Next construct a group G'' of $t_n + 1$ people by adding t_1 people who do not know any of the other $t_n - t_1 + 1$ people in G'. Finally, construct a group G by complementing the knowledge relation in G'': two persons play against each other in G if and only if they do not play against each other in G''.

By construction $t \in \mathcal{T}$ if and only if there exists a person in G'' that played against exactly $t_n - t$ other people (if $t = t_n$, choose one of the t_1 people added to G'). That person knows $t_n - (t_n - t) = t$ other students in G, completing the proof.

2. Solution

Let $\mathcal{T} = \{t_1, \ldots, t_n\}$. The proof proceeds by induction on $n = |\mathcal{T}|$. If $n = 1$ and $\mathcal{T} = \{t\}$, we choose a group of $t + 1$ people such that everyone plays with

everyone else. If $n = 2$ and $\mathcal{T} = \{t_1, t_2\}$ with $t_1 < t_2$, divide the $t_2 + 1$ people into groups A resp. B of size t_1 resp. $t_2 - t_1 + 1$ such that everyone from group A played with everyone else whereas people from group B only played with the people from group A. Then the people from group A resp. B played with exactly t_2 resp. t_1 other people. In the inductive step, suppose that T has $n > 2$ elements $t_1 < \ldots < t_n$. Consider the set

$$\mathcal{T}' = (\mathcal{T} \backslash \{t_1, t_n\}) - t_1 = \{t_{n-1} - t_1, t_{n-1} - t_1, \ldots, t_2 - t_1\}.$$

By the induction hypothesis there exists a group C of $t_{n-1} - t_1 + 1$ people that satisfies the conditions of the problem for T'. Next add groups D resp. E of t_1 resp. $t_n - t_{n-1}$ people such that people from group D played with everyone else whereas people from group E only played with the people from group D. Then the people from group C played with t other people if and only if they played with $t - t_1$ many people among C, i.e. if and only if $t \in \{t_2, \ldots, t_{n-1}\} = \mathcal{T} \backslash \{t_1, t_n\}$. People from group D resp. E played with t_n resp t_1 peoples, which completes the proof.

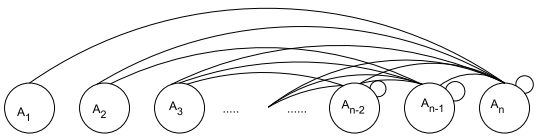
3. Solution

The proof proceeds by induction on $|t_n|$. If $t_n = 1$ we have $n = 1$ and we can consider two persons that play against each other. Then every player has played 1 game and the conditions of the problem are satisfied. If $t_n > 1$ we distinguish the two cases $t_1 > 1$ and $t_1 = 1$. If $t_1 > 1$ there exists, by the induction hypothesis, a group A of size t_n that satisfies the conditions of the problem for $t'_1 = t_1 - 1, \ldots, t'_n = t_n - 1$. Now add a new person to A and let him/her play against everyone from A. The new group will be of size $t_n + 1$ and there exists a person which has played t games if and only if there exists a person that has played $t - 1$ games within A, i.e. if and only if $t \in \{t_1, \ldots, t_n\}$. Hence the conditions of the problem are satisfied.

If $t_1 = 1$ there exists, by the induction hypothesis, a group B of size t_{n-1} that satisfies the conditions of the problem for $t'_1 = t_2 - 1, \ldots, t'_{n-2} = t_{n-1} - 1$. Now add a new person P and let him/her play with everyone from group B and a group C of size $t_n - t_{n-1} > 0$ and let them play with P. The new group will be of size $t_{n-1} + 1 + (t_n - t_{n-1}) + 1 = t_n + 1$. Since person P has played against everyone he will have played t_n games. The people in C will have played $1 = t_1$ games. There exists a person in B that has played t games if and only if there exist a person in B that has played $t - 1$ games within B, i.e. if and only if $t \in \{t_2, \ldots, t_{n-1}\}$. Hence the conditions of the problem are satisfied.

4. Solution

We generalize the construction for $\mathcal{T} = \{1, ..., n\}$
Construction
Take sets of people $A_1, ..., A_n$. Let all people of A_i play chess with all people in A_j with $j \geq n - i + 1$



Now the number of games played by anyone in A_i is
$(\sum_{j\geq n-i+1}|A_j|)$ or $(\sum_{j\geq n-i+1}|A_j|)-1$ if $i\geq n-i+1$.
Now if we start with one person in each A_i and two people in $A_{\lceil \frac{n}{2}\rceil}$. The number of played games for anyone in A_i is equal to i. In particular this is a construction for $\mathcal{T}=\{1,..,n\}$
Now to get to numbers of general sets $\mathcal{T}$ of size n we can change the sizes of A_i but keep the construction.

Variant 1

Observation 1 *Adding a person to a set A_i increases the number of games played in A_j for $j\geq n-i+1$, by exactly one.*

Start with the construction above and then add t_1-1 people to group A_n, making the new set of games played equal to $\{t_1,t_1+1,...,n+t_1-1\}$. Then add t_2-t_1-1 to A_{n-1} to get set of games played to $\{t_1,t_2,t_2+1,...,n+t_2-2\}$ and repeat until we get to the set $\mathcal{T}$ adding a total of $\sum_{j=1}^{n}t_j-t_{j-1}-1=t_n-n$ people (let $t_0=0$), so we get t_n+1 people in the end.
Clearly we can start by adding vertices to A_1 or any other set instead of A_n first and obtain an equivalent construction with the same number of people.

Variant 2

It is also possible to calculate the necessary sizes of A_i's all at once. We have by construction the number of games played in A_1 is less than the number of games played in A_2 etc. So we have that in the end we want the games played in A_i to be exactly t_i.
So $(t_t,t_2,t_3,...,t_n)\stackrel{!}{=}(|A_n|\ ,\ |A_n|+|A_{n-1}|\ ,\ ...\ ,\ (\sum_{j=2}^{n}|A_j|)-1\ ,\ (\sum_{j=1}^{n}|A_j|)-1)$.
This gives us by induction that $|A_n|\stackrel{!}{=}t_1$, $|A_{n-1}|\stackrel{!}{=}t_2-t_1$, ..., $|A_{\lceil\frac{n}{2}\rceil}|\stackrel{!}{=}t_{n-\lceil\frac{n}{2}\rceil+1}-t_{n-\lceil\frac{n}{2}\rceil}+1$, ..., $|A_1|\stackrel{!}{=}t_n-t_{n-1}$ and a quick calculation shows that the sum of all sets is exactly $1+\sum_{j=1}^{n}(t_j-t_{j-1})=t_n+1$. (where the $+1$ comes from the set $A_{\lceil\frac{n}{2}\rceil}$ and $t_0=0$.)

References

[1] Timothy A. Sipka. The orders of graphs with prescribed degree sets. *Journal of Graph Theory*, 4(3):301–307, 1980.

[2] Amitabha Tripathi and Sujith Vijay. A short proof of a theorem on degree sets of graphs. *Discrete Appl. Math.*, 155(5):670–671, 2007.

Problem 5

Let $n \geq 2$ be an integer. An n-tuple $(a_1, a_2, \ldots, a_n)$ of positive integers is *expensive* if there exists a positive integer k such that

$$(a_1 + a_2)(a_2 + a_3) \cdot \cdots \cdot (a_{n-1} + a_n)(a_n + a_1) = 2^{2k-1}.$$

a) Find all positive integers $n \geq 2$ for which there exists an expensive n-tuple.

b) Prove that for every positive integer m there exists an integer $n \geq 2$ such that m belongs to an expensive n-tuple.

There are exactly n factors in the product on the left hand side.

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Solution 1

a) Notice that for odd integers $n > 2$, the tuple $(1, 1, \ldots, 1)$ is expensive. We will prove that there are no expensive n-tuples for even n.

Lemma 0.1. *If an expensive n-tuple exists for some $n \geq 4$, then also an expensive $n-2$-tuple.*

Proof. In what follows all indices are considered modulo n. Let $(a_1, a_2, \ldots, a_n)$ be an expensive n-tuple and a_t the largest element of the tuple. We have the inequalities

$$a_{t-1} + a_t \leq 2a_t < 2(a_t + a_{t+1}) \tag{1}$$
$$a_t + a_{t+1} \leq 2a_t < 2(a_{t-1} + a_t). \tag{2}$$

Since both $a_{t-1} + a_t$ and $a_t + a_{t+1}$ are powers of 2 (they are divisors of a power of 2), we deduce from (1) and (2)

$$a_{t-1} + a_t = a_t + a_{t+1} = 2^r$$

for some positive integer r, and in particular $a_{t-1} = a_{t+1}$.
Consider now the $n-2$-tuple $(b_1, \ldots, b_{n-2})$ obtained by removing a_t and a_{t+1} from $(a_1, a_2, \ldots, a_n)$. By what we just said we have

$$\prod_{i=1}^{n-2}(b_i + b_{i+1}) = \frac{\prod_{i=1}^{n}(a_i + a_{i+1})}{(a_{t-1} + a_t)(a_t + a_{t+1})} = 2^{2(k-r)-1},$$

and hence $(b_1, \ldots, b_{n-2})$ is again expensive. □

From the lemma we now conclude that if there exists an expensive n-tuple for some even n, then also an expensive 2-tuple i.e.

$$(a_1 + a_2)^2 = 2^{2k-1}$$

for some positive integers a_1, a_2, which is impossible since the right hand side is not a square.

b) We prove this by induction. In a) we saw that 1 belongs to an expensive n-tuple. Assume now that all odd positive integers less that 2^k belong to an expensive n-tuple, for some $k \geq 1$. Hence for any odd $r < 2^k$ there is an integer n and an expensive n-tuple $(a_1, \ldots, r, \ldots, a_n)$. We notice that then also $(a_1, \ldots, r, 2^{k+1} - r, r, \ldots, a_n)$ is expensive. Since $2^{k+1} - r$ can take all odd values between 2^k and 2^{k+1} the induction step is complete.

Solution 2

a) For odd n the tuple $(1, 1, \ldots, 1)$ is a solution.
Now consider n even. Since the product $\prod(a_i + a_{i+1})$ is a power of two, every factor needs to be a power of two. We are going to prove that for all tuples $(a_1, \ldots, a_n)$ such that $a_i + a_{i+1}$ is always a power of two , it is the case that $\prod(a_i + a_{i+1})$ is equal to an even power of two. We are going to prove this with strong induction on $\sum a_i$. When all a_i are equal to one this is certainly the case. Since $a_i + a_{i+1} > 1$ it is even and we conclude that the a_i are either all odd or all even. In the case they are all even, then consider the tuple $(b_1, \ldots, b_n)$ with $b_i = a_i/2$. This tuple clearly satisfies the hypothesis as well and we have $\sum b_i < \sum a_i$. Furthermore we have $\prod(a_i + a_{i+1}) = 2^n \prod(b_i + b_{i+1})$ and since n is even we are done in this case.
Now all a_i are odd. Suppose none of are one, then consider the tuple $(b_1, \ldots, b_n)$ with $b_i = (a_i + (-1)^i)/2$. Since all a_i are odd and strictly larger than one, the b_i are positive integers and satisfy $b_i + b_{i+1} = (a_i + a_{i+1})/2$, a power of two. Since $\sum b_i < \sum a_i$ and $\prod(a_i + a_{i+1}) = 2^n \prod(b_i + b_{i+1})$ we are done in this case again. Now there is at least one a_i being one. We may assume $i = 1$, because the condition is cyclic. Moreover we may also assume that $a_2 > 1$ since not all of the a_i are equal to one. Let now k be the smallest index larger than one such that a_k is equal to one. We are not excluding the case $k = n + 1$, yet. Now for $i = 1, \ldots, k - 1$ we have $a_i + a_{i+1} > 2$ and thus divisible by four. By induction it easily follows that $a_i \equiv (-1)^{i+1} \mod (4)$ for $i = 1, \ldots, k - 1$. In particular, since $a_k = 1$ we find that k is odd and at least three. Now consider the tuple $(b_1, \ldots, b_n)$ with $b_i = (a_i - (-1)^i)/2$ for $i = 1, \ldots, k$ and $b_i = a_i$ otherwise. This is again a tuple that satisfies the hypothesis, since $b_1 = a_1 = 1 = b_k = a_k$. Moreover $b_2 < a_2$ and thus $\sum b_i < \sum a_i$. Finally we have $\prod(a_i + a_{i+1}) = 2^{k-1} \prod(b_i + b_{i+1})$ and since k is odd we conclude the proof.

b) We use some of the ideas from a). Consider the operators $T_\pm(n) = 2n \pm 1$. We claim that for every odd integer m there is an integer r and signs $\epsilon_i \in \{+, -\}$ for $i = 1, \ldots r$ such that $T_{\epsilon_r} \circ \cdots \circ T_{\epsilon_1}(1) = m$. This is certainly true for $m = 1$ and for $m > 1$ we find that $m = T_-((m + 1)/2)$ if $m \equiv 1 \mod (4)$ and $m = T_+((m - 1)/2)$ if $m \equiv 3 \mod (4)$. Note that both $(m + 1)/2$ and $(m - 1)/2$ are odd integers in their respective cases and $(m-1)/2 \leq (m+1)/2 < m$ for $m > 1$. Therefore iterating the procedure will eventually terminate in one.
For the construction it is most convenient to set $n = 2l + 1$ and label the tuple $(a_{-l}, a_{-l+1}, \ldots, a_l)$. For $m = 1$ we have the expensive tuple $(1, 1, 1)$. For $m > 1$ we will define operators $T_\pm$ on expensive tuples with the condition $a_{-l} = a_l = 1$ that give rise to a new expensive tuple $(b_{-l'}, \ldots, b_{l'})$ with $b_{-l'} = b_{l'} = 1$ and $b_0 = T_\pm a_0$. It is then clear that $T_{\epsilon_r} \circ \cdots \circ T_{\epsilon_1}((1, 1, 1))$ is an expensive tuple containing m. We define $T_\pm$ as follows: set $l' = l + 1$ and $b_{-l'} = b_{l'} = 1$ and $b_i = T_{\pm(-1)^i}(a_i)$ for $i = -l, \ldots, l$. Here we identify $+$ with $+1$ and $-$ with -1. We are left to prove that the new tuple is indeed expensive. If $\pm(-1)^l = -1$, then $\prod(b_i + b_{i+1}) = 4 \cdot 2^{2l} \prod(a_i + a_{i+1})$, and if $\pm(-1)^l = +1$, then $\prod(b_i + b_{i+1}) = 4 \cdot 2^{2l+2} \prod(a_i + a_{i+1})$. In both cases we end up with an expensive tuple again.

Problem 6

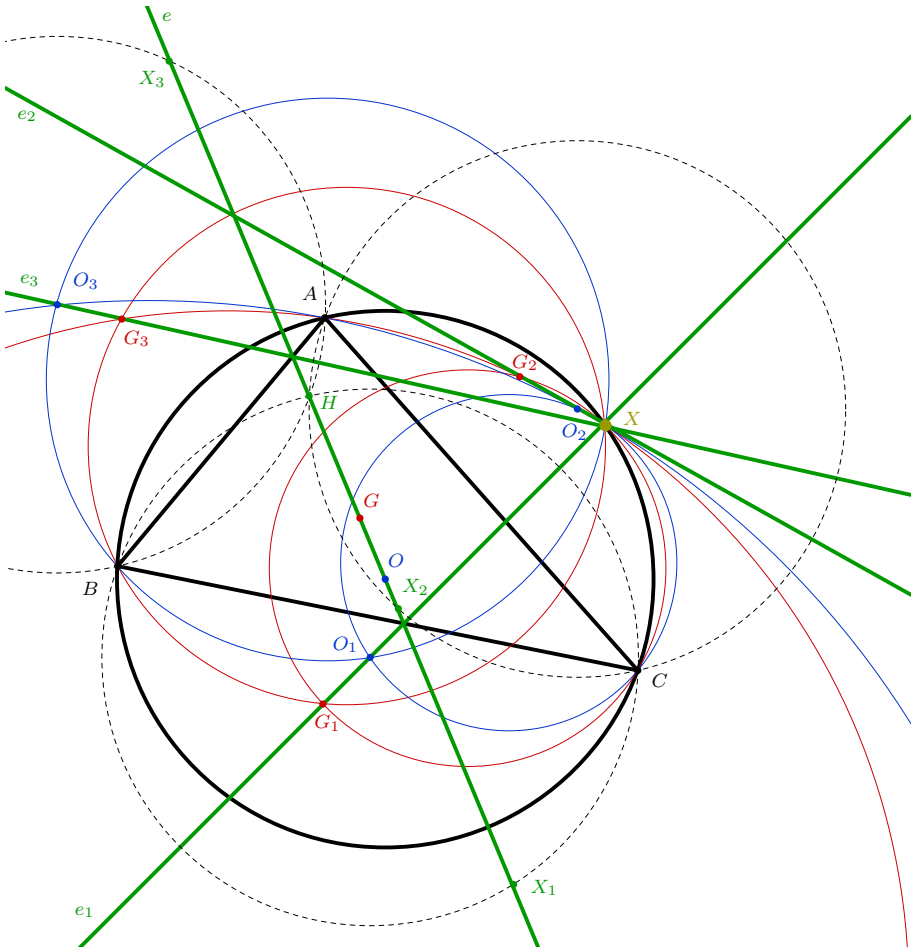
Let ABC be an acute-angled triangle in which no two sides have the same length. The reflections of the centroid G and the circumcentre O of ABC in its sides BC, CA, AB are denoted by G_1, G_2, G_3, and O_1, O_2, O_3, respectively. Show that the circumcircles of the triangles G_1G_2C, G_1G_3B, G_2G_3A, O_1O_2C, O_1O_3B, O_2O_3A and ABC have a common point.

The centroid of a triangle is the intersection point of the three medians. A median is a line connecting a vertex of the triangle to the midpoint of the opposite side.

Charles Leytem, Luxembourg

Solution 1 (Euler lines)

Let H denote the orthocenter of ABC, and let e denote its Euler line. Let e_1, e_2, e_3 denote the respective reflections of e in BC, CA, AB. The proof naturally divides into two parts: we first show that pairwise intersections of the circles in question correspond to pairwise intersections of e_1, e_2, e_3, and then prove that e_1, e_2, e_3 intersect in a single point on the circumcircle of ABC.

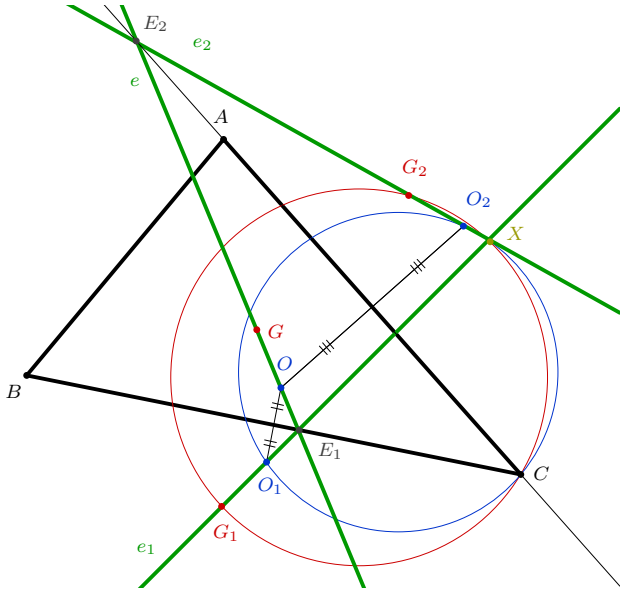


Now consider for example the circumcircles of O_1O_2C and G_1G_2C. By construction, it is clear that $\angle O_2CO_1 = \angle G_2CG_1 = 2\angle ACB$. Let G_1O_1 and G_2O_2 meet at X, and let e meet e_1, e_2 at E_1, E_2, respectively, as shown in the diagram below. Chasing angles,

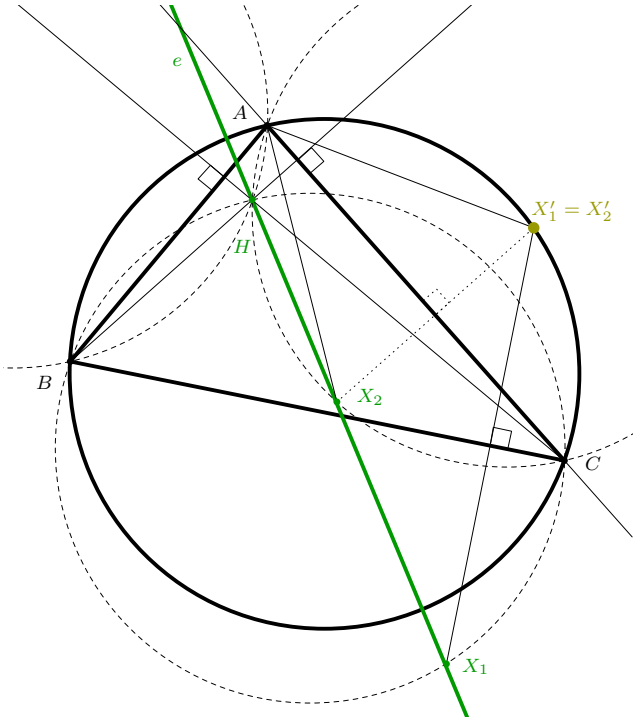
$$\begin{aligned}\angle G_2XG_1 = \angle O_2XO_1 = \angle E_2XE_1 &= 180^\circ - \angle E_1E_2X - \angle XE_1E_2 \\ &= 180^\circ - 2\angle E_1E_2C - \big(\angle CE_1E_2 - \angle CE_1X\big).\end{aligned}$$

But $\angle CE_1X = \angle BE_1O_1 = \angle BE_1O = 180^\circ - \angle CE_1E_2$, and thus

$$\angle G_2XG_1 = \angle O_2XO_1 = 2\big(180^\circ - \angle E_1E_2C - \angle CE_1E_2\big) = 2\angle ACB.$$



It follows from this that X lies on the circumcircles of G_1G_2C and O_1O_2C. In other words, the second point of intersection of the circumcircles of G_1G_2C and O_1O_2C is the intersection of e_1 and e_2. Similarly, the circumcircles of G_1G_3B and O_1O_3B meet again at the intersection of e_1 and e_3, and those of G_2G_3A and O_2O_3A meet again at the intersection of e_2 and e_3.



It thus remains to show that e_1, e_2, e_3 are concurrent, and intersect on the circumcircle of ABC. Let e meet the circumcircles of the triangles BCH, ACH, ABH at X_1, X_2, X_3, respectively. It is well known that the reflections of H in the sides of ABC lie on the circumcircle of ABC. For this reason, the circumcircles of BCH, ACH, ABH have the same radius as the circumcircle of ABC, and hence the reflections X_1', X_2', X_3' of X_1, X_2, X_3 in the sides $[BC], [CA], [AB]$ lie on the circumcircle of ABC. By definition, X_1', X_2', X_3' lie on e_1, e_2, e_3, respectively. It thus remains to show that they coincide.
To show that, for example, $X_1' = X_2'$, it will be sufficient to show that $\angle X_2AC = \angle X_1'AC$, since we have already shown that X_1' and X_2' lie on the circumcircle of ABC. But, chasing angles in the diagram above,

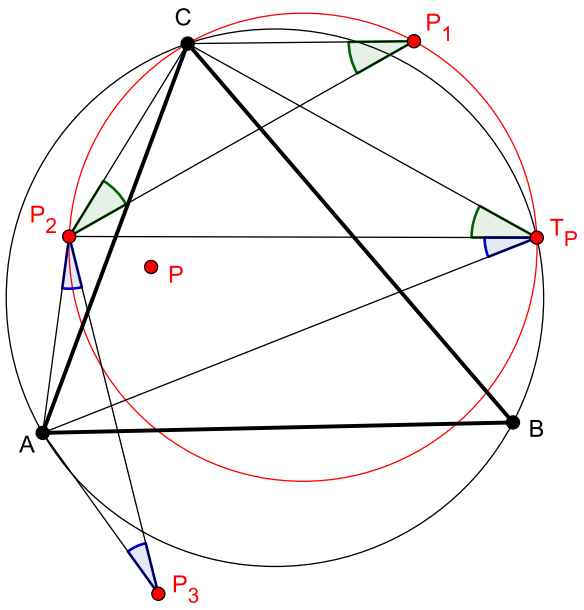
$$\begin{aligned}\angle X_1'AC &= \angle X_1'AB - \angle BAC = \big(180^\circ - \angle X_1'CB\big) - \angle BAC \\ &= 180^\circ - \angle X_1CB - \angle BAC = \big(180^\circ - \angle BAC\big) - \angle X_1HB \\ &= \angle BHC - \angle X_1HB = \angle X_2HC = \angle X_2AC,\end{aligned}$$

where we have used the fact that $BHCX_1$ and AHX_2C are cyclic by construction, and the fact that $\angle BHC = 180^\circ - \angle BAC$. This shows that $X_1' = X_2'$. Similarly, $X_1' = X_3'$, which completes the proof. □

Remark. The statement of the problem remains true if O and G are replaced with two points that are aligned with the orthocenter H of the triangle, and indeed, the proof above did not require any property of G and O other than the fact that they lie on a line through H, the Euler line.

Solution 2

The proof consists of two parts. First, we show that if P is any point inside the triangle ABC and P_1, P_2, P_3 are its reflections in the sides BC, CA, AB, then the circumcircles of the triangles P_1P_2C, P_1P_3B, P_2P_3A intersect in a point T_P on the circumcircle of the triangle ABC. In the second part, we show that T_G coincides with T_O.
Now let P be any point inside the triangle ABC and let P_1, P_2, P_3 be the reflections in the sides as above. Let T_P be the second intersection of the circumcircles of the triangles P_1P_2C and ABC. We want to show that T_P lies on the circumcircles of the triangles P_1P_3B and P_2P_3A.



By construction, we have $P_1C = P_2C$, hence

$$\angle CP_1P_2 = 90^\circ - \frac{1}{2}\angle P_2CP_1 = 90^\circ - \angle ACB.$$

Similarly, $\angle P_2P_3A = 90^\circ - \angle BAC$. This gives us

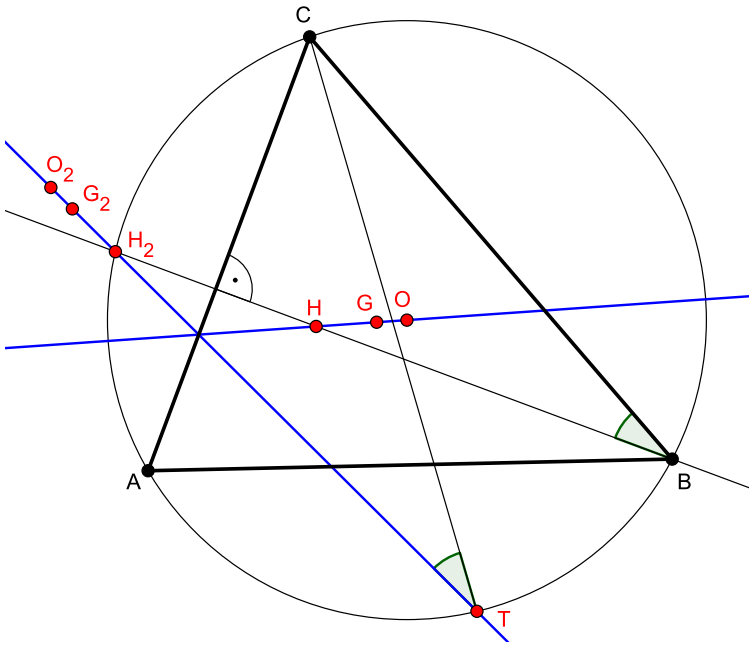
$$\begin{aligned}\angle P_2T_PA &= \angle CT_PA - \angle CT_PP_2 = \angle CBA - \angle CP_1P_2\\ &= \angle CBA - 90^\circ + \angle ACB = 90^\circ - \angle BAC = \angle P_2P_3A,\end{aligned}$$

so T_P lies on the circumcircle of the triangle P_2P_3A. Similarly, T_P lies on the circumcircle of the triangle P_1P_3B which completes the first part.
Note that if P_2 is given, then T_P is the unique point on the circumcircle of the triangle ABC with $\angle CT_PP_2 = 90^\circ - \angle ACB$. In the second part, we will use this as follows: If we can find a point T on the circumcircle of the triangle ABC with $\angle CTG_2 = \angle CTO_2 = 90^\circ - \angle ACB$,

then $T = T_G = T_O$ and we are done.
Let H be the orthocenter of the triangle ABC and let H_2 be the reflection in the side AC. It is known that H_2 lies on the circumcircle of the triangle ABC. G, O, H lie on the Euler line, so G_2, O_2, H_2 are collinear as well. Let T be the second intersection of G_2H_2 and the circumcircle of the triangle ABC. We can now complete the proof by seeing that

$$\angle CTG_2 = \angle CTO_2 = \angle CTH_2 = \angle CBH_2 = 90^\circ - \angle ACB.$$

□



Solution 3 (complex numbers)

For every point P, let p denote the corresponding complex number. Set O to be the origin, so $o = 0$, and without loss of generality we can assume that a, b and c lie on the unit circle. Then the centroid can be expressed as $g = \dfrac{a+b+c}{3}$.
The segments oo_1 and bc have a common midpoint, so $o_1 + o = b + c$, and then $o_1 = b + c$. Similarly $o_2 = a + c$ and $o_3 = a + b$. In order to compute g_1, define y to be the projection of g onto bc. Since b and c are on the unit circle, it is well known that y can be expressed as

$$y = \frac{1}{2}(b + c + g - bc\bar{g}).$$

By using $\bar{a} = \dfrac{1}{a}$, $\bar{b} = \dfrac{1}{b}$ and $\bar{c} = \dfrac{1}{c}$ (points on the unit circle), we obtain

$$g_1 = b + c - \frac{ab + bc + ca}{3a}.$$

Similarly, we get $g_2 = a + c - \dfrac{ab + bc + ca}{3b}$ and $g_3 = a + b - \dfrac{ab + bc + ca}{3c}$.
1) Proof that circumcircles of triangles abc, o_1o_2c, o_1o_3b and o_2o_3a have common point.
Let x be the point of intersection of circumcircles of triangles o_1o_2c and abc ($x \neq c$). We know that x, o_1, o_2 and c are concyclic if and only if $\dfrac{x-c}{o_1-c} : \dfrac{x-o_2}{o_1-o_2}$ is real number, which is equivalent to

$$\frac{x-c}{\bar{x}-\bar{c}} \cdot \frac{o_1-o_2}{\bar{o_1}-\bar{o_2}} = \frac{o_1-c}{\bar{o_1}-\bar{c}} \cdot \frac{x-o_2}{\bar{x}-\bar{o_2}}. \tag{1}$$

Since x and c are on the unit circle $\dfrac{x-c}{\bar{x}-\bar{c}} = -xc$. Also, $\dfrac{o_1-o_2}{\bar{o_1}-\bar{o_2}} = \dfrac{b-a}{\bar{b}-\bar{a}} = -ab$, and $\dfrac{o_1-c}{\bar{o_1}-\bar{c}} = \dfrac{b}{\bar{b}} = b^2$. Since $\bar{x} = \dfrac{1}{x}$, from (1) and previous relations, we have:

$$x = \frac{ab + bc + ca}{a + b + c}.$$

This formula is symmetric, so we conclude that x also belongs to circumcircles of o_1o_3b and o_2o_3a.
2) Proof that x belongs to circumcircles of g_1g_2c, g_1g_3b and g_2g_3a.
Because of symmetry, it is enough to prove that x belongs to circumcircle of g_1g_2c, i.e. to prove the following:

$$\frac{x-c}{\bar{x}-\bar{c}} \cdot \frac{g_1-g_2}{\bar{g_1}-\bar{g_2}} = \frac{g_1-c}{\bar{g_1}-\bar{c}} \cdot \frac{x-g_2}{\bar{x}-\bar{g_2}}. \tag{2}$$

Easy computations give that

$$g_1 - g_2 = (b-a)\frac{2ab - bc - ac}{3ab}, \quad \bar{g_1} - \bar{g_2} = (\bar{b}-\bar{a})\frac{2 - \frac{a}{c} - \frac{b}{c}}{3},$$

and then

$$\frac{g_1-g_2}{\bar{g_1}-\bar{g_2}} = \frac{c(bc + ac - 2ab)}{2c - a - b}.$$

On the other hand we have

$$g_1 - c = \frac{2ab - bc - ac}{3a}, \quad \bar{g_1} - \bar{c} = \frac{2c - a - b}{3bc}.$$

This implies

$$\frac{g_1 - c}{\bar{g_1} - \bar{c}} = \frac{2ab - bc - ac}{2c - a - b} \cdot \frac{bc}{a}.$$

Then (2) is equivalent to

$$\begin{aligned} -xc \cdot \frac{c(bc + ac - 2ab)}{2c - a - b} &= \frac{2ab - bc - ac}{2c - a - b} \cdot \frac{bc}{a} \cdot \frac{x - g_2}{\bar{x} - \bar{g_2}} \\ \iff xca(\bar{x} - \bar{g_2}) &= b(x - g_2), \end{aligned}$$

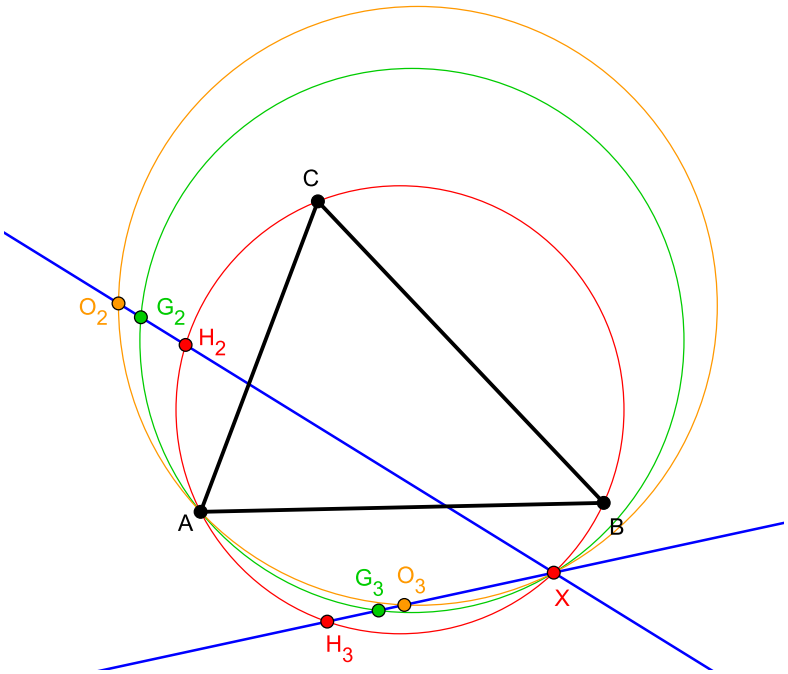
which is also equivalent to

$$\frac{ab + bc + ca}{a + b + c} \cdot ca \left(\frac{a + b + c}{ab + bc + ca} - \frac{1}{a} - \frac{1}{c} + \frac{a + b + c}{3ac} \right) = b \cdot \left(\frac{ab + bc + ca}{a + b + c} - a - c + \frac{ab + bc + ca}{3b} \right).$$

The last equality can easily be verified, which implies that x belongs to circumcircle of triangle $g_1 g_2 c$. This concludes our proof. □

Solution 4 (rotation)

The first part of the first solution and the second part of the second solution can also be done by the following rotation argument: The rotation through $2\angle BAC$ about A takes O_3 to O_2, G_3 to G_2 and H_3 to H_2 (again, H_2, H_3 are the reflections of the orthocenter H in the sides CA, AB). Let X be the intersection of the Euler line reflections e_2 (going through O_2, G_2, H_2) and e_3 (going through O_3, G_3, H_3). We now use the well-known fact that if a rotation about a point A takes a line l and a point P on l to the line l' and the point P', then the quadrilateral $APP'X$ is cyclic, where X is the intersection of l and l'. For this reason, AO_3O_2X, AG_3G_2X and AH_3H_2X are cyclic quadrilaterals. Since H_2 and H_3 lie on the circumcircle of the triangle ABC, the circumcircles of the triangles AH_3H_2 and ABC are the same, hence X lies on the circumcircle of ABC.
This proves the first part of the first solution as well as the second part of the second solution. □



Remark: Problem 6 is a special case of Corollary 3 in Darij Grinberg's paper *Anti-Steiner points with respect to a triangle*.

EGMO 2016, Day 1 – Solutions

Problem 1. Let n be an odd positive integer, and let x_1, x_2, ..., x_n be non-negative real numbers. Show that

$$\min_{i=1,\ldots,n}(x_i^2 + x_{i+1}^2) \leq \max_{j=1,\ldots,n}(2x_j x_{j+1}),$$

where $x_{n+1} = x_1$.

Solution. In what follows, indices are reduced modulo n. Consider the n differences $x_{k+1} - x_k$, $k = 1, \ldots, n$. Since n is odd, there exists an index j such that $(x_{j+1} - x_j)(x_{j+2} - x_{j+1}) \geq 0$. Without loss of generality, we may and will assume both factors non-negative, so $x_j \leq x_{j+1} \leq x_{j+2}$. Consequently,

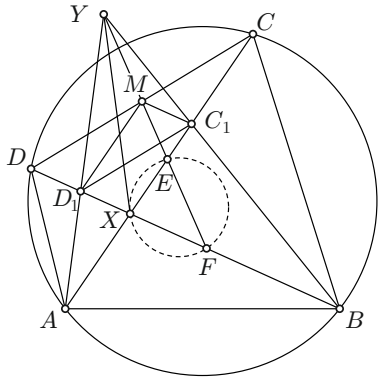
$$\min_{k=1,\ldots,n}(x_k^2 + x_{k+1}^2) \leq x_j^2 + x_{j+1}^2 \leq 2x_{j+1}^2 \leq 2x_{j+1}x_{j+2} \leq \max_{k=1,\ldots,n}(2x_k x_{k+1}).$$

Remark. If $n \geq 3$ is odd, and one of the x_k is negative, then the conclusion may no longer hold. This is the case if, for instance, $x_1 = -b$, and $x_{2k} = a$, $x_{2k+1} = b$, $k = 1, \ldots, (n-1)/2$, where $0 \leq a < b$, so the string of numbers is

$$-b, b, a, b, a, \ldots, b, a.$$

If n is even, the conclusion may again no longer hold, as shown by any string of alternate real numbers: $a, b, a, b, \ldots, a, b$, where $a \neq b$.

Problem 2. Let $ABCD$ be a cyclic quadrilateral, and let diagonals AC and BD intersect at X. Let C_1, D_1 and M be the midpoints of segments CX, DX and CD, respectively. Lines AD_1 and BC_1 intersect at Y, and line MY intersects diagonals AC and BD at different points E and F, respectively. Prove that line XY is tangent to the circle through E, F and X.



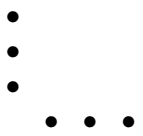
Solution. We are to prove that $\angle EXY = \angle EFX$; alternatively, but equivalently, $\angle AYX + \angle XAY = \angle BYF + \angle XBY$.

Since the quadrangle $ABCD$ is cyclic, the triangles XAD and XBC are similar, and since AD_1 and BC_1 are corresponding medians in these triangles, it follows that $\angle XAY = \angle XAD_1 = \angle XBC_1 = \angle XBY$.

Finally, $\angle AYX = \angle BYF$, since X and M are corresponding points in the similar triangles ABY and C_1D_1Y: indeed, $\angle XAB = \angle XDC = \angle MC_1D_1$, and $\angle XBA = \angle XCD = \angle MD_1C_1$.

Problem 3. Let m be a positive integer. Consider a $4m \times 4m$ array of square unit cells. Two different cells are *related* to each other if they are in either the same row or in the same column. No cell is related to itself. Some cells are coloured blue, such that every cell is related to at least two blue cells. Determine the minimum number of blue cells.

Solution 1 (Israel). The required minimum is $6m$ and is achieved by a diagonal string of m 4×4 blocks of the form below (bullets mark centres of blue cells):



In particular, this configuration shows that the required minimum does not exceed $6m$.

We now show that any configuration of blue cells satisfying the condition in the statement has cardinality at least $6m$.

Fix such a configuration and let m_1^r be the number of blue cells in rows containing exactly one such, let m_2^r be the number of blue cells in rows containing exactly two such, and let m_3^r be the number of blue cells in rows containing at least three such; the numbers m_1^c, m_2^c and m_3^c are defined similarly.

Begin by noticing that $m_3^c \geq m_1^r$ and, similarly, $m_3^r \geq m_1^c$. Indeed, if a blue cell is alone in its row, respectively column, then there are at least two other blue cells in its column, respectively row, and the claim follows.

Suppose now, if possible, the total number of blue cells is less than $6m$. We will show that $m_1^r > m_3^r$ and $m_1^c > m_3^c$, and reach a contradiction by the preceding: $m_1^r > m_3^r \geq m_1^c > m_3^c \geq m_1^r$.

We prove the first inequality; the other one is dealt with similarly. To this end, notice that there are no empty rows — otherwise, each column would contain at least two blue cells, whence a total of at least $8m > 6m$ blue cells, which is a contradiction. Next, count rows to get $m_1^r + m_2^r/2 + m_3^r/3 \geq 4m$,

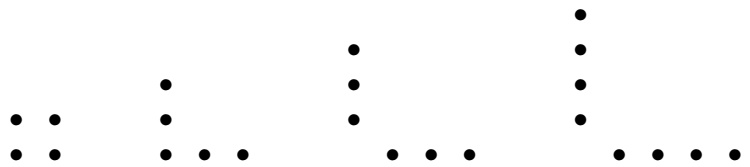
and count blue cells to get $m_1^r + m_2^r + m_3^r < 6m$. Subtraction of the latter from the former multiplied by $3/2$ yields $m_1^r - m_3^r > m_2^r/2 \geq 0$, and the conclusion follows.

Solution 2. To prove that a minimal configuration of blue cells satisfying the condition in the statement has cardinality at least $6m$, consider a bipartite graph whose vertex parts are the rows and the columns of the array, respectively, a row and a column being joined by an edge if and only if the two cross at a blue cell. Clearly, the number of blue cells is equal to the number of edges of this graph, and the relationship condition in the statement reads: for every row r and every column c, $\deg r + \deg c - \epsilon(r, c) \geq 2$, where $\epsilon(r, c) = 2$ if r and c are joined by an edge, and $\epsilon(r, c) = 0$ otherwise.

Notice that there are no empty rows/columns, so the graph has no isolated vertices. By the preceding, the cardinality of every connected component of the graph is at least 4, so there are at most $2 \cdot 4m/4 = 2m$ such and, consequently, the graph has at least $8m - 2m = 6m$ edges. This completes the proof.

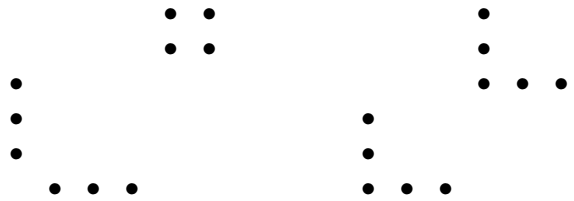
Remarks. The argument in the first solution shows that equality to $6m$ is possible only if $m_1^r = m_3^r = m_1^c = m_3^c = 3m$, $m_2^r = m_2^c = 0$, and there are no rows, respectively columns, containing four blue cells or more.

Consider the same problem for an $n \times n$ array. The argument in the second solution shows that the corresponding minimum is $3n/2$ if n is divisible by 4, and $3n/2 + 1/2$ if n is odd; if $n \equiv 2 \pmod 4$, the minimum in question is $3n/2 + 1$. To describe corresponding minimal configurations C_n, refer to the minimal configurations C_2, C_3, C_4, C_5 below:



The case $n \equiv 0 \pmod 4$ was dealt with above: a C_n consists of a diagonal string of $n/4$ blocks C_4. If $n \equiv r \pmod 4$, $r = 2, 3$, a C_n consists of a diagonal string of $\lfloor n/4 \rfloor$ blocks C_4 followed by a C_r, and if $n \equiv 1 \pmod 4$, a C_n consists of a diagonal string of $\lfloor n/4 \rfloor - 1$ blocks C_4 followed by a C_5.

Minimal configurations are not necessarily unique (two configurations being equivalent if one is obtained from the other by permuting the rows and/or the columns). For instance, if $n = 6$, the configurations below are both minimal:



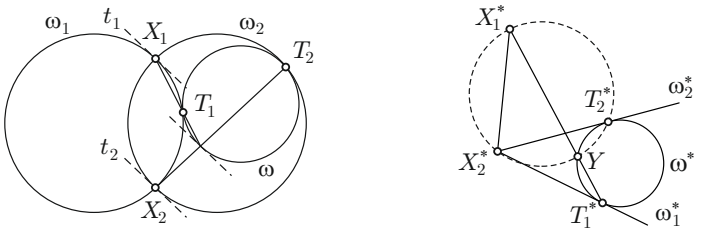
4

EGMO 2016, Day 2 – Solutions

Problem 4. Two circles, ω_1 and ω_2, of equal radius intersect at different points X_1 and X_2. Consider a circle ω externally tangent to ω_1 at a point T_1, and internally tangent to ω_2 at a point T_2. Prove that lines X_1T_1 and X_2T_2 intersect at a point lying on ω.

Solution 1. Let the line X_kT_k and ω meet again at X_k', $k = 1, 2$, and notice that the tangent t_k to ω_k at X_k and the tangent t_k' to ω at X_k' are parallel. Since the ω_k have equal radii, the t_k are parallel, so the t_k' are parallel, and consequently the points X_1' and X_2' coincide (they are not antipodal, since they both lie on the same side of the line T_1T_2). The conclusion follows.

Solution 2. The circle ω is the image of ω_k under a homothety h_k centred at T_k, $k = 1, 2$. The tangent to ω at $X_k' = h_k(X_k)$ is therefore parallel to the tangent t_k to ω_k at X_k. Since the ω_k have equal radii, the t_k are parallel, so $X_1' = X_2'$; and since the points X_k, T_k and X_k' are collinear, the conclusion follows.



Solution 3. Invert from X_1 and use an asterisk to denote images under this inversion. Notice that ω_k^* is the tangent from X_2^* to ω^* at T_k^*, and the pole X_1 lies on the bisectrix of the angle formed by the ω_k^*, not containing ω^*. Letting $X_1T_1^*$ and ω^* meet again at Y, standard angle chase shows that Y lies on the circle $X_1X_2^*T_2^*$, and the conclusion follows.

Remarks. The product h_1h_2 of the two homotheties in the first solution is reflexion across the midpoint of the segment X_1X_2, which lies on the line T_1T_2.

Various arguments, involving similarities, radical axes, and the like, work equally well to prove the required result.

Problem 5. Let k and n be integers such that $k \geq 2$ and $k \leq n \leq 2k - 1$. Place rectangular tiles, each of size $1 \times k$ or $k \times 1$, on an $n \times n$ chessboard so that each tile covers exactly k cells, and no two tiles overlap. Do this until

no further tile can be placed in this way. For each such k and n, determine the minimum number of tiles that such an arrangement may contain.

Solution. The required minimum is n if $n = k$, and it is $\min(n, 2n-2k+2)$ if $k < n < 2k$.

The case $n = k$ being clear, assume henceforth $k < n < 2k$. Begin by describing maximal arrangements on the board $[0, n] \times [0, n]$, having the above mentioned cardinalities.

If $k < n < 2k-1$, then $\min(n, 2n-2k+2) = 2n-2k+2$. To obtain a maximal arrangement of this cardinality, place four tiles, $[0, k] \times [0, 1]$, $[0, 1] \times [1, k+1]$, $[1, k+1] \times [k, k+1]$ and $[k, k+1] \times [0, k]$ in the square $[0, k] \times [0, k]$, stack $n-k-1$ horizontal tiles in the rectangle $[1, k+1] \times [k+1, n]$, and erect $n-k-1$ vertical tiles in the rectangle $[k+1, n] \times [1, k+1]$.

If $n = 2k-1$, then $\min(n, 2n-2k+2) = n = 2k-1$. A maximal arrangement of $2k-1$ tiles is obtained by stacking $k-1$ horizontal tiles in the rectangle $[0, k] \times [0, k-1]$, another $k-1$ horizontal tiles in the rectangle $[0, k] \times [k, 2k-1]$, and adding the horizontal tile $[k-1, 2k-1] \times [k-1, k]$.

The above examples show that the required minimum does not exceed the mentioned values.

To prove the reverse inequality, consider a maximal arrangement and let r, respectively c, be the number of rows, respectively columns, not containing a tile.

If $r = 0$ or $c = 0$, the arrangement clearly contains at least n tiles.

If r and c are both positive, we show that the arrangement contains at least $2n-2k+2$ tiles. To this end, we will prove that the rows, respectively columns, not containing a tile are consecutive. Assume this for the moment, to notice that these r rows and c columns cross to form an $r \times c$ rectangular array containing no tile at all, so $r < k$ and $c < k$ by maximality. Consequently, there are $n - r \geq n - k + 1$ rows containing at least one horizontal tile each, and $n - c \geq n - k + 1$ columns containing at least one vertical tile each, whence a total of at least $2n - 2k + 2$ tiles.

We now show that the rows not containing a tile are consecutive; columns are dealt with similarly. Consider a horizontal tile T. Since $n < 2k$, the nearest horizontal side of the board is at most $k-1$ rows away from the row containing T. These rows, if any, cross the k columns T crosses to form a rectangular array no vertical tile fits in. Maximality forces each of these rows to contain a horizontal tile and the claim follows.

Consequently, the cardinality of every maximal arrangement is at least $\min(n, 2n-2k+2)$, and the conclusion follows.

Remarks. **(1)** If $k \geq 3$ and $n = 2k$, the minimum is $n + 1 = 2k + 1$

and is achieved, for instance, by the maximal arrangement consisting of the vertical tile $[0,1]\times[1,k+1]$ along with $k-1$ horizontal tiles stacked in $[1,k+1]\times[0,k-1]$, another $k-1$ horizontal tiles stacked in $[1,k+1]\times[k+1,2k]$, and two horizontal tiles stacked in $[k,2k]\times[k-1,k+1]$. This example shows that the corresponding minimum does not exceed $n+1<2n-2k+2$. The argument in the solution also applies to the case $n=2k$ to infer that for a maximal arrangement of minimal cardinality either $r=0$ or $c=0$, and the cardinality is at least n. Clearly, we may and will assume $r=0$. Suppose, if possible, such an arrangement contains exactly n tiles. Then each row contains exactly one tile, and there are no vertical tiles. Since there is no room left for an additional tile, some tile T must cover a cell of the leftmost column, so it covers the k leftmost cells along its row, and there is then room for another tile along that row — a contradiction.

(2) For every pair (r,c) of integers in the range $2k-n,\ldots,k-1$, at least one of which is positive, say $c>0$, there exists a maximal arrangement of cardinality $2n-r-c$.

Use again the board $[0,n]\times[0,n]$ to stack $k-r$ horizontal tiles in each of the rectangles $[0,k]\times[0,k-r]$ and $[k-c,2k-c]\times[k,2k-r]$, erect $k-c$ vertical tiles in each of the rectangles $[0,k-c]\times[k-r,2k-r]$ and $[k,2k-c]\times[0,k]$, then stack $n-2k+r$ horizontal tiles in the rectangle $[k-c,2k-c]\times[2k-r,n]$, and erect $n-2k+c$ vertical tiles in the rectangle $[2k-c,n]\times[1,k+1]$.

Problem 6. Let S be the set of all positive integers n such that n^4 has a divisor in the range $n^2+1,\ n^2+2,\ \ldots,\ n^2+2n$. Prove that there are infinitely many elements of S of each of the forms $7m$, $7m+1$, $7m+2$, $7m+5$, $7m+6$ and no elements of S of the form $7m+3$ or $7m+4$, where m is an integer.

Solution. The conclusion is a consequence of the lemma below which actually provides a recursive description of S. The proof of the lemma is at the end of the solution.

Lemma. *The fourth power of a positive integer n has a divisor in the range $n^2+1,\ n^2+2,\ \ldots,\ n^2+2n$ if and only if at least one of the numbers $2n^2+1$ and $12n^2+9$ is a perfect square.*

Consequently, a positive integer n is a member of S if and only if $m^2-2n^2=1$ or $m^2-12n^2=9$ for some positive integer m.

The former is a Pell equation whose solutions are $(m_1,n_1)=(3,2)$ and

$$(m_{k+1},n_{k+1})=(3m_k+4n_k,2m_k+3n_k),\quad k=1,2,3,\ldots.$$

In what follows, all congruences are modulo 7. Iteration shows that $(m_{k+3},n_{k+3})\equiv(m_k,n_k)$. Since $(m_1,n_1)\equiv(3,2)$, $(m_2,n_2)\equiv(3,-2)$, and

$(m_3, n_3) \equiv (1, 0)$, it follows that S contains infinitely many integers from each of the residue classes 0 and ± 2 modulo 7.

The other equation is easily transformed into a Pell equation, $m'^2 - 12n'^2 = 1$, by noticing that m and n are both divisible by 3, say $m = 3m'$ and $n = 3n'$. In this case, the solutions are $(m_1, n_1) = (21, 6)$ and

$$(m_{k+1}, n_{k+1}) = (7m_k + 24n_k, 2m_k + 7n_k), \quad k = 1, 2, 3, \ldots .$$

This time iteration shows that $(m_{k+4}, n_{k+4}) \equiv (m_k, n_k)$. Since $(m_1, n_1) \equiv (0, -1)$, $(m_2, n_2) \equiv (-3, 0)$, $(m_3, n_3) \equiv (0, 1)$, and $(m_4, n_4) \equiv (3, 0)$, it follows that S contains infinitely many integers from each of the residue classes 0 and ± 1 modulo 7.

Finally, since the n_k from the two sets of formulae exhaust S, by the preceding no integer in the residue classes ± 3 modulo 7 is a member of S.

We now turn to the proof of the lemma. Let n be a member of S, and let $d = n^2 + m$ be a divisor of n^4 in the range $n^2 + 1, n^2 + 2, \ldots, n^2 + 2n$, so $1 \le m \le 2n$. Consideration of the square of $n^2 = d - m$ shows m^2 divisible by d, so m^2/d is a positive integer. Since $n^2 < d < (n + 1)^2$, it follows that d is not a square; in particular, $m^2/d \ne 1$, so $m^2/d \ge 2$. On the other hand, $1 \le m \le 2n$, so $m^2/d = m^2/(n^2 + m) \le 4n^2/(n^2 + 1) < 4$. Consequently, $m^2/d = 2$ or $m^2/d = 3$; that is, $m^2/(n^2 + m) = 2$ or $m^2/(n^2 + m) = 3$. In the former case, $2n^2 + 1 = (m - 1)^2$, and in the latter, $12n^2 + 9 = (2m - 3)^2$.

Conversely, if $2n^2 + 1 = m^2$ for some positive integer m, then $1 < m^2 < 4n^2$, so $1 < m < 2n$, and $n^4 = (n^2 + m + 1)(n^2 - m + 1)$, so the first factor is the desired divisor.

Similarly, if $12n^2 + 9 = m^2$ for some positive integer m, then m is odd, $n \ge 6$, and $n^4 = (n^2 + m/2 + 3/2)(n^2 - m/2 + 3/2)$, and again the first factor is the desired divisor.

The EGMO 2015 Problem Committee thanks the following countries for submitting problem proposals:

- Ireland
- Japan
- Luxembourg
- Macedonia
- Mexico
- Republic of Moldova
- Netherlands
- Poland
- Romania
- Slovenia
- Turkey
- Ukraine
- United States of America

The Members of the Problem Committee:

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Mikhail Karpuk

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Igor Voronovich

PROBLEMS

Day 1

Problem 1. Let $\triangle ABC$ be an acute-angled triangle, and let D be the foot of the altitude from C. The angle bisector of $\angle ABC$ intersects CD at E and meets the circumcircle ω of triangle $\triangle ADE$ again at F. If $\angle ADF = 45^\circ$, show that CF is tangent to ω.

Problem 2. A *domino* is a 2×1 or 1×2 tile. Determine in how many ways exactly n^2 dominoes can be placed without overlapping on a $2n \times 2n$ chessboard so that every 2×2 square contains at least two uncovered unit squares which lie in the same row or column.

Problem 3. Let n, m be integers greater than 1, and let $a_1, a_2, \ldots, a_m$ be positive integers not greater than n^m. Prove that there exist positive integers $b_1, b_2, \ldots, b_m$ not greater than n, such that

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) < n,$$

where $\gcd(x_1, x_2, \ldots, x_m)$ denotes the greatest common divisor of $x_1, x_2, \ldots, x_m$.

Day 2

Problem 4. Determine whether there exists an infinite sequence $a_1, a_2, a_3, \ldots$ of positive integers which satisfies the equality

$$a_{n+2} = a_{n+1} + \sqrt{a_{n+1} + a_n}$$

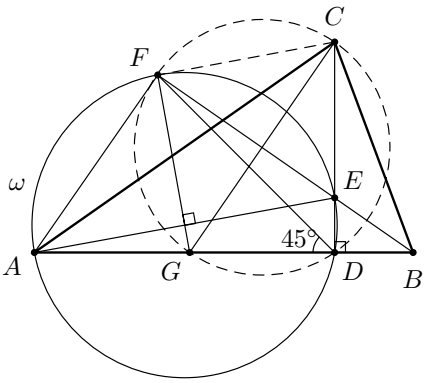
for every positive integer n.

Problem 5. Let m, n be positive integers with $m > 1$. Anastasia partitions the integers $1, 2, \ldots, 2m$ into m pairs. Boris then chooses one integer from each pair and finds the sum of these chosen integers. Prove that Anastasia can select the pairs so that Boris cannot make his sum equal to n.

Problem 6. Let H be the orthocentre and G be the centroid of acute-angled triangle $\triangle ABC$ with $AB \neq AC$. The line AG intersects the circumcircle of $\triangle ABC$ at A and P. Let P' be the reflection of P in the line BC. Prove that $\angle CAB = 60^\circ$ if and only if $HG = GP'$.

Problem 1. *Let $\triangle ABC$ be an acute-angled triangle, and let D be the foot of the altitude from C. The angle bisector of $\angle ABC$ intersects CD at E and meets the circumcircle ω of triangle $\triangle ADE$ again at F. If $\angle ADF = 45^\circ$, show that CF is tangent to ω.*

(Luxembourg)



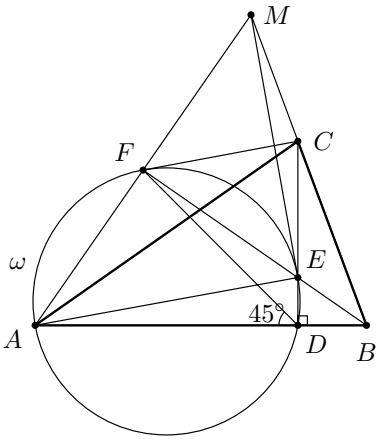
Solution 1: Since $\angle CDF = 90^\circ - 45^\circ = 45^\circ$, the line DF bisects $\angle CDA$, and so F lies on the perpendicular bisector of segment AE, which meets AB at G. Let $\angle ABC = 2\beta$. Since $ADEF$ is cyclic, $\angle AFE = 90^\circ$, and hence $\angle FAE = 45^\circ$. Further, as BF bisects $\angle ABC$, we have $\angle FAB = 90^\circ - \beta$, and thus

$$\angle EAB = \angle AEG = 45^\circ - \beta, \quad \text{and} \quad \angle AED = 45^\circ + \beta,$$

so $\angle GED = 2\beta$. This implies that right-angled triangles $\triangle EDG$ and $\triangle BDC$ are similar, and so we have $|GD|/|CD| = |DE|/|DB|$. Thus the right-angled triangles $\triangle DEB$ and $\triangle DGC$ are similar, whence $\angle GCD = \angle DBE = \beta$. But $\angle DFE = \angle DAE = 45^\circ - \beta$, then $\angle GFD = 45^\circ - \angle DFE = \beta$. Hence $GDCF$ is cyclic, so $\angle GFC = 90^\circ$, whence CF is perpendicular to the radius FG of ω. It follows that CF is a tangent to ω, as required.

Solution 2: As $\angle ADF = 45^\circ$ line DF is an exterior bisector of $\angle CDB$. Since BF bisects $\angle DBC$ line CF is an exterior bisector of $\angle BCD$. Let $\angle ABC = 2\beta$, so $\angle ECF = (\angle DBC + \angle CDB)/2 = 45^\circ + \beta$. Hence $\angle CFE = 180^\circ - \angle ECF - \angle BCE - \angle EBC = 180^\circ - (45^\circ + \beta + 90^\circ - 2\beta + \beta) = 45^\circ$. It follows that $\angle FDC = \angle CFE$, then CF is tangent to ω.

Solution 3: Note that AE is diameter of circumcircle of $\triangle ABC$ since $\angle CDF = 90^\circ$. From $\angle AEF = \angle ADF = 45^\circ$ it follows that triangle $\triangle AFE$ is right-angled and isosceles. Without loss of generality, let points A, E and F have coordinates $(-1, 0)$, $(1, 0)$ and $(0, 1)$ respectively. Points F, E, B are collinear, hence B have coordinates $(b, 1-b)$ for some $b \neq -1$. Let point C' be intersection of line tangent to circumcircle of $\triangle AFE$ at F with line ED. Thus C' have coordinates $(c, 1)$ and from $\overline{C'E} \perp \overline{AB}$ we get $c = 2b/(b+1)$. Now vector $\overline{BC'} = (2b/(b+1) - b, b) = b/(b+1) \cdot (1-b, b+1)$, vector $\overline{BF} = (-b, b) = (-1, 1) \cdot b$ and vector $\overline{BA} = (-(b+1), -(1-b))$. Its clear that $(1-b, b+1)$ and $(-(b+1), -(1-b))$ are symmetric with respect to $\overline{FE} = (-1, 1)$, hence BF bisects $\angle C'BA$ and $C' = C$ which completes the proof.



Solution 4: Again F lies on the perpendicular bisector of segment AE, so $\triangle AFE$ is right-angled and isosceles. Let M be an intersection of BC and AF. Note that $\triangle AMB$ is isosceles since BF is a bisector and altitude in this triangle. Thus BF is a symmetry line of $\triangle AMB$. Then $\angle FDA = \angle FEA = \angle MEF = 45^\circ$, $AF = FE = FM$ and $\angle DAE = \angle EMC$. Let us show that $EC = CM$. Indeed,

$$\angle CEM = 180^\circ - (\angle AED + \angle FEA + \angle MEF) = 90^\circ - \angle AED = \\ = \angle DAE = \angle EMC.$$

It follows that $FMCE$ is a kite, since $EF = FM$ and $MC = CE$. Hence $\angle EFC = \angle CFM = \angle EDF = 45^\circ$, so FC is tangent to ω.

Solution 5: Let the tangent to ω at F intersect CD at C'. Let $\angle ABF = \angle FBC = \beta$. It follows that $\angle C'FE = 45^\circ$ since $C'F$ is tangent. We have

$$\frac{\sin\angle BDC}{\sin\angle CDF} \cdot \frac{\sin\angle DFC'}{\sin\angle C'FB} \cdot \frac{\sin\angle FBC}{\sin\angle CBD} = \frac{\sin 90^\circ}{\sin 45^\circ} \cdot \frac{\sin(90^\circ - \beta)}{\sin 45^\circ} \cdot \frac{\sin\beta}{\sin 2\beta} = \frac{2\sin\beta\cos\beta}{\sin 2\beta} = 1.$$

So by trig Ceva on triangle $\triangle BDF$, lines FC', DC and BC are concurrent (at C), so $C = C'$. Hence CF is tangent to ω.

Problem 2. *A domino is a 2×1 or 1×2 tile. Determine in how many ways exactly n^2 dominoes can be placed without overlapping on a $2n \times 2n$ chessboard so that every 2×2 square contains at least two uncovered unit squares which lie in the same row or column.*

(Turkey)

Solution: The answer is $\binom{2n}{n}^2$.

Divide the schessboard into 2×2 squares. There are exactly n^2 such squares on the chessboard. Each of these squares can have at most two unit squares covered by the dominos. As the dominos cover exactly $2n^2$ squares, each of them must have exactly two unit squares which are covered, and these squares must lie in the same row or column.

We claim that these two unit squares are covered by the same domino tile. Suppose that this is not the case for some 2×2 square and one of the tiles covering one of its unit squares sticks out to the left. Then considering one of the leftmost 2×2 squares in this division with this property gives a contradiction.

Now consider this $n \times n$ chessboard consisting of 2×2 squares of the original board. Define A, B, C, D as the following configurations on the original chessboard, where the gray squares indicate the domino tile, and consider the covering this $n \times n$ chessboard with the letters A, B, C, D in such a

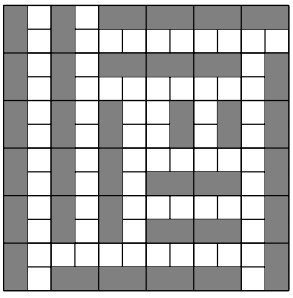


way that the resulting configuration on the original chessboard satisfies the condition of the question.

Note that then a square below or to the right of one containing an A or B must also contain an A or B. Therefore the (possibly empty) region consisting of all squares containing an A or B abuts the lower right corner of the chessboard and is separated from the (possibly empty) region consisting of all squares containing a C or D by a path which goes from the lower left corner to the upper right corner of this chessboard and which moves up or right at each step.

A similar reasoning shows that the (possibly empty) region consisting of all squares containing an A or D abuts the lower left corner of the chessboard and is separated from the (possibly empty) region consisting of all squares containing a B or C by a path which goes from the upper left corner to the lower right corner of this chessboard and which moves down or right at each step.

D	D	C	C	C	C
D	D	C	C	C	B
D	D	D	B	B	B
D	D	D	A	A	B
D	D	D	A	A	B
D	A	A	A	A	B



Therefore the $n \times n$ chessboard is divided by these two paths into four (possibly empty) regions that consist respectively of all squares containing A or B or C or D. Conversely, choosing two such paths and filling the four regions separated by them with As, Bs, Cs and Ds counterclockwise starting at the bottom results in a placement of the dominos on the original board satisfying the condition of the question.

As each of these paths can be chosen in $\binom{2n}{n}$ ways, there are $\binom{2n}{n}^2$ ways the dominos can be placed.

Problem 3. *Let n, m be integers greater than 1, and let $a_1, a_2, \ldots, a_m$ be positive integers not greater than n^m. Prove that there exist positive integers $b_1, b_2, \ldots, b_m$ not greater than n, such that*

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) < n,$$

where $\gcd(x_1, x_2, \ldots, x_m)$ denotes the greatest common divisor of $x_1, x_2, \ldots, x_m$.

(USA)

Solution 1: Suppose without loss of generality that a_1 is the smallest of the a_i. If $a_1 \geq n^m - 1$, then the problem is simple: either all the a_i are equal, or $a_1 = n^m - 1$ and $a_j = n^m$ for some j. In the first case, we can take (say) $b_1 = 1$, $b_2 = 2$, and the rest of the b_i can be arbitrary, and we have

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) \leq \gcd(a_1 + b_1, a_2 + b_2) = 1.$$

In the second case, we can take $b_1 = 1$, $b_j = 1$, and the rest of the b_i arbitrary, and again

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) \leq \gcd(a_1 + b_1, a_j + b_j) = 1.$$

So from now on we can suppose that $a_1 \leq n^m - 2$.

Now, let us suppose the desired $b_1, \ldots, b_m$ do not exist, and seek a contradiction. Then, for any choice of $b_1, \ldots, b_m \in \{1, \ldots, n\}$, we have

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) \geq n.$$

Also, we have

$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) \leq a_1 + b_1 \leq n^m + n - 2.$$

Thus there are at most $n^m - 1$ possible values for the greatest common divisor. However, there are n^m choices for the m-tuple $(b_1, \ldots, b_m)$. Then, by the pigeonhole principle, there are two m-tuples that yield the same values for the greatest common divisor, say d. But since $d \geq n$, for each i there can be at most one choice of $b_i \in \{1, 2, \ldots, n\}$ such that $a_i + b_i$ is divisible by d — and therefore there can be at most one m-tuple $(b_1, b_2, \ldots, b_m)$ yielding d as the greatest common divisor. This is the desired contradiction.

Solution 2: Similarly to Solution 1 suppose that $a_1 \leq n^m - 2$. The gcd of $a_1+1, a_2+1, a_3+1, \ldots, a_m+1$ is co-prime with the gcd of $a_1 + 1, a_2 + 2, a_3 + 1, \ldots, a_m + 1$, thus $a_1 + 1 \geq n^2$. Now change another 1 into 2 and so on. After $m - 1$ changes we get $a_1 + 1 \geq n^m$ which gives us a contradiction.

Solution 3: We will prove stronger version of this problem:

For $m, n > 1$, let $a_1, \ldots, a_m$ be positive integers with at least one $a_i \leq n^{2^{m-1}}$. Then there are integers $b_1, \ldots, b_m$, each equal to 1 or 2, such that $\gcd(a_1 + b_1, \ldots, a_m + b_m) < n$.

Proof: Suppose otherwise. Then the 2^{m-1} integers $\gcd(a_1 + b_1, \ldots, a_m + b_m)$ with $b_1 = 1$ and $b_i = 1$ or 2 for $i > 1$ are all pairwise coprime, since for any two of them, there is some $i > 1$ with $a_i + 1$ appearing in one and $a_i + 2$ in the other. Since each of these 2^{m-1} integers divides $a_1 + 1$, and each is $\geq n$ with at most one equal to n, it follows that $a_1 + 1 \geq n(n + 1)^{2^{m-1}-1}$ so $a_1 \geq n^{2^{m-1}}$. The same is true for each a_i, $i = 1, \ldots, n$, a contradiction.

Remark: Clearly the $n^{2^{m-1}}$ bound can be strengthened as well.

Problem 4. *Determine whether there exists an infinite sequence $a_1, a_2, a_3, \ldots$ of positive integers which satisfies the equality*

$$a_{n+2} = a_{n+1} + \sqrt{a_{n+1} + a_n}$$

for every positive integer n.

(Japan)

Solution 1: The answer is no.

Suppose that there exists a sequence (a_n) of positive integers satisfying the given condition. We will show that this will lead to a contradiction.

For each $n \geq 2$ define $b_n = a_{n+1} - a_n$. Then, by assumption, for $n \geq 2$ we get $b_n = \sqrt{a_n + a_{n-1}}$ so that we have

$$b_{n+1}^2 - b_n^2 = (a_{n+1} + a_n) - (a_n + a_{n-1}) = (a_{n+1} - a_n) + (a_n - a_{n-1}) = b_n + b_{n-1}.$$

Since each a_n is a positive integer we see that b_n is positive integer for $n \geq 2$ and the sequence (b_n) is strictly increasing for $n \geq 3$. Thus $b_n + b_{n-1} = (b_{n+1} - b_n)(b_{n+1} + b_n) \geq b_{n+1} + b_n$, whence $b_{n-1} \geq b_{n+1}$ – a contradiction to increasing of the sequence (b_i).

Thus we conclude that there exists no sequence (a_n) of positive integers satisfying the given condition of the problem.

Solution 2: Suppose that such a sequence exists. We will calculate its members one by one and get a contradiction.

From the equality $a_3 = a_2 + \sqrt{a_2 + a_1}$ it follows that $a_3 > a_2$. Denote positive integers $\sqrt{a_3 + a_2}$ by b and a_3 by a, then we have $\sqrt{2a} > b$. Since $a_4 = a + b$ and $a_5 = a + b + \sqrt{2a + b}$ are positive integers, then $\sqrt{2a + b}$ is positive integer.

Consider $a_6 = a + b + \sqrt{2a + b} + \sqrt{2a + 2b + \sqrt{2a + b}}$. Number $c = \sqrt{2a + 2b + \sqrt{2a + b}}$ must be positive integer, obviously it is greater than $\sqrt{2a + b}$. But

$$(\sqrt{2a + b} + 1)^2 = 2a + b + 2\sqrt{2a + b} + 1 = 2a + 2b + \sqrt{2a + b} + (\sqrt{2a + b} - b) + 1 > c^2.$$

So $\sqrt{2a + b} < c < \sqrt{2a + b} + 1$ which is impossible.

Solutions 3: We will show that there is no sequence (a_n) of positive integers which consists of $N > 5$ members and satisfies

$$a_{n+2} = a_{n+1} + \sqrt{a_{n+1} + a_n} \tag{1}$$

for all $n = 1, \ldots, N - 2$. Moreover, we will describe all such sequences with five members.

Since every a_i is a positive integer it follows from (1) that there exists such positive integer k (obviously k depends on n) that

$$a_{n+1} + a_n = k^2. \tag{2}$$

From (1) we have $(a_{n+2} - a_{n+1})^2 = a_{n+1} + a_n$, consider this equality as a quadratic equation with respect to a_{n+1}:

$$a_{n+1}^2 - (2a_{n+2} + 1)a_{n+1} + a_{n+2}^2 - a_n = 0.$$

Obviously its solutions are $(a_{n+1})_{1,2} = \dfrac{2a_{n+2} + 1 \pm \sqrt{D}}{2}$, where

$$D = 4(a_n + a_{n+2}) + 1. \tag{3}$$

Since $a_{n+2} > a_{n+1}$ we have

$$a_{n+1} = \frac{2a_{n+2} + 1 - \sqrt{D}}{2}.$$

From the last equality, using that a_{n+1} and a_{n+2} are positive integers, we conclude that D is a square of some odd number i.e. $D = (2m+1)^2$ for some positive integer $m \in \mathbb{N}$, substitute this into (3):

$$a_n + a_{n+2} = m(m+1). \tag{4}$$

Now adding a_n to both sides of (1) and using (2) and (4) we get $m(m+1) = k^2 + k$ whence $m = k$. So

$$\begin{cases} a_n + a_{n+1} = k^2, \\ a_n + a_{n+2} = k^2 + k \end{cases} \tag{5}$$

for some positive integer k (recall that k depends on n).

Write equations (5) for $n = 2$ and $n = 3$, then for some positive integers k and ℓ we get

$$\begin{cases} a_2 + a_3 = k^2, \\ a_2 + a_4 = k^2 + k, \\ a_3 + a_4 = \ell^2, \\ a_3 + a_5 = \ell^2 + \ell. \end{cases} \tag{6}$$

Solution of this linear system is

$$a_2 = \frac{2k^2 - \ell^2 + k}{2}, \quad a_3 = \frac{\ell^2 - k}{2}, \quad a_4 = \frac{\ell^2 + k}{2}, \quad a_5 = \frac{\ell^2 + 2\ell + k}{2}. \tag{7}$$

From $a_2 < a_4$ we obtain $k^2 < \ell^2$ hence $k < \ell$.

Consider a_6:

$$a_6 = a_5 + \sqrt{a_5 + a_4} = a_5 + \sqrt{\ell^2 + \ell + k}\,.$$

Since $0 < k < l$ we have $\ell^2 < \ell^2 + \ell + k < (\ell+1)^2$. So a_6 cannot be integer i.e. there is no such sequence with six or more members.

To find all required sequences with five members we must find positive integers a_2, a_3, a_4 and a_5 which satisfy (7) for some positive integers $k < \ell$. Its clear that k and ℓ must be of the same parity. Vise versa, let positive integers k, ℓ be of the same parity and satisfy $k < \ell$ then from (7) we get integers a_2, a_3, a_4 and a_5 then $a_1 = (a_3 - a_2)^2 - a_2$ and it remains to verify that a_1 and a_2 are positive i.e. $2k^2 + k > \ell^2$ and $2(\ell^2 - k^2 - k)^2 > 2k^2 - \ell^2 + k$.

Solution 4: It is easy to see that (a_n) is increasing for large enough n. Hence

$$a_{n+1} < a_n + \sqrt{2a_n} \tag{1}$$

and

$$a_n < a_{n-1} + \sqrt{2a_{n-1}}. \tag{2}$$

Lets define $b_n = a_n + a_{n-1}$. Using AM-QM inequality we have

$$\frac{\sqrt{2a_n} + \sqrt{2a_{n-1}}}{2} \le \sqrt{\frac{2a_n + 2a_{n-1}}{2}}. \tag{3}$$

Adding (1), (2) and using (3):

$$b_{n+1} < b_n + \sqrt{2a_n} + \sqrt{2a_{n-1}} \le b_n + 2\sqrt{b_n}.$$

Let $b_n = m^2$. Since (b_n) is increasing for large enough n, we have:

$$m^2 < b_{n+1} < m^2 + 2m < (m+1)^2.$$

So, b_{n+1} can't be a perfect square, so we get contradiction.

Problem 5. *Let m, n be positive integers with $m > 1$. Anastasia partitions the integers $1, 2, \ldots, 2m$ into m pairs. Boris then chooses one integer from each pair and finds the sum of these chosen integers. Prove that Anastasia can select the pairs so that Boris cannot make his sum equal to n.*

(Netherlands)

Solution 1A: Define the following ordered partitions:

$$P_1 = (\{1, 2\}, \{3, 4\}, \ldots, \{2m-1, 2m\}),$$
$$P_2 = (\{1, m+1\}, \{2, m+2\}, \ldots, \{m, 2m\}),$$
$$P_3 = (\{1, 2m\}, \{2, m+1\}, \{3, m+2\}, \ldots, \{m, 2m-1\}).$$

For each P_j we will compute the possible values for the expression $s = a_1 + \ldots + a_m$, where $a_i \in P_{j,i}$ are the chosen integers. Here, $P_{j,i}$ denotes the i-th coordinate of the ordered partition P_j.
We will denote by σ the number $\sum_{i=1}^{m} i = (m^2 + m)/2$.

- Consider the partition P_1 and a certain choice with corresponding sum s. We find that

$$m^2 = \sum_{i=1}^{m} (2i - 1) \leq s \leq \sum_{i=1}^{m} 2i = m^2 + m.$$

  Hence, if $n < m^2$ or $n > m^2 + m$, this partition gives a positive answer.

- Consider the partition P_2 and a certain choice with corresponding s. We find that

$$s \equiv \sum_{i=1}^{m} i \equiv \sigma \pmod{m}.$$

  Hence, if $m^2 \leq n \leq m^2 + m$ and $n \not\equiv \sigma \pmod{m}$, this partition solves the problem.

- Consider the partition P_3 and a certain choice with corresponding s. We set

$$d_i = \begin{cases} 0 & \text{if } a_i = i \\ 1, & \text{if } a_i \neq i. \end{cases}$$

  We also put $d = \sum_{i=1}^{m} d_i$, and note that $0 \leq d \leq m$. Note also that if $a_i \neq i$, then $a_i \equiv i - 1 \pmod{m}$. Hence, for all $a_i \in P_{3,i}$ it holds that

$$a_i \equiv i - d_i \pmod{m}.$$

  Hence,

$$s \equiv \sum_{i=1}^{m} a_i \equiv \sum_{i=1}^{m} (i - d_i) \equiv \sigma - d \pmod{m},$$

  which can only be congruent to σ modulo m if all d_i are equal, which forces $s = (m^2 + m)/2$ or $s = (3m^2 + m)/2$. Since $m > 1$, it holds that

$$\frac{m^2 + m}{2} < m^2 < m^2 + m < \frac{3m^2 + m}{2}.$$

  Hence if $m^2 \leq n \leq m^2 + m$ and $n \equiv \sigma \pmod{m}$, then s cannot be equal to n, so partition P_3 suffices for such n.

Note that all n are treated in one of the cases above, so we are done.

Common notes for solutions 1B and 1C: Given the analysis of P_1 and P_2 as in the solution 1A, we may conclude (noting that $\sigma \equiv m(m+1)/2 \pmod m$) that if m is odd then m^2 and m^2+m are the only candidates for counterexamples n, while if m is even then $m^2+\frac{m}{2}$ is the only candidate.

There are now various ways to proceed as alternatives to the partition P_3.

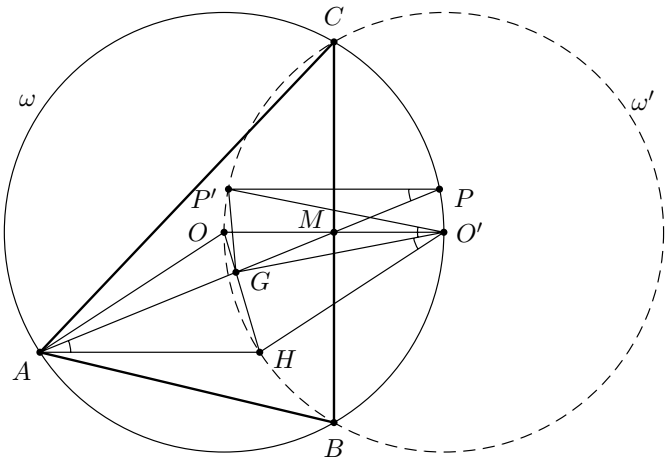
Solution 1B: Consider the partition $(\{1, m+2\}, \{2, m+3\}, \ldots, \{m-1, 2m\}, \{m, m+1\})$. We consider possible sums mod $m+1$. For the first $m-1$ pairs, the elements of each pair are congruent mod $m+1$, so the sum of one element of each pair is (mod $m+1$) congruent to $\frac{1}{2}m(m+1)-m$, which is congruent to 1 if $m+1$ is odd and $1+\frac{m+1}{2}$ if $m+1$ is even. Now the elements of the last pair are congruent to -1 and 0, so any achievable value of n is congruent to 0 or 1 if $m+1$ is odd, and to 0 or 1 plus $\frac{m+1}{2}$ if $m+1$ is even. If m is even then $m^2+\frac{m}{2} \equiv 1+\frac{m}{2}$, which is not congruent to 0 or 1. If m is odd then $m^2 \equiv 1$ and $m^2+m \equiv 0$, neither of which can equal 0 or 1 plus $\frac{m+1}{2}$.

Solution 1C: Similarly, consider the partition $(\{1, m\}, \{2, m+1\}, \ldots, \{m-1, 2m-2\}, \{2m-1, 2m\})$, this time considering sums of elements of pairs mod $m-1$. If $m-1$ is odd, the sum is congruent to 1 or 2; if $m-1$ is even, to 1 or 2 plus $\frac{m-1}{2}$. If m is even then $m^2+\frac{m}{2} \equiv 1+\frac{m}{2}$, and this can only be congruent to 1 or 2 when $m=2$. If m is odd, m^2 and m^2+m are congruent to 1 and 2, and these can only be congruent to 1 or 2 plus $\frac{m-1}{2}$ when $m=3$. Now the cases of $m=2$ and $m=3$ need considering separately (by finding explicit partitions excluding each n).

Solution 2: This solution does not use modulo arguments. Use only P_1 from the solution 1A to conclude that $m^2 \le n \le m^2+m$. Now consider the partition $(\{1, 2m\}, \{2, 3\}, \{4, 5\}, \ldots, \{2m-2, 2m-1\})$. If 1 is chosen from the first pair, the sum is at most m^2; if $2m$ is chosen, the sum is at least m^2+m. So either $n=m^2$ or $n=m^2+m$. Now consider the partition $(\{1, 2m-1\}, \{2, 2m\}, \{3, 4\}, \{5, 6\}, \ldots, \{2m-3, 2m-2\})$. Sums of one element from each of the last $m-2$ pairs are in the range from $(m-2)m = m^2-2m$ to $(m-2)(m+1) = m^2-m-2$ inclusive. Sums of one element from each of the first two pairs are 3, $2m+1$ and $4m-1$. In the first case we have $n \le m^2-m+1 < m^2$, in the second $m^2+1 \le n \le m^2+m-1$ and in the third $n \ge m^2+2m-1 > m^2+m$. So these three partitions together have eliminated all n.

Problem 6. *Let H be the orthocenter and G be the centroid of acute-angled triangle $\triangle ABC$ with $AB \neq AC$. The line AG intersects the circumcircle of $\triangle ABC$ at A and P. Let P' be the reflection of P in the line BC. Prove that $\angle CAB = 60^\circ$ if and only if $HG = GP'$.*

(Ukraine)



Solution 1: Let ω be the circumcircle of $\triangle ABC$. Reflecting ω in line BC, we obtain circle ω' which, obviously, contains points H and P'. Let M be the midpoint of BC. As triangle $\triangle ABC$ is acute-angled, then H and O lie inside this triangle.

Let us assume that $\angle CAB = 60^\circ$. Since

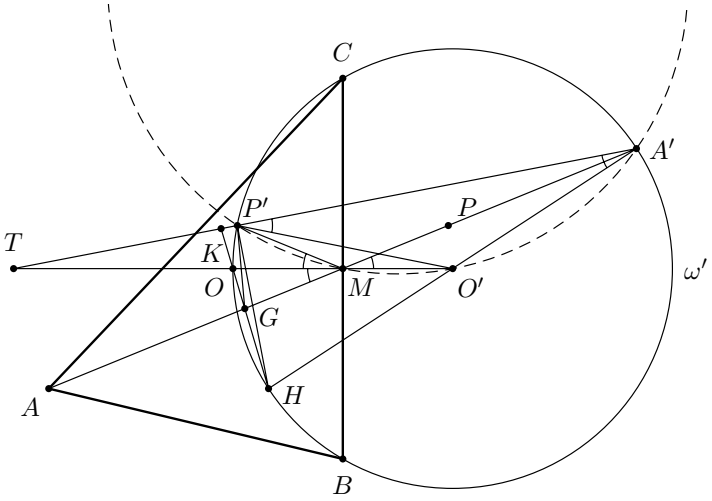
$$\angle COB = 2\angle CAB = 120^\circ = 180^\circ - 60^\circ = 180^\circ - \angle CAB = \angle CHB,$$

hence O lies on ω'. Reflecting O in line BC, we obtain point O' which lies on ω and this point is the center of ω'. Then $OO' = 2OM = 2R\cos\angle CAB = AH$, so $AH = OO' = HO' = AO = R$, where R is the radius of ω and, naturally, of ω'. Then quadrilateral $AHO'O$ is a rhombus, so A and O' are symmetric to each other with respect to HO. As H, G and O are collinear (Euler line), then $\angle GAH = \angle HO'G$. Diagonals of quadrilateral $GOPO'$ intersects at M. Since $\angle BOM = 60^\circ$, so

$$OM = MO' = \operatorname{ctg} 60^\circ \cdot MB = \frac{MB}{\sqrt{3}}.$$

As $3MO \cdot MO' = MB^2 = MB \cdot MC = MP \cdot MA = 3MG \cdot MP$, then $GOPO'$ is a cyclic. Since BC is a perpendicular bisector of OO', so the circumcircle of quadrilateral $GOPO'$ is symmetrical with respect to BC. Thus P' also belongs to the circumcircle of $GOPO'$, hence $\angle GO'P' = \angle GPP'$. Note that $\angle GPP' = \angle GAH$ since $AH \| PP'$. And as it was proved $\angle GAH = \angle HO'G$, then $\angle HO'G = \angle GO'P'$. Thus triangles $\triangle HO'G$ and $\triangle GO'P'$ are equal and hence $HG = GP'$.

Now we will prove that if $HG = GP'$ then $\angle CAB = 60^\circ$. Reflecting A with respect to M, we get A'. Then, as it was said in the first part of solution, points B, C, H and P' belong to ω'. Also it is clear that A' belongs to ω'. Note that $HC \perp CA'$ since $AB \| CA'$ and hence HA' is a diameter of ω'. Obviously, the center O' of circle ω' is midpoint of HA'. From $HG = GP'$ it follows that $\triangle HGO'$ is equal to $\triangle P'GO'$. Therefore H and P' are symmetric with respect to GO'. Hence $GO' \perp HP'$ and $GO' \| A'P'$. Let HG intersect $A'P'$ at K and $K \not\equiv O$ since $AB \neq AC$. We conclude that $HG = GK$, because line GO' is midline of the triangle $\triangle HKA'$. Note that $2GO = HG$. since HO is Euler line of triangle ABC. So O is midpoint of segment GK. Because of $\angle CMP = \angle CMP'$, then $\angle GMO = \angle OMP'$. Line OM, that passes through O', is an external angle bisector of $\angle P'MA'$. Also we know that $P'O' = O'A'$, then O' is the midpoint of arc $P'MA'$ of the circumcircle of triangle $\triangle P'MA'$. It



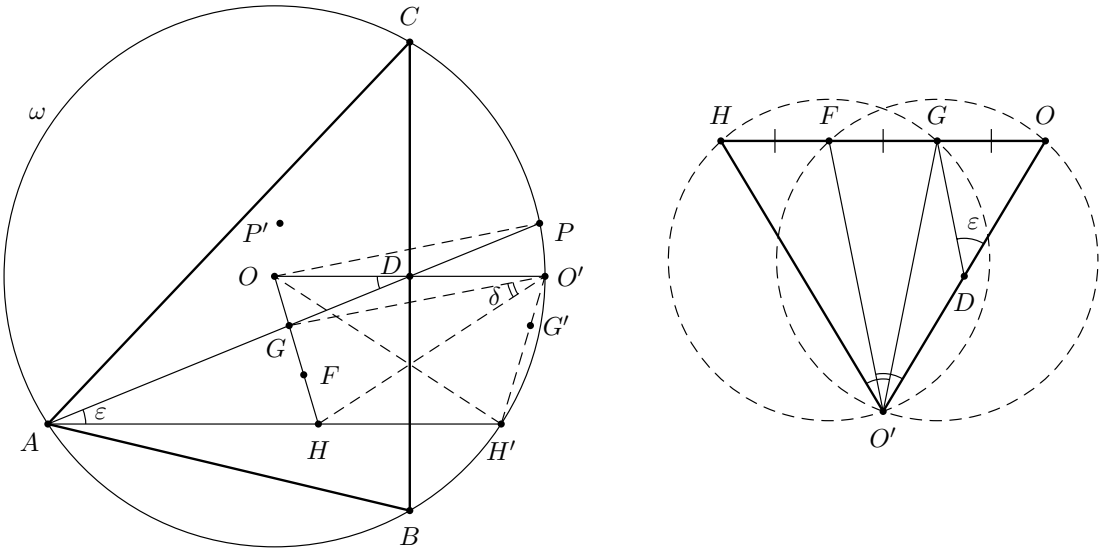
follows that quadrilateral $P'MO'A'$ is cyclic, then $\angle O'MA' = \angle O'P'A' = \angle O'A'P'$. Let OM and $P'A'$ intersect at T. Triangles $\triangle TO'A'$ and $\triangle A'O'M$ are similar, hence $O'A'/O'M = O'T/O'A'$. In the other words, $O'M \cdot O'T = O'A'^2$. Using Menelaus' theorem for triangle $\triangle HKA'$ and line TO', we obtain that

$$\frac{A'O'}{O'H} \cdot \frac{HO}{OK} \cdot \frac{KT}{TA'} = 3 \cdot \frac{KT}{TA'} = 1.$$

It follows that $KT/TA' = 1/3$ and $KA' = 2KT$. Using Menelaus' theorem for triangle $TO'A'$ and line HK we get that

$$1 = \frac{O'H}{HA'} \cdot \frac{A'K}{KT} \cdot \frac{TO}{OO'} = \frac{1}{2} \cdot 2 \cdot \frac{TO}{OO'} = \frac{TO}{OO'}.$$

It means that $TO = OO'$, so $O'A'^2 = O'M \cdot O'T = OO'^2$. Hence $O'A' = OO'$ and, consequently, $O \in \omega'$. Finally we conclude that $2\angle CAB = \angle BOC = 180^\circ - \angle CAB$, so $\angle CAB = 60^\circ$.



Solution 2: Let O' and G' denote the reflection of O and G, respectively, with respect to the line BC. We then need to show $\angle CAB = 60^\circ$ iff $G'H' = G'P$. Note that $\triangle H'OP$ is isosceles and hence

$G'H' = G'P$ is equivalent to G' lying on the bisector $\angle H'OP$. Let $\angle H'AP = \varepsilon$. By the assumption $AB \neq AC$, we have $\varepsilon \neq 0$. Then $\angle H'OP = 2\angle H'AP = 2\varepsilon$, hence $G'H' = G'P$ iff $\angle G'OH' = \varepsilon$. But $\angle GO'H = \angle G'OH'$. Let D be the midpoint of OO'. It is known that $\angle GDO = \angle GAH = \varepsilon$. Let F be the midpoint of HG. Then $HG = FO$ (Euler line). Let $\angle GO'H = \delta$. We then have to show $\delta = \varepsilon$ iff $\angle CAB = 60^\circ$. But by similarity ($\triangle GDO \sim \triangle FO'O$) we have $\angle FO'O = \varepsilon$. Consider the circumcircles of the triangles $FO'O$ and $GO'H$. By the sine law and since the segments HG and FO are of equal length we deduce that the circumcircles of the triangles $FO'O$ and $GO'H$ are symmetric with respect to the perpendicular bisector of the segment FG iff $\delta = \varepsilon$. Obviously, O' is the common point of these two circles. Hence O' must be fixed after the symmetry about the perpendicular bisector of the segment FG iff $\delta = \varepsilon$ so we have $\varepsilon = \delta$ iff $\triangle HOO'$ is isosceles. But $HO' = H'O = R$, and so

$$\varepsilon = \delta \iff OO' = R \iff OD = \frac{R}{2} \iff \cos\angle CAB = \frac{1}{2} \iff \angle CAB = 60^\circ.$$

Solution 3: Let H' and G' denote the reflection of points H and G with respect to the line BC. It is known that H' belongs to the circumcircle of $\triangle ABC$. The equality $HG = GP'$ is equivalent to $H'G' = G'P$. As in the **Solution 2**, it is equivalent to the statement that point G' belongs to the perpendicular bisector of $H'P$, which is equivalent to $OG' \perp H'P$, where O is the circumcenter of $\triangle ABC$.

Let points $A(a)$, $B(b)$, and $C(c = -\overline{b})$ belong to the unit circle in the complex plane. Point G have coordinate $g = (a + b - \overline{b})/3$. Since BC is parallel to the real axis point H' have coordinate $h' = \overline{a} = 1/a$.

Point $P(p)$ belongs to the unit circle, so $\overline{p} = 1/p$. Since a, p, g are collinear we have $\dfrac{p-a}{g-a} = \overline{\left(\dfrac{p-a}{g-a}\right)}$. After computation we get $p = \dfrac{g-a}{1-\overline{g}a}$. Since $G'(g)$ is the reflection of G with respect to the chord BC, we have $g' = b + (-\overline{b}) - b(-\overline{b})\overline{g} = b - \overline{b} + \overline{g}$. Let $b - \overline{b} = d$. We have $\overline{d} = -d$. So

$$g = \frac{a+d}{3}, \quad \overline{g} = \frac{\overline{a}-d}{3}, \quad g' = d + \overline{g} = \frac{\overline{a}+2d}{3}, \quad \overline{g'} = \frac{a-2d}{3} \quad \text{and} \quad p = \frac{g-a}{1-\overline{g}a} = \frac{d-2a}{2+ad}. \qquad (1)$$

It is easy to see that $OG' \perp H'P'$ is equivalent to

$$\frac{g'}{h'-p} = -\overline{\left(\frac{g'}{h'-p}\right)} = -\frac{g'}{\frac{1}{h'} - \frac{1}{p}} = \frac{\overline{g'}h'p}{h'-p}$$

since h' and p belong to the unit circle (note that $H' \neq P$ because $AB \neq AC$). This is equivalent to $g' = \overline{g'}h'p$ and from (1), after easy computations, this is equivalent to $a^2g^2 + a^2 + d^2 + 1 = (a^2+1)(d^2+1) = 0$.

We cannot have $a^2 + 1 = 0$, because then $a = \pm i$, but $AB \neq AC$. Hence $d = b - \overline{b} = \pm i$, and the pair $\{b, c = -\overline{b}\}$ is either $\{-\sqrt{3}/2 + i/2, \sqrt{3}/2 + i/2\}$ or $\{-\sqrt{3}/2 - i/2, \sqrt{3}/2 - i/2\}$. Both cases are equivalent to $\angle BAC = 60^\circ$ which completes the proof.

EGMO 2013

Problems with Solutions

Problem Selection Committee:

Charles Leytem (chair), Pierre Haas, Jingran Lin,
Christian Reiher, Gerhard Woeginger.

The Problem Selection Committee gratefully acknowledges the receipt of 38 problems proposals from 9 countries:

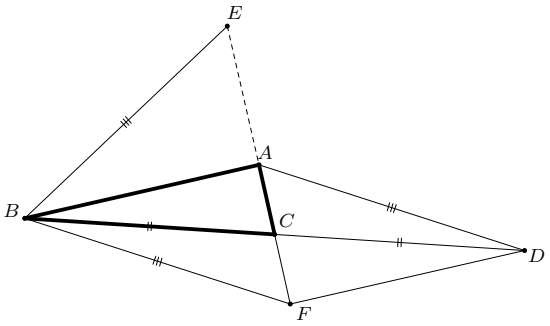
Belarus,	the Netherlands,	Slovenia,
Bulgaria,	Poland,	Turkey,
Finland,	Romania,	the United Kingdom.

Problem 1. (Proposed by David Monk, United Kingdom)

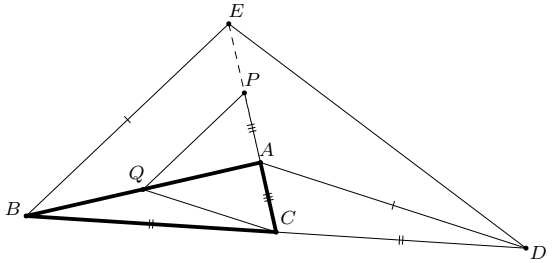
The side BC of the triangle ABC is extended beyond C to D so that $CD = BC$. The side CA is extended beyond A to E so that $AE = 2CA$.

Prove that if $AD = BE$, then the triangle ABC is right-angled.

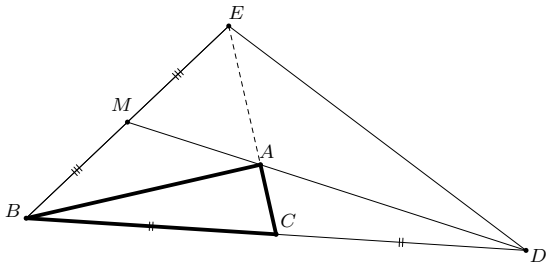
Solution 1: Define F so that $ABFD$ is a parallelogram. Then E, A, C, F are collinear (as diagonals of a parallelogram bisect each other) and $BF = AD = BE$. Further, A is the midpoint of EF, since $AF = 2AC$, and thus AB is an altitude of the isosceles triangle EBF with apex B. Therefore $AB \perp AC$.



A Variant. Let P be the midpoint of $[AE]$, so that $AP = AB$ because $AE = 2AB$. Let Q be the midpoint of $[AB]$. Then $PQ = \frac{1}{2}BE = \frac{1}{2}AD = CQ$. Hence PA is a median of the isosceles triangle CPQ. In other words, $PA \perp AB$, which completes the proof.



Solution 2: Notice that A is the centroid of triangle BDE, since C is the midpoint of $[BD]$ and $AE = 2CA$. Let M be the midpoint of $[BE]$. Then M, A, D lie on a line, and further, $AM = \frac{1}{2}AD = \frac{1}{2}BE$. This implies that $\angle EAB = 90°$.



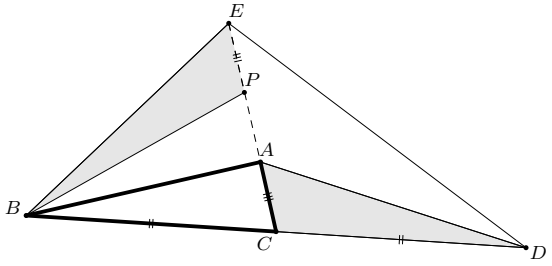
Solution 3: Let P be the midpoint $[AE]$. Since C is the midpoint of $[BD]$, and, moreover, $AC = EP$, we have

$$[ACD] = [ABC] = [EBP].$$

But $AD = BE$, and, as mentioned previously, $AC = EP$, so this implies that

$$\angle BEP = \angle CAD \quad \text{or} \quad \angle BEP = 180^\circ - \angle CAD.$$

But $\angle BEP < \angle BAC$ and $\angle BAC + \angle CAD = \angle BAD < 180^\circ$, so we must be in the first case, i.e. $\angle BEP = \angle CAD$. It follows that triangles BEP and DAC are congruent, and thus $\angle BPA = \angle ACB$. But $AP = AC$, so BA is a median of the isosceles triangle BCP. Thus $AB \perp PC$, completing the proof.



Solution 4: Write $\beta = \angle ECB$, and let $x = AC$, $y = BC = CD$, $z = BE = AD$. Notice that $EC = 3x$. Then, using the cosine theorem,

$$\begin{aligned} z^2 &= x^2 + y^2 + 2xy\cos\beta \quad \text{in triangle } ACD; \\ z^2 &= 9x^2 + y^2 - 6xy\cos\beta \quad \text{in triangle } BCE. \end{aligned}$$

Hence $4z^2 = 12x^2 + 4y^2$ or $z^2 - y^2 = 3x^2$. Let H be the foot of the perpendicular through B to AC, and write $h = BH$. Then

$$y^2 - h^2 = CH^2,\ z^2 - h^2 = EH^2.$$

Hence $z^2 - y^2 = EH^2 - CH^2$. Substituting from the above,

$$EH^2 - CH^2 = 3x^2 = EA^2 - CA^2.$$

Thus $H = A$, and hence the triangle ABC is right-angled at A.

Remark. It is possible to conclude directly from $z^2 - y^2 = 3x^2 = (2x)^2 - x^2$ using Carnot's theorem.

Solution 5: Writing $a = BC$, $b = CA$, $c = AB$, we have

$$\left.\begin{aligned} a^2 &= b^2 + c^2 - 2bc\cos\angle A \\ c^2 &= a^2 + b^2 - 2ab\cos\angle C \end{aligned}\right\} \text{ in triangle } ABC;$$
$$EB^2 = 4b^2 + c^2 + 4bc\cos\angle A \quad \text{in triangle } AEB;$$
$$AD^2 = a^2 + b^2 + 2ab\cos\angle C \quad \text{in triangle } ACD.$$

Thus

$$\begin{aligned} 6b^2 + 3c^2 - 2a^2 &= 4b^2 + c^2 + 4bc\cos\angle A = EB^2 = AD^2 \\ &= a^2 + b^2 + 2ab\cos\angle C = 2a^2 + 2b^2 - c^2, \end{aligned}$$

which gives $a^2 = b^2 + c^2$. Therefore $\angle BAC$ is a right angle by the converse of the theorem of Pythagoras.

Solution 6: Let $\overrightarrow{AC} = \vec{x}$ and $\overrightarrow{AB} = \vec{y}$. Now $\overrightarrow{AD} = 2\vec{x} - \vec{y}$ and $\overrightarrow{EB} = 2\vec{x} + \vec{y}$. Then

$$\overrightarrow{BE} \cdot \overrightarrow{BE} = \overrightarrow{AD} \cdot \overrightarrow{AD} \quad \Longleftrightarrow \quad (2\vec{x} + \vec{y})^2 = (2\vec{x} - \vec{y})^2 \quad \Longleftrightarrow \quad \vec{x} \cdot \vec{y} = 0$$

and thus $AC \perp AB$, whence triangle ABC is right-angled at A.

Remark. It is perhaps more natural to introduce $\overrightarrow{CA} = \vec{a}$ and $\overrightarrow{CB} = \vec{b}$. Then we have the equality

$$\left(3\vec{a} - \vec{b}\right)^2 = \left(\vec{a} + \vec{b}\right)^2 \quad \Longrightarrow \quad \vec{a} \cdot \left(\vec{a} - \vec{b}\right) = 0.$$

Solution 7: Let a, b, c, d, e denote the complex co-ordinates of the points A, B, C, D, E and take the unit circle to be the circumcircle of ABC. We have

$$d = b + 2(c - b) = 2c - b \quad \text{and} \quad e = c + 3(a - c) = 3a - 2c\,.$$

Thus $b - e = (d - a) + 2(b - a)$, and hence

$$\begin{aligned} BE = AD &\iff (b-e)(\overline{b-e}) = (d-a)(\overline{d-a}) \\ &\iff 2(d-a)(\overline{b-a}) + 2(\overline{d-a})(b-a) + 4(b-a)(\overline{b-a}) = 0 \\ &\iff 2(d-a)(a-b) + 2(\overline{d-a})(b-a)ab + 4(b-a)(a-b) = 0 \\ &\iff (d-a) - (\overline{d-a})ab + 2(b-a) = 0 \\ &\iff 2c - b - a - 2\overline{c}ab + a + b + 2(b-a) = 0 \\ &\iff c^2 - ab + bc - ac = 0 \\ &\iff (b+c)(c-a) = 0, \end{aligned}$$

implying $c = -b$ and that triangle ABC is right-angled at A.

Solution 8: We use areal co-ordinates with reference to the triangle ABC. Recall that if (x_1, y_1, z_1) and (x_2, y_2, z_2) are points in the plane, then the square of the distance between these two points is $-a^2vw - b^2wu - c^2uv$, where $(u, v, w) = (x_1 - x_2, y_1 - y_2, z_1 - z_2)$. In our case $A = (1, 0, 0)$, $B = (0, 1, 0)$, $C = (0, 0, 1)$, so $E = (3, 0, 2)$ and, introducing point F as in the first solution, $F = (-1, 0, 2)$. Then

$$BE^2 = AD^2 \quad \iff \quad -2a^2 + 6b^2 + 3c^2 = 2a^2 + 2b^2 - c^2,$$

and thus $a^2 = b^2 + c^2$. Therefore $\angle BAC$ is a right angle by the converse of the theorem of Pythagoras.

Problem 2. (Proposed by Matti Lehtinen, Finland)

Determine all integers m for which the $m \times m$ square can be dissected into five rectangles, the side lengths of which are the integers $1, 2, 3, \ldots, 10$ in some order.

Solution: The solution naturally divides into three different parts: we first obtain some bounds on m. We then describe the structure of possible dissections, and finally, we deal with the few remaining cases.

In the first part of the solution, we get rid of the cases with $m \leqslant 10$ or $m \geqslant 14$. Let $\ell_1, \ldots, \ell_5$ and $w_1, \ldots, w_5$ be the lengths and widths of the five rectangles. Then the rearrangement inequality yields the lower bound

$$\begin{aligned}
&\ell_1 w_1 + \ell_2 w_2 + \ell_3 w_3 + \ell_4 w_4 + \ell_5 w_5 \\
&\quad = \frac{1}{2}\Big(\ell_1 w_1 + \ell_2 w_2 + \ell_3 w_3 + \ell_4 w_4 + \ell_5 w_5 + w_1 \ell_1 + w_2 \ell_2 + w_3 \ell_3 + w_3 \ell_4 + w_5 \ell_5\Big) \\
&\quad \geqslant \frac{1}{2}\Big(1 \cdot 10 + 2 \cdot 9 + 3 \cdot 8 + \cdots + 8 \cdot 3 + 9 \cdot 2 + 10 \cdot 1\Big) \;=\; 110,
\end{aligned}$$

and the upper bound

$$\begin{aligned}
&\ell_1 w_1 + \ell_2 w_2 + \ell_3 w_3 + \ell_4 w_4 + \ell_5 w_5 \\
&\quad = \frac{1}{2}\Big(\ell_1 w_1 + \ell_2 w_2 + \ell_3 w_3 + \ell_4 w_4 + \ell_5 w_5 + w_1 \ell_1 + w_2 \ell_2 + w_3 \ell_3 + w_3 \ell_4 + w_5 \ell_5\Big) \\
&\quad \leqslant \frac{1}{2}\Big(1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + \cdots + 8 \cdot 8 + 9 \cdot 9 + 10 \cdot 10\Big) \;=\; 192.5,
\end{aligned}$$

As the area of the square is sandwiched between 110 and 192.5, the only possible candidates for m are 11, 12, and 13.

In the second part of the solution, we show that a dissection of the square into five rectangles must consist of a single inner rectangle and four outer rectangles that each cover one of the four corners of the square. Indeed, if one of the sides the square had three rectangles adjacent to it, removing these three rectangles would leave a polygon with eight vertices, which is clearly not the union of two rectangles. Moreover, since $m > 10$, each side of the square has at least two adjacent rectangles. Hence each side of the square has precisely two adjacent rectangles, and thus the only way of partitionning the square into five rectangles is to have a single inner rectangle and four outer rectangles each covering of the four corners of the square, as claimed.

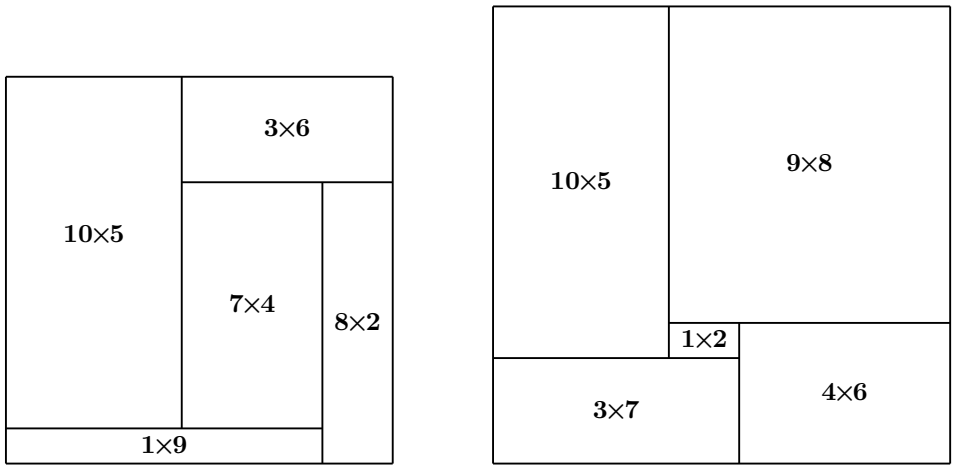
Let us now show that a square of size 12×12 cannot be dissected in the desired way. Let R_1, R_2, R_3 and R_4 be the outer rectangles (in clockwise orientation along the

boundary of the square). If an outer rectangle has a side of length s, then some adjacent outer rectangle must have a side of length $12 - s$. Therefore, neither of $s = 1$ or $s = 6$ can be sidelengths of an outer rectangle, so the inner rectangle must have dimensions 1×6. One of the outer rectangles (say R_1) must have dimensions $10 \times x$, and an adjacent rectangle (say R_2) must thus have dimensions $2 \times y$. Rectangle R_3 then has dimensions $(12 - y) \times z$, and rectangle R_4 has dimensions $(12 - z) \times (12 - x)$. Note that exactly one of the three numbers x, y, z is even (and equals 4 or 8), while the other two numbers are odd. Now, the total area of all five rectangles is

$$144 \;=\; 6 + 10x + 2y + (12 - y)\,z + (12 - z)(12 - x),$$

which simplifies to $(y - x)(z - 2) = 6$. As exactly one of the three numbers x, y, z is even, the factors $y - x$ and $z - 2$ are either both even or both odd, so their product cannot equal 6, and thus there is no solution with $m = 12$.

Finally, we handle the cases $m = 11$ and $m = 13$, which indeed are solutions. The corresponding rectangle sets are 10×5, 1×9, 8×2, 7×4 and 3×6 for $m = 11$, and 10×5, 9×8, 4×6, 3×7 and 1×2 for $m = 13$. These sets can be found by trial and error. The corresponding partitions are shown in the figure below.



Remark. The configurations for $m = 11$ and $m = 13$ given above are not unique.

A Variant for Obtaining Bounds. We first exclude the cases $m \leqslant 9$ by the observation that one of the small rectangles has a side of length 10 and must fit into the square; hence $m \geqslant 10$.

To exclude the cases $m \geqslant 14$, we work via the perimeter: as every rectangle has at most two sides on the boundary of the $m \times m$ square, the perimeter $4m$ of the square is bounded by $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$; hence $m \leqslant 13$.

We are left to deal with the case $m = 10$: clearly, the rectangle with side length 10 must have one its sides of length 10 along the boundary of the square. The remaining rectangle R of dimensions $10 \times s$, say, would have to be divided into four rectangles with different sidelengths strictly less than 10. If there were at least two rectangles adjacent to one of the sides of length s of R, removing these two rectangles from R would leave a polygon with at least six vertices (since the sidelengths of the rectangles partitioning R are strictly less than 10). It is clearly impossible to partition such a polygon into no more than two rectangles with different sidelengths. Hence, given a side of length s of R, there is only one rectangle adjacent to that side, so the rectangles adjacent to the sides of length s of R would have to have the same length s, a contradiction.

Remark. Note that the argument of the second part of the main solution cannot be directly applied to the case $m = 10$.

A Variant for Dealing with $m = 12$. As in the previous solution, we show that the inner rectangle must have dimensions 1×6. Since the area of the square and the area of the inner rectangle are even, the areas of the four outer rectangles must sum to an even number. Now the four sides of the square are divided into segments of lengths 2 and 10, 3 and 9, 4 and 8, and 5 and 7. Hence the sides with adjacent segments of lengths 3 and 9, and 5 and 7 must be opposite sides of the square (otherwise, exactly one of the outer rectangles would have odd area). However, the difference of two rectangle side lengths on opposite sides of the square must be 1 or 6 (in order to accomodate the inner rectangle). This is not the case, so there is no solution with $m = 12$.

Remark. In the case $m = 12$, having shown that the inner rectangle must have dimensions 1×6, this case can also be dealt with by listing the remaining configurations one by one.

Problem 3. (Proposed by Dan Schwarz, Romania)

Let n be a positive integer.

(a) Prove that there exists a set S of $6n$ pairwise different positive integers, such that the least common multiple of any two elements of S is no larger than $32n^2$.

(b) Prove that every set T of $6n$ pairwise different positive integers contains two elements the least common multiple of which is larger than $9n^2$.

Solution: (a) Let the set A consist of the $4n$ integers $1, 2, \ldots, 4n$ and let the set B consist of the $2n$ even integers $4n+2, 4n+4, \ldots, 8n$. We claim that the $6n$-element set $S = A \cup B$ has the desired property.

Indeed, the least common multiple of two (even) elements of B is no larger than $8n \cdot (8n/2) = 32n^2$, and the least common multiple of some element of A and some element of $A \cup B$ is at most their product, which is at most $4n \cdot 8n = 32n^2$.

(b) We prove the following lemma: *"If a set U contains $m+1$ integers, where $m \geqslant 2$, that are all not less than m, then some two of its elements have least common multiple strictly larger than m^2."*

Let the elements of U be $u_1 > u_2 > \cdots > u_{m+1} \geqslant m$. Note that $1/u_1 \leqslant 1/u_i \leqslant 1/m$ for $1 \leqslant i \leqslant m+1$. We partition the interval $[1/u_1;\, 1/m]$ into m subintervals of equal length. By the pigeonhole principle, there exist indices i, j with $1 \leqslant i < j \leqslant m+1$ such that $1/u_i$ and $1/u_j$ belong to the same subinterval. Hence

$$0 \;<\; \frac{1}{u_j} - \frac{1}{u_i} \;\leqslant\; \frac{1}{m}\left(\frac{1}{m} - \frac{1}{u_1}\right) \;<\; \frac{1}{m^2}.$$

Now $1/u_j - 1/u_i$ is a positive fraction with denominator $\operatorname{lcm}(u_i, u_j)$. The above thus yields the lower bound $\operatorname{lcm}(u_i, u_j) > m^2$, completing the proof of the lemma.

Applying the lemma with $m = 3n$ to the $3n+1$ largest elements of T, which are all not less than $3n$, we arrive at the desired statement.

A Variant. Alternatively, for part (b), we prove the following lemma: *"If a set U contains $m \geqslant 2$ integers that all are greater than m, then some two of its elements have least common multiple strictly larger than m^2."*

Let $u_1 > u_2 > \cdots > u_m$ be the elements of U. Since $u_m > m = m^2/m$, there exists a smallest index k such that $u_k > m^2/k$. If $k = 1$, then $u_1 > m^2$, and the least common multiple of u_1 and u_2 is strictly larger than m^2. So let us suppose $k > 1$ from now on, so that we have $u_k > m^2/k$ and $u_{k-1} \leqslant m^2/(k-1)$. The greatest common divisor d of u_{k-1} and u_k satisfies

$$d \;\leqslant\; u_{k-1} - u_k \;<\; \frac{m^2}{k-1} - \frac{m^2}{k} \;=\; \frac{m^2}{(k-1)k}.$$

This implies $m^2/(dk) > k-1$ and $u_k/d > k-1$, and hence $u_k/d \geqslant k$. But then the least common multiple of u_{k-1} and u_k equals

$$\frac{u_{k-1}u_k}{d} \;\geqslant\; u_k \cdot \frac{u_k}{d} \;>\; \frac{m^2}{k} \cdot k \;=\; m^2,$$

and the proof of the lemma is complete.

If we remove the $3n$ smallest elements from set T and apply the lemma with $m = 3n$ to the remaining elements, we arrive at the desired statement.

Problem 4. (Proposed by Vesna Iršič, Slovenia)

Find all positive integers a and b for which there are three consecutive integers at which the polynomial

$$P(n) = \frac{n^5 + a}{b}$$

takes integer values.

Solution 1: Denote the three consecutive integers by $x-1$, x, and $x+1$, so that

$$(x-1)^5+a \equiv 0 \pmod b, \qquad x^5+a \equiv 0 \pmod b, \qquad (x+1)^5+a \equiv 0 \pmod b. \quad (1)$$

By computing the differences of the equations in (1) we get

$$A := (x+1)^5 - (x-1)^5 = 10x^4 + 20x^2 + 2 \equiv 0 \pmod b, \quad (2)$$

$$B := (x+1)^5 - x^5 = 5x^4 + 10x^3 + 10x^2 + 5x + 1 \equiv 0 \pmod b. \quad (3)$$

Adding the first and third equation in (1) and subtracting twice the second equation yields

$$C := (x+1)^5 + (x-1)^5 - 2x^5 = 20x^3 + 10x \equiv 0 \pmod b. \quad (4)$$

Next, (2) and (4) together yield

$$D := 4xA - (2x^2+3)C = -22x \equiv 0 \pmod b. \quad (5)$$

Finally we combine (3) and (5) to derive

$$22B + (5x^3 + 10x^2 + 10x + 5)D = 22 \equiv 0 \pmod b.$$

As the positive integer b divides 22, we are left with the four cases $b = 1$, $b = 2$, $b = 11$ and $b = 22$.

If b is even (i.e. $b = 2$ or $b = 22$), then we get a contradiction from (3), because the integer $B = 2(5x^3 + 5x^2) + 5(x^4 + x) + 1$ is odd, and hence not divisible by any even integer.

For $b = 1$, it is trivial to see that a polynomial of the form $P(n) = n^5 + a$, with a any positive integer, has the desired property.

For $b = 11$, we note that

$$n \equiv 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \pmod{11}$$
$$\Longrightarrow \quad n^5 \equiv 0, 1, -1, 1, 1, 1, -1, -1, -1, 1, -1 \pmod{11}.$$

Hence a polynomial of the form $P(n) = (n^5 + a)/11$ has the desired property if and only if $a \equiv \pm 1 \pmod{11}$. This completes the proof.

A Variant. We start by following the first solution up to equation (4). We note that $b = 1$ is a trivial solution, and assume from now on that $b \geqslant 2$. As $(x-1)^5 + a$ and $x^5 + a$ have different parity, b must be odd. As B in (3) is a multiple of b, we conclude that (i) b is not divisible by 5 and that (ii) b and x are relatively prime. As $C = 10x(2x^2+1)$ in (4) is divisible by b, we altogether derive

$$E \;:=\; 2x^2 + 1 \;\equiv\; 0 \pmod{b}.$$

Together with (2) this implies that

$$5E^2 + 10E - 2A \;=\; 11 \;\equiv\; 0 \pmod{b}.$$

Hence $b = 11$ is the only remaining candidate, and it is handled as in the first solution.

Solution 2: Let p be a prime such that p divides b. For some integer x, we have

$$(x-1)^5 \equiv x^5 \equiv (x+1)^5 \pmod{p}.$$

Now, there is a primitive root g modulo p, so there exist u, v, w such that

$$x - 1 \equiv g^u \pmod{p}, \quad x \equiv g^v \pmod{p}, \quad x + 1 \equiv g^w \pmod{p}. \tag{6}$$

The condition of the problem is thus

$$g^{5u} \equiv g^{5v} \equiv g^{5w} \pmod{p} \quad \Longrightarrow \quad 5u \equiv 5v \equiv 5w \pmod{p-1}.$$

If $p \not\equiv 1 \pmod 5$, then 5 is invertible modulo $p-1$ and thus $u \equiv v \equiv w \pmod{p-1}$, i.e. $x - 1 \equiv x \equiv x + 1 \pmod p$. This is a contradiction. Hence $p \equiv 1 \pmod 5$ and thus $u \equiv v \equiv w \pmod{\frac{p-1}{5}}$. Thus, from (6), there exist integers k, ℓ such that

$$\left.\begin{aligned} x - 1 &\equiv g^{v + k\frac{p-1}{5}} \equiv x t^k \pmod p \\ x + 1 &\equiv g^{v + \ell\frac{p-1}{5}} \equiv x t^\ell \pmod p \end{aligned}\right\} \quad \text{where } t = g^{\frac{p-1}{5}}.$$

Let $r = t^k$ and $s = t^\ell$. In particular, the above yields $r, s \not\equiv 1 \pmod p$, and thus

$$x \equiv -(r-1)^{-1} \equiv (s-1)^{-1} \pmod p.$$

It follows that

$$(r-1)^{-1} + (s-1)^{-1} \equiv 0 \pmod p \quad \Longrightarrow \quad r + s \equiv 2 \pmod p.$$

Now $t^5 \equiv 1 \pmod p$, so r and s must be congruent, modulo p, to some of the non-trivial fifth roots of unity t, t^2, t^3, t^4. Observe that, for any pair of these non-trivial roots of unity,

either one is the other's inverse, or one is the other's square. In the first case, we have $r + r^{-1} \equiv 2 \pmod{p}$, implying $r \equiv 1 \pmod{p}$, a contradiction. Hence

$$r + r^2 \equiv 2 \pmod{p} \quad \Longrightarrow \quad (r-1)(r+2) \equiv 0 \pmod{p},$$

or

$$s^2 + s \equiv 2 \pmod{p} \quad \Longrightarrow \quad (s-1)(s+2) \equiv 0 \pmod{p}.$$

Thus, since $r, s \not\equiv 1 \pmod{p}$, we have $r \equiv -2 \pmod{p}$ or $s \equiv -2 \pmod{p}$, and thus $1 \equiv r^5 \equiv -32 \pmod{p}$ or an analogous equation obtained from s. Hence $p \mid 33$. Since $p \equiv 1 \pmod{5}$, it follows that $p = 11$, i.e. b is a power of 11.

Examining the fifth powers modulo 11, we see that $b = 11$ is indeed a solution with $a \equiv \pm 1 \pmod{11}$ and, correspondingly, $x \equiv \pm 4 \pmod{11}$. Now suppose, for the sake of contradiction, that 11^2 divides b. Then, for some integer m, we must have

$$\Big(x-1, x, x+1\Big) \equiv \pm\Big(3 + 11m, 4 + 11m, 5 + 11m\Big) \pmod{121},$$

and thus, substituting into the condition of the problem,

$$\begin{aligned} 3^5 + 55 \cdot 3^4 m \equiv 4^5 + 55 \cdot 4^4 m \equiv 5^5 + 55 \cdot 5^4 m \pmod{121} \\ \Longrightarrow \quad 1 - 22m \equiv 56 + 44m \equiv -21 + 11m \pmod{121}. \end{aligned}$$

Hence $33m \equiv 22 \pmod{121}$ and $33m \equiv 44 \pmod{121}$, so $22 \equiv 0 \pmod{121}$, a contradiction. It follows that $b \mid 11$.

Finally, we conclude that the positive integers satisfying the original condition are $b = 11$, with $a \equiv \pm 1 \pmod{11}$, and $b = 1$, for any positive integer a.

Solution 3: Denote the three consecutive integers by $x-1$, x, and $x+1$ as in Solution 1. By computing the differences in (1), we find

$$\begin{aligned} F &:= (x+1)^5 - x^5 = 5x^4 + 10x^3 + 10x^2 + 5x + 1 \equiv 0 \pmod{b}, \\ G &:= x^5 - (x-1)^5 = 5x^4 - 10x^3 + 10x^2 - 5x + 1 \equiv 0 \pmod{b}. \end{aligned}$$

By determining the polynomial greatest divisor of $F(x)$ and $G(x)$ using the Euclidean algorithm, we find that

$$p(x)F(x) + q(x)G(x) = 22, \tag{7}$$

where

$$\begin{aligned} p(x) &= -15x^3 + 30x^2 - 28x + 11, \\ q(x) &= 15x^3 + 30x^2 + 28x + 11. \end{aligned}$$

Since $b \mid F(x)$ and $b \mid G(x)$, it follows from (7) that $b \mid 22$. We now finish off the problem as in Solution 1.

Problem 5. (Proposed by Waldemar Pompe, Poland)

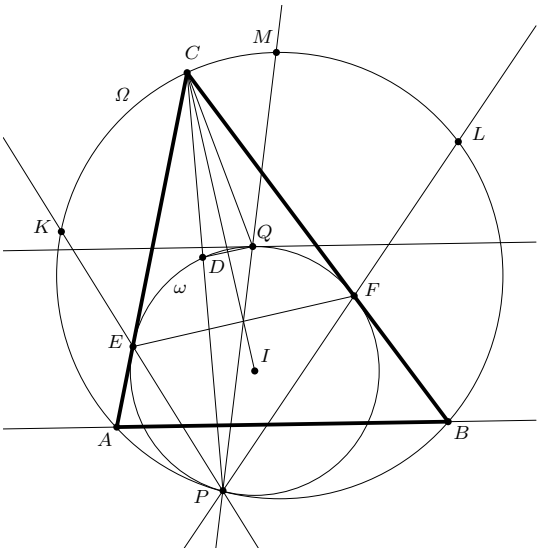
Let Ω be the circumcircle of the triangle ABC. The circle ω is tangent to the sides AC and BC, and it is internally tangent to Ω at the point P. A line parallel to AB and intersecting the interior of triangle ABC is tangent to ω at Q.

Prove that $\angle ACP = \angle QCB$.

Solution 1: Assume that ω is tangent to AC and BC at E and F, respectively and let PE, PF, PQ meet Ω at K, L, M, respectively. Let I and O denote the respective centres of ω and Ω, and consider the homethety $\mathscr{H}$ that maps ω onto Ω. Now K is the image of E under $\mathscr{H}$, and $EI \perp AC$. Hence $OK \perp AC$, and thus K is the midpoint of the arc CA. Similarly, L is the midpoint of the arc BC and M is the midpoint of the arc BA. It follows that arcs LM and CK are equal, because

$$\begin{aligned} \widehat{BM} = \widehat{MA} \quad &\Longrightarrow \quad \widehat{BL} + \widehat{LM} = \widehat{MK} + \widehat{KA} \quad \Longrightarrow \quad \widehat{LC} + \widehat{LM} = \widehat{MK} + \widehat{CK} \\ &\Longrightarrow \quad 2\widehat{LM} + \widehat{MC} = \widehat{MC} + 2\widehat{CK} \quad \Longrightarrow \quad \widehat{LM} = \widehat{CK}. \end{aligned}$$

Thus arcs FQ and DE of ω are equal, too, where D is the intersection of CP with ω. Since CE and CF are tangents to ω, this implies that $\angle DEC = \angle CFQ$. Further, $CE = CF$, and thus triangles CED and CFQ are congruent. In particular, $\angle ECD = \angle QCF$, as required.



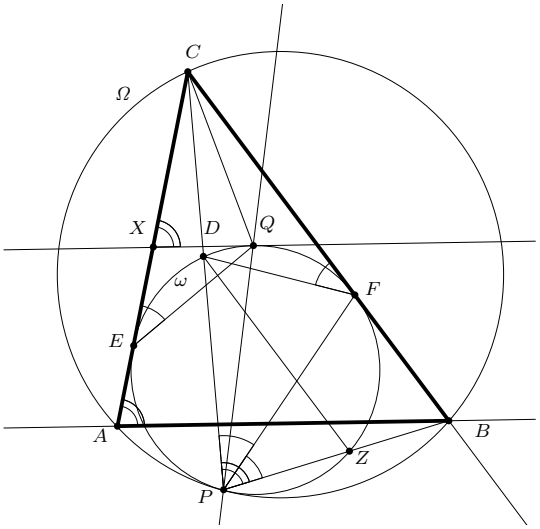
A Variant. As above, we show that arcs FQ and DE of ω are equal, which implies that $DEFQ$ is an isoceles trapezoid, and so we have $\angle FED = \angle QFE$. Together with $|FQ| = |DE|$, this implies that, since E and F are images of each other under reflection in the angle bisector CI of $\angle C$, so are the segments $[EQ]$ and $[FD]$, and, in particular, D and Q. In turn, this yields $\angle ECD = \angle QCF$, as required.

Remark. Let J denote the incentre of ABC. By Sawayama's theorem, J is the midpoint of $[EF]$, i.e. PJ is a median of PFE. Since C is the intersection of the tangents AC and BC to the circumcircle of PFE at E and F, respectively, PC is a symmedian of PFE. Thus $\angle CPE = \angle FPJ$. But, since the arcs FQ and DE of ω are equal, $\angle CPE = \angle FPQ$. This shows that J lies on the line PQ.

Another Variant. We show that arcs QE and FD are equal, and then finish as in the main solution. Let BP meet ω again at Z. Consider the homothety $\mathscr{H}$ that maps ω onto Ω. Under $\mathscr{H}$, $D \mapsto C$ and $Z \mapsto B$, so $DZ \parallel CB$. (*This also follows by considering the common tangent to ω and Ω, and tangential angles.*) Now, by power of a point,

$$BF^2 = BZ \cdot BP, \quad CF^2 = CD \cdot CP.$$

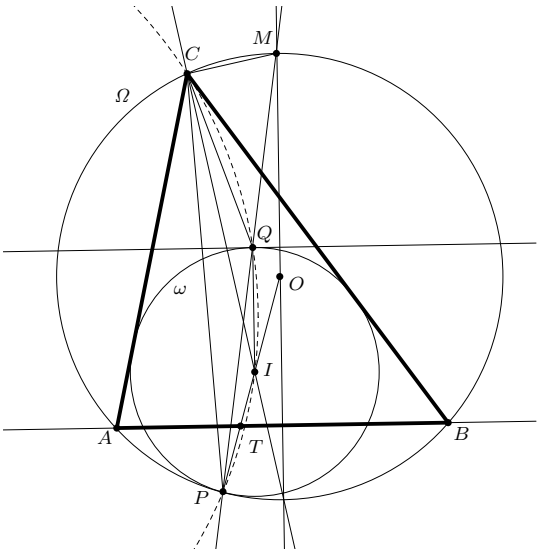
Now $DZ \parallel CB$ implies $BZ/BP = CD/CP$, and so, dividing the two previous equations by each other, and taking square roots, $BF/CF = BP/CP$. Hence PF bissects angle $\angle BPC$. Now let $\angle BPF = \angle FPC = \beta$. By tangential angles, it follows that $\angle CFD = \beta$. Further, $\angle BAC = \angle BPC = 2\beta$. Let the tangent to ω through Q and parallel to AB meet AC at X. Then $\angle QXC = 2\beta$, so, since $XQ = XE$ by tangency, $\angle QEX = \beta$. By tangential angles, it follows that arcs FD and QE are equal, as claimed.



Solution 2: Let I and O denote the respective centres of ω and Ω. Observe that CI is the angle bisector of angle $\angle C$, because ω is tangent to AC and BC. Consider the homethety $\mathscr{H}$ that maps ω onto Ω. Let M be the image of Q under $\mathscr{H}$. By construction, $IQ \perp AB$, so $OM \perp AB$. Thus the diameter OM of Ω passes through the midpoint of the arc AB of Ω, which also lies on the angle bisector CI. This implies that $\angle ICM = 90^\circ$. We next show that P, I, Q, C lie on a circle. Notice that

$$\begin{aligned}\angle PQI &= 90^\circ - \tfrac{1}{2}\angle QIP = 90^\circ - \tfrac{1}{2}\angle MOP = 90^\circ - \big(180^\circ - \angle PCM\big) \\ &= \big(\angle PCI + \angle ICM\big) - 90^\circ = \angle PCI.\end{aligned}$$

Hence P, I, Q, C lie on a circle. But $PI = IQ$, so CI is the angle bisector of $\angle PCQ$. Since CI is also the angle bisector of angle $\angle C$, it follows that $\angle ACP = \angle QCB$, as required.



A Variant. We show that $PIQC$ is cyclic by chasing angles. Define $\alpha = \angle BAC$, $\beta = \angle CBA$ and $\gamma = \angle ACP$. For convenience, we consider the configuration where A and P lie one the same side of the angle bisector CI of $\angle C$. In this configuration,

$$\angle PCI = \tfrac{1}{2}\angle ACB - \angle ACP = 90^\circ - \tfrac{1}{2}\alpha - \tfrac{1}{2}\beta - \gamma.$$

Now notice that $\angle PBA = \angle ACP = \gamma$, and therefore $\angle CAP = 180^\circ - \beta - \gamma$, whence $\angle PAB = 180^\circ - \alpha - \beta - \gamma$. Further, PO is a diameter of Ω, and therefore $\angle APO = 90^\circ - \gamma$. Let AB and PO intersect at T. Then

$$\angle BTO = 180^\circ - \angle PAB - \angle APO = \alpha + \beta + 2\gamma - 90^\circ.$$

But $QI \perp AB$ by construction, and thus

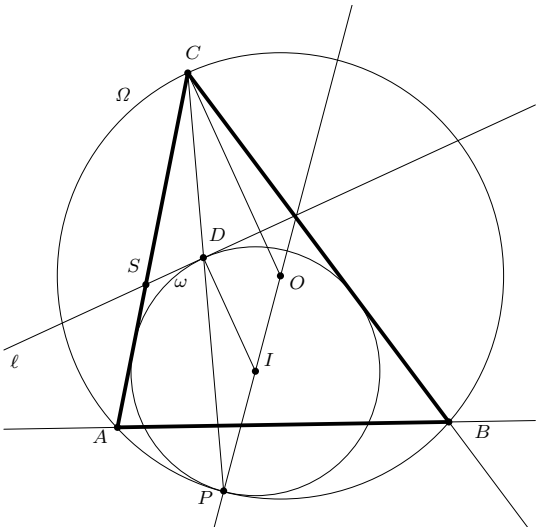
$$\begin{aligned} \angle OIQ = 90^\circ - \angle BTO &= 180^\circ - \alpha - \beta - 2\gamma \\ \implies \quad \angle QIP &= 180^\circ - \angle OIQ = \alpha + \beta + 2\gamma \\ \implies \quad \angle PQI &= 90^\circ - \tfrac{1}{2}\alpha - \tfrac{1}{2}\beta - \gamma. \end{aligned}$$

Hence $\angle ICQ = \angle PQI$, and thus $PIQC$ is cyclic. Since $PI = QI$, it follows that CI is the angle bisector of $\angle PCQ$, which completes the proof.

Solution 3: Let I and O denote the respective centres of ω and Ω. Let D be the second point of intersection of CP with ω, and let ℓ denote the tangent to ω at D, which meets AC at S. Hence $ID \perp \ell$. By construction, P, I, O lie one a line, and hence the isosceles triangles PID and POC are similar. In particular, it follows that $OC \perp \ell$, so C is the midpoint of the arc of Ω defined by the points of intersection of ℓ with Ω. It is easy to see that this implies that

$$\angle DSC = \angle ABC.$$

Under reflection in the angle bisector CI of $\angle C$, ℓ is thus mapped to a tangent to ω parallel to AB and intersecting the interior of ABC, since ω is mapped to itself under this reflection. In particular, D is mapped to Q, and thus $\angle QCB = \angle ACD$, as required.



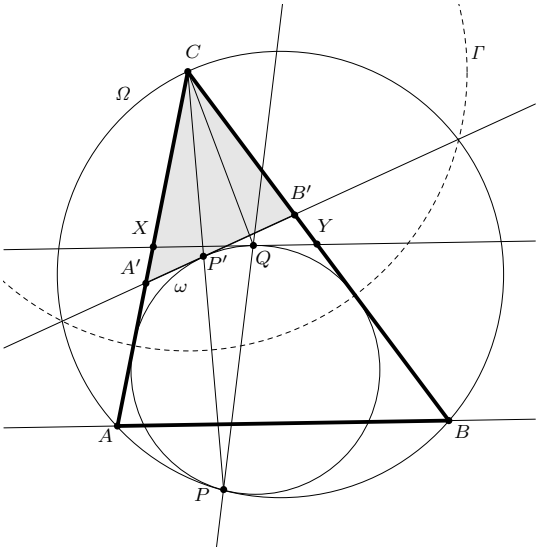
Remark. Conceptually, this solution is similar to Solution 1, but here, we proceed more directly via the reflectional symmetry. Therefore, this solution links Solution 1 to Solution 4, in which we use an inversion.

Solution 4: Let the tangent to ω at Q meet AC and BC at X and Y, respectively. Then $AC/XC = BC/YC$, and thus there is a radius r such that $r^2 = AC \cdot YC = BC \cdot XC$. Let Γ denote the circle with centre C and radius r, and consider the inversion $\mathscr{I}$ in the circle Γ. Under $\mathscr{I}$,

$A \longmapsto A'$, the point on the ray CA satisfying $CA' = CY$;
$B \longmapsto B'$, the point on the ray CB satisfying $CB' = CX$;
$\Omega \longmapsto$ the line $A'B'$;
$\omega \longmapsto \omega'$, the excircle of $CA'B'$ opposite C;
$P \longmapsto P'$, the point where ω' touches $A'B'$;

In particular, ω', the image of ω, is a circle tangent to AC, BC and $A'B'$, so it is either the excircle of $CA'B'$ opposite C, or the incircle of $CA'B'$. Let ω be tangent to BC at F, and let F' be the image of F under $\mathscr{I}$. Then $CF \cdot CF' = BC \cdot XC$. Now $CF < BC$, so $CF' > CX = CB'$. Hence ω' cannot be the incircle, so ω' is indeed the excircle of $CA'B'$ opposite C.

Now note that ω is the excircle of CXY opposite C. The reflection about the angle bisector of $\angle C$ maps X to B', Y to A'. It thus maps the triangle CXY to $CB'A'$, ω to ω' and, finally, Q to P'. It follows that $\angle ACP = \angle ACP' = \angle QCB$, as required.



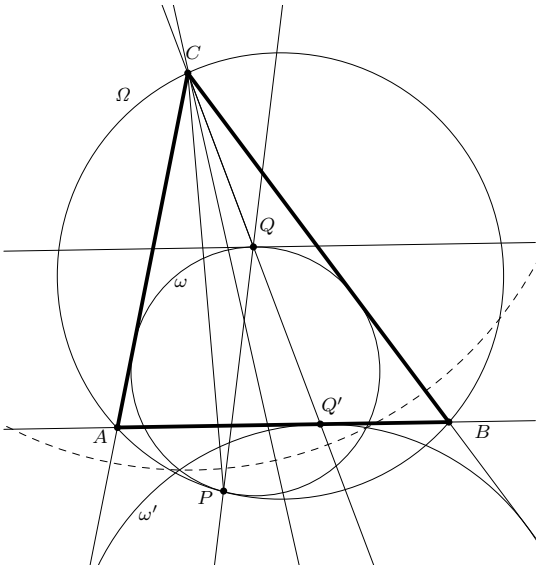
Solution 5: Let r be the radius such that $r^2 = AC \cdot BC$. Let $\mathscr{J}$ denote the composition of the inversion $\mathscr{I}$ in the circle of centre C and radius r, followed by the reflection in the

angle bisector of $\angle C$. Under $\mathscr{J}$,

$A \longmapsto B,\ B \mapsto A$;
$\Omega \longmapsto$ the line AB;
$\omega \longmapsto \omega'$, the excircle of ABC opposite the vertex C;
$P \longmapsto Q'$, the point where ω' touches AB;

In particular, note that the image ω' of ω under $\mathscr{J}$ is a circle tangent to AC, BC and AB, so it is either the incircle of ABC, or the excircle opposite vertex C. Observe that $r \geqslant \min\{AC, BC\}$, so the image of the points of tangency of ω must lie outside ABC, and thus ω' cannot be the incircle. Thus ω' is the excircle opposite vertex C as claimed. Further, the point of tangency P is mapped to Q'.

Now, since the line CP is mapped to itself under the inversion $\mathscr{I}$, and mapped onto CQ' under $\mathscr{J}$, CP and CQ' are images of each other under reflection in the angle bisector of $\angle C$. But C, Q, Q' lie on a line for there is a homothety with centre C that maps ω onto the excircle ω'. This completes the proof.



Solution 6: Assume that ω is tangent to AC and BC at E and F, respectively. Assume that CP meets ω at D. Let I and O denote the respective centres of ω and Ω. To set up a solution in the complex plane, we take the circle ω as the unit circle centered at the origin of the complex plane, and let PO be the real axis with $o > 0$, where we use the convention that lowercase letters denote complex coordinates of corresponding points in the plane denoted by uppercase letters.

Now, a point Z on the circle Ω satisfies

$$|z-o|^2=(o+1)^2 \quad\Longleftrightarrow\quad zz^*-o(z+z^*)-2o-1=0.$$

The triangle ABC is defined by the points E and F on ω, the intersection C of the corresponding tangents lying on Ω. Thus $c=2ef\big/(e+f)$, and further

$$|c-o|^2=(o+1)^2 \quad\Longleftrightarrow\quad cc^*-o(c+c^*)-2o-1=0, \tag{1}$$

and this is the equality defining o. The points A and B are the second intersection points of Ω with the tangents to ω at E and F respectively. A point Z on the tangent through E is given by $z=2e-e^2z^*$, and thus A and C satisfy

$$\begin{aligned}\Big(2e-e^2z^*\Big)z^*-o\Big(2e-e^2z^*+z^*\Big)-2o-1&=0\\ \Longleftrightarrow\quad -e^2z^{*2}+\Big(2e+oe^2-o\Big)z^*-\Big(2eo+2o+1\Big)&=0\\ \Longleftrightarrow\quad z^{*2}-\Big(2e^*+o-oe^{*2}\Big)z^*+\Big(2e^*o+2oe^{*2}+e^{*2}\Big)&=0,\end{aligned}$$

since $|e|=1$. Thus

$$a^*+c^*=2e^*+o-oe^{*2} \quad\Longrightarrow\quad a^*=\frac{2e^*f}{e+f}+o\Big(1-e^{*2}\Big),$$

and similarly

$$b^*=\frac{2f^*e}{f+e}+o\Big(1-f^{*2}\Big).$$

Then

$$\begin{aligned}b^*-a^*&=\frac{2(ef^*-e^*f)}{e+f}+o\Big(e^{*2}-f^{*2}\Big)\\ &=\frac{2ef\Big(f^{*2}-e^{*2}\Big)}{e+f}+o\Big(e^{*2}-f^{*2}\Big)\\ &=\Big(f^{*2}-e^{*2}\Big)\left(\frac{2ef}{e+f}-o\right)=\Big(f^{*2}-e^{*2}\Big)(c-o).\end{aligned}$$

Now let Z be a point on the tangent to ω parallel to AB passing through Q. Then

$$z=2q-q^2z^* \quad\Longleftrightarrow\quad z-q=q-q^2z^*=-q^2(z^*-q^*),$$

for $|q|=1$, and thus

$$\frac{b-a}{b^*-a^*}=\frac{z-q}{z^*-q^*}=\frac{-q^2(z^*-q^*)}{z^*-q^*}=-q^2.$$

It follows that

$$q^2 = -\frac{b-a}{b^*-a^*} = -\frac{\left(f^2-e^2\right)(c^*-o)}{\left(f^{*2}-e^{*2}\right)(c-o)} = e^2f^2\frac{c^*-o}{c-o}$$
$$= e^2f^2\frac{(c^*-o)^2}{|c-o|^2} = e^2f^2\frac{(c^*-o)^2}{(1+o)^2},$$

where we have used (1). In particular,

$$q = ef\frac{c^*-o}{1+o},$$

where the choice of sign is to be justified *a posteriori*. Further, the point D satisfies

$$-dp = \frac{d-p}{d^*-p^*} = \frac{c-p}{c^*-p^*} \quad\Longrightarrow\quad d = -\frac{c-p}{c^*p-1} = \frac{c+1}{c^*+1},$$

using $p=-1$ to obtain the final equality.

Now, it suffices to show that (i) $DQ \parallel EF \perp CI$ and (ii) the midpoint of $[DQ]$ is on CI. The desired equality then follows by symmetry with respect to the angle bisector of the angle $\angle ACB$. Notice that (i) is equivalent with

$$\frac{d-q}{d^*-q^*} = \frac{e-f}{e^*-f^*} \quad\Longleftrightarrow\quad dq = ef.$$

for $[DQ]$ and $[EF]$ are chords of ω. But

$$dq = ef \quad\Longleftrightarrow\quad \frac{c+1}{c^*+1}ef\frac{c^*-o}{1+o} = ef \quad\Longleftrightarrow\quad (c+1)(c^*-o) = (c^*+1)(1+o)$$
$$\Longleftrightarrow\quad cc^* - o(c+c^*) - 2o - 1 = 0.$$

The last equality is precisely the defining relation for o, (1). This proves (i). Further, the midpoint of $[DQ]$ is $\frac{1}{2}(d+q)$, so it remains to check that

$$dq = \frac{d+q}{d^*+q^*} = \frac{c}{c^*} = ef,$$

where the first equality expresses that $[DQ]$ is a chord of ω (obviously) containing its midpoint, the second equality expresses the alignment of the midpoint of $[DQ]$, C and I, and the third equality follows from the expression for c. But we have just shown that $dq = ef$. This proves (ii), justifies the choice of sign for q *a posteriori*, and thus completes the solution of the problem.

Problem 6. (Proposed by Emil Kolev, Bulgaria)

Snow White and the Seven Dwarves are living in their house in the forest. On each of 16 consecutive days, some of the dwarves worked in the diamond mine while the remaining dwarves collected berries in the forest. No dwarf performed both types of work on the same day. On any two different (not necessarily consecutive) days, at least three dwarves each performed both types of work. Further, on the first day, all seven dwarves worked in the diamond mine.

Prove that, on one of these 16 days, all seven dwarves were collecting berries.

Solution 1: We define V as the set of all 128 vectors of length 7 with entries in $\{0, 1\}$. Every such vector encodes the work schedule of a single day: if the i-th entry is 0 then the i-th dwarf works in the mine, and if this entry is 1 then the i-th dwarf collects berries. The 16 working days correspond to 16 vectors $d_1, \ldots, d_{16}$ in V, which we will call *day-vectors*. The condition imposed on any pair of distinct days means that any two distinct day-vectors d_i and d_j differ in at least three positions.

We say that a vector $x \in V$ *covers* some vector $y \in V$, if x and y differ in at most one position; note that every vector in V covers exactly eight vectors. For each of the 16 day-vectors d_i we define $B_i \subset V$ as the set of the eight vectors that are covered by d_i. As, for $i \neq j$, the day-vectors d_i and d_j differ in at least three positions, their corresponding sets B_i and B_j are disjoint. As the sets $B_1, \ldots, B_{16}$ together contain $16 \cdot 8 = 128 = |V|$ distinct elements, they form a partition of V; in other words, every vector in V is covered by precisely one day-vector.

The *weight* of a vector $v \in V$ is defined as the number of 1-entries in v. For $k = 0, 1, \ldots, 7$, the set V contains $\binom{7}{k}$ vectors of weight k. Let us analyse the 16 day-vectors $d_1, \ldots, d_{16}$ by their weights, and let us discuss how the vectors in V are covered by them.

1. As all seven dwarves work in the diamond mine on the first day, the first day-vector is $d_1 = (0000000)$. This day-vector covers all vectors in V with weight 0 or 1.

2. No day-vector can have weight 2, as otherwise it would differ from d_1 in at most two positions. Hence each of the $\binom{7}{2} = 21$ vectors of weight 2 must be covered by some day-vector of weight 3. As every vector of weight 3 covers three vectors of weight 2, exactly $21/3 = 7$ day-vectors have weight 3.

3. How are the $\binom{7}{3} = 35$ vectors of weight 3 covered by the day-vectors? Seven of them are day-vectors, and the remaining 28 ones must be covered by day-vectors of weight 4. As every vector of weight 4 covers four vectors of weight 3, exactly $28/4 = 7$ day-vectors have weight 4.

To summarize, one day-vector has weight 0, seven have weight 3, and seven have weight 4. None of these 15 day-vectors covers any vector of weight 6 or 7, so that the eight heavy-weight vectors in V must be covered by the only remaining day-vector; and this remaining vector must be (1111111). On the day corresponding to (1111111) all seven dwarves are collecting berries, and that is what we wanted to show.

Solution 2: If a dwarf X performs the same type of work on three days D_1, D_2, D_3, then we say that this triple of days is *monotonous* for X. We claim that the following configuration cannot occur: There are three dwarves X_1, X_2, X_3 and three days D_1, D_2, D_3, such that the triple (D_1, D_2, D_3) is monotonous for each of the dwarves X_1, X_2, X_3.

(Proof: Suppose that such a configuration occurs. Then among the remaining dwarves there exist three dwarves Y_1, Y_2, Y_3 that performed both types of work on day D_1 and on day D_2; without loss of generality these three dwarves worked in the mine on day D_1 and collected berries on day D_2. On day D_3, two of Y_1, Y_2, Y_3 performed the same type of work, and without loss of generality Y_1 and Y_2 worked in the mine. But then on days D_1 and D_3, each of the five dwarves X_1, X_2, X_3, Y_1, Y_2 performed only one type of work; this is in contradiction with the problem statement.)

Next we consider some fixed triple X_1, X_2, X_3 of dwarves. There are eight possible working schedules for X_1, X_2, X_3 (like mine-mine-mine, mine-mine-berries, mine-berries-mine, etc). As the above forbidden configuration does not occur, each of these eight working schedules must occur on exactly two of the sixteen days. In particular this implies that every dwarf worked exactly eight times in the mine and exactly eight times in the forest.

For $0 \leqslant k \leqslant 7$ we denote by $d(k)$ the number of days on which exactly k dwarves were collecting berries. Since on the first day all seven dwarves were in the mine, on each of the remaining days at least three dwarves collected berries. This yields $d(0) = 1$ and $d(1) = d(2) = 0$. We assume, for the sake of contradiction, that $d(7) = 0$ and hence

$$d(3) + d(4) + d(5) + d(6) \;=\; 15. \tag{1}$$

As every dwarf collected berries exactly eight times, we get that, further,

$$3\,d(3) + 4\,d(4) + 5\,d(5) + 6\,d(6) \;=\; 7 \cdot 8 \;=\; 56. \tag{2}$$

Next, let us count the number q of quadruples (X_1, X_2, X_3, D) for which X_1, X_2, X_3 are three pairwise distinct dwarves that all collected berries on day D. As there are $7 \cdot 6 \cdot 5 = 210$ triples of pairwise distinct dwarves, and as every working schedule for three fixed dwarves occurs on exactly two days, we get $q = 420$. As every day on which k dwarves collect berries contributes $k(k-1)(k-2)$ such quadruples, we also have

$$3 \cdot 2 \cdot 1 \cdot d(3) + 4 \cdot 3 \cdot 2 \cdot d(4) + 5 \cdot 4 \cdot 3 \cdot d(5) + 6 \cdot 5 \cdot 4 \cdot d(6) \;=\; q \;=\; 420,$$

which simplifies to

$$d(3) + 4\,d(4) + 10\,d(5) + 20\,d(6) \;=\; 70. \tag{3}$$

Finally, we count the number r of quadruples (X_1, X_2, X_3, D) for which X_1, X_2, X_3 are three pairwise distinct dwarves that all worked in the mine on day D. Similarly as above we see that $r = 420$ and that

$$7 \cdot 6 \cdot 5 \cdot d(0) + 4 \cdot 3 \cdot 2 \cdot d(3) + 3 \cdot 2 \cdot 1 \cdot d(4) \;=\; r \;=\; 420,$$

which simplifies to

$$4\,d(3) + d(4) \;=\; 35. \tag{4}$$

Multiplying (1) by -40, multiplying (2) by 10, multiplying (3) by -1, multiplying (4) by 4, and then adding up the four resulting equations yields $5d(3) = 30$ and hence $d(3) = 6$. Then (4) yields $d(4) = 11$. As $d(3) + d(4) = 17$, the total number of days cannot be 16. We have reached the desired contradiction.

A Variant. We follow the second solution up to equation (3). Multiplying (1) by 8, multiplying (2) by -3, and adding the two resulting equations to (3) yields

$$3\,d(5) + 10\,d(6) \;=\; 22. \tag{5}$$

As $d(5)$ and $d(6)$ are positive integers, (5) implies $0 \leqslant d(6) \leqslant 2$. Only the case $d(6) = 1$ yields an integral value $d(5) = 4$. The equations (1) and (2) then yield $d(3) = 10$ and $d(4) = 0$.

Now let us look at the $d(3) = 10$ special days on which exactly three dwarves were collecting berries. One of the dwarves collected berries on at least five special days (if every dwarf collected berries on at most four special days, this would allow at most $7 \cdot 4/3 < 10$ special days); we call this dwarf X. On at least two out of these five special days, some dwarf Y must have collected berries together with X. Then these two days contradict the problem statement. We have reached the desired contradiction.

Comment. Up to permutations of the dwarves, there exists a unique set of day-vectors (as introduced in the first solution) that satisfies the conditions of the problem statement:

0000000	1110000	1001100	1000011	0101010	0100101	0010110	0011001
1111111	0001111	0110011	0111100	1010101	1011010	1101001	1100110

XVIII МАТЕМАТИЧЕСКАЯ ОЛИМПИАДА ИМЕНИ ЛЕОНАРДА ЭЙЛЕРА

Решения заданий заключительного этапа, 1 день

1. *Назовем натуральное число k-хорошим, если оно представимо в виде суммы k последовательных натуральных чисел. Учитель попросил Васю придумать число n, и обещал поставить по пятерке за каждое k, большее 1 и меньшее 7, при котором n окажется k-хорошим. Какое наибольшее число пятерок мог получить Вася?* (Методсовет)

Ответ. 4. **Решение**. Оценка. Заметим, что сумма двух идущих подряд натуральных чисел всегда нечетна, а четырех идущих подряд — всегда четна. Поэтому число n не может одновременно быть 2-хорошим и 4-хорошим, и Вася не мог получить больше четырех пятерок. Пример. Сумма двух идущих подряд чисел m и m+1 равна 2m+1, трех идущих подряд, начиная с l — 3l+3 = 3(l+1), пяти идущих подряд, начиная с s — 5s+10 = 5(s+2), шести идущих подряд, начиная с t — 6t+15 = 3(2t+5). Поэтому если Вася примет за n достаточно большое нечетное число, делящееся на 3 и 5, то он сможет найти подходящие m, l, s и t. Например, при n = 45 имеем m = 22, l = 14, s = 7, t = 5.

2. *В классе учится больше 6, но меньше 60 учеников, и в нём организовано семь кружков. Каждый ученик класса посещает одинаковое количество кружков. Известно, что для любых двух кружков найдутся ровно три ученика, которые посещают их оба. Сколько учеников может быть в таком классе?* (Ф. Фот)

Ответ. 21. **Решение**. Пусть в классе u учеников, и каждый посещает k кружков. Подсчитаем количество T троек, составленных из ученика и двух кружков, которые он посещает. С одной стороны, для каждой пары кружков есть три таких тройки, поэтому $T = 3\cdot(7\cdot6/2) = 63$. С другой стороны, для каждого ученика есть $k(k–1)/2$ таких троек. Значит, $uk(k–1) = 126$. По условию $u \ge 7$, поэтому $k(k–1) \le 18$, откуда $k \le 4$. При $k = 4$ получаем нецелое u, при $k = 2$ $u > 60$. При $k = 3$ получаем $u = 21$. **Замечание**. Примером может служить класс, в котором кружкам соответствуют вершины, середины сторон и центр равностороннего треугольника, а тройкам учеников, посещающим три данных кружка — стороны, медианы и вписанная окружность треугольника. Но построение примера в решении не обязательно, поскольку его существование постулировано в условии задачи.

3. *Неотрицательные числа a, b и c таковы, что $a^2+b^2+c^2 = a+b+c$. Докажите, что*

$$\frac{(a-1)^2}{b+c+1}+\frac{(b-1)^2}{c+a+1}+\frac{(c-1)^2}{a+b+1}\geqslant\frac{3}{1+a+b+c}.$$ (А. Кузнецов)

Первое решение. Положим для краткости $s = a+b+c$. Так как $(a–1)^2+(b–1)^2+(c–1)^2 = 3–s$, нам надо доказать неравенство

$$\frac{(a-1)^2}{b+c+1}+\frac{(b-1)^2}{c+a+1}+\frac{(c-1)^2}{a+b+1}\le\frac{3}{a+b+c+1}=\frac{(a-1)^2+(b-1)^2+(c-1)^2+s}{1+s}.$$

Сгруппируем

$$\left(\frac{(a-1)^2}{b+c+1}-\frac{(a-1)^2}{s+1}\right)+\left(\frac{(b-1)^2}{c+a+1}-\frac{(b-1)^2}{s+1}\right)+\left(\frac{(c-1)^2}{a+b+1}-\frac{(c-1)^2}{s+1}\right)\le\frac{s}{1+s}=\frac{a+b+c}{1+s}.$$

Приведем разность в первой скобке к общему знаменателю:

$$\frac{(a-1)^2}{b+c+1}-\frac{(a-1)^2}{s+1}=\frac{a(a-1)^2}{(b+c+1)(1+s)}$$

и докажем, что эта дробь не превосходит $\frac{a}{1+s}$:

$$\frac{(a-1)^2}{b+c+1}\le 1 \Leftrightarrow b+c+1-(a-1)^2 = 2a+b+c-a^2 = a+b^2+c^2 \ge 0.$$

Аналогично, две другие скобки не превосходят $\frac{b}{1+s}$ и $\frac{c}{1+s}$. Осталось сложить три полученных неравенства.

Второе решение. $a^2+b^2+c^2 = a+b+c \Leftrightarrow (2a-1)^2+(2b-1)^2+(2c-1)^2 = 3$, откуда $a < 2$ и $(a-1)^2 < 1$. Поэтому $\frac{(a-1)^2}{1+b+c}<\frac{(a-1)^2+a}{1+a+b+c}$. В самом деле, $(a-1)^2(1+a+b+c) < ((a-1)^2+a)(1+b+c) \Leftrightarrow a(a-1)^2 < a(1+b+c) \Leftrightarrow (a-1)^2 < 1+b+c$, что верно, так как $(a-1)^2 < 1$, а $1+b+c \ge 1$. Аналогично $\frac{(b-1)^2}{1+c+a}<\frac{(b-1)^2+b}{1+a+b+c}$ и $\frac{(c-1)^2}{1+c+a}<\frac{(c-1)^2+c}{1+a+b+c}$. Складывая три полученных неравенства, находим, что $\frac{(a-1)^2}{b+c+1}+\frac{(b-1)^2}{c+a+1}+\frac{(c-1)^2}{a+b+1}$ меньше, чем $\frac{a^2-a+1+b^2-b+1+c^2-c+1}{1+a+b+c}=\frac{3}{1+a+b+c}$, что и требовалось доказать.

Третье решение. Докажем, что $\frac{(a-1)^2}{b+c+1}\leqslant\frac{1}{1+a+b+c}$ (*). Складывая это неравенство с двумя аналогичными для b и c, получим требуемое. Умножив неравенство (*) на произведение знаменателей, после приведения подобных членов получим равносильное неравенство $a+(a^2-2a)(1+a+b+c) \le 0$. При $a = 0$ оно верно. При $a \ne 0$, поделив обе его части на a, получим $1+(a-2)(1+a+b+c) \le 0$. Теперь достаточно доказать, что $(2-a)(1+a) \ge 1 \Leftrightarrow a^2-a \le 1$. Для этого заметим, что $(a^2-a)+(b^2-b)+(c^2-c) = 0$ и каждая из двух последних скобок не меньше, чем $-1/4$, так что a^2-a не превосходит даже $1/2$.

4. *Дана замкнутая тысячезвенная ломаная, в которой длины всех звеньев равны, никакие два конца несоседних звеньев не совпадают и никакой конец одного из звеньев не лежит внутри другого звена. При каком наибольшем k может случиться, что каждое звено пересекает хотя бы k из оставшихся звеньев под прямым углом?* (А. Кузнецов)

Ответ. При $k = 250$. **Решение**. Оценка. Допустим, все звенья ломаной параллельны двум взаимно перпендикулярным прямым. Тогда все звенья ломаной являются сторонами квадратов некоторой квадратной сетки, и никакие два звена не могут

пересекаться в точках, отличных от своих концов, то есть $k = 0$. В противном случае существуют по крайней мере четыре попарно не параллельные прямые, каждой из которых параллельно некоторое количество звеньев ломаной. Среди этих прямых найдется такая, которой параллельны не более 250 звеньев. Тогда каждое звено, перпендикулярное этой прямой, пересечено перпендикулярным звеном не более 250 раз. Таким образом, $k \leq 250$. Пример. Возьмем квадрат со стороной 125. Каждую его сторону разделим на 125 отрезков длины 1 и на каждом отрезке, как на основании, построим равнобедренный треугольник с высотой 125, направленный внутрь квадрата. Ломаная, образованная боковыми сторонами этих треугольников — искомая.

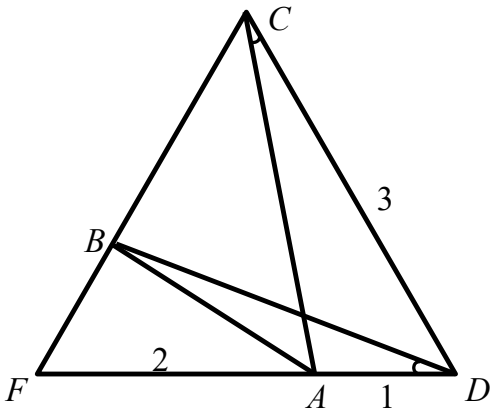
XVIII МАТЕМАТИЧЕСКАЯ ОЛИМПИАДА ИМЕНИ ЛЕОНАРДА ЭЙЛЕРА

Решения заданий заключительного этапа, 2 день

5. *На шахматной доске размера 30×30 стоит 220 не бьющих друг друга королей. Докажите, что в любом квадрате 9×9 стоит не менее 11 королей. Напомним, что король бьет клетки, соседние со своей по горизонтали, вертикали или диагонали.* (С. Берлов)

Решение. Разобьем доску 30×30 на 225 квадратиков 2×2. В каждом из них не более одного короля. Значит, в 220 квадратиках по одному королю, а в пяти королей нет. В квадрат 9×9 можно поместить 16 квадратиков 2×2 из нашего разбиения, не имеющих общих клеточек. Из них может не быть королей максимум в пяти. Значит, не менее, чем в 11 из них, короли есть.

6. *В выпуклом четырёхугольнике ABCD диагонали равны, $\angle ADB = \angle ACD$, $CD = 3$, $AD = 1$ и $\angle BAD = 150°$. Найдите BC.* (С. Берлов)



Ответ. $BC = 2$. **Первое решение**. На продолжении отрезка DA за точку A отметим такую точку F, что $DF = 3$. Треугольники ACD и BDF равны по двум сторонам и углу между ними ($AC = BD$, $CD = DF$ и $\angle ACD = \angle BDF$). Следовательно, $BF = AD = 1$. В треугольнике ABF $AF = 2$ и $\angle BAF = 180° - \angle BAD = 30°$. Опустим перпендикуляр FG на прямую AB. Из прямоугольного треугольника AFG имеем $FG = AF/2 = 1 = FB$, откуда $G = B$ и $\angle CDA = \angle BFA = 60°$. Таким образом, равнобедренный треугольник CDF является равносторонним, и $\angle BFD = \angle CFD$. Тем самым, точка B лежит на отрезке FC, и потому $BC = FC - FB = 3 - 1 = 2$. **Второе решение**. Отметим точку K на стороне CD так, чтобы $CK = 1$. Тогда треугольники CAK и DBA равны по двум сторонам и углу между ними, следовательно, $150° = \angle BAD = \angle AKC$, $\angle AKD = 30°$. Но $KD = 2AD$, поэтому треугольник AKD — прямоугольный. Кроме того, $AK = AB$, $\angle BAK = 150° - 90° = 60°$, значит треугольник ABK — равносторонний и $\angle BKC = 150° - 60° = 90°$, поэтому треугольники KAD и KBC равны по двум катетам, откуда $BC = KD = 2$.

7. *Натуральные числа a, b, c таковы, что число $(ab+a+1)(bc+b+1)(ca+c+1)$ делится на $abc+1$. Может ли их частное быть квадратом простого числа?* (А. Кузнецов)

Ответ. Не может. **Решение**. Пусть частное равно p^2. Произведение любых двух скобок больше, чем $abc+1$, поэтому две скобки кратны p — пусть это $ab+a+1$ и $bc+b+1$. Тогда $ca+c+1 = (c+1)(ab+a+1) - a(bc+b+1)$ тоже делится на p. Значит, $(ab+a+1)(bc+b+1)(ca+c+1)$ делится на p^3, и потому $abc+1$ делится на p. Но тогда и $2 = (abc+1) - c(ab+a+1) + (ca+c+1)$ делится на p, откуда $p = 2$. Однако в этом случае abc нечетно, следовательно, нечетно и произведение скобок из условия, и потому его частное от деления на $abc+1$ не может равняться 2^2.

8. *На столе стоит 50 гирь попарно различных положительных весов. Может ли случиться, что для любых 24 из этих гирь среди оставшихся гирь можно выбрать несколько такого же суммарного веса?* (С. Берлов)

Ответ. Не может. **Решение**. Пусть гири имеют веса $a_1 > a_2 > a_3 \ldots > a_{50}$. Рассмотрим 24 набора $\{a_1, a_2, \ldots, a_{24}\}$, $\{a_1, a_2, \ldots, a_{23}, a_{25}\}$, ..., $\{a_1, a_2, \ldots, a_{23}, a_{47}\}$ и 22 набора $\{a_1, a_2, \ldots, a_{22}, a_{24}, a_{47}\}$, $\{a_1, a_2, \ldots, a_{22}, a_{25}, a_{47}\}$, ..., $\{a_1, a_2, \ldots, a_{22}, a_{45}, a_{47}\}$ — всего 46 наборов. Для каждого из этих наборов если расположить по убыванию веса наибольшие 24 из оставшихся гирь, то каждая будет весить меньше, чем соответствующая гиря набора. Поэтому для каждого из них нужно выбрать 26 или 25 из оставшихся 26 гирь, чтобы получить набор такого же веса. Заметим, что в рассматриваемых наборах суммарные веса гирь монотонно убывают. Поэтому для уравнивания можно выбрать не более одной группы из 26 гирь и не более 28 групп из 25 гирь, так как из оставшихся гирь можно удалять одну, вес этой удаляемой гири должен монотонно возрастать, и это не может быть гиря с номерами от 1 до 22. Противоречие.